

ANOMALOUS SCALING LIMIT OF A BROWNIAN PARTICLE IN A LOG-CORRELATED POTENTIAL

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ABSTRACT. We study a Brownian particle in $\mathbb{R}^d$, $d \geq 2$, with drift given by the gradient of a log-correlated Gaussian potential. At weak disorder, we prove convergence to a scaling limit whose law is singular with respect to Brownian motion and whose invariant measure is given by Gaussian multiplicative chaos. The proof uses a renormalization group induction: at each scale, the elliptic operator is well approximated by a Laplacian with effective diffusivity decaying as a power of scale. We compute this exponent exactly in dimension two.

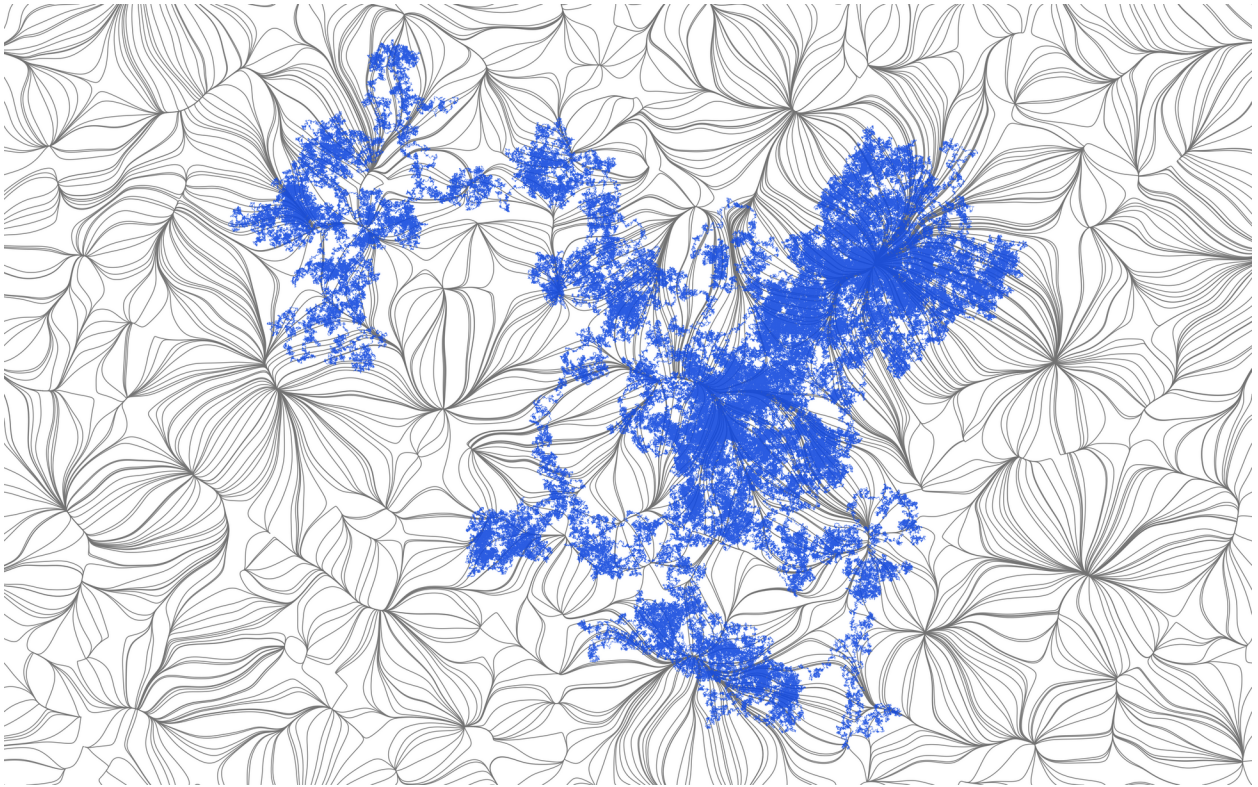


FIGURE 1. A Brownian particle advected by the gradient of a log-correlated random potential.

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1. INTRODUCTION

1.1. The diffusion and its scaling limit. We study the long-time behavior of a Brownian particle in $\mathbb{R}^d$, $d \geq 2$, whose drift is the gradient of a log-correlated Gaussian potential $\log \mathbf{a}$. The particle position X_t is described by the stochastic differential equation

$$(1.1) \quad dX_t = \nabla \log \mathbf{a}(X_t) dt + \sqrt{2} dW_t, \quad X_0 = x \in \mathbb{R}^d,$$

where W is a standard Brownian motion on $\mathbb{R}^d$.

For a bounded stationary potential with sufficiently rapid decay of correlations, homogenization theory gives a Brownian scaling limit with a deterministic positive effective diffusivity. In the model considered here, the potential has independent fluctuations of equal strength at every scale. As more scales are included, their effects accumulate and the effective diffusivity tends to zero.

In this paper, we prove that, at weak disorder, the effective diffusivity decays at least as a power of the length scale. The particle is therefore **subdiffusive**: the time needed to travel a large distance grows faster than the square of that distance. We also prove convergence to a unique scaling limit and show that its path law is singular with respect to Brownian motion.

We construct the potential from independent fields $\mathbf{g}_0, \mathbf{g}_1, \mathbf{g}_2, \dots$ at scales $1, 3, 3^2, \dots$. The field $\mathbf{g}_0$ is smooth, stationary, centered and Gaussian, with finite range of dependence. The field $\mathbf{g}_k$ has the law of $\mathbf{g}_0(3^{-k} \cdot)$: it is a copy of $\mathbf{g}_0$ stretched by the factor 3^k . We denote the size of these fluctuations by the **disorder strength** $\bar{\delta} \in (0, 1/2]$ and set

$$(1.2) \quad \mathbf{a}(x) := \exp \left(\sum_{k=0}^{\infty} (\mathbf{g}_k(x) - \mathbf{g}_k(0)) \right).$$

Subtracting the values at the origin makes the series converge locally in $C^{1,1}$.

The logarithmic correlations come from counting scales. Fields with scale much smaller than $|x-y|$ take independent values at x and y , whereas fields with much larger scale change little between the two points. The number of contributing scales grows logarithmically with the distance. Thus, for $|x-y| \geq 2$, the variance of $\log \mathbf{a}(x) - \log \mathbf{a}(y)$ lies between two positive multiples of $\log |x-y|$, depending only on d and the law of $\mathbf{g}_0$.

There are two sources of randomness. We write $\mathbb{P}$ and $\mathbb{E}$ for the law and expectation of the environment ω , a realization of the fields. The law of the trajectories in a fixed environment is called **quenched**; its average over the environment is called **annealed**.

Our main result holds when $\bar{\delta}$ is smaller than a threshold depending only on d . To state it informally, let $\bar{\mathbf{a}}_N$ be the effective diffusivity obtained by retaining only $\mathbf{g}_0, \dots, \mathbf{g}_N$ and normalizing the coefficient to have mean one; the precise definition is given after (1.7). Let $\tau^2 := \log \mathbb{E}[\exp(\mathbf{g}_0(0))]$. For the particle started at the origin, Theorem A and our estimate of $\log \bar{\mathbf{a}}_N$ give, roughly,

$$(1.3) \quad (3^{-N} X_{3^{2N} t / \bar{\mathbf{a}}_N})_{t \geq 0} \longrightarrow (Z_t)_{t \geq 0} \quad \text{with} \quad \bar{\mathbf{a}}_N \approx e^{-2\tau^2(N+1)/d} \quad \text{as } N \rightarrow \infty,$$

where the approximation describes the logarithm of $\bar{\mathbf{a}}_N$; its error is given in (1.9). The limit Z is a continuous, conservative, strong Markov diffusion.

The assumptions below also allow non-Gaussian fields. Fix $d \geq 2$ and $\bar{\delta} \in (0, 1/2]$, and let $\mathbf{g}_0, \mathbf{g}_1, \mathbf{g}_2, \dots$ be independent random fields such that, for every $k \in \mathbb{N}_0$,

$$(1.4) \quad \mathbf{g}_k \quad \text{and} \quad \mathbf{g}_0(3^{-k} \cdot) \quad \text{have the same law.}$$

For $m \in \mathbb{Z}$, let $\square_m := (-\frac{1}{2}3^m, \frac{1}{2}3^m)^d$ be the cube of side 3^m ; in particular, $\square_0 = (-1/2, 1/2)^d$. We use the maximum norm for spatial cubes, balls and their geometry. In (g2), the gradient has the dual norm, and the Hessian norm is the Lipschitz constant of the derivative on $\square_0$ with respect to these norms.

Quadratic energies and variational formulas use the Euclidean pairing $p \cdot q = \sum_{i=1}^d p_i q_i$ and magnitude $|p|_2 = (p \cdot p)^{1/2}$, including for gradients and fluxes; unit vectors in these formulas are Euclidean unit vectors. Matrix magnitudes in the coarse-graining formulas are Euclidean operator norms. The range-of-dependence distance in (g1) and the fractional Sobolev and Hölder conventions below are also Euclidean. We assume the following.

- (g1) The field $\mathbf{g}_0$ is stationary, has mean zero, and has range of dependence $\sqrt{d}$, measured in Euclidean distance: the sigma-fields generated by its restrictions to measurable sets $U, V \subseteq \mathbb{R}^d$ are independent whenever $|x - y|_2 \geq \sqrt{d}$ for every $x \in U$ and $y \in V$.
- (g2) Almost surely, $\mathbf{g}_0 \in C_{\text{loc}}^{1,1}(\mathbb{R}^d)$, and

$$(1.5) \quad \mathbb{E} \left[\exp \left(\bar{\delta}^{-2} (\|\mathbf{g}_0\|_{L^\infty(\square_0)} + \|\nabla \mathbf{g}_0\|_{L^\infty(\square_0)} + \|\nabla^2 \mathbf{g}_0\|_{L^\infty(\square_0)})^2 \right) \right] \leq 2.$$

- (g3) The law of $\mathbf{g}_0$ is invariant under signed permutations of the coordinates and under $\mathbf{g}_0 \mapsto -\mathbf{g}_0$.

- (g4) The number $\tau^2 = \log \mathbb{E}[\exp(\mathbf{g}_0(0))]$ is positive.

Assumption (g2) says that the suprema of $|\mathbf{g}_0|$, $|\nabla \mathbf{g}_0|$ and $|\nabla^2 \mathbf{g}_0|$ on a unit cube are of order $\bar{\delta}$, with Gaussian tails. Together with the symmetry $\mathbf{g}_0 \mapsto -\mathbf{g}_0$ in (g3), it gives $C(d) < \infty$ such that

$$(1.6) \quad \tau^2 \leq C \bar{\delta}^2,$$

and $|\tau^2 - \frac{1}{2} \mathbb{E}[|\mathbf{g}_0(0)|^2]| \leq C \bar{\delta}^4$. We say that a result holds **at weak disorder** if it holds whenever $\bar{\delta} \leq \bar{\delta}_0$, for some $\bar{\delta}_0 > 0$ depending only on d and the indicated parameters.

Throughout, X denotes the continuous Feller diffusion with generator $\mathbf{a}^{-1} \nabla \cdot (\mathbf{a} \nabla)$. We construct it from its resolvent in Section 9.1 and show that, from every starting point, its law is that of the solution of (1.1). Its reversible measure is $\mathbf{a}(x) dx$.

For $L \in \mathbb{N}_0$, we introduce an **infrared cutoff** by retaining the fields with scales at most 3^L . The number 3^L is the **cutoff scale**, and the corresponding coefficient is

$$(1.7) \quad \mathbf{a}_L(x) := \exp \left(\sum_{k=0}^L (\mathbf{g}_k(x) - \tau^2) \right).$$

This coefficient is stationary, has mean one and has range of dependence of order 3^L . Subtracting τ^2 normalizes the mean. As in (1.2), subtracting constants from the fields leaves the drift unchanged. Let $X^{(L)}$ be the diffusion constructed with $\mathbf{a}_L$ in place of $\mathbf{a}$; its generator is

$$(1.8) \quad \mathbf{a}_L^{-1} \nabla \cdot (\mathbf{a}_L \nabla) = \Delta + \nabla \log \mathbf{a}_L \cdot \nabla.$$

The homogenized matrix of $\mathbf{a}_L$ describes the equation on domains much larger than 3^L : replacing $\mathbf{a}_L$ by this constant matrix gives nearly the same solutions. By (g3), the matrix is a scalar multiple of the identity. We denote the scalar by $\bar{\mathbf{a}}_L$ and call it the **effective diffusivity**.

At weak disorder, this diffusivity also describes motion on the cutoff scale itself. The annealed mean time for $X^{(L)}$ to leave a cube of side 3^L , when started at its center, is of order $3^{2L}/\bar{\mathbf{a}}_L$. The same is true of X without cutoff (Lemma 10.2). This explains the time scale $3^{2N}/\bar{\mathbf{a}}_N$ in (1.3).

Theorem 5.1 shows that $\bar{\mathbf{a}}_m$ is nonincreasing, is bounded below by $e^{-(m+1)\tau^2}$, and is bounded above by a negative power of 3^{m+1} whose exponent depends only on d and the law of $\mathbf{g}_0$. At weak disorder we identify the leading decay rate: there exist $\bar{\delta}_0(d) > 0$ and $C(d) < \infty$ such that, if $\bar{\delta} \leq \bar{\delta}_0$,

then, for every $m \in \mathbb{N}_0$,

$$(1.9) \quad \left| \log \bar{\mathbf{a}}_m + \frac{2\tau^2(m+1)}{d} \right| \leq C\bar{\delta}^2 |\log \bar{\delta}| (1 + m\bar{\delta}).$$

In dimension two, planar duality gives the exact formula $\bar{\mathbf{a}}_m = e^{-(m+1)\tau^2}$ for every $m \in \mathbb{N}_0$. Rotation by a right angle exchanges gradients and divergence-free fields, so the homogenized matrices of $\mathbf{a}_m$ and $\mathbf{a}_m^{-1}$ are inverse to each other. The symmetry $\mathbf{g}_0 \mapsto -\mathbf{g}_0$ gives $\mathbf{a}_m^{-1}$ the law of $e^{2(m+1)\tau^2} \mathbf{a}_m$, which determines $\bar{\mathbf{a}}_m$. The time scale in (1.3) is then exactly $3^{2N} e^{(N+1)\tau^2}$.

When $m\bar{\delta} \geq 1$ and τ^2 is at least a fixed positive multiple of $\bar{\delta}^2$, the error in (1.9) is smaller than the leading term by a factor of order $|\bar{\delta}| \log \bar{\delta}|$.

Theorem A (Scaling limit). *Assume (g1)–(g4). There exists $\bar{\delta}_0(d) > 0$ such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then there is a continuous strong Markov process Z on $\mathbb{R}^d$ in a random environment, with quenched laws $\mathbf{P}_x$ and expectations $\mathbf{E}_x$ when started at x , such that the following statements hold.*

- (i) Convergence: For every $x \in \mathbb{R}^d$, as the integer N tends to infinity, the annealed law of the process $(3^{-N} X_{3^{2N}t/\bar{\mathbf{a}}_N})_{t \geq 0}$ with $X_0 = 3^N x$ converges to the annealed law of Z started at x .
- (ii) Reversible measure: Almost surely, Z is reversible with respect to a measure μ that is singular with respect to Lebesgue measure.
- (iii) Anomalous scaling: There exist $C < \infty$ and $\eta > 0$, depending only on d and the law of $\mathbf{g}_0$, such that, if σ_r denotes the first time at which Z leaves the cube of side r centered at its starting point, then, for every $x \in \mathbb{R}^d$ and $r \in (0, 1]$,

$$\mathbf{E}[\mathbf{E}_x[\sigma_r]] \leq Cr^{2+\eta}.$$

For every $x \in \mathbb{R}^d$, almost surely, $\mathbf{P}_x$ -almost every path satisfies

$$\limsup_{t \downarrow 0} t^{-\gamma} |Z_t - Z_0| = \infty \quad \text{for every } \gamma > \frac{1}{2+\eta}.$$

In dimension two, one can take $\eta = \tau^2 / \log 3$.

- (iv) Non-Gaussian marginals: Almost surely, for every $x \in \mathbb{R}^d$ and $t > 0$, the law of Z_t under $\mathbf{P}_x$ is absolutely continuous with respect to μ and is not Gaussian.

Theorem 7.1 gives the stronger, quenched version of (i). We realize the rescaled processes in one environment, and their quenched laws converge almost surely. Rescaling space by 3^{-N} moves the field at scale 3^k to scale 3^{k-N} . The common environment therefore contains independent fields at every scale 3^j , with $j \in \mathbb{Z}$.

The reversible measures of these rescaled processes, suitably normalized, converge almost surely to μ . For Gaussian fields, this is the usual construction of Gaussian multiplicative chaos: exponentiate the smoothed field, divide by its mean, and let the smoothing scale tend to zero. More precisely, μ is a stationary chaos measure multiplied by a positive continuous factor from the fields at scales larger than one.

The slow motion of X over large distances becomes rapid motion of Z over small distances. Brownian motion needs time of order r^2 to leave a cube of side r , whereas (iii) bounds the annealed mean exit time of Z by $Cr^{2+\eta}$. These rapid exits force its pointwise Hölder exponent at time zero to be at most $1/(2+\eta)$, below the Brownian value $1/2$.

By the law of the iterated logarithm, the quenched path law $\mathbf{P}_x$ is therefore singular with respect to every mixture of laws of Brownian motions. The same roughness bound also gives singularity of the annealed path law. For Gaussian $\mathbf{g}_0$, we have $\tau^2 = \frac{1}{2} \text{Var}[\mathbf{g}_0(0)]$; in dimension two the bound on the Hölder exponent is thus $(2 + \text{Var}[\mathbf{g}_0(0)]/(2 \log 3))^{-1}$.

1.2. Quantitative homogenization. We study the particle through its elliptic operator. After dividing the cutoff coefficient by its effective diffusivity, we compare the corresponding Dirichlet problem with the one for the Laplacian. At weak disorder, the error is small uniformly in the cutoff, even on a domain of the same size as the cutoff scale. For $0 \leq L \leq M$, define $A_{L,M}(x) := \bar{\mathbf{a}}_L^{-1} \mathbf{a}_L(3^M x)$ for $x \in \square_0$. The coefficient $A_{L,M}$ is stationary, has homogenized matrix equal to the identity, and has range of dependence of order 3^{L-M} . Given $f \in L^2(\square_0)$ and $h \in H^2(\square_0)$, let $u_{L,M}, u_{\text{hom}} \in H^1(\square_0)$ solve

$$\begin{cases} -\nabla \cdot (A_{L,M} \nabla u_{L,M}) = f & \text{in } \square_0, \\ u_{L,M} = h & \text{on } \partial \square_0, \end{cases} \quad \begin{cases} -\Delta u_{\text{hom}} = f & \text{in } \square_0, \\ u_{\text{hom}} = h & \text{on } \partial \square_0, \end{cases}$$

and measure the errors in the solution, its gradient and its flux by the quantity below. Here H^{-1} denotes the dual of H_0^1 , with the volume-normalized, componentwise zero-boundary norm of (2.2); since $|\square_0| = 1$, we omit the underline:

$$\mathcal{D}_{L,M} := \|u_{L,M} - u_{\text{hom}}\|_{L^2(\square_0)} + \|\nabla u_{L,M} - \nabla u_{\text{hom}}\|_{H^{-1}(\square_0)} + \|A_{L,M} \nabla u_{L,M} - \nabla u_{\text{hom}}\|_{H^{-1}(\square_0)}.$$

We suppress the dependence of $u_{L,M}$ and $\mathcal{D}_{L,M}$ on f and h . The first estimate below is uniform in the cutoff: for fixed $\vartheta \in (0, 1)$ and $q \in [1, \infty)$, at weak disorder, $\mathcal{D}_{L,M}$ is at most $\bar{\delta}^\vartheta$ times the norms of the data and a random factor whose $L^q(\mathbb{P})$ norm is bounded uniformly in L and M . The second gives, up to a random factor, the algebraic rate $3^{-\alpha_{\text{hom}}(M-L)}$, for some $\alpha_{\text{hom}}(d) > 0$, when the cutoff is fixed and the domain grows.

Theorem B (Quantitative homogenization of the Dirichlet problem). *Assume (g1)–(g4). The negative norms in $\mathcal{D}_{L,M}$ are the duals of H_0^1 , with the componentwise zero-boundary convention of (2.2). The following statements hold.*

- (i) Estimate uniform in the cutoff: *For every $\vartheta \in (0, 1)$ and $q \in [1, \infty)$, there exist $\bar{\delta}_0(\vartheta, q, d) > 0$ and $C(\vartheta, q, d) < \infty$ such that, if $\bar{\delta} \leq \bar{\delta}_0$, then there exist random variables $\mathcal{Z}_{L,M} \geq 1$ satisfying*

$$\sup_{0 \leq L \leq M} \|\mathcal{Z}_{L,M}\|_{L^q(\mathbb{P})} \leq C$$

such that, almost surely, for every L, M, f and h ,

$$\mathcal{D}_{L,M} \leq \mathcal{Z}_{L,M} \bar{\delta}^\vartheta (\|f\|_{L^2(\square_0)} + \|h\|_{H^2(\square_0)}).$$

If $f = 0$, then the factor $\bar{\delta}^\vartheta$ may be replaced by $C\bar{\delta}$.

- (ii) Algebraic rate at a fixed cutoff: *There exists $\alpha_{\text{hom}}(d) > 0$ such that, for every $L \in \mathbb{N}_0$ and $q \in [1, \infty)$, there exists a random variable $\mathcal{Y}_{L,q} \geq 1$ with $\|\mathcal{Y}_{L,q}\|_{L^q(\mathbb{P})} \leq C(L, q, d, \bar{\delta})$ such that, almost surely, for every $M \geq L$ and every f and h ,*

$$\mathcal{D}_{L,M} \leq \mathcal{Y}_{L,q} 3^{-\alpha_{\text{hom}}(M-L)} (\|f\|_{L^2(\square_0)} + \|h\|_{H^2(\square_0)}).$$

The estimate in (i) includes $M = L$, when the range of dependence is comparable to the size of the domain. Its small parameter is the disorder strength, and its constants are uniform in the cutoff. In (ii), the cutoff is fixed and enlarging the domain gives an algebraic rate; weak disorder is not required. These estimates are proved in Theorems 6.19 and 6.20 of Section 6.

There is a probabilistic interpretation. For smooth data, $u_{L,M}$ describes the diffusion with generator $\nabla \cdot (A_{L,M} \nabla)$, a time change of $X^{(L)}$ with space rescaled by 3^{-M} . Taking $f = 1$ and $h = 0$ gives its expected exit time from $\square_0$; taking $f = 0$ gives the expected value of h at the exit position. Theorem B compares these quantities with their Brownian counterparts in $L^2(\square_0)$.

For the original particle, the generator also contains the factor $\mathbf{a}^{-1}$. Its law is determined by the energy $\int \mathbf{a} |\nabla u|^2 dx$ together with the reversible measure $\mathbf{a} dx$. Constructing the scaling limit requires convergence of both. The bound in Theorem B(i) remains small but does not tend to zero with

the cutoff, so it does not establish convergence of the energies. The proof of Theorem A proceeds through the regularity and uniqueness arguments described below; it does not use Theorem B.

1.3. Large-scale regularity and a Liouville theorem. The next result controls the oscillation of $\mathbf{a}_L$ -harmonic functions, that is, solutions of $\nabla \cdot (\mathbf{a}_L \nabla u) = 0$. We obtain Hölder estimates with exponents up to $1 - C_0 \bar{\delta} |\log \bar{\delta}|^{1/2}$, where C_0 depends only on d . The estimates are uniform in the cutoff, and the exponents approach one as $\bar{\delta} \rightarrow 0$. The largest of these exponents also gives a Liouville theorem: almost surely, for every cutoff L , an $\mathbf{a}_L$ -harmonic function on $\mathbb{R}^d$ that grows more slowly than $|x|^{1-C_0 \bar{\delta} |\log \bar{\delta}|^{1/2}}$ is constant.

For each exponent, a random number of scales must separate the cube where the function is harmonic from the subcubes where it is estimated. This number has an exponential tail, uniformly in L . We use the notation $(u)_U := \oint_U u$ and $\|u\|_{\underline{L}^p(U)} := (\oint_U |u|^p)^{1/p}$ for a measurable set U of finite positive volume and $p \in [1, \infty)$.

Theorem C (Large-scale Hölder regularity and a Liouville theorem). *Assume (g1)–(g4). There exist $\bar{\delta}_0(d) > 0$, $C_0(d) < \infty$ and $C(d) < \infty$ such that, if $\bar{\delta} \leq \bar{\delta}_0$, then the following statements hold with*

$$\gamma_{\text{reg}} := 1 - C_0 \bar{\delta} |\log \bar{\delta}|^{1/2}.$$

In both statements, L ranges over $\mathbb{N}_0 \cup \{\infty\}$, and $\mathbf{a}_L$ is read as $\mathbf{a}$ when $L = \infty$.

(i) Large-scale Hölder estimate: *Let $\gamma \in [1/2, \gamma_{\text{reg}}]$. For every L and $m \in \mathbb{N}$, there exists an $\mathbb{N}$ -valued random variable $\mathcal{L}_L(\gamma, m)$ such that, for every $k \in \mathbb{N}$,*

$$(1.10) \quad \mathbb{P}[\mathcal{L}_L(\gamma, m) > k] \leq C \exp\left(-\frac{(1-\gamma)^2(k-C)_+}{C \bar{\delta}^2 |\log \bar{\delta}|}\right).$$

Moreover, almost surely, for every L and m , if $u \in H^1(\square_m)$ is $\mathbf{a}_L$ -harmonic in $\square_m$, then, for every $n \in \mathbb{N}_0$ satisfying $n \leq m - \mathcal{L}_L(\gamma, m)$ and every $z \in 3^n \mathbb{Z}^d$ with $z + \square_n \subseteq \square_{m-1}$,

$$(1.11) \quad \|u - (u)_{z+\square_n}\|_{\underline{L}^2(z+\square_n)} \leq C 3^{-\gamma(m-n)} \|u - (u)_{\square_m}\|_{\underline{L}^2(\square_m)}$$

and

$$(1.12) \quad \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(z+\square_n)} \leq C 3^{(1-\gamma)(m-n)} \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(\square_m)}.$$

(ii) Liouville theorem: *Almost surely, for every L , if $u \in H_{\text{loc}}^1(\mathbb{R}^d)$ satisfies*

$$-\nabla \cdot (\mathbf{a}_L \nabla u) = 0 \quad \text{in } \mathbb{R}^d \quad \text{and} \quad \liminf_{R \rightarrow \infty} R^{-\gamma_{\text{reg}}} \inf_{c \in \mathbb{R}} \|u - c\|_{\underline{L}^2(B_R)} = 0,$$

then u is constant.

The energy estimate prevents solutions from concentrating their energy on too few small cubes. If the energy in $\square_m$ is positive, then a family of cubes $z + \square_n$, with n and z as in (1.12), carrying a fixed positive proportion of that energy must contain at least a positive multiple of $3^{(d-2(1-\gamma))(m-n)}$ cubes. The multiple depends on d and on the prescribed proportion. For $\gamma = \gamma_{\text{reg}}$, the exponent approaches d as $\bar{\delta} \rightarrow 0$.

This control of energy on small cubes is used both to obtain subsequential limits and to prove that they coincide. Theorem C is proved in Section 6, as Theorem 6.22.

1.4. Comparison with previous work. Diffusion in random force fields was studied in the physics literature by Fisher [1984], Kravtsov, Lerner, and Yudson [1985], Fisher, Friedan, Qiu, Shenker, and Shenker [1985] and Bouchaud, Comtet, Georges, and Le Doussal [1987]; see Bouchaud and Georges [1990] for a review. For gradient drifts with log-correlated Gaussian potentials, renormalization group computations predicted subdiffusion with an exponent depending on the disorder strength [Bouchaud and Georges, 1990, Section 4.3.2.3].

In our normalization, the Gaussian potential has increment variance asymptotic to $4\tau^2 \log|x - y|/\log 3$ as $|x - y| \rightarrow \infty$. The weak-disorder formula of Castillo and Le Doussal [2001, equation (3)] therefore predicts a time of order $r^{2+2\tau^2/(d \log 3)}$ to travel distance r . Estimate (1.9) gives this leading exponent for the time scale $3^{2N}/\bar{\mathbf{a}}_N$. In dimension two, the identity $\bar{\mathbf{a}}_N = e^{-(N+1)\tau^2}$ gives exactly the predicted exponent $2 + \tau^2/\log 3$.

The planar lattice analogue is a random walk whose edge conductances are exponentials of the discrete Gaussian free field. For this model, Biskup, Ding, and Goswami [2020] proved recurrence, return-probability estimates and exponents for expected exit times. At small disorder, Ding and Wang [2026] proved tightness of the random quenched path laws for a version using a smooth approximation to the field.

In these walks, the probability of choosing a neighbor is proportional to the conductance of the joining edge, and the reversible measure at a vertex is the sum of the incident conductances. Thus the field affects both the transition probabilities and the reversible measure, as it does in (1.1).

For the walk of Biskup et al. [2020], Biskup [2020, Problem 16.13] formulated the scaling-limit problem through a continuum diffusion driven by the gradient of a smoothed planar Gaussian free field. The question asks for joint convergence of the field and the diffusion, after a suitable change of time, with continuous, nonconstant limiting paths. We prove existence and uniqueness of the scaling limit for the continuum model (1.1) under (g1)–(g4), at weak disorder and in every dimension $d \geq 2$. Identifying it with a scaling limit of either lattice model remains a separate question.

For Gaussian fields, the limiting reversible measure is obtained from Gaussian multiplicative chaos, introduced by Kahane [1985]. We use standard facts from this theory; see Rhodes and Vargas [2014] and Berestycki [2017] for accounts of the construction, and Duplantier, Rhodes, Sheffield, and Vargas [2017] for log-correlated fields.

Liouville Brownian motion, constructed by Garban, Rhodes, and Vargas [2016] and Berestycki [2015], uses the chaos measure of the planar Gaussian free field to change the time parametrization of Brownian motion. Its Dirichlet form is the unweighted energy $\int |\nabla u|^2 dx$ on the L^2 space of that measure. For a smooth positive density $\mathbf{a}$, its generator is $\mathbf{a}^{-1}\Delta$.

Our diffusion has energy $\int \mathbf{a}|\nabla u|^2 dx$ on $L^2(\mathbf{a} dx)$ and generator $\mathbf{a}^{-1}\nabla \cdot (\mathbf{a}\nabla)$. The field enters the energy as well as the reversible measure. Identifying the limit of these weighted energies is the main step in constructing our limiting process.

The renormalization argument is closely related to Armstrong, Bou-Rabee, and Kuusi [2026], who proved power-law superdiffusion in a self-similar incompressible random drift with sufficiently small positive Hurst exponent. Both papers follow an induction across scales for the elliptic operator, using variationally defined coarse-grained matrices in place of pointwise ellipticity bounds. In the incompressible model the effective diffusivity increases with scale; here it decreases. The present paper also constructs a scaling limit.

1.5. Overview of the proofs. We next outline the proof. The argument has three main parts; the central step is the identification of a unique limiting energy.

- (1) The coarse-graining estimates and the induction across scales give control of the effective diffusivities and large-scale regularity (Sections 2–6).

- (2) We use this regularity to obtain subsequential limits of the energies, compare any two such limits, and show that they coincide (Section 8).
- (3) The limiting energy and reversible measure determine the diffusion. We prove convergence of the path laws and then derive the properties of the limit (Sections 9–10).

The dependence of the main results is shown in Figure 1.1. The appendices collect moment bounds, concentration inequalities and comparisons of Besov norms.

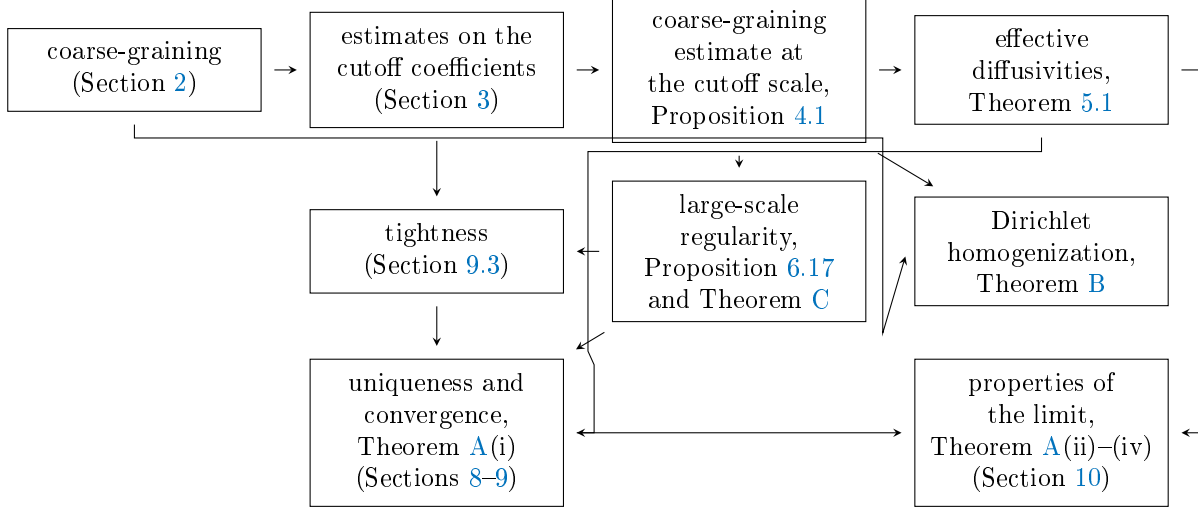


FIGURE 1.1. Logical dependence of the main results. An arrow means that the first result is used in the proof of the second.

1.5.1. *Coarse-grained matrices.* Pointwise ellipticity deteriorates as the cutoff grows. Following [Armstrong and Kuusi \[2025\]](#) and [Armstrong et al. \[2026\]](#), we work with the two **coarse-grained matrices** of a cube. One records the minimal energy for affine boundary values. The inverse of the other records the energy of the Neumann solution with the boundary flux of a constant vector field. Both equal the coefficient when it is constant.

Their difference measures the error in approximating the equation by a constant-coefficient equation. The matrices on subcubes also control Poincaré and Caccioppoli inequalities and harmonic approximation. Section 2 recalls this deterministic theory. Normalizing the matrices by the effective diffusivity gives estimates that remain useful as that diffusivity tends to zero.

1.5.2. *Induction across scales and the effective diffusivities.* We prove that both coarse-grained matrices of $\mathbf{a}_m$ on $\square_m$ stay close to $\bar{\mathbf{a}}_m \mathbf{I}_d$. More precisely, their relative error is of order $\bar{\delta}$ in $L^p(\mathbb{P})$, uniformly in m , for each fixed $p < \infty$ and $\bar{\delta}$ sufficiently small depending on p (Proposition 4.1).

The induction has two operations. First, we keep the cutoff L fixed and pass from cubes of side 3^L to a larger cube $\square_m$. The restrictions of $\mathbf{a}_L$ to well-separated subcubes are independent, so averaging reduces the gap between the coarse-grained matrices. As in [Armstrong, Kuusi, and Mourrat \[2019\]](#), the variational estimate has two terms: spatial averages of the gradient of an optimizer, controlled using independence, and the decrease of the expected gap between scales. These decreases have a bounded sum. The estimates of [Armstrong and Kuusi \[2025\]](#) control both terms using coarse-grained ellipticity.

Second, we add $\mathbf{g}_{L+1}, \dots, \mathbf{g}_m$. These fields need not be nearly constant on $\square_m$, but their effect is a small relative perturbation at weak disorder. Fields much coarser than a subcube are nearly constant there and multiply both coarse-grained matrices by nearly the same factor. We choose $m - L$ so

that the contraction from averaging absorbs the error from the added fields. Independence and the scaling law (1.4) then let us repeat the argument with the same constants.

The same induction lets us follow the effective diffusivity. To first order, adding a block of fields changes the variational optimizer by the solution of a Poisson equation forced by those fields. Its energy gives the leading decrease $2\tau^2/d$ per scale in the logarithm of the diffusivity. Summing over blocks yields (1.9); planar duality gives the exact formula in dimension two.

Power-law decay requires a separate argument, since the perturbative error need not be smaller than the leading term for every law satisfying (g1)–(g4). Proposition 5.6 proves this decay without weak disorder. Since $\tau^2 > 0$, one field lowers the expected minimal energy with affine boundary values below the energy of the affine function by a fixed factor smaller than one. Such factors multiply over scales separated by a fixed gap.

1.5.3. Large-scale regularity and homogenization. The regularity argument compares solutions with harmonic functions. When the coarse-grained matrices are close to a constant matrix, a solution is close to a harmonic function and hence to an affine function on a smaller cube. Its error of affine approximation decreases across such a scale. On the remaining scales the error can increase, but only by a bounded factor.

The difficulty is that approximation errors on different scales depend on the same fields. We define a scale to be **good** by bounds on the fields and on the variational energies, with tolerances growing geometrically with the distance from that scale. Independence then shows that only a small proportion of scales fail these bounds, outside an event of exponentially small probability. The improvement on good scales absorbs the losses on the others. This gives both the Hölder exponent in Theorem C and the tail estimate (1.10).

Above the cutoff scale, no new fields enter, and averaging continues to improve the approximation. The estimates are therefore uniform in the cutoff. Combined with Caccioppoli's inequality, (1.11) shows that small cubes, and in particular thin strips along their faces, carry little energy. We use this repeatedly in the compactness and uniqueness arguments. Theorem B follows directly from the coarse-graining estimates on the whole domain.

1.5.4. Uniqueness of the limit form. For the rescaled process, write μ_N for the reversible measure and $\int A_N |\nabla u|^2 dx$ for its Dirichlet form on $L^2(\mu_N)$. The coefficient A_N is the density of μ_N divided by $\bar{\mathbf{a}}_N$. Apart from a positive factor from the fields at scales larger than one, independent of N , it has the law of $\bar{\mathbf{a}}_N^{-1} \mathbf{a}_N(3^N \cdot)$. Thus N is the cutoff in these coordinates. The energy measure distributes a function's energy over space; before taking the limit it is $A_N |\nabla u|^2 dx$.

The measures μ_N converge by the multiplicative chaos construction of Kahane [1985]. After removing the factor from the fields at scales larger than one, their masses on a fixed cube form a nonnegative martingale: each new field multiplies the density by an independent factor of mean one. At weak disorder these masses have a bounded moment of order greater than one, so no mass is lost. We must identify the limit of the energies.

Compactness comes from resampling. A field at a much smaller scale than a cube has little effect on its minimal energy. We divide the cube into boxes and remove this field from thin strips along their faces. The remaining parts of the boxes are farther apart than its range of dependence. Their restrictions of the field are independent, and the Efron–Stein inequality bounds the fluctuations of the modified energy. Regularity controls both this bound and the energy discarded in the strips.

Very coarse fields are nearly constant on the cube. Thus each minimal energy is close, uniformly in the cutoff, to its conditional expectation given finitely many fields. These conditional expectations form a compact family in $L^p(\mathbb{P})$ (Section 8.3). Limits of the Dirichlet Green operators then define random Dirichlet forms.

Let E and F be two subsequential limit forms. We first prove that they bound each other and have the same domain (Theorem 8.42). After replacing fields coarser than a small cube by their values at its center, rescaling the cube multiplies both forms by the same random factor. Their deterministic normalizations can still differ. If their ratio were too small on a positive proportion of scales, harmonic replacement on cubes of those scales would make one energy too small, contradicting the positive and negative moment bounds. Interchanging the forms excludes large ratios. This adapts Ding and Gwynne [2023a, Section 3.4].

We next consider the optimal lower and upper comparison constants. Resampling one field multiplies both energy measures by the same weight and leaves the constants unchanged. Independence of the fields makes these constants deterministic.

If the constants differ, we improve one of them. On a small cube with affine boundary values, the minimal F -energy is close to a deterministic multiple of the minimal E -energy. This multiple lies between the constants and may depend on the scale. On a positive proportion of scales it lies on the same side of their midpoint.

Suppose it lies below the midpoint. On those scales, the minimal F -energy for affine boundary values falls below the upper bound by a fixed fraction of the gap between the constants. The upper constant times the minimal E -energy minus the minimal F -energy is a nonnegative quadratic form in the boundary values, bounded by the gap times the minimal E -energy. Consequently, the same improvement holds, with a controlled error, for boundary values close to affine.

Replacing an E -harmonic function by F -harmonic functions on suitable cubes and using lower semicontinuity improves the upper comparison constant. This contradicts its optimality. If the multiple lies above the midpoint, we interchange the forms. Thus the forms are proportional (Theorem 8.52).

Two estimates make this argument work. Every error must be small relative to the gap between the comparison constants. We therefore apply Efron–Stein directly to a difference of the two minimal energies. Lemma 8.44 bounds the energy of the difference of the minimizers by a multiple of the squared gap times the minimal E -energy. This gives the required concentration estimate (Proposition 8.46).

The replacement cubes must also carry a fixed fraction of the solution’s energy. The energy and the events on which regularity holds depend on the same fields. We control this dependence by bounding the number of failed scales along every chain of nested cubes, on one event valid for all solutions. Integrating against a solution’s energy measure then supplies enough cubes for the replacements. The estimates required on each cube are collected in Lemma 8.34.

Finally, the proportionality factor is fixed by homogenization. Once the factor from the fields at scales larger than one is removed, A_N is stationary with homogenized matrix equal to the identity. Its coarse-grained matrices approach the identity on large cubes uniformly in N (Theorem 8.53). Both limits inherit this normalization, so the proportionality factor is one (Theorem 8.54).

The comparison of optimal constants follows Gwynne and Miller [2021, Theorem 1.9], who proved uniqueness up to a deterministic factor for weak subcritical Liouville quantum gravity metrics. In their argument, a shortest path corresponds to our minimizer, path length to energy, and replacement of a path segment to harmonic replacement on a cube.

They controlled local events along a shortest path through confluence of geodesics [Gwynne and Miller, 2021, Section 4]: geodesics from a common point share an initial segment. For critical and supercritical metrics, Ding and Gwynne [2023b, Section 4.2] used modifications of the field in annuli. Our estimate along chains of cubes serves this purpose for energy measures and works in every dimension $d \geq 2$. We also prove the properties needed for the comparison: locality, consistency under restriction to subcubes, and compatibility with multiplication of the coefficient by a positive continuous function.

1.5.5. Construction of the limiting diffusion. The unique limit form and the limiting reversible measure μ determine the limiting diffusion. To prove convergence, we begin with resolvents of the processes killed on leaving a cube. Although the densities of μ_N have no uniform bound, their masses on balls are bounded by a random constant times a power of the radius, with exponent approaching d from below as the disorder strength decreases. These estimates are uniform in N and control integrals against μ_N by energy.

We can therefore pass to the limit in the variational problems for the killed resolvents. Smoothing their measure-valued sources and applying the regularity estimates gives convergence uniformly in the starting point.

Tightness follows from bounds on the probability of leaving a small cube in a short time. We obtain them from a resolvent estimate for the Laplace transform of the exit time and a local maximum principle with respect to the reversible measure. The latter follows from Moser iteration (Section 9.3). The same estimates confine paths to a large cube over bounded time intervals with high probability.

This confinement lets us pass from killed to whole-space resolvents. Iterating the resolvents determines the joint laws at finitely many times. Together with tightness, this gives convergence in probability on continuous path space, locally uniformly in the starting point.

For almost-sure convergence, we compare two finite cutoffs. We adapt the comparison argument of Ding and Gwynne [2023a], in the form strengthened by Devlin [2024, Lemma 3.11]. If the ratio of the deterministic energy normalizations stayed below one by a fixed amount on a positive proportion of scales, harmonic replacements on disjoint cubes would force one energy below its positive limit. This contradicts convergence in probability.

Excluding these ratios, and interchanging the cutoffs, gives comparisons with summable error probabilities. Borel–Cantelli yields almost-sure convergence of the energies. Bounds uniform in the cutoff then transfer it to the killed resolvents and the path laws.

The properties of the limiting paths follow from power-law decay of the effective diffusivities between arbitrary scales. This gives rapid exit from small cubes and hence singularity with respect to Brownian motion, both quenched and annealed.

Singularity of μ follows from the usual change of measure in multiplicative chaos. Weighting the joint law of the field and a point by the stationary chaos measure exponentially tilts the fields at that point. The strong law shows that the approximating densities diverge at μ -almost every point and vanish at Lebesgue-almost every point. Finally, the transition laws are absolutely continuous with respect to μ . Since μ is singular and charges no hyperplane, these laws cannot be Gaussian.

Use of artificial intelligence. The authors developed all the main ideas of this paper in 2024. Versions of GPT were used to check the arguments, and Claude was used to help with the writing. The authors take responsibility for the mathematical content and presentation of the paper.

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2. COARSE-GRAINING THEORY

In this section we recall a deterministic coarse-graining theory for uniformly elliptic coefficient fields $\mathbf{a} : \mathbb{R}^d \rightarrow \mathbb{R}_{\text{sym},+}^{d \times d}$. The two coarse-grained matrices of a bounded domain give two-sided bounds on the energy of solutions in that domain, and their difference measures how far the equation is from being homogenized. The main argument of this paper tracks this difference from one scale to the next.

Throughout this section, and in contrast to those that follow, no probabilistic structure is used: the coefficient field is arbitrary and fixed. We define the matrices and record their subadditivity under partitions of a cube, show that solutions satisfy Poincaré and Caccioppoli inequalities with constants given by the coarse-grained matrices, estimate how the matrices change under a perturbation of the coefficient field, and combine these into the general coarse-graining estimate used later. The definitions and estimates are from [Armstrong and Kuusi \[2025\]](#), arXiv version 3, and [Armstrong et al. \[2026\]](#), arXiv version 2. We indicate below where a result is adapted to the present setting.

2.1. Notation. For a matrix A , the symbol $|A|$ denotes its Euclidean operator norm, $|A| = \sup_{p \neq 0} |Ap|_2 / |p|_2$. In quadratic and variational formulas we write $|p|$ for $|p|_2$. We identify a positive scalar coefficient with the corresponding scalar multiple of $\mathbf{I}_d$. For the coarse-grained matrices $\mathbf{a}(U)$ and $\mathbf{a}_*(U)$ defined in (2.11), superscripts denote matrix powers; for example, $\mathbf{a}_*^{-1}(U) = (\mathbf{a}_*(U))^{-1}$. A coefficient field used to construct a coarse-grained matrix is written after a semicolon, as in $\mathbf{a}(U; \mathbf{a}^{-1})$.

For a permutation π of $\{1, \dots, d\}$, $n \in \mathbb{Z}$, and $z \in \mathbb{R}^d$, define

$$\Delta_n^\pi(z) := z + 3^n \left\{ (x_1, \dots, x_d) \in \mathbb{R}^d : -1/2 < x_{\pi(1)} < x_{\pi(2)} < \dots < x_{\pi(d)} < 1/2 \right\}.$$

We call $\Delta_n^\pi(z)$ a **Kuhn simplex** of the cube $z + \square_n$; these $d!$ simplices form its **Kuhn triangulation**.

For $n \leq m$, each Kuhn simplex of a cube of side 3^m is partitioned, up to null sets, by the scale- n Kuhn simplices contained in it. We retain these fine simplices even when their supporting cubes cross the diagonal boundary of the parent simplex. Moreover, the cube $z + \square_n$ is the union of the simplices $\Delta_n^\pi(z)$ over all permutations π , up to null sets, and $|\Delta_n^\pi(z)| = 3^{nd}/d!$.

Function spaces. The coarse-graining estimates of this section are stated in volume-normalized norms on cubes. The positive-order seminorms measure the oscillation of a function at all scales up to the size of the cube, the negative-order norms measure a gradient or a flux through its averages against test functions or over subcubes, and the spaces of solenoidal fields at the end of this subsection enter the variational formulas for the coarse-grained matrices. For $s \in (0, 1)$ and $p \in [1, \infty)$, the fractional Sobolev seminorm on the cube $\square_m$ is defined by

$$[u]_{\underline{W}^{s,p}(\square_m)} := \left(s \int_{\square_m} \int_{\square_m} \frac{|u(x) - u(y)|^p}{|x - y|^{d+sp}} dx dy \right)^{1/p}.$$

In this fractional Sobolev seminorm, distance and vector magnitude are Euclidean. At $s = 1$, the seminorm is the classical first-order one, $\|\nabla u\|_{\underline{L}^p(\square_m)}$, with Euclidean gradient magnitude. For a bounded Lipschitz set U , the same formulas define $[\cdot]_{\underline{W}^{s,p}(U)}$. We set

$$\|u\|_{\underline{W}^{s,p}(U)} := [u]_{\underline{W}^{s,p}(U)} + |U|^{-s/d} \|u\|_{\underline{L}^p(U)}.$$

For $p = \infty$ we write $[u]_{\underline{W}^{s,\infty}(U)} := \sup_{x \neq y \in U} |u(x) - u(y)| |x - y|^{-s}$ for the Hölder seminorm of order s , and define the norm by the same formula with $\|u\|_{\underline{L}^\infty(U)}$. For $s \in (0, 1]$, $q \in (1, \infty)$, $N \in \mathbb{N}$, and $F : U \rightarrow \mathbb{R}^N$, define the inhomogeneous volume-normalized negative Sobolev norm by

$$\|F\|_{\widehat{W}^{-s,q}(U)} := \sup_{\varphi \in C^\infty(\bar{U}; \mathbb{R}^N), \varphi \neq 0} \frac{|\int_U F \cdot \varphi|}{[\varphi]_{\underline{W}^{s,q'}(U)} + |U|^{-s/d} \|\varphi\|_{\underline{L}^{q'}(U)}}.$$

When $q = 2$ and $N = d$, we write

$$(2.1) \quad \|F\|_{\widehat{H}^{-s}(U)} := \sup_{\varphi \in C^\infty(\bar{U}; \mathbb{R}^d), \varphi \neq 0} \frac{|\int_U F \cdot \varphi|}{[\varphi]_{\underline{H}^s(U)} + |U|^{-s/d} \|\varphi\|_{\underline{L}^2(U)}}.$$

Here and below, $\underline{H}^s = \underline{W}^{s,2}$. If $F \in (H^s(U; \mathbb{R}^d))^*$, then the numerator in (2.1) is interpreted as $|U|^{-1}|\langle F, \varphi \rangle|$; by density of smooth functions, the supremum is then the corresponding dual norm. For the unadorned negative norm in Theorem B we use zero-boundary tests:

$$(2.2) \quad \|F\|_{\underline{H}^{-1}(U)} := \sum_{i=1}^d \sup_{\substack{\varphi \in C_c^\infty(U) \\ \|\nabla \varphi\|_{\underline{L}^2(U; \ell^2)} \leq 1}} \left| \oint_U F_i \varphi \right|.$$

Each summand is the volume-normalized norm of the component F_i in the dual of $H_0^1(U)$. The hatted negative norm continues to use the unrestricted smooth tests of (2.1). On $\square_0$ the volume-normalized and unnormalized conventions coincide. For every nonempty finite set A , write $\sum_{z \in A} c_z := \frac{1}{\#A} \sum_{z \in A} c_z$. In particular, if a cube U is the disjoint union of equal-volume cubes $(Q_z)_{z \in A}$, then

$$(2.3) \quad \sum_{z \in A} \|f\|_{\underline{L}^p(Q_z)}^p = \|f\|_{\underline{L}^p(U)}^p.$$

We work with the (volume-normalized) Besov-type seminorm defined in the cube $\square_m$ for exponents $s \in (0, 1)$ and $p, q, r \in [1, \infty)$ by

$$(2.4) \quad [u]_{\underline{B}_{p,q,r}^s(\square_m)} := \left(s \sum_{n=-\infty}^m 3^{-nsq} \left(\sum_{z \in 3^{n-1}\mathbb{Z}^d \cap \square_m, z+\square_n \subseteq \square_m} \|u - (u)_{z+\square_n}\|_{\underline{L}^p(z+\square_n)}^r \right)^{q/r} \right)^{1/q}.$$

For $r = p$, the seminorm reads

$$[u]_{\underline{B}_{p,q,p}^s(\square_m)} = \left(s \sum_{n=-\infty}^m 3^{-nsq} \left(\sum_{z \in 3^{n-1}\mathbb{Z}^d \cap \square_m, z+\square_n \subseteq \square_m} \|u - (u)_{z+\square_n}\|_{\underline{L}^p(z+\square_n)}^p \right)^{q/p} \right)^{1/q}.$$

When the third index is omitted, it equals the first: we write $\underline{B}_{p,q}^s := \underline{B}_{p,q,p}^s$. The corresponding norm is $\|f\|_{\underline{B}_{p,q,r}^s(\square_m)} := [f]_{\underline{B}_{p,q,r}^s(\square_m)} + 3^{-sm}|(f)_{\square_m}|$. By Lemma C.2, there exists $C(d) < \infty$ such that, for every $s \in (0, 1)$ and $p \in [1, \infty)$,

$$(2.5) \quad C^{-1}[u]_{\underline{W}^{s,p}(\square_m)} \leq [u]_{\underline{B}_{p,p,p}^s(\square_m)} \leq C[u]_{\underline{W}^{s,p}(\square_m)}.$$

The corresponding norms are equivalent for each fixed s, p ; unlike the seminorm comparison, their constants need not be uniform in these parameters.

We allow each of p, q, r to equal ∞ . If $p = \infty$, the volume-normalized L^p norm is replaced by the L^∞ norm; if $r = \infty$, the average over cube centers is replaced by a supremum; and if $q = \infty$, the sum over scales is replaced by a supremum. We also allow $s = 1$ when $q = \infty$.

We use Euclidean distance and vector magnitude in the Hölder seminorms. When $q = r = \infty$, the Besov seminorm is equivalent to the Hölder seminorm of order s for every p . More precisely,

$$(2.6) \quad \begin{aligned} C^{-1}s \sup_{x,y \in \square_m, x \neq y} \frac{|u(x) - u(y)|}{|x - y|^s} &\leq [u]_{\underline{B}_{1,\infty,\infty}^s(\square_m)} \leq [u]_{\underline{B}_{\infty,\infty,\infty}^s(\square_m)} \\ &\leq C \sup_{x,y \in \square_m, x \neq y} \frac{|u(x) - u(y)|}{|x - y|^s}. \end{aligned}$$

The first inequality of (2.6) is the only nontrivial one, and it is the Campanato characterization of Hölder spaces.

For $s \in (0, 1]$ and $q, r \in [1, \infty]$, define

$$(2.7) \quad \|f\|_{\underline{B}_{r,q}^{s-}(\square_m)} := \left(s \sum_{n=-\infty}^m 3^{nsq} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |(f)_{z+\square_n}|^r \right)^{q/r} \right)^{1/q}.$$

If $q^{-1} + (q')^{-1} = r^{-1} + (r')^{-1} = 1$ and either $s < 1$ or $(s, q) = (1, 1)$, then Lemma C.1 gives, for every f, g for which the right side is finite and $fg \in L^1(\square_m)$,

$$(2.8) \quad \left| \int_{\square_m} fg \right| \leq Cs^{-1} \|g\|_{\dot{B}_{1,q',r'}^s(\square_m)} \|f\|_{\dot{B}_{r,q}^{-s}(\square_m)}.$$

For a bounded convex domain $U \subseteq \mathbb{R}^d$, set

$$(2.9) \quad L_{\text{sol}}^2(U) := (\nabla H_0^1(U))^\perp \quad \text{and} \quad L_{\text{sol},0}^2(U) := (\nabla H^1(U))^\perp.$$

All orthogonal complements in (2.9) are taken in $L^2(U; \mathbb{R}^d)$.

2.2. Coarse-grained matrices. We denote the linear space of solutions u of $-\nabla \cdot \mathbf{a} \nabla u = 0$ in an open set $U \subseteq \mathbb{R}^d$ by

$$\mathcal{A}(U; \mathbf{a}) := \{u \in H_{\text{loc}}^1(U) : -\nabla \cdot \mathbf{a} \nabla u = 0 \text{ in } U\}.$$

For every $p, q \in \mathbb{R}^d$ and bounded convex domain $U \subseteq \mathbb{R}^d$, we define the quantity

$$(2.10) \quad J(U, p, q; \mathbf{a}) := \sup_{v \in \mathcal{A}(U; \mathbf{a}) \cap H^1(U)} \int_U \left(-\frac{1}{2} \nabla v \cdot \mathbf{a} \nabla v - p \cdot \mathbf{a} \nabla v + q \cdot \nabla v \right).$$

The supremum is attained in $H^1(U)$, and the maximizer is unique up to an additive constant. We denote its mean-zero representative by $v(\cdot, U, p, q; \mathbf{a})$, so that $(v(\cdot, U, p, q; \mathbf{a}))_U = 0$. For a bounded convex domain $U \subseteq \mathbb{R}^d$, define the **coarse-grained matrices** $\mathbf{a}(U)$ and $\mathbf{a}_*(U)$ by the identity, valid for every $p, q \in \mathbb{R}^d$,

$$(2.11) \quad J(U, p, q; \mathbf{a}) =: \frac{1}{2} p \cdot \mathbf{a}(U) p + \frac{1}{2} q \cdot \mathbf{a}_*^{-1}(U) q - p \cdot q.$$

The identity (2.11) records both matrices in one quadratic function of (p, q) , which is subadditive under partitions of U (see (2.16) below). The matrices also have variational characterizations [Armstrong et al., 2019, Lemma 10.5], using the version dated 7 May 2019, p. 384, which express them as minimal energies; we use these to compare the coarse-grained matrices of two coefficient fields in Section 2.4. For $p, q \in \mathbb{R}^d$,

$$(2.12) \quad \frac{1}{2} p \cdot \mathbf{a}(U) p = \min_{u \in p \cdot x + H_0^1(U)} \int_U \frac{1}{2} \nabla u \cdot \mathbf{a} \nabla u = \max_{\mathbf{g} \in L_{\text{sol}}^2(U)} \int_U \left(-\frac{1}{2} \mathbf{g} \cdot \mathbf{a}^{-1} \mathbf{g} + p \cdot \mathbf{g} \right),$$

and

$$(2.13) \quad \frac{1}{2} q \cdot \mathbf{a}_*^{-1}(U) q = \max_{w \in H^1(U)} \int_U \left(-\frac{1}{2} \nabla w \cdot \mathbf{a} \nabla w + q \cdot \nabla w \right) = \min_{\mathbf{g} \in q + L_{\text{sol},0}^2(U)} \int_U \frac{1}{2} \mathbf{g} \cdot \mathbf{a}^{-1} \mathbf{g}.$$

2.2.1. Properties of the coarse-grained matrices. We use the following identities and inequalities; see Armstrong and Kuusi [2024, 2025]. Throughout, $U \subseteq \mathbb{R}^d$ is a bounded convex domain.

The coarse-grained matrices are bounded by averages of the field, in the Loewner order:

$$(2.14) \quad \left(\int_U \mathbf{a}^{-1} \right)^{-1} \leq \mathbf{a}_*(U) \leq \mathbf{a}(U) \leq \int_U \mathbf{a}.$$

The coarse-grained matrices are **subadditive**: for every partition $\{U_i\}_{i=1}^N$ of U (up to a Lebesgue null set)

$$(2.15) \quad \mathbf{a}(U) \leq \sum_{i=1}^N \frac{|U_i|}{|U|} \mathbf{a}(U_i) \quad \text{and} \quad \mathbf{a}_*(U) \geq \left(\sum_{i=1}^N \frac{|U_i|}{|U|} \mathbf{a}_*^{-1}(U_i) \right)^{-1},$$

which generalizes the first and third inequalities of (2.14). Equivalently, for every $p, q \in \mathbb{R}^d$,

$$(2.16) \quad J(U, p, q; \mathbf{a}) \leq \sum_{i=1}^N \frac{|U_i|}{|U|} J(U_i, p, q; \mathbf{a}).$$

The spatial averages of the gradient and the flux of the maximizer can be written explicitly in terms of the coarse-grained matrices: for $p, q \in \mathbb{R}^d$,

$$(2.17) \quad \begin{cases} \oint_U \nabla v(\cdot, U, p, q; \mathbf{a}) = -p + \mathbf{a}_*^{-1}(U)q \\ \oint_U \mathbf{a} \nabla v(\cdot, U, p, q; \mathbf{a}) = q - \mathbf{a}(U)p. \end{cases}$$

For every $e \in \mathbb{R}^d$, the maximizer $v(\cdot, U, 0, \mathbf{a}_*(U)e; \mathbf{a})$ satisfies, by (2.17),

$$\begin{pmatrix} (\nabla v(\cdot, U, 0, \mathbf{a}_*(U)e; \mathbf{a}))_U \\ (\mathbf{a} \nabla v(\cdot, U, 0, \mathbf{a}_*(U)e; \mathbf{a}))_U \end{pmatrix} = \begin{pmatrix} e \\ \mathbf{a}_*(U)e \end{pmatrix}.$$

The identity (2.23) below gives $J(U, e, \mathbf{a}_*(U)e; \mathbf{a}) = \frac{1}{2}e \cdot (\mathbf{a}(U) - \mathbf{a}_*(U))e$. For every $w \in \mathcal{A}(U; \mathbf{a}) \cap H^1(U)$ and $p, q \in \mathbb{R}^d$, the first variation of (2.10) gives

$$(2.18) \quad q \cdot \oint_U \nabla w - p \cdot \oint_U \mathbf{a} \nabla w = \oint_U \nabla w \cdot \mathbf{a} \nabla v(\cdot, U, p, q; \mathbf{a}).$$

The second variation gives

$$(2.19) \quad \begin{aligned} J(U, p, q; \mathbf{a}) - \oint_U \left(-\frac{1}{2} \nabla w \cdot \mathbf{a} \nabla w - p \cdot \mathbf{a} \nabla w + q \cdot \nabla w \right) \\ = \oint_U \frac{1}{2} (\nabla v(\cdot, U, p, q; \mathbf{a}) - \nabla w) \cdot \mathbf{a} (\nabla v(\cdot, U, p, q; \mathbf{a}) - \nabla w). \end{aligned}$$

It follows that J can be written as the energy of the maximizer: for every $p, q \in \mathbb{R}^d$,

$$(2.20) \quad J(U, p, q; \mathbf{a}) = \oint_U \frac{1}{2} \nabla v(\cdot, U, p, q; \mathbf{a}) \cdot \mathbf{a} \nabla v(\cdot, U, p, q; \mathbf{a}).$$

The Euler equation also gives

$$J(U, p, q; \mathbf{a}) = \frac{1}{2} \left(q \cdot \oint_U \nabla v(\cdot, U, p, q; \mathbf{a}) - p \cdot \oint_U \mathbf{a} \nabla v(\cdot, U, p, q; \mathbf{a}) \right).$$

The coarse-grained matrix $\mathbf{a}_*(U)$ bounds the energy of an arbitrary $H^1(U)$ function from below in terms of the spatial average of its gradient: for every $u \in H^1(U)$,

$$(2.21) \quad \frac{1}{2} (\nabla u)_U \cdot \mathbf{a}_*(U) (\nabla u)_U \leq \oint_U \frac{1}{2} \nabla u \cdot \mathbf{a} \nabla u.$$

The dual inequality is, for every $\mathbf{h} \in L_{\text{sol}}^2(U)$,

$$\frac{1}{2} (\mathbf{h})_U \cdot \mathbf{a}^{-1}(U) (\mathbf{h})_U \leq \oint_U \frac{1}{2} \mathbf{h} \cdot \mathbf{a}^{-1} \mathbf{h}.$$

For every constant positive-definite matrix $\mathbf{b} \in \mathbb{R}_{\text{sym},+}^{d \times d}$ and every $u \in \mathcal{A}(U; \mathbf{a}) \cap H^1(U)$,

$$(2.22) \quad |((\mathbf{a} - \mathbf{b}) \nabla u)_U| \leq 2 \left(\max_{|e|=1} J(U, e, \mathbf{b}e; \mathbf{a}) \right)^{1/2} \left(\oint_U \frac{1}{2} \nabla u \cdot \mathbf{a} \nabla u \right)^{1/2},$$

in particular, with the coarse-graining error $\mathcal{E}_{s,\infty,1}$ of Definition 2.3 below, for every $m \in \mathbb{Z}$ and every $u \in \mathcal{A}(\square_m; \mathbf{a}) \cap H^1(\square_m)$,

$$3^{-sm} \|(\mathbf{a} - \mathbf{b}) \nabla u\|_{\dot{B}_{2,1}^{-s}(\square_m)} \leq 4s^{-1} |\mathbf{b}|^{1/2} \|\mathbf{a}^{1/2} \nabla u\|_{L^2(\square_m)} \mathcal{E}_{s,\infty,1}(\square_m; \mathbf{a}, \mathbf{b}).$$

The bound (2.22) follows from (2.18), (2.20), and the Cauchy–Schwarz inequality.

The identity (2.11) can be rewritten for every symmetric matrix $\mathbf{b} \in \mathbb{R}_{\text{sym}}^{d \times d}$ as

$$(2.23) \quad J(U, e, \mathbf{b}e; \mathbf{a}) = \frac{1}{2} e \cdot (\mathbf{a}(U) - \mathbf{a}_*(U))e + \frac{1}{2} e \cdot (\mathbf{b} - \mathbf{a}_*(U)) \mathbf{a}_*^{-1}(U) (\mathbf{b} - \mathbf{a}_*(U))e.$$

If $\mathbf{b} \in \mathbb{R}_{\text{sym},+}^{d \times d}$, then completing the square in the first argument of J instead of the second gives

$$(2.24) \quad J(U, \mathbf{b}^{-1}e, e; \mathbf{a}) = \frac{1}{2}e \cdot (\mathbf{a}_*^{-1}(U) - \mathbf{a}^{-1}(U))e + \frac{1}{2}e \cdot (\mathbf{b}^{-1} - \mathbf{a}^{-1}(U))\mathbf{a}(U)(\mathbf{b}^{-1} - \mathbf{a}^{-1}(U))e.$$

Applying (2.23) and (2.24) with $\mathbf{b}^{-1/2}e$ and $\mathbf{b}^{1/2}e$, respectively, in place of e yields

$$(2.25) \quad \begin{aligned} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a}) &= \frac{1}{2}e \cdot \mathbf{b}^{-1/2}(\mathbf{a}(U) - \mathbf{a}_*(U))\mathbf{b}^{-1/2}e \\ &\quad + \frac{1}{2}e \cdot (\mathbf{I}_d - \mathbf{b}^{-1/2}\mathbf{a}_*(U)\mathbf{b}^{-1/2})\mathbf{b}^{1/2}\mathbf{a}_*^{-1}(U)\mathbf{b}^{1/2}(\mathbf{I}_d - \mathbf{b}^{-1/2}\mathbf{a}_*(U)\mathbf{b}^{-1/2})e \\ &= \frac{1}{2}e \cdot \mathbf{b}^{1/2}(\mathbf{a}_*^{-1}(U) - \mathbf{a}^{-1}(U))\mathbf{b}^{1/2}e \end{aligned}$$

$$(2.26) \quad + \frac{1}{2}e \cdot (\mathbf{I}_d - \mathbf{b}^{1/2}\mathbf{a}^{-1}(U)\mathbf{b}^{1/2})\mathbf{b}^{-1/2}\mathbf{a}(U)\mathbf{b}^{-1/2}(\mathbf{I}_d - \mathbf{b}^{1/2}\mathbf{a}^{-1}(U)\mathbf{b}^{1/2})e.$$

Lemma 2.1. *For every bounded convex domain $U \subseteq \mathbb{R}^d$ and $\mathbf{b} \in \mathbb{R}_{\text{sym},+}^{d \times d}$,*

$$(2.27) \quad \begin{aligned} \max_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a}) &\leq \frac{1}{2}|\mathbf{b}^{-1/2}(\mathbf{a}(U) - \mathbf{a}_*(U))\mathbf{b}^{-1/2}| + \frac{1}{2}|\mathbf{b}^{-1/2}\mathbf{a}_*^{1/2}(U) - \mathbf{b}^{1/2}\mathbf{a}_*^{-1/2}(U)|^2 \\ &\leq 2 \max_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a}), \end{aligned}$$

and

$$(2.28) \quad \begin{aligned} \max_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a}) &\leq \frac{1}{2}|\mathbf{b}^{1/2}(\mathbf{a}^{-1}(U) - \mathbf{a}_*^{-1}(U))\mathbf{b}^{1/2}| + \frac{1}{2}|\mathbf{b}^{1/2}\mathbf{a}^{-1/2}(U) - \mathbf{b}^{-1/2}\mathbf{a}^{1/2}(U)|^2 \\ &\leq 2 \max_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a}). \end{aligned}$$

In particular, there exists a universal constant $C < \infty$ such that

$$(2.29) \quad \begin{aligned} &|\mathbf{b}^{-1/2}(\mathbf{a}(U) - \mathbf{a}_*(U))\mathbf{b}^{-1/2}| + |\mathbf{b}^{1/2}\mathbf{a}_*^{-1}(U)\mathbf{b}^{1/2} - \mathbf{I}_d|^2 + |\mathbf{b}^{-1/2}\mathbf{a}(U)\mathbf{b}^{-1/2} - \mathbf{I}_d|^2 \\ &\leq C \max_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a}) (1 + \max_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a})). \end{aligned}$$

Proof. For positive-semidefinite matrices M and N , we have $\max\{|M|, |N|\} \leq |M + N| \leq |M| + |N| \leq 2|M + N|$. Applying this inequality to the two summands in (2.25) proves (2.27); applying it to the two summands in (2.26) proves (2.28).

For every positive-definite matrix A and $J \geq 0$,

$$(2.30) \quad \frac{1}{2}|(\mathbf{I}_d - A)A^{-1}(\mathbf{I}_d - A)| \leq J \implies |\mathbf{I}_d - A|^2 \leq 4J(1 + J).$$

Indeed, the hypothesis implies

$$|\mathbf{I}_d - A|^2 \leq 2J|A| \leq 2J|\mathbf{I}_d - A| + 2J \leq \frac{1}{2}|\mathbf{I}_d - A|^2 + 2J^2 + 2J.$$

Since $(\mathbf{I}_d - A)A^{-1}(\mathbf{I}_d - A) = (\mathbf{I}_d - A^{-1})A(\mathbf{I}_d - A^{-1})$, apply (2.30) with $A = (\mathbf{b}^{-1/2}\mathbf{a}_*(U)\mathbf{b}^{-1/2})^{-1}$ in (2.25) and with $A = (\mathbf{b}^{1/2}\mathbf{a}^{-1}(U)\mathbf{b}^{1/2})^{-1}$ in (2.26). This gives

$$(2.31) \quad \begin{aligned} &\max\left\{|\mathbf{b}^{-1/2}\mathbf{a}(U)\mathbf{b}^{-1/2} - \mathbf{I}_d|^2, |\mathbf{b}^{1/2}\mathbf{a}_*^{-1}(U)\mathbf{b}^{1/2} - \mathbf{I}_d|^2\right\} \\ &\leq 10 \sup_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a}) (1 + \sup_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a})). \end{aligned}$$

Combining (2.31) with (2.27) gives (2.29). $\square$

For every $\lambda > 0$, bounded convex domain $U \subseteq \mathbb{R}^d$ and $p, q \in \mathbb{R}^d$,

$$(2.32) \quad J(U, p, q; \mathbf{a}) = J(U, \lambda^{-1/2}p, \lambda^{1/2}q; \lambda\mathbf{a}).$$

2.3. Coarse-grained elliptic and functional inequalities. Following [Armstrong and Kuusi \[2025, Definition 2.1, \(2.39\), p. 25\]](#), we attach to each cube $\square_m$ the coarse-grained ellipticity constants $\Lambda_{s,q}(\square_m; \mathbf{a})$ and $\lambda_{s,q}(\square_m; \mathbf{a})$. The resulting Poincaré and Caccioppoli inequalities depend on these constants rather than on the pointwise ellipticity ratio.

The weights $(1 - 3^{-s})3^{-ns}$ sum to one over $n \geq 0$. For $0 < s \leq 2$, their normalizing factor satisfies $\frac{4}{9}s \leq 1 - 3^{-s} \leq (\log 3)s$.

Definition 2.2 (Coarse-grained ellipticity constants). For $s > 0$, $q \in [1, \infty)$, $m \in \mathbb{Z}$, and a uniformly elliptic field $\mathbf{a} : \square_m \rightarrow \mathbb{R}_{\text{sym},+}^{d \times d}$, define the **coarse-grained ellipticity constants** by

$$(2.33) \quad \begin{cases} \Lambda_{s,q}(\square_m; \mathbf{a}) := \left((1 - 3^{-sq}) \sum_{k=-\infty}^m 3^{-sq(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{a}(z + \square_k; \mathbf{a})|^{q/2} \right)^{2/q}, \\ \lambda_{s,q}(\square_m; \mathbf{a}) := \left((1 - 3^{-sq}) \sum_{k=-\infty}^m 3^{-sq(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{a}_*^{-1}(z + \square_k; \mathbf{a})|^{q/2} \right)^{-2/q}, \end{cases}$$

and, for $q = \infty$,

$$(2.34) \quad \begin{cases} \Lambda_{s,\infty}(\square_m; \mathbf{a}) := \sup_{k \in (-\infty, m] \cap \mathbb{Z}} 3^{-2s(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{a}(z + \square_k; \mathbf{a})|, \\ \lambda_{s,\infty}(\square_m; \mathbf{a}) := \left(\sup_{k \in (-\infty, m] \cap \mathbb{Z}} 3^{-2s(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{a}_*^{-1}(z + \square_k; \mathbf{a})| \right)^{-1}. \end{cases}$$

When the exponent q is omitted, it is understood that $q = 1$, that is, $\Lambda_s(\square_m; \mathbf{a}) = \Lambda_{s,1}(\square_m; \mathbf{a})$ and $\lambda_s(\square_m; \mathbf{a}) = \lambda_{s,1}(\square_m; \mathbf{a})$. Equation (2.34) is the limit of (2.33) as $q \rightarrow \infty$. On $y + \square_m$, every subcube $z + \square_k$ in these definitions is replaced by $y + z + \square_k$, with $z \in 3^k \mathbb{Z}^d \cap \square_m$.

For every $m \in \mathbb{Z}$, $q \in [1, \infty]$ and $t, s \in (0, \infty)$ with $t < s$, we have

$$(2.35) \quad \lambda_{t,q}(\square_m; \mathbf{a}) \leq \lambda_{s,q}(\square_m; \mathbf{a}) \leq |\mathbf{a}_*^{-1}(\square_m; \mathbf{a})|^{-1} \leq |\mathbf{a}(\square_m; \mathbf{a})| \leq \Lambda_{s,q}(\square_m; \mathbf{a}) \leq \Lambda_{t,q}(\square_m; \mathbf{a}).$$

There exists $C(d) < \infty$ such that, for every $s \in (0, 1]$ and $p, q \in [1, \infty]$ with $p \leq q$,

$$\Lambda_{s,q}(\square_m; \mathbf{a}) \leq C s^{2/q-2/p} \Lambda_{s,p}(\square_m; \mathbf{a}) \leq C s^{2/q-2} \Lambda_s(\square_m; \mathbf{a}),$$

and

$$\lambda_{s,q}^{-1}(\square_m; \mathbf{a}) \leq C s^{2/q-2/p} \lambda_{s,p}^{-1}(\square_m; \mathbf{a}) \leq C s^{2/q-2} \lambda_s^{-1}(\square_m; \mathbf{a}).$$

For every $k \in (-\infty, m] \cap \mathbb{Z}$, applying (2.35) on the subcubes and comparing the definitions gives

$$(2.36) \quad \begin{cases} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{a}(z + \square_k; \mathbf{a})| \leq \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \Lambda_{s,q}(z + \square_k; \mathbf{a}) \leq 3^{2s(m-k)} \Lambda_{s,q}(\square_m; \mathbf{a}), \\ \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{a}_*^{-1}(z + \square_k; \mathbf{a})| \leq \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \lambda_{s,q}^{-1}(z + \square_k; \mathbf{a}) \leq 3^{2s(m-k)} \lambda_{s,q}^{-1}(\square_m; \mathbf{a}). \end{cases}$$

We measure the error in replacing $\mathbf{a}$ by a constant matrix $\mathbf{a}_0$ through the quantities $J(z + \square_l, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a})$ on the subcubes $z + \square_l$.

Definition 2.3 (Coarse-graining error). For $s \in (0, 1]$, $p, q \in [1, \infty]$, $m, n \in \mathbb{Z}$ with $n \leq m$, a uniformly elliptic field $\mathbf{a} : \square_m \rightarrow \mathbb{R}_{\text{sym},+}^{d \times d}$, and a matrix $\mathbf{a}_0 \in \mathbb{R}_{\text{sym},+}^{d \times d}$, define the **coarse-graining**

error by

$$\mathcal{E}_{s,p,q}(\square_m, n; \mathbf{a}, \mathbf{a}_0) = \begin{cases} \left((1 - 3^{-sq}) \sum_{l=-\infty}^n 3^{-sq(n-l)} \left(\sum_{z \in 3^l \mathbb{Z}^d \cap \square_m} \max_{|e|=1} (J(z + \square_l, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a}))^{p/2} \right)^{q/p} \right)^{1/q}, & p, q < \infty, \\ \left((1 - 3^{-sq}) \sum_{l=-\infty}^n 3^{-sq(n-l)} \max_{z \in 3^l \mathbb{Z}^d \cap \square_m} \max_{|e|=1} (J(z + \square_l, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a}))^{q/2} \right)^{1/q}, & p = \infty, q < \infty, \\ \sup_{l \in (-\infty, n] \cap \mathbb{Z}} 3^{-s(n-l)} \left(\sum_{z \in 3^l \mathbb{Z}^d \cap \square_m} \max_{|e|=1} (J(z + \square_l, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a}))^{p/2} \right)^{1/p}, & p < \infty, q = \infty, \\ \sup_{l \in (-\infty, n] \cap \mathbb{Z}} 3^{-s(n-l)} \max_{z \in 3^l \mathbb{Z}^d \cap \square_m} \max_{|e|=1} (J(z + \square_l, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a}))^{1/2}, & p = q = \infty. \end{cases}$$

On $y + \square_m$, every subcube $z + \square_l$ in this definition is replaced by $y + z + \square_l$, with $z \in 3^l \mathbb{Z}^d \cap \square_m$. We use the abbreviation

$$\mathcal{E}_{s,p,q}(\square_m; \mathbf{a}, \mathbf{a}_0) := \mathcal{E}_{s,p,q}(\square_m, m; \mathbf{a}, \mathbf{a}_0).$$

When the exponent q is omitted, it is understood that $q = 1$.

The definition gives, for $m \in \mathbb{Z}$, $p \in [2, \infty]$, $q \in [1, \infty]$, and $0 < t < s \leq 1$,

$$\max_{|e|=1} J(\square_m, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a})^{1/2} \leq \mathcal{E}_{s,p,q}(\square_m; \mathbf{a}, \mathbf{a}_0) \leq \mathcal{E}_{t,p,q}(\square_m; \mathbf{a}, \mathbf{a}_0),$$

and, for every $k \in (-\infty, m] \cap \mathbb{Z}$,

$$(2.37) \quad \begin{cases} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \mathcal{E}_{s,\infty,q}(z + \square_k; \mathbf{a}, \mathbf{a}_0) \leq 3^{s(m-k)} \mathcal{E}_{s,\infty,q}(\square_m; \mathbf{a}, \mathbf{a}_0), \\ \sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} \mathcal{E}_{s,p,q}(z + \square_k; \mathbf{a}, \mathbf{a}_0) \leq 3^{(s+d(\frac{1}{p}-\frac{1}{q})_+)(m-k)} \mathcal{E}_{s,p,q}(\square_m; \mathbf{a}, \mathbf{a}_0). \end{cases}$$

For a positive scalar $\mathbf{a}_0$, $s \in (0, 1]$ and $q \in [1, \infty]$,

$$(2.38) \quad \begin{aligned} \frac{1}{2} \mathcal{E}_{s,\infty,q}(\square_m; \mathbf{a}, \mathbf{a}_0)^2 &\leq \max\{|\mathbf{a}_0|^{-1} \Lambda_{s,q}(\square_m; \mathbf{a}), |\mathbf{a}_0| \lambda_{s,q}^{-1}(\square_m; \mathbf{a})\} \\ &\leq (1 + \sqrt{2} \mathcal{E}_{s,\infty,q}(\square_m; \mathbf{a}, \mathbf{a}_0))^2. \end{aligned}$$

To see the upper bound, put $J_U = \max_{|e|=1} J(U, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a})$. The ordering $\mathbf{a}_*(U) \leq \mathbf{a}(U)$ and (2.11) imply that each eigenvalue t of $\mathbf{a}_0^{-1} \mathbf{a}(U)$ or $\mathbf{a}_0 \mathbf{a}_*^{-1}(U)$ satisfies $t + t^{-1} - 2 \leq 2J_U$. Thus $\sqrt{t} \leq 1 + \sqrt{2J_U}$ and $t \leq 1 + 2J_U + \sqrt{2J_U}$. The first inequality, followed by the weighted ℓ^q triangle inequality, gives the upper bound in (2.38). For the lower bound, use $\sqrt{J_U} \leq 2^{-1/2} (|\mathbf{a}_0^{-1} \mathbf{a}(U)|^{1/2} + |\mathbf{a}_0 \mathbf{a}_*^{-1}(U)|^{1/2})$ and the same triangle inequality. The second eigenvalue bound also gives

$$\max\{\mathbf{a}_0^{-1} \Lambda_{s,2}(\square_m; \mathbf{a}), \mathbf{a}_0 \lambda_{s,2}^{-1}(\square_m; \mathbf{a})\} \leq 1 + 2\mathcal{E}_{s,\infty,2}^2(\square_m; \mathbf{a}, \mathbf{a}_0) + 2\mathcal{E}_{s,\infty,2}(\square_m; \mathbf{a}, \mathbf{a}_0).$$

Moreover, in the case $q = 2$,

$$1 + \frac{1}{2} \mathcal{E}_{s,\infty,2}(\square_m; \mathbf{a}, \mathbf{a}_0)^2 \leq \max\{|\mathbf{a}_0|^{-1} \Lambda_{s,2}(\square_m; \mathbf{a}), |\mathbf{a}_0| \lambda_{s,2}^{-1}(\square_m; \mathbf{a})\}.$$

By Definition 2.3, for every $p, q \in [1, \infty]$ and $s \in (d/p, 1)$,

$$\mathcal{E}_{s,\infty,q}(\square_m; \mathbf{a}, \mathbf{a}_0) \leq \left(\frac{(1 - 3^{-sq})}{(1 - 3^{-q(s-\frac{d}{p})})} \right)^{1/q} \mathcal{E}_{s-d/p,p,q}(\square_m; \mathbf{a}, \mathbf{a}_0).$$

For $1 \leq q \leq p < \infty$ and $s > 0$,

$$\mathcal{E}_{s,\infty,q}(\square_m; \mathbf{a}, \mathbf{a}_0) \leq \mathcal{E}_{sq/p,\infty,p}(\square_m; \mathbf{a}, \mathbf{a}_0).$$

The same definitions give

$$(2.39) \quad \Lambda_{s,q}(\square_m; \mathbf{a}) \leq \Lambda_{sq/p,p}(\square_m; \mathbf{a}) \quad \text{and} \quad \lambda_{s,q}^{-1}(\square_m; \mathbf{a}) \leq \lambda_{sq/p,p}^{-1}(\square_m; \mathbf{a}).$$

By Hölder's inequality, there exists a universal constant $C < \infty$ such that, for $0 < s < t \leq 1$,

$$(2.40) \quad \mathcal{E}_{t,\infty,1}(\square_m, n; \mathbf{a}, \mathbf{a}_0) \leq \frac{Ct}{(t-s)^{1/2} s^{1/2}} \mathcal{E}_{s,\infty,2}(\square_m, n; \mathbf{a}, \mathbf{a}_0).$$

The coarse-grained ellipticity constants satisfy the following stability estimate, which is [Armstrong et al. \[2026, Lemma 2.5, p. 33\]](#).

Lemma 2.4 (Stability of the coarse-grained ellipticity constants). *There exists a constant $C(d) < \infty$ such that, for every $q \in [1, \infty]$, $s, t \in (0, 1/2]$ with $s < t$, and $x \in \square_0$ such that $x + \square_{-1} \subseteq \square_0$, we have the estimates*

$$(2.41) \quad \lambda_{t,q}^{-1}(x + \square_{-1}; \mathbf{a}) \leq \frac{C}{1-2s} \left(\frac{t}{t-s} \right)^{2/q} \lambda_{s,q}^{-1}(\square_0; \mathbf{a}),$$

$$(2.42) \quad \Lambda_{t,q}(x + \square_{-1}; \mathbf{a}) \leq \frac{C}{1-2s} \left(\frac{t}{t-s} \right)^{2/q} \Lambda_{s,q}(\square_0; \mathbf{a}),$$

and, for every $\mathbf{a}_0 \in \mathbb{R}_{\text{sym},+}^{d \times d}$ (see [Armstrong et al. \[2026, \(2.62\), p. 33\]](#)),

$$\mathcal{E}_{t,\infty,2}(x + \square_{-1}; \mathbf{a}, \mathbf{a}_0) \leq \frac{C}{(1-2s)^{1/2}} \left(\frac{t}{t-s} \right)^{1/2} \mathcal{E}_{s,\infty,2}(\square_0; \mathbf{a}, \mathbf{a}_0).$$

The next proposition is the weak-norm estimate of [Armstrong and Kuusi \[2025, Lemma 2.2, p. 27\]](#), stated there for the gradient and flux of a solution. The same argument uses (2.21) and its dual on each subcube, so it applies to every $u \in H^1(\square_m)$ and every $\mathbf{h} \in L_{\text{sol}}^2(\square_m)$. In the flux case, a zero-trace test function on a subcube extends by zero to one on $\square_m$, so the restriction of $\mathbf{h}$ is still solenoidal. The positive-order estimate (2.46) below corresponds to [Armstrong and Kuusi \[2025, Lemma 2.3, p. 27\]](#).

Proposition 2.5 (Coarse-grained Poincaré inequality). *For every $s \in (0, 1]$, $q \in [1, \infty]$, $m \in \mathbb{Z}$, every $u \in H^1(\square_m)$ and $\mathbf{h} \in L_{\text{sol}}^2(\square_m)$,*

$$(2.43) \quad 3^{-sm} [\nabla u]_{\dot{B}_{2,q}^{-s}(\square_m)} \leq (1 - 3^{-sq})^{-1/q} \lambda_{s,q}^{-1/2}(\square_m; \mathbf{a}) \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)},$$

and

$$(2.44) \quad 3^{-sm} [\mathbf{h}]_{\dot{B}_{2,q}^{-s}(\square_m)} \leq (1 - 3^{-sq})^{-1/q} \Lambda_{s,q}^{1/2}(\square_m; \mathbf{a}) \|\mathbf{a}^{-1/2} \mathbf{h}\|_{\underline{L}^2(\square_m)}.$$

We combine the functional inequality

$$(2.45) \quad [u]_{\dot{B}_{2,\infty}^{1-s}(\square_0)} \leq C(d) [\nabla u]_{\dot{B}_{2,1}^{-s}(\square_0)},$$

with (2.43) and the bound $\|u - (u)_{\square_0}\|_{\underline{L}^2(\square_0)} \leq C \liminf_{s \uparrow 1} [u]_{\dot{B}_{2,\infty}^{1-s}(\square_0)}$, and let $s \uparrow 1$. This gives, for some $C(d) < \infty$ and every $u \in H^1(\square_0)$,

$$(2.46) \quad \|u - (u)_{\square_0}\|_{\underline{L}^2(\square_0)} \leq C \lambda_{1,1}^{-1/2}(\square_0; \mathbf{a}) \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)}.$$

For zero boundary data we also have the following estimate. Let $w \in H_0^1(\square_0)$ be the **torsion function** of $\square_0$, that is, the solution of

$$\begin{cases} -\Delta w = 1 & \text{in } \square_0, \\ w = 0 & \text{on } \partial \square_0. \end{cases}$$

Then, for every $u \in H_0^1(\square_0)$,

$$|(u)_{\square_0}| = \left| \int_{\square_0} \nabla w \cdot \nabla u \right| \leq C [\nabla w]_{\dot{B}_{2,\infty}^1(\square_0)} [\nabla u]_{\dot{B}_{2,1}^{-1}(\square_0)} \leq C [\nabla u]_{\dot{B}_{2,1}^{-1}(\square_0)}.$$

Combining this with (2.46) yields, for some $C(d) < \infty$ and every $u \in H_0^1(\square_0)$,

$$(2.47) \quad \|u\|_{\underline{L}^2(\square_0)} \leq C \lambda_{1,1}^{-1/2}(\square_0; \mathbf{a}) \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)}.$$

The homogeneous case of the next proposition is [Armstrong and Kuusi \[2025, Proposition 2.5, p. 29\]](#); the forcing term follows from [Lemma 2.10](#) below.

Proposition 2.6 (Coarse-grained Caccioppoli inequality). *There exists a constant $C(d) < \infty$ such that, for every $s \in (0, 1)$, $t \in (0, 1/2)$ with $s + t < 1$, every coefficient field $\mathbf{a} \in L^\infty(\square_0; \mathbb{R}_{\text{sym},+}^{d \times d})$ and every $u \in H^1(\square_0)$ and $\mathbf{f} \in H^{2t}(\square_0; \mathbb{R}^d)$ satisfying*

$$-\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{f} \quad \text{in } \square_0,$$

we have the estimate

$$\begin{aligned} \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2(\square_{-1})}^2 &\leq \left(\frac{C}{1-s-t} \right)^{2+4s/(1-s-t)} \left(\frac{\Lambda_{s,1}(\square_0; \mathbf{a})}{\lambda_{t,1}(\square_0; \mathbf{a})} \right)^{s/(1-s-t)} \Lambda_{s,1}(\square_0; \mathbf{a}) \|u\|_{\underline{L}^2(\square_0)}^2 \\ &\quad + t^{-11} \left(\frac{C}{1-s-t} \right)^{2+4s/(1-s-t)} \left(\frac{\Lambda_{s,1}(\square_0; \mathbf{a})}{\lambda_{t,1}(\square_0; \mathbf{a})} \right)^{(1-t)/(1-s-t)} \lambda_{t,1}^{-1}(\square_0; \mathbf{a}) \|\mathbf{f}\|_{\underline{H}^{2t}(\square_0)}^2. \end{aligned}$$

Proof. Apply [Lemma 2.10](#) with zero boundary datum on each cube of a fixed finite cover of $\square_{-1}$ by cubes of side 3^{-2} compactly contained in $\square_0$, and sum the resulting estimates. The forcing term has the displayed exponent since $(1-t)/(1-s-t) = 1 + s/(1-s-t)$. $\square$

2.4. Sensitivity estimates. We quantify the dependence of J , $\mathbf{a}(U)$, and $\mathbf{a}_*(U)$ on a scalar coefficient field.

Lemma 2.7. *Let U be a bounded convex domain and let $\mathbf{a}, \mathbf{b} : U \rightarrow (0, \infty)$ be scalar, uniformly elliptic coefficient fields. Then*

$$|\mathbf{b}_*^{-1/2}(U) \mathbf{a}_*(U) \mathbf{b}_*^{-1/2}(U) - \text{Id}| \leq \|\mathbf{a} \mathbf{b}^{-1} - 1\|_{L^\infty(U)} \vee \|\mathbf{a}^{-1} \mathbf{b} - 1\|_{L^\infty(U)},$$

and

$$|\mathbf{b}^{-1/2}(U) \mathbf{a}(U) \mathbf{b}^{-1/2}(U) - \text{Id}| \leq \|\mathbf{a} \mathbf{b}^{-1} - 1\|_{L^\infty(U)} \vee \|\mathbf{a}^{-1} \mathbf{b} - 1\|_{L^\infty(U)}.$$

Proof. Both inequalities follow from the more precise bounds

$$(2.48) \quad \begin{cases} (\inf_U (\mathbf{a} \mathbf{b}^{-1})) \mathbf{b}(U) \leq \mathbf{a}(U) \leq (\sup_U (\mathbf{a} \mathbf{b}^{-1})) \mathbf{b}(U), \\ (\inf_U (\mathbf{a}^{-1} \mathbf{b})) \mathbf{b}_*^{-1}(U) \leq \mathbf{a}_*^{-1}(U) \leq (\sup_U (\mathbf{a}^{-1} \mathbf{b})) \mathbf{b}_*^{-1}(U). \end{cases}$$

For the upper bound in the first line of (2.48), the variational formula (2.12) gives, for every $e \in \mathbb{R}^d$,

$$\begin{aligned} \frac{1}{2} e \cdot \mathbf{a}(U) e &= \inf_{v \in e \cdot x + H_0^1(U)} \int_U \frac{1}{2} \nabla v \cdot \mathbf{a} \nabla v \leq \sup_U (\mathbf{a} \mathbf{b}^{-1}) \inf_{v \in e \cdot x + H_0^1(U)} \int_U \frac{1}{2} \nabla v \cdot \mathbf{b} \nabla v \\ &= \sup_U (\mathbf{a} \mathbf{b}^{-1}) \frac{1}{2} e \cdot \mathbf{b}(U) e. \end{aligned}$$

The dual variational formula (2.13) gives, for every $e \in \mathbb{R}^d$,

$$\begin{aligned} \frac{1}{2} e \cdot \mathbf{a}_*^{-1}(U) e &= \min_{\mathbf{h} \in e + L_{\text{sol},0}^2(U)} \int_U \frac{1}{2} \mathbf{h} \cdot \mathbf{a}^{-1} \mathbf{h} \leq \sup_U (\mathbf{a}^{-1} \mathbf{b}) \min_{\mathbf{h} \in e + L_{\text{sol},0}^2(U)} \int_U \frac{1}{2} \mathbf{h} \cdot \mathbf{b}^{-1} \mathbf{h} \\ &= \sup_U (\mathbf{a}^{-1} \mathbf{b}) \frac{1}{2} e \cdot \mathbf{b}_*^{-1}(U) e. \end{aligned}$$

Interchanging $\mathbf{a}$ and $\mathbf{b}$ gives the lower bounds in (2.48). $\square$

Lemma 2.8. *Let $U \subseteq \mathbb{R}^d$ be a bounded convex domain and $\mathbf{a}, \mathbf{b} : U \rightarrow (0, \infty)$ be scalar, uniformly elliptic coefficient fields. Then, for every $\lambda \in (0, \infty)$, $p, q \in \mathbb{R}^d$ and $\delta \in (0, 1]$,*

$$(2.49) \quad \begin{aligned} & J(U, \lambda^{-1/2}p, \lambda^{1/2}q; \mathbf{b}) \\ & \leq (1 + \delta)J(U, p, q; \mathbf{a}) + \frac{3}{\delta} \left(\|\lambda \mathbf{b}^{-1} \mathbf{a} - 1\|_{L^\infty(U)}^2 + \|\lambda^{-1} \mathbf{b} \mathbf{a}^{-1} - 1\|_{L^\infty(U)}^2 \right) (J(U, p, q; \mathbf{a}) + p \cdot q). \end{aligned}$$

Proof. By (2.32), we may reduce to the case $\lambda = 1$. Let v and u be the maximizers of $J(U, p, 0; \mathbf{a})$ and $J(U, 0, q; \mathbf{a})$, respectively. Then $-v + c \in p \cdot x + H_0^1(U)$ for an additive constant c , and $\mathbf{a} \nabla v \in q + L_{\text{sol},0}^2(U)$. Therefore, by (2.12) and (2.13),

$$\begin{aligned} J(U, p, 0; \mathbf{b}) &= \min_{w \in p \cdot x + H_0^1(U)} \int_U \frac{1}{2} \nabla w \cdot \mathbf{b} \nabla w \leq \int_U \frac{1}{2} \nabla v \cdot \mathbf{b} \nabla v \\ &= J(U, p, 0; \mathbf{a}) + \int_U \frac{1}{2} \nabla v \cdot (\mathbf{b} \mathbf{a}^{-1} - 1) \mathbf{a} \nabla v, \end{aligned}$$

and

$$\begin{aligned} J(U, 0, q; \mathbf{b}) &= \min_{\mathbf{h} \in q + L_{\text{sol},0}^2(U)} \int_U \frac{1}{2} \mathbf{h} \cdot \mathbf{b}^{-1} \mathbf{h} \leq \int_U \frac{1}{2} \mathbf{a} \nabla u \cdot \mathbf{b}^{-1} \mathbf{a} \nabla u \\ &= J(U, 0, q; \mathbf{a}) + \int_U \frac{1}{2} \nabla u \cdot (\mathbf{a} \mathbf{b}^{-1} - 1) \mathbf{a} \nabla u. \end{aligned}$$

Hence,

$$J(U, p, q; \mathbf{b}) \leq J(U, p, q; \mathbf{a}) + \int_U \frac{1}{2} \nabla v \cdot (\mathbf{b} \mathbf{a}^{-1} - 1) \mathbf{a} \nabla v + \int_U \frac{1}{2} \nabla u \cdot (\mathbf{a} \mathbf{b}^{-1} - 1) \mathbf{a} \nabla u.$$

Since

$$\mathbf{a} \mathbf{b}^{-1} - 1 = -\mathbf{a} \mathbf{b}^{-1} (\mathbf{b} \mathbf{a}^{-1} - 1) = -(\mathbf{b} \mathbf{a}^{-1} - 1) + \mathbf{a} \mathbf{b}^{-1} (\mathbf{b} \mathbf{a}^{-1} - 1)^2,$$

we have

$$\int_U \frac{1}{2} \nabla u (\mathbf{a} \mathbf{b}^{-1} - 1) \cdot \mathbf{a} \nabla u \leq - \int_U \frac{1}{2} \nabla u (\mathbf{b} \mathbf{a}^{-1} - 1) \cdot \mathbf{a} \nabla u + \|\mathbf{a} \mathbf{b}^{-1} (\mathbf{b} \mathbf{a}^{-1} - 1)^2\|_{L^\infty(U)} J(U, 0, q; \mathbf{a}).$$

Pointwise, $\mathbf{a} \mathbf{b}^{-1} (\mathbf{b} \mathbf{a}^{-1} - 1)^2 \leq 2(\mathbf{a} \mathbf{b}^{-1} - 1)^2 + 2(\mathbf{b} \mathbf{a}^{-1} - 1)^2$, and, hence,

$$\begin{aligned} & \|\mathbf{a} \mathbf{b}^{-1} (\mathbf{b} \mathbf{a}^{-1} - 1)^2\|_{L^\infty(U)} J(U, 0, q; \mathbf{a}) \\ & \leq 2 \left(\|\mathbf{b} \mathbf{a}^{-1} - 1\|_{L^\infty(U)}^2 + \|\mathbf{b}^{-1} \mathbf{a} - 1\|_{L^\infty(U)}^2 \right) (J(U, p, q; \mathbf{a}) + p \cdot q). \end{aligned}$$

By Hölder's and Young's inequalities, for every $\delta > 0$,

$$\begin{aligned} & \int_U \frac{1}{2} \nabla v (\mathbf{b} \mathbf{a}^{-1} - 1) \cdot \mathbf{a} \nabla v - \int_U \frac{1}{2} \nabla u (\mathbf{b} \mathbf{a}^{-1} - 1) \cdot \mathbf{a} \nabla u \\ &= \int_U \frac{1}{2} \mathbf{a}^{1/2} \nabla (v + u) (\mathbf{b} \mathbf{a}^{-1} - 1) \cdot \mathbf{a}^{1/2} \nabla (v - u) \\ &\leq \|\mathbf{b} \mathbf{a}^{-1} - 1\|_{L^\infty(U)} \left(J(U, p, q; \mathbf{a}) \right)^{1/2} 2^{-1/2} \|\mathbf{a}^{1/2} \nabla (v - u)\|_{\underline{L}^2(U)} \\ &\leq \delta J(U, p, q; \mathbf{a}) + \frac{1}{8\delta} \|\mathbf{b} \mathbf{a}^{-1} - 1\|_{L^\infty(U)}^2 \|\mathbf{a}^{1/2} \nabla (v - u)\|_{\underline{L}^2(U)}^2 \\ &\leq \delta J(U, p, q; \mathbf{a}) + \frac{1}{2\delta} \|\mathbf{b} \mathbf{a}^{-1} - 1\|_{L^\infty(U)}^2 (J(U, p, 0; \mathbf{a}) + J(U, 0, q; \mathbf{a})). \end{aligned}$$

Combining these displays gives (2.49). □

2.5. General coarse-graining estimates. Proposition 2.11 compares heterogeneous and constant-coefficient equations with the same data. We use the special case for symmetric coefficients of Armstrong et al. [2026, Lemmas 2.11–2.12 and Proposition 2.21] (version 2); for such coefficients, the boundary estimate below has a direct proof.

The first lemma is taken from Armstrong et al. [2026, Lemma 2.11, p. 39].

Lemma 2.9 (Coarse-grained estimates with a divergence-form right-hand side). *There exists $C(d) < \infty$ such that the following holds. Fix $m \in \mathbb{N}$ and $s \in (0, 1)$. Let $\mathbf{a} : \square_m \rightarrow \mathbb{R}_{\text{sym},+}^{d \times d}$ be uniformly elliptic and $\mathbf{g} \in H^s(\square_m; \mathbb{R}^d)$. If $u \in H^1(\square_m)$ solves*

$$-\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{g} \quad \text{in } \square_m,$$

then we have the estimates

$$3^{-sm} \|\nabla u\|_{\dot{B}_{2,2}^{-s}(\square_m)} \leq C s^{-3/2} \lambda_{s/2,2}^{-1/2}(\square_m; \mathbf{a}) \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} + C s^{-3} \lambda_{s/2,2}^{-1}(\square_m; \mathbf{a}) 3^{sm} [\mathbf{g}]_{\underline{H}^s(\square_m)},$$

$$3^{-sm} \|\mathbf{a} \nabla u\|_{\dot{B}_{2,2}^{-s}(\square_m)} \leq C s^{-3/2} \Lambda_{s/2,2}^{1/2}(\square_m; \mathbf{a}) \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} + C s^{-9/2} \frac{\Lambda_{s/2,2}^{1/2}(\square_m; \mathbf{a})}{\lambda_{s/2,2}^{1/2}(\square_m; \mathbf{a})} 3^{sm} [\mathbf{g}]_{\underline{H}^s(\square_m)}.$$

Moreover, for every $h \in H^{1+s}(\square_m)$, the solutions $v, w \in H^1(\square_m)$ of the Dirichlet and Neumann problems

$$\begin{cases} -\nabla \cdot \mathbf{a} \nabla v = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ v = h & \text{on } \partial \square_m, \end{cases} \quad \text{and} \quad \begin{cases} -\nabla \cdot \mathbf{a} \nabla w = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ \mathbf{n} \cdot (\mathbf{a} \nabla w + \mathbf{g} - (\mathbf{g})_{\square_m}) = 0 & \text{on } \partial \square_m \end{cases}$$

satisfy the estimate

$$(2.50) \quad \begin{aligned} & \|\mathbf{a}^{1/2} \nabla v\|_{\underline{L}^2(\square_m)} + \|\mathbf{a}^{1/2} \nabla w\|_{\underline{L}^2(\square_m)} \\ & \leq C s^{-3} \lambda_{s/2,2}^{-1/2}(\square_m; \mathbf{a}) 3^{sm} [\mathbf{g}]_{\underline{H}^s(\square_m)} + C s^{-3/2} \Lambda_{s/2,2}^{1/2}(\square_m; \mathbf{a}) 3^{sm} \|\nabla h\|_{\underline{H}^s(\square_m)}. \end{aligned}$$

The same estimates may be used at scale zero without changing the positive-scale statement of the lemma. Given data on $\square_0$, apply the lemma on $\square_1$ to

$$\mathbf{a}'(y) = \mathbf{a}(y/3), \quad u'(y) = 3u(y/3), \quad \mathbf{g}'(y) = \mathbf{g}(y/3), \quad h'(y) = 3h(y/3).$$

The weak equations and boundary conditions are preserved. The gradients and fluxes are unchanged under this dilation, the coarse-grained ellipticity constants have the same values, and each order- s seminorm or negative norm gains precisely the factor canceled by 3^{sm} or 3^{-sm} in the estimates. Thus changing variables gives the estimates on $\square_0$ used below.

The next lemma is a symmetric-coefficient adaptation of the boundary Caccioppoli estimate in Armstrong et al. [2026, Lemma 2.12, p. 42]. Its direct proof below gives the boundary factor t^{-3} and does not require the additional factor $(1-2t)^{-1}$ in that source.

Lemma 2.10 (Coarse-grained Caccioppoli inequality with right-hand side). *There exists $C(d) < \infty$ such that, for every $s \in (0, 1)$, $t \in (0, 1/2)$ with $\sigma := 1-s-t > 0$, every $x \in \square_0$, every $\mathbf{a} \in L^\infty(\square_0; \mathbb{R}_{\text{sym},+}^{d \times d})$, $u \in H^1(\square_0)$, $\mathbf{g} \in H^{2t}(\square_0; \mathbb{R}^d)$ and $h \in H^{1+2t}(\square_0)$ with $(h)_{\square_0} = 0$ satisfying*

$$\begin{cases} -\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{g} & \text{in } \square_0, \\ u = h & \text{on } (\partial \square_0) \cap (x + \square_{-1}), \end{cases}$$

we have the estimate

$$\begin{aligned}
 & \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2((x+\square_{-2}) \cap \square_0)}^2 \\
 & \leq \left(\frac{C}{\sigma}\right)^{2+4s/\sigma} \left(\frac{\Lambda_s(\square_0; \mathbf{a})}{\lambda_t(\square_0; \mathbf{a})}\right)^{(s+\sigma)/\sigma} \left(\lambda_t(\square_0; \mathbf{a}) \|u\|_{\underline{L}^2(\square_0)}^2 + t^{-11} \lambda_t^{-1}(\square_0; \mathbf{a}) \|\mathbf{g}\|_{\underline{H}^{2t}(\square_0)}^2 \right. \\
 (2.51) \quad & \left. + t^{-3} \Lambda_t(\square_0; \mathbf{a}) \|\nabla h\|_{\underline{H}^{2t}(\square_0)}^2\right).
 \end{aligned}$$

Proof. In this proof, all norms and coarse-grained ellipticity constants are taken on $\square_0$ unless otherwise indicated, and $\|\cdot\|_2$ denotes $\|\cdot\|_{\underline{L}^2(\square_0)}$. Let v solve the Dirichlet problem in $\square_0$ with forcing $\nabla \cdot \mathbf{g}$ and boundary values h on all of $\partial \square_0$, and put $V = \|\mathbf{a}^{1/2} \nabla v\|_2$. By subadditivity, $\Lambda_{t,2} \leq \Lambda_t$ and $\lambda_{t,2}^{-1} \leq \lambda_t^{-1}$. Consequently, (2.50), applied with $2t < 1$ in place of s , gives

$$V \leq Ct^{-3} \lambda_t^{-1/2} \|\mathbf{g}\|_{\underline{H}^{2t}} + Ct^{-3/2} \Lambda_t^{1/2} \|\nabla h\|_{\underline{H}^{2t}}.$$

Since $v - h \in H_0^1(\square_0)$ and $(h)_{\square_0} = 0$, testing $v - h$ against the torsion function of $\square_0$, as in the argument preceding (2.47), controls $(v)_{\square_0}$ by the negative Besov seminorms of ∇v and ∇h . Combining this bound with (2.46) and with (2.43) for $s = q = 1$, we obtain $\|v\|_2 \leq C \lambda_t^{-1/2} V + C \|\nabla h\|_2$. Apply the homogeneous boundary Caccioppoli inequality [Armstrong et al., 2026, Proposition 2.9, p. 36] to $u - v$, which has zero trace on the part of $\partial \square_0$ where the boundary condition is imposed. Put

$$B = \left(\frac{C}{\sigma}\right)^{2+4s/\sigma} \left(\frac{\Lambda_s}{\lambda_t}\right)^{s/\sigma} \Lambda_s.$$

Put $D = (x + \square_{-2}) \cap \square_0$. Since $x \in \square_0$, each side of D has length at least $1/18$, so $|\square_0|/|D| \leq 18^d$. Hence $\|\mathbf{a}^{1/2} \nabla v\|_{\underline{L}^2(D)}^2 \leq 18^d V^2$. The triangle inequality now bounds the left side of (2.51) by $CB(\|u\|_2^2 + \|v\|_2^2) + C(d)V^2$. Insert the preceding estimates and absorb this dimensional multiple of V^2 to obtain (2.51). $\square$

The next proposition is the general coarse-graining estimate from Armstrong et al. [2026, Proposition 2.21, p. 54].

Proposition 2.11 (General coarse-graining estimate). *Let $m, n \in \mathbb{Z}$ with $n < m$, let $\mathbf{a} \in L^\infty(\square_m; \mathbb{R}_{\text{sym},+}^{d \times d})$ and $\mathbf{a}_0 \in (0, \infty)$. Fix $p \in [2, \infty)$, $s, s_1, s_2 \in (0, 1)$ with $s_1 < s < s_2 < 1$ and $sp' < 1$, where $p' = p/(p-1)$. There exists $C(p, d) < \infty$ such that, for every $\mathbf{g} \in W^{s_2, p}(\square_m; \mathbb{R}^d)$ and $u, v \in H^1(\square_m)$ satisfying*

$$\begin{cases} -\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ -\nabla \cdot \mathbf{a}_0 \nabla v = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ u - v \in H_0^1(\square_m), \end{cases}$$

we have the estimate

$$\begin{aligned}
 & 3^{-ms} \mathbf{a}_0 \|\nabla u - \nabla v\|_{\widehat{W}^{-s, p}(\square_m)} + 3^{-ms} \|\mathbf{a} \nabla u - \mathbf{a}_0 \nabla v\|_{\widehat{W}^{-s, p}(\square_m)} \\
 & \leq Cs^{-1-1/p'} \mathbf{a}_0^{1/2} \mathcal{E}_{s_1, \infty, 1}(\square_m, n; \mathbf{a}, \mathbf{a}_0) \left(\sum_{k=-\infty}^n 3^{-(s-s_1)p(n-k)} \sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_k)}^p \right)^{1/p} \\
 (2.52) \quad & + Cs^{-11/2} (s_2 - s)^{-1} (1 + \mathcal{E}_{s_1/2, \infty, 2}^2(\square_m, n; \mathbf{a}, \mathbf{a}_0)) 3^{s_2 n} [\mathbf{g}]_{\underline{W}^{s_2, p}(\square_m)}.
 \end{aligned}$$

To explain the normalization in this estimate, write subscripts here and ASD only in this paragraph to distinguish the two conventions. The fractional seminorm in Armstrong et al. [2026] omits the leading factor s used here, so

$$[g]_{\underline{W}^{s_2, p}, \text{here}} = s_2^{1/p} [g]_{\underline{W}^{s_2, p}, \text{ASD}}, \quad \|F\|_{\underline{W}^{-s, p}, \text{here}} \leq s^{-1/p'} \|F\|_{\underline{W}^{-s, p}, \text{ASD}}.$$

The source uses a mean-zero dual norm for the gradient difference. Since $u - v \in H_0^1(\square_m)$, this gradient has zero mean; subtracting the mean of a test function therefore leaves its pairing unchanged and increases its positive full norm by at most a factor of two. The flux uses the full dual norm in both conventions. Consequently the source's factors s^{-1} and $s^{-9/2}(s_2 - s)^{-1}$ become, respectively,

$$s^{-1-1/p'}, \quad s^{-9/2-1/p'} s_2^{-1/p} (s_2 - s)^{-1} \leq s^{-11/2} (s_2 - s)^{-1}.$$

The last step uses $s < s_2$ and $1/p + 1/p' = 1$. The error quantity also agrees in the symmetric case: if $j(e) = J(U, \mathbf{a}_0^{-1/2}e, \mathbf{a}_0^{1/2}e; \mathbf{a})$, the doubled response in the source splits as $\frac{1}{2}j(e - f) + \frac{1}{2}j(e + f) = j(e) + j(f)$. Maximizing over $|e|^2 + |f|^2 = 1$ gives precisely the directional maximum used in $\mathcal{E}$. These observations give the powers and norms displayed in (2.52).

By Sobolev embedding (Appendix C), if $p - 2$ is small in terms of s , then the left side of (2.52) controls the quantity $\mathbf{a}_0 3^{-m} \|u - v\|_{L^2(\square_m)}$.

3. ESTIMATES FOR THE CUTOFF COEFFICIENTS

In this section we prove the probabilistic estimates on the cutoff coefficients used in the rest of the paper. They are of two kinds. Sensitivity bounds (Lemmas 3.1 and 3.3), the first of which underlies the induction of Section 4, measure how much the coefficient and its coarse-grained matrices change when the fields of a range of scales are added to the coefficient. Negative Besov bounds (Lemma 3.2) control the ratio $\mathbf{a}_m \mathbf{a}_n^{-1}$ of two cutoff coefficients. We also prove, in Proposition 3.5, the planar identity $\bar{\mathbf{a}}_m = e^{-\tau^2(m+1)}$, which gives the exact two-dimensional exponents.

Throughout this section, $\mathbb{P}$ is a probability measure satisfying Assumptions (g1)–(g4). On a given cube, a single field has little effect in two opposite situations. A field whose scale is much smaller than the cube oscillates on it, and is therefore small in a negative Besov norm on the cube. A field whose scale is much larger than the cube is nearly constant on it, and multiplying the coefficient by a constant does not change the solutions of the equation.

Stochastic bounds are written in the Orlicz notation: for $s, A > 0$ we write $X \leq \mathcal{O}_{\Gamma_s}(A)$ if $\mathbb{E}[\exp((A^{-1} \max\{X, 0\})^s)] \leq 2$, and $X = \mathcal{O}_{\Gamma_s}(A)$ if this holds for both X and $-X$; Appendix A collects the properties we use.

Recall from (1.7) that, for $L \in \mathbb{N}_0$, the cutoff coefficient is $\mathbf{a}_L(x) = \prod_{k=0}^L \exp(\mathbf{g}_k(x) - \tau^2)$. By (1.6), $\tau \leq C\bar{\delta}$. We also set $\mathbf{a}_{-1} \equiv 1$. For every $m \in \mathbb{N}_0$ and $n \in \mathbb{Z}$ with $-1 \leq n < m$, we have $\mathbb{E}[(\mathbf{a}_m \mathbf{a}_n^{-1})(0)] = 1$ and $\mathbb{E}[(\mathbf{a}_m^{-1} \mathbf{a}_n)(0)] = \exp(2\tau^2(m - n))$. Moreover,

$$(3.1) \quad \text{the fields } \mathbf{a}_m \mathbf{a}_n^{-1} \text{ and } \exp(-2\tau^2(m - n)) \mathbf{a}_m^{-1} \mathbf{a}_n \text{ have the same law.}$$

3.1. Besov space estimates. We first estimate the oscillation of the ratios $\mathbf{a}_m \mathbf{a}_n^{-1}$.

Lemma 3.1. *There exists a constant $C(d) < \infty$ such that, for every $n \in \mathbb{N}_0$ and $k \in \mathbb{Z}$,*

$$(3.2) \quad \sup_{m \geq n} \sup_{x, y \in \square_k} \log \left(\frac{(\mathbf{a}_m \mathbf{a}_n^{-1})(x)}{(\mathbf{a}_m \mathbf{a}_n^{-1})(y)} \right) \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta} 3^{k-n}),$$

and, for every $\xi \in [1, \infty)$,

$$(3.3) \quad \mathbb{E} \left[\left| \sup_{m \geq n} \sup_{x, y \in \square_k} \frac{(\mathbf{a}_m \mathbf{a}_n^{-1})(x)}{(\mathbf{a}_m \mathbf{a}_n^{-1})(y)} - 1 \right|^\xi \right] \leq C \xi^{1/2} \bar{\delta} 3^{k-n} \exp(C \xi \bar{\delta}^2 3^{2(k-n)}).$$

Moreover, for every $m \in \mathbb{N}_0$ with $m > n$,

$$(3.4) \quad \|\log(\mathbf{a}_m \mathbf{a}_n^{-1}) + (m - n)\tau^2\|_{L^\infty(\square_k)} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}(|m - n|^{1/2} + (k - n)_+)),$$

and, for every $\xi \in [1, \infty)$,

$$(3.5) \quad \mathbb{E} \left[\left\| \mathbf{a}_m \mathbf{a}_n^{-1} - 1 \right\|_{L^\infty(\square_k)}^\xi \right]^{1/\xi} + \mathbb{E} \left[\left\| \mathbf{a}_m^{-1} \mathbf{a}_n - 1 \right\|_{L^\infty(\square_k)}^\xi \right]^{1/\xi} \leq (C\xi^{1/2} \bar{\delta} ((m-n)^{1/2} + (k-n)_+)) \exp(C\xi \bar{\delta}^2 (m-n + (k-n)_+^2)).$$

Finally, for every bounded domain $U \subseteq \mathbb{R}^d$, every $p \in [1, \infty)$, and every $m \in \mathbb{N}_0$ and $n \in \mathbb{Z}$ with $-1 \leq n < m$,

$$(3.6) \quad \mathbb{E} \left[\left\| \mathbf{a}_m \mathbf{a}_n^{-1} - \text{Id} \right\|_{\underline{L}^p(U)}^p \right]^{1/p} \leq (Cp^{1/2} \bar{\delta} (m-n)^{1/2}) \exp(Cp \bar{\delta}^2 (m-n)),$$

and

$$(3.7) \quad \mathbb{E} \left[\left\| \mathbf{a}_m^{-1} \mathbf{a}_n - \text{Id} \right\|_{\underline{L}^p(U)}^p \right]^{1/p} \leq (Cp^{1/2} \bar{\delta} (m-n)^{1/2}) \exp(Cp \bar{\delta}^2 (m-n)).$$

Proof. We bound the logarithm $\log(\mathbf{a}_m \mathbf{a}_n^{-1}) = \sum_{j=n+1}^m (\mathbf{g}_j - \tau^2)$ on $\square_k$ and then exponentiate. Its oscillation gives (3.2) and (3.3). The main step is the bound (3.4) on its size: we split the sum at the scale of $\square_k$, use that the fields of larger scale are nearly constant on $\square_k$, and control the remaining fields on the subcubes of side 3^n by a bound on the maximum of many random variables. The moment bounds (3.5) and (3.6) then follow by exponentiating, and (3.7) follows from (3.1).

We first prove (3.2). By (1.5),

$$\sup_{m \geq n} \sup_{x, y \in \square_k} \log \frac{(\mathbf{a}_m \mathbf{a}_n^{-1})(x)}{(\mathbf{a}_m \mathbf{a}_n^{-1})(y)} \leq \sum_{j=n+1}^{\infty} \sup_{x, y \in \square_k} |\mathbf{g}_j(x) - \mathbf{g}_j(y)| \leq \mathcal{O}_{\Gamma_2}(C3^{k-n} \bar{\delta}).$$

Indeed, for $j \geq k$ the mean value theorem bounds the oscillation by $C3^k \|\nabla \mathbf{g}_j\|_{L^\infty(\square_k)} \leq \mathcal{O}_{\Gamma_2}(C3^{k-j} \bar{\delta})$. For $j < k$, cover $\square_k$ by the $3^{d(k-j)}$ cubes $z + \square_j$ with $z \in 3^j \mathbb{Z}^d \cap \square_k$ and use (A.4) to bound the oscillation by $2 \|\mathbf{g}_j\|_{L^\infty(\square_k)} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta} \sqrt{1+k-j}) \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta} 3^{k-j})$. Then (3.3) follows from (A.8), applied to the supremum in (3.2). Since $\log(\mathbf{a}_m \mathbf{a}_n^{-1}) + (m-n)\tau^2 = \sum_{j=n+1}^m \mathbf{g}_j$, the bound (3.4) follows once we prove, for a fixed $m \in \mathbb{N}_0$ with $m > n$, that

$$(3.8) \quad \sup_{x \in \square_k} \left| \sum_{j=n+1}^m \mathbf{g}_j(x) \right| \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}((m-n)^{1/2} + (k-n)_+)).$$

By (1.5), (A.5), and (A.3),

$$(3.9) \quad \sup_{x \in \square_k} \left| \sum_{j=k \vee (n+1)}^m \mathbf{g}_j(x) \right| \leq \underbrace{\left| \sum_{j=k \vee (n+1)}^m \mathbf{g}_j(0) \right|}_{\leq \mathcal{O}_{\Gamma_2}(\bar{\delta}(m-n)^{1/2})} + \sum_{j=k \vee (n+1)}^m \underbrace{\|\mathbf{g}_j - \mathbf{g}_j(0)\|_{L^\infty(\square_k)}}_{\leq \mathcal{O}_{\Gamma_2}(C3^{k-j} \bar{\delta})} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}(m-n)^{1/2}).$$

This proves (3.8) when $k \leq n$. Suppose that $k > n$. The same three estimates, now on $\square_n$ and with the sums stopped at $m \wedge k$, give

$$\sup_{x \in \square_n} \left| \sum_{j=n+1}^{m \wedge k} \mathbf{g}_j(x) \right| \leq \underbrace{\left| \sum_{j=n+1}^{m \wedge k} \mathbf{g}_j(0) \right|}_{\mathcal{O}_{\Gamma_2}(\bar{\delta}((m \wedge k) - n)^{1/2})} + \sum_{j=n+1}^{m \wedge k} \underbrace{\|\mathbf{g}_j - \mathbf{g}_j(0)\|_{L^\infty(\square_n)}}_{\mathcal{O}_{\Gamma_2}(C3^{n-j} \bar{\delta})} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}((m \wedge k) - n)^{1/2}).$$

The maximum of $N \geq 2$ random variables, each of which is $\mathcal{O}_{\Gamma_2}(A)$, is $\mathcal{O}_{\Gamma_2}((3 \log N)^{1/2} A)$; this is [Armstrong et al. \[2024, Lemma 2.3\]](#). Applying it with $N = 3^{d(k-n)}$ gives

$$\sup_{x \in \square_k} \left| \sum_{j=n+1}^{m \wedge k} \mathbf{g}_j(x) \right| = \max_{z \in 3^n \mathbb{Z}^d \cap \square_k} \left\| \sum_{j=n+1}^{m \wedge k} \mathbf{g}_j \right\|_{L^\infty(z + \square_n)} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}(k-n)).$$

If $m > k$, then, by (3.9), the remaining sum $\sum_{j=k+1}^m \mathbf{g}_j$ contributes at most $\mathcal{O}_{\Gamma_2}(C\bar{\delta}(m-n)^{1/2})$ on $\square_k$. This proves (3.8).

To prove (3.5), we use

$$\|\mathbf{a}_m \mathbf{a}_n^{-1} - 1\|_{L^\infty(\square_k)} \vee \|\mathbf{a}_m^{-1} \mathbf{a}_n - 1\|_{L^\infty(\square_k)} \leq \|\log(\mathbf{a}_m \mathbf{a}_n^{-1})\|_{L^\infty(\square_k)} \exp(\|\log(\mathbf{a}_m \mathbf{a}_n^{-1})\|_{L^\infty(\square_k)}).$$

Equations (1.6) and (3.4), together with (A.8) and (A.6) at exponent 2ξ , bound the $L^{2\xi}(\mathbb{P})$ norms of the random variable $\|\log(\mathbf{a}_m \mathbf{a}_n^{-1})\|_{L^\infty(\square_k)}$ and of its exponential. Hölder's inequality then gives (3.5).

At a fixed point, the pointwise version of (3.4) and the exponential moment bound used above give

$$\|\mathbf{a}_m(x) \mathbf{a}_n^{-1}(x) - 1\|_{L^p(\mathbb{P})} \leq Cp^{1/2} \bar{\delta}(m-n)^{1/2} \exp(Cp\bar{\delta}^2(m-n)).$$

For $n = -1$, separate the independent factor $e^{\mathbf{g}_0 - \tau^2}$ and apply the estimate with $n = 0$. Stationarity now gives (3.6). Finally, (3.1) and (1.6) give (3.7). $\square$

Lemma 3.2. *There exist $C(d) < \infty$ and $c(d) > 0$ such that, for every $s \in (0, 1]$ and $p \in [2, \infty)$ satisfying*

$$(3.10) \quad s \geq Cp\bar{\delta}^2 \quad \text{and} \quad p\bar{\delta}^2 \log(2+p) \leq c,$$

and every $m \in \mathbb{N}_0$ and $n \in \mathbb{Z}$ with $-1 \leq n < m$, we have the estimates

$$(3.11) \quad 3^{-sm} \mathbb{E} \left[\|\mathbf{a}_m \mathbf{a}_n^{-1} - 1\|_{\dot{B}_{p,p}^{-s}(\square_m)}^p \right]^{1/p} \leq Cs^{-1} \bar{\delta} p^{1/2} (\log p),$$

and

$$(3.12) \quad 3^{-sm} \mathbb{E} \left[\|\exp(-2\tau^2(m-n)) \mathbf{a}_m^{-1} \mathbf{a}_n - 1\|_{\dot{B}_{p,p}^{-s}(\square_m)}^p \right]^{1/p} \leq Cs^{-1} \bar{\delta} p^{1/2} (\log p).$$

Proof. The estimates (3.11) and (3.12) are equivalent by (3.1); we prove (3.11). By (2.7), its left side is controlled by the p th moments of the averages of $\mathbf{a}_m \mathbf{a}_n^{-1} - 1$ on the cubes of size 3^j , $j \leq m$, and (3.14) below splits each such average into the contributions of the fields $\mathbf{g}_{k+1}$ with $n \leq k < m$. The proof proceeds in three steps. Step 1 bounds the contribution of one field $\mathbf{g}_{k+1}$ on a cube of size 3^j with $k \leq j \leq m$, Step 2 sums these bounds over $k < j$ and bounds the terms with $k \geq j$ directly, to bound the average on a single cube, and Step 3 sums over the scales j . The heart of the matter is the decay in $j - k$ obtained in Step 1, which comes from the independence of $\mathbf{g}_{k+1}$ from $\mathbf{a}_m \mathbf{a}_{k+1}^{-1}$ and keeps the sum over $k < j$ in Step 2 of order $\log(2+p)$.

When $n = -1$, the summand with $k = -1$ in the decomposition (3.14) below contains $\mathbf{g}_0$ and $\mathbf{a}_m \mathbf{a}_0^{-1}$. Assumptions (g1)–(g2), independence, and Lemma 3.1 with $n = 0$ give the bound required for this summand. Every remaining summand has $k \geq 0$. Here and below, Lemma 3.1 is applied to a ratio $\mathbf{a}_m \mathbf{a}_{k+1}^{-1}$ only when $k+1 < m$; when $k+1 = m$ (for instance, when $m = 0$ in the summand just treated), the ratio is identically 1 and the bounds used for it hold trivially.

By (2.7),

$$(3.13) \quad 3^{-sm} \mathbb{E} \left[\|\mathbf{a}_m \mathbf{a}_n^{-1} - 1\|_{\dot{B}_{p,p}^{-s}(\square_m)}^p \right]^{1/p} = \mathbb{E} \left[\sum_{j=-\infty}^m s 3^{sp(j-m)} \sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |(\mathbf{a}_m \mathbf{a}_n^{-1})_{z+\square_j} - 1|^p \right]^{1/p}.$$

For each $j \leq m$, we write

$$(3.14) \quad \oint_{\square_j} \mathbf{a}_m \mathbf{a}_n^{-1} - 1 = \sum_{k=n}^{m-1} \oint_{\square_j} (\mathbf{a}_m \mathbf{a}_k^{-1} - \mathbf{a}_m \mathbf{a}_{k+1}^{-1}) = \sum_{k=n}^{m-1} \oint_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1).$$

Step 1. A moment bound on the cubes $\square_j$ with $k \leq j \leq m$. We show the existence of $C(d) < \infty$ such that, for every $j \in \{k, \dots, m\}$,

$$(3.15) \quad \mathbb{E} \left[\left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \right]^{1/p} \leq C \bar{\delta} p^{1/2} \left(3^{-\frac{d}{2}(j-k)} + \min\{p 3^{-d(1-\frac{1}{p})(j-k)}, 1\} \right) \exp(C p \bar{\delta}^2 (m-k)).$$

For each $k \leq j$, we have

$$\int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) = \sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \int_{z + \square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1).$$

Condition on $\mathbf{a}_m \mathbf{a}_{k+1}^{-1}$, which is independent of the field $\mathbf{g}_{k+1}$. Since $\mathbb{E}[\exp(\mathbf{g}_{k+1}(x) - \tau^2)] = 1$ for every x , each summand then has conditional mean zero. We bound the conditional moments of one summand and then apply Lemma B.1 to the sum. By the Cauchy-Schwarz inequality, (3.5), (3.10), and (1.6),

$$(3.16) \quad \mathbb{E} \left[\left| \int_{\square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \leq \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(\square_k)} \mathbb{E} \left[\left(\int_{\square_k} |\exp(\mathbf{g}_{k+1} - \tau^2) - 1|^2 \right)^{p/2} \right]^{1/p} \leq C p^{1/2} \bar{\delta} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(\square_k)}.$$

In particular,

$$\begin{aligned} & \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \mathbb{E} \left[\left| \int_{z + \square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^2 \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right] \right)^{1/2} \\ & \leq C \bar{\delta} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(z + \square_k)}^2 \right)^{1/2} = C \bar{\delta} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(\square_j)}, \end{aligned}$$

and

$$\begin{aligned} & \mathbb{E} \left[\max_{z \in 3^k \mathbb{Z}^d \cap \square_j} \left| \int_{z + \square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \\ & \leq \mathbb{E} \left[\sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \left| \int_{z + \square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \\ & \leq C p^{1/2} \bar{\delta} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(z + \square_k)}^p \right)^{1/p} \leq C 3^{\frac{d}{p}(j-k)} p^{1/2} \bar{\delta} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^p(\square_j)}. \end{aligned}$$

Conditionally on $\mathbf{a}_m \mathbf{a}_{k+1}^{-1}$, the field $\mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1)$ has range of dependence at most $\sqrt{d} 3^{k+1}$. The summands indexed by $z \in 3^k \mathbb{Z}^d \cap \square_j$ can therefore be partitioned into $C(d)$

classes within each of which they are independent. Applying Lemma B.1 to each class, we obtain

$$\begin{aligned}
& \mathbb{E} \left[\left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \int_{z + \square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \\
& \leq Cp^{1/2} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \mathbb{E} \left[\left| \int_{z + \square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^2 \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right] \right)^{1/2} \\
& \quad + Cp \mathbb{E} \left[\max_{z \in 3^k \mathbb{Z}^d \cap \square_j} \left| \int_{z + \square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \\
& \leq C3^{\frac{d}{2}(j-k)} \bar{\delta} p^{1/2} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(\square_j)} + Cp^{3/2} \bar{\delta} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(z + \square_k)}^p \right)^{1/p}.
\end{aligned}$$

Dividing by $3^{d(j-k)}$ gives

$$\begin{aligned}
& \mathbb{E} \left[\left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \int_{z + \square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \\
& \leq C3^{-\frac{d}{2}(j-k)} \bar{\delta} p^{1/2} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(\square_j)} + C3^{-d(1-\frac{1}{p})(j-k)} p^{3/2} \bar{\delta} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^p(\square_j)}.
\end{aligned}$$

Separately, the triangle inequality and (3.16) give the cruder bound

$$\mathbb{E} \left[\left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \int_{z + \square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \leq Cp^{1/2} \bar{\delta} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^p(\square_j)}.$$

Combining the previous two displays yields

$$\begin{aligned}
& \mathbb{E} \left[\left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \\
& \leq C \bar{\delta} p^{1/2} 3^{-\frac{d}{2}(j-k)} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(\square_j)} + C \bar{\delta} p^{1/2} \min\{p3^{-d(1-\frac{1}{p})(j-k)}, 1\} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^p(\square_j)}.
\end{aligned}$$

Taking $L^p(\mathbb{P})$ norms and using (3.6) gives

$$\begin{aligned}
& \mathbb{E} \left[\left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \right]^{1/p} \\
& \leq C \bar{\delta} p^{1/2} \left(3^{-\frac{d}{2}(j-k)} + \min\{p3^{-d(1-\frac{1}{p})(j-k)}, 1\} \right) \mathbb{E} \left[\|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^p(\square_j)}^p \right]^{1/p} \\
& \leq C \bar{\delta} p^{1/2} \left(3^{-\frac{d}{2}(j-k)} + \min\{p3^{-d(1-\frac{1}{p})(j-k)}, 1\} \right) \exp(Cp \bar{\delta}^2(m-k)).
\end{aligned}$$

This is (3.15).

Step 2. A moment bound on a single cube. We show that, for every $j \leq m$,

$$(3.17) \quad \mathbb{E} \left[\left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_n^{-1} - 1 \right|^p \right]^{1/p} \leq C \bar{\delta} p^{1/2} (\log p) (1 + m - j) \exp(Cp \bar{\delta}^2(m - j)).$$

By the triangle inequality and (3.14) we have

$$\begin{aligned} & \left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_n^{-1} - 1 \right| \\ & \leq \sum_{k=n}^{j-1} \left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right| + \sum_{k=j}^{m-1} \left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|. \end{aligned}$$

We bound the first term on the right side by summing (3.15) over $k \in \{n, \dots, j-1\}$. The resulting sum, after factoring out $\exp(Cp\bar{\delta}^2(m-j))$, is at most $C \log(2+p)$ by (3.10). Thus

$$\begin{aligned} \mathbb{E} \left[\left| \sum_{k=n}^{j-1} \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \right]^{1/p} & \leq \sum_{k=n}^{j-1} \mathbb{E} \left[\left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \right]^{1/p} \\ & \leq C\bar{\delta}p^{1/2}(\log p) \exp(Cp\bar{\delta}^2(m-j)). \end{aligned}$$

For the second sum over $k \geq j$ we use the cruder bound coming from (3.5) and $\tau \leq C\bar{\delta}$, to obtain

$$\begin{aligned} & \mathbb{E} \left[\left(\sum_{k=j}^{m-1} \left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right| \right)^p \right]^{1/p} \\ & \leq \sum_{k=j}^{m-1} \mathbb{E} \left[\|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{L^\infty(\square_j)}^{2p} \right]^{1/2p} \mathbb{E} \left[\|(\exp(\mathbf{g}_{k+1} - \tau^2) - 1)\|_{L^\infty(\square_j)}^{2p} \right]^{1/2p} \\ & \leq \sum_{k=j}^{m-1} C\bar{\delta}p^{1/2} \exp(Cp\bar{\delta}^2(m-k)) \leq C\bar{\delta}p^{1/2}(m-j) \exp(Cp\bar{\delta}^2(m-j)). \end{aligned}$$

Combining these yields (3.17) when $j \geq n$.

Suppose now that $j < n$. By (3.14), the triangle inequality, independence of $\mathbf{g}_{k+1}$ from $\mathbf{a}_m \mathbf{a}_{k+1}^{-1}$, Assumption (g2), and Lemma 3.1,

$$\begin{aligned} \mathbb{E} \left[\left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_n^{-1} - 1 \right|^p \right]^{1/p} & \leq \sum_{k=n}^{m-1} \mathbb{E} \left[\left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \right]^{1/p} \\ & \leq C\bar{\delta}p^{1/2}(m-n) \exp(Cp\bar{\delta}^2(m-n)) \\ & \leq C\bar{\delta}p^{1/2}(1+m-j) \exp(Cp\bar{\delta}^2(m-j)). \end{aligned}$$

This proves (3.17) for every $j \leq m$.

Step 3. We sum over the scales to prove (3.11). Returning to (3.13) and applying (3.17) and the triangle inequality, we obtain

$$\begin{aligned} & 3^{-spm} \mathbb{E} \left[\|\mathbf{a}_m \mathbf{a}_n^{-1} - 1\|_{\underline{B}_{p,p}^{-s}(\square_m)}^p \right] \\ & \leq C \sum_{j=-\infty}^m s 3^{sp(j-m)} \mathbb{E} \left[\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |(\mathbf{a}_m \mathbf{a}_n^{-1})_{z+\square_j} - 1|^p \right] \\ (3.18) \quad & \leq s(C\bar{\delta}p^{1/2}(\log p))^p \sum_{j=-\infty}^m (1+m-j)^p \exp((\log 3)sp - Cp^2\bar{\delta}^2)(j-m). \end{aligned}$$

By (3.10), after increasing the constant there, the exponent in the last line of (3.18) satisfies $(\log 3)sp - Cp^2\bar{\delta}^2 \geq \frac{1}{2}(\log 3)sp$, so we obtain

$$\begin{aligned} 3^{-sm} \mathbb{E} \left[\left\| \mathbf{a}_m \mathbf{a}_n^{-1} - 1 \right\|_{\dot{B}_{p,p}^{-s}(\square_m)}^p \right]^{1/p} &\leq Cs^{1/p} \bar{\delta} p^{1/2} (\log p) \left(\sum_{j=-\infty}^m (1+m-j)^p \exp\left(\frac{1}{2}sp(j-m)\right) \right)^{1/p} \\ &\leq Cs^{-1} \bar{\delta} p^{1/2} (\log p). \end{aligned}$$

This is (3.11), which is equivalent to (3.12), as noted at the beginning of the proof. $\square$

3.2. Sensitivity estimates for coarse-grained matrices.

Lemma 3.3. *There exists a constant $C(d) < \infty$ such that, for every $n \in \mathbb{N}_0$ and $k \in \mathbb{Z}$, every $\xi \in [1, \infty)$, and every convex domain $U \subseteq \square_k$,*

$$\begin{aligned} (3.19) \quad \mathbb{E} \left[\sup_{L \geq n} \left(\left| \frac{\mathbf{a}_{L,*}(U)^{-1/2} \mathbf{a}_{n,*}(U) \mathbf{a}_{L,*}(U)^{-1/2}}{(\mathbf{a}_n \mathbf{a}_L^{-1})(0)} - \mathbf{I}_d \right| + \left| \frac{\mathbf{a}_L(U)^{-1/2} \mathbf{a}_n(U) \mathbf{a}_L(U)^{-1/2}}{(\mathbf{a}_n \mathbf{a}_L^{-1})(0)} - \mathbf{I}_d \right| \right)^\xi \right]^{1/\xi} \\ + \mathbb{E} \left[\sup_{L \geq n} \left(\left| \frac{\mathbf{a}_{n,*}(U)^{-1/2} \mathbf{a}_{L,*}(U) \mathbf{a}_{n,*}(U)^{-1/2}}{(\mathbf{a}_L \mathbf{a}_n^{-1})(0)} - \mathbf{I}_d \right| + \left| \frac{\mathbf{a}_n(U)^{-1/2} \mathbf{a}_L(U) \mathbf{a}_n(U)^{-1/2}}{(\mathbf{a}_L \mathbf{a}_n^{-1})(0)} - \mathbf{I}_d \right| \right)^\xi \right]^{1/\xi} \\ \leq C \xi^{1/2} \bar{\delta} 3^{k-n} \exp(C \xi \bar{\delta}^2 3^{2(k-n)}), \end{aligned}$$

and, for every $m \in \mathbb{N}_0$ with $m > n$,

$$\begin{aligned} (3.20) \quad \mathbb{E} \left[\left| \mathbf{a}_{m,*}(U)^{-1/2} \mathbf{a}_{n,*}(U) \mathbf{a}_{m,*}(U)^{-1/2} - \mathbf{I}_d \right|^\xi + \left| \mathbf{a}_m(U)^{-1/2} \mathbf{a}_n(U) \mathbf{a}_m(U)^{-1/2} - \mathbf{I}_d \right|^\xi \right]^{1/\xi} \\ + \mathbb{E} \left[\left| \mathbf{a}_{n,*}(U)^{-1/2} \mathbf{a}_{m,*}(U) \mathbf{a}_{n,*}(U)^{-1/2} - \mathbf{I}_d \right|^\xi + \left| \mathbf{a}_n(U)^{-1/2} \mathbf{a}_m(U) \mathbf{a}_n(U)^{-1/2} - \mathbf{I}_d \right|^\xi \right]^{1/\xi} \\ \leq (C \xi^{1/2} \bar{\delta} ((m-n)^{1/2} + (k-n)_+)) \exp(C \xi \bar{\delta}^2 (m-n + (k-n)_+^2)). \end{aligned}$$

Proof. Apply Lemmas 3.1 and 2.7 on $U \subseteq \square_k$, first to $(\mathbf{a}_n, (\mathbf{a}_n \mathbf{a}_L^{-1})(0) \mathbf{a}_L)$ and then with the two coefficients interchanged. Taking the supremum over $L \geq n$ and using (3.3) gives (3.19). For (3.20), apply Lemma 2.7 directly to the unnormalized pair $(\mathbf{a}_n, \mathbf{a}_m)$, in both orders, and use (3.5). $\square$

For $m \in \mathbb{N}_0$, take the expectation of $\mathbf{a}_m(U)$ and the inverse of the expectation of $\mathbf{a}_{m,*}^{-1}(U)$:

$$(3.21) \quad \bar{\mathbf{a}}_m(U) := \mathbb{E}[\mathbf{a}_m(U)] \quad \text{and} \quad \bar{\mathbf{a}}_{m,*}(U) := \mathbb{E}[\mathbf{a}_{m,*}^{-1}(U)]^{-1}.$$

By subadditivity, the matrices $\bar{\mathbf{a}}_m(\square_n)$ and $\bar{\mathbf{a}}_{m,*}^{-1}(\square_n)$ are nonincreasing in n in the Loewner order. Using (2.14) and (2.16), the limits of $\bar{\mathbf{a}}_m(\square_n)$ and $\bar{\mathbf{a}}_{m,*}(\square_n)$ as $n \rightarrow \infty$ exist. We define

$$(3.22) \quad \bar{\mathbf{a}}_m := \lim_{n \rightarrow \infty} \bar{\mathbf{a}}_m(\square_n).$$

For each fixed m , these two limits agree: the ratio b_n/c_n below is the renormalized ellipticity ratio of $\mathbf{a}_m$ at scale 3^n in the sense of Armstrong and Kuusi [2025, (2.106), p. 41], and tends to one by Armstrong and Kuusi [2025, Theorem 3.1, pp. 52–53]. We verify the hypotheses of that theorem for the fixed-cutoff field. After a fixed spatial rescaling the field is stationary with finite range of dependence, and the supremum of $|\log \mathbf{a}_m|$ on each fixed cube has a sub-Gaussian bound by Assumption (g2) and the finiteness of the sum defining $\mathbf{a}_m$. Thus Proposition D.3 of Armstrong and Kuusi [2025, pp. 145–146], with exponent $\gamma = 1/4$, supplies the coarse-grained ellipticity assumption (P2 $\dagger$) of that theorem [Armstrong and Kuusi, 2025, Section 2.6, p. 33]. The theorem's stationarity and mixing assumptions (P1) and (P3) [Armstrong and Kuusi, 2025, Section 1.2, pp. 7–9] hold with $\beta = 0$ and $\nu = d/2 > \gamma$. All constants in this application may depend on the fixed cutoff m . The same verification, for the normalized coefficient, is used in the proof of Theorem 6.20 in Section 6.

By the symmetries of Assumption (g3), write $\bar{\mathbf{a}}_m(\square_n) = b_n \mathbf{I}_d$ and $\bar{\mathbf{a}}_{m,*}(\square_n) = c_n \mathbf{I}_d$. Theorem 3.1 of Armstrong and Kuusi [2025] gives $b_n/c_n \rightarrow 1$ after undoing the fixed spatial rescaling. Subadditivity gives decreasing b_n and increasing c_n , and the constant competitors and Cauchy–Schwarz give $\mathbb{E}[\mathbf{a}_m^{-1}(0)]^{-1} \leq c_n \leq b_n \leq \mathbb{E}[\mathbf{a}_m(0)]$. Their limits are therefore positive and equal, so $\bar{\mathbf{a}}_{m,*}(\square_n) \rightarrow \bar{\mathbf{a}}_m$. This argument concerns the annealed matrices themselves: the dual quantity throughout is $\mathbb{E}[\mathbf{a}_{m,*}^{-1}(\square_n)]^{-1}$, with the inverse taken after the expectation. Its admissible fields are $q + L_{\text{sol},0}^2(\square_n)$, as in the dual variational formula. No interchange of a samplewise limit and expectation is needed.

Lemma 3.4. *For every bounded convex domain $U \subseteq \mathbb{R}^d$ and $m, n \in \mathbb{N}_0$ with $n < m$, we have*

$$(3.23) \quad \begin{cases} \bar{\mathbf{a}}_m(U) \leq \bar{\mathbf{a}}_n(U) \leq \mathbb{E} \left[\sup_U (\mathbf{a}_m^{-1} \mathbf{a}_n) \right] \bar{\mathbf{a}}_m(U), \\ \exp(-2\tau^2(m-n)) \bar{\mathbf{a}}_{m,*}^{-1}(U) \leq \bar{\mathbf{a}}_{n,*}^{-1}(U) \leq \mathbb{E} \left[\sup_U (\mathbf{a}_m \mathbf{a}_n^{-1}) \right] \bar{\mathbf{a}}_{m,*}^{-1}(U). \end{cases}$$

In particular, for every $m, n \in \mathbb{N}_0$ with $n < m$, we have

$$(3.24) \quad \bar{\mathbf{a}}_m \leq \bar{\mathbf{a}}_n \leq \exp(2\tau^2(m-n)) \bar{\mathbf{a}}_m.$$

Proof. Since $\mathbf{a}_m \mathbf{a}_n^{-1}$ is independent of $\mathbf{a}_n$, it is independent of $v(\cdot, U, e, 0; \mathbf{a}_n)$. Moreover, for a constant c , the function $-v(\cdot, U, e, 0; \mathbf{a}_n) + c$ belongs to $e \cdot x + H_0^1(U)$. Hence

$$\begin{aligned} e \cdot \bar{\mathbf{a}}_m(U) e &= \mathbb{E} \left[\min_{w \in e \cdot x + H_0^1(U)} \int_U \nabla w \cdot \mathbf{a}_m \nabla w \right] \\ &\leq \mathbb{E} \left[\int_U \nabla v(\cdot, U, e, 0; \mathbf{a}_n) \cdot \mathbf{a}_m \nabla v(\cdot, U, e, 0; \mathbf{a}_n) \right] \\ &= \int_U \mathbb{E} [\nabla v(x, U, e, 0; \mathbf{a}_n) \cdot \mathbf{a}_m(x) \nabla v(x, U, e, 0; \mathbf{a}_n)] dx \\ &= \int_U \underbrace{\mathbb{E} [\mathbf{a}_m(x) \mathbf{a}_n^{-1}(x)]}_{=1} \mathbb{E} [\nabla v(x, U, e, 0; \mathbf{a}_n) \cdot \mathbf{a}_n(x) \nabla v(x, U, e, 0; \mathbf{a}_n)] dx = e \cdot \bar{\mathbf{a}}_n(U) e. \end{aligned}$$

For the other direction, we have

$$\begin{aligned} e \cdot \bar{\mathbf{a}}_m(U) e &= \mathbb{E} \left[\min_{w \in e \cdot x + H_0^1(U)} \int_U \nabla w \cdot \mathbf{a}_m \nabla w \right] \\ &\geq \mathbb{E} \left[\inf_U (\mathbf{a}_m \mathbf{a}_n^{-1}) \min_{w \in e \cdot x + H_0^1(U)} \int_U \nabla w \cdot \mathbf{a}_n \nabla w \right] \\ &= \mathbb{E} \left[\inf_U (\mathbf{a}_m \mathbf{a}_n^{-1}) \right] \mathbb{E} \left[\min_{w \in e \cdot x + H_0^1(U)} \int_U \nabla w \cdot \mathbf{a}_n \nabla w \right] = \mathbb{E} \left[\inf_U (\mathbf{a}_m \mathbf{a}_n^{-1}) \right] (e \cdot \bar{\mathbf{a}}_n(U) e). \end{aligned}$$

Since $t \mapsto t^{-1}$ is convex on $(0, \infty)$ and $(\inf_U (\mathbf{a}_m \mathbf{a}_n^{-1}))^{-1} = \sup_U (\mathbf{a}_m^{-1} \mathbf{a}_n)$, Jensen's inequality gives $\mathbb{E}[\inf_U (\mathbf{a}_m \mathbf{a}_n^{-1})]^{-1} \leq \mathbb{E}[\sup_U (\mathbf{a}_m^{-1} \mathbf{a}_n)]$; together with the previous two displays, this gives the first line of (3.23). For the second line of (3.23), use the dual variational formula with the

minimizing flux for $\mathbf{a}_n$:

$$\begin{aligned}
e \cdot \bar{\mathbf{a}}_{m,*}^{-1}(U)e &= \mathbb{E} \left[\min_{\mathbf{h} \in e + L_{\text{sol},0}^2(U)} \int_U \mathbf{h} \cdot \mathbf{a}_m^{-1} \mathbf{h} \right] \\
&\leq \mathbb{E} \left[\int_U (\mathbf{a}_n \nabla v(\cdot, U, 0, e; \mathbf{a}_n)) \cdot \mathbf{a}_m^{-1} (\mathbf{a}_n \nabla v(\cdot, U, 0, e; \mathbf{a}_n)) \right] \\
&= \int_U \mathbb{E} [(\mathbf{a}_n(x) \nabla v(x, U, 0, e; \mathbf{a}_n)) \cdot \mathbf{a}_m^{-1}(x) (\mathbf{a}_n(x) \nabla v(x, U, 0, e; \mathbf{a}_n))] dx \\
&= \int_U \underbrace{\mathbb{E} [\mathbf{a}_m^{-1}(x) \mathbf{a}_n(x)]}_{=\exp(2\tau^2(m-n))} \mathbb{E} [\nabla v(x, U, 0, e; \mathbf{a}_n) \cdot \mathbf{a}_n(x) \nabla v(x, U, 0, e; \mathbf{a}_n)] dx \\
&= \exp(2\tau^2(m-n)) (e \cdot \bar{\mathbf{a}}_{n,*}^{-1}(U)e),
\end{aligned}$$

and

$$\begin{aligned}
e \cdot \bar{\mathbf{a}}_{m,*}^{-1}(U)e &= \mathbb{E} \left[\min_{\mathbf{h} \in e + L_{\text{sol},0}^2(U)} \int_U \mathbf{h} \cdot \mathbf{a}_m^{-1} \mathbf{h} \right] \\
&\geq \mathbb{E} \left[\inf_U (\mathbf{a}_m^{-1} \mathbf{a}_n) \min_{\mathbf{h} \in e + L_{\text{sol},0}^2(U)} \int_U \mathbf{h} \cdot \mathbf{a}_n^{-1} \mathbf{h} \right] \\
&= \mathbb{E} \left[\inf_U (\mathbf{a}_m^{-1} \mathbf{a}_n) \right] \mathbb{E} \left[\min_{\mathbf{h} \in e + L_{\text{sol},0}^2(U)} \int_U \mathbf{h} \cdot \mathbf{a}_n^{-1} \mathbf{h} \right] = \mathbb{E} \left[\inf_U (\mathbf{a}_m^{-1} \mathbf{a}_n) \right] (e \cdot \bar{\mathbf{a}}_{n,*}^{-1}(U)e).
\end{aligned}$$

In the same way, $\mathbb{E}[\inf_U (\mathbf{a}_m^{-1} \mathbf{a}_n)]^{-1} \leq \mathbb{E}[\sup_U (\mathbf{a}_m \mathbf{a}_n^{-1})]$, and the last two displays give the second line of (3.23). $\square$

By (3.24) and (1.6), there exists $C(d) < \infty$ such that, for every $m, n \in \mathbb{N}_0$ with $n \leq m$,

$$(3.25) \quad |\bar{\mathbf{a}}_m \bar{\mathbf{a}}_n^{-1} - \text{Id}_d| + |\bar{\mathbf{a}}_m^{-1} \bar{\mathbf{a}}_n - \text{Id}_d| \leq C\bar{\delta}^2(m-n) \exp(C\bar{\delta}^2(m-n)).$$

In dimension two, planar duality determines $\bar{\mathbf{a}}_m$ exactly.

Proposition 3.5. *In dimension $d = 2$, for every $m \in \mathbb{N}_0$, we have $\bar{\mathbf{a}}_m = \exp(-\tau^2(m+1))$.*

Proof. Let R be the matrix with rows $(0, 1)$ and $(-1, 0)$. On the simply connected square $\square_n$, the divergence-free fields with zero normal component on the boundary are exactly the rotated gradients of functions vanishing on the boundary: $L_{\text{sol},0}^2(\square_n) = R \nabla H_0^1(\square_n)$. Substituting this into the dual minimum in (2.13) therefore gives, for every scalar coefficient a bounded above and away from zero on $\square_n$, the identity $a_*^{-1}(\square_n; a) = R a(\square_n; a^{-1}) R^T$. By the invariance of the law of $\mathbf{g}_0$ under $\mathbf{g}_0 \mapsto -\mathbf{g}_0$ in (g3), $\mathbf{a}_m^{-1}$ has the same law as $e^{2(m+1)\tau^2} \mathbf{a}_m$. Taking expectations in the preceding identity and using homogeneity and the invariance of the law under signed permutations, we obtain $\bar{\mathbf{a}}_{m,*}^{-1}(\square_n) = e^{2(m+1)\tau^2} \bar{\mathbf{a}}_m(\square_n)$. Let $n \rightarrow \infty$ and use that $\bar{\mathbf{a}}_{m,*}(\square_n) \rightarrow \bar{\mathbf{a}}_m$, as shown after (3.22). The positive scalar $\bar{\mathbf{a}}_m$ satisfies $\bar{\mathbf{a}}_m^{-1} = e^{2(m+1)\tau^2} \bar{\mathbf{a}}_m$, hence $\bar{\mathbf{a}}_m = e^{-(m+1)\tau^2}$. This argument is a finite-volume variational form of planar duality [Dykhne, 1971]. $\square$

Lemma 3.6. *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{N}_0$, every $\xi \in [1, \infty)$, and every convex domain $U \subseteq \square_m$,*

$$(3.26) \quad \mathbb{E} \left[\max_{|e|=1} J(U, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m)^\xi \right]^{1/\xi} \leq C\xi(m+1)\bar{\delta}^2 \exp(C\xi(m+1)\bar{\delta}^2).$$

Proof. By (2.11) and (2.14),

$$\begin{aligned} \max_{|e|=1} J(U, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m) &\leq \frac{1}{2} \int_U \left(\frac{\mathbf{a}_m}{\bar{\mathbf{a}}_m} + \frac{\bar{\mathbf{a}}_m}{\mathbf{a}_m} - 2 \right) = \frac{1}{2} \int_U \left(\frac{\mathbf{a}_m}{\bar{\mathbf{a}}_m} - 1 \right) \left(1 - \frac{\bar{\mathbf{a}}_m}{\mathbf{a}_m} \right) \\ &\leq \frac{1}{4} \int_U \left(\left(\frac{\mathbf{a}_m}{\bar{\mathbf{a}}_m} - 1 \right)^2 + \left(\frac{\bar{\mathbf{a}}_m}{\mathbf{a}_m} - 1 \right)^2 \right). \end{aligned}$$

Taking expectations in (2.14) also gives $e^{-2(m+1)\tau^2} \leq \bar{\mathbf{a}}_m \leq 1$. Consequently, Lemma 3.1, with $n = -1$ and exponent 2ξ , and (1.6) bound the $L^\xi(\mathbb{P})$ norm of the right side of the last display by the right side of (3.26). $\square$

4. INDUCTIVE COARSE-GRAINING UP THE SCALES

In this section we prove the coarse-graining estimate at the cutoff scale, stated below in Proposition 4.1. For a bounded convex domain U and $m \in \mathbb{N}_0$, abbreviate

$$(4.1) \quad \mathcal{J}_m(U) := \max_{|e|=1} J(U, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m).$$

On the cube $\square_m$, whose size matches the cutoff scale, the quantity $\mathcal{J}_m(\square_m)$ is bounded in $L^\xi(\mathbb{P})$ by $C\xi \log(2 + \xi) \bar{\delta}^2$ whenever $\xi \leq C^{-1} \bar{\delta}^{-2} |\log \bar{\delta}|^{-1}$. In particular, the bound does not deteriorate as further scales are added.

The quantity $\mathcal{J}_m(\square_m)$ is the gap between the two variational principles that define the coarse-grained matrices. It measures the error in replacing the coefficient on a cube by a constant matrix: by Lemma 2.1, it controls $\mathbf{a}_m(\square_m) - \mathbf{a}_{m,*}(\square_m)$ and the distance of these matrices from $\bar{\mathbf{a}}_m$, and it vanishes if and only if $\mathbf{a}_m(\square_m) = \mathbf{a}_{m,*}(\square_m) = \bar{\mathbf{a}}_m \mathbf{I}_d$. We argue by induction up the scales: the inductive hypothesis of Definition 4.2 bounds $\mathcal{J}_m(\square_m)$ in $L^\xi(\mathbb{P})$ for every m up to some m_0 , and the induction step extends this bound to the next scale. The field added at each step has a scale larger than the cubes already treated, so, by Section 3, it acts on them almost as a constant factor; the gain comes from estimating its effect on the larger cube in negative norms.

Proposition 4.1 (Coarse-graining estimate at the cutoff scale). *There are constants $c(d) > 0$ and $C(d) < \infty$ such that, for every $\xi \in [1, \infty)$, if the parameter $\bar{\delta}$ in Assumption (g2) satisfies*

$$(4.2) \quad \xi \leq C^{-1} \bar{\delta}^{-2} |\log \bar{\delta}|^{-1},$$

then, for every $m \in \mathbb{N}_0$, we have

$$(4.3) \quad \mathbb{E} \left[\max_{|e|=1} (J(\square_m, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m))^\xi \right]^{1/\xi} \leq C\xi \log(2 + \xi) \bar{\delta}^2.$$

Moreover, for every $s \in (0, 1]$ satisfying

$$4ds^{-1} \leq \xi, \quad C\xi \log(2 + \xi) \bar{\delta}^2 \leq cs \quad \text{and} \quad s \log(2 + \xi) \geq c,$$

we have

$$(4.4) \quad \mathbb{E} [\mathcal{E}_{s,\infty}(\square_m; \mathbf{a}_m, \bar{\mathbf{a}}_m)^\xi]^{1/\xi} \leq Cs^{-1} (\xi \log(2 + \xi) \bar{\delta}^2)^{1/2}.$$

After reducing the constant in (4.2), apply (4.3) with moment 2ξ and use Lemma 2.1 to obtain

$$\mathbb{E} [|\bar{\mathbf{a}}_m^{-1} \mathbf{a}_m(\square_m) - \mathbf{I}_d|^{2\xi}]^{1/\xi} + \mathbb{E} [|\bar{\mathbf{a}}_m \mathbf{a}_{m,*}^{-1}(\square_m) - \mathbf{I}_d|^{2\xi}]^{1/\xi} \leq C\xi \log(2 + \xi) \bar{\delta}^2.$$

The dependence on $\bar{\delta}$ is optimal: the scale- m field alone produces fluctuations of order $\bar{\delta}$ in the coarse-grained matrices.

Definition 4.2 (Inductive hypothesis $\mathcal{S}(m_0, \xi, \delta_1)$). Given $m_0 \in \mathbb{N}_0$, an exponent $\xi \in [1, \infty)$, and $\delta_1 \in (0, 1)$, let $\mathcal{S}(m_0, \xi, \delta_1)$ denote the statement

$$(4.5) \quad \max_{m \in \{0, \dots, m_0\}} \left\| \max_{|e|=1} J(\square_m, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m) \right\|_{L^\xi(\mathbb{P})} \leq \delta_1.$$

Let $\mathcal{S}(\infty, \xi, \delta_1)$ denote the statement

$$\sup_{m \in \mathbb{N}_0} \left\| \max_{|e|=1} J(\square_m, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m) \right\|_{L^\xi(\mathbb{P})} \leq \delta_1.$$

The bound (4.3) is the assertion that $\mathcal{S}(\infty, \xi, C\xi \log(2 + \xi)\bar{\delta}^2)$ holds throughout the stated range of ξ . The bound (4.4) on $\mathcal{E}_{s, \infty}$ then follows from Lemma 4.4 with $\delta_1 = C\xi \log(2 + \xi)\bar{\delta}^2$.

Proposition 4.1 is used in every later section: in Section 5 to derive the decay of $\bar{\mathbf{a}}_m$, in Section 6 to show that the scales with a large homogenization error are rare, and in Sections 8–10 to bound the coarse-grained ellipticity of $\mathbf{a}_L$. The proof has three parts and uses two inductions, the first of which supplies the low-moment estimate used in the second.

- In Section 4.1 we use $\mathcal{S}(m_0, \xi, \delta_1)$ to bound the coarse-graining error $\mathcal{E}_{s, \infty}$ of $\mathbf{a}_L$ on cubes below the cutoff scale (Lemma 4.4) and above it (Lemma 4.5).
- In Section 4.2 we prove the contraction of Proposition 4.10: under $\mathcal{S}(m_0, \xi, \delta_1)$, the quantity $\mathcal{J}_L(\square_m)$ with $m \geq L + C \log(2 + \xi)$ is at most $\frac{1}{4}\delta_1$ in $L^\xi(\mathbb{P})$.
- At the end of Section 4.2 we add the omitted fields $\mathbf{g}_{L+1}, \dots, \mathbf{g}_m$ using Lemma 2.8 and close the inductions.

4.1. A multiscale estimate. The statement $\mathcal{S}(m_0, \xi, \delta_1)$ bounds, for every $m \leq m_0$, the $L^\xi(\mathbb{P})$ norm of the maximum in (4.5). We now deduce multiscale bounds from this estimate.

Definition 4.3 (The coefficient $\mathbf{b}_{L, m}$). For $L, m \in \mathbb{N}_0$, define

$$\mathbf{b}_{L, m} := \bar{\mathbf{a}}_{m \wedge L} \frac{\mathbf{a}_L}{\mathbf{a}_{m \wedge L}}.$$

Thus, if $m \leq L$, then $\mathbf{b}_{L, m} = \bar{\mathbf{a}}_m \exp(\sum_{j=m+1}^L (\mathbf{g}_j - \tau^2))$, whereas $\mathbf{b}_{L, m} = \bar{\mathbf{a}}_L$ if $m \geq L$.

Lemma 4.4. Let $m_0 \in \mathbb{N}_0$, $\xi \in [6, \infty)$ and $\delta_1 \in [\bar{\delta}^2, 1)$, and assume that $\mathcal{S}(m_0, \xi, \delta_1)$ holds. There exist constants $c(d) \in (0, 1]$ and $C(d) \in [1, \infty)$ such that, for every $s \in (0, 1]$ satisfying

$$(4.6) \quad 4ds^{-1} \leq \xi \leq cs\bar{\delta}^{-2}\delta_1,$$

and every $m \in \mathbb{N}_0$ and $n \in \mathbb{Z} \cap (-\infty, m_0]$ with $m \geq n$, we have

$$(4.7) \quad \mathbb{E} \left[\sup_{L \geq m} \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} (\mathcal{E}_{s, \infty}(z + \square_n; \mathbf{a}_L, (\mathbf{b}_{L, m})_{\square_m}))^\xi \right]^{1/\xi} \leq Cs^{-1} \delta_1^{1/2} 3^{(\frac{d}{\xi} + \frac{s}{2})(m-n)},$$

and, for every $L \in \mathbb{N}_0$ with $L \leq m_0$,

$$(4.8) \quad \mathbb{E} [\mathcal{E}_{s, \infty}(\square_L; \mathbf{a}_L, \bar{\mathbf{a}}_L)^\xi]^{1/\xi} \leq Cs^{-1} \delta_1^{1/2}.$$

Proof. Let $s \in (0, 1]$ satisfy (4.6). By Definition 2.3, the coarse-graining error in (4.7) is controlled by the values of J on the subcubes of $z + \square_n$ of every size 3^k with $k \leq n$. The proof proceeds in three steps. Step 1 bounds the moments of these values, Step 2 sums these bounds over the scales, and Step 3 specializes to $m = n = L$. The heart of the matter is the case $k \geq 0$ of Step 1, where Lemma 2.8 transfers the induction hypothesis from the coefficient $\mathbf{a}_k$ to $\mathbf{a}_L$; the case $k < 0$ is estimated directly from the coefficient.

Step 1. We first treat the scales at which the induction hypothesis applies. We show the existence of $C(d) < \infty$ such that for every $m, k \in \mathbb{N}_0$ with $k \leq m \wedge m_0$ and every $z \in 3^k \mathbb{Z}^d \cap \square_m$,

$$(4.9) \quad \mathbb{E} \left[\sup_{L \geq m} \max_{|e|=1} \left(J(z + \square_k, (\mathbf{b}_{L,m})_{\square_m}^{-1/2} e, (\mathbf{b}_{L,m})_{\square_m}^{1/2} e; \mathbf{a}_L) \right)^\xi \right]^{1/\xi} \leq C \delta_1 3^{s(m-k)}.$$

We apply Lemma 2.8 with $\delta = 1$ and $\lambda = (\mathbf{a}_L \mathbf{a}_m^{-1})_{\square_m} \bar{\mathbf{a}}_m \bar{\mathbf{a}}_k^{-1}$, and with $\mathbf{b} = \mathbf{a}_L$ and $\mathbf{a} = \mathbf{a}_k$. We obtain, for every $L, m, k \in \mathbb{N}_0$ and $z \in 3^k \mathbb{Z}^d \cap \square_m$ with $L \geq m \geq k$, and $e \in \mathbb{R}^d$ with $|e| = 1$,

$$\begin{aligned} & J(z + \square_k, (\mathbf{b}_{L,m})_{\square_m}^{-1/2} e, (\mathbf{b}_{L,m})_{\square_m}^{1/2} e; \mathbf{a}_L) \\ & \leq 2J(z + \square_k, \bar{\mathbf{a}}_k^{-1/2} e, \bar{\mathbf{a}}_k^{1/2} e; \mathbf{a}_k) \\ & \quad + 3 \left(\sup_{L \geq m} \sup_{x \in \square_m, y \in z + \square_k} (|\bar{\mathbf{a}}_m \bar{\mathbf{a}}_k^{-1} (\mathbf{a}_L \mathbf{a}_m^{-1})(x) (\mathbf{a}_L^{-1} \mathbf{a}_k)(y) - 1| \right. \\ & \quad \left. + |\bar{\mathbf{a}}_m^{-1} \bar{\mathbf{a}}_k (\mathbf{a}_L^{-1} \mathbf{a}_m)(x) (\mathbf{a}_L \mathbf{a}_k^{-1})(y) - 1|) \right)^2 (J(z + \square_k, \bar{\mathbf{a}}_k^{-1/2} e, \bar{\mathbf{a}}_k^{1/2} e; \mathbf{a}_k) + 1). \end{aligned}$$

We first bound the moments of the supremum in this display:

$$(4.10) \quad \mathbb{E} \left[\left(\sup_{L \geq m} \sup_{x \in \square_m, y \in z + \square_k} (|\bar{\mathbf{a}}_m \bar{\mathbf{a}}_k^{-1} (\mathbf{a}_L \mathbf{a}_m^{-1})(x) (\mathbf{a}_L^{-1} \mathbf{a}_k)(y) - 1| \right. \right. \\ \left. \left. + |\bar{\mathbf{a}}_m^{-1} \bar{\mathbf{a}}_k (\mathbf{a}_L^{-1} \mathbf{a}_m)(x) (\mathbf{a}_L \mathbf{a}_k^{-1})(y) - 1|) \right)^{2\xi} \right]^{1/\xi} \leq C \delta_1 3^{s(m-k)}.$$

Given (4.10), the induction hypothesis $\mathcal{S}(m_0, \xi, \delta_1)$ and the independence of this supremum from the field $\mathbf{a}_k$ give (4.9).

To prove (4.10), we write

$$|\bar{\mathbf{a}}_m \bar{\mathbf{a}}_k^{-1} (\mathbf{a}_L \mathbf{a}_m^{-1})(x) (\mathbf{a}_L^{-1} \mathbf{a}_k)(y) - 1| = \left| \bar{\mathbf{a}}_m \bar{\mathbf{a}}_k^{-1} \frac{(\mathbf{a}_L \mathbf{a}_m^{-1})(x)}{(\mathbf{a}_L \mathbf{a}_m^{-1})(y)} (\mathbf{a}_m^{-1} \mathbf{a}_k)(y) - 1 \right|,$$

and use the general bound

$$\begin{aligned} |uvw - 1| & \leq |u - 1| + |v - 1| + |w - 1| + |(u - 1)(v - 1)| \\ & \quad + |(u - 1)(w - 1)| + |(v - 1)(w - 1)| + |(u - 1)(v - 1)(w - 1)|, \end{aligned}$$

together with the independence of the fields $\mathbf{a}_L \mathbf{a}_m^{-1}$ and $\mathbf{a}_m^{-1} \mathbf{a}_k$ and the following three inequalities, invoked at moment 2ξ , which follow from (3.25), (3.3), (3.5) and (4.6). For the deterministic factor,

$$|\bar{\mathbf{a}}_m \bar{\mathbf{a}}_k^{-1} - 1| \leq C \bar{\delta}^2 (m - k) \exp(C \bar{\delta}^2 (m - k)) \leq C \delta_1^{1/2} 3^{\frac{s}{6}(m-k)},$$

for the ratio of the values of $\mathbf{a}_L \mathbf{a}_m^{-1}$ at two points,

$$\mathbb{E} \left[\left(\sup_{L \geq m} \sup_{x, y \in \square_m} \frac{(\mathbf{a}_L \mathbf{a}_m^{-1})(x)}{(\mathbf{a}_L \mathbf{a}_m^{-1})(y)} - 1 \right)^{2\xi} \right]^{1/\xi} \leq C \xi \bar{\delta}^2 \exp(C \xi \bar{\delta}^2 (m - k)) \leq C \delta_1 3^{\frac{s}{3}(m-k)},$$

and for the factor $\mathbf{a}_m^{-1} \mathbf{a}_k$,

$$\mathbb{E} \left[\|\mathbf{a}_m^{-1} \mathbf{a}_k - 1\|_{L^\infty(\square_k)}^{2\xi} \right]^{1/\xi} \leq C \xi \bar{\delta}^2 (m - k) \exp(C \xi \bar{\delta}^2 (m - k)) \leq C \delta_1 3^{\frac{s}{3}(m-k)}.$$

Each of the three deviations thus has $L^{2\xi}$ norm at most $C \delta_1^{1/2} 3^{s(m-k)/6}$. The two random factors depend on disjoint sets of fields, and the remaining factor is deterministic, so the $L^{2\xi}$ norm of a product of deviations is the product of their $L^{2\xi}$ norms. Since $\delta_1 < 1$, even the product of all

three is bounded by $C\delta_1^{1/2}3^{s(m-k)/2}$. The bound on $|XYZ - 1|$ above therefore proves (4.10), and hence (4.9).

We next estimate the terms in Definition 2.3 with cube size smaller than one. Equations (2.11) and (2.14) imply that, for every positive scalar coefficient $\mathbf{a}$, positive scalar b , bounded convex domain V , and $|e| = 1$,

$$(4.11) \quad J(V, b^{-1/2}e, b^{1/2}e; \mathbf{a}) \leq \frac{1}{2} \int_V \left(\frac{\mathbf{a}}{b} + \frac{b}{\mathbf{a}} - 2 \right).$$

Indeed, (2.14) bounds the two quadratic terms in (2.11) by the corresponding averages of $\mathbf{a}$ and $\mathbf{a}^{-1}$. To estimate the right side with $\mathbf{a} = \mathbf{a}_L$ and $b = (\mathbf{b}_{L,m})_{\square_m}$, we factor

$$\frac{\mathbf{a}_L}{(\mathbf{b}_{L,m})_{\square_m}} = \frac{\mathbf{a}_m}{\bar{\mathbf{a}}_m} \cdot \frac{\mathbf{a}_L \mathbf{a}_m^{-1}}{(\mathbf{a}_L \mathbf{a}_m^{-1})_{\square_m}}.$$

The elementary inequality $r + r^{-1} - 2 \leq (r - 1)^2 + (r^{-1} - 1)^2$, valid for every $r > 0$, allows us to estimate the two factors separately. We first estimate $\mathbf{a}_m/\bar{\mathbf{a}}_m$. For each unit cube $z + \square_0$ meeting $\square_m$, independence and Assumptions (g1)–(g2) give, for every $p \geq 1$,

$$\left\| \sum_{i=0}^m \mathbf{g}_i(z) \right\|_{L^p(\mathbb{P})} \leq C\sqrt{p}\bar{\delta}\sqrt{m+1},$$

and, for the oscillations of the fields on the same cube,

$$\left\| \sum_{i=0}^m \text{osc}_{z+\square_0} \mathbf{g}_i \right\|_{L^p(\mathbb{P})} \leq C\sqrt{p}\bar{\delta}.$$

There are at most $C3^{dm}$ such unit cubes. We apply (A.8) on each cube and then take a union bound over these cubes. The variational bounds (2.14), together with $\mathbb{E}[\mathbf{a}_0(0)] = 1$ and $\mathbb{E}[\mathbf{a}_0(0)^{-1}] = e^{2\tau^2}$, give $e^{-2\tau^2} \leq \bar{\mathbf{a}}_0 \leq 1$. Combining this with (3.25) controls the deterministic factor $\bar{\mathbf{a}}_m$. This yields

$$\mathbb{E} \left[\left\| \frac{\mathbf{a}_m}{\bar{\mathbf{a}}_m} + \frac{\bar{\mathbf{a}}_m}{\mathbf{a}_m} - 2 \right\|_{L^\infty(\square_m)}^\xi \right]^{1/\xi} \leq C\xi\bar{\delta}^2(m+1) \exp \left(\frac{d(\log 3)m + C}{\xi} + C\xi\bar{\delta}^2(m+1) + C\bar{\delta}^2m \right).$$

For the second factor, the average in the denominator lies between the minimum and the maximum of $\mathbf{a}_L \mathbf{a}_m^{-1}$ on $\square_m$. Hence (3.3), applied with $n = k = m$, gives the same bound with $m+1$ replaced by a dimensional constant. The two factors are independent. Hölder's inequality and $|uv - 1| \leq |u - 1||v| + |v - 1|$, applied also to the inverses, therefore give

$$\begin{aligned} & \mathbb{E} \left[\sup_{L \geq m} \left\| \frac{\mathbf{a}_L}{(\mathbf{b}_{L,m})_{\square_m}} + \frac{(\mathbf{b}_{L,m})_{\square_m}}{\mathbf{a}_L} - 2 \right\|_{L^\infty(\square_m)}^\xi \right]^{1/\xi} \\ & \leq C\xi\bar{\delta}^2(m+1) \exp \left(\frac{d(\log 3)m + C}{\xi} + C\xi\bar{\delta}^2(m+1) + C\bar{\delta}^2m \right) \leq C\delta_1 3^{sm}. \end{aligned}$$

The last inequality follows from (4.6). Thus, by (4.11), for every $k < 0$,

$$(4.12) \quad \mathbb{E} \left[\sup_{L \geq m} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \max_{|e|=1} J(z + \square_k, (\mathbf{b}_{L,m})_{\square_m}^{-1/2} e, (\mathbf{b}_{L,m})_{\square_m}^{1/2} e; \mathbf{a}_L)^\xi \right]^{1/\xi} \leq C\delta_1 3^{sm}.$$

Step 2. We prove (4.7). Split the sum in Definition 2.3 into $k \geq 0$ and $k < 0$. Minkowski's inequality, (4.9), (4.12), and the union bound over the $O(3^{d(m-k)})$ cubes of scale k give

$$\begin{aligned} \left\| \sup_{L \geq m} \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \mathcal{E}_{s,\infty}(z + \square_n; \mathbf{a}_L, (\mathbf{b}_{L,m})_{\square_m}) \right\|_{L^\xi(\mathbb{P})} &\leq C \delta_1^{1/2} 3^{\frac{d}{\xi}(m-n)} \sum_{k=0}^n 3^{-(\frac{s}{2}-\frac{d}{\xi})(n-k)+\frac{s}{2}(m-n)} \mathbb{1}_{\{n \geq 0\}} \\ &\quad + C \delta_1^{1/2} 3^{\frac{s}{2}m} \sum_{k=-\infty}^{n \wedge (-1)} 3^{-s(n-k)} \\ &\leq C s^{-1} \delta_1^{1/2} 3^{(\frac{d}{\xi}+\frac{s}{2})(m-n)}. \end{aligned}$$

Here we used $\xi \geq 4ds^{-1}$ from (4.6).

Step 3. We prove (4.8). Specializing (4.7) to $m = n = L$ and using $\mathbf{b}_{L,L} = \bar{\mathbf{a}}_L$ gives (4.8). $\square$

Lemma 4.5 (Coarse-graining error on cubes above the cutoff scale). *There exist $c(d) > 0$ and $C(d) < \infty$ such that the following holds. Let $L, m_0 \in \mathbb{N}_0$ satisfy $L \leq m_0$, let $s \in (0, 1]$ and $\delta_1 \in [\bar{\delta}^2, 1)$, and let ξ satisfy $4ds^{-1} \leq \xi \leq cs\bar{\delta}^{-2}\delta_1$. If $\mathcal{S}(m_0, \xi, \delta_1)$ is valid, then the following estimate holds for every $M \geq L$, every translate U of $\square_M$, and $r \in \{1, 2\}$:*

$$(4.13) \quad \|\mathcal{E}_{s,\infty,r}(U; \mathbf{a}_L, \bar{\mathbf{a}}_L)\|_{L^\xi(\mathbb{P})} \leq C s^{-1/r} \delta_1^{1/2}.$$

Proof. It is enough, by stationarity, to take $U = \square_M$. Recall the notation $\mathcal{J}_L(U)$ of (4.1). Definition 2.3 and Minkowski's inequality give

$$(4.14) \quad \|\mathcal{E}_{s,\infty,r}(\square_M; \mathbf{a}_L, \bar{\mathbf{a}}_L)\|_{L^\xi}^r \leq C \sum_{k=-\infty}^M 3^{-sr(M-k)} \left\| \max_{z \in 3^k \mathbb{Z}^d \cap \square_M} \mathcal{J}_L(z + \square_k)^{1/2} \right\|_{L^\xi}^r.$$

We split the sum into three ranges.

If $L \leq k \leq M$, then subadditivity (2.16) on the partition into cubes of scale L , the induction hypothesis, and Jensen's inequality yield $\|\mathcal{J}_L(z + \square_k)\|_{L^\xi} \leq C \delta_1$. Taking the maximum over at most $C3^{d(M-k)}$ cubes gives a contribution to (4.14) bounded by

$$C \delta_1^{r/2} \sum_{j=0}^{M-L} 3^{-r(s-\frac{d}{\xi})j} \leq C s^{-1} \delta_1^{r/2}.$$

If $0 \leq k < L$, then (4.9) with $m = L$ and cutoff L gives $\|\mathcal{J}_L(z + \square_k)\|_{L^\xi} \leq C \delta_1 3^{s(L-k)}$. After taking the maximum over the cubes of scale k in $\square_M$, we bound the corresponding summand by

$$C \delta_1^{r/2} 3^{-r(s-\frac{d}{\xi})(M-L)} 3^{-r(\frac{s}{2}-\frac{d}{\xi})(L-k)}.$$

The sum of these bounds over k is at most $C s^{-1} \delta_1^{r/2}$.

Finally, if $k < 0$, apply (4.12) with $m = L$ on each scale- L parent cube. The maximum over the at most $C3^{d(M-L)}$ such cubes gives

$$\begin{aligned} &\sum_{k=-\infty}^{-1} 3^{-sr(M-k)} \left\| \max_{z \in 3^k \mathbb{Z}^d \cap \square_M} \mathcal{J}_L(z + \square_k)^{1/2} \right\|_{L^\xi}^r \\ &\leq C \delta_1^{r/2} 3^{-r(s-\frac{d}{\xi})(M-L)} 3^{-r(\frac{s}{2}-\frac{d}{\xi})L} \sum_{j=1}^{\infty} 3^{-srj} \leq C s^{-1} \delta_1^{r/2}. \end{aligned}$$

Inserting the estimates into (4.14) and taking the r th root proves the lemma. $\square$

4.2. The contraction step. The coefficient fields in Lemmas 4.6, 4.7 and 4.8 are understood in the class used to define the coarse-grained matrices in Section 2: they are measurable and symmetric, and on every bounded cube in their domain there are constants $0 < \lambda \leq \Lambda < \infty$ such that $\lambda \mathbf{I}_d \leq \mathbf{a} \leq \Lambda \mathbf{I}_d$ almost everywhere. These constants may depend on the cube. For a random field this condition holds almost surely, and the field and its coarse-grained matrices are measurable random variables. Variances and nonnegative expectations are allowed to be infinite, with the variance convention stated before Lemma 4.8; no additional square-integrability assumption is imposed.

We now prove the contraction step: if $\mathcal{J}_m(\square_m)$ is at most δ_1 in $L^\xi(\mathbb{P})$ for every scale m up to m_0 , then, for $L \leq m_0$ and a cube $\square_m$ of scale $m \geq L + C \log(2 + \xi)$, the quantity $\mathcal{J}_L(\square_m)$, which measures $\mathbf{a}_L$ against $\bar{\mathbf{a}}_L$, is at most $\frac{1}{4}\delta_1$ in $L^\xi(\mathbb{P})$. This is Proposition 4.10, and the factor $\frac{1}{4}$ compensates for the losses incurred elsewhere in the induction. We prove it by the multiscale variational argument of Armstrong et al. [2019], using the estimates of Armstrong and Kuusi [2025], whose constants depend on coarse-grained ellipticity constants rather than on pointwise bounds on the coefficient. We first bound J by a negative Sobolev norm of the gradient of its maximizer and by the amount by which the values of J on the subcubes exceed its value on the cube (Lemma 4.6). Throughout this subsection, for $p, q \in \mathbb{R}^d$, a coefficient field $\mathbf{a}$, $n \in \mathbb{Z}$ and $z \in 3^n \mathbb{Z}^d$, we write $v_{n,z} := v(\cdot, z + \square_n, p, q; \mathbf{a})$ for the maximizer of $J(z + \square_n, p, q; \mathbf{a})$, and we abbreviate $v_n := v_{n,0}$.

Lemma 4.6 (Estimate of J by a negative Sobolev norm). *There exists $C(d) < \infty$ such that, for every $m \in \mathbb{N}$, coefficient field $\mathbf{a} : \square_m \rightarrow \mathbb{R}_{\text{sym},+}^{d \times d}$ and $p, q \in \mathbb{R}^d$,*

$$(4.15) \quad \begin{aligned} J(\square_{m-1}, p, q; \mathbf{a}) &\leq C \Lambda_{1/4}(\square_m; \mathbf{a}) \left(\frac{\Lambda_{1/4}(\square_m; \mathbf{a})}{\lambda_{1/4}(\square_m; \mathbf{a})} \right)^{1/2} 3^{-2m} \|\nabla v(\cdot, \square_m, p, q; \mathbf{a})\|_{\underline{H}^{-1}(\square_m)}^2 \\ &\quad + \sum_{z \in 3^{m-1} \mathbb{Z}^d \cap \square_m} 2(J(z + \square_{m-1}, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a})). \end{aligned}$$

Proof. We compute, using the triangle inequality, (2.19) and (2.20),

$$\begin{aligned} J(\square_{m-1}, p, q; \mathbf{a}) &= \int_{\square_{m-1}} \frac{1}{2} \nabla v_{m-1} \cdot \mathbf{a} \nabla v_{m-1} \\ &\leq \int_{\square_{m-1}} \nabla v_m \cdot \mathbf{a} \nabla v_m + \int_{\square_{m-1}} (\nabla v_m - \nabla v_{m-1}) \cdot \mathbf{a} (\nabla v_m - \nabla v_{m-1}) \\ &\leq \int_{\square_{m-1}} \nabla v_m \cdot \mathbf{a} \nabla v_m + \sum_{z \in 3^{m-1} \mathbb{Z}^d \cap \square_m} \int_{z + \square_{m-1}} (\nabla v_m - \nabla v_{m-1,z}) \cdot \mathbf{a} (\nabla v_m - \nabla v_{m-1,z}) \\ &= \int_{\square_{m-1}} \nabla v_m \cdot \mathbf{a} \nabla v_m + \sum_{z \in 3^{m-1} \mathbb{Z}^d \cap \square_m} 2(J(z + \square_{m-1}, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a})). \end{aligned}$$

The coarse-grained Caccioppoli inequality (Proposition 2.6) yields

$$\int_{\square_{m-1}} \nabla v_m \cdot \mathbf{a} \nabla v_m \leq C \Lambda_{1/4}(\square_m; \mathbf{a}) \left(\frac{\Lambda_{1/4}(\square_m; \mathbf{a})}{\lambda_{1/4}(\square_m; \mathbf{a})} \right)^{1/2} 3^{-2m} \int_{\square_m} |v_m - (v_m)_{\square_m}|^2.$$

Combining these and using $\|v_m - (v_m)_{\square_m}\|_{\underline{L}^2(\square_m)} \leq C \|\nabla v_m\|_{\underline{H}^{-1}(\square_m)}$, we obtain (4.15). $\square$

Lemma 4.7 (Negative-norm estimates). *There exists $C(d) < \infty$ such that, for every $m \in \mathbb{N}$, coefficient field $\mathbf{a} : \square_m \rightarrow \mathbb{R}_{\text{sym},+}^{d \times d}$, exponent $s \in (0, 1)$, integer $L_0 \leq m$, and $p, q \in \mathbb{R}^d$, we have*

$$\begin{aligned}
& 3^{-m} \|\nabla v(\cdot, \square_m, p, q; \mathbf{a}) - (\mathbf{a}_*^{-1}(\square_m)q - p)\|_{\hat{H}^{-1}(\square_m)} \\
& \leq C \lambda_s^{-1/2}(\square_m; \mathbf{a}) \sum_{k=L_0}^m 3^{-(1-s)(m-k)} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} J(z + \square_k, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right)^{1/2} \\
& \quad + C \sum_{n=L_0}^m 3^{-(m-n)} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |\mathbf{a}_*^{-1}(\square_m)q - \mathbf{a}_*^{-1}(z + \square_n)q|^2 \right)^{1/2} \\
& \quad + \frac{C \lambda_s^{-1/2}(\square_m; \mathbf{a})}{1-s} 3^{-(1-s)(m-L_0)} \|\mathbf{a}^{1/2} \nabla v(\cdot, \square_m, p, q; \mathbf{a})\|_{\underline{L}^2(\square_m)} \\
& \quad + C 3^{-(m-L_0)} |\mathbf{a}_*^{-1}(\square_m)q - p|.
\end{aligned} \tag{4.16}$$

Proof. The duality inequality (2.8), applied to $\nabla v_m - (\nabla v_m)_{\square_m}$ and an H^1 test function, together with the Poincaré inequality on cubes, gives the first inequality below. For the second, we split the sum at L_0 and use the triangle inequality:

$$\begin{aligned}
& 3^{-m} \|\nabla v_m - (\mathbf{a}_*^{-1}(\square_m)q - p)\|_{\hat{H}^{-1}(\square_m)} \\
& \leq C \sum_{n=-\infty}^m 3^{-(m-n)} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z+\square_n} - (\mathbf{a}_*^{-1}(\square_m)q - p)|^2 \right)^{1/2} \\
& \leq C \sum_{n=L_0}^m 3^{-(m-n)} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z+\square_n} - (\mathbf{a}_*^{-1}(z + \square_n)q - p)|^2 \right)^{1/2} \\
& \quad + C \sum_{n=L_0}^m 3^{-(m-n)} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |\mathbf{a}_*^{-1}(\square_m)q - \mathbf{a}_*^{-1}(z + \square_n)q|^2 \right)^{1/2} \\
& \quad + C \sum_{n=-\infty}^{L_0-1} 3^{-(m-n)} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(|(\nabla v_m)_{z+\square_n}|^2 + |\mathbf{a}_*^{-1}(\square_m)q - p|^2 \right) \right)^{1/2}.
\end{aligned} \tag{4.17}$$

For $n \in \mathbb{Z}$ with $n \leq L_0 - 1$, we have, by (2.21) and (2.36),

$$\begin{aligned}
\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z+\square_n}|^2 & \leq \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} |\mathbf{a}_*^{-1}(z + \square_n)| \|\mathbf{a}^{1/2} \nabla v_m\|_{\underline{L}^2(\square_m)}^2 \\
& \leq 3^{2s(m-n)} \lambda_s^{-1}(\square_m; \mathbf{a}) \|\mathbf{a}^{1/2} \nabla v_m\|_{\underline{L}^2(\square_m)}^2.
\end{aligned}$$

This gives the following bound for the third term on the right side of (4.17):

$$\begin{aligned}
& \sum_{n=-\infty}^{L_0-1} 3^{-(m-n)} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(|(\nabla v_m)_{z+\square_n}|^2 + |\mathbf{a}_*^{-1}(\square_m)q - p|^2 \right) \right)^{1/2} \\
& \leq \frac{C}{1-s} 3^{-(1-s)(m-L_0)} \lambda_s^{-1/2}(\square_m; \mathbf{a}) \|\mathbf{a}^{1/2} \nabla v_m\|_{\underline{L}^2(\square_m)} + C 3^{-(m-L_0)} |\mathbf{a}_*^{-1}(\square_m)q - p|.
\end{aligned} \tag{4.18}$$

Next we use (2.17) and (2.21) to obtain, for each $n \leq m$,

$$\begin{aligned}
|\mathbf{a}_*^{1/2}(z + \square_n)((\nabla v_m)_{z+\square_n} - (\mathbf{a}_*^{-1}(z + \square_n)q - p))|^2 & = |\mathbf{a}_*^{1/2}(z + \square_n)(\nabla v_m - \nabla v_{n,z})_{z+\square_n}|^2 \\
& \leq \int_{z+\square_n} (\nabla v_m - \nabla v_{n,z}) \cdot \mathbf{a}(\nabla v_m - \nabla v_{n,z}).
\end{aligned}$$

Summing this over $z \in 3^n \mathbb{Z}^d \cap \square_m$ and applying (2.19) yields, for each $n \leq m$,

$$\begin{aligned} & \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left| \mathbf{a}_*^{1/2}(z + \square_n) \left((\nabla v_m)_{z + \square_n} - (\mathbf{a}_*^{-1}(z + \square_n)q - p) \right) \right|^2 \\ & \leq \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \int_{z + \square_n} (\nabla v_m - \nabla v_{n,z}) \cdot \mathbf{a}(\nabla v_m - \nabla v_{n,z}) \\ & = 2 \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(J(z + \square_n, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right). \end{aligned}$$

Using that, by (2.36), for every $P \in \mathbb{R}^d$, every $n \leq m$ and every $z \in 3^n \mathbb{Z}^d \cap \square_m$,

$$|P|^2 \leq |\mathbf{a}_*^{-1}(z + \square_n)| |\mathbf{a}_*^{1/2}(z + \square_n)P|^2 \leq 3^{2s(m-n)} \lambda_s^{-1}(\square_m; \mathbf{a}) |\mathbf{a}_*^{1/2}(z + \square_n)P|^2,$$

we obtain the following bound for the first term on the right side of (4.17):

$$\begin{aligned} & \sum_{n=L_0}^m 3^{-(m-n)} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z + \square_n} - (\mathbf{a}_*^{-1}(z + \square_n)q - p)|^2 \right)^{1/2} \\ (4.19) \quad & \leq C \lambda_s^{-1/2}(\square_m; \mathbf{a}) \sum_{k=L_0}^m 3^{-(1-s)(m-k)} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} J(z + \square_k, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right)^{1/2}. \end{aligned}$$

Combining (4.17), (4.18) and (4.19) yields (4.16). $\square$

For a square-integrable random matrix M , we write $\text{var}[M] := \mathbb{E}[|M - \mathbb{E}M|^2]$, with the matrix norm fixed in Section 2.1. If $\mathbb{E}[|M|^2] = +\infty$, we set $\text{var}[M] := +\infty$.

Lemma 4.8 (Variance estimate). *For every $\mathbb{Z}^d$ -stationary random coefficient field $\mathbf{a} : \mathbb{R}^d \rightarrow \mathbb{R}_{\text{sym},+}^{d \times d}$, every $n, k \in \mathbb{Z}$ with $k < n$, and every positive-definite matrix $\mathbf{b} \in \mathbb{R}_{\text{sym},+}^{d \times d}$, we have*

$$\begin{aligned} \text{var}[\mathbf{a}_*^{-1}(\square_n)] & \leq 2 \text{var} \left[\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k) \right] \\ (4.20) \quad & + 32 \mathbb{E} \left[\left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_k, \mathbf{b}^{-1}e_i, e_i; \mathbf{a}) \right)^2 \right]. \end{aligned}$$

Proof. If the right side of (4.20) is infinite, the assertion is immediate. Otherwise the average of the positive-definite matrices $\mathbf{a}_*^{-1}(z + \square_k)$ is in $L^2(\mathbb{P})$ by the variance convention. Subadditivity gives

$$0 \leq \mathbf{a}_*^{-1}(\square_n) \leq \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k).$$

Hence $\mathbf{a}_*^{-1}(\square_n)$ is also in $L^2(\mathbb{P})$. Each subcube matrix is bounded in the Loewner order by the number of subcubes times their average, so it too is in $L^2(\mathbb{P})$. The centered matrices and their differences below are therefore well defined and square-integrable. For the operator norm, the triangle inequality and Jensen's inequality give $\text{var}[M] \leq 4\mathbb{E}[|M|^2]$ for every square-integrable random matrix M . We apply this to the difference between $\mathbf{a}_*^{-1}(\square_n)$ and the average of $\mathbf{a}_*^{-1}(z + \square_k)$ over $z \in 3^k \mathbb{Z}^d \cap \square_n$, and obtain, for every $k < n$,

$$(4.21) \quad \text{var}[\mathbf{a}_*^{-1}(\square_n)] \leq 2 \text{var} \left[\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k) \right] + 8 \mathbb{E} \left[\left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k) - \mathbf{a}_*^{-1}(\square_n) \right|^2 \right].$$

We compare the arithmetic and harmonic means. Let $\mathbf{b}_1, \dots, \mathbf{b}_N$ be symmetric positive-definite matrices, and write $\mathbf{h} := (\sum_{i=1}^N \mathbf{b}_i^{-1})^{-1}$ for their harmonic mean. For every positive-definite matrix $\mathbf{b}$,

$$(4.22) \quad \sum_{i=1}^N \mathbf{b}_i = \mathbf{h} + \sum_{i=1}^N (\mathbf{b}_i - \mathbf{b}) \mathbf{b}_i^{-1} (\mathbf{b}_i - \mathbf{b}) - \underbrace{(\mathbf{h} - \mathbf{b}) \mathbf{h}^{-1} (\mathbf{h} - \mathbf{b})}_{\geq 0} \leq \mathbf{h} + \sum_{i=1}^N (\mathbf{b}_i - \mathbf{b}) \mathbf{b}_i^{-1} (\mathbf{b}_i - \mathbf{b}).$$

Apply (4.22) with $\{\mathbf{a}^{-1}(z + \square_k) : z \in 3^k \mathbb{Z}^d \cap \square_n\}$ in place of the $\mathbf{b}_i$ and combine this with (2.24) and (2.14) to obtain, for every positive-definite matrix $\mathbf{b}$,

$$\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k) - \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}(z + \square_k) \right)^{-1} \leq 2 \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_k, \mathbf{b}^{-1} e_i, e_i; \mathbf{a}) \mathbf{I}_d.$$

Using the previous display and the subadditivity of the coarse-grained matrices $\mathbf{a}_*^{-1}$ and $\mathbf{a}$, we find

$$(4.23) \quad \begin{aligned} \mathbf{a}^{-1}(\square_n) &\leq \mathbf{a}_*^{-1}(\square_n) \leq \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k) \\ &\leq \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}(z + \square_k) \right)^{-1} + 2 \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_k, \mathbf{b}^{-1} e_i, e_i; \mathbf{a}) \right) \mathbf{I}_d \\ &\leq \mathbf{a}^{-1}(\square_n) + 2 \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_k, \mathbf{b}^{-1} e_i, e_i; \mathbf{a}) \right) \mathbf{I}_d. \end{aligned}$$

We deduce from (4.23) that

$$0 \leq \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k) - \mathbf{a}_*^{-1}(\square_n) \leq 2 \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_k, \mathbf{b}^{-1} e_i, e_i; \mathbf{a}) \right) \mathbf{I}_d,$$

and therefore

$$\mathbb{E} \left[\left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k) - \mathbf{a}_*^{-1}(\square_n) \right|^2 \right] \leq 4 \mathbb{E} \left[\left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_k, \mathbf{b}^{-1} e_i, e_i; \mathbf{a}) \right)^2 \right].$$

Combining this estimate with (4.21) yields (4.20). $\square$

We next combine the previous four lemmas into an upper bound on $\mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)]$.

Proposition 4.9. *There exist constants $c(d) > 0$ and $C(d) < \infty$ such that the following holds. Let $L \in \mathbb{N}_0$, $\xi \in [16d, \infty)$ and $\delta_1 \in [C\xi^{\bar{d}}, c]$ be such that $\mathcal{S}(L, \xi, \delta_1)$ is valid. For every $m \in \mathbb{N}$ with $m \geq L$ and $C3^{-(m-L)} \leq 1/4$, and every $p, q \in \mathbb{R}^d$ satisfying*

$$(4.24) \quad q = \bar{\mathbf{a}}_L p \quad \text{and} \quad |q|^2 \leq \bar{\mathbf{a}}_L,$$

we have the estimate

$$(4.25) \quad \begin{aligned} &\mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)] \\ &\leq C\delta_1(\delta_1 + 3^{-(m-L)}) + C \sum_{n=L}^m 3^{-(m-n)} \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L) - J(\square_m, p, q; \mathbf{a}_L)]. \end{aligned}$$

Proof. The proof proceeds in five steps. Step 1 bounds $\mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)]$ by the sum on the right side of (4.25) and three further terms: the distances of the expectations $\bar{\mathbf{a}}_{L,*}^{-1}(\square_n)$ of $\mathbf{a}_{L,*}^{-1}(\square_n)$ from $\bar{\mathbf{a}}_L^{-1}$, the variances of $\mathbf{a}_{L,*}^{-1}(\square_n)$, and the contribution of the complement of the event $\mathbf{G}$ defined below, on which the coarse-grained ellipticity constants of $\mathbf{a}_L$ on $\square_m$ lie within a factor 2 of $\bar{\mathbf{a}}_L$. Steps 2, 3 and 4 bound these three terms using $\mathcal{S}(L, \xi, \delta_1)$, and Step 5 combines the bounds. The

heart of the matter is Step 3: there the independence of $\mathbf{a}_L$ on distant cubes of scale L makes the variance of the average of $\mathbf{a}_{L,*}^{-1}(z + \square_L)$ over $z \in 3^L \mathbb{Z}^d \cap \square_n$ decay like $3^{-d(n-L)}$, which gives the term $\delta_1 3^{-(m-L)}$ in (4.25).

Define the event $\mathbf{G}$ by

$$(4.26) \quad \mathbf{G} := \left\{ \frac{1}{2} \bar{\mathbf{a}}_L \leq \lambda_{1/4}(\square_m; \mathbf{a}_L) \leq \Lambda_{1/4}(\square_m; \mathbf{a}_L) \leq 2 \bar{\mathbf{a}}_L \right\}.$$

Step 1. We combine the previous lemmas to obtain an upper bound for $\mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)]$. We show that there exists $C(d) < \infty$ such that, if $3^{-\frac{3}{2}(m-L)}C \leq 1/4$, then

$$(4.27) \quad \begin{aligned} \mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)] &\leq C \sum_{n=L}^m 3^{-(m-n)} \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L) - J(\square_m, p, q; \mathbf{a}_L)] \\ &\quad + C \bar{\mathbf{a}}_L^2 \sum_{n=L}^m 3^{-(m-n)} \left(\text{var}[\mathbf{a}_{L,*}^{-1}(\square_n)] + |\bar{\mathbf{a}}_{L,*}^{-1}(\square_n) - \bar{\mathbf{a}}_L^{-1}|^2 \right) \\ &\quad + C \mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L) \mathbf{1}_{\mathbf{G}^c}]. \end{aligned}$$

Taking expectations in (4.15) and using the subadditivity of J and stationarity, we obtain

$$\begin{aligned} \mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)] &\leq \mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L)] \\ &= \mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L) \mathbf{1}_{\mathbf{G}}] + \mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L) \mathbf{1}_{\mathbf{G}^c}] \\ &\leq C \mathbb{E} \left[\Lambda_{1/4}(\square_m; \mathbf{a}_L) \left(\frac{\Lambda_{1/4}(\square_m; \mathbf{a}_L)}{\lambda_{1/4}(\square_m; \mathbf{a}_L)} \right)^{1/2} 3^{-2m} \|\nabla v(\cdot, \square_m, p, q; \mathbf{a}_L)\|_{\underline{H}^{-1}(\square_m)}^2 \mathbf{1}_{\mathbf{G}} \right] \\ &\quad + C \mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L) - J(\square_m, p, q; \mathbf{a}_L)] \\ &\quad + C \mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L) \mathbf{1}_{\mathbf{G}^c}]. \end{aligned}$$

Applying Lemma 4.7, squaring both sides of (4.16), taking expectations and using that

$$\left(\frac{\Lambda_{1/4}(\square_m; \mathbf{a}_L)}{\lambda_{1/4}(\square_m; \mathbf{a}_L)} \right)^{3/2} \mathbf{1}_{\mathbf{G}} \leq 8,$$

and that $\Lambda_{1/4}(\square_m; \mathbf{a}_L) \mathbf{1}_{\mathbf{G}} \leq 2 \bar{\mathbf{a}}_L$, we obtain

$$\begin{aligned} &\mathbb{E} \left[\Lambda_{1/4}(\square_m; \mathbf{a}_L) \left(\frac{\Lambda_{1/4}(\square_m; \mathbf{a}_L)}{\lambda_{1/4}(\square_m; \mathbf{a}_L)} \right)^{1/2} 3^{-2m} \|\nabla v(\cdot, \square_m, p, q; \mathbf{a}_L)\|_{\underline{H}^{-1}(\square_m)}^2 \mathbf{1}_{\mathbf{G}} \right] \\ &\leq C \sum_{n=L}^m 3^{-(m-n)} \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L) - J(\square_m, p, q; \mathbf{a}_L)] \\ &\quad + C \bar{\mathbf{a}}_L \sum_{n=L}^m 3^{-(m-n)} \mathbb{E} \left[\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |\mathbf{a}_{L,*}^{-1}(\square_m) q - \mathbf{a}_{L,*}^{-1}(z + \square_n) q|^2 \right] \\ &\quad + C \bar{\mathbf{a}}_L \mathbb{E} \left[|\mathbf{a}_{L,*}^{-1}(\square_m) q - p|^2 \right] \\ &\quad + C 3^{-\frac{3}{2}(m-L)} \mathbb{E} \left[\|\mathbf{a}_L^{1/2} \nabla v\|_{\underline{L}^2(\square_m)}^2 \right]. \end{aligned}$$

Under the condition $C3^{-3(m-L)/2} \leq 1/4$, we use (2.20) to bound the last term by $\frac{1}{2}\mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)]$, which we absorb into the left side. By stationarity and the triangle inequality,

$$\begin{aligned} & \mathbb{E} \left[\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |\mathbf{a}_{L,*}^{-1}(\square_m) - \mathbf{a}_{L,*}^{-1}(z + \square_n)|^2 \right] \\ & \leq 3 \text{var}[\mathbf{a}_{L,*}^{-1}(\square_m)] + 3 \text{var}[\mathbf{a}_{L,*}^{-1}(\square_n)] + 3|\bar{\mathbf{a}}_{L,*}^{-1}(\square_m) - \bar{\mathbf{a}}_{L,*}^{-1}(\square_n)|^2. \end{aligned}$$

Combining the previous two displays and using (4.24), we obtain (4.27).

Step 2. Comparison of $\bar{\mathbf{a}}_{L,*}(\square_n)$ with $\bar{\mathbf{a}}_L$ for $n \geq L$. Assuming $\mathcal{S}(L, \xi, \delta_1)$, we prove that, for every $n \geq L$,

$$(4.28) \quad |\bar{\mathbf{a}}_{L,*}^{-1}(\square_n) - \bar{\mathbf{a}}_L^{-1}|^2 \leq C\bar{\mathbf{a}}_L^{-2}\delta_1^2.$$

Indeed, subadditivity on the partition into cubes of scale L and the induction hypothesis give, for every $|e| = 1$,

$$\mathbb{E}[J(\square_n, \bar{\mathbf{a}}_L^{-1/2}e, \bar{\mathbf{a}}_L^{1/2}e; \mathbf{a}_L)] \leq \delta_1.$$

On the other hand, (2.11), the definition of the expected coarse-grained matrices, and their monotonicity imply

$$2\mathbb{E}[J(\square_n, \bar{\mathbf{a}}_L^{-1/2}e, \bar{\mathbf{a}}_L^{1/2}e; \mathbf{a}_L)] \geq \bar{\mathbf{a}}_L \bar{\mathbf{a}}_{L,*}^{-1}(\square_n) - 1 \geq 0.$$

This proves (4.28).

Step 3. Using the variance estimates in Lemma 4.8, the independence assumption, and the hypothesis $\mathcal{S}(L, \xi, \delta_1)$, we show that, for every $n \geq L$,

$$(4.29) \quad \text{var}[\mathbf{a}_{L,*}^{-1}(\square_n)] \leq C\bar{\mathbf{a}}_L^{-2}\delta_1(\delta_1 + 3^{-d(n-L)}).$$

For $n = L$, (4.29) follows directly from $\mathcal{S}(L, \xi, \delta_1)$ and (2.29). By Lemma 4.8, for every $n > L$ we have

$$\text{var}[\mathbf{a}_{L,*}^{-1}(\square_n)] \leq 2 \text{var} \left[\sum_{z \in 3^L \mathbb{Z}^d \cap \square_n} \mathbf{a}_{L,*}^{-1}(z + \square_L) \right] + 32 \mathbb{E} \left[\left(\sum_{z \in 3^L \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_L, \bar{\mathbf{a}}_L^{-1}e_i, e_i; \mathbf{a}_L) \right)^2 \right].$$

By assumption (g1) and the definition of $\mathbf{a}_L$, for every $y, z \in 3^L \mathbb{Z}^d$ with $|y - z| > d3^{L+1}$, the random variables $\mathbf{a}_{L,*}^{-1}(y + \square_L)$ and $\mathbf{a}_{L,*}^{-1}(z + \square_L)$ are independent. Consequently, by $\mathcal{S}(L, \xi, \delta_1)$ and (2.29),

$$\begin{aligned} \text{var} \left[\sum_{z \in 3^L \mathbb{Z}^d \cap \square_n} \mathbf{a}_{L,*}^{-1}(z + \square_L) \right] & \leq C3^{-d(n-L)} \text{var}[\mathbf{a}_{L,*}^{-1}(\square_L)] \leq C3^{-d(n-L)} \mathbb{E}[|\mathbf{a}_{L,*}^{-1}(\square_L) - \bar{\mathbf{a}}_L^{-1}|^2] \\ & \leq C\bar{\mathbf{a}}_L^{-2}3^{-d(n-L)}\delta_1. \end{aligned}$$

The identical calculation for the averages over z of $\sum_{i=1}^d J(z + \square_L, \bar{\mathbf{a}}_L^{-1}e_i, e_i; \mathbf{a}_L)$ gives

$$\text{var} \left[\sum_{z \in 3^L \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_L, \bar{\mathbf{a}}_L^{-1}e_i, e_i; \mathbf{a}_L) \right] \leq C\bar{\mathbf{a}}_L^{-2}3^{-d(n-L)}\delta_1^2.$$

Finally, using $\mathcal{S}(L, \xi, \delta_1)$ again, we have

$$\mathbb{E}[J(z + \square_L, \bar{\mathbf{a}}_L^{-1}e_i, e_i; \mathbf{a}_L)]^2 = \bar{\mathbf{a}}_L^{-2} \mathbb{E}[J(\square_L, \bar{\mathbf{a}}_L^{-1/2}e_i, \bar{\mathbf{a}}_L^{1/2}e_i; \mathbf{a}_L)]^2 \leq \bar{\mathbf{a}}_L^{-2}\delta_1^2.$$

Combining the previous three displays yields (4.29).

Step 4. The contribution of the complement of $\mathbf{G}$. We prove that

$$(4.30) \quad \mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L) \mathbb{1}_{\mathbf{G}^c}] \leq C\delta_1^2.$$

By (4.26) and (2.38),

$$\mathbf{G}^c \subseteq \{\mathcal{E}_{1/4, \infty}(\square_m; \mathbf{a}_L, \bar{\mathbf{a}}_L) \geq 1/5\}.$$

Apply the bound (4.13) of Lemma 4.5 with $s = 1/4$ and the same exponent ξ . This gives, for a constant $C_B(d) < \infty$,

$$(4.31) \quad \mathbb{P}[\mathbf{G}^c] \leq \mathbb{P}[\mathcal{E}_{1/4, \infty}(\square_m; \mathbf{a}_L, \bar{\mathbf{a}}_L) \geq 1/5] \leq (C_B \delta_1)^{\xi/2}.$$

We have $\mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L)^2] \leq C\delta_1^2$ by subadditivity down to scale L , the induction hypothesis, and (4.24). After decreasing $c(d)$, we may suppose that $C_B \delta_1 \leq 1$. The Cauchy–Schwarz inequality, (4.31), and $\xi \geq 4$ give

$$\mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L) \mathbb{1}_{\mathbf{G}^c}] \leq C\delta_1 (C_B \delta_1)^{\xi/4} \leq C\delta_1^2.$$

Step 5. We conclude. Combining (4.27), (4.28), (4.29), and (4.30) gives (4.25). $\square$

Proposition 4.10 (Contraction above the cutoff scale). *There exist constants $c(d) > 0$ and $C(d) < \infty$ such that the following holds. Let $m_0 \in \mathbb{N}$, $\xi \in [16d, \infty)$ and $\delta_1 \in [C\xi\bar{\delta}^2, c]$ be such that $\mathcal{S}(m_0, \xi, \delta_1)$ is valid. Then, for every $m \in \mathbb{N}$ and $L \in \mathbb{N}_0$ with $L \leq m_0$ and $m \geq L + C \log(2 + \xi)$, we have*

$$(4.32) \quad \mathbb{E} \left[\max_{|e|=1} (J(\square_m, \bar{\mathbf{a}}_L^{-1/2} e, \bar{\mathbf{a}}_L^{1/2} e; \mathbf{a}_L))^\xi \right]^{1/\xi} \leq \frac{1}{4} \delta_1.$$

Proof. Fix $p, q \in \mathbb{R}^d$ satisfying (4.24). By subadditivity on the partition into cubes of scale L , the induction hypothesis and (4.24), for every $n \geq L$,

$$(4.33) \quad \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L)^\xi]^{1/\xi} \leq \delta_1.$$

Proposition 4.9 gives (4.25) at the scales $m \geq L$ with $C3^{-(m-L)} \leq 1/4$; at each of the at most $C(d)$ smaller scales $n \geq L$, the bound (4.34) below follows from (4.33) and Jensen's inequality once its constant is large enough. Iterating (4.25) over the larger scales, starting from (4.33), we obtain an exponent $\alpha(d) > 0$ and a constant $C(d) < \infty$ such that, for every $n \in \mathbb{N}$ with $n \geq L$,

$$(4.34) \quad \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L)] \leq C\delta_1 (\delta_1 + 3^{-\alpha(n-L)}).$$

Using subadditivity and the triangle inequality, we have, for every $m \geq n \geq L$,

$$\begin{aligned} & \mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)^\xi]^{1/\xi} \\ & \leq \mathbb{E} \left[\left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} J(z + \square_n, p, q; \mathbf{a}_L) \right)^\xi \right]^{1/\xi} \\ & \leq \mathbb{E} \left[\left| \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} (J(z + \square_n, p, q; \mathbf{a}_L) - \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L)]) \right|^\xi \right]^{1/\xi} + \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L)]. \end{aligned}$$

Since $J(z + \square_n, p, q; \mathbf{a}_L)$ depends only on $\mathbf{a}_L|_{z + \square_n}$ and $\mathbf{a}_L$ has range of dependence at most $\sqrt{d}3^n$, we may group the summands into $C(d)$ families of independent random variables. Applying Rosenthal's inequality (Lemma B.1) to each family and using (4.33), we find that

$$\begin{aligned} & \mathbb{E} \left[\left| \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} (J(z + \square_n, p, q; \mathbf{a}_L) - \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L)]) \right|^\xi \right]^{1/\xi} \\ & \leq C3^{-\frac{d}{2}(m-n)} \xi^{1/2} \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L)^2]^{1/2} + C3^{-\frac{d(\xi-1)}{\xi}(m-n)} \xi \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L)^\xi]^{1/\xi} \\ & \leq C3^{-\frac{d}{2}(m-n)} \xi^{1/2} \delta_1 + C3^{-\frac{d(\xi-1)}{\xi}(m-n)} \xi \delta_1 \\ (4.35) \quad & \leq C3^{-\frac{d}{2}(m-n)} \xi \delta_1. \end{aligned}$$

Combining the previous two displays with (4.34), we obtain

$$\mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)^\xi]^{1/\xi} \leq C3^{-\frac{d}{2}(m-n)} \xi \delta_1 + C\delta_1 (\delta_1 + 3^{-\alpha(n-L)}).$$

Selecting $n := L + \lceil \frac{1}{2}(m - L) \rceil$ yields

$$\mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)^\xi]^{1/\xi} \leq C 3^{-\frac{d}{4}(m-L)} \xi \delta_1 + C \delta_1 (\delta_1 + 3^{-\frac{1}{2}\alpha(m-L)}).$$

If $m \geq L + C \log(2 + \xi)$, then the right side is at most $c(d)\delta_1$. To pass from a fixed direction to the supremum in (4.32), we use a fixed $1/4$ -net of the unit sphere. Since the map $e \mapsto J(\square_m, \bar{\mathbf{a}}_L^{-1/2}e, \bar{\mathbf{a}}_L^{1/2}e; \mathbf{a}_L)$ is a nonnegative quadratic form, its supremum on the sphere is at most twice its maximum on the net. Bounding the maximum over the net by the sum over its points, whose number depends only on d , we obtain

$$\mathbb{E} \left[\max_{|e|=1} J(\square_m, \bar{\mathbf{a}}_L^{-1/2}e, \bar{\mathbf{a}}_L^{1/2}e; \mathbf{a}_L)^\xi \right]^{1/\xi} \leq \frac{1}{4} \delta_1.$$

This completes the proof. $\square$

To complete the proof of Proposition 4.1, we compare the coefficient at each scale with the coefficient h scales below it using Lemma 2.8, and choose h so that the resulting error is absorbed by the gain in Proposition 4.10.

Proof of Proposition 4.1. We use two inductions. The first proves $\mathcal{S}(\infty, \xi, C\xi \log^2(2 + \xi)\bar{\delta}^2)$; the second removes one factor of $\log(2 + \xi)$.

Step 1. The base case. For both inductions, it is the assertion that, for some constant $C(d) < \infty$ and every $\xi \in [1, \infty)$ and $h \in \mathbb{N}$ satisfying $C\xi\bar{\delta}^2 h \leq 1$,

$$(4.36) \quad \mathcal{S}(3h, \xi, C\xi\bar{\delta}^2 h) \text{ is valid.}$$

Lemma 3.6 implies that, for every $n \in \{0, \dots, 3h\}$,

$$\|\mathcal{J}_n(\square_n)\|_{L^\xi(\mathbb{P})} \leq C\xi\bar{\delta}^2 h \exp(C\xi\bar{\delta}^2 h).$$

This implies (4.36).

Step 2. First induction. We prove that there exists $C(d) < \infty$ such that $\mathcal{S}(\infty, \xi, C\xi \log^2(2 + \xi)\bar{\delta}^2)$ is valid for every $\xi \in [16d, C^{-1}\bar{\delta}^{-2}|\log \bar{\delta}|^{-2}]$. We assume that

$$(4.37) \quad h \geq K \log(2 + \xi) \quad \text{and} \quad \xi\bar{\delta}^2 h^2 \leq K^{-1}\delta_1 \leq K^{-2},$$

where $K(d)$ is a large constant to be determined below. We iterate in blocks of h scales. For parameters satisfying (4.37) and every $m \in \mathbb{N} \cap [3h, \infty)$, we prove

$$(4.38) \quad \mathcal{S}(m, \xi, \delta_1) \implies \mathcal{S}(m + h, \xi, \delta_1).$$

Equations (4.36) and (4.38) imply by induction that $\mathcal{S}(\infty, \xi, \delta_1)$ holds for every choice of parameters satisfying (4.37), after enlarging K .

To prove (4.38), we fix $m \in \mathbb{N}$ with $m \geq 3h$ such that $\mathcal{S}(m, \xi, \delta_1)$ is valid. By Proposition 4.10, if K is taken sufficiently large, then

$$(4.39) \quad \mathbb{E} \left[\max_{|e|=1} J(\square_{m+h}, \bar{\mathbf{a}}_m^{-1/2}e, \bar{\mathbf{a}}_m^{1/2}e; \mathbf{a}_m)^\xi \right]^{1/\xi} \leq \frac{1}{4} \delta_1.$$

We apply Lemma 2.8 with $\delta = 1$ and $\lambda = \bar{\mathbf{a}}_{m+h}\bar{\mathbf{a}}_m^{-1}$, and with $\mathbf{b} = \mathbf{a}_{m+h}$ and $\mathbf{a} = \mathbf{a}_m$. This gives, for every bounded convex domain $U \subseteq \mathbb{R}^d$ and $|e| = 1$,

$$(4.40) \quad J(U, \bar{\mathbf{a}}_{m+h}^{-1/2}e, \bar{\mathbf{a}}_{m+h}^{1/2}e; \mathbf{a}_{m+h}) \leq 2J(U, \bar{\mathbf{a}}_m^{-1/2}e, \bar{\mathbf{a}}_m^{1/2}e; \mathbf{a}_m) + 3\mathcal{X}_{m,h}(U)^2 (J(U, \bar{\mathbf{a}}_m^{-1/2}e, \bar{\mathbf{a}}_m^{1/2}e; \mathbf{a}_m) + 1),$$

where we have defined

$$\mathcal{X}_{m,h}(U) := \|\bar{\mathbf{a}}_{m+h}\bar{\mathbf{a}}_m^{-1}(\mathbf{a}_{m+h}^{-1}\mathbf{a}_m) - 1\|_{L^\infty(U)} + \|\bar{\mathbf{a}}_{m+h}^{-1}\bar{\mathbf{a}}_m(\mathbf{a}_{m+h}\mathbf{a}_m^{-1}) - 1\|_{L^\infty(U)}.$$

Taking the maximum over $|e| = 1$ in (4.40) gives

$$(4.41) \quad \mathcal{J}_{m+h}(U) \leq 2\mathcal{J}_m(U) + 3\mathcal{X}_{m,h}(U)^2(\mathcal{J}_m(U) + 1).$$

The fields $\mathbf{a}_{m+h}^{-1}\mathbf{a}_m$ and $\mathbf{a}_m$ are independent; hence so are $\mathcal{X}_{m,h}(\square_{m+h})$ and $\mathcal{J}_m(\square_{m+h})$. The triangle inequality therefore gives

$$(4.42) \quad \begin{aligned} \|\mathcal{J}_{m+h}(\square_{m+h})\|_{L^\xi(\mathbb{P})} &\leq 2\|\mathcal{J}_m(\square_{m+h})\|_{L^\xi(\mathbb{P})} \\ &\quad + 3\|\mathcal{X}_{m,h}(\square_{m+h})^2\|_{L^\xi(\mathbb{P})}(1 + \|\mathcal{J}_m(\square_{m+h})\|_{L^\xi(\mathbb{P})}). \end{aligned}$$

By (3.25) and (3.5), applied with moment 2ξ , together with (4.37), we have the estimate

$$\mathbb{E}[\mathcal{X}_{m,h}(\square_{m+h})^{2\xi}]^{1/\xi} \leq C\xi\bar{\delta}^2 h^2 \leq \frac{1}{12}\delta_1.$$

Inserting this and (4.39) into (4.42) gives

$$\|\mathcal{J}_{m+h}(\square_{m+h})\|_{L^\xi(\mathbb{P})} \leq \frac{1}{2}\delta_1 + C\xi\bar{\delta}^2 h^2 \leq \delta_1.$$

The same argument, with $(m, m+h)$ replaced by $(k-h, k)$, gives the same bound on $\square_k$ for each $k \in \{m+1, \dots, m+h\}$; together with $\mathcal{S}(m, \xi, \delta_1)$, this proves (4.38). Hence $\mathcal{S}(\infty, \xi, C\xi \log^2(2+\xi)\bar{\delta}^2)$ holds for all $\xi \geq 16d$ in the stated range. The remaining range $1 \leq \xi < 16d$ follows from monotonicity of $L^\xi(\mathbb{P})$ norms.

Step 3. Second induction. We prove (4.3). We now consider parameters ξ, δ_1 and h satisfying

$$(4.43) \quad h \geq K \log(2 + \xi) \quad \text{and} \quad \xi\bar{\delta}^2 h \leq K^{-1}\delta_1 \leq K^{-2},$$

where $K(d)$ is a large constant to be determined below. We also assume $\xi \geq 4$.

For every $m \in \mathbb{N} \cap [3h, \infty)$, we prove the implication

$$(4.44) \quad \mathcal{S}(m, \xi, \delta_1) \implies \mathcal{S}(m+h, \xi, \delta_1).$$

By subadditivity, (4.41), Hölder's inequality, and the independence of the fields defining $\mathcal{X}_{m,h}$ and $\mathcal{J}_m$, we have

$$\begin{aligned} \|\mathcal{J}_{m+h}(\square_{m+h})\|_{L^\xi(\mathbb{P})} &\leq \mathbb{E} \left[\left(\sum_{z \in 3^m \mathbb{Z}^d \cap \square_{m+h}} \mathcal{J}_{m+h}(z + \square_m) \right)^\xi \right]^{1/\xi} \\ &\leq 3\mathbb{E} \left[\sum_{z \in 3^m \mathbb{Z}^d \cap \square_{m+h}} (\mathcal{X}_{m,h}(z + \square_m))^{2\xi} \right]^{1/\xi} \\ &\quad + 3\mathbb{E} \left[\sum_{z \in 3^m \mathbb{Z}^d \cap \square_{m+h}} (1 + \mathcal{X}_{m,h}(z + \square_m)^2)^\xi \right]^{1/\xi} \\ &\quad \times \mathbb{E} \left[\left(\sum_{z \in 3^m \mathbb{Z}^d \cap \square_{m+h}} \mathcal{J}_m(z + \square_m)^2 \right)^{\xi/2} \right]^{1/\xi} \\ &\leq C\xi\bar{\delta}^2 h + 3(1 + C\xi\bar{\delta}^2 h) \mathbb{E} \left[\left(\sum_{z \in 3^m \mathbb{Z}^d \cap \square_{m+h}} \mathcal{J}_m(z + \square_m)^2 \right)^{\xi/2} \right]^{1/\xi}. \end{aligned}$$

Each summand depends only on the restriction of $\mathbf{a}_m$ to its own cube, and $\mathbf{a}_m$ has range of dependence at most $\sqrt{d}3^m$. We can therefore color the cubes with $C(d)$ colors so that the summands of each color are independent. Applying Rosenthal's inequality (Lemma B.1) to each color as in (4.35),

with $\mathcal{J}_m^2$ in place of J and $\xi/2$ in place of ξ , and using $\mathcal{S}(m, \xi, \delta_1)$ and the estimate $\mathcal{S}(\infty, 2, C\bar{\delta}^2)$ proved in Step 2, we obtain

$$\begin{aligned} & \mathbb{E} \left[\left(\sum_{z \in 3^m \mathbb{Z}^d \cap \square_{m+h}} \mathcal{J}_m(z + \square_m)^2 \right)^{\xi/2} \right]^{2/\xi} \\ & \leq C 3^{-\frac{d}{2}h} \xi^{1/2} \mathbb{E}[\mathcal{J}_m(\square_m)^4]^{1/2} + C 3^{-\frac{d(\xi-2)}{\xi}h} \xi \mathbb{E}[\mathcal{J}_m(\square_m)^\xi]^{2/\xi} \\ & \quad + \mathbb{E}[\mathcal{J}_m(\square_m)^2] \\ & \leq C 3^{-\frac{d}{2}h} \xi^{1/2} \delta_1^2 + C 3^{-\frac{d}{2}h} \xi \delta_1^2 + C \bar{\delta}^4 \\ & \leq \frac{1}{36} \delta_1^2. \end{aligned}$$

Combining the previous two displays and using $\delta_1 \leq 1$, we obtain

$$\|\mathcal{J}_{m+h}(\square_{m+h})\|_{L^\xi(\mathbb{P})} \leq C \xi \bar{\delta}^2 h + \frac{1}{2} (1 + C \xi \bar{\delta}^2 h) \delta_1 \leq \frac{1}{2} \delta_1 + C \xi \bar{\delta}^2 h \leq \delta_1.$$

The last inequality follows from (4.43) after fixing $K(d)$ sufficiently large. The same calculation, with $(m, m+h)$ replaced by $(k-h, k)$, gives the same bound on $\square_k$ for each $k \in \{m+1, \dots, m+h\}$; together with $\mathcal{S}(m, \xi, \delta_1)$, this proves (4.44).

The base case (4.36) and (4.44) imply that $\mathcal{S}(\infty, \xi, C \xi \log(2 + \xi) \bar{\delta}^2)$ holds for $\xi \geq 4$ whenever $\xi \log(2 + \xi) \bar{\delta}^2 \leq C^{-1}$. The range $1 \leq \xi < 4$ follows by monotonicity of $L^\xi(\mathbb{P})$ norms. This proves (4.3).

Step 4. We prove (4.4). It follows from (4.8), applied with $\delta_1 = C \xi \log(2 + \xi) \bar{\delta}^2$. $\square$

5. EFFECTIVE DIFFUSIVITIES

In this section we determine how fast the effective diffusivities $\bar{\mathbf{a}}_m$ decay. We prove that they are nonincreasing in m , that $e^{-(m+1)\tau^2} \leq \bar{\mathbf{a}}_m \leq 3^{-\eta_{\text{sub}}(m+1)}$ for every $m \in \mathbb{N}_0$, with an exponent $\eta_{\text{sub}} > 0$ depending only on d and the law of $\mathbf{g}_0$, and that, at weak disorder,

$$\left| \log \bar{\mathbf{a}}_m + \frac{2\tau^2(m+1)}{d} \right| \leq C \bar{\delta}^2 |\log \bar{\delta}| (1 + m \bar{\delta}).$$

In dimension two, planar duality gives the exact identity $\bar{\mathbf{a}}_m = e^{-\tau^2(m+1)}$. These statements are collected in Theorem 5.1. The decay of $\bar{\mathbf{a}}_m$ gives the time scale $3^{2N}/\bar{\mathbf{a}}_N$ in Theorem A and the exponent η in Theorem A(iii).

We prove the asymptotics at weak disorder through matching upper and lower bounds on $\bar{\mathbf{a}}_m/\bar{\mathbf{a}}_{m-h}$ for integers $h \leq \bar{\delta}^{-1}$. We compare the corrector of $\mathbf{a}_m$ with the solution of the linearized corrector equation. The corrector is the function added to an affine function to make it solve the equation with coefficient $\mathbf{a}_m$. The linearized corrector solves a Poisson equation forced by the added fields, and we compute its energy exactly in Lemma 5.3. The error term in these asymptotics exceeds the main term unless τ^2 is bounded below by a power of $\bar{\delta}$. We therefore prove the two-sided bound $e^{-(m+1)\tau^2} \leq \bar{\mathbf{a}}_m \leq 3^{-\eta_{\text{sub}}(m+1)}$ separately, by a nonperturbative argument which assumes no smallness of $\bar{\delta}$. Its upper bound makes the process subdiffusive in every dimension, and the asymptotics identify the exponent at weak disorder.

Theorem 5.1 (Bounds on the effective diffusivities). *There exists $\eta_{\text{sub}} = \eta_{\text{sub}}(d, \text{law}(\mathbf{g}_0)) > 0$ such that, for every $0 \leq n \leq m$,*

$$e^{-(m+1)\tau^2} \leq \bar{\mathbf{a}}_m \leq \bar{\mathbf{a}}_n \leq \min\{1, e^{2\tau^2(m-n)} \bar{\mathbf{a}}_n\}, \quad \bar{\mathbf{a}}_m \leq 3^{-\eta_{\text{sub}}(m+1)}.$$

In dimension two, for every $m \in \mathbb{N}_0$,

$$(5.1) \quad \bar{\mathbf{a}}_m = e^{-(m+1)\tau^2}.$$

Moreover, there exist $\bar{\delta}_0(d) > 0$ and $C(d) < \infty$ such that, if $\bar{\delta} \leq \bar{\delta}_0(d)$, then, for every $m \in \mathbb{N}_0$,

$$\left| \log \bar{\mathbf{a}}_m + \frac{2\tau^2(m+1)}{d} \right| \leq C\bar{\delta}^2 |\log \bar{\delta}| (1 + m\bar{\delta}).$$

Proof. The monotonicity in m and the comparison between two scales are (3.24), and the affine competitor gives $\bar{\mathbf{a}}_0 \leq 1$. The remaining assertions are Propositions 3.5, 5.2, 5.6, and 5.7. $\square$

Proposition 5.2 (Asymptotics of the effective diffusivities at weak disorder). *There exist constants $\bar{\delta}_0(d) > 0$ and $C(d) < \infty$ such that, if $\bar{\delta} \leq \bar{\delta}_0(d)$, then, for every $m \in \mathbb{N}_0$,*

$$(5.2) \quad \left| \log \bar{\mathbf{a}}_m + \frac{2\tau^2(m+1)}{d} \right| \leq C\bar{\delta}^2 |\log \bar{\delta}| (1 + m\bar{\delta}).$$

Lemma 5.3 (Energy of the linearized corrector). *Let $m \in \mathbb{N}_0$ and $h \in \mathbb{N}$ satisfy $h \leq m+1$ and $h \leq \bar{\delta}^{-1}$, let $X := \prod_{k=m-h+1}^m \exp(\mathbf{g}_k - \tau^2) - 1$, and let $p \in \mathbb{R}^d$ be a unit vector. If $w_p^{(K)} \in H_0^1(\square_K)$ solves*

$$-\Delta w_p^{(K)} = \nabla \cdot (Xp) \quad \text{in } \square_K,$$

then, with an implicit constant that depends only on d ,

$$(5.3) \quad \lim_{K \rightarrow \infty} \mathbb{E}[\|\nabla w_p^{(K)}\|_{\underline{L}^2(\square_K)}^2] = \frac{2\tau^2 h}{d} + O(\bar{\delta}^4 h^2).$$

The same conclusion holds if $w_p^{(K)}$ is the mean-zero solution of

$$-\Delta w_p^{(K)} = -\nabla \cdot (Xp) \quad \text{in } \square_K, \quad \partial_{\mathbf{n}} w_p^{(K)} = Xp \cdot \mathbf{n} \quad \text{on } \partial \square_K.$$

Proof. We show that the limit equals $\mathbb{E}[|\nabla \phi(0)|^2]$, where $\nabla \phi$ is the orthogonal projection of Xp onto the stationary gradients, and then compute this quantity. First, the spectral representation of X gives the decomposition $Xp = \nabla \phi + \nabla \cdot S$ into a scalar and a skew potential, and both potentials grow sublinearly. Next, the potentials, cut off near $\partial \square_K$, serve as competitors in the variational formulas for the Dirichlet and the Neumann problems; the main point is that, by the sublinearity, the cutoffs change the energy by an amount that vanishes as $K \rightarrow \infty$ and then $L \rightarrow \infty$. Finally, the symmetry of the spectral measure gives $\mathbb{E}[|\nabla \phi(0)|^2] = \frac{1}{d} \mathbb{E}[X(0)^2]$, and an expansion of $\mathbb{E}[X(0)^2]$ in the last h fields completes the proof.

The vector products and unit probes in this proof are Euclidean, and every constant C and every implicit constant depends only on d . We construct the scalar and skew potentials, with row divergence $(\nabla \cdot S)_i = \sum_{j=1}^d \partial_j S_{ij}$. The field X is centered, stationary, mean-square continuous and square integrable. Its finite range of dependence makes its covariance compactly supported and integrable. Its spectral measure μ is therefore finite and has no atom at zero. Write the real-field spectral representation as $X(x) = \int e^{ik \cdot x} Z(dk)$, with $Z(-dk) = \overline{Z(dk)}$, where $\mathbb{E}|Z(dk)|^2 = \mu(dk)$ and $\mu(\mathbb{R}^d) = \mathbb{E}[X(0)^2]$. Define potentials anchored at the origin by

$$\phi(x) = \int (e^{ik \cdot x} - 1) \frac{-i p \cdot k}{|k|_2^2} Z(dk), \quad S_{ij}(x) = \int (e^{ik \cdot x} - 1) \frac{-i(p_i k_j - p_j k_i)}{|k|_2^2} Z(dk).$$

The multipliers are understood to be zero at $k = 0$. Their squared magnitudes, after multiplication by $|e^{ik \cdot x} - 1|^2$, are bounded by $C|p|_2^2 \min\{|x|_2^2, |k|_2^{-2}\}$. The derivative multipliers are bounded. Taking jointly measurable versions, we thus have $\phi, S \in L^2(\mathbb{P}; H_{\text{loc}}^1)$, with stationary square-integrable gradients. Differentiation gives the multipliers

$$\widehat{\nabla \phi} = \frac{k(p \cdot k)}{|k|_2^2} \widehat{X}, \quad \widehat{\nabla \cdot S} = \left(p - \frac{k(p \cdot k)}{|k|_2^2} \right) \widehat{X}.$$

Consequently $Xp = \nabla\phi + \nabla \cdot S$, $S = -S^T$, and the two vector fields are orthogonal in $L^2(\mathbb{P}; \mathbb{R}^d)$. Cutting the scalar multiplier off near $k = 0$ approximates $\nabla\phi$ by stationary gradients. The complementary multiplier is divergence free and is orthogonal to stationary gradients by stationary integration by parts. Thus $\nabla\phi$ is the orthogonal projection of Xp onto their closure.

For a centered cube Q_R of side R , put $a_R(k) = \int_{Q_R} e^{ik \cdot x} dx$. Since

$$\int_{Q_R} |e^{ik \cdot x} - a_R(k)|^2 dx = 1 - |a_R(k)|^2 \leq \min\{1, R^2|k|_2^2/12\},$$

the spectral isometry and $\sum_{i,j} |p_i k_j - p_j k_i|^2 \leq 2|p|_2^2 |k|_2^2$ imply

$$R^{-2} \mathbb{E} \int_{Q_R} (|\phi - (\phi)_{Q_R}|^2 + |S - (S)_{Q_R}|^2) \leq 3|p|_2^2 \int \min\{R^{-2}|k|_2^{-2}, 1/12\} \mu(dk) \longrightarrow 0.$$

Here the squared spectral matrix norm is bounded by the sum of the squared entries used in the calculation. Dominated convergence applies because μ is finite and $\mu(\{0\}) = 0$. At $R = 3^K$ this proves

$$(5.4) \quad \lim_{K \rightarrow \infty} 3^{-2K} \mathbb{E} \left[\int_{\square_K} (|\phi - (\phi)_{\square_K}|^2 + |S - (S)_{\square_K}|^2) \right] = 0.$$

The argument applies in dimension two as well: it requires stationary gradients and anchored increments, and makes no finite-variance stationarity assertion about the values of the potentials.

Fix $L \in \mathbb{N}$. Choose $\eta_{K,L} \in C_c^\infty(\square_K)$ equal to one at distance at least 3^{K-L} from $\partial\square_K$ and satisfying $0 \leq \eta_{K,L} \leq 1$ and $|\nabla \eta_{K,L}| \leq C3^{L-K}$. The support of $1 - \eta_{K,L}$ has relative volume at most $C3^{-L}$. Consequently, by stationarity and (5.4), the function $-\eta_{K,L}(\phi - (\phi)_{\square_K}) \in H_0^1(\square_K)$ satisfies

$$\limsup_{K \rightarrow \infty} \mathbb{E} [\|\nabla\phi - \nabla(\eta_{K,L}(\phi - (\phi)_{\square_K}))\|_{\underline{L}^2(\square_K)}^2] \leq C3^{-L} \mathbb{E}[|\nabla\phi(0)|^2],$$

and, for the same reasons, the field

$$\mathbf{j}_{K,L} := \nabla \cdot (\eta_{K,L}(S - (S)_{\square_K})) \in L_{\text{sol},0}^2(\square_K)$$

satisfies

$$(5.5) \quad \limsup_{K \rightarrow \infty} \mathbb{E} [\|\mathbf{j}_{K,L} - \nabla \cdot S\|_{\underline{L}^2(\square_K)}^2] \leq C3^{-L} \mathbb{E}[|\nabla \cdot S(0)|^2].$$

For the Dirichlet problem, the variational identity

$$\frac{1}{2} \|\nabla w_p^{(K)}\|_{\underline{L}^2(\square_K)}^2 = \sup_{\varphi \in H_0^1(\square_K)} \int_{\square_K} \left(-Xp \cdot \nabla \varphi - \frac{1}{2} |\nabla \varphi|^2 \right)$$

and the orthogonality of $\nabla \cdot S$ to compactly supported gradients give

$$\mathbb{E} [\|\nabla w_p^{(K)}\|_{\underline{L}^2(\square_K)}^2] \leq \mathbb{E}[|\nabla\phi(0)|^2].$$

Using $\varphi = -\eta_{K,L}(\phi - (\phi)_{\square_K})$ in the variational identity, then sending $K \rightarrow \infty$ and $L \rightarrow \infty$ in this order, proves the reverse inequality.

For the Neumann problem, let $w_p^{(K)}$ now denote the mean-zero solution. Its weak formulation says that $\nabla w_p^{(K)}$ is the orthogonal projection of Xp onto $\nabla H^1(\square_K)$. Using the restriction of ϕ to $\square_K$ as a competitor in the variational formula for the norm of this projection, and using $\mathbb{E}[(Xp)(0) \cdot \nabla\phi(0)] = \mathbb{E}[|\nabla\phi(0)|^2]$, we obtain

$$\mathbb{E} [\|\nabla w_p^{(K)}\|_{\underline{L}^2(\square_K)}^2] \geq \mathbb{E}[|\nabla\phi(0)|^2].$$

On the other hand, the field $\mathbf{j}_{K,L} \in L_{\text{sol},0}^2(\square_K)$ is orthogonal to $\nabla w_p^{(K)}$, and hence

$$\|\nabla w_p^{(K)}\|_{\underline{L}^2(\square_K)}^2 \leq \|Xp - \mathbf{j}_{K,L}\|_{\underline{L}^2(\square_K)}^2.$$

Sending first $K \rightarrow \infty$ and then $L \rightarrow \infty$ and using (5.5) proves the reverse inequality. Consequently, in both cases,

$$\lim_{K \rightarrow \infty} \mathbb{E}[\|\nabla w_p^{(K)}\|_{L^2(\square_K)}^2] = \mathbb{E}[|\nabla \phi(0)|^2].$$

The spectral measure of the scalar stationary field X is invariant under coordinate reflections and permutations. The off-diagonal entries of the Fourier multiplier of the projection therefore integrate to zero against this measure, while the diagonal entries have equal integrals, which sum to $\mathbb{E}[X(0)^2]$. Hence $\mathbb{E}[|\nabla \phi(0)|^2] = \frac{1}{d} \mathbb{E}[X(0)^2]$. Independence of the fields and the definition $\tau^2 = \log \mathbb{E}[\exp(\mathbf{g}_0(0))]$ yield $\mathbb{E}[X(0)^2] = \exp(h(\log \mathbb{E}[\exp(2\mathbf{g}_0(0))] - 2\tau^2)) - 1$. The symmetry and sub-Gaussian bound in Assumptions (g2)–(g3) give $\log \mathbb{E}[\exp(2\mathbf{g}_0(0))] - 2\tau^2 = 2\tau^2 + O(\bar{\delta}^4)$. Since $h\bar{\delta}^2 \leq \bar{\delta}$, expansion of the exponential proves (5.3). $\square$

Lemma 5.4 (Upper bound on $\bar{\mathbf{a}}_m/\bar{\mathbf{a}}_{m-h}$). *There exists $C(d) < \infty$ such that, for every $m, h \in \mathbb{N}$ with*

$$h \leq \bar{\delta}^{-1} \quad \text{and} \quad m - h \geq 16\lceil \log_3 \bar{\delta} \rceil,$$

we have

$$(5.6) \quad \bar{\mathbf{a}}_m \leq \bar{\mathbf{a}}_{m-h} \left(1 - \frac{2\tau^2 h}{d} + C\bar{\delta}^4 h^2 + C\bar{\delta}^2 |\log \bar{\delta}| \right).$$

Proof. We first record a bound on τ^2 , settle the planar case by the exact formula, fix the moment orders, and treat the range $\bar{\delta} > \bar{\delta}_*$, where the ordering (3.24) suffices. The main argument, for $d \geq 3$ and $\bar{\delta} \leq \bar{\delta}_*$, bounds $\bar{\mathbf{a}}_m$ from above by the Dirichlet principle, with a competitor built from the linearized corrector; it runs in four steps, outlined before Step 1, and its heart is Step 3, where Lemma 5.3 gives the main term $-2\tau^2 h/d$.

Symmetry and Assumption (g2), followed by Jensen with exponent $\bar{\delta}^2/2 \leq 1$, give

$$e^{\tau^2} = \mathbb{E} \cosh(\mathbf{g}_0(0)) \leq \mathbb{E} e^{\mathbf{g}_0(0)^2/2} \leq \left(\mathbb{E} e^{\mathbf{g}_0(0)^2/\bar{\delta}^2} \right)^{\bar{\delta}^2/2} \leq 2^{\bar{\delta}^2/2}.$$

Thus $\tau^2 \leq (\log 2/2)\bar{\delta}^2$. First consider $d = 2$. Proposition 3.5 gives $\bar{\mathbf{a}}_m/\bar{\mathbf{a}}_{m-h} = e^{-\tau^2 h}$. The bound $\tau^2 \leq (\log 2/2)\bar{\delta}^2$ and $h\bar{\delta} \leq 1$ permit the estimate $e^{-x} \leq 1 - x + x^2$ with $x = \tau^2 h$, which proves (5.6). Henceforth $d \geq 3$.

We fix all moment orders before restricting the disorder. Let c_P, C_P be constants for Proposition 4.1, set $s = 1/4$ and $\xi_* := \lceil \max\{16d, e^{4c_P}\} \rceil$. Choose a dimensional $\bar{\delta}_* \in (0, e^{-1}]$ so that

$$C_P \xi_* \bar{\delta}_* \leq 1, \quad C_P \xi_* \log(2 + \xi_*) \bar{\delta}_*^2 \leq c_P/4, \quad 16C_P^2 \xi_* \log(2 + \xi_*) \bar{\delta}_*^2 \leq 1.$$

For $0 < \bar{\delta} \leq \bar{\delta}_*$ these ensure every condition of the proposition at (ξ_*, s) : indeed $\bar{\delta}^2 |\log \bar{\delta}| \leq \bar{\delta}$, $4d/s \leq \xi_*$ and $s \log(2 + \xi_*) \geq c_P$. Its first part also applies at moment 1. The field moments used below have fixed orders at most 16 and follow from Assumption (g2); they impose no scale-dependent restriction.

For the complementary range $\bar{\delta} > \bar{\delta}_*$, the ordering (3.24) gives $\bar{\mathbf{a}}_m \leq \bar{\mathbf{a}}_{m-h}$. Since $h \geq 1$,

$$\bar{\delta}^2 h \leq \bar{\delta}_*^{-2} \bar{\delta}^4 h^2, \quad \frac{2\tau^2 h}{d} \leq \frac{\log 2}{d} \bar{\delta}_*^{-2} \bar{\delta}^4 h^2.$$

Increasing $C(d)$ proves the conclusion in this range. We may therefore assume $\bar{\delta} \leq \bar{\delta}_*$ throughout the remaining proof.

We bound $\mathbb{E}[p \cdot \mathbf{a}_m(\square_K)p]$, for a unit vector p and large K , from above by the Dirichlet principle. The competitor is $p \cdot x + w$, where w solves a Poisson equation forced by the last h fields, corrected on each interior cell of the scale 3^n fixed in Step 1 by solutions of cell problems for $\mathbf{a}_m$. Step 1 sets up w and the cells. Step 2 bounds the energy of the competitor by the sum of the cell energies, up to an error $C\bar{\delta}^{15}\bar{\mathbf{a}}_{m-h}$. Step 3 uses the independence of the last h fields from $\mathbf{a}_{m-h}$ to bound this sum, up to a further error of the same size, by $\bar{\mathbf{a}}_{m-h}(\square_n)$ times an expectation involving only these

fields, and evaluates this expectation. This is the heart of the matter: Lemma 5.3 gives the main term $-2\tau^2 h/d$, and reversing the signs of the last h fields shows that the remaining error is $O(\bar{\delta}^4 h^2)$. Step 4 concludes.

Step 1. We set up the Dirichlet problem and record the interior estimates on the **cells** of scale 3^n , the cubes $z + \square_n$ with $z \in 3^n \mathbb{Z}^d$. Fix $p \in \mathbb{R}^d$ with $|p|_2 = 1$, and set

$$(5.7) \quad \mathbf{h} := \sum_{k=m-h+1}^m (\mathbf{g}_k - \tau^2).$$

Choose

$$(5.8) \quad n := m - h - 16 \lceil |\log_3 \bar{\delta}| \rceil,$$

so that $3^{-(m-h-n)} \leq \bar{\delta}^{16}$. For $K > m$, let $w \in H_0^1(\square_K)$ solve

$$-\Delta w = \nabla \cdot ((e^{\mathbf{h}} - 1)p) \quad \text{in } \square_K.$$

Testing the equation with w and applying the gradient Calderón–Zygmund estimate at exponent 8 on the cube (see the reflection argument below), followed by Hölder’s inequality for the lower exponents, give, for $1 \leq \xi \leq 8$,

$$\|\nabla w\|_{L^\xi(\mathbb{P}; \underline{L}^\xi(\square_K))} \leq C \bar{\delta} h^{1/2}, \quad \|\nabla w\|_{\underline{L}^2(\square_K)}^2 = - \int_{\square_K} (e^{\mathbf{h}} - 1)p \cdot \nabla w.$$

Lemma 5.3 yields

$$\lim_{K \rightarrow \infty} \mathbb{E} \|\nabla w\|_{\underline{L}^2(\square_K)}^2 = \frac{2\tau^2 h}{d} + O(\bar{\delta}^4 h^2).$$

For every scale- 3^n cell $Q_z = z + \square_n$ satisfying $z + \square_{m-h} \subseteq \square_K$, define

$$p_z = (p + \nabla w)_{Q_z}, \quad F_z = \nabla w - (\nabla w)_{Q_z}, \quad A_z = \|F_z\|_{\underline{L}^2(Q_z)}, \quad B_z = 3^n \|\nabla F_z\|_{\underline{L}^2(Q_z)}.$$

Odd reflection of the zero-Dirichlet solution across the cube faces, with the forcing field $(e^{\mathbf{h}} - 1)p$ reflected so that its normal component is even and its tangential components are odd, reduces the equation to a periodic Poisson equation. Its right-hand side $\nabla \cdot ((e^{\mathbf{h}} - 1)p) = p \cdot \nabla e^{\mathbf{h}}$ is a scalar function, and the reflected equation has the odd reflection of that function as right-hand side. The periodic Calderón–Zygmund estimates, followed by scaling, give the gradient estimate above and the following Hessian estimate with constants independent of K . In particular, the weak Hessian of w is in $L^4(\square_K)$ and

$$\|D^2 w\|_{\underline{L}^4(\square_K)} \leq C \|\nabla e^{\mathbf{h}}\|_{\underline{L}^4(\square_K)}.$$

At each point, the geometric scaling of the derivatives of the last h fields gives $\|\nabla \mathbf{h}\|_{L^8(\mathbb{P})} \leq C \bar{\delta} 3^{-(m-h)}$, while $\mathbb{E} e^{8\mathbf{h}} \leq C$ because $h \bar{\delta}^2 \leq \bar{\delta}$. Hölder’s inequality and stationarity consequently give $(\mathbb{E} \int_{\square_K} |\nabla e^{\mathbf{h}}|^4)^{1/4} \leq C \bar{\delta} 3^{-(m-h)}$. We use this estimate on the finite family of cells, with no spatial maximum of the Hessian. Jensen’s inequality on each cell and the partition into scale- 3^n cubes give

$$\limsup_{K \rightarrow \infty} \sum_z \mathbb{E}[B_z^4] \leq C 3^{4n} \sup_K \mathbb{E} \int_{\square_K} |D^2 w|^4 \leq C \bar{\delta}^{68}.$$

The passage from all cells to the retained cells contributes their volume ratio, which tends to one. Mean-zero Poincaré on Q_z gives $A_z \leq C B_z$. In particular,

$$\limsup_{K \rightarrow \infty} \left(\sum_z \mathbb{E}[A_z^4] \right)^{1/4} + \limsup_{K \rightarrow \infty} \left(\sum_z \mathbb{E}[B_z^4] \right)^{1/4} \leq C \bar{\delta}^{17} \leq C \bar{\delta}^{15}.$$

The Hessians and the cell norms have measurable versions: the Poisson solution depends continuously on its datum, and difference quotients converge to its weak derivatives. Here and below the

averages run over the retained interior cells. The omitted cells lie within distance $C3^{m-h}$ of $\partial\Box_K$, and their relative volume tends to zero as $K \rightarrow \infty$.

Step 2. We solve cell problems on the interior cells, patch their solutions into a competitor for the Dirichlet principle, and bound the energy by a sum over cells. On Q_z , let $\chi_z, \phi_z \in H_0^1(Q_z)$ solve

$$-\nabla \cdot (\mathbf{a}_m \nabla \chi_z) = \nabla \cdot (\mathbf{a}_m p_z), \quad -\nabla \cdot (\mathbf{a}_m \nabla \phi_z) = \nabla \cdot (\mathbf{a}_m F_z),$$

and set

$$S_z = p_z + \nabla \chi_z, \quad T_z = F_z + \nabla \phi_z.$$

Then $\nabla \cdot (\mathbf{a}_m S_z) = \nabla \cdot (\mathbf{a}_m T_z) = 0$, $(T_z)_{Q_z} = 0$, and $\int_{Q_z} |S_z|^2 \mathbf{a}_m = p_z \cdot \mathbf{a}_m(Q_z) p_z$. Testing the Dirichlet principle with the function equal to $p \cdot x + w + \chi_z + \phi_z$ on each interior cell Q_z and to $p \cdot x + w$ on the omitted cells gives

$$\frac{1}{2} \mathbb{E}[p \cdot \mathbf{a}_m(\Box_K) p] \leq \sum_z \mathbb{E} \left[\frac{1}{2} p_z \cdot \mathbf{a}_m(Q_z) p_z + \int_{Q_z} S_z \cdot \mathbf{a}_m T_z + \frac{1}{2} \int_{Q_z} |T_z|^2 \mathbf{a}_m \right] + o_K(1).$$

Since ϕ_z minimizes the energy of $F_z + \nabla \phi$ and $(F_z)_{Q_z} = 0$, we have $\int_{Q_z} |T_z|^2 \mathbf{a}_m = \int_{Q_z} F_z \cdot \mathbf{a}_m T_z$. Apply (2.8), (2.44), and

$$3^{n/4} [F_z]_{\underline{B}_{2,\infty}^{1/4}(Q_z)} \leq C(A_z + A_z^{3/4} B_z^{1/4})$$

to obtain

$$\int_{Q_z} |T_z|^2 \mathbf{a}_m \leq C \Lambda_{1/4,1}(Q_z; \mathbf{a}_m) (A_z^2 + A_z^{3/2} B_z^{1/2}).$$

We verify the coefficient moment used here. The fixed choice (ξ_*, s) above and stationarity give $\|\mathcal{E}_{1/4,\infty}(Q_z; \mathbf{a}_n, \bar{\mathbf{a}}_n)\|_{L^8(\mathbb{P})} \leq 1$. The squared bound (2.38), with $q = 1$, therefore gives

$$\|\Lambda_{1/4,1}(Q_z; \mathbf{a}_n)\|_{L^4(\mathbb{P})} \leq C \bar{\mathbf{a}}_n, \quad \|\lambda_{1/4,1}^{-1}(Q_z; \mathbf{a}_n)\|_{L^4(\mathbb{P})} \leq C \bar{\mathbf{a}}_n^{-1}.$$

For $i \geq j$, write $R_{i,j}^\pm = \sup_{Q_z} (\mathbf{a}_i \mathbf{a}_j^{-1})^{\pm 1}$. Applying the pathwise comparison (2.48) on every subcube in the definitions of the coarse ellipticities gives

$$\Lambda_{1/4,1}(Q_z; \mathbf{a}_m) \leq R_{m,m-h}^+ R_{m-h,n}^+ \Lambda_{1/4,1}(Q_z; \mathbf{a}_n),$$

and the same comparison for $\lambda_{1/4,1}^{-1}$ with the minus signs. The field estimate (3.5), on a cube of scale n , bounds both ratios in $L^8(\mathbb{P})$ by $C(d)$: their block lengths are $h \leq \bar{\delta}^{-1}$ and $m-h-n \leq C|\log \bar{\delta}|$. Hölder with $1/8 + 1/8 + 1/4 = 1/2$ and (3.24) yield

$$\|\Lambda_{1/4,1}(Q_z; \mathbf{a}_m)\|_{L^2(\mathbb{P})} \leq C \bar{\mathbf{a}}_{m-h}, \quad \|\lambda_{1/4,1}^{-1}(Q_z; \mathbf{a}_m)\|_{L^2(\mathbb{P})} \leq C \bar{\mathbf{a}}_{m-h}^{-1}.$$

This comparison is simultaneous over the subcubes and requires no maximum of separate moment estimates. To combine the coefficient and cell bounds, take Hölder on the product of $\mathbb{P}$ with the uniform measure on the finite retained family. In particular,

$$\sum_z \mathbb{E}[\Lambda_z(A_z^2 + A_z^{3/2} B_z^{1/2})] \leq \left(\sum_z \mathbb{E}[\Lambda_z^2] \right)^{1/2} \left(\sum_z \mathbb{E}[(A_z^2 + A_z^{3/2} B_z^{1/2})^2] \right)^{1/2},$$

where $\Lambda_z = \Lambda_{1/4,1}(Q_z; \mathbf{a}_m)$. The last factor is bounded by $C(\sum_z \mathbb{E}[A_z^4 + B_z^4])^{1/2}$. Thus the estimates of Step 1 imply

$$\limsup_{K \rightarrow \infty} \sum_z \mathbb{E} \int_{Q_z} |T_z|^2 \mathbf{a}_m \leq C \bar{\delta}^{30} \bar{\mathbf{a}}_{m-h}.$$

The same coefficient bounds give $\sum_z \mathbb{E} \int_{Q_z} |S_z|^2 \mathbf{a}_m \leq C \bar{\mathbf{a}}_{m-h}$. Cauchy–Schwarz therefore yields

$$\limsup_{K \rightarrow \infty} \sum_z \mathbb{E} \left| \int_{Q_z} S_z \cdot \mathbf{a}_m T_z \right| \leq C \bar{\delta}^{15} \bar{\mathbf{a}}_{m-h}.$$

With the bounds on the T_z terms and on the cross terms, the estimate obtained from the Dirichlet principle above bounds the energy of the competitor by the sum of the cell energies, up to an error $C\bar{\delta}^{15}\bar{\mathbf{a}}_{m-h}$ in the limit $K \rightarrow \infty$.

Step 3. We estimate the sum of the cell energies $p_z \cdot \mathbf{a}_m(Q_z)p_z$. Since $\mathbf{a}_m = e^{\mathbf{h}}\mathbf{a}_{m-h}$,

$$p_z \cdot \mathbf{a}_m(Q_z)p_z \leq \left(\sup_{Q_z} e^{\mathbf{h}} \right) p_z \cdot \mathbf{a}_{m-h}(Q_z)p_z.$$

We have the deterministic estimate

$$\left| \left(\sup_{Q_z} e^{\mathbf{h}} \right) |p_z|^2 - \int_{Q_z} e^{\mathbf{h}} |p + \nabla w|^2 \right| \leq C e^{\|\mathbf{h}\|_{L^\infty(Q_z)}} \left(\operatorname{osc}_{Q_z} \mathbf{h} |p_z|^2 + (|p_z| + A_z) A_z \right).$$

The choice of n gives $\|\operatorname{osc}_{Q_z} \mathbf{h}\|_{L^8(\mathbb{P})} \leq C\bar{\delta}^{17}$. The exponential factor and p_z have bounded eighth moments, while Step 1 controls A_z . Combining these bounds with the independence of $\mathbf{h}$ and $\mathbf{a}_{m-h}$, we obtain

$$\limsup_{K \rightarrow \infty} \sum_z \mathbb{E}[p_z \cdot \mathbf{a}_m(Q_z)p_z] \leq \bar{\mathbf{a}}_{m-h}(\square_n) \limsup_{K \rightarrow \infty} \mathbb{E} \int_{\square_K} e^{\mathbf{h}} |p + \nabla w|^2 + C\bar{\delta}^{15}\bar{\mathbf{a}}_{m-h}.$$

It remains to evaluate the expectation in the last display. The equation for w , the zero trace of w , and $\mathbb{E}[e^{\mathbf{h}}] = 1$ give

$$\mathbb{E} \int_{\square_K} e^{\mathbf{h}} |p + \nabla w|^2 = 1 - \mathbb{E} \|\nabla w\|_{\underline{L}^2(\square_K)}^2 + \mathbb{E} \int_{\square_K} |\nabla w|^2 (e^{\mathbf{h}} - 1).$$

We now show that the last term is $O(\bar{\delta}^4 h^2)$, uniformly in K . Decompose $w = v + \bar{w}$, where

$$-\Delta v = \nabla \cdot (\mathbf{h}p), \quad -\Delta \bar{w} = \nabla \cdot ((e^{\mathbf{h}} - 1 - \mathbf{h})p), \quad v, \bar{w} \in H_0^1(\square_K).$$

The Calderón–Zygmund and exponential-moment estimates give

$$\|\nabla v\|_{L^4(\mathbb{P}; \underline{L}^4)} \leq C\bar{\delta} h^{1/2}, \quad \|\nabla \bar{w}\|_{L^4(\mathbb{P}; \underline{L}^4)} \leq C\bar{\delta}^2 h.$$

Since constants have zero divergence, v is linear in the sum $\mathbf{h} + h\tau^2 = \sum_{k=m-h+1}^m \mathbf{g}_k$ of the last h fields. Reversing the signs of these fields simultaneously therefore gives

$$\mathbb{E} \int_{\square_K} |\nabla v|^2 (\mathbf{h} + h\tau^2) = 0, \quad \mathbb{E} \int_{\square_K} |\nabla v|^2 \mathbf{h} = -h\tau^2 \mathbb{E} \|\nabla v\|_{\underline{L}^2}^2 = O(\bar{\delta}^4 h^2).$$

The preceding moment bounds control the terms containing $\bar{w}$, and $|e^s - 1 - s| \leq \frac{1}{2}s^2 e^{|s|}$ controls the nonlinear remainder. Hölder's inequality and $\bar{\delta}h \leq 1$ thus give

$$\sup_K \left| \mathbb{E} \int_{\square_K} |\nabla w|^2 (e^{\mathbf{h}} - 1) \right| \leq C\bar{\delta}^4 h^2.$$

Lemma 5.3 now yields

$$\limsup_{K \rightarrow \infty} \mathbb{E} \int_{\square_K} e^{\mathbf{h}} |p + \nabla w|^2 \leq 1 - \frac{2\tau^2 h}{d} + C\bar{\delta}^4 h^2.$$

Step 4. We conclude the proof of (5.6). Combining Steps 2 and 3 and passing to the thermodynamic limit gives

$$\frac{1}{2}\bar{\mathbf{a}}_m \leq \left(\frac{1}{2} - \frac{\tau^2 h}{d} + C\bar{\delta}^4 h^2 \right) \bar{\mathbf{a}}_{m-h}(\square_n) + C\bar{\delta}^{15}\bar{\mathbf{a}}_{m-h}.$$

For the finite-volume readout we use the first moment, rather than a bound on the expected square root of the defect. For a Euclidean unit vector e , (2.11) in expectation gives

$$2\mathbb{E}J(\square_n, \bar{\mathbf{a}}_n^{-1/2}e, \bar{\mathbf{a}}_n^{1/2}e; \mathbf{a}_n) = \left(\frac{\bar{\mathbf{a}}_n(\square_n)}{\bar{\mathbf{a}}_n} - 1 \right) + (\bar{\mathbf{a}}_n \bar{\mathbf{a}}_{n,*}^{-1}(\square_n) - 1).$$

Both summands are nonnegative by the monotone limits (3.22). Proposition 4.1 at moment 1 bounds each by $C\bar{\delta}^2$. The finite-volume ordering (3.23) and (3.24) then give

$$\bar{\mathbf{a}}_{m-h} \leq \bar{\mathbf{a}}_{m-h}(\square_n) \leq \bar{\mathbf{a}}_n(\square_n) \leq (1 + C\bar{\delta}^2)\bar{\mathbf{a}}_n \leq (1 + C\bar{\delta}^2)e^{2\tau^2(m-h-n)}\bar{\mathbf{a}}_{m-h}.$$

Since $\tau^2(m-h-n) \leq C\bar{\delta}^2|\log \bar{\delta}|$, this proves

$$\bar{\mathbf{a}}_{m-h} \leq \bar{\mathbf{a}}_{m-h}(\square_n) \leq (1 + C\bar{\delta}^2|\log \bar{\delta}|)\bar{\mathbf{a}}_{m-h}.$$

The preceding bracket is bounded in absolute value by $C(d)$. For substitution, one may use $Bx \leq Ba + |B|(x-a)$ for $x \geq a$; no sign assumption on the bracket is needed. Also $\bar{\delta}^{15} \leq \bar{\delta}^2|\log \bar{\delta}|$ in the present range. Substituting this into the previous display and using $\tau^2 \leq C\bar{\delta}^2$ and $\bar{\delta}h \leq 1$ proves (5.6). $\square$

Lemma 5.5 (Lower bound on $\bar{\mathbf{a}}_m/\bar{\mathbf{a}}_{m-h}$). *There exists $C(d) < \infty$ such that, for every $m, h \in \mathbb{N}$ with*

$$h \leq \bar{\delta}^{-1} \quad \text{and} \quad m-h \geq 16[|\log_3 \bar{\delta}|],$$

we have

$$(5.9) \quad \bar{\mathbf{a}}_m^{-1} \leq \bar{\mathbf{a}}_{m-h}^{-1} \left(1 + \frac{2\tau^2 h}{d} + C\bar{\delta}^4 h^2 + C\bar{\delta}^2 |\log \bar{\delta}| \right).$$

Proof. In dimension two, Proposition 3.5 gives $\bar{\mathbf{a}}_{m-h}/\bar{\mathbf{a}}_m = e^{\tau^2 h}$. With $x = \tau^2 h \leq (\log 2/2)\bar{\delta} \leq 1/2$, the estimate $e^x \leq 1 + x + 2x^2$ proves (5.9). Henceforth $d \geq 3$. Fix the moment orders and dimensional threshold $\bar{\delta}_*$ as in the upper-bound proof. If $\bar{\delta} > \bar{\delta}_*$, (3.24) gives $\bar{\mathbf{a}}_{m-h}/\bar{\mathbf{a}}_m \leq e^{2\tau^2 h}$. For $t = \bar{\delta}^2 h \in [0, 1]$, convexity gives $e^{(\log 2)t} \leq 1 + t$. Since $t \leq \bar{\delta}_*^{-2}\bar{\delta}^4 h^2$, increasing $C(d)$ proves the conclusion in this range. We may therefore assume $\bar{\delta} \leq \bar{\delta}_*$ without adding a hypothesis to the lemma.

The proof is dual to that of Lemma 5.4. We bound $\mathbb{E}[q \cdot \mathbf{a}_{m,*}^{-1}(\square_K)q]$, for a unit vector q and large K , from above by the dual variational principle. The competitor is built from a divergence-free field $\mathbf{z}$, defined through the solution W of a Neumann problem forced by the last h fields: on each interior cell of the scale 3^n fixed in Step 1, it is the flux of the solution of the Neumann problem for $\mathbf{a}_m$ whose normal flux on the cell boundary is that of $\mathbf{z}$. Step 1 sets up W , $\mathbf{z}$, and the cells. Step 2 bounds the dual energy of the competitor by the sum of the cell energies, up to an error $C\bar{\delta}^{15}\bar{\mathbf{a}}_{m-h}^{-1}$. Step 3 bounds this sum, up to a further error of the same size, by $\bar{\mathbf{a}}_{m-h,*}^{-1}(\square_n)$ times an expectation involving only the last h fields, and evaluates this expectation. As in the upper bound, this is the heart of the matter: Lemma 5.3 gives the main term $2\tau^2 h/d$, and reversing the signs of the last h fields shows that the remaining error is $O(\bar{\delta}^4 h^2)$. Step 4 concludes.

Step 1. We set up the Neumann problem and record the interior estimates on the scale- 3^n cells. Fix $q \in \mathbb{R}^d$ with $|q|_2 = 1$, and define $\mathbf{h}$ and n by (5.7) and (5.8). For $K > m$, let W be the mean-zero solution of

$$-\Delta W = -\nabla \cdot ((e^{\mathbf{h}} - 1)q) \quad \text{in } \square_K, \quad \partial_{\mathbf{n}} W = (e^{\mathbf{h}} - 1)q \cdot \mathbf{n} \quad \text{on } \partial \square_K,$$

and set $\mathbf{z} = e^{\mathbf{h}} q - \nabla W$. Then $\nabla \cdot \mathbf{z} = 0$ and $\mathbf{z} \cdot \mathbf{n} = q \cdot \mathbf{n}$ on $\partial \square_K$. Even reflection of W , with odd reflection of the normal forcing component and even reflection of the tangential components, preserves its weak Neumann equation and reduces it to the periodic divergence-form Poisson equation. Apply the periodic gradient estimate at exponent 8 and then Hölder's inequality for the lower exponents. Its constant is independent of K by scaling, and the same estimate applies on each parent cube P below. This estimate and Lemma 5.3 give, for $1 \leq \xi \leq 8$,

$$\mathbb{E}[\|\nabla W\|_{\underline{L}^\xi(\square_K)}^\xi]^{1/\xi} \leq C\bar{\delta}h^{1/2}, \quad \lim_{K \rightarrow \infty} \mathbb{E}[\|\nabla W\|_{\underline{L}^2(\square_K)}^2] = \frac{2\tau^2 h}{d} + O(\bar{\delta}^4 h^2).$$

We retain a finite family of interior cells on which the weak Hessian satisfies a fourth-moment bound. To make the choice explicit, set $N = 1 + \max\{2, 68\lceil \log_3 \bar{\delta} \rceil\}$. For K sufficiently large, take parents P of side 3^{n+N} , centered on the grid of spacing 3^{n+N-1} , and contained in $\square_K$. Their concentric inner thirds tile the retained region. Retain all scale- 3^n cells in these inner thirds. The parents have overlap bounded by a dimensional constant, and the omitted cells lie within distance $C3^{n+N}$ of $\partial\square_K$. Their relative volume tends to zero with $m, h, \bar{\delta}, N$ fixed.

Here is the two-radius decomposition giving the required fourth-moment bound. On each parent P , let $U_P \in H_0^1(P)$ solve $-\Delta U_P = -\nabla \cdot ((e^{\mathbf{h}} - 1)q)$, and let W_P solve the corresponding mean-zero Neumann problem with normal derivative $(e^{\mathbf{h}} - 1)q \cdot \mathbf{n}$. In the interior of P , we decompose $W = U_P + (W_P - U_P) + (W - W_P)$. The last two terms are harmonic, as follows by using compactly supported tests in the Neumann equations. The Dirichlet weak-Hessian estimate used in the upper proof gives a cell-averaged fourth-moment bound $C\bar{\delta}^{68}$ for the first term. For each harmonic term, the interior Hessian estimate on the inner third gives its scale- 3^n Hessian norm bounded by $C3^{-N}$ times its normalized gradient norm on P . The local Dirichlet and Neumann gradient estimates bound their fourth moments uniformly, since $(\mathbb{E}_{\mathfrak{f}_P} |e^{\mathbf{h}} - 1|^4)^{1/4} \leq C\bar{\delta}h^{1/2} \leq C$. For the restriction of ∇W to the parents, Jensen and their bounded overlap reduce the averaged fourth moments to the global bound already proved. Consequently the averages over the finite family of the fourth moments of the two harmonic contributions are at most $C3^{-4N}$. In particular they are at most $C3^{-3(N-1)} \leq C\bar{\delta}^{68}$. The inequality $(a + b + c)^4 \leq 64(a^4 + b^4 + c^4)$ now gives the same fourth-moment bound for the weak Hessian of W on the retained cells. This argument uses the interior scalar equation and does not replace the global Neumann boundary condition by a scalar Dirichlet condition.

For every retained cell put

$$q_z = (\mathbf{z})_{Q_z}, \quad G_z = \mathbf{z} - q_z, \quad A_z = \|G_z\|_{L^2(Q_z)}, \quad B_z = 3^n \|\nabla G_z\|_{L^2(Q_z)}.$$

The additional derivative of $e^{\mathbf{h}}q$ in ∇G_z satisfies the fourth-moment bound $C\bar{\delta}^{68}$ by the field estimate in the upper proof. Mean-zero Poincaré gives $A_z \leq CB_z$. Thus $\limsup_{K \rightarrow \infty} \sum_z \mathbb{E}[A_z^4 + B_z^4] \leq C\bar{\delta}^{68}$, and, in particular, both fourth-moment roots are at most $C\bar{\delta}^{15}$. All averages in this proof run over the retained cells. Measurability follows from continuous dependence of the weak solutions and their weak-derivative carriers, as in the upper proof. The cells are fixed by their geometry and the parameters, independently of the realized fields.

Step 2. We solve Neumann problems on the interior cells, patch the resulting fluxes into an admissible flux, and bound the dual energy by a sum over cells. On every interior cell, let ψ_z, η_z be the mean-zero solutions of

$$-\nabla \cdot (\mathbf{a}_m \nabla \psi_z) = 0, \quad \mathbf{a}_m \nabla \psi_z \cdot \mathbf{n} = q_z \cdot \mathbf{n},$$

and

$$-\nabla \cdot (\mathbf{a}_m \nabla \eta_z) = 0, \quad \mathbf{a}_m \nabla \eta_z \cdot \mathbf{n} = G_z \cdot \mathbf{n},$$

respectively. These problems are compatible because G_z is divergence free. The normal trace of the flux $\mathbf{a}_m \nabla(\psi_z + \eta_z)$ agrees with that of $\mathbf{z}$ on every cell boundary. Therefore the field equal to $\mathbf{a}_m \nabla(\psi_z + \eta_z)$ on each interior cell and to $\mathbf{z}$ on the omitted cells belongs to $q + L_{\text{sol},0}^2(\square_K)$, and the contribution of the omitted cells to its dual energy vanishes in expectation as $K \rightarrow \infty$. Indeed, if O_K is their union, Hölder with exponents $(4, 4, 2)$ on probability times normalized volume gives

$$\mathbb{E} \int_{\square_K} \mathbb{1}_{O_K} |\mathbf{z}|^2 \mathbf{a}_m^{-1} \leq C \left(\frac{|O_K|}{|\square_K|} \right)^{1/4} \longrightarrow 0.$$

Here C may depend on m and h but not on K ; it is finite by the eighth-moment gradient estimate and the second moment of $\mathbf{a}_m^{-1}$. Passing from the volume-weighted sum to the average over retained

cells introduces another deterministic remainder tending to zero in expectation. Taking expectations in the dual variational inequality therefore gives

$$\frac{1}{2}\mathbb{E}[q \cdot \mathbf{a}_{m,*}^{-1}(\square_K)q] \leq \sum_z \mathbb{E} \left[\frac{1}{2}q_z \cdot \mathbf{a}_{m,*}^{-1}(Q_z)q_z + \int_{Q_z} \mathbf{a}_m \nabla \psi_z \cdot \nabla \eta_z + \frac{1}{2} \int_{Q_z} \mathbf{a}_m |\nabla \eta_z|^2 \right] + o_K(1).$$

The argument of Step 2 of the proof of Lemma 5.4, applied to fluxes instead of gradients and with (2.43) in place of (2.44), gives

$$\int_{Q_z} \mathbf{a}_m |\nabla \eta_z|^2 \leq C \lambda_{1/4,1}^{-1}(Q_z; \mathbf{a}_m) (A_z^2 + A_z^{3/2} B_z^{1/2}).$$

The fixed-moment squared-error calculation and the simultaneous pathwise subcube comparison in Step 2 of the upper proof, now with inverse ratios, give the required coefficient bound. In particular, using (3.24) to compare $\bar{\mathbf{a}}_n^{-1}$ with $\bar{\mathbf{a}}_{m-h}^{-1}$,

$$\mathbb{E}[\lambda_{1/4,1}^{-2}(\square_n; \mathbf{a}_m)]^{1/2} \leq C \bar{\mathbf{a}}_{m-h}^{-1}.$$

The preceding three displays and Cauchy–Schwarz thus yield

$$\limsup_{K \rightarrow \infty} \sum_z \mathbb{E} \int_{Q_z} \mathbf{a}_m |\nabla \eta_z|^2 \leq C \bar{\delta}^{30} \bar{\mathbf{a}}_{m-h}^{-1},$$

$$\limsup_{K \rightarrow \infty} \sum_z \mathbb{E} \left| \int_{Q_z} \mathbf{a}_m \nabla \psi_z \cdot \nabla \eta_z \right| \leq C \bar{\delta}^{15} \bar{\mathbf{a}}_{m-h}^{-1}.$$

With these two bounds, the dual variational inequality above bounds the dual energy of the competitor by the sum of the cell energies, up to an error $C \bar{\delta}^{15} \bar{\mathbf{a}}_{m-h}^{-1}$ in the limit $K \rightarrow \infty$.

Step 3. We estimate the sum of the cell energies $q_z \cdot \mathbf{a}_{m,*}^{-1}(Q_z)q_z$. The pair (q_z, G_z) is a function of the last h fields and is independent of $\mathbf{a}_{m-h}$. Since $\mathbf{a}_m^{-1} = e^{-\mathbf{h}} \mathbf{a}_{m-h}^{-1}$, the deterministic estimate in Step 3 of the proof of Lemma 5.4, with $\mathbf{h}$ replaced by $-\mathbf{h}$, $p + \nabla w$ by $\mathbf{z}$, and p_z by q_z , gives

$$\limsup_{K \rightarrow \infty} \sum_z \mathbb{E}[q_z \cdot \mathbf{a}_{m,*}^{-1}(Q_z)q_z] \leq \bar{\mathbf{a}}_{m-h,*}^{-1}(\square_n) \limsup_{K \rightarrow \infty} \mathbb{E} \int_{\square_K} |\mathbf{z}|^2 e^{-\mathbf{h}} + C \bar{\delta}^{15} \bar{\mathbf{a}}_{m-h}^{-1}.$$

We next show that

$$(5.10) \quad \limsup_{K \rightarrow \infty} \left| \mathbb{E} \int_{\square_K} |\nabla W|^2 (e^{-\mathbf{h}} - 1) \right| \leq C \bar{\delta}^4 h^2.$$

Indeed, decompose $W = V + \bar{W}$, where V solves the Neumann problem defining W with $e^{\mathbf{h}} - 1$ replaced by its linearization $\mathbf{h}$. If V_0 is the solution of the same problem with $\mathbf{h}$ replaced by the sum $\mathbf{h} + h\tau^2 = \sum_{k=m-h+1}^m \mathbf{g}_k$ of the last h fields, then

$$\nabla V = \nabla V_0 - h\tau^2 q, \quad \mathbb{E} \|\nabla V_0\|_{L^2(\square_K)}^2 \leq C \bar{\delta}^2 h.$$

Reversing the signs of the last h fields simultaneously gives

$$\mathbb{E} \int_{\square_K} |\nabla V_0|^2 (\mathbf{h} + h\tau^2) = 0, \quad \mathbb{E} \int_{\square_K} q \cdot \nabla V_0 = 0,$$

while Cauchy–Schwarz gives $|\mathbb{E} \int_{\square_K} (q \cdot \nabla V_0)(\mathbf{h} + h\tau^2)| \leq C \bar{\delta}^2 h$. Expanding $|\nabla V|^2 \mathbf{h}$ with the identities for ∇V and $\mathbf{h}$ therefore yields $\mathbb{E} \int_{\square_K} |\nabla V|^2 \mathbf{h} = O(\bar{\delta}^4 h^2)$. The Neumann Calderón–Zygmund estimate gives $\mathbb{E}[\|\nabla \bar{W}\|_{L^4}^4]^{1/4} \leq C \bar{\delta}^2 h$. The Hölder estimates used in Step 3 of the upper-bound proof, together with $|e^{-s} - 1 + s| \leq \frac{1}{2} s^2 e^{|s|}$, control every term containing $\bar{W}$ and the nonlinear remainder by $C \bar{\delta}^4 h^2$. This proves (5.10).

Since $\mathbf{z} = e^{\mathbf{h}}q - \nabla W$,

$$\int_{\square_K} |\mathbf{z}|^2 e^{-\mathbf{h}} = \int_{\square_K} e^{\mathbf{h}} - 2 \int_{\square_K} q \cdot \nabla W + \int_{\square_K} e^{-\mathbf{h}} |\nabla W|^2.$$

The expectation of the first term is one. Taking expectations in the mean-zero Neumann problem gives $\mathbb{E}W = 0$, so the expectation of the second term vanishes. Lemma 5.3 and (5.10) consequently give

$$\limsup_{K \rightarrow \infty} \mathbb{E} \int_{\square_K} |\mathbf{z}|^2 e^{-\mathbf{h}} \leq 1 + \frac{2\tau^2 h}{d} + C\bar{\delta}^4 h^2.$$

Step 4. We conclude the proof of (5.9). Combining Steps 2 and 3, letting $K \rightarrow \infty$, and recalling that $\bar{\mathbf{a}}_{m,*}(\square_K) \rightarrow \bar{\mathbf{a}}_m$, we obtain

$$\frac{1}{2} \bar{\mathbf{a}}_m^{-1} \leq \left(\frac{1}{2} + \frac{\tau^2 h}{d} + C\bar{\delta}^4 h^2 \right) \bar{\mathbf{a}}_{m-h,*}^{-1}(\square_n) + C\bar{\delta}^{15} \bar{\mathbf{a}}_{m-h}^{-1}.$$

The first-moment identity in Step 4 of the upper proof gives $\bar{\mathbf{a}}_{n,*}^{-1}(\square_n) \leq (1 + C\bar{\delta}^2) \bar{\mathbf{a}}_n^{-1}$. The finite-volume dual ordering (3.23) and the monotone limit (3.22) yield

$$\bar{\mathbf{a}}_{m-h}^{-1} \leq \bar{\mathbf{a}}_{m-h,*}^{-1}(\square_n) \leq e^{2\tau^2(m-h-n)} \bar{\mathbf{a}}_{n,*}^{-1}(\square_n) \leq (1 + C\bar{\delta}^2) e^{2\tau^2(m-h-n)} \bar{\mathbf{a}}_{m-h}^{-1}.$$

Since $\tau^2(m-h-n) \leq C\bar{\delta}^2 |\log \bar{\delta}|$, we obtain

$$\bar{\mathbf{a}}_{m-h}^{-1} \leq \bar{\mathbf{a}}_{m-h,*}^{-1}(\square_n) \leq (1 + C\bar{\delta}^2 |\log \bar{\delta}|) \bar{\mathbf{a}}_{m-h}^{-1}.$$

The preceding bracket is positive and bounded by a dimensional constant. Absorbing $\bar{\delta}^{15}$ into $\bar{\delta}^2 |\log \bar{\delta}|$ and using (1.6) and $\bar{\delta}h \leq 1$ proves (5.9). $\square$

Proof of Proposition 5.2. It suffices to consider $\bar{\delta} \leq c(d)$. Set $J := 16 \lceil |\log_3 \bar{\delta}| \rceil$. The ordering (3.24) gives

$$(5.11) \quad \left| \log \bar{\mathbf{a}}_J + \frac{2\tau^2(J+1)}{d} \right| \leq C\bar{\delta}^2(J+1) \leq C\bar{\delta}^2 |\log \bar{\delta}|.$$

By Lemmas 5.4 and 5.5 and the inequality $\log t \leq t - 1$ for $t > 0$, applied to $\bar{\mathbf{a}}_m/\bar{\mathbf{a}}_{m-h}$ and to $\bar{\mathbf{a}}_{m-h}/\bar{\mathbf{a}}_m$, whenever $1 \leq h \leq \bar{\delta}^{-1}$ and $m-h \geq J$,

$$(5.12) \quad \left| \log \bar{\mathbf{a}}_m - \log \bar{\mathbf{a}}_{m-h} + \frac{2\tau^2 h}{d} \right| \leq C\bar{\delta}^4 h^2 + C\bar{\delta}^2 |\log \bar{\delta}|.$$

We may assume $m > J$. Starting at m , use (5.12) repeatedly with $h = \lfloor \bar{\delta}^{-1} \rfloor$, each time lowering the scale by h , for as long as the new scale stays at least J , and then once more with the remaining $h < \bar{\delta}^{-1}$, if any, to arrive at J . This takes at most $1 + (m-J)/\lfloor \bar{\delta}^{-1} \rfloor \leq C(1 + m\bar{\delta})$ steps, and each step has $h \leq \bar{\delta}^{-1}$ and therefore contributes an error at most $C\bar{\delta}^2 + C\bar{\delta}^2 |\log \bar{\delta}|$, because $\bar{\delta}^4 h^2 \leq \bar{\delta}^2$. Together with (5.11), this gives

$$\left| \log \bar{\mathbf{a}}_m + \frac{2\tau^2(m+1)}{d} \right| \leq C\bar{\delta}^2 |\log \bar{\delta}| (1 + m\bar{\delta}),$$

which is (5.2). $\square$

5.1. Further bounds on the effective diffusivities. The error term in Proposition 5.2 dominates $2\tau^2 m/d$ unless τ^2 is bounded below by a multiple of $\bar{\delta}^3 |\log \bar{\delta}|$, and at $m = 0$ the proposition gives no upper bound. The following decay estimate assumes no such lower bound.

Proposition 5.6 (Power-law decay of the effective diffusivities). *There exists $\eta_{\text{sub}}(d, \text{law}(\mathbf{g}_0)) > 0$ satisfying*

$$(5.13) \quad \eta_{\text{sub}} = \frac{\tau^2}{\log 3} \quad \text{if } d = 2, \quad 0 < \eta_{\text{sub}} \leq \frac{\tau^2}{d \log 3} \quad \text{if } d \geq 3,$$

and, for every $m \in \mathbb{N}_0$,

$$(5.14) \quad \bar{\mathbf{a}}_m \leq 3^{-\eta_{\text{sub}}(m+1)}.$$

Proof. In dimension $d = 2$, Proposition 3.5 proves the result with $\eta_{\text{sub}} = \tau^2 / \log 3$. We may therefore suppose that $d \geq 3$. We then compare $\bar{\mathbf{a}}_m$ with the coarse-grained matrices of the product of the factors $\exp(\mathbf{g}_k - \tau^2)$ over the multiples k of a fixed integer R . Step 1 shows that, with the weight $\exp(\mathbf{g}_0 - \tau^2)$, the expected minimal energy of affine boundary data on a simplex is strictly smaller than that of the affine function. Step 2 replaces this minimal energy by a discretized one, which is still smaller than one and can be iterated. Step 3 iterates it over the scales $0, R, 2R, \dots$, using the independence of the fields; this is the heart of the matter. Step 4 reinserts the omitted factors, which have mean one and are independent of the others, and concludes.

Put

$$B_k := \exp(\mathbf{g}_k - \tau^2), \quad A_N^{(R)} := \prod_{j=0}^N B_{jR} \quad (N \in \mathbb{N}_0),$$

where the integer $R \geq 1$ will be fixed below. Thus $\mathbb{E}[B_k(x)] = 1$ for every k and x . Define, with the maximum over all permutations π of $\{1, \dots, d\}$,

$$(5.15) \quad q_* := \max_{\pi} \sup_{|p|=1} \mathbb{E} \left[\min_{u \in p \cdot x + H_0^1(\Delta_0^\pi)} \int_{\Delta_0^\pi} B_0 |\nabla u|^2 \right].$$

Step 1. We prove that $q_* < 1$. The affine competitor gives $q_* \leq 1$. If equality held, there would be a permutation π and $|p| = 1$ for which the expected minimum in (5.15) equals $\mathbb{E}[\int_{\Delta_0^\pi} B_0 |p|^2] = 1$. The energy of $p \cdot x$ minus the minimum is nonnegative and would then have zero expectation, so it would vanish almost surely. Hence $p \cdot x$ would be a minimizer, and the Euler–Lagrange equation would give $p \cdot \nabla B_0 = 0$ in Δ_0^π almost surely. By stationarity and the almost-sure C^1 regularity of B_0 , $p \cdot \nabla B_0 = 0$ on $\mathbb{R}^d$. Consequently, $B_0(0) = B_0(tp)$ for every t . Choosing t with $|tp|_2 > \sqrt{d}$, we see that the random variables $B_0(0)$ and $B_0(tp)$, which are independent by Assumption (g1), coincide almost surely; hence each is deterministic. It follows that $\mathbf{g}_0(0)$ is deterministic and, since it has mean zero, that $\tau^2 = 0$, contrary to Assumption (g4). Thus

$$(5.16) \quad q_* < 1.$$

Step 2. We show that q_R converges to q_* , and fix R with $q_R < 1$. For a permutation π and $p \in \mathbb{R}^d$, let T run over all Kuhn simplices contained in Δ_0^π whose supporting cubes have side 3^{-R} , and let v run over the continuous functions on $\bar{\Delta}_0^\pi$ which are affine on every such T and equal to $p \cdot x$ on $\partial \Delta_0^\pi$. Set

$$(5.17) \quad Q_{R,\pi}(p) := \min_v \frac{1}{|\Delta_0^\pi|} \sum_T |T| (\sup_T B_0) |\nabla v|_T^2, \quad q_R := \max_{\pi} \sup_{|p|=1} \mathbb{E}[Q_{R,\pi}(p)].$$

Since B_0 is almost surely continuous and conforming piecewise-affine functions are dense in $H_0^1(\Delta_0^\pi)$, we have, almost surely as $R \rightarrow \infty$,

$$Q_{R,\pi}(p) \longrightarrow \min_{u \in p \cdot x + H_0^1(\Delta_0^\pi)} \int_{\Delta_0^\pi} B_0 |\nabla u|^2.$$

The affine competitor bounds $Q_{R,\pi}(p)$ by $|p|^2 \sup_{\Delta_0^\pi} B_0$, whose expectation is finite by Assumption (g2). Dominated convergence and (5.16) show that $q_R \rightarrow q_* < 1$. Fix R such that

$$(5.18) \quad q_R < 1.$$

Step 3. We iterate (5.18) over scales separated by R . We show by induction on N that, for every $N \in \mathbb{N}_0$, every permutation π and every $p \in \mathbb{R}^d$,

$$(5.19) \quad \mathbb{E}[p \cdot A_N^{(R)}(\Delta_{NR}^\pi)p] \leq q_R^{N+1}|p|^2.$$

For $N = 0$, this follows from (5.17), because restricting the minimization to piecewise-affine functions and replacing B_0 by its supremum on each simplex can only increase the energy. Suppose that (5.19) holds with $N - 1$ in place of N . Fix a parent simplex $U := \Delta_{NR}^\pi$ and a vector p . The Kuhn simplices T contained in U and supported on cubes of side $3^{(N-1)R}$ form, after rescaling, the mesh used in (5.17). On this mesh, we consider the continuous function with boundary values $p \cdot x$ that is affine on each simplex and minimizes the energy in (5.17) with the weights $\sup_{\bar{T}} B_{NR}$ in place of $\sup_{\bar{T}} B_0$, and let p_T be its slope on T . On every T , we then take the minimizer of $\int_T A_{N-1}^{(R)} |\nabla u|^2$ among the functions whose boundary values on ∂T agree with this piecewise-affine function. These minimizers glue to a competitor in $p \cdot x + H_0^1(U)$, and the variational formula gives

$$p \cdot A_N^{(R)}(U)p \leq \sum_T \frac{|T|}{|U|} \left(\sup_{\bar{T}} B_{NR} \right) p_T \cdot A_{N-1}^{(R)}(T) p_T.$$

Condition on B_{NR} . The slopes p_T and the displayed weights are then fixed, whereas $A_{N-1}^{(R)}$ is independent of B_{NR} . Stationarity and the induction hypothesis bound the conditional expectation of the right side by

$$q_R^N \sum_T \frac{|T|}{|U|} \left(\sup_{\bar{T}} B_{NR} \right) |p_T|^2.$$

Taking the remaining expectation and using the scaling relation $\mathbf{g}_{NR} \stackrel{\text{law}}{=} \mathbf{g}_0(3^{-NR} \cdot)$ and (5.17), we obtain

$$\mathbb{E}[p \cdot A_N^{(R)}(U)p] \leq q_R^{N+1}|p|^2,$$

which proves (5.19).

Step 4. We pass from the products $A_N^{(R)}$, which contain only every R -th field, to $\bar{\mathbf{a}}_m$ and conclude. Fix $m \in \mathbb{N}_0$ and put $N := \lfloor m/R \rfloor$. The field $\mathbf{a}_m/A_N^{(R)}$ is the product of the factors B_k with $0 \leq k \leq m$ and $R \nmid k$. It is independent of $A_N^{(R)}$ and satisfies $\mathbb{E}[(\mathbf{a}_m/A_N^{(R)})(x)] = 1$. If u minimizes $\int_V A_N^{(R)} |\nabla u|^2$ among the functions on a bounded convex domain V with boundary values $p \cdot x$, then u is a function of $A_N^{(R)}$, so conditioning on $A_N^{(R)}$ and using its independence from $\mathbf{a}_m/A_N^{(R)}$ give

$$(5.20) \quad \mathbb{E}[p \cdot \mathbf{a}_m(V)p] \leq \mathbb{E}\left[\int_V \mathbf{a}_m |\nabla u|^2\right] = \mathbb{E}[p \cdot A_N^{(R)}(V)p].$$

For $K \geq NR$, partition $\square_K$ into the Kuhn simplices of the cubes of side 3^{NR} . Using on each simplex the minimizer for $A_N^{(R)}$ with boundary values $p \cdot x$ as a competitor, we deduce from (5.19) that

$$\mathbb{E}[p \cdot A_N^{(R)}(\square_K)p] \leq q_R^{N+1}|p|^2.$$

Letting $K \rightarrow \infty$ in (5.20) and using (3.22), we obtain $\bar{\mathbf{a}}_m \leq q_R^{\lfloor m/R \rfloor + 1}$. Since $R(\lfloor m/R \rfloor + 1) \geq m + 1$, the choice $\eta_{\text{sub}} := -\log q_R/(R \log 3) > 0$ proves (5.14). Replacing this value by its minimum with $\tau^2/(d \log 3)$ proves (5.13) as well. $\square$

Proposition 5.2 controls $\bar{\mathbf{a}}_m$ only up to its error term. The following lower bound has no error term and holds for every $\bar{\delta}$.

Proposition 5.7 (Lower bound on the effective diffusivities). *For every $m \in \mathbb{N}_0$,*

$$(5.21) \quad \bar{\mathbf{a}}_m \geq e^{-(m+1)\tau^2}.$$

Proof. Fix $m, n \in \mathbb{N}_0$ and put $c_m := e^{(m+1)\tau^2} \mathbf{a}_m = \exp(\sum_{k=0}^m \mathbf{g}_k)$. Assumption **(g3)** gives $c_m^{-1} \stackrel{\text{law}}{=} c_m$. Since coarse-grained matrices are homogeneous of degree one in the coefficient field, it follows that

$$(5.22) \quad \mathbb{E}[c_m(\square_n)] = \mathbb{E}[(c_m^{-1})(\square_n)] = e^{(m+1)\tau^2} \bar{\mathbf{a}}_m(\square_n).$$

Here $(c_m^{-1})(\square_n)$ denotes the coarse-grained matrix associated with the reciprocal coefficient field $x \mapsto c_m(x)^{-1}$, and not the inverse of the matrix $c_m(\square_n)$.

For $i \in \{1, \dots, d\}$, let u_i and v_i be the minimizers in (2.12), with $p = e_i$, for the respective coefficients c_m and c_m^{-1} on $\square_n$. Let E and F be the Jacobian matrices of $x \mapsto (u_i(x))_i$ and $x \mapsto (v_i(x))_i$. Since u_i and v_i have boundary values $e_i \cdot x$, both maps have trace x on $\partial\square_n$. For two d -by- d matrices, set

$$Q(E, F) := \text{tr}(E) \text{tr}(F) - \text{tr}(EF), \quad |M|_{\mathbb{F}} := \text{tr}(M^T M)^{1/2},$$

where $|\cdot|_{\mathbb{F}}$ is the Frobenius norm. The bilinear form Q is the polarization of the sum of the second-order principal minors. These minors are null Lagrangians, and $(u_i)_i$ and $(v_i)_i$ have the same trace, so

$$(5.23) \quad \oint_{\square_n} Q(E, F) = Q(\text{Id}, \text{Id}) = d(d-1).$$

Equivalently, this follows by integration by parts from

$$Q(E, F) = \sum_{i,j=1}^d (\partial_i(u_i \partial_j v_j) - \partial_j(u_i \partial_i v_j)).$$

For arbitrary matrices E, F , Cauchy–Schwarz and $\text{tr}(F)^2 \leq d|F|_{\mathbb{F}}^2$ give

$$\begin{aligned} Q(E, F) &= \text{tr}(E^T(\text{tr}(F)\text{Id} - F^T)) \leq (d-1)|E|_{\mathbb{F}}|F|_{\mathbb{F}}, \\ |\text{tr}(F)\text{Id} - F^T|_{\mathbb{F}}^2 &= |F|_{\mathbb{F}}^2 + (d-2)\text{tr}(F)^2 \leq (d-1)^2|F|_{\mathbb{F}}^2. \end{aligned}$$

Consequently, pointwise on $\square_n$,

$$c_m|E|_{\mathbb{F}}^2 + c_m^{-1}|F|_{\mathbb{F}}^2 \geq 2|E|_{\mathbb{F}}|F|_{\mathbb{F}} \geq \frac{2}{d-1}Q(E, F).$$

Integrating, taking expectations, and using (5.22), (2.12), and (5.23), we obtain

$$2e^{(m+1)\tau^2} \text{tr} \bar{\mathbf{a}}_m(\square_n) = \mathbb{E} \left[\oint_{\square_n} (c_m|E|_{\mathbb{F}}^2 + c_m^{-1}|F|_{\mathbb{F}}^2) \right] \geq 2d.$$

The matrix $\bar{\mathbf{a}}_m(\square_n)$ is scalar by Assumption **(g3)**. Dividing by $2d$ and letting $n \rightarrow \infty$ proves (5.21). $\square$

6. LARGE-SCALE REGULARITY AND DIRICHLET HOMOGENIZATION

In this section we prove the large-scale Hölder estimate (Theorem C) and the homogenization error bound for the Dirichlet problem (Theorem B). The regularity estimates are used in Sections 8 and 9.

Heuristically, a solution of an equation with a rapidly oscillating coefficient cannot be regular on small scales, but it can be regular on large ones: if on every scale it is close to a solution of the homogenized equation, then it inherits the regularity of that solution down to the scale at which the approximation fails. The quantity we iterate is the excess, the L^2 distance from a solution to the affine functions on a cube. On a scale where the coarse-graining error is small, the excess contracts

by a fixed factor; if such scales are frequent enough, the excess therefore decays geometrically and a Hölder estimate follows.

The scales on which the error fails to be small occur in a proportion that we can make as small as we wish by lowering the Hölder exponent; here randomness and independence between the scales enter through a concentration inequality. The cutoff coefficient $\mathbf{a}_L$ contains no fields of scale larger than the cutoff scale 3^L , so averaging over cubes larger than the cutoff scale improves its comparison with a constant coefficient. For this reason no relation between the size of the domain and the cutoff scale is needed, and the estimates are uniform in L and pass to the limiting coefficient.

The section is organized as follows.

- In Section 6.1 we define the good event, on which the coarse-graining error at a given scale is deterministically bounded (Proposition 6.3), and we show that, outside an event of exponentially small probability, the proportion of scales on which the good event fails is small (Proposition 6.4).
- In Section 6.2 we show that, on a good event, a solution is close to a harmonic function and its excess contracts by a fixed factor (Lemmas 6.9 and 6.10). The iteration of Lemma 6.11 combines this contraction on the good scales with a loss of at most a bounded factor on each of the other scales.
- In Section 6.3 we define the minimal scale and prove Proposition 6.1, the large-scale Hölder estimate in $\square_m$ for $L \geq m$.
- In Section 6.4 we remove the restriction $L \geq m$ (Proposition 6.17) and show that the estimates are stable when the coefficient is multiplied by a positive continuous function that is uniformly close to a constant (Proposition 6.18).
- Theorems 6.19 and 6.20 are the two parts of Theorem B: they bound the homogenization error of the Dirichlet problem uniformly in the cutoff and, at a fixed cutoff, with an algebraic rate. They follow from the general coarse-graining estimate of Proposition 2.11, applied in the whole domain, and do not use the regularity estimates.
- In the last subsection we extend the regularity estimates to the limiting coefficient $\mathbf{a}$, which proves Theorem C in the form of Theorem 6.22.

In (6.1) and (6.2) below, ℓ denotes an affine function on $\mathbb{R}^d$, and the minimum and the minimizer are taken over all affine functions. For $m, n \in \mathbb{Z}$ with $n \leq m$ and $x \in \square_m$, we define the **excess** of a function $u \in L^2((x + \square_n) \cap \square_m)$ by

$$(6.1) \quad E(u, (x + \square_n) \cap \square_m) := 3^{-n} \min_{\ell} \|u - \ell\|_{\underline{L}^2((x + \square_n) \cap \square_m)}$$

and its best affine approximation by

$$(6.2) \quad \ell(u, (x + \square_n) \cap \square_m) := \operatorname{argmin}_{\ell} \|u - \ell\|_{\underline{L}^2((x + \square_n) \cap \square_m)}.$$

Recall that $\mathbf{b}_{L,m}$ is defined in Definition 4.3.

Proposition 6.1 (Large-scale Hölder estimate for $L \geq m$). *There exists a constant $C(d) < \infty$ such that, if $\bar{\delta} \leq C^{-1}$, then, for every $\gamma \in [1/2, 1 - C\bar{\delta}|\log \bar{\delta}|^{1/2}]$ and $m \in \mathbb{N}_0$, there exists an $\mathbb{N}$ -valued random variable $\mathcal{L}_L(\gamma, m)$ (the same for every $L \geq m$) satisfying*

$$(6.3) \quad \mathbb{P}[\mathcal{L}_L(\gamma, m) > k] \leq C \exp\left(-\frac{(1 - \gamma)^2(k - C)_+}{C\bar{\delta}^2|\log \bar{\delta}|}\right) \quad (k \in \mathbb{N})$$

such that, for every $L \in \mathbb{N}_0$ with $L \geq m$ and every $\mathbf{g} \in C^{0,1/2}(\square_m; \mathbb{R}^d)$, $h \in C^{1,1/2}(\square_m)$ and $u \in H^1(\square_m)$ solving

$$(6.4) \quad \begin{cases} -\nabla \cdot \mathbf{a}_L \nabla u = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ u = h & \text{on } \partial \square_m, \end{cases}$$

we have the following estimates:

- Large-scale Hölder estimate: for every $n \in \mathbb{N}_0$ with $n \leq m - \mathcal{L}_L(\gamma, m)$ and $x \in \square_m$,

$$(6.5) \quad \begin{aligned} & \sup_{\substack{j \in \mathbb{N}_0 \\ j \leq n}} \max_{y \in 3^j \mathbb{Z}^d \cap ((x + \square_n) \cap \square_m)} 3^{\gamma(n-j)} \|u - (u)_{(y + \square_j) \cap \square_m}\|_{\underline{L}^2((y + \square_j) \cap \square_m)} \\ & \leq C 3^{-\gamma(m-n)} \left(\|u - (u)_{\square_m}\|_{\underline{L}^2(\square_m)} + (\mathbf{b}_{L,m})_{\square_m}^{-1} 3^{\frac{3}{2}m} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \right) \\ & \quad + C 3^{-\gamma(m-n)} 3^{\frac{3}{2}m} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)} \mathbb{1}_{\{x \notin \square_{m-1}\}}; \end{aligned}$$

- Energy estimate: for every $n \in \mathbb{N}_0$ with $n \leq m - \mathcal{L}_L(\gamma, m)$ and $x \in \square_m$,

$$(6.6) \quad \begin{aligned} \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2((x + \square_n) \cap \square_m)} & \leq C 3^{(1-\gamma)(m-n)} \left(\|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} + (\mathbf{b}_{L,m})_{\square_m}^{-1/2} 3^{\frac{1}{2}m} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \right) \\ & \quad + C 3^{(1-\gamma)(m-n)} (\mathbf{b}_{L,m})_{\square_m}^{1/2} 3^{\frac{1}{2}m} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)} \mathbb{1}_{\{x \notin \square_{m-1}\}}; \end{aligned}$$

- Excess decay estimate: if $n \in \mathbb{N}_0$ is such that $\mathcal{L}_L(\gamma, m) \leq m - n \leq (1 - \gamma)^{-1}$, then for every $l \in \mathbb{N}_0$ satisfying $n \leq l \leq m - 5$ and every $x \in \square_m$,

$$(6.7) \quad \begin{aligned} & E(u, (x + \square_n) \cap \square_m) \\ & \leq C 3^{-\frac{1}{4}(l-n)} E(u, (x + \square_l) \cap \square_m) + C(1 - \gamma)^{1/2} 3^{-l} \|u - (u)_{(x + \square_l) \cap \square_m}\|_{\underline{L}^2((x + \square_l) \cap \square_m)} \\ & \quad + C(\mathbf{b}_{L,m})_{\square_m}^{-1} 3^{\frac{1}{2}l} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \\ & \quad + C \left((1 - \gamma)(m - n) \|\nabla h\|_{L^\infty(\square_m)} + 3^{\frac{1}{2}l} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)} \right) \mathbb{1}_{\{x \notin \square_{m-1}\}}. \end{aligned}$$

At scale 3^m , Proposition 2.11 compares an $\mathbf{a}_L$ -harmonic function with a harmonic function when the coarse-graining error $\mathcal{E}_{1/4,\infty}(\square_m; \mathbf{a}_L, (\mathbf{b}_{L,m})_{\square_m})$ is small.

6.1. Harmonic approximation and independence between scales. We now introduce the good event $\mathcal{G}_m(\varepsilon, s)$ at scale m , on which the coarse-graining error at that scale is deterministically bounded, and estimate the number of scales at which it fails. Proposition 6.3 bounds the coarse-graining error on a scale where the good event holds, and Proposition 6.4 shows that, among consecutive scales, the proportion of those where it fails is small. These two estimates are the input to the iteration of Lemma 6.11 in Section 6.3, and the second uses the independence of the fields across scales.

Definition 6.2 (Good event). For $s, \varepsilon \in (0, 1]$ and $m \in \mathbb{N}_0$, let the **good event** $\mathcal{G}_m(\varepsilon, s)$ be the event on which the following inequalities hold. The gradient magnitude in (6.8) is $|\nabla \mathbf{g}_i|_2$, and the unit vectors in (6.10) are Euclidean:

$$(6.8) \quad \sup_{j \geq 0} 3^{-\frac{1}{8}sj} \sum_{i=(m-j) \vee 0}^{m+j} \left(|\mathbf{g}_i| + 3^i |\nabla \mathbf{g}_i| \right)_{L^\infty(\square_{m+1+j})} \leq \varepsilon,$$

$$(6.9) \quad \sup_{j \geq 0} 3^{-\frac{1}{8}sj} \left\| \prod_{i=(m-j) \vee 0}^{m+j} \exp(|\mathbf{g}_i|) + \prod_{i=m+j}^{\infty} \exp(4|\mathbf{g}_i - \mathbf{g}_i(0)|) \right\|_{L^\infty(\square_{m+1+j})} \leq 6,$$

$$(6.10) \quad \sup_{0 \leq j \leq m} \sup_{0 \leq n \leq j-2} 3^{-\frac{1}{8}sj(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) \leq \varepsilon^2.$$

For $y \in \mathbb{R}^d$, we let $\mathcal{G}_{m,y}(\varepsilon, s)$ denote the event obtained by translating the displayed quantities by y . The supremum in (6.10) is defined to be zero when there is no pair (j, n) satisfying the displayed constraints.

We call a scale m **good** if the event $\mathcal{G}_m(\varepsilon, s)$ holds: the fields at scales $m \pm j$ and the quantities J on subcubes of scale $m - j$ are small, up to weights $3^{-sj/8}$ that decay with j . A scale that is not good is called **bad**. We use the same words for the translated events $\mathcal{G}_{m,y}(\varepsilon, s)$, with respect to the center y .

Throughout this section, we use the following notation. For $m, j \in \mathbb{N}_0$ with $j \leq m$, we denote

$$(6.11) \quad \mathbf{g}_{m,j} := \sum_{i=j+1}^m \mathbf{g}_i.$$

We also note that, for $0 \leq j \leq m$,

$$(6.12) \quad \frac{\mathbf{a}_m}{\mathbf{a}_j} = \exp(\mathbf{g}_{m,j} - (m-j)\tau^2).$$

For $s, \varepsilon \in (0, 1]$, condition (6.8) implies

$$(6.13) \quad \left(\sup_{0 \leq j \leq m} 3^{-\frac{1}{8}s(m-j)} \|\mathbf{g}_{m,j}\|_{L^\infty(\square_m)} + \sum_{j=m}^{\infty} 3^m \|\nabla \mathbf{g}_j\|_{L^\infty(\square_m)} \right) \mathbb{1}_{\mathcal{G}_m(\varepsilon, s)} \leq C\varepsilon.$$

Proposition 6.3 (Coarse-graining on good scales). *There is a constant $C(d) < \infty$ such that, for every $s \in [64\bar{\delta}^2, 1/2]$ and $L, m \in \mathbb{N}_0$ with $L \geq m$, we have*

$$(6.14) \quad \begin{aligned} & \mathcal{E}_{s,\infty,2}(\square_m; \mathbf{a}_L, (\mathbf{b}_{L,m})_{\square_m}) \mathbb{1}_{\mathcal{G}_m(1,s)} \\ & \leq C \sup_{0 \leq j \leq m} \sup_{0 \leq n \leq j-2} 3^{-\frac{1}{2}s(m-n)} \max_{\substack{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1}) \\ |e|=1}} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n)^{1/2} \\ & \quad + C s^{-1} \bar{\delta}^2 \\ & \quad + C \sup_{0 \leq j \leq m} 3^{-\frac{1}{8}s(m-j)} \|\mathbf{g}_{m,j}\|_{L^\infty(\square_m)} \\ & \quad + C 3^{-\frac{1}{8}sm} \left\| \sum_{i=0}^m \mathbf{g}_i \right\|_{L^\infty(\square_m)} \\ & \quad + C \sum_{j=m}^{\infty} 3^m \|\nabla \mathbf{g}_j\|_{L^\infty(\square_m)}. \end{aligned}$$

In particular, if $\varepsilon \in [s^{-1}\bar{\delta}^2, 1]$, then (6.13) yields, on $\mathcal{G}_m(\varepsilon, s)$,

$$\mathcal{E}_{s,\infty,2}(\square_m; \mathbf{a}_L, (\mathbf{b}_{L,m})_{\square_m}) \leq C\varepsilon.$$

We will also use the following elementary inequality. If $\varepsilon \in (0, 1]$, $s > 0$, and $D \geq 0$, then every $x \in [0, \varepsilon 3^{sD/16}]$ satisfies $3^{-sD/2}x \leq 3^{-sD/2} \min\{x, 1\} + \varepsilon^8$. Indeed, the second term on the right is needed only when $x > 1$, and then $3^{-sD/16} < \varepsilon$, so that $3^{-sD/2}x \leq \varepsilon 3^{-7sD/16} \leq \varepsilon^8$.

Proposition 6.4 (Proportion of good scales). *There exists $C(d) < \infty$ such that, for every $s, \theta, \varepsilon \in (0, 1]$ satisfying*

$$(6.15) \quad \theta \geq C s^{-6} \varepsilon^{-2} \bar{\delta}^2 |\log \bar{\delta}|,$$

and every $m_0 \in \mathbb{N}_0$ and $M \in \mathbb{N}_0$, we have the estimate

$$(6.16) \quad \mathbb{P} \left[\sum_{m=m_0}^{m_0+M} \mathbb{1}_{\{\mathcal{G}_m(\varepsilon, s)\}} \leq 1 - \theta \right] \leq \exp \left(- \frac{s^6 \varepsilon^2 \theta}{C \bar{\delta}^2 |\log \bar{\delta}|} (M+1) \right).$$

Proof of Proposition 6.3. Fix $s \in [64\bar{\delta}^2, 1/2]$. Throughout the proof, we denote $\tilde{\mathbf{a}}_{L,m} := (\mathbf{b}_{L,m})_{\square_m}^{-1} \mathbf{a}_L$. The proof compares each quantity $J(z + \square_n, e, e; \tilde{\mathbf{a}}_{L,m})$ entering $\mathcal{E}_{s,\infty,2}$ with the same quantity for the coefficient $\bar{\mathbf{a}}_n^{-1} \mathbf{a}_n$. Step 1 reduces $\mathcal{E}_{s,\infty,2}$ to the cubes lying in the annuli $\square_j \setminus \square_{j-1}$. Step 2 is the heart of the matter: it treats the cubes of side at most one directly and, on the larger cubes, bounds the ratio of the two coefficients by means of conditions (6.8) and (6.9). Step 3 combines the bounds.

Step 1. An annular decomposition. It bounds the error by the contributions of cubes lying in the annuli $\square_j \setminus \square_{j-1}$: there exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{N}_0$ and $s \in (0, 1/2]$,

$$(6.17) \quad \begin{aligned} \mathcal{E}_{s,\infty,2}(\square_m; \mathbf{a}_L, (\mathbf{b}_{L,m})_{\square_m})^2 &= (1 - 3^{-2s}) \sum_{n=-\infty}^m 3^{-2s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \max_{|e|=1} J(z + \square_n, e, e; \tilde{\mathbf{a}}_{L,m}) \\ &\leq C \sup_{j \leq m} \sup_{n \leq j-2} 3^{-\frac{3}{2}s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \tilde{\mathbf{a}}_{L,m}). \end{aligned}$$

For a domain U and $e \in \mathbb{R}^d$, write $J(U) := J(U, e, e; \tilde{\mathbf{a}}_{L,m})$. By decomposing into annular regions and interchanging the order of summation, we obtain

$$\begin{aligned} s \sum_{n=-\infty}^m 3^{-2s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} J(z + \square_n) \\ \leq s \sum_{n=-\infty}^m 3^{-2s(m-n)} \max_{n+1 \leq j \leq m} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} J(z + \square_n) + s \sum_{n=-\infty}^m 3^{-2s(m-n)} J(\square_n) \\ \leq C \sup_{j \leq m} \sup_{n \leq j-1} 3^{-\frac{3}{2}s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} J(z + \square_n) + s \sum_{n=-\infty}^m 3^{-2s(m-n)} J(\square_n). \end{aligned}$$

For the second term on the right, we use subadditivity. For every $n \leq m$, we have

$$J(\square_n) \leq \sum_{k=-\infty}^{n-1} \sum_{z \in 3^k \mathbb{Z}^d \cap (\square_{k+1} \setminus \square_k)} \frac{|\square_k|}{|\square_n|} J(z + \square_k) \leq \sum_{k=-\infty}^{n-1} 3^{-d(n-1-k)} \max_{z \in 3^k \mathbb{Z}^d \cap (\square_{k+1} \setminus \square_k)} J(z + \square_k).$$

Substituting back and interchanging the order of summation yields

$$\begin{aligned} \sum_{n=-\infty}^m 3^{-2s(m-n)} J(\square_n) &\leq \sum_{n=-\infty}^m 3^{-2s(m-n)} \sum_{k=-\infty}^{n-1} 3^{-d(n-1-k)} \max_{z \in 3^k \mathbb{Z}^d \cap (\square_{k+1} \setminus \square_k)} J(z + \square_k) \\ &\leq 3^{2s} \sum_{k=-\infty}^{m-1} 3^{-2s(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap (\square_{k+1} \setminus \square_k)} J(z + \square_k) \sum_{n=k+1}^m 3^{-(d-2s)(n-1-k)} \\ &\leq \frac{3}{2} 3^{2s} \sum_{k=-\infty}^{m-1} 3^{-2s(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap (\square_{k+1} \setminus \square_k)} J(z + \square_k). \end{aligned}$$

After relabeling $k \rightarrow n$ and $k+1 \rightarrow j$, the sum on the right becomes

$$\sum_{j=-\infty}^m 3^{-2s(m-j)} \max_{z \in 3^{j-1} \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} J(z + \square_{j-1}) \leq \frac{C}{s} \sup_{j \leq m} 3^{-\frac{3}{2}s(m-j)} \max_{z \in 3^{j-1} \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} J(z + \square_{j-1}),$$

which is of the form appearing on the right side of (6.17) with $n = j-1$. We obtain (6.17) by combining the previous displays with subadditivity in the form

$$\max_{z \in 3^{j-1} \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} J(z + \square_{j-1}) \leq \max_{z \in 3^{j-2} \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} J(z + \square_{j-2}).$$

Step 2. We first treat the cubes of side at most one. By (4.11), with $b = (\mathbf{b}_{L,m})_{\square_m}$, the contribution of the indices $n \leq 0$ to the left side of (6.17) is at most

$$(6.18) \quad C3^{-2sm} \left\| \frac{\mathbf{a}_L}{(\mathbf{b}_{L,m})_{\square_m}} + \frac{(\mathbf{b}_{L,m})_{\square_m}}{\mathbf{a}_L} - 2 \right\|_{L^\infty(\square_m)}.$$

On $\mathcal{G}_m(1, s)$, the relations (6.21), (6.22), and (6.23) established below, together with (6.9), bound (6.18) by the last four terms on the right side of (6.19) below. It remains to bound the contribution of the cubes of side larger than one.

We estimate only $n \leq j-2$, the range used in Step 3. The scale- $j-1$ cubes were subdivided at the end of Step 1 with the coefficient $\tilde{\mathbf{a}}_{L,m}$ unchanged; sensitivity is applied only after that subdivision. We show that there exists $C(d) < \infty$ such that

$$(6.19) \quad \begin{aligned} & \sup_{1 \leq j \leq m} \sup_{1 \leq n \leq j-2} 3^{-\frac{3}{2}s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \tilde{\mathbf{a}}_{L,m}) \mathbb{1}_{\mathcal{G}_m(1,s)} \\ & \leq C \sup_{0 \leq j \leq m} \sup_{0 \leq n \leq j-2} 3^{-s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) \\ & \quad + Cs^{-2} \bar{\delta}^4 \\ & \quad + C \left(\sum_{n=m}^{\infty} 3^m \|\nabla \mathbf{g}_n\|_{L^\infty(\square_m)} \right)^2 \\ & \quad + C \sup_{0 \leq n \leq m} 3^{-s(m-n)} \|\mathbf{g}_{m,n}\|_{L^\infty(\square_m)}^2 \\ & \quad + C3^{-sm} \left\| \sum_{i=0}^m \mathbf{g}_i \right\|_{L^\infty(\square_m)}^2. \end{aligned}$$

We apply Lemma 2.8 with $\mathbf{a} = \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n$ and $\mathbf{b} = \tilde{\mathbf{a}}_{L,m}$. For every $n \in \mathbb{N}$ with $n < m$, this gives

$$(6.20) \quad \begin{aligned} & J(z + \square_n, e, e; \tilde{\mathbf{a}}_{L,m}) \\ & \leq 2J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) \\ & \quad + 3 \left(\left\| \frac{\bar{\mathbf{a}}_n^{-1} \mathbf{a}_n}{\tilde{\mathbf{a}}_{L,m}} - 1 \right\|_{L^\infty(z + \square_n)}^2 + \left\| \frac{\tilde{\mathbf{a}}_{L,m}}{\bar{\mathbf{a}}_n^{-1} \mathbf{a}_n} - 1 \right\|_{L^\infty(z + \square_n)}^2 \right) (J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) + 1). \end{aligned}$$

Put $\rho_{m,n} := \log(\bar{\mathbf{a}}_n/\bar{\mathbf{a}}_m) - (m-n)\tau^2$. By (6.12),

$$(6.21) \quad \frac{\tilde{\mathbf{a}}_{L,m}}{\bar{\mathbf{a}}_n^{-1} \mathbf{a}_n} = \frac{\mathbf{a}_L \mathbf{a}_m^{-1}}{(\mathbf{a}_L \mathbf{a}_m^{-1})_{\square_m}} \exp(\mathbf{g}_{m,n} + \rho_{m,n}).$$

By (3.24),

$$(6.22) \quad |\rho_{m,n}| \leq \tau^2(m-n).$$

Set $S_m := \sum_{j=m}^{\infty} 3^m \|\nabla \mathbf{g}_j\|_{L^\infty(\square_m)}$. Conditions (6.8) and (6.9) in the definition of $\mathcal{G}_m(1, s)$, the mean-value theorem, and $|\exp(r) - 1| \leq |r| \exp(|r|)$ give

$$(6.23) \quad \left(\left\| \frac{\mathbf{a}_L \mathbf{a}_m^{-1}}{(\mathbf{a}_L \mathbf{a}_m^{-1})_{\square_m}} - 1 \right\|_{L^\infty(\square_m)} + \left\| \frac{(\mathbf{a}_L \mathbf{a}_m^{-1})_{\square_m}}{\mathbf{a}_L \mathbf{a}_m^{-1}} - 1 \right\|_{L^\infty(\square_m)} \right) \mathbb{1}_{\mathcal{G}_m(1,s)} \leq C \min\{1, S_m\},$$

and also $S_m \mathbb{1}_{\mathcal{G}_m(1,s)} \leq C$. Moreover, (6.9), (6.22), $\tau^2 \leq 4\bar{\delta}^2$, and $s \geq 64\bar{\delta}^2$ imply

$$\exp(\|\mathbf{g}_{m,n}\|_{L^\infty(z + \square_n)}) \mathbb{1}_{\mathcal{G}_m(1,s)} \leq C3^{s(m-n)/8}, \quad \exp(|\rho_{m,n}|) \leq 3^{s(m-n)/16}.$$

Applying these bounds to (6.21) and its reciprocal yields

$$\begin{aligned} & \left\| \frac{\bar{\mathbf{a}}_n^{-1} \mathbf{a}_n}{\tilde{\mathbf{a}}_{L,m}} - 1 \right\|_{L^\infty(z+\square_n)} + \left\| \frac{\tilde{\mathbf{a}}_{L,m}}{\bar{\mathbf{a}}_n^{-1} \mathbf{a}_n} - 1 \right\|_{L^\infty(z+\square_n)} \\ & \leq C 3^{\frac{3}{16}s(m-n)} \min \left\{ 1, S_m + \|\mathbf{g}_{m,n}\|_{L^\infty(z+\square_n)} + \tau^2(m-n) \right\}. \end{aligned}$$

On $\mathcal{G}_m(1, s)$, condition (6.10) bounds the term $J(z+\square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n)$ in (6.20) by $3^{s(m-n)/8}$. Hence the square of the right side of the last display, multiplied by $J(z+\square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) + 1$ and by $3^{-3s(m-n)/2}$, is bounded by

$$C 3^{-s(m-n)} \left(S_m^2 + \|\mathbf{g}_{m,n}\|_{L^\infty(z+\square_n)}^2 + (\tau^2(m-n))^2 \right).$$

Finally, $\sup_{r \geq 0} 3^{-sr} (\tau^2 r)^2 \leq C s^{-2} \bar{\delta}^4$. Combining these estimates with (6.20) proves (6.19).

Step 3. We conclude. Combining (6.17), the estimate (6.18) for cubes of side at most one, and (6.19) gives (6.14). $\square$

For each of the three conditions (6.8), (6.9), and (6.10) separately, we estimate the probability that it fails on a given proportion of the scales.

Lemma 6.5. *There exists a constant $C(d) < \infty$ such that, for every $\theta, s, \varepsilon \in (0, 1]$ satisfying $\theta \geq C s^{-6} \varepsilon^{-2} \bar{\delta}^2$, we have, for every $m_0 \in \mathbb{N}_0$ and $M \in \mathbb{N}_0$,*

$$(6.24) \quad \mathbb{P} \left[\sum_{m=m_0}^{m_0+M} \mathbb{1} \left\{ \sup_{j \geq 0} 3^{-sj} \sum_{i=(m-j) \vee 0}^{m+j} \|\mathbf{g}_i\| + 3^i |\nabla \mathbf{g}_i| \|_{L^\infty(\square_{m+1+j})} > \varepsilon \right\} \geq \theta \right] \leq \exp \left(-\frac{s^6 \varepsilon^2 \theta}{C \bar{\delta}^2} (M+1) \right).$$

Lemma 6.6. *There exists a constant $C(d) < \infty$ such that, for every $m_0, M \in \mathbb{N}_0$ and $s, \theta \in (0, 1]$ satisfying $\theta \geq C s^{-2} \bar{\delta}^2$, we have the estimate*

$$(6.25) \quad \mathbb{P} \left[\sum_{m=m_0}^{m_0+M} \mathbb{1} \left\{ \sup_{j \geq 0} 3^{-sj} \left\| \prod_{i=(m-j) \vee 0}^{m+j} \exp(|\mathbf{g}_i|) + \prod_{i=m+j}^{\infty} \exp(4|\mathbf{g}_i - \mathbf{g}_i(0)|) \right\|_{L^\infty(\square_{m+1+j})} > 6 \right\} \geq \theta \right] \leq \exp \left(-\frac{s^2 \theta}{C \bar{\delta}^2} (M+1) \right).$$

Lemma 6.7. *There exists a constant $C(d) < \infty$ such that, if $\varepsilon, s, \theta \in (0, 1]$ satisfy the conditions*

$$\bar{\delta}^2 |\log \bar{\delta}| \leq C^{-1} s^3 \varepsilon^2 \quad \text{and} \quad \theta \geq C s^{-3} \varepsilon^{-2} \bar{\delta}^2 |\log \bar{\delta}|,$$

then we have, for every $m_0 \in \mathbb{N}_0$ and $M \in \mathbb{N}_0$, the estimate

$$(6.26) \quad \mathbb{P} \left[\sum_{m=m_0}^{m_0+M} \mathbb{1} \left\{ \sup_{0 \leq j \leq m} \sup_{0 \leq n \leq j-2} 3^{-s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) > \varepsilon^2 \right\} > \theta \right] \leq \exp \left(-\frac{s^3 \varepsilon^2 \theta}{C |\log \bar{\delta}| \bar{\delta}^2} (M+1) \right).$$

Proof of Proposition 6.4. Fix $s, \theta, \varepsilon \in (0, 1]$ and assume (6.15) with its constant C large enough that Lemmas 6.5, 6.6, and 6.7 are applicable with parameters $(s/8, \theta/4, \varepsilon)$ in place of (s, θ, ε) . If the good event fails on at least a proportion θ of the scales, then one of the three conditions (6.8)–(6.10) fails on at least a proportion $\theta/3$ of them. Since $\theta/3 > \theta/4$, this case is covered by the strict inequality in Lemma 6.7. A union bound over the three conditions gives (6.16). $\square$

Proof of Lemma 6.5. Interchanging the order of summation gives

$$\begin{aligned} \sup_{j \geq 0} 3^{-sj} \sum_{i=(m-j) \vee 0}^{m+j} \|\mathbf{g}_i| + 3^i |\nabla \mathbf{g}_i|\|_{L^\infty(\square_{m+1+j})} &\leq \sum_{j=0}^{\infty} 3^{-sj} \sum_{i=(m-j) \vee 0}^{m+j} \|\mathbf{g}_i| + 3^i |\nabla \mathbf{g}_i|\|_{L^\infty(\square_{m+1+j})} \\ &= \sum_{i \in \mathbb{N}_0} 3^{-s|i-m|} \left(\sum_{j=|i-m|}^{\infty} 3^{-s(j-|i-m|)} \|\mathbf{g}_i| + 3^i |\nabla \mathbf{g}_i|\|_{L^\infty(\square_{m+1+j})} \right). \end{aligned}$$

For a universal constant C_2 to be fixed and $i \in \mathbb{N}_0$, set

$$X_{m,i} := C_2 s^{-1} \varepsilon^{-1} 3^{-\frac{1}{2}s|m-i|} \sum_{j=|i-m|}^{\infty} 3^{-s(j-|i-m|)} \|\mathbf{g}_i| + 3^i |\nabla \mathbf{g}_i|\|_{L^\infty(\square_{m+1+j})},$$

and set $X_{m,i} = 0$ if $m < 0$ or $i < 0$. We shall apply Proposition B.2 to this extension to all indices in $\mathbb{Z}$. We have

$$\sup_{j \geq 0} 3^{-sj} \sum_{i=(m-j) \vee 0}^{m+j} \|\mathbf{g}_i| + 3^i |\nabla \mathbf{g}_i|\|_{L^\infty(\square_{m+1+j})} \leq C_2^{-1} s \varepsilon \sum_{i \in \mathbb{N}_0} 3^{-\frac{1}{2}s|m-i|} X_{m,i}.$$

By (A.4), (1.4), and (1.5),

$$\|\mathbf{g}_i| + 3^i |\nabla \mathbf{g}_i|\|_{L^\infty(\square_{m+1+j})} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}(|m-i| + j + 1)^{1/2}).$$

By (A.6), there exists a constant $C_1(d)$ such that, for every $p \in [1, \infty)$,

$$\mathbb{E}[X_{m,i}^p]^{1/p} \leq C_1 C_2 s^{-5/2} \varepsilon^{-1} \bar{\delta} p^{1/2}.$$

Moreover, for $m, m', i, i' \in \mathbb{N}_0$ with $|i - i'| \geq 1$, the variables $X_{m,i}$ and $X_{m',i'}$ are independent. We fix $p := C_1^{-2} C_2^{-2} s^5 \varepsilon^2 \bar{\delta}^{-2}$. With this choice of p , if $C(d)$ is large enough in terms of C_1 and C_2 , the assumption $\theta \geq C s^{-6} \varepsilon^{-2} \bar{\delta}^2$, which since $\theta \leq 1$ also gives $\bar{\delta} \leq C^{-1/2} s^3 \varepsilon$, implies that $\theta \geq p^{-1}$ and $s \geq 2p^{-1}$. By the inequality displayed after the definition of $X_{m,i}$, we then have

$$\left\{ \sup_{j \geq 0} 3^{-sj} \sum_{i=(m-j) \vee 0}^{m+j} \|\mathbf{g}_i| + 3^i |\nabla \mathbf{g}_i|\|_{L^\infty(\square_{m+1+j})} > \varepsilon \right\} \subseteq \left\{ \sum_{i \in \mathbb{N}_0} 3^{-\frac{1}{2}s|m-i|} X_{m,i} > C_2 s^{-1} \right\}.$$

For C_2 large enough, Proposition B.2 gives (6.24). $\square$

Proof of Lemma 6.6. We apply Propositions B.3 and B.4. For sufficiently large $C(d)$, set $X_n := C 3^n \|\nabla \mathbf{g}_n\|_{L^\infty(\square_{n+1})}$ for $n \in \mathbb{N}_0$ and $X_n := 0$ for $n < 0$. On $\square_{k+1+j}$,

$$\prod_{i=k+j}^{\infty} \exp(4|\mathbf{g}_i - \mathbf{g}_i(0)|) \leq \prod_{i=k+j}^{\infty} \exp\left(3^{k+j-i} (C 3^i \|\nabla \mathbf{g}_i\|_{L^\infty(\square_{i+1})})\right) \leq \prod_{i=0}^{\infty} \exp(3^{-i} X_{k+i+j}).$$

The sequence (X_n) satisfies the hypotheses of Proposition B.3 with $\delta_0 := C\bar{\delta}$. If $\bar{\delta} \leq C^{-1} s \theta^{1/2}$, then that proposition gives

$$\mathbb{P} \left[\sum_{m=m_0}^{m_0+M} \mathbb{1} \left\{ \sup_{j \geq 0} 3^{-sj} \left\| \prod_{i=m+j}^{\infty} \exp(4|\mathbf{g}_i - \mathbf{g}_i(0)|) \right\|_{L^\infty(\square_{m+1+j})} > 3 \right\} \geq \theta \right] \leq \exp \left(-\frac{s^2 \theta}{C \bar{\delta}^2} (M+1) \right).$$

If $\bar{\delta} \leq C^{-1} s \theta^{1/2}$ for sufficiently large $C(d)$, then Proposition B.4, applied with $\mathbf{g}_i$ in place of X_i for $i \in \mathbb{N}_0$ and with 0 in place of X_i for $i < 0$, yields

$$\mathbb{P} \left[\sum_{k=m_0}^{m_0+M} \mathbb{1} \left\{ \sup_{j \geq 0} 3^{-sj} \left\| \prod_{i=(k-j) \vee 0}^{k+j} \exp(|\mathbf{g}_i|) \right\|_{L^\infty(\square_{k+1+j})} > 3 \right\} \geq \theta \right] \leq \exp \left(-\frac{s^2 \theta}{C \bar{\delta}^2} (M+1) \right).$$

Apply the previous two displays with $\theta/2$ in place of θ . If the sum of the two products exceeds six, then at least one exceeds three, so a union bound gives (6.25). $\square$

Proof of Lemma 6.7. First,

$$\begin{aligned} & \sup_{0 \leq j \leq m} \sup_{0 \leq n \leq j-2} 3^{-s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) \\ & \leq \sum_{j=0}^m 3^{-s(m-j)} \sum_{n=0}^{j-2} 3^{-s(j-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n). \end{aligned}$$

We set, for a universal constant $K \geq 1$ to be fixed,

$$X_{m,j} := K s^{-1} \varepsilon^{-2} \sum_{n=0}^{j-2} 3^{-s(j-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) \mathbb{1}_{\{j \leq m\}}.$$

Then

$$\begin{aligned} & \left\{ \sup_{0 \leq j \leq m} \sup_{0 \leq n \leq j-2} 3^{-s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) \geq \varepsilon^2 \right\} \\ & \subseteq \left\{ \sum_{j=0}^m 3^{-s(m-j)} X_{m,j} \geq K s^{-1} \right\}. \end{aligned}$$

There is an integer $r(d) < \infty$ such that the columns $(X_{m,j})_m$, indexed by j , are $r(d)$ -dependent in the sense of (B.1). To verify this, fix one of the fields $\mathbf{g}_i$. Column j depends on $\mathbf{g}_i$ only if $i \leq j-2$, and then only through its values on the annulus $\square_j \setminus \square_{j-1}$. If $j' > j$ and $j' - j \geq r(d)$, then the distance between these annuli is at least $(3^{j'-1} - 3^j)/2 > \sqrt{d} 3^{j-2} \geq \sqrt{d} 3^i$, provided that $r(d)$ is large enough. Thus, by Assumption (g1), the restrictions of $\mathbf{g}_i$ to the regions used by two sets of columns at distance at least $r(d)$ are independent. Since the fields are mutually independent, so are the two sets of columns. Moreover, by the triangle inequality, stationarity, and Proposition 4.1, there is a constant $C_1(d) < \infty$ such that, if p satisfies $2ds^{-1} \leq p \leq C_1^{-1} \bar{\delta}^{-2} |\log \bar{\delta}|^{-1}$, then

$$\begin{aligned} \mathbb{E}[X_{m,j}^p]^{1/p} & \leq K s^{-1} \varepsilon^{-2} \sum_{n=0}^{j-2} 3^{-s(j-n)} \mathbb{E} \left[\max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n)^p \right]^{1/p} \\ & \leq K s^{-1} \varepsilon^{-2} \sum_{n=0}^{j-2} 3^{-(s-\frac{d}{p})(j-n)} \mathbb{E} \left[\max_{|e|=1} J(\square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n)^p \right]^{1/p} \leq C_1 K s^{-2} \varepsilon^{-2} \bar{\delta}^2 |\log \bar{\delta}| p. \end{aligned}$$

Taking $p := C_1^{-1} K^{-1} s^2 \varepsilon^2 \bar{\delta}^{-2} |\log \bar{\delta}|^{-1}$ and assuming that $\bar{\delta}$ satisfies

$$\bar{\delta}^2 |\log \bar{\delta}| \leq \frac{s^3 \varepsilon^2}{2dC_1 K} \implies 2ds^{-1} \leq p = C_1^{-1} K^{-1} s^2 \varepsilon^2 \bar{\delta}^{-2} |\log \bar{\delta}|^{-1},$$

we deduce that $\mathbb{E}[X_{m,j}^p]^{1/p} \leq 1$. Assuming also $\theta \geq p^{-1}$ and applying Proposition B.2 with $r = r(d)$, we obtain (6.26) for a sufficiently large universal constant K . $\square$

6.2. Excess decay on good scales. Throughout this subsection, for $m, n \in \mathbb{Z}$ with $n \leq m$ and $x \in \square_m$, we denote $U_{m,n}(x) := (x + \square_n) \cap \square_m$.

Lemma 6.8 (Centered energy at a good scale). *Let $s, L, m, n, z, x, u, h, \mathbf{g}$ satisfy the hypotheses of Lemma 6.9, and put $U = U_{m,n}(x)$ and $\mathbf{a}_0 = (\mathbf{b}_{L,n+2})_{z+\square_{n+2}}$. On $\mathcal{G}_{n+2,z}(1, s/8)$, for every $q \in$*

$U_{m,n-1}(x),$

$$(6.27) \quad \begin{aligned} \mathbf{a}_0^{-1} \int_{U_{m,n-4}(q)} \mathbf{a}_L |\nabla u|_2^2 &\leq C 3^{-2n} \|u - (u)_U\|_{\underline{L}^2(U)}^2 + C |(\nabla h)_U|_2^2 \mathbb{1}_{\{\bar{U} \cap \partial \square_m \neq \emptyset\}} \\ &+ C s^{-12} \mathbf{a}_0^{-2} 3^{2sn} [\mathbf{g}]_{\underline{H}^s(U)}^2 + C s^{-4} 3^{2sn} [\nabla h]_{\underline{H}^s(U)}^2 \mathbb{1}_{\{\bar{U} \cap \partial \square_m \neq \emptyset\}}. \end{aligned}$$

Here C depends only on d . The same estimate holds with the cutoff modification of the good event and with the hypothesis $L \geq m$ removed.

Proof. We give the boundary argument with its centering scalar retained. The proof tracks the energy $F(r)$ of u on the windows W_r around x defined below. We begin with ellipticity bounds for every cube contained in $z + \square_{n+2}$, including cubes off the triadic grid. Next, for each cell of a cover scale $k \leq n - 4$ whose parent cube P has a face on $\partial \square_m$, we compare u on P with the solution w_P of the same equation with boundary data h ; this gives an energy estimate in which the energy of u on P has the small coefficient θ . Interior cells use the coarse-grained Caccioppoli inequality of Proposition 2.6. The heart of the matter is a parent with a face on $\partial \square_m$: there the Caccioppoli estimate for $u - w_P$ contains the mean μ of $u - w_P$ on P , which we bound through slabs along that face, at a depth chosen so that the coefficient of the energy of u on P is at most θ . Finally, we sum over the cells, whose overlap is bounded, and iterate the resulting inequality for F . All energies in the proof are for $\mathbf{a}_L$; the comparison scalar is always the fixed $\mathbf{a}_0$ above.

First we record the ellipticity bounds needed below the scale n . For every translated cube $T = y + \square_k$ contained in $z + \square_{n+2}$, including $k < 0$, stability gives

$$(6.28) \quad \mathcal{E}_{s/6, \infty, 2}(T; \mathbf{a}_L, \mathbf{a}_0) \leq C 3^{(s/8)(n+2-k)} \mathcal{E}_{s/8, \infty, 2}(z + \square_{n+2}; \mathbf{a}_L, \mathbf{a}_0).$$

For completeness, the arbitrary-depth form of stability follows by partitioning an off-grid cube into the maximal cubes of the anchor's triadic grid that it contains. At depth j , their total volume is at most $C(d)3^{-j}$ times the volume of the off-grid cube: a maximal cube has its parent meeting the boundary, so it lies in a boundary layer of width $C3^{-j}$ times the side length. Subadditivity of J and (2.37) bound the response of an off-grid cube of scale $k - \ell$ by

$$\frac{C(d)}{1 - 2t'} 3^{2t'(n+2-k+\ell)} \mathcal{E}_{t', \infty, 2}(z + \square_{n+2}; \mathbf{a}_L, \mathbf{a}_0)^2, \quad 0 < t' < 1/2.$$

The maximal cubes cover up to a null set. To justify countable subadditivity, restrict each admissible harmonic field in the variational definition of J to these open subcubes. Its response integrand is integrable, since the fixed finite cutoff is uniformly elliptic on the anchor. The integral equals the sum over the subcubes, and each summand is bounded by the volume of that subcube times its response. The displayed summable majorant permits this comparison before taking the supremum over the admissible fields. Apply this to both scalar parts of the doubled response and then to its unit-sphere maximum. Summing with the normalized weights $(1 - 3^{-2t})3^{-2t\ell}$ gives

$$\mathcal{E}_{t, \infty, 2}(T; \mathbf{a}_L, \mathbf{a}_0)^2 \leq \frac{C(d)t}{(t - t')(1 - 2t')} 3^{2t'(n+2-k)} \mathcal{E}_{t', \infty, 2}(z + \square_{n+2}; \mathbf{a}_L, \mathbf{a}_0)^2.$$

Taking $(t, t') = (s/6, s/8)$ proves (6.28), with a dimensional constant and no inverse power of s . On the stated event, the anchor error is bounded by C . By (2.38), monotonicity in the order, and (2.39), this gives

$$(6.29) \quad \max \left\{ \mathbf{a}_0^{-1} \Lambda_{s/6, 2}(T), \mathbf{a}_0 \lambda_{s/6, 2}^{-1}(T), \mathbf{a}_0^{-1} \Lambda_{1/2, 1}(T), \mathbf{a}_0 \lambda_{s/3, 1}^{-1}(T) \right\} \leq C 3^{(s/4)(n+2-k)}.$$

We work at a free cover scale $k \leq n - 4$. For a cell center q , let $P = \hat{q} + \square_k$ be the coordinatewise clamping of $q + \square_k$ inside $\square_m$. The cell is $U_{m, k-2}(q)$. Its scale- $(k-1)$ patch meets no artificial face of P . If this patch leaves $\square_m$, then P has a face on $\partial \square_m$; this is the boundary case. Write $\Delta = n + 2 - k$ and $E_P(w) = \int_P \mathbf{a}_L |\nabla w|_2^2$.

Let w_P solve the same forced equation as u in P , with boundary data h on all of ∂P . Its energy satisfies

$$(6.30) \quad E_P(w_P) \leq C 3^{s\Delta/4} \left(\mathbf{a}_0 |(\nabla h)_P|_2^2 + \mathbf{a}_0 s^{-4} 3^{2sk} [\nabla h]_{\underline{H}^s(P)}^2 + \mathbf{a}_0^{-1} s^{-12} 3^{2sk} [\mathbf{g}]_{\underline{H}^s(P)}^2 \right).$$

Here is the separation of the three terms. Split h into an affine function with slope $(\nabla h)_P$ and a remainder whose gradient has mean zero. The affine Dirichlet minimum is bounded by $|(\nabla h)_P|_2^2 \Lambda_{s/6,2}(P)$, by the definition of the coarse matrix and its ordering. The centered remainder is controlled by the Dirichlet energy estimate [Armstrong et al., 2026, Lemma 2.11, (2.82)]; its squared datum term is at most $C s^{-3} \Lambda_{s/2,2}(P)$ times the squared scale-normalized positive norm of the remainder's gradient. This norm is bounded by $C 3^{sk} s^{-1/2} [\nabla h]_{\underline{H}^s(P)}$, giving the s^{-4} term. Add the forced zero-boundary solution. The same estimate gives the squared forcing term $C s^{-7} \lambda_{s/2,2}^{-1}(P) 3^{2sk} [\mathbf{g}]_{\underline{H}^s(P)}^2$, which is bounded by the displayed s^{-12} term. The factor $s^{-1/2}$ in both conversions comes from the factor s in this paper's squared fractional seminorm. All constants in these comparisons are uniform for $0 < s \leq 1/4$. The energy triangle inequality and (6.29) prove (6.30). In particular, the affine slope has no inverse power of s .

Apply Lemma 2.10, rescaled to P , to $u - w_P$, with zero forcing and zero boundary datum, and with orders $1/2$ and $s/2$ at the cell center. This difference has zero trace on the physical faces needed by the estimate. Equation (2.46), rescaled to P , bounds the oscillation of w_P by its own energy. Here the Caccioppoli parameter is $\sigma = (1 - s)/2$, and its ellipticity factor satisfies

$$\left(\frac{\Lambda_{1/2,1}(P)}{\lambda_{s/2,1}(P)} \right)^{(2-s)/(1-s)} \lambda_{s/2,1}(P) \leq C \mathbf{a}_0 3^{17s\Delta/12}.$$

Indeed, $(2-s)/(1-s) \leq 7/3$ for $s \leq 1/4$; the powers of the bound in (6.29) are thus at most $17/3 \leq 7$ in the solution oscillation term and $20/3 \leq 8$ after the extra inverse ellipticity factor from Poincaré in the competitor term. Thus, for any scalar c_0 and with $\mu := (u - w_P)_P$,

$$(6.31) \quad \int_{U_{m,k-2}(q)} \mathbf{a}_L |\nabla u|_2^2 \leq C \left(3^{7s\Delta/4} \mathbf{a}_0 3^{-2k} (\|u - c_0\|_{\underline{L}^2(P)}^2 + \mu^2) + 3^{2s\Delta} E_P(w_P) \right).$$

Indeed, $\|u - (w_P)_P\|_{\underline{L}^2(P)}^2 \leq \|u - c_0\|_{\underline{L}^2(P)}^2 + \mu^2$, by the identity for variance and the minimizing property of $(u)_P$.

To estimate μ , extend $w_P - h \in H_0^1(P)$ by zero to $\square_m$ and put $\rho = (u - h) - \text{ext}_P(w_P - h) \in H_0^1(\square_m)$. On P this is $u - w_P$. Its zero extension across a met face vanishes on the outer half of every cube centered on that face. For such a tile T of side 3^κ , the rescaled inequality (2.46) and the vanishing half give

$$|(\rho)_T|^2 \leq C 3^{(s/4)(n+2-\kappa)} \mathbf{a}_0^{-1} 3^{2\kappa} \int_T \mathbf{a}_L |\nabla \rho|_2^2.$$

This applies to either sign of the met face. At a lower face the vanishing half is below the face; at an upper face it is above. The coefficient is kept in its original coordinates.

Tile the doubled slab adjoining one met face at depth j , so $\kappa = k - j$. Its inner half S lies in P , and $|P|/|S| \leq C(d)3^j$. Jensen on the equal-volume tiles bounds the slab mean by the tile energies. On the inner slab, the triangle inequality between its mean and the parent mean gives

$$(6.32) \quad \begin{aligned} \mu^2 &\leq C_f 3^{s(\Delta+j)/4} \mathbf{a}_0^{-1} 3^{2(k-j)} 3^j (E_P(u) + E_P(w_P)) \\ &\quad + C 3^j \left(\|u - c_0\|_{\underline{L}^2(P)}^2 + \|w_P - (w_P)_P\|_{\underline{L}^2(P)}^2 \right). \end{aligned}$$

The doubled slab's energy is at most $C 3^j$ times the normalized parent energy, since its exterior half contributes zero. The last competitor oscillation is at most $C 3^{s\Delta/4} \mathbf{a}_0^{-1} 3^{2k} E_P(w_P)$. This proves (6.32) while keeping the scalar μ .

After substitution in (6.31), the coefficient of $E_P(u)$ is at most $C'_f 3^{2s\Delta} 3^{-(1-s/4)j}$. Fix $\theta = (16 \cdot 28^d)^{-1}$ and choose

$$j = \left\lceil \frac{16}{15} (2s\Delta + \max\{0, \log_3(C'_f/\theta)\}) \right\rceil.$$

Since $1-s/4 \geq 15/16$, this coefficient is at most θ , and $3^j \leq C(d)3^{(32/15)s\Delta}$. The factor of the solution oscillation term is at most $3^{(7/4+32/15)s\Delta} = 3^{(233/60)s\Delta} \leq 3^{4s\Delta}$. The factors of the competitor terms, including the additional $3^{s\Delta/4}$, are at most $C3^{12s\Delta}$. Multiplying by the cell volume and using that it is at most $|P|$ gives the parent-local estimate

$$(6.33) \quad \begin{aligned} & \int_{U_{m,k-2}(q)} \mathbf{a}_L |\nabla u|_2^2 \\ & \leq \theta \int_P \mathbf{a}_L |\nabla u|_2^2 + C3^{4s\Delta} \mathbf{a}_0 3^{-2k} \int_P |u - c_0|^2 \\ & \quad + C3^{12s\Delta} \left(\mathbf{a}_0 \int_P |\nabla h|_2^2 + \mathbf{a}_0 s^{-4} 3^{2sk} |P| [\nabla h]_{\underline{H}^s(P)}^2 + \mathbf{a}_0^{-1} s^{-12} 3^{2sk} |P| [\mathbf{g}]_{\underline{H}^s(P)}^2 \right). \end{aligned}$$

We now sum before converting the parent-local terms into window terms. Put $N = 3^n$ and $W_r = \{p \in \square_m : |p - x|_\infty \leq r\}$, for $N/3 \leq r \leq 4N/9$. Denote by $\mathbf{B}$ the sum of the four terms on the right of (6.27), multiplied by $\mathbf{a}_0$ and with their constants omitted. For $N/3 \leq r < R \leq 4N/9$, choose $k \in \mathbb{Z}$ so that $9 \cdot 3^k \leq R - r \leq 27 \cdot 3^k$. The scale- $(k-2)$ cells meeting W_r have their projected parents inside W_R and U . Their overlap is at most 28^d : their centers have spacing 3^{k-2} and clamping moves a center by at most $3^k/2$ in each coordinate. Cells whose centers are outside $\square_m$ have zero intersection with $\square_m$, since the scales are integer scales of the triadic grid.

In the boundary class use (6.33) with $c_0 = (u)_U$; in the interior class use Proposition 2.6, rescaled to the patch $q + \square_{k-1}$, with the same c_0 : apply it to $u - c_0$ with the orders $1/2$ and $s/2$ and the forcing $\mathbf{g}$ minus its mean on the patch. In the boundary class, $\bar{U}$ meets $\partial\square_m$, so both datum terms have precisely the indicator in $\mathbf{B}$. Bounded overlap controls the integrals of $|u - c_0|^2$ and $|\nabla h|_2^2$. For any field F , positivity of the fractional kernel gives $\sum_P |P| [F]_{\underline{H}^s(P)}^2 \leq 28^d |U| [F]_{\underline{H}^s(U)}^2$. Finally,

$$\|\nabla h\|_{\underline{L}^2(U)}^2 \leq |(\nabla h)_U|_2^2 + C s^{-1} 3^{2sn} [\nabla h]_{\underline{H}^s(U)}^2.$$

This is the variance identity followed by the diameter bound in the fractional integral; every side of U is at least $3^n/2$. Its fractional term is bounded by the s^{-4} datum term.

Writing $\delta_k = n - k$, the oscillation transport is $3^{4s(\delta_k+2)+2\delta_k} \leq C3^{3\delta_k}$. The fractional transports are $3^{12s(\delta_k+2)-2s\delta_k} \leq C3^{3\delta_k}$, and the datum mean transport is $3^{12s(\delta_k+2)} \leq C3^{3\delta_k}$. All three inequalities use $s \leq 1/4$. Therefore, with $F(r) = \int_{W_r} \mathbf{a}_L |\nabla u|_2^2$,

$$(6.34) \quad F(r) \leq \frac{1}{16} F(R) + \frac{CN^{d+3}\mathbf{B}}{(R-r)^3}.$$

The total energy bounds F . At $r_i = N/3 + (N/9)(1 - 2^{-i})$, iterating (6.34) gives

$$F(N/3) \leq 16^{-M} F(r_M) + CN^{d+3}\mathbf{B}(N/9)^{-3} \sum_{i=0}^{M-1} 16^{-i} 8^{i+1} \leq 16^{-M} F(r_M) + CN^d \mathbf{B}.$$

Let $M \rightarrow \infty$. Every cell $U_{m,n-4}(q)$ in the lemma lies in $W_{N/3}$ and has volume at least $c(d)N^d$. Dividing proves (6.27). For the cutoff modification, the same argument uses Proposition 6.15 for the anchor error bound. All subsequent steps concern $\mathbf{a}_L$ and $\mathbf{a}_0$ deterministically and do not require $L \geq m$. $\square$

The next lemma adapts [Armstrong et al. \[2026, Lemma 4.10\]](#).

Lemma 6.9 (Harmonic approximation on good scales). *There exists $C(d) < \infty$ such that, for every $s \in [512\bar{\delta}^2, 1/4]$, every $L, m, n \in \mathbb{N}_0$ with $L \geq m \geq n + 5$, every $z \in \square_m$ and $x \in U_{m,n-3}(z)$, and every $u \in H^1(\square_m)$, $\mathbf{g} \in H^s(\square_m; \mathbb{R}^d)$, and $h \in H^{1+s}(\square_m)$ satisfying*

$$\begin{cases} -\nabla \cdot \mathbf{a}_L \nabla u = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ u = h & \text{on } \partial \square_m, \end{cases}$$

the following holds. Let $y \in \square_m$ be such that $U_{m,n-4}(x) \subseteq y + \square_{n-2} \subseteq U_{m,n-1}(x)$. Let v be the unique solution of the Dirichlet problem

$$\begin{cases} -\Delta v = 0 & \text{in } y + \square_{n-2}, \\ v = u & \text{on } \partial(y + \square_{n-2}). \end{cases}$$

Then

$$\begin{aligned} & \|u - v\|_{\underline{L}^2(U_{m,n-4}(x))} \mathbb{1}_{\mathcal{G}_{n+2,z}(1,s/8)} \\ & \leq C s^{-3/2} \mathcal{E}_{s/8,\infty,2}(z + \square_{n+2}; \mathbf{a}_L, (\mathbf{b}_{L,n+2})_{z+\square_{n+2}}) \left(\|u - (u)_{U_{m,n}(x)}\|_{\underline{L}^2(U_{m,n}(x))} \right. \\ & \quad \left. + s^{-3/2} 3^n |(\nabla h)_{U_{m,n}(x)}| \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}} \right) \\ & \quad + C s^{-15/2} (\mathbf{b}_{L,n+2})_{z+\square_{n+2}}^{-1} 3^{(1+s)n} [\mathbf{g}]_{\underline{H}^s(U_{m,n}(x))} \\ & \quad + C s^{-7/2} 3^{(1+s)n} [\nabla h]_{\underline{H}^s(U_{m,n}(x))} \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}}. \end{aligned} \tag{6.35}$$

Proof. Set $D = y + \square_{n-2}$, $U = U_{m,n}(x)$ and $\mathbf{a}_0 = (\mathbf{b}_{L,n+2})_{z+\square_{n+2}}$. The proof has three parts. First, the good event bounds the coarse-graining errors on D . Second, the coarse-graining estimate of [Armstrong et al. \[2026, Proposition 2.21\]](#) and a duality argument bound the $\underline{L}^2$ norm of $u - v$ by these errors times the energy norm of u on D , plus a forcing term, as in (6.38); most of the work is in tracking the powers of s in this part. Third, Lemma 6.8, applied on a partition of D , bounds this energy norm by the oscillation of u and the data on U .

Stability and Proposition 6.3 give, on $\mathcal{G}_{n+2,z}(1, s/8)$,

$$(6.36) \quad \mathcal{E}_{s/3,\infty,1}(D; \mathbf{a}_L, \mathbf{a}_0) + \mathcal{E}_{s/6,\infty,2}(D; \mathbf{a}_L, \mathbf{a}_0) \leq C \mathcal{E}_{s/8,\infty,2}(z + \square_{n+2}; \mathbf{a}_L, \mathbf{a}_0) \leq C.$$

By Definition 2.3, for $D = y + \square_{n-2}$ and every $t \in (0, 1]$ and $q \in [1, \infty]$,

$$(6.37) \quad \mathcal{E}_{t,\infty,q}(D, n-3; \mathbf{a}_L, \mathbf{a}_0) \leq 3^t \mathcal{E}_{t,\infty,q}(D; \mathbf{a}_L, \mathbf{a}_0).$$

This follows by dropping the nonnegative terms above $n-3$ and shifting the normalization by one scale; the supremum endpoint has the same comparison.

Let $v_{\mathbf{g}}$ solve $-\mathbf{a}_0 \Delta v_{\mathbf{g}} = \nabla \cdot \mathbf{g}$ in D with trace u . We apply [Armstrong et al. \[2026, Proposition 2.21, \(2.119\)\]](#) directly, retaining its unrenormalized fractional test norm during the following duality calculation. Its parameters are $p = 2$ and $(s_1, \vartheta, s_2) = (s/3, s/2, s)$; the sums stop at $n-3$. Thus (6.36) and (6.37) also bound the truncated errors in (2.119). For $w = u - v_{\mathbf{g}} \in H_0^1(D)$, solve $-\Delta \psi = w$ in D , with zero trace, and put $R = 3^{n-2}$. Cube H^2 regularity and rescaling give

$$\|\nabla \psi\|_{\underline{L}^2(D)} \leq CR \|w\|_{\underline{L}^2(D)}, \quad \|\nabla^2 \psi\|_{\underline{L}^2(D)} \leq C \|w\|_{\underline{L}^2(D)}.$$

For $0 < \vartheta \leq 1/8$, the test-function bound in ASD's norm is

$$\left(\int_D \int_D \frac{|\nabla \psi(p) - \nabla \psi(q)|_2^2}{|p - q|_2^{d+2\vartheta}} dp dq \right)^{1/2} + R^{-\vartheta} \|\nabla \psi\|_{\underline{L}^2(D)} \leq CR^{1-\vartheta} \|w\|_{\underline{L}^2(D)}.$$

To see uniformity, extend $\nabla \psi$ in H^1 from the unit cube. For displacements of size at most a fixed dimensional constant, its squared translation difference is bounded by the squared displacement times the H^1 norm. The radial integral is $C/(2-2\vartheta)$, bounded in the stated range. Smooth

approximation permits this vector field as a dual test. The identity $\oint_D w^2 = \oint_D \nabla w \cdot \nabla \psi$ therefore bounds $\|w\|_{\underline{L}^2(D)}$ by $CR^{1-\vartheta}$ times ASD's negative norm of ∇w , with no inverse power of ϑ .

In (2.119), the $p = 2$ local energy sum is at most

$$\left(\sum_{k=-\infty}^{n-3} 3^{-2(s/2-s/3)(n-3-k)} \right)^{1/2} \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(D)} \leq Cs^{-1/2} \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(D)}.$$

At each scale, the volume-weighted average of the squared local energies is the squared energy norm on D , since the subcubes partition D . Thus their number contributes no scale loss. Thus ASD's s^{-1} energy coefficient becomes $s^{-3/2}$. Its forcing coefficient is $s^{-9/2}(s - s/2)^{-1}$, and converting its unrenormalized H^s seminorm to this paper's seminorm introduces the factor $s^{-1/2}$. It gives Cs^{-6} . The factor $1 + \mathcal{E}_{s/6, \infty, 2}(D)^2$ is bounded on the event. The zero-trace difference $v_{\mathbf{g}} - v$ satisfies the ordinary energy and Poincaré estimates, giving $\|v_{\mathbf{g}} - v\|_{\underline{L}^2(D)} \leq Cs^{-1/2} \mathbf{a}_0^{-1} 3^{(1+s)n} [\mathbf{g}]_{\underline{H}^s(D)}$. Combining these bounds and passing to $U_{m, n-4}(x) \subseteq D$ by the bounded volume ratio yields

$$(6.38) \quad \|u - v\|_{\underline{L}^2(U_{m, n-4}(x))} \mathbb{1}_{\mathcal{G}_{n+2, z}(1, s/8)} \leq Cs^{-3/2} 3^n \mathcal{E}_{s/8, \infty, 2}(z + \square_{n+2}; \mathbf{a}_L, \mathbf{a}_0) \mathbf{a}_0^{-1/2} \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(D)} \\ + Cs^{-6} \mathbf{a}_0^{-1} 3^{(1+s)n} [\mathbf{g}]_{\underline{H}^s(U)}.$$

Partition D into its 9^d grandchildren and apply Lemma 6.8 at their centers. Taking square roots gives

$$(6.39) \quad \mathbf{a}_0^{-1/2} \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(D)} \mathbb{1}_{\mathcal{G}_{n+2, z}(1, s/8)} \leq C3^{-n} \|u - (u)_U\|_{\underline{L}^2(U)} + C|(\nabla h)_U|_2 \mathbb{1}_{\{\overline{U} \cap \partial \square_m \neq \emptyset\}} \\ + Cs^{-6} \mathbf{a}_0^{-1} 3^{sn} [\mathbf{g}]_{\underline{H}^s(U)} + Cs^{-2} 3^{sn} [\nabla h]_{\underline{H}^s(U)} \mathbb{1}_{\{\overline{U} \cap \partial \square_m \neq \emptyset\}}.$$

Insert (6.39) in (6.38). The forcing and trace powers are respectively $s^{-3/2}s^{-6} = s^{-15/2}$ and $s^{-3/2}s^{-2} = s^{-7/2}$. Their remaining error factors are bounded by C on the event; the direct forcing term s^{-6} is bounded by $s^{-15/2}$. The mean slope term $Cs^{-3/2} 3^n \mathcal{E}|(\nabla h)_U|_2$ is bounded by the mean slope term in (6.35), which allows an additional $s^{-3/2}$ inside the parentheses. This proves the unchanged conclusion for every admissible y . $\square$

The next lemma adapts Armstrong et al. [2026, Lemma 4.11].

Lemma 6.10 (Excess decay on good scales). *There exists a constant $C(d) < \infty$ such that, for every $s \in [512\overline{\delta}^2, 1/4]$ and $\varepsilon \in [8s^{-1}\overline{\delta}^2, 1]$, every $k \in \mathbb{N}$ and $L, m, n \in \mathbb{N}_0$ with $k \leq n$ and $L \geq m \geq n+5$, every $x, z \in \square_m$ such that $x \in U_{m, n-3}(z)$, and every $u \in H^1(\square_m)$, $\mathbf{g} \in H^s(\square_m; \mathbb{R}^d)$, and $h \in C^{1, 1/2}(\square_m)$ satisfying*

$$\begin{cases} -\nabla \cdot \mathbf{a}_L \nabla u = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ u = h & \text{on } \partial \square_m, \end{cases}$$

we have the estimate

$$(6.40) \quad E(u, U_{m, n-k}(x)) \mathbb{1}_{\mathcal{G}_{n+2, z}(\varepsilon, s/8)} \\ \leq C(3^{-\frac{1}{2}k} + 3^{(1+\frac{d}{2})k} s^{-3/2} \varepsilon) E(u, U_{m, n}(x)) \\ + C3^{(1+\frac{d}{2})k} s^{-3/2} \mathcal{E}_{s/8, \infty, 2}(z + \square_{n+2}; \mathbf{a}_L, (\mathbf{b}_{L, n+2})_{z+\square_{n+2}}) \left(|\nabla \ell(u, U_{m, n}(x))| \right. \\ \left. + s^{-3/2} |(\nabla h)_{U_{m, n}(x)}| \mathbb{1}_{\{\overline{U_{m, n}(x)} \cap \partial \square_m \neq \emptyset\}} \right) \\ + Cs^{-15/2} 3^{(1+\frac{d}{2})k} (\mathbf{b}_{L, n+2})_{z+\square_{n+2}}^{-1} 3^{sn} [\mathbf{g}]_{\underline{H}^s(U_{m, n}(x))} \\ + Cs^{-3} 3^{(1+\frac{d}{2})k} 3^{\frac{1}{2}n} [\nabla h]_{W^{1/2, \infty}(U_{m, n}(x))} \mathbb{1}_{\{\overline{U_{m, n}(x)} \cap \partial \square_m \neq \emptyset\}}.$$

Proof. We compare u with the harmonic function that has the same boundary values on a cube of side 3^{n-2} near x . Step 1 bounds the difference by Lemma 6.9, Step 2 records the excess decay of the harmonic function, and Step 3 transfers this decay to u . The heart of the matter is Step 3: on the good event the coarse-graining error is at most $C\varepsilon$, so that the error of the comparison contributes $C3^{(1+\frac{d}{2})k}s^{-3/2}\varepsilon E(u, U_{m,n}(x))$, plus terms involving the slope of u and the data.

Fix $x, z \in \square_m$ as in the statement. We may assume $k \geq 6$, since the case $k < 6$ follows by increasing $C(d)$. Choose $y \in \square_m$ so that $U_{m,n-4}(x) \subseteq y + \square_{n-2} \subseteq U_{m,n-1}(x)$.

Step 1. We approximate u by a harmonic function. Let v be the harmonic function in $y + \square_{n-2}$ with boundary data u :

$$\begin{cases} -\Delta v = 0 & \text{in } y + \square_{n-2}, \\ v = u & \text{on } \partial(y + \square_{n-2}). \end{cases}$$

In particular, the centered energy estimate used to prove the comparison is Lemma 6.8. It is valid at tolerance 1 for the same arbitrary comparison cube. The factors from the comparison are exactly $s^{-3/2}$, $s^{-15/2}$ and $s^{-7/2}$, as in (6.35); the smaller event here introduces no additional boundary restriction.

Since $\varepsilon \leq 1$, the event $\mathcal{G}_{n+2,z}(\varepsilon, s/8)$ implies $\mathcal{G}_{n+2,z}(1, s/8)$. Lemma 6.9, applied in $y + \square_{n-2}$ on the good event $\mathcal{G}_{n+2,z}(1, s/8)$, therefore yields (6.35) with $\mathbb{1}_{\mathcal{G}_{n+2,z}(\varepsilon, s/8)}$ in place of $\mathbb{1}_{\mathcal{G}_{n+2,z}(1, s/8)}$ on the left side.

Step 2. We use the excess decay of harmonic functions. The function v is harmonic in $U_{m,n-4}(x)$. If $\overline{U_{m,n-4}(x)} \cap \partial \square_m \neq \emptyset$, then moreover $v = u = h$ on $\partial(y + \square_{n-2}) \cap \partial \square_m$, with $h \in C^{1,1/2}$. The Schauder estimates for harmonic functions, combined with odd reflection across the flat parts of the boundary, therefore give $v \in C^{1,1/2}(U_{m,n-5}(x))$ with the estimate, where $[\cdot]_{C^{0,1/2}(U)}$ denotes the Hölder seminorm of order $1/2$ on U ,

$$[\nabla v]_{C^{0,1/2}(U_{m,n-5}(x))} \leq C3^{-\frac{1}{2}n}E(v, U_{m,n-4}(x)) + C[\nabla h]_{C^{0,1/2}(U_{m,n-4}(x))} \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}}.$$

It follows that, for every $k \geq 6$,

$$E(v, U_{m,n-k}(x)) \leq C3^{-\frac{1}{2}k}E(v, U_{m,n-4}(x)) + C3^{\frac{1}{2}(n-k)}[\nabla h]_{C^{0,1/2}(U_{m,n-2}(x))} \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}}.$$

Step 3. We conclude. By the definition of the excess (6.1) and the triangle inequality, for $k \geq 6$,

$$\begin{aligned} E(u, U_{m,n-k}(x)) &\leq E(v, U_{m,n-k}(x)) + 3^{-(n-k)}\|u - v\|_{\underline{L}^2(U_{m,n-k}(x))} \\ &\leq C3^{-\frac{1}{2}k}E(v, U_{m,n-4}(x)) + 3^{-(n-k)}\|u - v\|_{\underline{L}^2(U_{m,n-k}(x))} \\ &\quad + C3^{\frac{1}{2}(n-k)}[\nabla h]_{C^{0,1/2}(U_{m,n}(x))} \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}}. \end{aligned}$$

For the $\|u - v\|$ term, since $U_{m,n-k}(x) \subseteq U_{m,n-4}(x)$ and $|U_{m,n}(x)|/|U_{m,n-k}(x)| \leq C3^{dk}$, we have

$$3^{-(n-k)}\|u - v\|_{\underline{L}^2(U_{m,n-k}(x))} \leq C3^{(\frac{d}{2}+1)k}3^{-n}\|u - v\|_{\underline{L}^2(U_{m,n-4}(x))}.$$

The excess of v also satisfies $E(v, U_{m,n-4}(x)) \leq CE(u, U_{m,n}(x)) + C3^{-n}\|u - v\|_{\underline{L}^2(U_{m,n-4}(x))}$. Thus

$$\begin{aligned} E(u, U_{m,n-k}(x)) &\leq C3^{-\frac{1}{2}k}E(u, U_{m,n}(x)) + C3^{(\frac{d}{2}+1)k}3^{-n}\|u - v\|_{\underline{L}^2(U_{m,n-4}(x))} \\ &\quad + C3^{\frac{1}{2}(n-k)}[\nabla h]_{C^{0,1/2}(U_{m,n}(x))} \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}}. \end{aligned}$$

To estimate the middle term, we use the bound from Step 1 together with the inequality

$$3^{-n}\|u - (u)_{U_{m,n}(x)}\|_{\underline{L}^2(U_{m,n}(x))} \leq E(u, U_{m,n}(x)) + C|\nabla \ell(u, U_{m,n}(x))|.$$

We multiply the resulting inequality by $\mathbb{1}_{\mathcal{G}_{n+2,z}(\varepsilon, s/8)}$. By (6.14) and the definition of $\mathcal{G}_{n+2,z}(\varepsilon, s/8)$, we have $\mathcal{E}_{s/8, \infty, 2}(z + \square_{n+2}; \mathbf{a}_L, (\mathbf{b}_L, n+2)z + \square_{n+2})\mathbb{1}_{\mathcal{G}_{n+2,z}(\varepsilon, s/8)} \leq C\varepsilon$, so the part of the bound on $\|u - v\|$

involving $E(u, U_{m,n}(x))$ contributes $C3^{(1+\frac{d}{2})k}s^{-3/2}\varepsilon E(u, U_{m,n}(x))$, the second term in the coefficient of $E(u, U_{m,n}(x))$ in (6.40). In the part involving $|\nabla\ell(u, U_{m,n}(x))|$, we keep the factor $\mathcal{E}_{s/8,\infty,2}(z + \square_{n+2}; \mathbf{a}_L, (\mathbf{b}_{L,n+2})_{z+\square_{n+2}})$ instead of replacing it by ε ; this produces the second and third lines of (6.40). Finally, the averaged seminorm satisfies

$$3^{sn}[\nabla h]_{\underline{H}^s(U_{m,n}(x))} \leq Cs^{1/2}3^{\frac{1}{2}n}[\nabla h]_{W^{1/2,\infty}(U_{m,n}(x))}.$$

The forcing and boundary terms from Step 1 therefore give the last two lines of (6.40). $\square$

The following lemma iterates the excess decay over a range of scales. For the sets U_j used in it, the scale-normalized excess and a best affine approximation are

$$E(u, U_j) := 3^{-j} \min_{\ell \text{ affine}} \|u - \ell\|_{\underline{L}^2(U_j)}, \quad \ell(u, U_j) \in \operatorname{argmin}_{\ell \text{ affine}} \|u - \ell\|_{\underline{L}^2(U_j)}.$$

The cube contained in U_j makes the quadratic form on affine functions nondegenerate, so the minimizer exists and is unique. The minimizing constant satisfies $(\ell(u, U_j))_{U_j} = (u)_{U_j}$.

Lemma 6.11. *Let $k \in \mathbb{N}$ and $\theta \in (0, 1)$ with $\theta^k \in (0, 3/5)$. Let $n, m \in \mathbb{Z}$ with $n < m$. Let $\{U_j\}_{j \in \mathbb{Z}}$ be bounded measurable sets such that, for every $j \leq m$, we have $U_{j-1} \subseteq U_j$ and U_j contains a translate of $\square_{j-2}$ and is contained in a translate of $\square_j$. Let $u \in L^2(U_m)$. Assume that $\mathcal{B} \subseteq \mathbb{Z} \cap [n, m]$ and that $\{\varepsilon_j\}_{j \in \mathbb{Z} \cap [n, m]}$ and $\{\delta_j\}_{j \in \mathbb{Z} \cap [n, m]}$ are sequences of nonnegative numbers such that*

$$(6.41) \quad E(u, U_{j-k}) \leq \theta^k E(u, U_j) + \varepsilon_j |\nabla \ell(u, U_j)| + \delta_j, \quad \forall j \in \mathbb{Z} \cap [n, m] \setminus \mathcal{B}.$$

Then there exists a constant $C(d) < \infty$ such that, with $A := C(k+1)(|\mathcal{B}|+1) + C \sum_{j=n}^m \varepsilon_j$, we have

$$(6.42) \quad 3^{-n} \|u - (u)_{U_n}\|_{\underline{L}^2(U_n)} \leq e^A \left(3^{-m} \|u - (u)_{U_m}\|_{\underline{L}^2(U_m)} + \sum_{j=n}^m \delta_j \right).$$

Moreover, for every $M \geq 0$ such that $\varepsilon_j \leq M$ for every $j \in [n, m] \cap \mathbb{Z}$,

$$(6.43) \quad E(u, U_n) \leq \theta^{-C(k+1)(|\mathcal{B}|+1)} e^A \left(\theta^{m-n} E(u, U_m) + M 3^{-m} \|u - (u)_{U_m}\|_{\underline{L}^2(U_m)} + (1+M) \sum_{j=n}^m \delta_j \right).$$

Proof. Write $E_j := E(u, U_j)$ and $\ell_j := \ell(u, U_j)$. The inclusions assumed for U_{j-1} and U_j , together with the definition of ℓ_j , give, for every $j \leq m$,

$$(6.44) \quad |\nabla \ell_j - \nabla \ell_{j-1}| \leq C(E_j + E_{j-1}).$$

Let $[a, b] \cap \mathbb{Z}$ be a maximal integer interval contained in $([n, m] \cap \mathbb{Z}) \setminus \mathcal{B}$, and partition it into residue classes modulo k . Summing (6.41) in each class and using $\theta^k < 3/5$ gives, for $i \in [a, b] \cap \mathbb{Z}$,

$$(6.45) \quad \sum_{j=i}^b E_j \leq C 3^{(3+\frac{3d}{2})k} E_b + C \sum_{j=i+k}^b \varepsilon_j |\nabla \ell_j| + C \sum_{j=i+k}^b \delta_j.$$

To obtain (6.45), apply (6.41) at $j+k$ for $i \leq j \leq b-k$. The remaining at most k terms satisfy $E_{b-r} \leq C 3^{(1+d/2)r} E_b$ by nesting and the cube sandwich. Since $1 - \theta^k > 2/5$, the excess sum on the right can be absorbed. The sums are empty when $i+k > b$.

Put $Y_r = E_r + |\nabla \ell_r|$. For a fixed $i \in [a, b]$, telescoping (6.44) and using the shifted sum at every $r \in [i, b]$ give

$$Y_r \leq B_i + C \sum_{j=r+1}^b \varepsilon_j Y_j, \quad B_i = C 3^{Ck} \left(E_b + |\nabla \ell_b| + \sum_{j=i}^b \delta_j \right).$$

There is no $\varepsilon_r Y_r$ term, because $k \geq 1$. Backward induction therefore gives the product form of the discrete Gronwall inequality,

$$Y_i \leq B_i \prod_{j=i+1}^b (1 + C\varepsilon_j) \leq B_i \exp\left(C \sum_{j=i+1}^b \varepsilon_j\right).$$

Only nonnegativity of ε_j is used. In particular,

$$(6.46) \quad E_i + |\nabla \ell_i| \leq \exp\left(C(k+1) + C \sum_{j=i}^b \varepsilon_j\right) \left(E_b + |\nabla \ell_b| + \sum_{j=i}^b \delta_j\right).$$

For every $j \leq m$, inequality (6.44) and the definition of the excess imply $E_{j-1} + |\nabla \ell_{j-1}| \leq C(E_j + |\nabla \ell_j|)$. There are at most $|\mathcal{B}| + 1$ maximal integer intervals contained in $([n, m] \cap \mathbb{Z}) \setminus \mathcal{B}$, and each interval has at most k incomplete residue classes. Iterating (6.46) over these intervals and across the indices in $\mathcal{B}$ yields

$$(6.47) \quad E_n + |\nabla \ell_n| \leq e^A \left(E_m + |\nabla \ell_m| + \sum_{j=n}^m \delta_j\right).$$

The definition of the best affine approximation gives

$$E_m + |\nabla \ell_m| \leq C3^{-m} \|u - (u)_{U_m}\|_{\underline{L}^2(U_m)},$$

and, conversely,

$$3^{-n} \|u - (u)_{U_n}\|_{\underline{L}^2(U_n)} \leq C(E_n + |\nabla \ell_n|).$$

Together with (6.47), these inequalities prove (6.42).

To prove (6.43), we again sum (6.41) separately over the residue classes modulo k , but now keep track of the powers of θ . For every maximal integer interval $[a, b] \cap \mathbb{Z}$ contained in $([n, m] \cap \mathbb{Z}) \setminus \mathcal{B}$ and every $i \in [a, b] \cap \mathbb{Z}$,

$$E_i \leq e^{C(k+1)} \theta^{-C(k+1)} \left(\theta^{b-i} E_b + M \max_{i \leq j \leq b} |\nabla \ell_j| + \sum_{j=i}^b \delta_j \right).$$

The gradient term contributes at most $M \max_{i \leq j \leq b} |\nabla \ell_j| \sum_{r \geq 0} \theta^{kr} \leq \frac{5}{2} M \max_{i \leq j \leq b} |\nabla \ell_j|$. At a bad index, quasi-monotonicity gives $E_{j-1} \leq C\theta^{-1}(\theta E_j)$; each incomplete residue class loses at most $\theta^{-Ck} e^{Ck}$. There are at most $|\mathcal{B}| + 1$ good intervals. Composing these inequalities therefore introduces a factor of at most $\theta^{-C(k+1)(|\mathcal{B}|+1)} e^{C(k+1)(|\mathcal{B}|+1)}$. The first conclusion, applied on every subinterval, bounds all the gradients by $e^A (3^{-m} \|u - (u)_{U_m}\|_{\underline{L}^2(U_m)} + \sum_{j=n}^m \delta_j)$, with its exponent no larger than A . Increasing the dimensional constant in A absorbs the preceding interval losses and gives (6.43). $\square$

6.3. The large-scale Hölder estimate. In this subsection we prove Proposition 6.1. Lemma 6.11 iterates the excess decay estimate (6.40) of Lemma 6.10 from scale m down to scale n , up to a factor controlled by the number of bad scales between n and m and by the sum of the coarse-graining errors on the good scales. The minimal scale $\mathcal{L}_L(\gamma, m)$ is built from the random variables M and N of Lemmas 6.12 and 6.13. For every $n \leq m - \mathcal{L}_L(\gamma, m)$, both the number of bad scales and the sum of the coarse-graining errors are at most C plus a small multiple of $(1 - \gamma)(m - n)$, and hence the factor is at most $C3^{\frac{1}{4}(1-\gamma)(m-n)}$.

6.3.1. *The minimal scale.* The iteration requires bounds for

$$(6.48) \quad \sum_{k=n}^m \mathcal{E}_{1/16, \infty, 2}(\square_k; \mathbf{a}_L, (\mathbf{b}_{L,k})_{\square_k}) \mathbb{1}_{\mathcal{G}_k(\varepsilon, 1/16)}$$

and for the number of bad scales between n and m , $\sum_{k=n}^m (1 - \mathbb{1}_{\mathcal{G}_k(\varepsilon, 1/16)})$. Propositions 6.3 and 6.4 control (6.48) and this number, respectively.

For $k \in \mathbb{N}_0$, $z \in \mathbb{R}^d$, and $s \in (0, 1/2]$, define

$$(6.49) \quad \begin{aligned} \mathbf{G}_{k,z}(s) := & \sup_{0 \leq j \leq k} \sup_{0 \leq l \leq j-2} 3^{-\frac{1}{2}s(k-l)} \max_{z' \in z + (3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1}))} \min \left\{ \max_{|e|=1} J(z' + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \\ & + \sup_{0 \leq j \leq k} 3^{-\frac{1}{8}s(k-j)} \|\mathbf{g}_{k,j}\|_{L^\infty(z + \square_k)} + 3^{-\frac{1}{8}sk} \|\mathbf{g}_0\|_{L^\infty(z + \square_k)} + \sum_{j=k}^{\infty} 3^k \|\nabla \mathbf{g}_j\|_{L^\infty(z + \square_k)}. \end{aligned}$$

Here and below, a supremum over an empty set of integer indices is zero.

Lemma 6.12. *Let $s \in (0, 1/2]$. There exists $C(d) < \infty$ such that, if $s \geq C\bar{\delta}^2 |\log \bar{\delta}|$, then, for every $m, n \in \mathbb{N}_0$ with $n \leq m$ and $z \in \mathbb{R}^d$,*

$$(6.50) \quad \sum_{k=n}^m \mathbf{G}_{k,z}(s) \leq C s^{-7/2} \bar{\delta} + \mathcal{O}_{\Gamma_2}(C s^{-7/2} \bar{\delta} |\log \bar{\delta}|^{1/2} (m - n + 1)^{-1/2}).$$

Moreover, for every $\lambda > 0$ satisfying $\lambda \geq C s^{-7/2} \bar{\delta} |\log \bar{\delta}|^{1/2}$ and every $m \in \mathbb{N}_0$, there exists a random variable $M(\lambda, s, m) \in \{-1, 0, \dots, m\}$ such that

$$(6.51) \quad 0 \leq m - M(\lambda, s, m) \leq 1 + \mathcal{O}_{\Gamma_1}(C s^{-7} \bar{\delta}^2 |\log \bar{\delta}| \lambda^{-2})$$

and, for every $n \in \mathbb{N}_0$ satisfying $n \leq M(\lambda, s, m)$, we have

$$(6.52) \quad \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m \mathbf{G}_{j,z}(s) \leq \lambda(m - n).$$

Proof. The proof has four steps. Step 1 bounds the sum over the scales of the terms of $\mathbf{G}_{k,z}(s)$ involving the fields $\mathbf{g}_j$. Step 2 bounds the sum of the term involving J by a sum over the annuli $\square_j \setminus \square_{j-1}$ and estimates the moments of each summand. Step 3 sums these estimates and proves (6.50). In Steps 1 and 3, the main point is the independence of the summands at different scales (in Step 3, within each residue class of scales): it makes the random part of each sum of order $(m - n + 1)^{1/2}$ rather than $m - n + 1$. Step 4 defines $M(\lambda, s, m)$ and deduces (6.51) from (6.50) by a union bound over centers and scales.

Step 1. We estimate the terms of $\mathbf{G}_{k,z}(s)$ involving the fields $\mathbf{g}_j$. By stationarity it suffices to prove (6.50) for $z = 0$. The triangle inequality gives

$$\begin{aligned}
& \sum_{k=n}^m \left(\sup_{0 \leq j \leq k} 3^{-\frac{1}{8}s(k-j)} \|\mathbf{g}_{k,j}\|_{L^\infty(\square_k)} + 3^{-\frac{1}{8}sk} \|\mathbf{g}_0\|_{L^\infty(\square_k)} \right) \\
& \leq 2 \sum_{k=n}^m \sup_{0 \leq j \leq k} 3^{-\frac{1}{8}s(k-j)} \sum_{i=j}^k \|\mathbf{g}_i\|_{L^\infty(\square_k)} \\
& \leq 2 \sum_{k=n}^m \sum_{i=0}^k \sup_{0 \leq j \leq i} 3^{-\frac{1}{8}s(k-j)} \|\mathbf{g}_i\|_{L^\infty(\square_k)} \\
& = 2 \sum_{k=n}^m \sum_{i=0}^k 3^{-\frac{1}{8}s(k-i)} \|\mathbf{g}_i\|_{L^\infty(\square_k)} \\
& = 2 \left(\sum_{i=n}^m \sum_{k=i}^m + \sum_{k=n}^m \sum_{i=0}^{n-1} \right) 3^{-\frac{1}{8}s(k-i)} \|\mathbf{g}_i\|_{L^\infty(\square_k)}.
\end{aligned}$$

By (1.4) and (1.5), together with (A.4), we obtain

$$\|\mathbf{g}_i\|_{L^\infty(\square_k)} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}(k-i+1)^{1/2})$$

and therefore, by independence, also

$$\begin{aligned}
\sum_{k=n}^m \left(\sup_{0 \leq j \leq k} 3^{-\frac{1}{8}s(k-j)} \|\mathbf{g}_{k,j}\|_{L^\infty(\square_k)} + 3^{-\frac{1}{8}sk} \|\mathbf{g}_0\|_{L^\infty(\square_k)} \right) & \leq C s^{-3/2} \bar{\delta} (m-n+1) \\
& + \mathcal{O}_{\Gamma_2}(C s^{-3/2} \bar{\delta} (m-n+1)^{1/2}).
\end{aligned}$$

For the gradient term,

$$\begin{aligned}
\sum_{k=n}^m \sum_{l=k}^{\infty} 3^k \|\nabla \mathbf{g}_l\|_{L^\infty(\square_k)} & = \sum_{l=n}^{\infty} \sum_{k=n}^{m \wedge l} 3^k \|\nabla \mathbf{g}_l\|_{L^\infty(\square_k)} \\
& \leq \frac{3}{2} \sum_{l=n}^{\infty} 3^{m \wedge l} \|\nabla \mathbf{g}_l\|_{L^\infty(\square_l)} \\
& = \frac{3}{2} \sum_{l=n}^{m-1} 3^l \|\nabla \mathbf{g}_l\|_{L^\infty(\square_l)} + \frac{3}{2} \sum_{l=m}^{\infty} 3^m \|\nabla \mathbf{g}_l\|_{L^\infty(\square_l)}.
\end{aligned}$$

Again by (1.4), (1.5), and independence,

$$\sum_{k=n}^m \sum_{l=k}^{\infty} 3^k \|\nabla \mathbf{g}_l\|_{L^\infty(\square_k)} \leq C \bar{\delta} (m-n+1) + \mathcal{O}_{\Gamma_2}(C \bar{\delta} (m-n+1)^{1/2}).$$

Thus

$$\begin{aligned}
(6.53) \quad & \sum_{k=n}^m \sum_{l=k}^{\infty} 3^k \|\nabla \mathbf{g}_l\|_{L^\infty(\square_k)} + \sum_{k=n}^m \left(\sup_{0 \leq j \leq k} 3^{-\frac{1}{8}s(k-j)} \|\mathbf{g}_{k,j}\|_{L^\infty(\square_k)} + 3^{-\frac{1}{8}sk} \|\mathbf{g}_0\|_{L^\infty(\square_k)} \right) \\
& \leq C s^{-3/2} \bar{\delta} (m-n+1) + \mathcal{O}_{\Gamma_2}(C s^{-3/2} \bar{\delta} (m-n+1)^{1/2}).
\end{aligned}$$

Step 2. We estimate the term of $\mathbf{G}_{k,z}(s)$ involving J . Bounding the supremum over j by a sum, interchanging the sums, and summing the geometric series in k , we obtain

$$\begin{aligned}
& \sum_{k=n}^m \sup_{0 \leq j \leq k} \sup_{0 \leq l \leq j-2} 3^{-\frac{1}{2}s(k-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \\
& \leq \sum_{j=0}^m \sum_{k=n \vee j}^m \sup_{0 \leq l \leq j-2} 3^{-\frac{1}{4}s(k-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \\
& \leq 4s^{-1} \sum_{j=n}^m \sup_{0 \leq l \leq j-2} 3^{-\frac{1}{4}s(j-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \\
(6.54) \quad & + 4s^{-1} \sum_{j=0}^{n-1} \sup_{0 \leq l \leq j-2} 3^{-\frac{1}{4}s(n-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2}.
\end{aligned}$$

By Proposition 4.1, for $\xi \in [1, C^{-1}\bar{\delta}^{-2}|\log \bar{\delta}|^{-1}]$,

$$\mathbb{E} \left[\max_{|e|=1} (J(\square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l))^{\xi/2} \right]^{1/\xi} \leq C\bar{\delta} \log(2 + \xi)^{1/2} \xi^{1/2}.$$

Hence, for every $\xi \in [1, \infty)$,

$$\mathbb{E} \left[\min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{\xi/2} \right]^{1/\xi} \leq C\bar{\delta} |\log \bar{\delta}|^{1/2} \xi^{1/2},$$

which yields

$$\min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta} |\log \bar{\delta}|^{1/2}).$$

Moreover, by Hölder's inequality, the condition $s \geq C\bar{\delta}^2 |\log \bar{\delta}|$, a union bound, and stationarity,

$$\begin{aligned}
& \mathbb{E} \left[\sup_{0 \leq l \leq j-2} 3^{-\frac{1}{4}s(j-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \right] \\
& \leq \mathbb{E} \left[\sup_{0 \leq l \leq j-2} 3^{-2d(j-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{4d/s} \right]^{s/8d} \\
& \leq \left(\sum_{l=0}^{j-2} 3^{-d(j-l)} \mathbb{E} \left[\max_{|e|=1} J(\square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l)^{4d/s} \right] \right)^{s/8d} \\
& \leq Cs^{-1/2} |\log s|^{1/2} \bar{\delta},
\end{aligned}$$

and, using (A.4),

$$\sup_{0 \leq l \leq j-2} 3^{-\frac{1}{4}s(j-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \leq \mathcal{O}_{\Gamma_2}(Cs^{-3/2} \bar{\delta} |\log \bar{\delta}|^{1/2}).$$

Step 3. We prove (6.50). We split the indices j into the finitely many residue classes modulo the integer $r(d)$ from the proof of Lemma 6.7; as shown there, the summands with indices in the same class are independent. In each class, we apply (A.5) to the centered summands, which are $\mathcal{O}_{\Gamma_2}(Cs^{-3/2} \bar{\delta} |\log \bar{\delta}|^{1/2})$ by the last display, and combining the result with the preceding estimates we obtain

$$\begin{aligned}
& \sum_{j=n}^m \sup_{0 \leq l \leq j-2} 3^{-\frac{1}{4}s(j-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \\
& \leq Cs^{-1/2} |\log s|^{1/2} \bar{\delta} (m - n + 1) + \mathcal{O}_{\Gamma_2}(Cs^{-3/2} \bar{\delta} |\log \bar{\delta}|^{1/2} (m - n + 1)^{1/2}).
\end{aligned}$$

Applying the last display before Step 3 to the annuli with $j < n$ and summing the resulting geometric series gives

$$\sum_{j=0}^{n-1} \sup_{0 \leq l \leq j-2} 3^{-\frac{1}{4}s(n-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \leq \mathcal{O}_{\Gamma_2}(Cs^{-5/2} \bar{\delta} |\log \bar{\delta}|^{1/2}).$$

The last two displays, together with (6.54) and (6.53), give (6.50).

Step 4. We define $M(\lambda, s, m)$ and prove (6.51). Define

$$M(\lambda, s, m) := \min \left(\left\{ n \in \mathbb{N}_0 : n \leq m-1, \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m \mathbf{G}_{j,z}(s) > \lambda(m-n) \right\} \cup \{m\} \right) - 1.$$

The averaged estimate (6.50), stationarity, and the sub-Gaussian tail in that estimate show that, for every $0 \leq k \leq m-1$,

$$\mathbb{P} \left[\max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \sum_{j=k}^m \mathbf{G}_{j,z}(s) > \lambda(m-k) \right] \leq \exp \left(C(m-k) - \frac{s^7 \lambda^2 (m-k)}{C \bar{\delta}^2 |\log \bar{\delta}|} \right).$$

Indeed, there are at most $C3^{d(m-k)}$ centers, and we apply (6.50), translated, at each of them. The same bound is valid for $k = m$ because its right side is one. Consequently, a union bound gives, for every $\lambda \geq 2Cs^{-7/2} \bar{\delta} |\log \bar{\delta}|^{1/2}$ and $0 \leq n \leq m-1$,

$$\begin{aligned} \mathbb{P}[M(\lambda, s, m) \leq n] &\leq \sum_{k=0}^{n+1} \mathbb{P} \left[\max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \sum_{j=k}^m \mathbf{G}_{j,z}(s) > \lambda(m-k) \right] \\ (6.55) \quad &\leq \sum_{k=0}^{n+1} \exp \left(C(m-k) - \frac{s^7 \lambda^2 (m-k)}{C \bar{\delta}^2 |\log \bar{\delta}|} \right) \leq C \exp \left(- \frac{s^7 \lambda^2 (m-n-1)}{C \bar{\delta}^2 |\log \bar{\delta}|} \right). \end{aligned}$$

This implies (6.51) and completes the proof. $\square$

Lemma 6.13. *There exists $C(d) < \infty$ such that, for every $\lambda, \varepsilon, s \in (0, 1]$ satisfying*

$$(6.56) \quad \lambda \geq Cs^{-8} \varepsilon^{-2} \bar{\delta}^2 |\log \bar{\delta}|$$

and for every $m \in \mathbb{N}_0$, there exists a random variable $N(\lambda, \varepsilon, s, m) \in \{-1, 0, \dots, m\}$ such that

$$(6.57) \quad 0 \leq m - N(\lambda, \varepsilon, s, m) \leq 1 + \mathcal{O}_{\Gamma_1}(Cs^{-8} \varepsilon^{-2} \lambda^{-1} \bar{\delta}^2 |\log \bar{\delta}|)$$

and, for every $n \in \mathbb{N}_0$ with $n \leq N(\lambda, \varepsilon, s, m)$,

$$(6.58) \quad \min_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m \mathbb{1}\{\mathcal{G}_{j,z}(\varepsilon, s)\} > (1 - \lambda)(m - n).$$

Equivalently, every such n satisfies

$$\max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m (1 - \mathbb{1}\{\mathcal{G}_{j,z}(\varepsilon, s)\}) < 1 + \lambda(m - n).$$

Proof. Define the random variable $N(\lambda, \varepsilon, s, m)$ by

$$N(\lambda, \varepsilon, s, m) := \min \left(\left\{ n \in \mathbb{N}_0 : n \leq m, \quad \min_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m \mathbb{1}\{\mathcal{G}_{j,z}(\varepsilon, s)\} \leq (1-\lambda)(m-n) \right\} \cup \{m+1\} \right) - 1.$$

The definition of $N(\lambda, \varepsilon, s, m)$ gives (6.58). On the event in the next display, the proportion of bad scales among the $m-n+1$ scales from n to m is at least $\lambda/2$. Proposition 6.4, applied with $\lambda/2$ in place of θ , therefore gives

$$\mathbb{P} \left[\sum_{j=n}^m (1 - \mathbb{1}_{\mathcal{G}_{j,z}(\varepsilon, s)}) > \lambda(m-n) \right] \leq \exp \left(- \frac{s^8 \varepsilon^2 \lambda(m-n+1)}{C \bar{\delta}^2 |\log \bar{\delta}|} \right).$$

By a union bound, stationarity, and the condition (6.56) with sufficiently large C , we obtain

$$\begin{aligned} \mathbb{P} \left[\max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m (1 - \mathbb{1}_{\mathcal{G}_{j,z}(\varepsilon, s)}) > \lambda(m-n) \right] &\leq 3^{d(m-n)} \mathbb{P} \left[\sum_{j=n}^m (1 - \mathbb{1}_{\mathcal{G}_j(\varepsilon, s)}) > \lambda(m-n) \right] \\ &\leq \exp \left(2d(m-n) - \frac{s^8 \varepsilon^2 \lambda(m-n+1)}{C \bar{\delta}^2 |\log \bar{\delta}|} \right) \\ &\leq \exp \left(- \frac{s^8 \varepsilon^2 \lambda(m-n+1)}{C \bar{\delta}^2 |\log \bar{\delta}|} \right). \end{aligned}$$

By another union bound, for $0 \leq n \leq m-1$,

$$\begin{aligned} \mathbb{P}[N(\lambda, \varepsilon, s, m) \leq n] &\leq \sum_{k=0}^{n+1} \mathbb{P} \left[\min_{z \in 3^k \mathbb{Z}^d \cap \square_m} \sum_{j=k}^m \mathbb{1}_{\mathcal{G}_{j,z}(\varepsilon, s)} \leq (1-\lambda)(m-k) \right] \\ &\leq C \exp \left(- \frac{s^8 \varepsilon^2 \lambda(m-n)}{C \bar{\delta}^2 |\log \bar{\delta}|} \right). \end{aligned}$$

This implies (6.57), completing the proof. $\square$

Proof of Proposition 6.1. The proof has seven steps. Step 1 fixes the parameters, and Step 2 defines the minimal scale $\mathcal{L}_L(\gamma, m)$, proves (6.3), and records the bounds (6.61) for $n \leq m - \mathcal{L}_L(\gamma, m)$. Step 3 turns Lemma 6.10 into the one-step inequality (6.67) on the good scales, and Step 4 iterates it by Lemma 6.11. Steps 5, 6 and 7 deduce (6.5), (6.6) and (6.7), respectively. The heart of the matter is Step 4: above the minimal scale, the number of bad scales and the sum of the errors $\varepsilon_{j,z}$ are at most $C + C\lambda(m-n)$, so the factor produced by the iteration is at most $C \exp(C\lambda(m-n))$; the choice of λ in Step 1 makes this at most $C 3^{\frac{1}{4}(1-\gamma)(m-n)}$.

Step 1. We fix the parameters. Fix γ in the range allowed by the statement. Choose the dimensional constants in the order indicated below and set

$$(6.59) \quad s_0 := 1/32, \quad \lambda := C_1^{-1}(1-\gamma), \quad \varepsilon := C_2^{-1}(1-\gamma)^{1/2}, \quad k := \left\lceil 4 \log_3(4s_0^{-3/2} C_3) \right\rceil,$$

where $C_3(d)$ is the constant of Lemma 6.10. Choose $C_1(d) \geq 2$ sufficiently large for both (6.76) below and the absorption in the proof of (6.6) in Step 6, and then choose $C_2(d)$ so that $C_1 C_2^{-8} \leq 1$ and

$$C_3 \left(3^{-k/2} + 3^{(1+\frac{d}{2})k} s_0^{-3/2} C_2^{-1} \right) \leq \frac{1}{2} 3^{-k/4}.$$

Finally, choose the constant C of the statement large enough, depending on C_1 and C_2 . The lower bound on $1-\gamma$ then ensures, for all sufficiently small $\bar{\delta}$, that the hypotheses of Lemma 6.10, Proposition 6.3, Lemma 6.12, and Lemma 6.13 hold.

Step 2. We define the minimal scale. For $m \in \mathbb{N}_0$, define

$$(6.60) \quad \mathcal{L}_L(\gamma, m) := \max\{m - M(\lambda, s_0, m), m - N(\lambda, \varepsilon, s_0, m)\} + k + 5,$$

where M and N are supplied by Lemmas 6.12 and 6.13. They do not involve L , so $\mathcal{L}_L(\gamma, m)$ is the same random variable for every $L \geq m$. The summand $k + 5$ ensures that every scale used in Lemma 6.10 is at most $m - 5$. Estimates (6.51) and (6.57) yield

$$\begin{aligned} \mathcal{L}_L(\gamma, m) &\leq C + \mathcal{O}_{\Gamma_1}(Cs_0^{-7}\bar{\delta}^2|\log \bar{\delta}|\lambda^{-2} + Cs_0^{-8}\varepsilon^{-2}\lambda^{-1}\bar{\delta}^2|\log \bar{\delta}|) \\ &\leq C + \mathcal{O}_{\Gamma_1}(CC_1^2(1 - \gamma)^{-2}\bar{\delta}^2|\log \bar{\delta}|). \end{aligned}$$

This is (6.3). Finally, (6.52) and (6.58) imply, whenever $n \leq m - \mathcal{L}_L(\gamma, m)$,

$$(6.61) \quad \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m \mathbf{G}_{j,z}(s_0) \leq \lambda(m-n) \quad \text{and} \quad \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m (1 - \mathbb{1}\{\mathcal{G}_{j,z}(\varepsilon, s_0)\}) < 1 + \lambda(m-n).$$

Step 3. We prove the excess decay estimate (6.40) on good scales. Fix $m \in \mathbb{N}_0$ with $L \geq m$, and let $u \in H^1(\square_m)$ be a solution of (6.4). Fix $n \in \mathbb{N}_0$ with $n \leq m - \mathcal{L}_L(\gamma, m)$ and $z \in 3^n \mathbb{Z}^d \cap \square_m$. We apply Lemma 6.10 with $s = 8s_0 = 1/4$ at the scales $j \in \mathbb{N}_0 \cap [n + k, m - 5]$, for which its hypotheses hold. For each such j , the conclusion (6.40), with j in place of n and with $x = z$, gives

$$\begin{aligned} (6.62) \quad &E(u, U_{m,j-k}(z)) \mathbb{1}_{\mathcal{G}_{j+2,z}(\varepsilon, s_0)} \\ &\leq C_3(3^{-\frac{1}{2}k} + 3^{(1+\frac{d}{2})k}s_0^{-3/2}\varepsilon)E(u, U_{m,j}(z)) \\ &\quad + C_3 3^{(1+\frac{d}{2})k}s_0^{-3/2}\mathcal{E}_{s_0,\infty,2}(z + \square_{j+2}; \mathbf{a}_L, (\mathbf{b}_{L,j+2})_{z+\square_{j+2}}) \left(|\nabla \ell(u, U_{m,j}(z))| \right. \\ &\quad \left. + s_0^{-3/2}|(\nabla h)_{U_{m,j}(z)}| \mathbb{1}_{\{\overline{U_{m,j}(z)} \cap \partial \square_m \neq \emptyset\}} \right) \mathbb{1}_{\mathcal{G}_{j+2,z}(\varepsilon, s_0)} \\ &\quad + Cs_0^{-15/2} 3^{(1+\frac{d}{2})k}(\mathbf{b}_{L,j+2})_{z+\square_{j+2}}^{-1} 3^{8s_0j}[\mathbf{g}]_{\underline{H}^{8s_0}(U_{m,j}(z))} \\ &\quad + Cs_0^{-3} 3^{(1+\frac{d}{2})k} 3^{\frac{1}{2}j}[\nabla h]_{W^{1/2,\infty}(U_{m,j}(z))} \mathbb{1}_{\{\overline{U_{m,j}(z)} \cap \partial \square_m \neq \emptyset\}}. \end{aligned}$$

By our choice of k and C_2 , we have $C_3(3^{-k/2} + 3^{(1+\frac{d}{2})k}s_0^{-3/2}\varepsilon) \leq \frac{1}{2}3^{-k/4}$. Definition 4.3 and the fact that an average lies between the infimum and the supremum give

$$\begin{aligned} &\max\left\{ \frac{(\mathbf{b}_{L,j+2})_{z+\square_{j+2}}}{(\mathbf{b}_{L,m})_{\square_m}}, \frac{(\mathbf{b}_{L,m})_{\square_m}}{(\mathbf{b}_{L,j+2})_{z+\square_{j+2}}} \right\} \\ &\leq \max\left\{ \frac{\bar{\mathbf{a}}_{j+2}}{\bar{\mathbf{a}}_m}, \frac{\bar{\mathbf{a}}_m}{\bar{\mathbf{a}}_{j+2}} \right\} \exp\left(\sum_{i=j+3}^m \|\mathbf{g}_i - \tau^2\|_{L^\infty(z+\square_{j+2})} + \text{osc}_{\square_m \cup (z+\square_{j+2})} \sum_{i=m+1}^L \mathbf{g}_i \right). \end{aligned}$$

Equations (6.11) and (6.49) give

$$\sum_{i=j+3}^m \|\mathbf{g}_i - \tau^2\|_{L^\infty(z+\square_{j+2})} \leq \tau^2(m - j - 2) + C \sum_{i=j+2}^m \mathbf{G}_{i,z}(s_0).$$

By the mean-value theorem and (6.49),

$$\text{osc}_{\square_m \cup (z+\square_{j+2})} \sum_{i=m+1}^L \mathbf{g}_i \leq C \sum_{i=m+1}^\infty 3^m (\|\nabla \mathbf{g}_i\|_{L^\infty(z+\square_m)} + \|\nabla \mathbf{g}_i\|_{L^\infty(\square_m)}) \leq C(\mathbf{G}_{m,z}(s_0) + \mathbf{G}_{m,0}(s_0)).$$

Applying (6.61) at the centers z and 0 , we obtain

$$\sum_{i=j+2}^m \mathbf{G}_{i,z}(s_0) + \mathbf{G}_{m,0}(s_0) \leq C\lambda(m - n).$$

The parameter choice in Step 1 gives $\tau^2 \leq C\bar{\delta}^2 \leq C\lambda$. Together with (3.24), the preceding estimates show that, for every $j \in \mathbb{N}_0 \cap [n, m-5]$,

$$(6.63) \quad C^{-1} \exp(-C\lambda(m-n)) \leq (\mathbf{b}_{L,j+2})_{z+\square_{j+2}} (\mathbf{b}_{L,m})_{\square_m}^{-1} \leq C \exp(C\lambda(m-n)).$$

Moreover,

$$3^{8s_0j} [\mathbf{g}]_{\underline{H}^{8s_0}(U_{m,j}(z))} \leq C 3^{\frac{1}{2}j} [\mathbf{g}]_{W^{1/2,\infty}(U_{m,j}(z))} \leq C 3^{-\frac{1}{2}(m-j)} (3^{\frac{1}{2}m} [\mathbf{g}]_{W^{1/2,\infty}(\square_m)}),$$

so, using (6.63),

$$(6.64) \quad (\mathbf{b}_{L,j+2})_{z+\square_{j+2}}^{-1} 3^{8s_0j} [\mathbf{g}]_{\underline{H}^{8s_0}(U_{m,j}(z))} \leq C \exp(C\lambda(m-n)) 3^{-\frac{1}{2}(m-j)} ((\mathbf{b}_{L,m})_{\square_m}^{-1} 3^{\frac{1}{2}m} [\mathbf{g}]_{W^{1/2,\infty}(\square_m)}).$$

The boundary datum satisfies

$$(6.65) \quad 3^{\frac{1}{2}j} [\nabla h]_{W^{1/2,\infty}(U_{m,j}(z))} \leq C 3^{-\frac{1}{2}(m-j)} (3^{\frac{1}{2}m} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)}).$$

Furthermore, we use the following elementary inequality: if $D \geq 0$ and $0 \leq x \leq \varepsilon 3^{s_0 D/16}$, then $3^{-s_0 D/2} x \leq 3^{-s_0 D/2} \min\{x, 1\} + \varepsilon^8$. Indeed, the second term is needed only when $x > 1$, in which case $3^{-s_0 D/16} < \varepsilon$. By (6.10), on $\mathcal{G}_{j+2,z}(\varepsilon, s_0)$ it applies to the square root x of every quantity J on the right side of (6.14), with D the difference between the scale $j+2$ and the scale of that quantity. Proposition 6.3 and (6.49) therefore give

$$(6.66) \quad \mathcal{E}_{s_0,\infty,2}(z + \square_{j+2}; \mathbf{a}_L, (\mathbf{b}_{L,j+2})_{z+\square_{j+2}}) \mathbb{1}_{\mathcal{G}_{j+2,z}(\varepsilon, s_0)} \leq C \min\{\varepsilon, s_0^{-1} \bar{\delta}^2 + \varepsilon^8 + \mathbf{G}_{j+2,z}(s_0)\}.$$

Inserting these bounds, including (6.64), (6.65) and (6.66), into (6.62) yields, for every $z \in 3^n \mathbb{Z}^d \cap \square_m$ and $j \in \mathbb{N}_0 \cap [n+k, m-5]$,

$$(6.67) \quad E(u, U_{m,j-k}(z)) \mathbb{1}_{\mathcal{G}_{j+2,z}(\varepsilon, s_0)} \leq 3^{-k/4} E(u, U_{m,j}(z)) + \varepsilon_{j,z} |\nabla \ell(u, U_{m,j}(z))| + \delta_{j,z},$$

where we define

$$(6.68) \quad \varepsilon_{j,z} := C s_0^{-3/2} \min\{\varepsilon, \bar{\delta}^2 + \varepsilon^8 + \mathbf{G}_{j+2,z}(s_0)\} \mathbb{1}_{\mathcal{G}_{j+2,z}(\varepsilon, s_0)}$$

and

$$\begin{aligned} \delta_{j,z} &:= C 3^{-\frac{1}{2}(m-j)} \exp(C\lambda(m-n)) (\mathbf{b}_{L,m})_{\square_m}^{-1} 3^{\frac{1}{2}m} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \\ &\quad + C \left(\varepsilon_{j,z} \|\nabla h\|_{L^\infty(\square_m)} + 3^{-\frac{1}{2}(m-j)} 3^{\frac{1}{2}m} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)} \right) \mathbb{1}_{\{\overline{U_{m,j}(z)} \cap \partial \square_m \neq \emptyset\}}. \end{aligned}$$

The geometric factor $3^{-\frac{1}{2}(m-j)}$ in the last term comes from (6.65). Since the sets $U_{m,j}(z)$ are nested, for $j \leq m'$,

$$\mathbb{1}_{\{\overline{U_{m,j}(z)} \cap \partial \square_m \neq \emptyset\}} \leq \mathbb{1}_{\{\overline{U_{m,m'}(z)} \cap \partial \square_m \neq \emptyset\}}.$$

By (6.61), we have

$$(6.69) \quad \sum_{j=n}^{m-2} \varepsilon_{j,z} \leq C s_0^{-3/2} \left((\bar{\delta}^2 + \varepsilon^8)(m-n) + \sum_{j=n}^{m-2} \mathbf{G}_{j+2,z}(s_0) \right) \leq C\lambda(m-n).$$

Therefore, we also obtain, for every $m', n' \in \mathbb{N}_0$ with $n \leq n' \leq m' \leq m-3$,

$$(6.70) \quad \begin{aligned} \sum_{j=n'}^{m'} \delta_{j,z} &\leq C \exp(C\lambda(m-n)) (\mathbf{b}_{L,m})_{\square_m}^{-1} 3^{\frac{1}{2}m'} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \\ &\quad + C \left(\lambda(m-n) \|\nabla h\|_{L^\infty(\square_m)} + 3^{\frac{1}{2}m'} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)} \right) \mathbb{1}_{\{\overline{U_{m,m'}(z)} \cap \partial \square_m \neq \emptyset\}}. \end{aligned}$$

Step 4. We set up the iteration. The inequality (6.67) is the key hypothesis (6.41) of Lemma 6.11 (with step k and $\theta = 3^{-1/4}$). For $z \in 3^n \mathbb{Z}^d \cap \square_m$, we define the set $\mathcal{B}$ of indices by

$$(6.71) \quad \mathcal{B} := \{j \in \mathbb{N}_0 \cap [n, m-2] : j < n+k \text{ or } \mathcal{G}_{j+2,z}(\varepsilon, s_0) \text{ fails}\}.$$

By (6.61) and the definition (6.71),

$$(6.72) \quad |\mathcal{B}| \leq k + \sum_{j=n+k}^{m-2} (1 - \mathbb{1}\{\mathcal{G}_{j+2,z}(\varepsilon, s_0)\}) \leq k + 1 + \lambda(m-n).$$

This and (6.69) imply, for the constant A in Lemma 6.11, that $e^A \leq C \exp(C\lambda(m-n))$. Since k is a dimensional constant, (6.72) shows that $3^{C(k+1)(|\mathcal{B}|+1)/4} e^A$ satisfies the same bound, with a larger C . We apply (6.43) with $M = C\varepsilon$, which is admissible because $\varepsilon_{j,z} \leq C\varepsilon$ by (6.68). For every $n', m' \in \mathbb{N}_0$ with $n \leq n' \leq m' \leq m-5$, Lemma 6.11 gives the following two inequalities (for $n' = m'$ they are immediate). The first inequality, from (6.42), is

$$(6.73) \quad 3^{-n'} \|u - (u)_{U_{m,n'}(z)}\|_{\underline{L}^2(U_{m,n'}(z))} \leq C \exp(C\lambda(m-n)) \left(3^{-m'} \|u - (u)_{U_{m,m'}(z)}\|_{\underline{L}^2(U_{m,m'}(z))} + \sum_{j=n'}^{m'} \delta_{j,z} \right)$$

and the second inequality, the excess decay estimate, is

$$(6.74) \quad E(u, U_{m,n'}(z)) \leq C \exp(C\lambda(m-n)) \left(3^{-m'-n'/4} E(u, U_{m,m'}(z)) + \varepsilon 3^{-m'} \|u - (u)_{U_{m,m'}(z)}\|_{\underline{L}^2(U_{m,m'}(z))} + \sum_{j=n'}^{m'} \delta_{j,z} \right).$$

Step 5. We prove the large-scale Hölder estimate (6.75). We show that, for every $n, n', m' \in \mathbb{N}_0$ with $n \leq m - \mathcal{L}_L(\gamma, m)$ and $n \leq n' \leq m' \leq m-5$ and every $z \in 3^n \mathbb{Z}^d \cap \square_m$, we have the estimate

$$(6.75) \quad \begin{aligned} & 3^{-n'} \|u - (u)_{U_{m,n'}(z)}\|_{\underline{L}^2(U_{m,n'}(z))} \\ & \leq C 3^{\frac{1}{2}(1-\gamma)(m-n)} \left(3^{-m'} \|u - (u)_{U_{m,m'}(z)}\|_{\underline{L}^2(U_{m,m'}(z))} + (\mathbf{b}_{L,m})_{\square_m}^{-1} 3^{\frac{1}{2}m} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \right) \\ & + C 3^{\frac{1}{2}(1-\gamma)(m-n)} 3^{\frac{1}{2}m} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)} \mathbb{1}_{\{\overline{U_{m,m'}(z)} \cap \partial \square_m \neq \emptyset\}}. \end{aligned}$$

By the choice of C_1 and the identity $\lambda = C_1^{-1}(1-\gamma)$,

$$(6.76) \quad (1 + \lambda(m-n)) \exp(C\lambda(m-n)) \leq C \exp(2CC_1^{-1}(1-\gamma)(m-n)) \leq C 3^{\frac{1}{4}(1-\gamma)(m-n)}.$$

Combining (6.73), (6.70), and (6.76) gives (6.75). For each $j \in \mathbb{N}_0$ with $j \leq n$ and each $y \in 3^j \mathbb{Z}^d \cap ((x + \square_n) \cap \square_m)$, we apply this estimate with j in place of n , with $n' = j$, $m' = m-5$, and $z = y$. If $x \in \square_{m-1}$, then $y \in x + \square_n$ and $n \leq m - \mathcal{L}_L(\gamma, m) \leq m - k - 5$, so $\overline{U_{m,m-5}(y)} \subseteq \square_m$ and the boundary indicator in (6.75) vanishes; hence this indicator is always bounded by $\mathbb{1}_{\{x \notin \square_{m-1}\}}$. Together with $3^{-\frac{1}{2}(1+\gamma)(m-j)} \leq 3^{-\gamma(m-j)}$, this proves (6.5).

Step 6. We prove the energy estimate (6.6). We use Lemma 6.8 at the nearest good anchor to convert the large-scale Hölder estimate (6.75) into a gradient bound. The coarse-graining estimates of Lemma 2.9 bound the oscillation of u at the larger good scale by the energy and the forcing.

Fix $x \in \square_m$. If $m-n \leq 2k+14$, then $|(x + \square_n) \cap \square_m| \geq c3^{nd} \geq c|\square_m|$, and (6.6) follows directly from the energy in $\square_m$. We may therefore assume that $m-n > 2k+14$. Then, since $\lambda \leq 1/2$, the interval $[n+2, m-5]$ contains more than $k+1 + \lambda(m-n)$ integers, so by (6.72) it contains a scale outside the set $\mathcal{B}$ of Step 4, for the center z chosen below. Let y be the coordinatewise clamp of x to the centers of the scale- $(n+1)$ cubes contained in $\square_m$; if $x \in \square_{m-1}$, then no coordinate is clamped

and $y = x$. Let $z \in 3^n \mathbb{Z}^d \cap \square_m$ be a nearest neighbor of y in $3^n \mathbb{Z}^d \cap \square_m$, chosen coordinatewise. We define the elements of $[n+2, m-5] \setminus \mathcal{B}$ nearest to n and to m by

$$n' := \min\{j \in \mathbb{N}_0 : j \in [n+2, m-5] \setminus \mathcal{B}\} \quad \text{and} \quad m' := \max\{j \in \mathbb{N}_0 : j \in [n', m-5] \setminus \mathcal{B}\},$$

respectively. Since $|\mathcal{B}| \leq k+1 + \lambda(m-n)$ by (6.72), we have $\max\{n'-n, m-m'\} \leq C + \lambda(m-n)$. As in Step 5, the boundary indicator in (6.75) vanishes if $x \in \square_{m-1}$, so it is bounded by $\mathbb{1}_{\{x \notin \square_{m-1}\}}$. We also fix $y' \in \square_m \cap (z + \square_{m'+1})$ such that $U_{m,m'}(z) \subseteq y' + \square_{m'} \subseteq \square_m$. By (2.38) and the last assertion of Proposition 6.3, for every $j \in [n, m-5] \setminus \mathcal{B}$ we have

$$(6.77) \quad C^{-1}(\mathbf{b}_{L,j+2})_{z+\square_{j+2}} \leq \lambda_{s_0,2}(z + \square_{j+2}; \mathbf{a}_L) \leq \Lambda_{s_0,2}(z + \square_{j+2}; \mathbf{a}_L) \leq C(\mathbf{b}_{L,j+2})_{z+\square_{j+2}}.$$

By Lemma 2.4, (2.42), (6.77) at $j = m'$ and (6.63), the ellipticity quantities at the larger good scale satisfy, for $q \in \{1, 2\}$,

$$(6.78) \quad \Lambda_{1/4,q}(y' + \square_{m'}; \mathbf{a}_L)(\mathbf{b}_{L,m})_{\square_m}^{-1} + \lambda_{1/8,q}^{-1}(y' + \square_{m'}; \mathbf{a}_L)(\mathbf{b}_{L,m})_{\square_m} \leq C \exp(C\lambda(m-n)).$$

Moreover,

$$\left(\frac{|U_{m,m}(z)|}{|U_{m,m'}(z)|} \right)^{1/2} \leq C 3^{d(m-m')/2} \leq C \exp(C\lambda(m-n)).$$

Put $T = (x + \square_n) \cap \square_m$ and $\mathbf{a}'_0 = (\mathbf{b}_{L,n'+2})_{z+\square_{n'+2}}$. The estimate near the boundary is applied at the available good anchor $n' + 2$. We have $n' \geq n + 2$ and $\mathcal{G}_{n'+2,z}(\varepsilon, s_0) \subseteq \mathcal{G}_{n'+2,z}(1, 1/32)$. Moreover $T \subseteq U_{m,n+1}(z)$. In a coordinate where clamping to $\square_m$ occurs, the coordinate of y is $\pm(3^m - 3^{n+1})/2$, which belongs to $3^n \mathbb{Z}$, so that $z_i = y_i$. In an unclamped coordinate, $y_i = x_i$ and $|z_i - x_i| \leq 3^n/2$. These two observations give the inclusion.

Cover T by scale- $(n' - 4)$ cells with centers in $U_{m,n'-1}(z)$. Only $C(d)$ such cells are needed, since $n' - 4 \geq n - 2$. Apply Lemma 6.8 with $(n, x, z) = (n', z, z)$ and $s = 1/4$ on each cell. The normalized volumes introduce a factor of at most $C 3^{d(n'-n)}$. If $x \in \square_{m-1}$, every boundary indicator for these windows vanishes, since $n' \leq m - 5$ and $|z - x|_\infty \leq 3^n/2$. Consequently

$$\begin{aligned} \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(T)} &\leq C 3^{d(n'-n)/2} (\mathbf{a}'_0)^{1/2} 3^{-n'} \|u - (u)_{U_{m,n'}(z)}\|_{\underline{L}^2(U_{m,n'}(z))} \\ &\quad + C 3^{d(n'-n)/2} (\mathbf{a}'_0)^{-1/2} 3^{n'/4} [\mathbf{g}]_{\underline{H}^{1/4}(U_{m,n'}(z))} \\ &\quad + C 3^{d(n'-n)/2} (\mathbf{a}'_0)^{1/2} \left(|(\nabla h)_{U_{m,n'}(z)}|_2 + 3^{n'/4} [\nabla h]_{\underline{H}^{1/4}(U_{m,n'}(z))} \right) \mathbb{1}_{\{x \notin \square_{m-1}\}}. \end{aligned}$$

The s powers are dimensional constants at $s = 1/4$. By (6.63) and $n' - n \leq C + \lambda(m-n)$, the volume factor and both ratios $(\mathbf{a}'_0)^{\pm 1/2} (\mathbf{b}_{L,m})_{\square_m}^{\mp 1/2}$ are bounded by $C \exp(C\lambda(m-n))$. The scaled fractional embedding gives $3^{n'/4} [F]_{\underline{H}^{1/4}(U_{m,n'}(z))} \leq C 3^{m/2} [F]_{\underline{W}^{1/2,\infty}(\square_m)}$. Thus

$$\begin{aligned} e^{-C\lambda(m-n)} \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(T)} &\leq C (\mathbf{b}_{L,m})_{\square_m}^{1/2} 3^{-n'} \|u - (u)_{U_{m,n'}(z)}\|_{\underline{L}^2(U_{m,n'}(z))} \\ &\quad + C (\mathbf{b}_{L,m})_{\square_m}^{-1/2} 3^{m/2} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \\ &\quad + C (\mathbf{b}_{L,m})_{\square_m}^{1/2} 3^{m/2} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)} \mathbb{1}_{\{x \notin \square_{m-1}\}}. \end{aligned}$$

We apply (6.75) at n' and m' to the first term. Its large-scale input is $3^{m-m'}$ times the quantity controlled by the following detachment estimate, and $3^{m-m'} \leq C \exp(C\lambda(m-n))$ by the choice of m' . Furthermore, (2.45) and the fractional Poincaré inequality give

$$3^{-m} \|u - (u)_{U_{m,m'}(z)}\|_{\underline{L}^2(U_{m,m'}(z))} \leq C 3^{-m} \|u - (u)_{y'+\square_{m'}}\|_{\underline{L}^2(y'+\square_{m'})} \leq C 3^{-\frac{1}{4}m} \|\nabla u\|_{\dot{B}_{2,2}^{-1/4}(y'+\square_{m'})}.$$

We apply Lemma 2.9 and (6.78) to obtain

$$\begin{aligned} & 3^{-\frac{1}{4}m} \|\nabla u\|_{\dot{B}_{2,2}^{-1/4}(y' + \square_{m'})} \\ & \leq C \lambda_{1/8,2}^{-1/2}(y' + \square_{m'}; \mathbf{a}_L) \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(y' + \square_{m'})} + C \lambda_{1/8,2}^{-1}(y' + \square_{m'}; \mathbf{a}_L) 3^{\frac{1}{4}m} [\mathbf{g}]_{\underline{H}^{1/4}(y' + \square_{m'})} \\ & \leq C \exp(C\lambda(m-n)) \left((\mathbf{b}_{L,m})_{\square_m}^{-1/2} \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} + (\mathbf{b}_{L,m})_{\square_m}^{-1} 3^{\frac{1}{2}m} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \right). \end{aligned}$$

Combining these estimates with (6.75), applied at n' , and (6.76), and enlarging C_1 to include the volume and coefficient losses above, proves (6.6).

Step 7. We prove the excess decay estimate (6.7). Fix m, n, l satisfying $\mathcal{L}_L(\gamma, m) \leq m - n \leq (1 - \gamma)^{-1}$ and $n \leq l \leq m - 5$, and fix $x \in \square_m$. We may assume $l > n + 2$, since otherwise (6.7) follows, after enlarging C , by comparing the volumes of the sets involved. Choose $z \in 3^n \mathbb{Z}^d \cap \square_m$ such that

$$(x + \square_n) \cap \square_m \subseteq U_{m,n+1}(z) \subseteq U_{m,l-1}(z) \subseteq (x + \square_l) \cap \square_m.$$

As in Step 5, the boundary indicator in (6.70) vanishes when $x \in \square_{m-1}$, so it is bounded by $\mathbb{1}_{\{x \notin \square_{m-1}\}}$. Using $E(u, (x + \square_n) \cap \square_m) \leq CE(u, U_{m,n+1}(z))$, $E(u, U_{m,l-1}(z)) \leq CE(u, (x + \square_l) \cap \square_m)$, and

$$\|u - (u)_{U_{m,l-1}(z)}\|_{\underline{L}^2(U_{m,l-1}(z))} \leq C \|u - (u)_{(x + \square_l) \cap \square_m}\|_{\underline{L}^2((x + \square_l) \cap \square_m)},$$

we obtain (6.7) from (6.74), (6.70), and the assumption $\mathcal{L}_L(\gamma, m) \leq (1 - \gamma)^{-1}$, using that $\exp(C\lambda(m - n)) \leq C$, which follows from the condition $m - n \leq (1 - \gamma)^{-1} = C_1 \lambda^{-1}$. $\square$

6.4. Regularity uniformly in the cutoff. We remove the restriction $L \geq m$ from Proposition 6.1. Recall from Definition 4.3 that $\mathbf{b}_{L,n} = \bar{\mathbf{a}}_L$ for $n \geq L$. For $n \leq m$, we shall use the identity

$$(6.79) \quad \frac{\mathbf{a}_{m \wedge L}}{\mathbf{a}_{n \wedge L}} = \exp \left(\sum_{i=n+1}^m (\mathbf{g}_i - \tau^2) \mathbb{1}_{\{i \leq L\}} \right)$$

and, by (3.24),

$$(6.80) \quad 1 \leq \frac{\bar{\mathbf{a}}_{n \wedge L}}{\bar{\mathbf{a}}_{m \wedge L}} \leq \exp(2\tau^2((m \wedge L) - (n \wedge L))) \leq \exp(2\tau^2(m - n)).$$

In this subsection, for the cutoff L , we modify the good event $\mathcal{G}_m(\varepsilon, s)$ of Definition 6.2 by replacing only the quantity J in (6.10) by

$$J(z + \square_n, e, e; \bar{\mathbf{a}}_{n \wedge L}^{-1} \mathbf{a}_{n \wedge L}).$$

The conditions (6.8) and (6.9) are left unchanged, and translated events are defined as before. Likewise, we modify the quantity $\mathbf{G}_{k,z}(s)$ in (6.49), for the cutoff L , by replacing its term involving J with

$$\min \left\{ \max_{|e|=1} J(z' + \square_l, e, e; \bar{\mathbf{a}}_{l \wedge L}^{-1} \mathbf{a}_{l \wedge L}), 1 \right\}^{1/2}.$$

The terms involving the fields $\mathbf{g}_j$ are unchanged. We retain the original notation. The modified event agrees with the original event when $L \geq m$, and the modified quantity agrees with the original quantity when $L \geq k$. Thus the arguments in the previous subsections still apply.

Lemma 6.14 (Moments of J uniformly in the cutoff). *There exists $C(d) < \infty$ such that, for every $1 \leq p \leq C^{-1} \bar{\delta}^{-2} |\log \bar{\delta}|^{-1}$, and every $L, n \in \mathbb{N}_0$,*

$$(6.81) \quad \left\| \max_{|e|=1} J(\square_n, e, e; \bar{\mathbf{a}}_{n \wedge L}^{-1} \mathbf{a}_{n \wedge L}) \right\|_{L^p(\mathbb{P})} \leq Cp \log(2 + p) \bar{\delta}^2.$$

Proof. For $n \leq L$, the estimate follows from Proposition 4.1 and the homogeneity identity

$$J(U, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) = J(U, \bar{\mathbf{a}}_n^{-1/2} e, \bar{\mathbf{a}}_n^{1/2} e; \mathbf{a}_n).$$

If $n > L$, partition $\square_n$ into cubes P of side 3^L . By (2.16),

$$J(\square_n, e, e; \bar{\mathbf{a}}_L^{-1} \mathbf{a}_L) \leq \sum_P J(P, e, e; \bar{\mathbf{a}}_L^{-1} \mathbf{a}_L).$$

By Minkowski's inequality and stationarity, the L^p norm of the right side is at most that of $\max_{|e|=1} J(\square_L, e, e; \bar{\mathbf{a}}_L^{-1} \mathbf{a}_L)$, which is covered by the case $n \leq L$. The maximum over e is handled as in Proposition 4.1. $\square$

Proposition 6.15 (Good-scale estimates uniformly in the cutoff). *There exists $C(d) < \infty$ such that, for every $L \in \mathbb{N}_0$, the following assertions hold, with $\mathcal{G}_m$ and $\mathbf{G}_{k,z}$ modified as above for the cutoff L .*

First, if $s, \varepsilon, \theta \in (0, 1]$ and

$$(6.82) \quad \theta \geq Cs^{-6} \varepsilon^{-2} \bar{\delta}^2 |\log \bar{\delta}|,$$

then, for every $m_0, M \in \mathbb{N}_0$,

$$(6.83) \quad \mathbb{P} \left[\sum_{m=m_0}^{m_0+M} \mathbb{1}_{\mathcal{G}_m(\varepsilon, s)} \leq 1 - \theta \right] \leq \exp \left(- \frac{s^6 \varepsilon^2 \theta}{C \bar{\delta}^2 |\log \bar{\delta}|} (M+1) \right).$$

Second, if $s \in (0, 1/2]$ and $s \geq C \bar{\delta}^2 |\log \bar{\delta}|$, then, for every $0 \leq n \leq m$ and $z \in \mathbb{R}^d$,

$$(6.84) \quad \sum_{k=n}^m \mathbf{G}_{k,z}(s) \leq Cs^{-7/2} \bar{\delta} + \mathcal{O}_{\Gamma_2} \left(Cs^{-7/2} \bar{\delta} |\log \bar{\delta}|^{1/2} (m-n+1)^{-1/2} \right).$$

Consequently, if $\lambda \in (0, 1]$ and $\lambda \geq Cs^{-7/2} \bar{\delta} |\log \bar{\delta}|^{1/2}$, there exists $M(\lambda, s, m) \in \{-1, 0, \dots, m\}$ such that

$$(6.85) \quad 0 \leq m - M(\lambda, s, m) \leq 1 + \mathcal{O}_{\Gamma_1} (Cs^{-7} \bar{\delta}^2 |\log \bar{\delta}| \lambda^{-2})$$

and, for every $n \in \mathbb{N}_0$ with $n \leq M(\lambda, s, m)$,

$$(6.86) \quad \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m \mathbf{G}_{j,z}(s) \leq \lambda(m-n).$$

If, in addition, $\varepsilon \in (0, 1]$ and $\lambda \geq Cs^{-8} \varepsilon^{-2} \bar{\delta}^2 |\log \bar{\delta}|$, there is $N(\lambda, \varepsilon, s, m) \in \{-1, 0, \dots, m\}$ satisfying

$$(6.87) \quad 0 \leq m - N(\lambda, \varepsilon, s, m) \leq 1 + \mathcal{O}_{\Gamma_1} (Cs^{-8} \varepsilon^{-2} \lambda^{-1} \bar{\delta}^2 |\log \bar{\delta}|)$$

and, for every $n \in \mathbb{N}_0$ with $n \leq N(\lambda, \varepsilon, s, m)$,

$$(6.88) \quad \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m \left(1 - \mathbb{1}_{\mathcal{G}_{j,z}(\varepsilon, s)} \right) < 1 + \lambda(m-n).$$

Finally, for $s \in [64 \bar{\delta}^2, 1/2]$, $\varepsilon \in [s^{-1} \bar{\delta}^2, 1]$, every $m \in \mathbb{N}_0$, and every $z \in \mathbb{R}^d$,

$$(6.89) \quad \mathcal{E}_{s,\infty,2}(z + \square_m; \mathbf{a}_L, (\mathbf{b}_{L,m})_{z+\square_m}) \mathbb{1}_{\mathcal{G}_{m,z}(\varepsilon, s)} \leq C \min \{ \varepsilon, s^{-1} \bar{\delta}^2 + \varepsilon^8 + \mathbf{G}_{m,z}(s) \}.$$

In particular,

$$(6.90) \quad \mathcal{E}_{s,\infty,2}(z + \square_m; \mathbf{a}_L, (\mathbf{b}_{L,m})_{z+\square_m}) \mathbb{1}_{\mathcal{G}_{m,z}(\varepsilon, s)} \leq C \varepsilon.$$

Proof. The conditions (6.8) and (6.9) in $\mathcal{G}_m$ are those of Definition 6.2. For the modified condition (6.10), the proof of Lemma 6.7 applies, with the moments of J bounded by (6.81). Indeed, the coefficient $\bar{\mathbf{a}}_{n \wedge L}^{-1} \mathbf{a}_{n \wedge L}$ has range of dependence at most $C3^{n \wedge L} \leq C3^n$, so $J(z + \square_n, e, e; \bar{\mathbf{a}}_{n \wedge L}^{-1} \mathbf{a}_{n \wedge L})$ depends only on the fields in the $C3^n$ -neighborhood of $z + \square_n$. Since the quantities J enter only with $n \leq j - 2$, splitting the indices j into finitely many residue classes, as in that proof, gives the finite-range hypothesis of Proposition B.2. The moment exponent in that proof is at least Cd/s , and (6.82) ensures both the smallness condition required there and $\theta \geq p^{-1}$. This proves (6.83).

For (6.84), the terms involving the fields $\mathbf{g}_j$ are the same as those in the proof of Lemma 6.12. For the term involving J , we estimate the maximum over z as in Step 2 of that proof, using the moment of order $8d/s$ of $\min\{\max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_{n \wedge L}^{-1} \mathbf{a}_{n \wedge L}), 1\}^{1/2}$, that is, the moment of order $4d/s$ of the same quantity without the square root. If the dimensional constant in the lower bound on s is large enough, this order lies in the range allowed by (6.81). With the same residue classes as above, the sum of these terms is then bounded exactly as in Step 3 of that proof, and the rearrangement (6.54) and the bound for the annuli below scale n are unchanged. This proves (6.84). The union bound over translated centers used to prove (6.55) gives (6.85) and (6.86). Applying the proof of Lemma 6.13 to (6.83) gives (6.87) and (6.88).

It remains to prove (6.89). By (6.10), on $\mathcal{G}_{m,z}(\varepsilon, s)$ the elementary inequality of Step 3 of the proof of Proposition 6.1, with s in place of s_0 , applies to the square root of every quantity J in the annular decomposition. For $m \leq L$, that inequality and (6.14) give (6.89). Suppose that $m > L$. We use the annular decomposition (6.17) again and, at each nonnegative scale n , compare $\bar{\mathbf{a}}_L^{-1} \mathbf{a}_L$ with $\bar{\mathbf{a}}_{n \wedge L}^{-1} \mathbf{a}_{n \wedge L}$. The two coefficients agree if $n \geq L$. If $n < L$, we apply Lemma 2.8, with the ratios of the two coefficients given by (6.79); the conditions (6.8) and (6.9) in $\mathcal{G}_m$ bound these ratios exactly as in (6.20)–(6.19), and the ratios of effective diffusivities are bounded by (6.80). The contribution of the negative scales is bounded by (6.18), with the same proof. Combining these estimates with the preceding inequality for these quantities, as in Step 3 of the proof of Proposition 6.3, gives (6.89). The definition of $\mathcal{G}_m(\varepsilon, s)$ and (6.13) give (6.90). $\square$

Lemma 6.16 (Regularity at a good scale, uniformly in the cutoff). *Let $L \in \mathbb{N}_0$. Then the following hold.*

- Harmonic approximation: in the setting of Lemma 6.9, with its hypothesis $L \geq m$ removed, its conclusion holds with the good event $\mathcal{G}_{n+2,z}$ modified as above for the cutoff L .
- Excess decay: in the setting of Lemma 6.10, with its hypothesis $L \geq m$ removed, its conclusion holds with the same modification.

Proof. Use $\mathbf{a}_0 = (\mathbf{b}_{L,n+2})_{z+\square_{n+2}}$ and Proposition 6.15 for the bound on the anchor error. The arbitrary-depth off-grid bound (6.28) concerns the same coefficient $\mathbf{a}_L$ and the same scalar $\mathbf{a}_0$ on both sides. Consequently its ellipticity consequences, the full-datum competitor, both face-tile estimates and the window absorption in Lemma 6.8 apply at every cutoff, including $L < n + 2$. The direct ASD L^2 argument in Lemma 6.9 then gives the same three powers of s and the same powers of $\mathbf{a}_0$. The harmonic comparison remains on the arbitrary cube allowed by that lemma. Multiplying the Laplacian by $\mathbf{a}_0 > 0$ leaves its harmonic functions unchanged, so the interior and boundary regularity estimates for the comparison function and the transfer in Lemma 6.10 apply with the cutoff-modified event. $\square$

Proposition 6.17 (Cutoff-uniform large-scale regularity). *There exists $C(d) < \infty$ such that, if $\bar{\delta} \leq C^{-1}$, then for every $\gamma \in [1/2, 1 - C\bar{\delta}|\log \bar{\delta}|^{1/2}]$ and every $L, m \in \mathbb{N}_0$, there is an $\mathbb{N}$ -valued random variable $\mathcal{L}_L(\gamma, m)$ satisfying, for every $k \in \mathbb{N}$,*

$$(6.91) \quad \mathbb{P}[\mathcal{L}_L(\gamma, m) > k] \leq C \exp\left(-\frac{(1-\gamma)^2(k-C)_+}{C\bar{\delta}^2|\log \bar{\delta}|}\right).$$

For every $\mathbf{g} \in C^{0,1/2}(\square_m; \mathbb{R}^d)$, $h \in C^{1,1/2}(\square_m)$, and $u \in H^1(\square_m)$ satisfying (6.4), all three conclusions (6.5), (6.6), and (6.7) hold with $\mathcal{L}_L(\gamma, m)$ as the minimal scale. If $L \geq m$, one may take

$$(6.92) \quad \mathcal{L}_L(\gamma, m) = \mathcal{L}_m(\gamma, m).$$

The same statement holds on every translate $y + \square_m$, with the tail in (6.91) uniform in y .

Proof. If $L \geq m$, then this is Proposition 6.1. We may therefore suppose $L < m$. Choose $s_0, \lambda, \varepsilon$, and the step size k exactly as in Step 1 of the proof of Proposition 6.1. Define $\mathcal{L}_L(\gamma, m)$ by (6.60). Here M and N are the random variables of Proposition 6.15 for the cutoff L , not those of Lemmas 6.12 and 6.13. Equations (6.85) and (6.87), with the parameter choices in that proof, give (6.91). Equations (6.86) and (6.88) give the two conclusions in (6.61) for the modified $\mathbf{G}$ and $\mathcal{G}$.

For $n \leq j \leq m - 5$ and $z \in 3^n \mathbb{Z}^d \cap \square_m$, the calculation leading to (6.63), with all sums over the fields and all deterministic sums stopped at L , gives

$$(6.93) \quad C^{-1} \exp(-C\lambda(m-n)) \leq \frac{(\mathbf{b}_{L,j+2})_{z+\square_{j+2}}}{(\mathbf{b}_{L,m})_{\square_m}} \leq C \exp(C\lambda(m-n)).$$

Indeed, if $j+2 \geq L$, the numerator and the denominator both equal $\bar{\mathbf{a}}_L$. If $j+2 < L < m$, only the fields $\mathbf{g}_{j+3}, \dots, \mathbf{g}_L$ occur; they are bounded by (6.86), while (6.80) controls the ratio of effective diffusivities.

We use Lemma 6.16 in place of Lemma 6.10 in Steps 3–5. With (6.93) in place of (6.63), the forcing bounds and coefficient comparisons there are unchanged. In Step 6, apply the cutoff-uniform version of Lemma 6.8 at the nearest good anchor $n'+2$, as in Step 6 of the proof of Proposition 6.1. The volume loss is $C3^{d(n'-n)/2}$ and the cutoff coefficient ratios are bounded by $C \exp(C\lambda(m-n))$ by (6.93). Proposition 6.15 supplies the same error and stopping bounds. The Campanato estimate at n' and the detachment estimate at m' therefore prove the energy estimate. Step 7 uses this unchanged iteration and proves the excess estimate. This proves all three conclusions with $\mathcal{L}_L(\gamma, m)$ as the minimal scale. $\square$

Proposition 6.18 (Stability under multiplication by a nearly constant function). *There are $\varepsilon_*(d) > 0$, $c(d) > 0$, and $C(d) < \infty$ such that the following holds if $\bar{\delta} \leq c$. Let $L \in \mathbb{N}_0$, $m \in \mathbb{Z}$, $z \in \mathbb{R}^d$, and $j \in \mathbb{N}$. Put $B = z + \square_m$, and let B' be a cube of side 3^{m-j} contained in the concentric cube $B/2$ of half the side of B . Outside an event of probability at most*

$$(6.94) \quad C \exp\left(-\frac{c}{\bar{\delta}^2 |\log \bar{\delta}|^2}\right),$$

the following estimates hold for every positive $\vartheta \in C(B)$ satisfying

$$(6.95) \quad \inf_{b>0} \|b^{-1}\vartheta - 1\|_{L^\infty(B)} \leq \varepsilon_*$$

and every weakly $\mathbf{a}_L \vartheta$ -harmonic function $h \in H^1(B)$:

$$\operatorname{osc}_{B'} h \leq C 3^{-cj} \operatorname{osc}_B h,$$

and

$$\sup_{B'} |h - (h)_B| \leq C \|h - (h)_B\|_{\underline{L}^2(B)}.$$

Proof. We bound both conclusions by the normalized L^2 oscillation $\|h - (h)_B\|_{\underline{L}^2(B)}$, combining an estimate below the unit scale with one above it. Below the unit scale, comparison with harmonic functions on small balls bounds point values and Hölder quotients of h by powers of $1 + \|\nabla \log \mathbf{a}_L\|_{L^\infty(B)}$; no derivative of ϑ enters. Above the unit scale, we run the large-scale iteration of the proof of Proposition 6.17 for the coefficient $\mathbf{a}_L \vartheta$. The main point is that the multiplier changes the coarse-graining errors by at most $C\varepsilon_*^{1/2}$ per scale, which is absorbed by running the iteration with exponent $3/4$ and

concluding with exponent $1/2$. Finally, we bound $\|\nabla \log \mathbf{a}_L\|_{L^\infty(B)}$ and combine the two estimates; the exceptional events depend on neither ϑ nor h .

Multiplying ϑ by a positive constant does not change the equation. The infimum in (6.95) allows a normalization with $\|\vartheta - 1\|_\infty \leq 2\varepsilon_*$. We choose ε_* small enough for all the estimates below with this bound. Write $R = 3^m$ and $H = \|h - (h)_B\|_{\underline{L}^2(B)}$.

We first record a local estimate that works in every dimension $d \geq 2$. Suppose that a scalar coefficient a on a Euclidean ball V satisfies $\|a/a_0 - 1\|_\infty \leq \eta$, where $a_0 > 0$ and η is a sufficiently small dimensional constant. Let v be the harmonic function with boundary values h on V . Testing the equation for $h - v$, and using the Dirichlet principle for v , gives

$$\|\nabla(h - v)\|_{L^2(V)} \leq C\eta \|\nabla h\|_{L^2(V)}.$$

The mean-square gradient of v decreases on concentric balls. Consequently,

$$\|\nabla h\|_{\underline{L}^2(V/2)} \leq (1 + C2^{d/2}\eta) \|\nabla h\|_{\underline{L}^2(V)}.$$

Choose η so that the factor is at most $2^{1/2}$. Iteration, Poincaré and telescoping ball averages give Hölder exponent $1/2$ on the inner ball. Caccioppoli on a slightly larger ball controls the initial gradient by the normalized L^2 norm of $h - c$, for any constant c . This also bounds point values relative to c , and the telescoping averages define a continuous representative of h .

Put $D = \|\nabla \log \mathbf{a}_L\|_{L^\infty(B)}$. Around any point of $Q/2$, where Q has side one and $2Q \subseteq B$, choose a ball of radius $r_0 = c(d)/(1 + D)$. The mean-value theorem and $\|\vartheta - 1\|_\infty \leq 2\varepsilon_*$ make $\mathbf{a}_L \vartheta$, divided by its value of $\mathbf{a}_L$ at the center, within η of one on twice that ball. The preceding comparison and Caccioppoli estimate, followed by the volume comparison with Q , give

$$\sup_{Q/2} |h - (h)_Q| \leq C(1 + D)^{d/2} \|h - (h)_Q\|_{\underline{L}^2(Q)}.$$

For $x, y \in Q/2$ the same argument gives

$$|h(x) - h(y)| \leq C(1 + D)^{(d+1)/2} |x - y|^{1/2} \|h - (h)_Q\|_{\underline{L}^2(Q)}.$$

For distances larger than r_0 use the point-value bound and $(|x - y|/r_0)^{1/2} \geq 1$. Thus this estimate covers every distance below one. On a cube of side ℓ , the rescaled bounds have $1 + \ell D$ in place of $1 + D$ and $|x - y|/\ell$ in place of $|x - y|$. No derivative of ϑ is needed.

We next run the interior large-scale iteration for $\mathbf{a}_L \vartheta$. On every comparison cube $Q \subseteq B$, Lemma 2.8 and Minkowski's inequality in the weighted subcube sum give

$$(6.96) \quad \mathcal{E}_{s_0, \infty, 2}(Q; \mathbf{a}_L \vartheta, \mathbf{a}_0) \leq (1 + C\varepsilon_*) \mathcal{E}_{s_0, \infty, 2}(Q; \mathbf{a}_L, \mathbf{a}_0) + C\varepsilon_*^{1/2}, \quad s_0 = 1/32.$$

Here $\mathbf{a}_0$ is the scalar in Lemma 6.16. Cube averages of the long-scale coefficient change by factors between $1 - 2\varepsilon_*$ and $1 + 2\varepsilon_*$, and their ratios by factors $1 + O(\varepsilon_*)$. The proofs of Lemmas 6.8, 6.9 and 6.10 use the good event only to bound the anchor error and the resulting ellipticity ratios. Their interior estimates therefore apply to any positive scalar coefficient with these bounds; for $\mathbf{a}_L \vartheta$ they are supplied by (6.96). Apply the deterministic interior comparison of Proposition 2.11 to the product coefficient; its Caccioppoli factors are comparable to the original ones. In that comparison, therefore, the excess contraction and slope-error coefficient change by at most $C\varepsilon_*^{1/2}$. Fix the block length and other parameters of the interior proof of Proposition 6.17 with exponent $3/4$, before choosing ε_* . The perturbation preserves a strict contraction on good blocks and adds at most $C\varepsilon_*^{1/2}$ times the number of scales to the error sum. More explicitly, on an interval $[a, b]$ the new error sum is at most

$$(1 + C\varepsilon_*^{1/2}) \sum_{j=a}^b \varepsilon_j + C\varepsilon_*^{1/2}(b - a + 1).$$

Choose ε_* so small that this additional exponential loss uses less than the margin between exponents $3/4$ and $1/2$. Lemma 6.11 then gives exponent $1/2$ on the same good-scale event, uniformly in ϑ . All comparisons are interior; no boundary estimate is used.

Here is the covering needed for that iteration. In each coordinate use the four translates with shifts $0, R/4, R/2, 3R/4$. At every scale $3^n \leq R$, their relative phases are the four quarter periods, since powers of three permute these phases. A cube of sufficiently smaller side than 3^n is therefore contained in a cube of one of the 4^d grids at that scale, at distance at least a fixed fraction of 3^n from its boundary. Fix a dimensional integer $a \geq 4$. For each grid retain the top cubes of side 3^{m-a} meeting $B/2$ whose comparison neighborhoods lie in B ; their inner halves cover $B/2$, and every point of $B/2$ lies in one of them, at distance at least a fixed fraction of 3^{m-a} from its boundary. There are only dimensionally many such top cubes. Every smaller comparison cube used by the iteration stays in B . In particular, unit cubes with $2Q \subseteq B$ and their inner halves cover $B/2$ when m is sufficiently large.

Choose a fixed j_0 larger than the additive stopping constant. On each retained top cube require the minimal depth for exponent $3/4$ to be at most j_0 . The tail (6.91) and the finite union over the top cubes bound the failure probability by (6.94), after changing dimensional constants. The perturbed iteration, volume comparison at the top cubes, and telescoping cube averages give

$$\|h - (h)_Q\|_{L^2(Q)} \leq C(\ell/R)^{1/2} H$$

on the interior cubes of side $1 \leq \ell \leq 3^{m-a}$ used in the covering. The finitely many initial scales and the at most j_0 missing scales contribute only a dimensional constant. Differences of their averages obey the same Campanato bound, and their averages differ from $(h)_B$ by at most CH .

The local point-value factors are also controlled on an event independent of ϑ . Covering B at scale 3^k , Assumption (g2) and scaling give a sub-Gaussian bound for $\|\nabla \mathbf{g}_k\|_{L^\infty(B)}$ with mean at most $C\bar{\delta}3^{-k}\sqrt{(m-k)_+ + 1}$ and tail scale $C\bar{\delta}3^{-k}$. Summing these bounds gives

$$\mathbb{P}[D > C\bar{\delta}\sqrt{m_+ + 1} + t] \leq C \exp\left(-\frac{ct^2}{\bar{\delta}^2}\right).$$

Take $t = |\log \bar{\delta}|^{-1}$. Its failure probability has the form (6.94); on the complementary event, $(1 + D)^{(d+1)/2} \leq C3^{m/8}$ for $m \geq 0$. Thus the local bound on unit cubes and the Campanato estimate bound $|h - (h)_B|$ on $B/2$ by CH . For B' of side $\rho = R3^{-j} \geq 1$, comparisons of averages and the unit-scale point errors give

$$\operatorname{osc}_{B'} h \leq C(3^{-j/2} + 3^{-3m/8})H \leq C3^{-cj}H,$$

since $j \leq m$. A target cube of side below a fixed dimensional fraction of one lies in a single inner half of a unit cube from the covering. Larger targets of side below one are covered by dimensionally many overlapping inner halves; their point errors and the differences of the unit-cube averages obey the same bound because ρ is then bounded below. If $\rho < 1$, the local Hölder estimate therefore gives

$$\operatorname{osc}_{B'} h \leq C3^{-3m/8}\rho^{1/2}H = C3^{-3m/8}3^{-(j-m)/2}H \leq C3^{-cj}H.$$

One may take any fixed $c \leq 3/8$ in these estimates, and then decrease it to match the probability bound. The finitely many small values of j follow from $\operatorname{osc}_{B'} h \leq \operatorname{osc}_B h$ after increasing C .

For $m \leq a + j_0 + 6$, including all $m < 0$, use the rescaled local estimates directly on interior cubes of side comparable to R . On the same gradient event, RD is dimensionally bounded, so both their point-value constant and their Hölder constant are bounded. This gives the same two conclusions. Finally $H \leq \operatorname{osc}_B h$ whenever that oscillation is finite, and otherwise its asserted upper bound is automatic. Taking the union of the two exceptional events completes the proof. Neither event depends on ϑ or h . $\square$

6.5. Dirichlet homogenization. The estimates above compare a solution with an affine function. We now compare it instead with the solution of the homogenized Dirichlet problem in the unit cube, after rescaling the cutoff coefficient $\mathbf{a}_L$ from a cube at least as large as the cutoff scale.

Theorem 6.19 (Quantitative homogenization of the Dirichlet problem). *For $L, M \in \mathbb{N}_0$ with $L \leq M$, recall the rescaled coefficient of the introduction,*

$$(6.97) \quad A_{L,M}(x) := \bar{\mathbf{a}}_L^{-1} \mathbf{a}_L(3^M x), \quad x \in \square_0.$$

For every $\vartheta \in (0, 1)$ and $q \in [1, \infty)$, there are $\bar{\delta}_0(\vartheta, q, d) > 0$ and $C(\vartheta, q, d) < \infty$ such that, if $\bar{\delta} \leq \bar{\delta}_0$, then there are random variables $\mathcal{Z}_{L,M} \geq 1$, $0 \leq L \leq M$, satisfying

$$(6.98) \quad \sup_{M \in \mathbb{N}_0} \sup_{0 \leq L \leq M} \|\mathcal{Z}_{L,M}\|_{L^q(\mathbb{P})} \leq C.$$

For $f \in L^2(\square_0)$ and $h \in H^2(\square_0)$, recall that $u_{L,M}$ and u_{hom} are the unique weak solutions in $H^1(\square_0)$ of

$$(6.99) \quad \begin{cases} -\nabla \cdot (A_{L,M} \nabla u_{L,M}) = f & \text{in } \square_0, \\ u_{L,M} = h & \text{on } \partial \square_0, \end{cases} \quad \begin{cases} -\Delta u_{\text{hom}} = f & \text{in } \square_0, \\ u_{\text{hom}} = h & \text{on } \partial \square_0, \end{cases}.$$

Then, almost surely, for every such f and h , the homogenization error $\mathcal{D}_{L,M}$ of the introduction satisfies

$$(6.100) \quad \mathcal{D}_{L,M} \leq \mathcal{Z}_{L,M} \bar{\delta}^\vartheta (\|f\|_{L^2(\square_0)} + \|h\|_{H^2(\square_0)}).$$

If $f = 0$, then the factor $\bar{\delta}^\vartheta$ on the right side may be replaced by $C\bar{\delta}$, with the same random variable $\mathcal{Z}_{L,M}$.

Proof. We rescale the two Dirichlet problems to the cube $\square_M$ and apply the general coarse-graining estimate of Proposition 2.11 at the scale $M - k$. It bounds the errors of the gradient and of the flux in $\hat{H}^{-s}(\square_0)$ by $3^{s_1 k}$ times coarse-graining errors whose moments are of order $\bar{\delta}$, plus a term of order $3^{-s_2 k}$ coming from the source f . The main step is the choice of k that balances these two terms; by (6.101), the resulting power of $\bar{\delta}$ is at most $\bar{\delta}^\vartheta$. A duality argument with the Dirichlet Laplacian then gives the L^2 estimate. These fractional-dual estimates use the unrestricted tests of (2.1).

Fix $s_2 \in (0, 1)$, and then choose $0 < s_1 < s < \min\{s_2, 1/2\}$ so that

$$(6.101) \quad \frac{s_2}{s_1 + s_2} > \vartheta.$$

Fix a finite exponent $r \geq \max\{4q, 8d/s_1\}$ so large that $(s_1/2) \log(2 + r) \geq c_P$, where c_P is the logarithmic-condition constant in Proposition 4.1. We use input moments through $4r$, leaving room for squares and products before taking L^r norms. These orders depend on ϑ through s_1 , as well as on q and d , and are fixed before the disorder threshold.

To check applicability explicitly, put $\xi = 4r$. Let C_P be the constant in the first conclusion of Proposition 4.1, and c_L the moment-range constant in Lemma 4.5. Fix

$$K \geq \max \left\{ 1, C_P \xi \log(2 + \xi), \frac{2\xi}{c_L s_1} \right\}, \quad \delta_1 = K \bar{\delta}^2.$$

Decrease $\bar{\delta}_0(\vartheta, q, d) \leq e^{-1}$ so that $C_P \xi \bar{\delta}_0 \leq 1$ and $K \bar{\delta}_0^2 < \min\{1/2, c_P s_1/2\}$. Then $\xi \leq C_P^{-1} \bar{\delta}^{-2} |\log \bar{\delta}|^{-1}$, using $\bar{\delta}^2 |\log \bar{\delta}| \leq \bar{\delta}$. The proposition supplies $\mathcal{S}(\infty, \xi, \delta_1)$, while $\bar{\delta}^2 \leq \delta_1 < 1$. At both orders s_1 and $s_1/2$, the large-cube lemma has $4d/(s_1/2) \leq \xi$ and $\xi \leq c_L (s_1/2) \bar{\delta}^{-2} \delta_1$ by the choice of K . Its bound is $C\sqrt{\delta_1} = C\bar{\delta}$. The lower-order estimates follow by monotonicity of probability norms. Thus, uniformly in $0 \leq L \leq M$, these results give

$$(6.102) \quad \|\mathcal{E}_{s_1, \infty, 1}(\square_M; \mathbf{a}_L, \bar{\mathbf{a}}_L)\|_{L^r(\mathbb{P})} + \|\mathcal{E}_{s_1/2, \infty, 2}(\square_M; \mathbf{a}_L, \bar{\mathbf{a}}_L)\|_{L^r(\mathbb{P})} \leq C\bar{\delta}.$$

Write f as the divergence of a field in $H^1(\square_0; \mathbb{R}^d)$ of norm at most $C\|f\|_{L^2(\square_0)}$, for instance minus the gradient of the solution of the Dirichlet problem for the Laplacian in $\square_0$ with source f . Rescale to $\square_M$ by setting $U(y) := u_{L,M}(3^{-M}y)$ and $V(y) := u_{\text{hom}}(3^{-M}y)$, and let $G(y)$ be $\bar{\mathbf{a}}_L 3^{-M}$ times the field above evaluated at $3^{-M}y$. Then

$$-\nabla \cdot (\mathbf{a}_L \nabla U) = \nabla \cdot G, \quad -\bar{\mathbf{a}}_L \Delta V = \nabla \cdot G, \quad U - V \in H_0^1(\square_M).$$

Let $k \in \mathbb{N}$ and apply Proposition 2.11, with $p = 2$ and $n = M - k$. Definition 2.3 gives

$$\mathcal{E}_{s_1, \infty, 1}(\square_M, M - k; \mathbf{a}_L, \bar{\mathbf{a}}_L) \leq 3^{s_1 k} \mathcal{E}_{s_1, \infty, 1}(\square_M; \mathbf{a}_L, \bar{\mathbf{a}}_L),$$

and

$$\mathcal{E}_{s_1/2, \infty, 2}(\square_M, M - k; \mathbf{a}_L, \bar{\mathbf{a}}_L)^2 \leq 3^{s_1 k} \mathcal{E}_{s_1/2, \infty, 2}(\square_M; \mathbf{a}_L, \bar{\mathbf{a}}_L)^2.$$

The geometric weights in the sum over k in (2.52), together with (2.3), bound that sum by $C\|\mathbf{a}_L^{1/2} \nabla U\|_{L^2(\square_M)}$. Moreover, (2.50) and (2.38), using the fixed input orders chosen above, give a random variable $Y_{L,M} \geq 1$ with $\|Y_{L,M}\|_{L^r(\mathbb{P})} \leq C(\vartheta, q, d)$ uniformly in $L \leq M$, such that

$$(6.103) \quad \|A_{L,M}^{1/2} \nabla u_{L,M}\|_{L^2(\square_0)} \leq Y_{L,M} (\|f\|_{L^2(\square_0)} + \|h\|_{H^2(\square_0)}).$$

Rescaling (2.52) back to $\square_0$ therefore yields

$$(6.104) \quad \begin{aligned} & \|\nabla u_{L,M} - \nabla u_{\text{hom}}\|_{\hat{H}^{-s}(\square_0)} + \|A_{L,M} \nabla u_{L,M} - \nabla u_{\text{hom}}\|_{\hat{H}^{-s}(\square_0)} \\ & \leq C \left(3^{s_1 k} \mathcal{E}_1 Y_{L,M} + 3^{-s_2 k} (1 + 3^{s_1 k} \mathcal{E}_2^2) \right) (\|f\|_{L^2(\square_0)} + \|h\|_{H^2(\square_0)}), \end{aligned}$$

where $\mathcal{E}_1$ and $\mathcal{E}_2$ denote the two random variables in (6.102).

Take

$$k := \left\lfloor \frac{\log_3(\bar{\delta}^{-1})}{s_1 + s_2} \right\rfloor.$$

With $\mathcal{E}_1, \mathcal{E}_2$, and $Y_{L,M}$ as above, define

$$\mathcal{Z}_{L,M} := 1 + C\bar{\delta}^{-\vartheta} \left(3^{s_1 k} \mathcal{E}_1 Y_{L,M} + 3^{-s_2 k} (1 + 3^{s_1 k} \mathcal{E}_2^2) \right) + C\bar{\delta}^{-1} \mathcal{E}_1 Y_{L,M}.$$

Equations (6.101), (6.102), and (6.103), followed by Hölder's inequality, give (6.98). Substituting this choice of k and the definition of $\mathcal{Z}_{L,M}$ into (6.104) proves (6.100) for the two H^{-1} terms of $\mathcal{D}_{L,M}$: restriction to zero-boundary tests and the embedding $H_0^1(\square_0) \hookrightarrow H^s(\square_0)$ bound each coordinate dual, and the finite sum over coordinates changes only $C(d)$. If $f = 0$, we apply (6.104) again, with $k = 1$. The $3^{-s_2 k}$ term is then absent, and the last term in the definition of $\mathcal{Z}_{L,M}$ gives the factor $C\bar{\delta}$.

Finally, the difference $u_{L,M} - u_{\text{hom}}$ belongs to $H_0^1(\square_0)$. For every $w \in H_0^1(\square_0)$, duality and the spectral theorem for the Dirichlet Laplacian on $\square_0$ give $\|w\|_{L^2(\square_0)} \leq C\|\nabla w\|_{\hat{H}^{-s}(\square_0)}$: test with the gradient of the image of w under the inverse Dirichlet Laplacian, which is bounded in $H^s(\square_0)$ by $C\|w\|_{L^2(\square_0)}$. Applying the unrestricted fractional-dual estimate (6.104) to the difference gives the L^2 term and completes the proof. $\square$

Theorem 6.20 (Quantitative homogenization at a fixed cutoff). *There exists $\alpha_{\text{hom}}(d) > 0$ with the following property. Fix $L \in \mathbb{N}_0$. For $M \in \mathbb{N}_0$ with $M \geq L$, let $A_{L,M}$ be as in (6.97). For every $q \in [1, \infty)$, there is a random variable $\mathcal{Y}_{L,q} \geq 1$ satisfying*

$$(6.105) \quad \|\mathcal{Y}_{L,q}\|_{L^q(\mathbb{P})} \leq C(L, q, d, \bar{\delta})$$

such that, almost surely, for every $M \geq L$, $f \in L^2(\square_0)$ and $h \in H^2(\square_0)$, the homogenization error $\mathcal{D}_{L,M}$ of the unique weak solutions $u_{L,M}$ and u_{hom} of (6.99) satisfies

$$(6.106) \quad \mathcal{D}_{L,M} \leq \mathcal{Y}_{L,q} 3^{-\alpha_{\text{hom}}(M-L)} (\|f\|_{L^2(\square_0)} + \|h\|_{H^2(\square_0)}).$$

Proof. Consider the coefficient $A_{L,L}$. For fixed L , this is a stationary positive scalar coefficient with range of dependence at most $\sqrt{d}$, and its homogenized matrix is the identity. It satisfies the logarithmic coefficient bound

$$(6.107) \quad \|\log A_{L,L}\|_{L^\infty(\square_0)} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}(L+1)).$$

Indeed, this follows from (3.4), Assumption (g2), and the bound $|\log \bar{a}_L| \leq C\bar{\delta}^2(L+1)$ from Theorem 5.1.

We apply the results of [Armstrong and Kuusi \[2025\]](#), Theorem B, pp. 9–10; Theorem 4.1, pp. 73–74; and Proposition D.3, pp. 145–146] to $A_{L,L}$. Stationarity and finite range give the mixing assumption of that paper with $\beta = 0$ and $\nu = d/2$, and with Gaussian concentration. Proposition D.3 of that paper, applied to (6.107) with exponent $1/4$, gives the coarse-grained ellipticity assumption of that paper above a random scale. The probability that this scale exceeds $\sqrt{d}t$ is at most $\exp(C_L - c_L \log^2 t)$, with positive constants C_L and c_L that may depend on L ; this tail bound implies that all polynomial moments of this scale are finite.

[Armstrong and Kuusi \[2025\]](#), Corollary 4.3, p. 76], with exponent $5/16$, gives an estimate uniform over all subcubes. Together with Definition 2.3, it yields a number $\alpha_{\text{hom}}(d) > 0$ and, for every fixed $q < \infty$, a random variable $\mathcal{W}_{L,q}$ with finite q th moment such that, for all $N \in \mathbb{N}_0$, $r \in \{1, 2\}$, and $s \in \{3/16, 3/8\}$,

$$(6.108) \quad \mathcal{E}_{s,\infty,r}(\square_N; A_{L,L}, 1) \leq \mathcal{W}_{L,q} 3^{-2\alpha_{\text{hom}}N}.$$

Indeed, by the estimate of [Armstrong and Kuusi \[2025\]](#), Corollary 4.3, p. 76] for subcubes, the quantity $\max_{|e|=1} J(\cdot, e, e; A_{L,L})$ on a subcube at scale $k \leq N$ is bounded by a random multiple of $3^{-4\alpha_{\text{hom}}N} 3^{\frac{5}{16}(N-k)}$. After taking square roots, the weighted sum over scales in Definition 2.3 converges, since $s > 5/32$ for both values of s above. Below that random scale, we use (2.14) and the extrema of $A_{L,L}^{\pm 1}$ instead. By a union bound over unit cubes and (6.107), the moments of these extrema on $\square_N$ grow at most like $C(L, q) \exp(C(L, q)\sqrt{N+1})$. Hölder's inequality, the Gaussian-type tail bound for that scale, and the analogous tail for the scale in [Armstrong and Kuusi \[2025\]](#), Corollary 4.3] therefore allow us to absorb these scales into $\mathcal{W}_{L,q}$ while keeping its moments finite.

We use the estimate leading to (6.104), after rescaling $\square_M$ first by 3^L and then by 3^{M-L} . Apply Proposition 2.11 with $p = 2$, $s_1 = 3/8$, $s = 7/16$, and $s_2 = 3/4$, so that $sp' = 7/8 < 1$ and both s_1 and $s_1/2$ are covered by (6.108), and choose

$$k := \max \left\{ 1, \left\lfloor \frac{2\alpha_{\text{hom}}(M-L)}{s_1 + s_2} \right\rfloor \right\}.$$

In (6.104), we use (6.108), applied at the scale $M-L$, in place of (6.102), and we construct $Y_{L,M}$ in (6.103) from (2.50), (2.38), and now (6.108). These give $\sup_{M \geq L} \|Y_{L,M}\|_{L^r} < \infty$ for every fixed r , with constants allowed to depend on $L, r, d, \bar{\delta}$. Substituting (6.108) and the choice of k in (6.104), and using the bound $Y_{L,M} \leq 3^{\alpha_{\text{hom}}(M-L)/25} \sup_{M' \geq L} 3^{-\alpha_{\text{hom}}(M'-L)/25} Y_{L,M'}$, we obtain the negative-Sobolev part of (6.106) with the random factor

$$\mathcal{Y}_{L,q} := C(1 + \mathcal{W}_{L,4q})^2 \left(1 + \sup_{M \geq L} 3^{-\alpha_{\text{hom}}(M-L)/25} Y_{L,M} \right).$$

Since $Y_{L,M}$ is constructed with a moment larger than $4q$, the supremum has a finite $4q$ th moment, because its $4q$ th power is at most the sum over $M \geq L$ of $3^{-4q\alpha_{\text{hom}}(M-L)/25} Y_{L,M}^{4q}$. Hölder's inequality shows that $\mathcal{Y}_{L,q}$ has a finite q th moment, which gives (6.105). As in the proof of Theorem 6.19, restriction of the unrestricted fractional-dual estimate to zero-boundary tests gives the two negative-norm terms of $\mathcal{D}_{L,M}$, while duality with the Dirichlet Laplacian gives the L^2 term. $\square$

6.6. Limiting coefficient and regularity. The limiting coefficient field is the almost-sure local limit

$$(6.109) \quad \tilde{\mathbf{a}}_L(x) := \frac{\mathbf{a}_L(x)}{\mathbf{a}_L(0)}, \quad \mathbf{a}(x) := \lim_{L \rightarrow \infty} \tilde{\mathbf{a}}_L(x) = \exp\left(\sum_{k=0}^{\infty} (\mathbf{g}_k(x) - \mathbf{g}_k(0))\right).$$

Dividing by $\mathbf{a}_L(0)$ is necessary, since the strong law gives $(L+1)^{-1} \log \mathbf{a}_L(0) \rightarrow -\tau^2$. This division does not affect the generator of $X^{(L)}$, and the reversible measures $(\mathbf{a}_L/\mathbf{a}_L(0)) dx$ converge locally to $\mathbf{a} dx$. By Assumption (g2) and scaling, the coefficient ratios converge locally in $C^{1,1}$. The following lemma gives bounds uniform in L .

Lemma 6.21 (The coefficients divided by their value at the origin). *There is $\bar{\delta}_0(d) > 0$ such that, if $\bar{\delta} \leq \bar{\delta}_0(d)$, then there are $\kappa \in (0, 1)$ and an almost surely finite random variable C_ω such that, almost surely, for every $L \in \mathbb{N}_0$ and $x \in \mathbb{R}^d$,*

$$(6.110) \quad \sup_{\mathbf{b} \in \{\mathbf{a}\} \cup \{\tilde{\mathbf{a}}_L\}_{L \in \mathbb{N}_0}} (\mathbf{b}(x) + \mathbf{b}(x)^{-1} + |\nabla \mathbf{b}(x)|) \leq C_\omega(1 + |x|)^\kappa.$$

Moreover,

$$(6.111) \quad \tilde{\mathbf{a}}_L \longrightarrow \mathbf{a} \quad \text{locally in } C^{1,1}, \quad \nabla \log \tilde{\mathbf{a}}_L \longrightarrow \nabla \log \mathbf{a} \quad \text{locally in } C^{0,1},$$

and

$$(6.112) \quad \sup_{L \in \mathbb{N}_0} (|\nabla \log \tilde{\mathbf{a}}_L(x)| + |\nabla^2 \log \tilde{\mathbf{a}}_L(x)|) \leq C_\omega(1 + \sqrt{\log(2 + |x|)}).$$

Proof. Fix a sufficiently large dimensional constant C , and cover each ball B_{C2^n} by cubes of side 3^k . Stationarity, (1.4), (1.5), and the Borel–Cantelli lemma give a finite random variable C_ω such that, for all $n, k \in \mathbb{N}_0$ and $r \in \{1, 2\}$,

$$(6.113) \quad \|\nabla^r \mathbf{g}_k\|_{L^\infty(B_{C2^n})} \leq C_\omega 3^{-rk} \sqrt{n+k+1}.$$

Summing over k proves the local convergence (6.111) and the bound (6.112).

It remains to obtain the bound on the coefficient itself uniformly in the cutoff L . Let $K_n = \max\{k \in \mathbb{N}_0 : 3^k \leq C2^n\}$, so $K_n + 1 \leq C(d)(n+1)$. For a fixed lattice point z , the independence of the fields and Assumptions (g1)–(g2) give centered, independent sub-Gaussian variables $\mathbf{g}_k(z)$. Hence

$$\mathbb{E} \exp\left(t \sum_{k=0}^j \mathbf{g}_k(z)\right) \leq \exp(Ct^2 \bar{\delta}^2(j+1)), \quad \mathbb{P}\left[\left|\sum_{k=0}^j \mathbf{g}_k(z)\right| > u\right] \leq 2 \exp\left(-\frac{u^2}{C\bar{\delta}^2(j+1)}\right).$$

The second bound follows by optimizing the exponential Markov bound for both signs. There are at most $C2^{dn}(n+1)$ pairs (z, j) with $z \in \mathbb{Z}^d \cap B_{C2^n}$ and $0 \leq j \leq K_n$. With $u = A\bar{\delta}(n+1)$ their union therefore has probability at most $C(n+1)2^{dn} \exp(-cA^2(n+1))$. Fix a dimensional A so large that this is summable in n . Borel–Cantelli bounds every such unanchored sum by $A\bar{\delta}(n+1) + C_\omega$. Since 0 is one of the lattice points, subtracting its sum gives the anchored bound with twice this constant. Absorbing the finitely many initial rows and the additive deterministic constant into C_ω yields

$$(6.114) \quad \max_{z \in \mathbb{Z}^d \cap B_{C2^n}, 0 \leq j \leq K_n} \left| \sum_{k=0}^j (\mathbf{g}_k(z) - \mathbf{g}_k(0)) \right| \leq C\bar{\delta}n + C_\omega.$$

For $x \in B_{2^n}$, we apply (6.114) at a nearest lattice point and control the difference using (6.113); for the scales k with $3^k \leq C2^n$, this difference is bounded by $C_\omega \sqrt{n+1}$. For the other scales, the mean-value theorem along the segment from 0 to x gives

$$\sum_{3^k > C2^n} |\mathbf{g}_k(x) - \mathbf{g}_k(0)| \leq C_\omega 2^n \sum_{3^k > C2^n} 3^{-k} \sqrt{n+k+1} \leq C_\omega \sqrt{n+1}.$$

Both estimates hold for every partial sum of the series. Consequently,

$$\sup_{L \in \mathbb{N}_0} \sup_{x \in B_{2^n}} |\log \tilde{\mathbf{a}}_L(x)| \leq C\bar{\delta}n + C_\omega \sqrt{n+1} + C_\omega.$$

Choosing $\bar{\delta}_0(d)$ small, bounding the square-root term by an arbitrarily small multiple of n plus a random constant, and exponentiating, we obtain (6.110) for $\tilde{\mathbf{a}}_L$ and its reciprocal. Since $\nabla \tilde{\mathbf{a}}_L = \tilde{\mathbf{a}}_L \nabla \log \tilde{\mathbf{a}}_L$, the bound (6.112) gives the gradient bound after a final arbitrarily small increase of κ . Passing to the local limit proves the assertions for $\mathbf{a}$. $\square$

Recall that, for a large $C_0(d) < \infty$ and $\bar{\delta}$ small, the regularity exponent $\gamma_{\text{reg}} := 1 - C_0\bar{\delta}|\log \bar{\delta}|^{1/2}$ of the introduction lies in $(1/2, 1)$.

Theorem 6.22 (Large-scale Hölder regularity and a Liouville theorem). *There are $\bar{\delta}_0(d) > 0$ and $C(d) < \infty$ such that the following holds if $\bar{\delta} \leq \bar{\delta}_0(d)$. For $L = \infty$, the coefficient $\mathbf{a}_L$ stands for the limiting coefficient $\mathbf{a}$ of (6.109).*

- Large-scale Hölder estimate: Let $\gamma \in [1/2, \gamma_{\text{reg}}]$. For every $L \in \mathbb{N}_0 \cup \{\infty\}$ and $m \in \mathbb{N}$, there is an $\mathbb{N}$ -valued random variable $\mathcal{L}_L(\gamma, m)$ satisfying

$$(6.115) \quad \mathbb{P}[\mathcal{L}_L(\gamma, m) > k] \leq C \exp\left(-\frac{(1-\gamma)^2(k-C)_+}{C\bar{\delta}^2|\log \bar{\delta}|}\right), \quad k \in \mathbb{N}.$$

Moreover, almost surely, for every L and m , if $u \in H^1(\square_m)$ is $\mathbf{a}_L$ -harmonic in $\square_m$, then, for every $n \in \mathbb{N}_0$ satisfying $n \leq m - \mathcal{L}_L(\gamma, m)$ and every $z \in 3^n\mathbb{Z}^d$ such that $z + \square_n \subseteq \square_{m-1}$,

$$(6.116) \quad \|u - (u)_{z+\square_n}\|_{\underline{L}^2(z+\square_n)} \leq C3^{-\gamma(m-n)}\|u - (u)_{\square_m}\|_{\underline{L}^2(\square_m)}$$

and

$$(6.117) \quad \left\| \mathbf{a}_L^{1/2} \nabla u \right\|_{\underline{L}^2(z+\square_n)} \leq C3^{(1-\gamma)(m-n)} \left\| \mathbf{a}_L^{1/2} \nabla u \right\|_{\underline{L}^2(\square_m)}.$$

- Liouville theorem: Almost surely, for every $L \in \mathbb{N}_0 \cup \{\infty\}$, if $u \in H_{\text{loc}}^1(\mathbb{R}^d)$ satisfies

$$(6.118) \quad -\nabla \cdot (\mathbf{a}_L \nabla u) = 0 \quad \text{in } \mathbb{R}^d, \quad \liminf_{R \rightarrow \infty} R^{-\gamma_{\text{reg}}} \inf_{c \in \mathbb{R}} \|u - c\|_{\underline{L}^2(B_R)} = 0,$$

then u is constant.

Proof. Fix $\gamma \in [1/2, \gamma_{\text{reg}}]$. We take C_0 at least as large as the constant C of Proposition 6.17, and $\bar{\delta}_0 \leq C^{-1}$; then that proposition applies with exponent γ . For finite L , the two estimates are special cases of Proposition 6.17, for $\mathbf{a}_L$ -harmonic functions and interior cubes: we take $\mathbf{g} = 0$, $x = z$, and $j = n$ in (6.5), and use (6.6). Approximating the trace in $H^{1/2}(\partial \square_m)$ by smooth functions extends the conclusion from smooth to arbitrary H^1 boundary data.

For $L = \infty$, fix m and, for $N \geq m$, let $u_N \in u + H_0^1(\square_m)$ be $\tilde{\mathbf{a}}_N$ -harmonic. By (6.92), Proposition 6.17 supplies the same random variable $\mathcal{L}_N(\gamma, m)$ for every $N \geq m$, and multiplication of the coefficient by $\mathbf{a}_N(0)^{-1}$ changes neither the equation nor the estimate (6.117). Lemma 6.21 and the energy identity for $u_N - u$ give $\|\nabla u_N - \nabla u\|_{\underline{L}^2(\square_m)} \rightarrow 0$. The large-scale Hölder and energy estimates therefore pass to the limit. Taking $\mathcal{L}_\infty(\gamma, m) = \mathcal{L}_N(\gamma, m)$, for any $N \geq m$, also gives (6.115).

For the Liouville assertion, fix $L \in \mathbb{N}_0 \cup \{\infty\}$ and suppose that u satisfies (6.118). By (6.115) and Borel–Cantelli, $\mathcal{L}_L(\gamma_{\text{reg}}, m) \leq m$ for all sufficiently large m . Choose a sequence of radii $R \rightarrow \infty$ and constants $c = c(R) \in \mathbb{R}$ such that $R^{-\gamma_{\text{reg}}}\|u - c\|_{\underline{L}^2(B_R)} \rightarrow 0$, and let m be the largest integer such that $\square_m \subseteq B_R$. Since 3^m is comparable to R and $(u)_{\square_m}$ minimizes $c \mapsto \|u - c\|_{\underline{L}^2(\square_m)}$,

$$3^{-\gamma_{\text{reg}}m}\|u - (u)_{\square_m}\|_{\underline{L}^2(\square_m)} \rightarrow 0.$$

For every fixed $z \in \mathbb{Z}^d$, we apply (6.116) with $\gamma = \gamma_{\text{reg}}$ and $n = 0$ on $z + \square_0$, which is contained in $\square_{m-1}$ once R is large enough, and let $R \rightarrow \infty$. Thus $\nabla u = 0$ almost everywhere in $\mathbb{R}^d$, so u is constant. $\square$

7. THE SCALING LIMIT

In this section we state Theorem A in the precise form in which we prove it. Sections 8 and 9 prove part (i) of Theorem 7.1 and the convergence of the measures in part (ii), and Section 10 proves the remaining assertions.

To prove almost-sure convergence, we realize all rescaled processes using one family of independent fields $(\mathbf{g}_j)_{j \in \mathbb{Z}}$, with $\mathbf{g}_j$ having the law of $\mathbf{g}_0(3^{-j} \cdot)$. Rescaling space by the factor 3^{-N} turns the field with scale 3^k into a field with scale 3^{k-N} . Accordingly, we take the rescaled potential at cutoff N to be $H + \sum_{j=0}^N \mathbf{g}_{-j}$, where $H(x) := \sum_{j=1}^{\infty} (\mathbf{g}_j(x) - \mathbf{g}_j(0))$ collects the fields with scales larger than one; the series converges locally in $C^{1,1}$. Increasing N adds the field with scale 3^{-N} and leaves the others unchanged. Let $K_{N,x}^\omega$ be the law on continuous paths of the diffusion started at x and constructed from the elliptic resolvents of the generator $\bar{\mathbf{a}}_N^{-1}(\Delta + \nabla(H + \sum_{j=0}^N \mathbf{g}_{-j}) \cdot \nabla)$. Its stochastic differential description is

$$(7.1) \quad dY_t = \bar{\mathbf{a}}_N^{-1} \nabla \left(H + \sum_{j=0}^N \mathbf{g}_{-j} \right) (Y_t) dt + \sqrt{2/\bar{\mathbf{a}}_N} dW_t, \quad Y_0 = x.$$

For each N , the fields $\mathbf{g}_{k-N}$ with $k \in \mathbb{N}_0$ have the same joint law as the fields $\mathbf{g}_k(3^N \cdot)$. Thus $K_{N,x}^\omega$, viewed as a random probability measure, has the same distribution as the law of $(3^{-N} X_{3^{2N}t/\bar{\mathbf{a}}_N})_{t \geq 0}$ conditional on the environment, with $X_0 = 3^N x$. The diffusion (7.1) is reversible with respect to

$$\mu_N(dx) := \exp \left(H(x) + \sum_{j=0}^N (\mathbf{g}_{-j}(x) - \tau^2) \right) dx.$$

The measure $\mu_N^0 := e^{-H} \mu_N$ is stationary and has the law of $\mathbf{a}_N(3^N x) dx$. For Gaussian fields, it is the standard approximation of Gaussian multiplicative chaos: its density is obtained by exponentiating $\sum_{j=0}^N \mathbf{g}_{-j}$ and dividing by the expectation of that exponential. We write d_{weak} for a bounded complete metric on the probability measures on $C([0, \infty), \mathbb{R}^d)$ that induces weak convergence.

Theorem 7.1 (Scaling limit). *Assume (g1)–(g4). There exists $\bar{\delta}_0(d) > 0$ such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then there exists a continuous probability kernel $x \mapsto K_x^\omega$ such that the following statements hold.*

(i) Convergence of the path laws: *Almost surely, for every compact set $B \subseteq \mathbb{R}^d$,*

$$(7.2) \quad \lim_{N \rightarrow \infty} \sup_{x \in B} d_{\text{weak}}(K_{N,x}^\omega, K_x^\omega) = 0.$$

Almost surely, the laws $(K_x^\omega)_{x \in \mathbb{R}^d}$ define a continuous, conservative, strong Markov process Z , started at x under K_x^ω . On the same event, its semigroup maps bounded continuous functions to continuous functions, and, for every bounded open set U , the mean exit time of Z from U under K_x^ω is bounded uniformly in $x \in U$. For every $x \in \mathbb{R}^d$, the annealed law of $(3^{-N} X_{3^{2N}t/\bar{\mathbf{a}}_N})_{t \geq 0}$, with $X_0 = 3^N x$, converges to $\mathbb{E}[K_x^\omega]$.

(ii) The reversible measure: *Almost surely, the measures μ_N^0 and μ_N converge locally weakly to measures μ^0 and $\mu = e^H \mu^0$, respectively. Both limits are locally finite, nonatomic, of full support, and singular with respect to Lebesgue measure. The measure μ^0 is stationary and not deterministic, and it satisfies $\mathbb{E}[\mu^0(dx)] = dx$. Almost surely, the process Z is reversible with respect to μ , which is invariant: for every nonnegative Borel function f and every $t \geq 0$,*

$$\int \left(\int f(Z_t) dK_x^\omega \right) \mu(dx) = \int f d\mu.$$

- (iii) Anomalous scaling: There exist constants $C \geq 1$ and $\eta > 0$, depending only on d and the law of $\mathbf{g}_0$, such that, for all $\ell, m \in \mathbb{N}_0$ with $\ell \leq m$,

$$\frac{\bar{\mathbf{a}}_m}{\bar{\mathbf{a}}_\ell} \leq C 3^{-\eta(m-\ell)}.$$

For a physical side length $r > 0$, write $\sigma_r := \inf\{t \geq 0 : Z_t - Z_0 \notin r\Box_0\}$, as in Theorem A. At the discrete scales we write $\sigma_k := \sigma_{3^{-k}}$, with k denoting a depth. Thus the discrete σ_0 and σ_1 correspond to physical sides 1 and 3^{-1} , respectively; the physical-scale subscript $r = 1$ denotes the exit time from the unit cube. For every $x \in \mathbb{R}^d$ and every integer $k \geq 0$,

$$(7.3) \quad \mathbb{E} \left[\int \sigma_k dK_x^\omega \right] \leq C 3^{-(2+\eta)k}.$$

For every $x \in \mathbb{R}^d$, almost surely in ω , the law K_x^ω is mutually singular with every mixture of Brownian laws with arbitrary covariance matrices, and K_x^ω -almost every path satisfies

$$(7.4) \quad \sup\{\gamma \geq 0 : |Z_t - Z_0| = O(t^\gamma) \text{ as } t \downarrow 0\} \leq \frac{1}{2+\eta}.$$

In dimension two, one can take $\eta = \tau^2 / \log 3$.

- (iv) Non-Gaussian marginals: Almost surely, for every $x \in \mathbb{R}^d$ and every $t > 0$, the law of Z_t under K_x^ω is absolutely continuous with respect to μ , and it is not a Gaussian law.

Theorem A follows, with $\mathbf{P}_x = K_x^\omega$: for its exit-time bound, apply (7.3) with $3^{-k-1} < r \leq 3^{-k}$.

8. UNIQUENESS OF THE LIMIT FORM

In this section we prove that the energies $\int A_N |\nabla u|^2 dx$ have a unique limit. We show that all subsequential limits of the Dirichlet forms on cubes coincide (Theorem 8.54). Section 9 pairs this limit with the limit of the reversible measures μ_N to construct the limiting diffusion. The argument has four parts.

- (1) We bound, uniformly in N , the energy that solutions carry in small balls; this includes solutions of Neumann problems whose datum is approximated by a volume source (Sections 8.1–8.2). These bounds control the change in the energy on a fixed cube when a fine-scale field is resampled. We use this control to prove compactness of the finite-volume energies (Section 8.3).
- (2) Along subsequences, the Dirichlet Green operators converge, and their limits define Dirichlet forms. These limit forms are local and consistent under restriction to subcubes, and multiplying the coefficient by a positive continuous function multiplies their energy measures by the same function (Section 8.4).
- (3) On a single event that does not depend on the solution, we bound the number of cubes on which the regularity, extension and harmonic-approximation estimates of Lemma 8.34 fail along every chain of nested cubes (Section 8.5). We integrate this bound against the energy measure of a solution. This selects cubes carrying a fixed fraction of its energy, on which these estimates hold and the solution is close to affine.
- (4) Any two subsequential limit forms are comparable: the second lies between two deterministic multiples of the first. The optimal comparison constants agree, since otherwise replacing a solution by harmonic functions on the selected cubes would improve one of them (Section 8.6). Both the energy saved by a replacement and the errors it must absorb are proportional to the gap between the optimal comparison constants. To obtain errors of this size, we apply the Efron–Stein bound of Section 8.3 to the difference between the minimal energy of the second limit form and a multiple of the minimal energy of the first, rather than to each energy separately (Proposition 8.46). Hence the two limits are proportional.

The factor is one because the coarse-grained matrices of $e^{-H}A_N$ on large cubes approach the identity uniformly in N , and both limits inherit this normalization (Section 8.7).

Throughout, Assumptions (g1)–(g4) hold. Before reducing $\bar{\delta}$, we fix all exponents and tolerances and a finite list of moment orders. Bounds in $L^p(\mathbb{P})$ for the orders in this list are uniform in N . When a result below is stated for every finite p , the moment orders needed for that application, including the larger orders used in Hölder's inequality, spatial maxima and resampling, are fixed first; the disorder threshold then depends on this finite list. This convention applies in particular to Lemmas 8.9, 8.35 and 8.36, and Proposition 8.46. It does not assert all moments at one fixed positive disorder.

8.1. Rescaled coefficients and energies. At fixed N , relabel the fields by replacing $\mathbf{g}_i$ with $\mathbf{g}_{i-N}(3^{-N}\cdot)$, and let $\mathbf{a}_N$ be the cutoff coefficient of (1.7) formed from the relabelled fields. The diffusion with law $K_{N,x}^\omega$ has Dirichlet form $\int A_N |\nabla u|^2 dx$ on $L^2(\mu_N)$, where $A_N := \bar{\mathbf{a}}_N^{-1} d\mu_N/dx$ (see (7.1)). In terms of the relabelled fields,

$$(8.1) \quad A_N(x) = e^{H(x)} \frac{\mathbf{a}_N(3^N x)}{\bar{\mathbf{a}}_N}, \quad A_N^0(x) = e^{-H(x)} A_N(x).$$

If H is replaced by one of its partial sums, then A_N becomes a constant multiple of a stationary cutoff coefficient, so the results of Sections 4 and 6 apply.

Lemma 8.1 (Moments of the field H). *The field H and its partial sums $\sum_{j=1}^{L'} (\mathbf{g}_j - \mathbf{g}_j(0))$ satisfy, on every compact cube Q ,*

$$\|H\|_{C^{1,1}(Q)}\|_{L^p} \leq C_Q \bar{\delta} \sqrt{p}, \quad \mathbb{E} e^{\lambda \|H\|_{C^{1,1}(Q)}} \leq 2e^{C_Q \lambda^2 \bar{\delta}^2} \quad (p \geq 2, \lambda \geq 0),$$

with the same bounds for the partial sums, uniformly in the number L' of terms. Moreover, the partial sums converge to H locally in $C^{1,1}$ almost surely.

Proof. As in the proof of Lemma 6.21, scaling, the mean value theorem and (1.5) give $\|\mathbf{g}_j - \mathbf{g}_j(0)\|_{C^{1,1}(Q)}\|_{L^p} \leq C_Q \bar{\delta} \sqrt{p} 3^{-j}$ for $j \geq 1$. Sum over j and expand the exponential. Since the sum of these norms has finite mean, the partial sums also converge in $C^{1,1}$ almost surely. $\square$

On a bounded cube Q , let

$$(8.2) \quad E_N^Q(u, v) = \int_Q A_N \nabla u \cdot \nabla v dx, \quad \Gamma_N(u) = A_N |\nabla u|^2 dx.$$

The form E_N^Q has domain $H_0^1(Q)$, and we use the same integral for functions in $H^1(Q)$ when prescribing boundary values. Let $\Lambda_{N,Q}(g)$ be the minimum of $E_N^Q(u)$ over $u \in H^1(Q)$ with $u - g \in H_0^1(Q)$. Let G_N^Q be the Dirichlet Green operator, the inverse of $-\nabla \cdot (A_N \nabla)$ on $L^2(Q, dx)$ with zero Dirichlet boundary values, so that

$$(8.3) \quad \langle f, G_N^Q f \rangle = \sup_{v \in H_0^1(Q)} \left(2 \int_Q f v dx - E_N^Q(v) \right).$$

For the constant flux p , the quantity $|Q| p \cdot a_*^{-1}(Q; A_N) p$ is the energy $E_N^Q(v)$ of a solution v of $-\nabla \cdot (A_N \nabla v) = 0$ in Q with boundary flux $A_N \nabla v \cdot n = p \cdot n$, and it has the variational formula (2.13). Since $d\mu_N/dx = \bar{\mathbf{a}}_N A_N$, the same energy, regarded as a form on $L^2(\mu_N)$, has generator $u \mapsto \nabla \cdot (A_N \nabla u)/(\bar{\mathbf{a}}_N A_N)$, the generator of the diffusion (7.1); its resolvent is studied in Section 9.2.

8.2. Local estimates and approximation of the Neumann datum. We derive the local estimates from the coarse-grained inequalities of Section 2, the moments of Proposition 4.1, and the regularity estimates of Proposition 6.17. The constants in these estimates have the required moments, uniformly in N . These estimates also give the Morrey bound (8.10), the coercivity bound (8.5) and the bounds of Lemma 8.22 on cutoff functions, which are used in Section 9.3.

At radii above the smallest field scale 3^{-N} , we apply Proposition 6.17 after replacing H by a partial sum. At smaller radii, elliptic estimates for coefficients with small oscillation complete the bounds. At these small radii we use Lemma 8.6 for the coefficient and its logarithmic gradient. The moments of these bounds grow with N , and positive powers of 3^{-N} absorb this growth.

8.2.1. Coercivity and extension. The coarse-grained bounds of Section 4 also cover scales below the smallest field scale. We first apply them on a rescaled cube. We call a cube $z + 3^{-k}\square_0$ of a triadic mesh a **cell**.

Lemma 8.2 (Ellipticity moments on a cell). *Fix $s \in (0, 1]$, $q \in \{1, 2\}$ and a bounded cube Q . For every $p < \infty$ and all sufficiently small $\bar{\delta}$, uniformly for $0 \leq k \leq N$ and cubes $z + 3^{-k}\square_0 \subseteq Q$,*

$$(8.4) \quad \|\Lambda_{s,q}(z + 3^{-k}\square_0; A_N) + \lambda_{s,q}^{-1}(z + 3^{-k}\square_0; A_N)\|_{L^p} \leq C_{p,s,Q} \exp(C(p + p^2)\bar{\delta}^2 k).$$

In particular, on a fixed cell U these moments are bounded uniformly in N .

Proof. Choose a finite moment order $\xi \geq 4p$ large enough for Proposition 4.1 at exponent $s/2$, then decrease $\bar{\delta}$ so that its remaining hypotheses hold. The estimate (4.4), the comparison of the weighted sums at exponents $s/2$ and s , and (2.38) give

$$\sup_m \|\Lambda_{s,q}(\square_0; A_m^0) + \lambda_{s,q}^{-1}(\square_0; A_m^0)\|_{L^{2p}} < \infty.$$

By definition of A_N , the coefficient $A_N(z + 3^{-k}y)$ is the product of three factors: the scalar $\frac{\bar{a}_{N-k}}{\bar{a}_N} \exp(H(z) + \sum_{j < k} (\mathbf{g}_{-j}(z) - \tau^2))$, the ratio of $\exp(H + \sum_{j < k} \mathbf{g}_{-j})$ at $z + 3^{-k}y$ to its value at z , and a function of y with the law of A_{N-k}^0 . Summing the bounds on the gradients of the fields as in the proof of Lemma 8.1 bounds the oscillation of $H + \sum_{j < k} \mathbf{g}_{-j}$ over the cube $z + 3^{-k}\square_0$ by a sub-Gaussian variable of size $C_Q \bar{\delta}$. The scalar involves the values of k independent fields. By (3.25), for every fixed $v < \infty$ the sum of the scalar and its inverse, multiplied by the exponential of this oscillation, has L^v norm at most a constant depending on v and Q times $e^{C(v+v^2)\bar{\delta}^2 k}$. Monotonicity and homogeneity of the coarse-grained ellipticity constants, followed by Hölder's inequality, prove (8.4). $\square$

Lemma 8.3 (Coercivity of the form E_N^Q). *Let Q be a fixed cube of side at most one. For every $N \geq 0$ there is a random variable K_N such that, for every $v \in H_0^1(Q)$ and every $w \in H^1(Q)$,*

$$(8.5) \quad \|v\|_{H^{3/4}(Q)}^2 \leq K_N E_N^Q(v) \quad \text{and} \quad \|w - (w)_Q\|_{H^{3/4}(Q)}^2 \leq K_N E_N^Q(w).$$

For every $p < \infty$ there is $\bar{\delta}_0(d, p) > 0$ such that $\sup_N \|K_N\|_{L^p} < \infty$ if $\bar{\delta} \leq \bar{\delta}_0$.

Proof. On $\square_0$, take $s = 1/8$. By the functional inequalities (2.43), (2.45) and (2.47), followed by $B_{2,\infty}^{7/8} \subseteq H^{3/4}$, we obtain (8.5) with $K_N = C\lambda_{1/8,1}^{-1}(\square_0; A_N)$. Lemma 8.2 bounds its L^p norms uniformly in N . For a fixed smaller cube, rescale the same inequalities and use (8.4). If its side is not triadic, then the remaining dilation lies in $[1, 3]$. Apply the same argument after replacing $\bar{\delta}$ by $C_d \bar{\delta}$, with constants depending on Q . $\square$

Lemma 8.4 (Boundary extension and bounds on grid cubes). *Fix $\beta \in (1/2, 1)$ before choosing $\bar{\delta}$, and set $\sigma = \frac{1}{2}(\beta - 1/2)$. The following statements hold.*

(i) There is $C(d, \beta) < \infty$ such that, for every cube $q = z + r\Box_0$ with $r \leq 1$ and every $g \in C^\beta(\partial q)$,

$$(8.6) \quad \Lambda_{N,q}(g) \leq C \Lambda_{\sigma/2,2}(q; A_N) r^{d-2} \|g(z + r \cdot)\|_{C^\beta(\partial\Box_0)/\mathbb{R}}^2.$$

(ii) Let Q be a bounded cube, and fix finitely many grids, each consisting of the cubes $z + r\Box_0$ with $r = 3^{-k}$, $k \geq 0$, and z in $z_0 + r\mathbb{Z}^d$ for a point z_0 fixed for each grid; we call their cubes contained in Q grid cubes. For every $\eta > 0$ and $p < \infty$ there is $\bar{\delta}_0(d, \beta, \eta, p) > 0$ such that, if $\bar{\delta} \leq \bar{\delta}_0$, then there are random variables K_N with $\sup_N \|K_N\|_{L^p} < \infty$ such that, for every N and every grid cube q of side $r \geq 3^{-N}$,

$$(8.7) \quad \Lambda_{\sigma/2,2}(q; A_N) + \lambda_{\sigma/2,2}^{-1}(q; A_N) \leq K_N r^{-\eta}.$$

Proof. Step 1. We prove (8.6). The change of variables $x = z + ry$ multiplies the unnormalized energy by r^{d-2} and leaves the volume-normalized constant $\Lambda_{\sigma/2,2}$ unchanged, so we may take $q = \Box_0$. We construct an extension h of g directly. First extend g to a β -Hölder function $\tilde{g}$ on $\mathbb{R}^d$, and let $\tilde{g}_k$ be its mollification at scale 3^{-k} . Let $\rho(x) := \text{dist}(x, \partial\Box_0)$ and use a smooth Whitney partition $\{\chi_k\}$ with bounded overlap, subordinate to the layers where ρ is comparable to 3^{-k} , satisfying $|D^i \chi_k| \leq C3^{ik}$ for $i = 1, 2$ and $\sum_k \chi_k = 1$. Set $h := \sum_k \chi_k \tilde{g}_k$. The mollifications $\tilde{g}_k$ on overlapping layers differ by at most $C[g]_{C^\beta} 3^{-k\beta}$. Differentiating h and using $\sum_k D^i \chi_k = 0$ shows that h is smooth inside the cube, has continuous boundary values g , and satisfies

$$|\nabla h(x)| \leq C[g]_{C^\beta} \rho(x)^{\beta-1}, \quad |D^2 h(x)| \leq C[g]_{C^\beta} \rho(x)^{\beta-2}.$$

For the fractional norm of ∇h , split pairs at a distance comparable to their distance from the boundary. On the closer pairs use the Hessian bound; on the remaining pairs use the gradient bound. Both contributions are bounded by

$$C[g]_{C^\beta}^2 \int_{\Box_0} \rho(x)^{2\beta-2-2\sigma} dx \leq C(d, \beta) [g]_{C^\beta}^2,$$

since $2\beta - 2 - 2\sigma > -1$. The same gradient bound gives its L^2 norm. Thus $h \in H^{1+\sigma}(\Box_0)$ and $\|\nabla h\|_{H^\sigma} \leq C(d, \beta) [g]_{C^\beta}$. Applying (2.50) with zero right side, $s = \sigma$ and boundary datum h , we find that the A_N -harmonic function v with $v = h$ on $\partial\Box_0$ satisfies

$$\Lambda_{N,\Box_0}(g) = \|A_N^{1/2} \nabla v\|_{L^2}^2 \leq C\sigma^{-3} \Lambda_{\sigma/2,2}(\Box_0; A_N) \|\nabla h\|_{H^\sigma}^2.$$

Subtracting a constant from g does not change $\Lambda_{N,\Box_0}(g)$ or its Hölder seminorm. The extension just constructed therefore gives $\|\nabla h\|_{H^\sigma} \leq C[g]_{C^\beta(\partial\Box_0)/\mathbb{R}}$. Since σ depends only on β , this proves (8.6).

Step 2. We prove (8.7). Since L^p norms increase with p , we may assume that $p > 1$ and $p\eta > d$. Apply Lemma 8.2 with exponents $\sigma/2$ and 2. There are at most $C3^{dk}$ grid cubes at depth k , so taking the maximum over them introduces a factor $C3^{dk/p}$ in L^p . Reduce $\bar{\delta}$ so that $C(p + p^2)\bar{\delta}^2 < \frac{1}{2}(\eta - d/p) \log 3$. For K_N the supremum, over $k \leq N$ and grid cubes q of depth k , of $3^{-\eta k} (\Lambda_{\sigma/2,2}(q; A_N) + \lambda_{\sigma/2,2}^{-1}(q; A_N))$, Minkowski's inequality gives

$$\|K_N\|_{L^p} \leq C_p \sum_{k \geq 0} 3^{-(\eta - \frac{d}{p})k} e^{C(p+p^2)\bar{\delta}^2 k} \leq C_p.$$

The right side does not depend on N , so $\sup_N \|K_N\|_{L^p} < \infty$. By the definition of K_N , every grid cube q of side $r = 3^{-k} \geq 3^{-N}$ satisfies (8.7). $\square$

8.2.2. Morrey bounds for the Dirichlet problem. To apply the regularity estimates with a bounded volume source, we first write that source as a divergence.

Lemma 8.5 (The gradient of the Newtonian potential). *Let $f \in L^\infty$ be supported in a cube of side R , and let g be the gradient of the Newtonian potential of f , that is, the convolution of f*

with the gradient of the fundamental solution of the Laplacian. Then $\nabla \cdot g = f$ and, for all x and y with $|x - y| \leq R$,

$$(8.8) \quad |g(x) - g(y)| \leq C\|f\|_\infty|x - y|(1 + \log(R/|x - y|)) \leq CR^{1/2}\|f\|_\infty|x - y|^{1/2}.$$

In particular, $[g]_{W^{1/2,\infty}} \leq CR^{1/2}\|f\|_\infty$ on that cube.

Proof. The kernel of the convolution and its gradient are bounded by $C|z|^{1-d}$ and $C|z|^{-d}$. If x is farther than $2\sqrt{d}R$ from the supporting cube, the derivative bound gives $|g(x) - g(y)| \leq C\|f\|_\infty|x - y|$ directly. Otherwise the support lies in $B_{CR}(x)$. We split the convolution according to whether the integration variable z satisfies $|z - x| \leq 2|x - y|$. The part with $|z - x| \leq 2|x - y|$ is bounded by $C\|f\|_\infty|x - y|$, both at x and at y . For $|z - x| > 2|x - y|$, the mean value theorem bounds the difference of the kernel at $x - z$ and at $y - z$ by $C|x - y||x - z|^{-d}$. The integral over $2|x - y| < |z - x| < CR$ is at most $C|x - y|\|f\|_\infty \log(CR/|x - y|)$. The second inequality in (8.8) follows from $(|x - y|/R)^{1/2}(1 + \log(R/|x - y|)) \leq C$. $\square$

Lemma 8.6 (Pointwise extremes of the coefficient). *Let Q be a fixed cube, and set $D_N = 3^{-N}\|\nabla \log A_N\|_{L^\infty(Q)}$. For every p there are $C_p < \infty$ depending on p and Q , $C_d < \infty$ and $c_d > 0$ such that, if $\bar{\delta} \leq c_d/p$, then*

$$\|D_N\|_{L^p} \leq C_p(1 + N)^{1/2},$$

and

$$(8.9) \quad \|A_N\|_{L^\infty(Q)} + \|A_N^{-1}\|_{L^\infty(Q)}\|_{L^p} \leq C_p e^{(C_d\bar{\delta} + C_p\bar{\delta}^2)N}.$$

The linear term $C_d\bar{\delta}N$ in the exponent arises from taking the maximum over the cubes of side 3^{-N} in Q .

Proof. Step 1. We bound D_N . We have $\nabla \log A_N = \nabla H + \sum_{j=0}^N \nabla \mathbf{g}_{-j}$ and, in law, $\|\nabla \mathbf{g}_{-j}\|_{L^\infty(Q)}$ is at most 3^j times the maximum, over at most $C3^{dj}$ cells, of the regularity norms on unit cells from (1.5). The maximum of n sub-Gaussian variables with parameter $\bar{\delta}$ has L^p norm at most $C\bar{\delta}(\sqrt{\log n} + \sqrt{p})$. Therefore

$$\|D_N\|_{L^p} \leq 3^{-N} \|\nabla H\|_{L^\infty(Q)}\|_{L^p} + \sum_{j \leq N} 3^{j-N} C_p \bar{\delta} \sqrt{1 + j} \leq C_p \sqrt{1 + N}.$$

This is the first estimate of the lemma.

Step 2. We bound $\|A_N\|_{L^\infty(Q)}$ and $\|A_N^{-1}\|_{L^\infty(Q)}$. The deterministic factor $\bar{\mathbf{a}}_N^{-1}e^{-(N+1)\tau^2}$ of A_N satisfies $1 \leq e^{(N+1)\tau^2}\bar{\mathbf{a}}_N \leq e^{C\bar{\delta}^2(N+1)}$ by (2.14), Proposition 5.7 and (1.6), and $e^{\|H\|_\infty}$ has moments of every order by Lemma 8.1. By Hölder's inequality, it therefore suffices to bound the moments of $\exp(\sup_Q |\sum_{j \leq N} \mathbf{g}_{-j}|)$. Cover Q by at most $C3^{dN}$ cubes Q_i of side 3^{-N} with centers x_i . On each of them, $\sup_{Q_i} |\sum_{j \leq N} \mathbf{g}_{-j}| \leq |\sum_{j \leq N} \mathbf{g}_{-j}(x_i)| + \sum_{j \leq N} \text{osc}_{Q_i} \mathbf{g}_{-j}$, where the first term is sub-Gaussian with parameter $C\bar{\delta}\sqrt{N+1}$ and the second is sub-Gaussian with parameter $C\bar{\delta}$. Hölder's inequality gives $\mathbb{E} \exp(q \sup_{Q_i} |\sum_{j \leq N} \mathbf{g}_{-j}|) \leq C e^{Cq^2\bar{\delta}^2(N+1)}$. For $q \geq p$, the inequality $\exp(p \sup_Q |\sum_{j \leq N} \mathbf{g}_{-j}|) \leq (\sum_i \exp(q \sup_{Q_i} |\sum_{j \leq N} \mathbf{g}_{-j}|))^{p/q}$ and Jensen's inequality bound $\mathbb{E} \exp(p \sup_Q |\sum_{j \leq N} \mathbf{g}_{-j}|)$ by $(C3^{dN} e^{Cq^2\bar{\delta}^2(N+1)})^{p/q}$. We choose $q = \bar{\delta}^{-1}\sqrt{d \log 3/C}$, which is admissible when $\bar{\delta} \leq c_d/p$; the exponent is then at most $C_d p \bar{\delta} (N+1)$. $\square$

Proposition 8.7 (Morrey bounds at all radii). *Let $t \in (d-1, d)$ and $\alpha \in (0, 1)$, fix a finite list of moment orders, and let Q be a fixed cube of side at most one. There is $\bar{\delta}_0(d, t, \alpha, \text{list}) > 0$ such that, if $\bar{\delta} \leq \bar{\delta}_0$, then there are random variables K_N with $\sup_N \|K_N\|_{L^p} < \infty$ for every order p in the list, such that, almost surely, the following estimates hold for every $f \in L^\infty(Q)$ and $\varphi \in C^2(\bar{Q})$. Here $u_N \in H^1(Q)$ solves $-\nabla \cdot (A_N \nabla u_N) = f$ in Q with $u_N = \varphi$ on ∂Q .*

(i) Morrey bound: for every $x \in Q$ and $r \in (0, 1]$,

$$(8.10) \quad \Gamma_N(u_N)(B_r(x) \cap Q) \leq K_N(\|f\|_\infty + \|\varphi\|_{C^2})^2 r^t.$$

(ii) Hölder bound: $\|u_N\|_{C^\alpha(\bar{Q})} \leq K_N(\|f\|_\infty + \|\varphi\|_{C^2})$ and, if $\varphi = 0$,

$$(8.11) \quad \|G_N^Q f\|_{C^\alpha(\bar{Q})} \leq K_N \|f\|_\infty.$$

Proof. The proof proceeds in seven steps, and the quantity we track is the energy $\Gamma_N(u_N)(B_r(x) \cap Q)$ of small balls. Step 1 replaces H by a partial sum of its series, so that the coefficient becomes a multiple of a cutoff coefficient. Steps 2 and 3 prove the Morrey bound (8.10) at radii $r \geq 3^{-N}$ from the regularity estimates for cutoff coefficients, and Steps 4 and 5 prove it at radii $r \leq 3^{-N}$, away from ∂Q and near ∂Q , from estimates for coefficients with small oscillation. The main difficulty lies at these small radii: there the moments of the coefficient bounds of Lemma 8.6 grow with N , and positive powers of 3^{-N} absorb this growth. Step 6 removes the truncation, and Step 7 proves the Hölder bounds.

Rescale so that $Q = y_0 + \square_0$. In the coordinates $y = 3^N x$ of the relabelled fields, the domain is $3^N y_0 + \square_N$, a deterministic translate of $\square_N$. Throughout the proof, K_N denotes a random variable with $\sup_N \|K_N\|_{L^p} < \infty$ for every order p in the list, which may change from line to line. Choose the exponents as follows. Take $\alpha_M \in (1/2, 1)$ with $\alpha_M > \max\{\alpha, (t+2-d)/2\}$, and set $t_0 = d-2+2\alpha_M > t$. Choose t_1 with $\max\{t, d-2+2\alpha\} < t_1 < t_0$, and require $\bar{\delta}$ to be so small that $\alpha_M \leq 1 - C\bar{\delta}|\log \bar{\delta}|^{1/2}$. We also take $\bar{\delta}$ so small that the tail condition (8.13) of Step 3 holds for every order in the list and for the finitely many larger orders used in Hölder's inequality, and that the moment absorptions in Steps 4, 5 and 7 hold; once the exponents and the orders are fixed, each of these is a condition on $\bar{\delta}$ alone.

Step 1. We replace H by a partial sum $\sum_{j=1}^{L'}(\mathbf{g}_j - \mathbf{g}_j(0))$ of its series, which makes the coefficient a constant multiple of a cutoff coefficient. In Steps 1–5 we keep the notation H , A_N , u_N and Γ_N for the truncated objects; the truncation is removed in Step 6. By (8.1) and (1.7),

$$A_N(3^{-N}y) = \bar{\mathbf{a}}_N^{-1} e^{L'\tau^2} \exp\left(-\sum_{i=N+1}^{N+L'} \mathbf{g}_i(0)\right) \mathbf{a}_{N+L'}(y),$$

so $A_N(3^{-N}\cdot)$ is a random scalar multiple of the cutoff coefficient $\mathbf{a}_{N+L'}$. In the coordinates y , the rescaled function $u_N(3^{-N}\cdot)$ solves $-\nabla \cdot (\mathbf{a}_{N+L'} \nabla u_N(3^{-N}\cdot)) = \nabla \cdot g$, where g is the vector field constructed in Lemma 8.5 for the source $f(3^{-N}\cdot)$, divided by 3^{2N} and by this random scalar and extended by zero. Moreover, $u_N(3^{-N}\cdot) = \varphi(3^{-N}\cdot) \in C^{1,1/2}$ on the boundary. Since $N+L' \geq N$, the estimate (6.6) applies with $m = N$; we write $\mathcal{L} = \mathcal{L}_{N+L'}(\alpha_M, N)$ for the minimal scale in that estimate.

Step 2. We prove the Morrey bound (8.10) at the radii $3^{-N} \leq r \leq 3^{-\mathcal{L}}$. For $x \in Q$ and $0 \leq n \leq N - \mathcal{L}$, we have $B_{3^{n-N/2}}(x) \cap Q \subseteq 3^{-N}((3^N x + \square_n) \cap (3^N y_0 + \square_N))$. Substituting the definitions of the random scalar and of $\mathbf{b}_{N+L',N}$, the constants $L'\tau^2$ cancel, and the random scalar times the average $(\mathbf{b}_{N+L',N})_{3^N y_0 + \square_N}$ equals $\oint_Q e^H dx$. We square (6.6), multiply it by the volume, which is at most 3^{dn} , and return to the x -coordinates. Using Lemma 8.5 with $R = 3^N$ and bounding $\oint_Q e^H dx$ and its inverse by $e^{\|H\|_{L^\infty(Q)}}$, we obtain, for $r = 3^{n-N}/2$,

$$(8.12) \quad \Gamma_N(u_N)(B_r(x) \cap Q) \leq C r^{t_0} \left(\Gamma_N(u_N)(Q) + e^{\|H\|_{L^\infty(Q)}} (\|f\|_\infty^2 + \|\varphi\|_{C^2}^2) \right).$$

For windows meeting ∂Q , the energy estimate used here is the boundary version obtained in Step 6 of the proof of Proposition 6.1, from Lemma 6.8 at its nearest good anchor, together with its cutoff extension in Proposition 6.17. The minimal scale $\mathcal{L}$ constructed in Propositions 6.1 and 6.17

is independent of the window center. Thus the same calculation gives (8.12) near the boundary, with the single scalar cancellation displayed above.

The factor $e^{\|H\|_{L^\infty(Q)}}$ has moments bounded uniformly in N and L' by Lemma 8.1. The random variable $\mathcal{L}$ does not depend on x .

Step 3. We treat the remaining large radii $3^{-\mathcal{L}} < r \leq 1$ and bound the total energy. For these radii, $\Gamma_N(u_N)(B_r \cap Q) \leq \Gamma_N(u_N)(Q) \leq 3^{t_1 \mathcal{L} r^{t_1}} \Gamma_N(u_N)(Q)$. Together with (8.12), this gives a bound $K_N r^{t_1}$ at every radius $r \geq 3^{-N}$, also when $\mathcal{L} > N$. For an order p in the list and for the larger orders used in Hölder's inequality, (6.91) gives $\mathbb{E} 3^{pt_1 \mathcal{L}} < \infty$ as soon as

$$(8.13) \quad \frac{(1 - \alpha_M)^2}{C \bar{\delta}^2 |\log \bar{\delta}|} > pt_1 \log 3,$$

which is a condition on $\bar{\delta}$ alone once α_M , p and t_1 are fixed. The global energy has moments: by (8.3), (8.5) and (8.6),

$$\Gamma_N(u_N)(Q) \leq 2\Lambda_{N,Q}(\varphi) + 2\langle f, G_N^Q f \rangle \leq 2\Lambda_{N,Q}(\varphi) + 2K_N \|f\|_{L^2}^2,$$

uniformly in the truncation, since the truncated coefficient satisfies the same estimates.

Step 4. We prove the Morrey bound for $r \leq 3^{-N}$ on balls away from ∂Q . Let $p_1 < \infty$ satisfy $d - 2d/p_1 > t_1$ and $1 - d/p_1 > \alpha$, and let ℓ_N be $3^{-N}(1 + D_N)^{-1}$ times a small constant depending on d and p_1 , chosen below. For $x_0 \in Q$ with $B_{2\ell_N}(x_0) \subseteq Q$, the oscillation of $\log A_N$ on $B_{2\ell_N}(x_0)$ is at most $2\ell_N D_N / 3^{-N}$, hence at most twice the small constant. Dividing the equation by $A_N(x_0)$ gives an equation with coefficient $A_N/A_N(x_0)$, whose logarithm is bounded by twice the small constant on the ball, and source $f/A_N(x_0)$. By Meyers [1963, Theorem 2, equation (48)], the admissible gradient exponent tends to infinity as the ellipticity ratio tends to one. Choose the small constant so that the admissible exponent exceeds p_1 , subtract the mean of u_N , and apply that estimate with initial gradient exponent 2 and a finite source exponent greater than d . After scaling to the unit ball, this gives

$$\left(\int_{B_{\ell_N}(x_0)} |\nabla u_N|^{p_1} \right)^{1/p_1} \leq C \left(\int_{B_{2\ell_N}(x_0)} |\nabla u_N|^2 \right)^{1/2} + C \ell_N A_N(x_0)^{-1} \|f\|_\infty.$$

Hölder's inequality on $B_r(x_0)$, $r \leq \ell_N$, and the fact that A_N is within a fixed factor of $A_N(x_0)$ on $B_{2\ell_N}(x_0)$ yield, for every $r \leq \ell_N$,

$$(8.14) \quad \Gamma_N(u_N)(B_r) \leq C \left(\frac{r}{\ell_N} \right)^{d-2d/p_1} \Gamma_N(u_N)(B_{2\ell_N}) + C \|A_N^{-1}\|_{L^\infty(Q)} \|f\|_\infty^2 \ell_N^{2+2d/p_1} r^{d-2d/p_1}.$$

We divide by r^t . Since $d - 2d/p_1 > t_1 > t$, the right side is largest at $r = \ell_N$, so

$$\Gamma_N(u_N)(B_r) r^{-t} \leq C(1 + D_N)^t 3^{tN} \Gamma_N(u_N)(B_{C3^{-N}}) + C \|A_N^{-1}\|_{L^\infty(Q)} \|f\|_\infty^2 3^{-(d+2-t)N}.$$

For $\ell_N \leq r \leq 3^{-N}$, we use $\Gamma_N(u_N)(B_r) \leq \Gamma_N(u_N)(B_{C3^{-N}}) \leq (3^{-N}/\ell_N)^t r^t 3^{tN} \Gamma_N(u_N)(B_{C3^{-N}})$. Together with Step 3 at $r = C3^{-N}$, this gives, for every $r \leq 3^{-N}$,

$$\Gamma_N(u_N)(B_r) \leq C(1 + D_N)^t 3^{-(t_1-t)N} K_N r^t + C \|A_N^{-1}\|_{L^\infty(Q)} \|f\|_\infty^2 3^{-(d+2-t)N} r^t.$$

By (8.9), the positive powers $3^{-(t_1-t)N}$ and $3^{-(d+2-t)N}$ absorb the polynomially growing moments of D_N and the moments of $\|A_N^{-1}\|_{L^\infty(Q)}$, once $\bar{\delta}$ is small enough for the finitely many larger moment orders used in Hölder's inequality.

Step 5. We prove the Morrey bound for $r \leq 3^{-N}$ near the faces of Q , where the Dirichlet condition holds. Let x_0 be within distance $2\ell_N$ of a face, say $\{x_d = 0\}$. The function $u_N - \varphi$ vanishes on the face and solves $-\nabla \cdot (A_N \nabla (u_N - \varphi)) = f + \nabla \cdot (A_N \nabla \varphi)$, where $\|\nabla \cdot (A_N \nabla \varphi)\|_\infty \leq C A_N(x_0) (\|D^2 \varphi\|_\infty + \ell_N^{-1} \|\nabla \varphi\|_\infty)$ on the ball around x_0 on which $\log A_N$ has small oscillation, as in Step 4. We reflect $u_N - \varphi$ oddly, A_N evenly and the source oddly across the face; the reflected function solves the reflected equation, with no additional source concentrated on the face. Indeed,

for a compactly supported test function on the doubled ball, the difference of the test function and its reflection, restricted to Q , lies in H^1 , has zero trace on the face, and vanishes near the other faces, so it is an admissible test function for the original equation. The map that folds the reflected domain back onto Q is 1-Lipschitz, so the oscillation of the logarithm of the reflected coefficient on a ball B centered on the face is at most that of $\log A_N$ on $B \cap \overline{Q}$. Hence (8.14) applies to the reflected function and coefficient, since the proof of (8.14) uses only the boundedness and the small oscillation of the coefficient, not its differentiability across the face. Since the energy measure of the reflected function on a symmetric set B is twice the energy measure of $u_N - \varphi$ on $B \cap Q$, and

$$\Gamma_N(u_N - \varphi)(B_{C3^{-N}} \cap Q) \leq 2\Gamma_N(u_N)(B_{C3^{-N}} \cap Q) + C\|A_N\|_{L^\infty(Q)}\|\nabla\varphi\|_\infty^2 3^{-dN},$$

we obtain

$$\begin{aligned} \Gamma_N(u_N)(B_r \cap Q) &\leq C(r/\ell_N)^{d-2d/p_1}\Gamma_N(u_N)(B_{C3^{-N}} \cap Q) + C\|A_N^{-1}\|_{L^\infty(Q)}\|f\|_\infty^2 \ell_N^{2+2d/p_1} r^{d-2d/p_1} \\ &\quad + C\|A_N\|_{L^\infty(Q)}\|\varphi\|_{C^2}^2(1 + D_N)^d \ell_N^{2d/p_1} r^{d-2d/p_1}. \end{aligned}$$

Dividing the last term by r^t and maximizing at $r = \ell_N$ gives $C\|A_N\|_{L^\infty(Q)}\|\varphi\|_{C^2}^2 3^{-(d-t)N}(1 + D_N)^t$. The positive power $3^{-(d-t)N}$ absorbs its moments as in Step 4. Near an edge or a corner of Q , we reflect across each face meeting the neighborhood of x_0 . On a set invariant under these reflections, the energy of the reflected solution is two to the power of the number of these faces times the energy of the solution in the part of the set inside Q . Divide by this factor to obtain the estimates on the original cube.

Step 6. We remove the truncation. By Lemma 8.1, the truncated coefficients converge uniformly on $\overline{Q}$ to the untruncated A_N almost surely. For fixed N , they are therefore bounded away from zero once the truncation is sufficiently large. Testing the difference of the two weak equations with the difference of the solutions, which lies in $H_0^1(Q)$, bounds its energy by the uniform distance between the coefficients times the product of the L^2 norms of the gradient of the untruncated solution and of the gradient of the difference. The lower bound on the coefficients then shows that the gradients converge in $L^2(Q)$. Consequently, for every x and r , the truncated energies converge to $\Gamma_N(u_N)(B_r \cap Q)$. By Steps 2–5, these energies satisfy (8.10) with random constants whose L^p norms are bounded uniformly in N and in the truncation, for every order p in the list. Their limit inferior gives a constant for the untruncated estimate, and Fatou's lemma preserves these moment bounds.

Step 7. We prove the Hölder bounds. At radii above 3^{-N} , (6.5) is a Campanato bound at every $x \in Q$, with exponent $\alpha_M > \alpha$ and the same source norm; the large radii $3^{-L} < r \leq 1$ are handled as in Step 3, and the L^2 oscillation on Q is bounded by the energy through (8.5). At radii below 3^{-N} , the W^{1,p_1} bound of Step 4 and Morrey's inequality bound the oscillation on B_{ℓ_N} . After division by r^α , the energy on the ball of radius $C3^{-N}$, the volume source and the boundary datum contribute, respectively,

$$\begin{aligned} CK_N^{1/2}\|A_N^{-1}\|_{L^\infty(Q)}^{1/2} 3^{-((t_1-d+2)/2-\alpha)N}(1 + D_N)^C, \\ C\|A_N^{-1}\|_{L^\infty(Q)}\|f\|_\infty 3^{-(2-\alpha)N}(1 + D_N)^C, \\ C\|\varphi\|_{C^2} 3^{-(1-\alpha)N}(1 + D_N)^C. \end{aligned}$$

All three powers of 3^{-N} are positive by the choice of the parameters. Chaining balls of radius ℓ_N inside a cube of side 3^{-N} introduces another fixed power of $1 + D_N$. Above the smallest field scale, telescoping the cube averages produces the geometric sum $\sum_{s \leq R} s^{\alpha_M} \leq CR^{\alpha_M}$. With the same absorption of moments as in Step 4, this proves the Hölder bounds. Approximate a general bounded source by smooth sources. Remove the truncation as in Step 6: the equicontinuity just proved and the convergence in L^2 give uniform convergence of the truncated solutions. $\square$

Remark 8.8 (Morrey and Hölder bounds on larger cubes). For the coefficient A_N , both conclusions of Proposition 8.7 hold on every fixed cube Q , with moment constants depending on Q . The disorder threshold depends only on d, t, α and the prescribed moment orders. To see this, choose $j \geq 1$ so that the side R of Q satisfies $R \leq 3^j$, and set $\lambda = 3^j$. The scale shift

$$(S_j \omega)_\ell(y) = \mathbf{g}_{\ell+j}^\omega(3^j y)$$

preserves the law of the fields. Splitting the anchored infrared sum at j gives

$$(8.15) \quad A_N^\omega(\lambda y) = c_N A_{N+j}^{S_j \omega}(y), \quad c_N = \frac{\bar{\mathbf{a}}_{N+j}}{\bar{\mathbf{a}}_N} \exp\left(j\tau^2 - \sum_{i=1}^j \mathbf{g}_i^\omega(0)\right).$$

By (3.24), both c_N and c_N^{-1} are at most $\exp(j\tau^2 + |\sum_{i=1}^j \mathbf{g}_i(0)|)$, uniformly in N . This envelope has every finite moment by (g2).

If u_N solves the Dirichlet problem on Q , then $v(y) = u_N(\lambda y)$ solves it on $\lambda^{-1}Q$, with coefficient $A_{N+j}^{S_j \omega}$, source $\lambda^2 c_N^{-1} f(\lambda \cdot)$ and boundary datum $\varphi(\lambda \cdot)$. The weak equation follows by changing variables in its test integrals, and the energy satisfies

$$\Gamma_N^\omega(u_N)(B_r(x) \cap Q) = c_N \lambda^{d-2} \Gamma_{N+j}^{S_j \omega}(v)(B_{r/\lambda}(x/\lambda) \cap \lambda^{-1}Q).$$

Apply Proposition 8.7 on the cube $\lambda^{-1}Q$ of side at most one. Since $t > d - 2$ and $\lambda \geq 1$, the radius factor λ^{d-2-t} is at most one. Also $\|\varphi(\lambda \cdot)\|_{C^2} \leq (1 + \lambda + \lambda^2)\|\varphi\|_{C^2}$, and $c_N \max\{1, c_N^{-1}\}^2 = \max\{c_N, c_N^{-1}\}$. The energy and Hölder bounds are consequently transferred by multiplying the smaller-cube majorant at cutoff $N + j$ by $(1 + \lambda + \lambda^2)^2 \exp(j\tau^2 + |\sum_{i=1}^j \mathbf{g}_i(0)|)$. For each requested order p , use the smaller-cube proposition at order $2p$ before choosing the disorder threshold. Hölder's inequality and preservation of the sample law then bound the product in L^p , uniformly in N . This proves both the Morrey and Hölder estimates on Q for A_N ; their moment constants may depend on the fixed position of Q .

8.2.3. Approximating the Neumann datum by a volume source. We approximate the functional $v \mapsto \int_Q p \cdot \nabla v$, which encodes the Neumann datum $p \cdot n$, by a bounded volume source near the faces.

Let $Q = (0, 1)^d$, let $p \in \mathbb{R}^d$ with $|p| = 1$, and let

$$L_p(v) = \int_Q p \cdot \nabla v \, dx = \int_{\partial Q} (p \cdot n) \operatorname{Tr} v \, d\mathcal{H}^{d-1},$$

where n is the outer normal. For $0 < \epsilon < 1/8$, let f_ϵ be the sum over $i = 1, \dots, d$ of p_i times the difference of two smooth bumps of integral one, the first supported in the layer $\{\epsilon < 1 - x_i < 2\epsilon\}$ near the face $\{x_i = 1\}$ and the second in the layer $\{\epsilon < x_i < 2\epsilon\}$ near the face $\{x_i = 0\}$, both obtained from one fixed profile by dilation by ϵ , and let

$$(8.16) \quad L_{p,\epsilon}(v) = \int_Q f_\epsilon v \, dx.$$

Each summand of f_ϵ has integral zero, $f_\epsilon \in C^\infty(\bar{Q})$, $\|f_\epsilon\|_\infty \leq C\epsilon^{-1}$, and $L_{p,\epsilon}(1) = 0$.

Lemma 8.9 (Error of the volume-source approximation of the Neumann datum). *Let $v_N, v_{N,\epsilon}$ be the mean-zero Riesz representer of $L_p, L_{p,\epsilon}$ with respect to E_N^Q , so that $E_N^Q(v_N) = p \cdot a_*^{-1}(Q; A_N)p$. For every $v \in H^1(Q)$,*

$$(8.17) \quad |L_p(v) - L_{p,\epsilon}(v)| \leq C\epsilon^{1/4} \|v - (v)_Q\|_{H^{3/4}(Q)}.$$

With K_N from (8.5), this gives

$$(8.18) \quad E_N^Q(v_N - v_{N,\epsilon}) \leq CK_N \epsilon^{1/2}.$$

For every moment order q in the list, if $\bar{\delta}$ is small enough,

$$(8.19) \quad \|E_N^Q(v_N) - E_N^Q(v_{N,\epsilon})\|_{L^q} \leq C_q \epsilon^{1/4}, \quad \sup_{N,\epsilon} \|E_N^Q(v_{N,\epsilon})\|_{L^q} < \infty.$$

Proof. Subtract $(v)_Q$ and extend to $H^{3/4}(\mathbb{R}^d)$. Fourier transformation in the normal variable and Cauchy–Schwarz give, on each face,

$$\|v(\tau, \cdot) - v(0, \cdot)\|_{L^2(\mathbb{R}^{d-1})}^2 \leq \|v\|_{H^{3/4}(\mathbb{R}^d)}^2 \int_{\mathbb{R}} \frac{\min\{4, \tau^2 \xi^2\}}{(1 + \xi^2)^{3/4}} d\xi \leq C \tau^{1/2} \|v\|_{H^{3/4}(\mathbb{R}^d)}^2.$$

Integrate against the bump profile and sum over the faces to get (8.17). Test the difference of the two weak equations with $v_N - v_{N,\epsilon}$ and use coercivity to obtain (8.18). The reverse triangle inequality for the energy norm then gives

$$|E_N^Q(v_N) - E_N^Q(v_{N,\epsilon})| \leq CK_N^{1/2} \epsilon^{1/4} E_N^Q(v_N)^{1/2} + CK_N \epsilon^{1/2}.$$

The response $E_N^Q(v_N) = p \cdot a_*^{-1}(Q; A_N)p$ has the required moments uniformly in N by Lemma 8.2. Hölder’s inequality at the doubled orders proves (8.19).

We also record the transfer used on a fixed physical cube $Q = w + R(0, 1)^d$. For $p \neq 0$, set $\hat{p} = p/|p|$ and let $f_h^{\hat{p}}$ be the preceding unit-cube source. Define

$$f_{Q,h}(w + Ry) = \frac{|p|}{R} f_h^{\hat{p}}(y), \quad L_{Q,p}(u) = \int_Q p \cdot \nabla u, \quad L_{Q,p,h}(u) = \int_Q f_{Q,h} u.$$

For $p = 0$ take both loads and the source to be zero. With $\tilde{u}(y) = u(w + Ry)$, change of variables gives

$$L_{Q,p}(u) = |p| R^{d-1} L_{\hat{p}}(\tilde{u}), \quad L_{Q,p,h}(u) = |p| R^{d-1} L_{\hat{p},h}(\tilde{u}).$$

In particular, $\int_Q f_{Q,h} = 0$, $\|f_{Q,h}\|_{\infty} \leq C_{Q,p} h^{-1}$, and the fixed dilation of the fractional norm in (8.17) gives

$$|L_{Q,p}(u) - L_{Q,p,h}(u)| \leq C_{Q,p} h^{1/4} \|u - (u)_Q\|_{H^{3/4}(Q)}.$$

Thus the source factor in physical coordinates is R^{-1} .

The coercivity bound needed here also transfers to fixed larger cubes. For $R \leq 1$ use Lemma 8.3. For $R > 1$, choose $k \geq 1$ with $\rho = R/3^k \in (1/3, 1]$ and use the chart $x = z + 3^k y$ from the centered residual cube of side ρ , where z is the center of Q . The environment

$$(\Psi\omega)_j(y) = \omega_{j+k}(z + 3^k y)$$

has the original law. The coefficient at cutoff $N + k$ in this environment is bounded above by $F_Q(\omega) A_N(z + 3^k y)$, and the same bound holds with A_N^0 on the right, where we may take

$$F_Q = \exp\left(k\tau^2 + \sum_{i=1}^k \|\omega_i\|_{C(\bar{Q})} + \|H(\Psi\omega)\|_{C(\bar{Q}_\rho)} + \|H(\omega)\|_{C(\bar{Q})}\right).$$

Here Q_ρ is that residual cube. The normalization ratio is bounded by (3.24), namely $\bar{\mathbf{a}}_N \leq e^{2k\tau^2} \bar{\mathbf{a}}_{N+k}$; the other factors are precisely the retained coarse layers and the two infrared restrictions. Each has every fixed exponential moment on its fixed compact set. Pull back the mean-zero function, and likewise a function with zero trace (the H_0^1 clause of Lemma 8.3); use that lemma on Q_ρ and scale the energy by $3^{k(d-2)}$. The fixed scaling of the fractional norm gives a physical coercivity constant

$$K_{Q,N} \leq C_Q F_Q K_{Q_\rho, N+k}(\Psi\omega).$$

The constants in the moment bounds may depend on Q ; the disorder threshold is the one for the finite doubled moment orders on the residual cube. Hölder’s inequality and preservation of the environment law give the required uniform-in- N moments, for both infrared conventions. For a triadic R , $\rho = 1$. Only a first-moment residual bound is needed for a fixed polynomial tail: the

union bound at $\sqrt{T}$ gives $\mathbb{P}[K_{Q,N} > T] \leq B_Q T^{-1/2}$ after absorbing C_Q into F_Q . This uses no independence between F_Q and the residual coercivity constant.

Let Y_N and $Y_{N,h}$ be the variational responses to these two physical loads. Their Riesz representers and the preceding load estimate give

$$|Y_N - Y_{N,h}| \leq 2C_{Q,p} h^{1/4} K_{Q,N}^{1/2} Y_N^{1/2} + C_{Q,p}^2 h^{1/2} K_{Q,N}.$$

The trace bound $|L_{Q,p}(u)| \leq C_{Q,p} \|u - (u)_Q\|_{H^{3/4}(Q)}$ gives $Y_N \leq C_{Q,p} K_{Q,N}$ by the variational formula. Consequently, at any of the finite moment orders chosen before disorder,

$$\sup_N \|Y_N - Y_{N,h}\|_{L^q} \leq C_{q,Q,p} h^{1/4}, \quad \sup_{N, 0 < h < 1/8} \|Y_{N,h}\|_{L^q} < \infty.$$

These are the fixed-cube versions of (8.19). They use transported loads and coercivity; the deterministic response of the identity coefficient on Q is $|Q|p \cdot p$. $\square$

8.2.4. Reflection across the faces. Let a be a scalar coefficient, and let u be a solution with zero conormal derivative and bounded source f . Reflect u and f evenly across a set of faces. This gives a solution $u \circ T$ of the equation with coefficient $a \circ T$ and source $f \circ T$, where T folds the reflected domain onto the original cube. The reflected equation has no additional source concentrated on the folding planes. On a set invariant under these reflections, the energy of $u \circ T$ is two to the power of the number of reflected faces times the energy of u in the part of the set inside the original cube.

Lemma 8.10 (Effect of folding on $\mathcal{E}_{s,\infty,2}$). *Let T fold the cube $z + \square_M$ across coordinate planes through z , as above. For every continuous positive coefficient a on its closure, scalar $a_0 > 0$, $s \in (0, 1/2)$ and $n \leq M$,*

$$(8.20) \quad \mathcal{E}_{s,\infty,2}(z + \square_M, n; a \circ T, a_0)^2 \leq \left(1 + \frac{3d}{3^{1-2s} - 1}\right) \mathcal{E}_{s,\infty,2}(z + \square_M, n; a, a_0)^2.$$

Proof. For each n , let $\kappa_n(a)$ be the maximum of $\sup_{|e|=1} J(U, e, e; a/a_0)$ over the grid cubes $U \subseteq z + \square_M$ of side 3^n . On a cube meeting no folding plane, T reflects some of the coordinates, so changing variables by T shows that the energy for $a \circ T$ is at most $\kappa_n(a)$. Subdivide each cube that meets a folding plane triadically, and repeat the subdivision on the subcubes that still meet one. After j subdivisions, the subcubes that still meet a folding plane fill at most a fraction $d3^{-j}$ of the cube. By subadditivity (2.16), $\kappa_n(a \circ T) \leq \kappa_n(a) + \sum_{j \geq 1} 3d3^{-j} \kappa_{n-j}(a)$. Insert this inequality into Definition 2.3 and sum the geometric series $\sum_{j \geq 1} 3^{-(1-2s)j}$ to obtain (8.20). $\square$

Proposition 8.11 (Energy decay for the folded coefficient). *There are dimensional constants $C < \infty$ and $K \geq 1$ with the following property. Let $\bar{\delta} \leq C^{-1}$ and $\gamma \in [1/2, 1)$ satisfy $1 - \gamma \geq CK\bar{\delta} |\log \bar{\delta}|^{1/2}$, let $m, L \in \mathbb{N}_0$ with $m \leq L$, and let T fold the cube $z + \square_m$ across coordinate planes through z . Set $\gamma_* = 1 - (1 - \gamma)/K$. Define $\mathcal{L}_L(\gamma, m)$ by (6.60) for the cube $z + \square_m$, with all parameter choices in (6.59) evaluated at γ_* ; in particular, $\lambda_* = \lambda/K$ and $\varepsilon_* = \varepsilon/\sqrt{K}$, where λ and ε denote the choices at γ . If $u \in H^1(z + \square_m)$ solves $-\nabla \cdot ((\mathbf{a}_L \circ T) \nabla u) = \nabla \cdot \tilde{g}$, with $\tilde{g} \in C^{0,1/2}(z + \square_m; \mathbb{R}^d)$, then, for $0 \leq n \leq m - \mathcal{L}_L(\gamma, m)$,*

$$(8.21) \quad \|(\mathbf{a}_L \circ T)^{1/2} \nabla u\|_{\underline{L}^2(z + \square_n)} \leq C 3^{(1-\gamma)(m-n)} \left(\|(\mathbf{a}_L \circ T)^{1/2} \nabla u\|_{\underline{L}^2(z + \square_m)} + (\mathbf{b}_{L,m})_{z + \square_m}^{-1/2} 3^{m/2} [\tilde{g}]_{W^{1/2,\infty}(z + \square_m)} \right).$$

For every $h \geq 0$, this minimal scale satisfies

$$\mathbb{P}[\mathcal{L}_L(\gamma, m) > h] \leq C \exp\left(-\frac{(1-\gamma)^2(h-C)_+}{CK^2\bar{\delta}^2 |\log \bar{\delta}|}\right),$$

uniformly in L , m and the deterministic translate z .

Proof. Apply the argument proving (6.6) at z , using the original averages $(\mathbf{b}_{L,j+2})_{z+\square_{j+2}}$. The harmonic approximation and excess decay use the good event through $\mathcal{E}_{s,\infty,2}$, as in the proof of Lemma 6.16. By (8.20), folding multiplies this quantity by at most a dimensional constant. The bounds with $q = 1$ at the larger exponents follow from the bound with $q = 2$ at $s_0 = 1/32$ by weighted Cauchy–Schwarz. Choose a dimensional $K \geq 1$ large enough to absorb these factors, and evaluate the parameter choices (6.59) at $\gamma_* = 1 - (1 - \gamma)/K$, so that $\lambda_* = \lambda/K$ and $\varepsilon_* = \varepsilon/\sqrt{K}$, as in the statement. This uses the same good events at the stricter parameter ε_* . The error sum, the bound on bad scales and the ratios of the original averages are the bounds (6.69), (6.72) and (6.63) with these parameter choices. They supply the iteration of Lemma 6.11; (2.38) supplies its coarse-grained ellipticity inputs. All its cubes are centered at z and invariant under the reflections. The stricter choices give the asserted estimate with exponent γ .

We record the tail of the minimal scale with this reduced tolerance. Each of the two scales in (6.60) has tail length bounded by a constant times

$$s_0^{-7}\bar{\delta}^2|\log \bar{\delta}|\lambda_*^{-2} + s_0^{-8}\varepsilon_*^{-2}\lambda_*^{-1}\bar{\delta}^2|\log \bar{\delta}| \leq \frac{CK^2\bar{\delta}^2|\log \bar{\delta}|}{(1 - \gamma)^2}.$$

Indeed, both λ_*^{-2} and $\varepsilon_*^{-2}\lambda_*^{-1}$ gain the factor K^2 . Equations (6.51) and (6.57), or (6.85) and (6.87) for the cutoff version, therefore give

$$\mathbb{P}[\mathcal{L}_L(\gamma, m) > h] \leq C \exp\left(-\frac{(1 - \gamma)^2(h - C)_+}{CK^2\bar{\delta}^2|\log \bar{\delta}|}\right).$$

The constants are uniform in the cutoff, the root scale and its deterministic translate. The hypothesis on $1 - \gamma$ places γ_* in the range of Proposition 6.17, because C may be taken at least as large as the constant in that proposition. These are parameter choices in the existing estimates; the definitions of the good events are unchanged. $\square$

Proposition 8.12 (Bounded Neumann sources). *Let $\alpha \in (0, 1)$ and $t \in (d - 1, d)$, and fix a finite list of moment orders. There is $\bar{\delta}_0(d, \alpha, t, \text{list}) > 0$ such that, if $\bar{\delta} \leq \bar{\delta}_0$, then on every fixed cube Q there are random variables K_N with $\sup_N \|K_N\|_{L^p} < \infty$ for every order p in the list, such that the following holds for every $f \in L^\infty(Q)$ with $\int_Q f = 0$. If $u_N \in H^1(Q)$ is the mean-zero solution of*

$$\int_Q A_N \nabla u_N \cdot \nabla v = \int_Q f v \quad (v \in H^1(Q)),$$

then

$$\|u_N\|_{C^\alpha(\bar{Q})} \leq K_N \|f\|_\infty,$$

and, for every $x \in Q$ and $0 < r \leq 1$,

$$\Gamma_N(u_N)(B_r(x) \cap Q) \leq K_N \|f\|_\infty^2 r^t.$$

The same threshold and moment bounds hold with A_N replaced by A_N^0 or by the coefficient obtained by replacing H with $H_{L'}$, uniformly in L' , with the solution u_N and the energy Γ_N computed using the corresponding coefficient. For each fixed Q , a common K_N may be chosen for these infrared conventions and for all sources. For the stationary coefficient A_N^0 , the same bounds hold on each translate $w + Q$ of Q , with the corresponding translated random constants, and a fixed finite collection of such translates may be handled by taking the maximum of these constants.

Proof. We track the energy $\Gamma_N(u_N)(B_r(x) \cap Q)$ of small balls. Step 1 rescales the problem to the stationary coefficient. Step 2 joins each target ball to a ball of fixed radius by a chain of at most $d + 1$ applications of Proposition 8.11, reflecting evenly across each face that the chain comes close to. The random factors in these applications depend on the target ball; the main work, in Steps 3 and 4, is to bound them, up to small negative powers of the radii, by random variables that do not depend on the target and whose moments are bounded uniformly in N ; this is done by

summing over meshes of stopping depths and of reference averages. Together with the coercivity estimate (8.5) for the total energy, this gives the Morrey bound for $r \geq 3^{-N}$. Step 5 treats the radii below 3^{-N} and proves the Hölder bound; the last paragraphs remove the infrared truncation and pass to general fixed cubes.

We first work on $Q = (0, 1)^d$. Choose

$$\max\{t, d - 2 + 2\alpha\} < t_1 < t_0 < d, \quad \eta = t_0 - t_1, \quad 0 < \theta < d + 2 - t_0.$$

Fix all moment orders to be used below before choosing $\bar{\delta}$. These include the given list, the larger orders for products with coercivity constants and the auxiliary orders used for the two meshes. When this proposition is used in Proposition 8.17, include $3p$; when it is used at order P and at order $2P$ in Lemma 8.36, include $3P$ and $6P$, respectively. This is still a finite list. The auxiliary orders are v in Step 3 and q in Step 4, and the conditions on $\bar{\delta}$ imposed in those steps depend only on these orders and on quantities fixed before $\bar{\delta}$.

Step 1. We specify the physical normalization. Temporarily replace H by $H_{L'} = \sum_{i=1}^{L'} (\mathbf{g}_i - \mathbf{g}_i(0))$, including $L' = 0$. Reindexing the stationary fields gives

$$A_N^{(L')}(y) = c_{N,L'} \mathbf{a}_{N+L'}(3^N y), \quad c_{N,L'} = \bar{\mathbf{a}}_N^{-1} \exp\left(L' \tau^2 - \sum_{i=1}^{L'} \mathbf{g}_i(0)\right).$$

Here the stationary field on the right is taken in the reindexed environment. Choose a measurable representative of f bounded everywhere by its essential supremum, changing it on a null set if necessary; this preserves its integral and weak load. Under $X = 3^N y$ and $v(X) = 3^N u_N(3^{-N} X)$, its equation has source $(c_{N,L'} 3^N)^{-1} f(3^{-N} X)$. Even reflection replaces this source by its composition with the coordinate fold. On a reflection-invariant set, the stationary energy of v is $2^{|I|} 3^{dN} c_{N,L'}^{-1}$ times the physical energy in its intersection with Q , where I is the set of reflected coordinates.

Thus the physical reference in the source term is the stationary average multiplied by $c_{N,L'}$. At depth k it has the equivalent formula

$$b_{N,k}(z) = \int_{B_{3^{-k}/2}(z)} \frac{\bar{\mathbf{a}}_{N-k}}{\bar{\mathbf{a}}_N} \exp\left(H_{L'}(x) + \sum_{j=0}^{k-1} \mathbf{g}_{-j}(x) - k\tau^2\right) dx, \quad 0 \leq k \leq N.$$

Use the fixed shift of depth prescribed by the outer cube in the iteration: for a root of radius $R_i = 3^{-k_i}/2$ this reference is $b_{N,(k_i-J)_+}(z_i)$, where J is a fixed dimensional buffer. Its averaging radius is comparable to R_i with a fixed factor. We write b_i for this physical reference. Its inverse is b_i^{-1} ; the factor $c_{N,L'}$ has already been included.

Step 2. We construct the reflected chains. Fix J first, and then fix $L_* > 10$ with $L_* \geq 2\sqrt{d}3^J$. Fix a half-triadic $R_* > 0$ with $R_* < (100L_*)^{-1}$ and $R_* \leq 3^{-J}/2$. All three are fixed independently of N and L' . At a center y , put $\delta_i = \min\{y_i, 1 - y_i\}$ and let p_i be its nearer face, 0 or 1. If I is the set of activated coordinates, the current center z_I has coordinate p_i for $i \in I$ and coordinate y_i otherwise. Start with the least half-triadic radius s containing the target cube. Stop if $s \geq R_*/24$.

If an inactive face satisfies $\delta_i \leq 100L_*s$, activate i , project the center to that face, and replace s by the least half-triadic radius at least $s + \delta_i$. The old target lies in the new one, and the radius increases by at most $3(1 + 100L_*)$. Otherwise choose the largest half-triadic R such that $R \leq R_*$ and $4L_*R \leq \delta_i$ for every $i \notin I$. Then $8s < R$. Reflect across every activated face and apply Proposition 8.11 between radii comparable to s and R . If $R = R_*$, stop. Otherwise activate a nearest inactive face and take the least half-triadic radius at least $L_*R + \delta_i$ about the newly projected center. The clearance condition keeps every other face outside the outer cube of this application.

Each transition activates a new coordinate. There are at most d transitions and $d+1$ energy-decay applications, including at edges and corners. The intervening containments bound the enlargement

factors by fixed constants. Telescoping the ratios in the energy-decay applications bounds their product by $C(r/R_*)^{t_0}$. No assumption that the target meets only one face is made.

Set $\gamma = (t_0 - d + 2)/2$. In each application let B_i be the minimal stopping depth for the original, translated stationary field at the reduced tolerance of Proposition 8.11, with its fixed outer-scale buffers. If the requested depth is below that stopping depth, use energy monotonicity instead: increasing a fixed $c \geq t_0 \log 3$ makes $e^{cB_i}(s/R_i)^{t_0} \geq 1$ in that case. Thus the same one-step inequality holds at every requested depth. The local Newtonian primitive of the evenly reflected bounded source has half-Hölder seminorm bounded by $CR_i^{1/2}\|f\|_\infty$, by (8.8). If instead a primitive supported on the fixed outer cube is used, its local bound has the harmless extra factor $1 + \log(1/R_i)$. Keeping this larger bound, composition of the chain gives

$$\Gamma_N(u_N)(B_r(x) \cap Q) \leq Ce^{c\sum_i B_i} r^{t_0} \left[R_*^{-t_0} E_N^Q(u_N) + \|f\|_\infty^2 \sum_i b_i^{-1} R_i^{d+2-t_0} (1 + \log(1/R_i))^2 \right].$$

Here and below constants absorb the at most 2^d reflection multiplicity and the bounded number of enlargements. It remains to bound the random factors $e^{c\sum_i B_i}$ and b_i^{-1} , up to small negative powers of the radii, by random variables that do not depend on the chain and whose moments are bounded uniformly in N ; Steps 3 and 4 do this.

Step 3. We sum the stopping-depth mesh. At target depth n , use spacing $3^{-(n+J)}$. Enlarge the target by this spacing to contain every target centered in the corresponding mesh cell, and choose an interior mesh point even for targets touching the boundary. Catalogue all permutations of the coordinates and all $d+1$ root-depth slots between 0 and $n+J$. Activated sets are initial segments of the permutation, so every chain in Step 2 is represented. This catalogue has cardinality at most $C_J(n+J+1)^{d+1}3^{dn}$. The allowance in a slot of depth $k \leq N$ is the minimal stopping depth on the outer stationary cube at scale $N-k+J$, plus J and the fixed iteration buffer; put it equal to zero when $k > N$.

By the reduced-tolerance tail just proved, there are fixed A, C such that each allowance B satisfies

$$\mathbb{P}[B > h] \leq Ae^{-\beta(h-A)_+}, \quad \beta = \frac{(1-\gamma)^2}{CK^2\bar{\delta}^2|\log \bar{\delta}|}.$$

This is uniform in every deterministic translated center, cutoff and root depth. For $a > 0$ and $\beta \geq 2a$, summing the tail gives

$$\mathbb{E}e^{aB} \leq e^{a(A+1)} \left(1 + A \sum_{h \geq 0} (e^a - 1)e^{-(\beta-a)h} \right) \leq C(A, a).$$

Choose v larger than all the moment orders needed for the products below and such that $v\eta > d$, and then decrease $\bar{\delta}$ so that $\beta \geq 2(d+1)vc$. Hölder's inequality over the $d+1$ slots gives a uniform v th moment of $Z_{N,n,\pi} = C_0 e^{c\sum_i B_i}$. It requires no independence between the slots. Define

$$U_N = \sup_{n \geq 0, \pi} 3^{-\eta n} Z_{N,n,\pi}.$$

Then

$$\mathbb{E}U_N^v \leq C \sum_{n \geq 0} (n+J+1)^{d+1} 3^{(d-v\eta)n} < \infty.$$

The bound is uniform in N . In particular, the mesh factor for a target of size r is at most $CU_N r^{-\eta}$. We have summed minimal stopping depths, whose tail controls the entire unavailable prefix of scales.

Step 4. We sum the reference mesh at its own root depth. Write $s_{N,k}(x)$ for the integrand defining $b_{N,k}$. At spacing comparable to $R_k = 3^{-k}/2$, use the statistic

$$T_{N,k}(y) = e^{\text{osc}_{B_{3R_k}}(y)(H_{L'} + \sum_{j < k} \mathbf{g}_{-j})} (s_{N,k}(y) + s_{N,k}(y)^{-1}).$$

Every average centered within R_k of y and its inverse are bounded by $T_{N,k}(y)$. The pointwise positive and negative moment bounds and the bounded-scale oscillation estimate used in Lemma 8.2 give, at every fixed auxiliary order q ,

$$\|T_{N,k}(y)\|_{L^q} \leq C_q e^{C(q+q^2)\bar{\delta}^2 k}, \quad k \leq N.$$

These are bounds for the normalized physical reference; in particular, they include the ratio $\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N$. Choose q at least as large as v and with $q\theta > d$, and then decrease $\bar{\delta}$ until $C(q+q^2)\bar{\delta}^2 < \frac{1}{2}(\theta - d/q) \log 3$. There are at most $C3^{dk}$ points at depth k . Consequently, for

$$V_N = \sup_{k \leq N, y \text{ in the depth-}k \text{ mesh}} R_k^\theta T_{N,k}(y),$$

Minkowski's inequality bounds $\|V_N\|_{L^q}$ by

$$C_q \sum_{k \geq 0} 3^{-(\theta-d/q)k} e^{C(q+q^2)\bar{\delta}^2 k} < \infty,$$

uniformly in N and L' . Thus $b_i^{-1} \leq CV_N R_i^{-\theta}$, including the fixed depth shift in Step 1. The strict margin gives $\sup_{0 < R \leq R_*} R^{d+2-t_0-\theta} (1 + \log(1/R))^2 < \infty$. With at most $d+1$ source terms, Steps 2–4 therefore imply, for $r \geq 3^{-N}$,

$$\Gamma_N(u_N)(B_r(x) \cap Q) \leq CU_N r^{t_1} (E_N^Q(u_N) + V_N \|f\|_\infty^2).$$

The mean-zero coercivity estimate (8.5) bounds $E_N^Q(u_N)$ by $CK_N \|f\|_\infty^2$. Taking U_N, V_N and K_N at twice the requested orders bounds the resulting product in every requested L^p . Here K_N denotes the coercivity constant; the final constant in the statement is enlarged to include the resulting products.

The constants can be chosen uniformly in the infrared approximation. On the fixed compact enlargement used above, put $S = \sum_{i \geq 1} \|\mathbf{g}_i - \mathbf{g}_i(0)\|_{C^{1,1}}$. The proof of Lemma 8.1 gives $\|S\|_{L^a} \leq C\bar{\delta}\sqrt{a}$ and finite exponential moments at every fixed order. It bounds H , 0 and all $H_{L'}$ simultaneously. In particular, their normalized references and inverse ellipticity constants compare with the zero-infrared ones by e^S . Use the doubled orders in these products; the microscopic logarithmic gradients are bounded by the same envelope as well. The stopping-depth mesh is built from the reindexed stationary field. Every root actually used in a chain has $k \geq J$, hence its outer scale $N - k + J$ is at most $N \leq N + L'$. Equation (6.92) therefore uses the same stopping variable for all L' . Unused catalogue slots can use that same stationary stopping variable without a one-step application. Thus no union over infinitely many infrared truncations is needed. The root R_* and all geometric constants were fixed before their number of terms. For the stationary coefficient A_N^0 , the translates $w + Q$ use the law-preserving translation of this field and their own cutoff and reference. Any spatial maximum needed inside a fixed cube is included in the displayed mesh sums. A fixed finite collection of such translates can be included in their catalogues without changing the exponent conditions.

Step 5. We finish at the microscopic scales and pass to H . Reflect evenly across every face meeting the microscopic neighborhood. The coordinate fold is 1-Lipschitz, so it preserves the required small oscillation of $\log A_N$; the reflected weak equation has the evenly reflected bounded source. Take p_1 so large that $d - 2d/p_1 > t_1$ and $1 - d/p_1 > \alpha$, and apply (8.14) at $\ell_N = c3^{-N}(1 + D_N)^{-1}$. Step 4 at scale 3^{-N} supplies its outer energy. Dividing by r^t gives the two factors $3^{-(t_1-t)N}$ and $3^{-(d+2-t)N}$ from Step 4 of Proposition 8.7; their strictly positive exponents absorb the coefficient-extreme moments at the finitely many larger orders used in the products. Divide by the reflection multiplicity to return to Q .

For the Hölder estimate, the local coarse-grained Poincaré inequality applied to the same energy bound gives the Campanato exponent $t_1 + 2 - d$ minus an arbitrarily small mesh loss. Choose that loss so that the exponent is larger than 2α . Its inverse-ellipticity mesh has the same convergent moment summation as (8.7). Below 3^{-N} use the preceding W^{1,p_1} estimate and Morrey's inequality.

The first two factors in Step 7 of Proposition 8.7 have positive exponents because $t_1 > d - 2 + 2\alpha$; there is no boundary-datum term. Telescope the cube averages and use the mean-zero condition to control the supremum norm. Enlarging the random constant gives both asserted estimates. Every constant constructed above is a statistic of the coefficient, independent of f and of the chosen solution.

On the common infrared-convergence event the truncated coefficients converge uniformly on $\overline{Q}$. Compare the weak equations on the mean-zero space, testing with the difference of the solutions. Coercivity yields gradient convergence; the Hölder bound yields uniform convergence. Fatou's lemma preserves the moment bounds. One can use the same moment-bounded coefficient statistics for every source before this passage, so the final constant is source independent. Countable intersections make the event common to all cutoffs and the required cell families; the deterministic weak-equation estimates then apply to every bounded mean-zero source.

Finally consider a fixed cube $w + R\Box_0$. For A_N , choose its fixed triadic scale, transport the stationary environment by translation and that scale shift, and run the preceding argument on the remaining cube of side in a fixed bounded interval. For A_N^Q and the truncations $H_{L'}$, a shift of a cube with $R > 1$ to bounded size deletes the coarsest layers of the coefficient, which only the anchored coefficient retains. We therefore run the preceding argument directly on $w + R\Box_0$ in these cases. Its chains have roots of radius at most R_* , so their outer scales do not exceed the cutoff; the catalogue of Step 3 is larger by a factor depending only on R ; and the global energy is bounded by the coercivity of Lemma 8.9. In the anchored case, if $A_N(w + Ry) = c_N \tilde{A}(y)$ and $\tilde{u}(y) = u_N(w + Ry)$, its source is $R^2 c_N^{-1} f(w + Ry)$ and its physical energy is $R^{d-2} c_N$ times the transported energy. Thus the transported Morrey constant is multiplied by $R^{d+2-t} c_N^{-1}$, and the Hölder constant by $R^2 \max\{1, R^{-\alpha}\} c_N^{-1}$. The reference inverse c_N^{-1} is bounded by an exponential of the fixed anchor field and finitely many retained layers; it has every required moment uniformly in N and in the infrared truncation. Use twice the requested orders in this final product. If the shifted cutoff is negative, only finitely many original cutoffs remain, and the microscopic argument covers them. The exponent conditions and the disorder threshold are the same; the final moment bounds may depend on the fixed cube. $\square$

Applying Proposition 8.12 to the mean-zero source f_ϵ of (8.16), with $\|f_\epsilon\|_\infty \leq C\epsilon^{-1}$, gives, with one random constant for all ϵ ,

$$(8.22) \quad \Gamma_N(v_{N,\epsilon})(B_r(x) \cap Q) \leq K_N \epsilon^{-2} r^t.$$

8.3. Compactness of the finite-volume energies. We approximate each finite-volume energy by its conditional expectation given finitely many fields, and we bound the error by resampling one field at a time. These conditional expectations form a compact family in $L^p(\mathbb{P})$.

On a fixed cube Q , write $E_N = E_N^Q$ for the form in (8.2). By a *finite-volume energy* $\mathcal{R}_N$ we mean either the minimal energy $\Lambda_{N,Q}(\varphi)$ with a nonconstant smooth boundary datum φ , or the quadratic form $\langle f, G_N^Q f \rangle$ of the Dirichlet Green operator for a nonzero smooth source f . Write $\mathcal{R}(g)$ for the same energy computed with the coefficient whose logarithm is g , so that $\mathcal{R}_N = \mathcal{R}(\log A_N)$.

8.3.1. Local perturbations of the coefficient.

Lemma 8.13 (Multiplicative perturbations). *Let A be bounded above and below by positive constants, let g be bounded and supported in $B \subseteq Q$. Write $E^g(v) = \int_Q e^g A |\nabla v|^2$ and $E = E^0$. Let u and u^g be the minimizers for the coefficients A and $e^g A$ with the same Dirichlet boundary values, or the solutions for these coefficients with the same linear right-hand side and zero Dirichlet or zero conormal boundary condition; in the conormal case, the right-hand side vanishes on constants and*

the solutions have mean zero. Denote their energies by $\mathcal{R}, \mathcal{R}^g$. Then

$$(8.23) \quad \begin{aligned} E^g(u^g - u) &\leq \|g\|_\infty^2 e^{3\|g\|_\infty} \Gamma_E(u)(B), & \Gamma_{E^g}(u^g)(B) &\leq 4e^{\|g\|_\infty} \Gamma_E(u)(B), \\ |\mathcal{R}^g - \mathcal{R}| &\leq 2\|g\|_\infty e^{4\|g\|_\infty} \Gamma_E(u)(B). \end{aligned}$$

For two log coefficients g, h and any of these energies,

$$(8.24) \quad e^{-\|g-h\|_\infty} \mathcal{R}(h) \leq \mathcal{R}(g) \leq e^{\|g-h\|_\infty} \mathcal{R}(h).$$

Proof. Set $v = u^g - u$. Subtracting the Euler equations gives $E^g(v, \phi) = -\int_B (e^g - 1) d\Gamma_E(u, \phi)$. Test with v and apply Cauchy–Schwarz to obtain

$$(8.25) \quad E^g(v)^{1/2} \leq (e^{\|g\|_\infty/2} - e^{-\|g\|_\infty/2}) \Gamma_E(u)(B)^{1/2}.$$

This proves the first bound in (8.23); the triangle inequality in $L^2(B, e^g A dx)$ gives the second estimate. The Euler equations also give

$$\mathcal{R}^g - \mathcal{R} = \pm \left[\int_B (e^g - 1) d\Gamma_E(u) - E^g(v) \right],$$

with plus sign when $\mathcal{R}$ is a minimal energy with given boundary values and minus sign when it is the energy of a solution with given right-hand side. Using $|e^g - 1| \leq \|g\|_\infty e^{\|g\|_\infty}$ proves the third bound. Monotonicity in the coefficient and scaling by constants give (8.24). These arguments use only the Euler equations and the Cauchy–Schwarz inequality for energy measures, so they also apply to the regular strongly local forms constructed below. $\square$

8.3.2. Resampling one random field. We bound the change in a finite-volume energy when a field $\mathbf{g}_{-j}$ with $j \geq 0$ is replaced by an independent copy. The field has scale $r = 3^{-j}$ and range of dependence $\sqrt{d}r$, so its restrictions to boxes at mutual distance larger than $\sqrt{d}r$ are independent. We partition Q into boxes of side ℓ with $r \ll \ell \ll 1$. Delete the field on strips of width of order r around the faces of the boxes, and apply the Efron–Stein inequality to the independent restrictions of the field to the remaining interiors of the boxes. The energy in the strips controls the error caused by deleting the field there. The largest energy in one box controls the differences $X - X^{(i)}$ in the Efron–Stein inequality (8.27) below. By the Morrey bound (8.10), both are bounded by a positive power of the scale r . Exponential moments of the sup norm of the field enter through Lemma 8.13; they grow more slowly than any negative power of the scale r , so this decay absorbs them.

Lemma 8.14 (Mass near the box faces and in a single box). *Let $t > d - 1$ and $K < \infty$, and let ν be a finite positive measure on a fixed cube Q such that $\nu(B_\rho(x) \cap Q) \leq K\rho^t$ for every $x \in Q$ and $0 < \rho \leq 1$. For $4\sqrt{d}r < \ell \leq 1$, partition Q into boxes of side ℓ , and call **strips** the points within distance $4\sqrt{d}r$ of a box face. There is $C(d, t, Q) < \infty$ such that*

$$\nu(\text{strips}) \leq CK\ell^{-1}r^{t-d+1},$$

and

$$\max_{B \text{ box}} \nu(B) \leq CK\ell^t.$$

With $\ell = r^{(t-d+1)/(t+1)}$ and sufficiently small r , both bounds are CKr^b , where $b = t(t-d+1)/(t+1) > 0$.

Proof. There are at most $C_Q\ell^{-1}$ face hyperplanes. Each strip is covered by C_Qr^{1-d} balls of radius $4\sqrt{d}r$, and each box by a dimensional number of balls of radius ℓ . Apply the bound $\nu(B_\rho(x) \cap Q) \leq K\rho^t$ and use $\ell^{-1}r^{t-d+1} = \ell^t = r^b$ for the stated choice of ℓ . $\square$

The sup norm on Q of a fine-scale field grows only logarithmically in the inverse of its scale, and its exponential moments grow more slowly than any power of the inverse scale. By (1.5), the variable $\|\mathbf{g}_0\|_{L^\infty(\square_0)}$ is sub-Gaussian with parameter $C\bar{\delta}$. Let $j \geq 0$ and $r = 3^{-j}$, and let $S =$

$\|\mathbf{g}_{-j}\|_{L^\infty(Q)}$. The field $\mathbf{g}_{-j}$ has the law of $\mathbf{g}_0(3^j \cdot)$. Cover $3^j Q$ by at most Cr^{-d} unit cubes to obtain $\mathbb{P}[S > s] \leq 2Cr^{-d} \exp(-s^2/C\bar{\delta}^2)$. The union bound and $(s+a)^2 \geq s^2 + a^2$ show that the excess of S over $\bar{\delta}\sqrt{C(1+|\log r|)}$ has the tail $\exp(-s^2/C\bar{\delta}^2)$. Thus the excess is sub-Gaussian with parameter $C\bar{\delta}$, uniformly in r . Consequently, for all $k \geq 0$, $\lambda \geq 0$, $p < \infty$ and $\gamma > 0$, and every $0 < \bar{\delta} \leq 1$,

$$(8.26) \quad \|S^k e^{\lambda S}\|_{L^p} \leq C(p, k, \lambda) \bar{\delta}^k (1 + |\log r|)^{k/2} \exp(C\lambda \bar{\delta} \sqrt{1 + |\log r|} + C_p \lambda^2 \bar{\delta}^2) \leq C(p, k, \lambda, \gamma) \bar{\delta}^k r^{-\gamma}.$$

The splitting of S into a deterministic logarithmic part and a sub-Gaussian excess is needed: a union bound applied directly to the exponential moment would produce the factor Cr^{-d} .

The fluctuations are estimated with the following moment form of the Efron–Stein inequality. Let $\xi_1, \dots, \xi_m$ be independent random elements, let $p \geq 2$, and let $X = X(\xi_1, \dots, \xi_m) \in L^p$ be measurable. Write $X^{(i)}$ for the variable obtained by replacing ξ_i with an independent copy, and $\mathbb{E}_i$ for the integral over this copy alone. Then

$$(8.27) \quad \|X - \mathbb{E}X\|_{L^p} \leq C_p \left\| \left(\sum_i \mathbb{E}_i[(X - X^{(i)})^2] \right)^{1/2} \right\|_{L^p}.$$

This is a consequence of [Boucheron, Bousquet, Lugosi, and Massart \[2005, Theorem 2\]](#), who proved two one-sided bounds in terms of variance proxies; both proxies are dominated by the sum on the right side of (8.27). The inequality also holds conditionally on every σ -field independent of the random elements and their copies.

Lemma 8.15 (Resampling one random field). *Let $0 < \bar{\delta} \leq 1$, $t \in (d-1, d)$ and $p \geq 2$, and let $\mathcal{R}_N$ be a finite-volume energy. Assume that, for every N , the function u_N defining $\mathcal{R}_N$ (the minimizer with the given boundary values, or the solution with zero Dirichlet boundary values) satisfies (8.10) on Q with the constant K_N , and that*

$$\sup_N \|K_N\|_{L^{3p}} + \sup_N \|\mathcal{R}_N\|_{L^{3p}} < \infty.$$

For $0 \leq j \leq N$, let $\mathcal{R}_N^{(j)}$ be the finite-volume energy obtained after replacing the field $\mathbf{g}_{-j}$ by an independent copy. Let b be the exponent of Lemma 8.14. Then there exists $C_p < \infty$, depending on p, d, t, Q , on the boundary datum or source defining $\mathcal{R}_N$ and on the two suprema in the previous display, such that, for all integers $0 \leq j \leq N$,

$$(8.28) \quad \|\mathcal{R}_N - \mathcal{R}_N^{(j)}\|_{L^p} \leq C_p \bar{\delta} 3^{-bj/8}.$$

Proof. Put $r = 3^{-j}$ and let $S = \|\mathbf{g}_{-j}\|_\infty$ on Q , as in (8.26). A prime denotes the corresponding quantity after this field is resampled. Throughout, expectations over this field are conditional on the other fields. For the finitely many scales at which the strip construction of Lemma 8.14 does not apply, the bound $|\mathcal{R}_N - \mathcal{R}'_N| \leq (S + S')e^{S+S'}\mathcal{R}_N$ and (8.26) suffice.

At the remaining scales, partition Q into boxes of side $\ell = r^{(t-d+1)/(t+1)}$. Choose smooth cutoff functions χ_i , one for each box, supported on interior subboxes that are separated by more than $\sqrt{d}r$, such that $\chi = \sum_i \chi_i$ equals one outside the strips of width $4\sqrt{d}r$. Let $\widehat{\mathcal{R}}_N$ be the energy after deleting $(1 - \chi)\mathbf{g}_{-j}$. Let $\widehat{\nu}$ be the energy measure of the corresponding solution. The restrictions of the field to these subboxes are independent. Lemmas 8.13 and 8.14 give $\widehat{\nu}(Q) \leq e^S \mathcal{R}_N$,

$$|\mathcal{R}_N - \widehat{\mathcal{R}}_N| \leq C S e^{CS} K_N r^b,$$

and

$$\max_B \widehat{\nu}(B) \leq C e^{CS} K_N r^b,$$

where the maximum is over the interior subboxes B . For the last bound, apply (8.25) on the strips and the inequality that $\widehat{\nu}(B)$ is at most $2e^S \Gamma_N(u_N)(B)$ plus twice the energy, for the form with the

deleted field, of the difference between the minimizers before and after the deletion. The second term bounds the change in the minimizer when replacing $\mathbf{g}_{-j}$ by $\chi\mathbf{g}_{-j}$.

Resampling the field on an interior subbox B changes $\widehat{\mathcal{R}}_N$ by at most $C(S + S')e^{C(S+S')}\widehat{\nu}(B)$. Since $\sum_B \widehat{\nu}(B)^2 \leq \max_B \widehat{\nu}(B)\widehat{\nu}(Q)$, the conditional Efron–Stein inequality (8.27) and conditional Jensen give

$$\|\widehat{\mathcal{R}}_N - \mathbb{E}_j \widehat{\mathcal{R}}_N\|_p \leq C_p r^{b/2} \|(S + S')e^{C(S+S')} K_N^{1/2} \mathcal{R}_N^{1/2}\|_p \leq C_p \bar{\delta} r^{3b/8}.$$

The last step uses Hölder’s inequality with three factors and (8.26), at a total loss of $r^{-b/8}$. The errors $\mathcal{R}_N - \widehat{\mathcal{R}}_N$ and $\widehat{\mathcal{R}}'_N - \mathcal{R}'_N$ from deleting the field on the strips are at most $C_p \bar{\delta} r^{7b/8}$ in L^p . The energy $\widehat{\mathcal{R}}'_N$, computed with the cutoff function applied to the resampled field, has the same conditional mean as $\widehat{\mathcal{R}}_N$, so

$$(8.29) \quad \mathcal{R}_N - \mathcal{R}'_N = (\mathcal{R}_N - \widehat{\mathcal{R}}_N) + (\widehat{\mathcal{R}}_N - \mathbb{E}_j \widehat{\mathcal{R}}_N) - (\widehat{\mathcal{R}}'_N - \mathbb{E}_j \widehat{\mathcal{R}}'_N) + (\widehat{\mathcal{R}}'_N - \mathcal{R}'_N).$$

Since $r^{3b/8} \leq 3^{-bj/8}$, this proves (8.28). $\square$

The following martingale bound adds up the effects of the fields on which we do not condition.

Lemma 8.16 (Summing resampling bounds). *Let $(\xi_j)_{j \in I}$ be a countable independent family and let $X \in L^p$, $1 \leq p < \infty$, be measurable with respect to this family. Write $X^{(j)}$ for the result of replacing ξ_j by an independent copy. For every $J \subseteq I$,*

$$\|X - \mathbb{E}[X \mid \sigma(\xi_j : j \in J)]\|_p \leq \sum_{j \notin J} \|X - X^{(j)}\|_p.$$

Proof. Condition first on the retained variables, and then reveal the omitted ones one at a time, in some enumeration. At the step that reveals ξ_j , the martingale difference is the conditional expectation of $X - X^{(j)}$ given the variables revealed so far. Conditional Jensen bounds its L^p norm by $\|X - X^{(j)}\|_p$. Sum and use L^p martingale convergence. $\square$

We apply this bound to approximate an energy by conditioning on finitely many fields.

Proposition 8.17 (Approximation by conditioning on finitely many fields). *Let $\ell \geq 0$ be an integer, and let b be the exponent of Lemma 8.14. Let $\mathcal{R}_N$ be a finite-volume energy satisfying the hypotheses of Lemma 8.15 for every $j \geq 0$. Then there exists $C_p < \infty$ such that, for every $N \geq \ell$,*

$$(8.30) \quad \|\mathcal{R}_N - \mathbb{E}[\mathcal{R}_N \mid \mathbf{g}_{-j}, |j| \leq \ell]\|_{L^p} \leq C_p \bar{\delta} 3^{-b\ell/8}.$$

On $Q = (0, 1)^d$, if instead $\mathcal{R}_N = |Q|e \cdot a_^{-1}(Q; A_N)e$ is the Neumann energy for a Euclidean unit vector e , fix $p \geq 2$ and take $\bar{\delta} \leq \bar{\delta}_0(d, t, p)$. Assume (8.22), with K_N bounded in L^{3p} , and the smoothed-response bound in (8.19) at order $3p$, and the approximation-error bound in (8.19) at order p . The threshold is chosen so that the moment range of Proposition 4.1 contains $4p$ and so that Lemma 8.2 applies at order $4p$; that lemma supplies the uniform L^{4p} bound for the unsmoothed response. Under these conditions, (8.30) holds with right side $C_p \bar{\delta}^{1/2} 3^{-b\ell/256}$.*

Proof. For $j \geq 0$, (8.28) bounds the resampling error by $C_p \bar{\delta} 3^{-bj/8}$, and the error is zero for $j > N$. For $j < 0$, the field H changes by the difference of the expressions $\mathbf{g}_{-j} - \mathbf{g}_{-j}(0)$ for the field and for its independent copy. This difference has norm of size $C \bar{\delta} 3^j$ and sub-Gaussian tails by Lemma 8.1. Equation (8.24) bounds the resampling error by $C_p \bar{\delta} 3^j$. Sum both geometric series using Lemma 8.16.

For the Neumann energy on $Q = (0, 1)^d$, first use the volume source f_ϵ . The proof of Lemma 8.15 applies because Lemma 8.13 also covers zero-conormal problems with a given right-hand side. Here C_p depends on the uniform L^{3p} bound for the source-independent K_N of (8.22), the uniform L^{3p} bounds for the smoothed energies, and the uniform L^p bound for the approximation error in (8.19) after division by $\epsilon^{1/4}$. The coefficient-comparison and centered-energy estimates use their

finite moment orders as well. All these orders are fixed before decreasing $\bar{\delta}$; the moment orders are those specified in the statement.

Choose a numerical bound B for the L^{3p} norms of K_N and the smoothed responses, the L^{4p} norms of the unsmoothed response, and the L^p approximation norm after division by $\epsilon^{1/4}$. The last two inputs are supplied by Lemma 8.2 and (8.19). Let $C_{\text{resp}} = C(d)$ be the constant in Proposition 4.1, and decrease $\bar{\delta}$ so that their admissible range contains $4p$, that is, $4p \leq C_{\text{resp}}^{-1} \bar{\delta}^{-2} |\log \bar{\delta}|^{-1}$. The constants and disorder threshold are chosen after these bounds and the fixed bump profile, and before the cutoff and field window. Here $p \geq 2$ as in Lemma 8.15; C_p depends on $d, t, p, B, C_{\text{resp}}$ and the fixed data and bump profile.

By (8.22), the Morrey constant for the solution with source f_ϵ is $K_N \epsilon^{-2}$. This factor enters deletion of the field on strips linearly and the Efron–Stein estimate through its square root. Together with (8.19) and Hölder’s inequality at the stated orders, this gives

$$\|\mathcal{R}_N - \mathbb{E}[\mathcal{R}_N \mid \mathbf{g}_{-j}, |j| \leq \ell]\|_p \leq C_p (\epsilon^{1/4} + \bar{\delta} \epsilon^{-2} 3^{-b\ell/8} + \bar{\delta} 3^{-\ell}).$$

Choose $\epsilon = 3^{-b\ell/32}/16$ to obtain $C_p 3^{-b\ell/128}$.

For the other bound use the deterministic reference $c_Q = |Q|e \cdot e = 1$. With H removed, the physical dilation $X = 3^N x$ maps Q to the translated cube of side 3^N at its matching stationary cutoff N . Reindex the stationary environment by this dilation and translation. Write J_N^{ref} for its normalized defect J . Proposition 4.1, at the finite orders up to $4p$, gives $\|J_N^{\text{ref}}\|_{L^{4p}} \leq C_p \bar{\delta}^2$. The matrix-deviation estimate (2.29) gives

$$|\mathcal{R}_N^0 - 1|^2 \leq C J_N^{\text{ref}} (1 + J_N^{\text{ref}}), \quad \|\mathcal{R}_N^0 - 1\|_p \leq C_p \bar{\delta}.$$

Restoring H changes the response by at most $(e^{\|H\|_{L^\infty(Q)}} - 1) \mathcal{R}_N^0$; Hölder’s inequality and Lemma 8.1 bound this change by $C_p \bar{\delta}$. Conditional expectation is an L^p contraction, so

$$\|\mathcal{R}_N - \mathbb{E}[\mathcal{R}_N \mid \mathbf{g}_{-j}, |j| \leq \ell]\|_p \leq 2 \|\mathcal{R}_N - c_Q\|_p \leq C_p \bar{\delta}.$$

Taking the geometric mean with the preceding decay bound proves the asserted $C_p \bar{\delta}^{1/2} 3^{-b\ell/256}$ estimate. The norm of e here is Euclidean: $|e|^2 = \sum_i e_i^2$.

For the cell terms in Lemma 8.36, apply this estimate to the normalized response on the unit cube in the translated environment at the cell’s own cutoff. Translation between $\square_0$ and $(0, 1)^d$ preserves the affine load and energy. More generally, if $A_N(w + Ry) = s_N \widetilde{A}_m(y)$, the physical constant-flux response is $R^d s_N^{-1}$ times the unit-cube response for $\widetilde{A}_m$. The conditioning estimate applies after removing this scalar and at cutoff m . Its finite moment inputs are chosen at $3P$ and $6P$ when the response is used at P and $2P$. For general fixed physical cubes we use the transported volume loads in the proof of Lemma 8.9, followed by Proposition 9.21. $\square$

8.3.3. Compactness of conditional expectations.

Proposition 8.18 (Strong compactness of the finite-volume energies). *Let Q be a cube, and let $\mathcal{R}_N$ be the minimal energy $\Lambda_{N,Q}(\varphi)$ with boundary values φ , the quadratic form $\langle f, G_N^Q f \rangle$ of the Dirichlet Green operator with source f , or the Neumann energy $|Q|e \cdot a_*^{-1}(Q; A_N)e$ for a constant flux $e \in \mathbb{R}^d$. Assume $1 \leq p < q < \infty$ and $\sup_N \|\mathcal{R}_N\|_{L^q} < \infty$. Assume also that, for some $a > 0$ and $C_Q < \infty$, every pair of integers $0 \leq \ell \leq N$ satisfies*

$$\|\mathcal{R}_N - \mathbb{E}[\mathcal{R}_N \mid \mathbf{g}_{-j}, |j| \leq \ell]\|_{L^p} \leq C_Q \bar{\delta} 3^{-a\ell}.$$

Here C_Q may depend on the fixed cube, data and disorder, but not on N or ℓ . For a countable collection the exponent a is common and C_Q and the moment bound may depend on its member. For the minimal energy and the Dirichlet Green quadratic form the approximation is supplied by Proposition 8.17; for the constant-flux Neumann response on a fixed cube it follows from the load approximation proved below. Then the following statements hold.

- (i) The sequence $(\mathcal{R}_N)_N$ is relatively compact in $L^p(\mathbb{P})$.
- (ii) The same holds for $\langle f, G_N^Q g \rangle$ with two sources f and g .
- (iii) For any countable collection of cubes, sources, boundary data and fixed integer shifts $N \mapsto N+j$ of the cutoff N , a common subsequence can be chosen along which all the corresponding energies converge.

Proof. Fix ℓ and $N \geq \ell$. The log coefficient on $\bar{Q}$ is the sum of the retained potential $\sum_{-\ell \leq j < 0} (\mathbf{g}_{-j} - \mathbf{g}_{-j}(0)) + \sum_{0 \leq j \leq \ell} \mathbf{g}_{-j}$ and a remaining potential independent of the retained fields. As a function of $v \in C(\bar{Q})$, take the expectation of the energy for the log coefficient v plus that remaining potential. Independence identifies its value at the retained potential with the conditional expectation of $\mathcal{R}_N$. By (8.24), values at v, w compare by $e^{\|v-w\|_\infty}$. Choose a fixed radius which bounds the retained potential with positive probability. Integration on this event bounds the value at 0 by $\mathbb{E}\mathcal{R}_N$ times the exponential of the radius divided by that probability. The functions are therefore uniformly bounded and equicontinuous on every bounded set. A diagonal extraction on a countable dense subset of $C(\bar{Q})$ gives pointwise convergence everywhere. Conditional Jensen gives a uniform L^q bound, so convergence at the retained random potential is in L^p .

For any $\varepsilon > 0$, choose ℓ so that $C_Q \bar{\delta} 3^{-a\ell} < \varepsilon/2$. A finite $\varepsilon/2$ -net of the conditional expectations is an ε -net of the energies with $N \geq \ell$. The omitted finite prefix is compact. This proves (i).

For the constant-flux Neumann response on a general fixed Q , use $Y_{N,h}$ from the transferred loads in the proof of Lemma 8.9. The source-independent estimate of Proposition 8.12 has Morrey constant at most $C_{Q,p} K_N h^{-2}$ for this source, while the smoothed responses have uniformly bounded prescribed moments by the load comparison. Lemma 8.13, the proof of Lemma 8.15 and the resampling sum therefore give

$$\|Y_{N,h} - \mathbb{E}[Y_{N,h} \mid \mathbf{g}_{-j}, |j| \leq \ell]\|_{L^p} \leq C(p, Q) \bar{\delta} (h^{-2} 3^{-b\ell/8} + 3^{-\ell}).$$

The required finite moment orders are fixed before disorder, independently of h . The load approximation and conditional-expectation contraction yield

$$\|Y_N - \mathbb{E}[Y_N \mid \mathbf{g}_{-j}, |j| \leq \ell]\|_{L^p} \leq C(p, Q) (h^{1/4} + \bar{\delta} h^{-2} 3^{-b\ell/8} + \bar{\delta} 3^{-\ell}).$$

Take $h = 3^{-b\ell/32}/16$. Since $0 < b < 1$ and $0 < \bar{\delta} \leq 1$, this is at most $C(p, Q) 3^{-b\ell/128}$. Thus the finite-band hypothesis holds with $a = b/128$ and $C_Q = C(p, Q)/\bar{\delta}$, or with any smaller positive a . No bound on C_Q uniform as $\bar{\delta} \downarrow 0$ is asserted here. The retained-potential argument applies to this fixed linear load vanishing on constants. This proves the general fixed-cube Neumann case; for $p < 2$ apply the resampling estimates at order two and downgrade the norm.

For a joint family, choose a common exponent $a = b/256$. Apply the assumptions to the two diagonal responses with data $f + g$ and $f - g$; the chosen countable linear source family contains both. Polarization gives (ii), and a diagonal extraction over the countable collection gives (iii). Translation and any fixed cutoff shift are included in that collection. The centered unit-cube estimate of Proposition 8.17 is used for the normalized cell responses; the general fixed-cube argument here only needs the transferred load approximation. $\square$

8.4. Subsequential limit forms. We construct limit forms on $L^2(Q, dx)$ along a subsequence and show that they are local, compatible under restriction to subcubes, and multiplied by a continuous weight when the coefficient is.

Let $\mathcal{C}$ be the set of cubes with rational centers and triadic side lengths. It contains the cubes $\square_j$, $j \geq 0$, and all their triadic subcubes. Fix $1/2 < \beta < \alpha < 1$, $d-1 < t < d$ and $0 < \eta < 2\alpha - 1$, with α so close to 1 that (8.48) of Lemma 8.48 holds for some $\gamma \in (0, 1)$ and $\zeta > 0$, and with $2(1 - \alpha) + \eta < 1/24$. Choose a countable $\mathbb{Q}$ -linear family $\mathcal{D} \subseteq C_c^\infty(\mathbb{R}^d)$ such that, for each cube, the functions of $\mathcal{D}$ supported in the cube are dense in L^2 and in C_c^∞ on that cube. This also gives density in C_c^α for the C^β norm. For each finite union of closed rational cubes

contained in a finite union of open rational cubes, include in $\mathcal{D}$ a smooth cutoff function, that is, a function with values in $[0, 1]$ equal to one near the first set and zero outside the second. Also include the smooth functions used to build the cutoff functions $\chi_{N,r}$ of Lemma 8.22(iii).

The **harmonic interpolant** of a function on a mesh is the function equal, on each cell, to the A_N -harmonic extension of the restriction of that function to the cell boundary.

We extract a subsequence jointly for the finite-volume energies for these cubes and data, the solutions, the environment and the random constants in (8.5)–(8.11), together with the coercivity constants $K_{Q,N}$ in the proof of Lemma 8.9 for the cubes $Q \in \mathcal{C}$. Include as single random variables each constant of (8.7), which is valid on all grid cubes at once, the constants of Proposition 8.7 and Remark 8.8, which do not depend on the source, the corresponding constants for A_N^0 on the cells of $\mathcal{C}$ (see the discussion after Proposition 8.31), and $3^{-\eta N} \|A_N\|_{L^\infty(Q)}$, whose moments are bounded by Lemma 8.6. Proposition 8.18 gives L^p convergence of the energies along a subsequence. The moment and Hölder bounds give tightness of all the remaining coordinates.

Apply Prokhorov's theorem and Skorokhod's representation theorem [Kallenberg, 2002, Theorems 16.3 and 4.30] in the countable product of the Polish coordinate spaces. We obtain a further subsequence and a representation on a probability space where all coordinates converge almost surely. The energy limits retain their joint law with the environment and their measurability with respect to the random fields.

We work on the event of probability one on which all coordinates converge. The constants just included are bounded along the sequence in the Skorokhod representation. The bounds (8.7) hold for all meshes, and the bounds (8.10) hold for all sources and boundary data. Fix $Q \in \mathcal{C}$ and abbreviate $E_N = E_N^Q$, $G_N = G_N^Q$.

8.4.1. The Dirichlet Green operator and the limit form.

Proposition 8.19 (Convergence of the Dirichlet Green operators and the limit form). *Along the sequence in the Skorokhod representation, the following statements hold.*

- (i) Norm convergence: the operators G_N converge in operator norm on $L^2(Q, dx)$ to a compact, positive, self-adjoint, injective operator G .
- (ii) Limit form: the form

$$(8.31) \quad E(u) = \sup_{f \in L^2(Q)} \{2\langle f, u \rangle - \langle f, Gf \rangle\}, \quad D(E) = \{u : E(u) < \infty\} = \text{Ran } G^{1/2},$$

is a densely defined closed Dirichlet form on $L^2(Q, dx)$.

- (iii) Mosco convergence: if $u_N \rightharpoonup u$ weakly in $L^2(Q)$, then $E(u) \leq \liminf E_N(u_N)$ (the *liminf inequality*). Every $u \in D(E)$ has a **recovery sequence**, that is, a sequence $u_N \in H_0^1(Q)$ with $u_N \rightarrow u$ in $L^2(Q)$ and $E_N(u_N) \rightarrow E(u)$.
- (iv) Density: the functions Gf with $f \in C_c^\infty(Q)$ are dense in $D(E)$ for the form norm.

Proof. Step 1. We show that G_N converges in operator norm. By (8.5) and the energy identity,

$$\|G_N f\|_{H^{3/4}}^2 \leq K_N E_N(G_N f) = K_N \langle f, G_N f \rangle \leq K_N \|f\|_2 \|G_N f\|_2,$$

so $\|G_N\|_{L^2 \rightarrow H^{3/4}} \leq \sup_N K_N < \infty$ along the sequence in the Skorokhod representation. By Rellich's theorem the images of the unit ball under all G_N lie in a common precompact subset of L^2 . Given $\varepsilon > 0$, choose a finite ε -net for these images consisting of elements of $\mathcal{D}$, and let P be the orthogonal projection onto its span. Then $\|(I - P)G_N\| \leq \varepsilon$. By self-adjointness, $\|G_N(I - P)\| = \|(I - P)G_N\| \leq \varepsilon$. Hence $\|G_N - G_M\| \leq \|P(G_N - G_M)P\| + 4\varepsilon$. The operator PG_NP is determined by the finite matrix of the bilinear form $\langle \cdot, G_N \cdot \rangle$ on the elements of the net. Polarization expresses each entry as a real linear combination of quadratic forms $\langle h, G_N h \rangle$ with $h \in \mathcal{D}$, which converge by the choice of the sequence in the Skorokhod representation. Hence (G_N) is Cauchy in operator norm.

Step 2. We show that G is injective. For $w \in H_0^1(Q)$ and $s \in \mathbb{R}$, the variational formula (8.3) gives $\langle f, G_N f \rangle \geq 2s\langle f, w \rangle - s^2 E_N(w)$, and optimizing in s gives $|\langle f, w \rangle|^2 \leq \langle f, G_N f \rangle E_N(w)$. Let $Gf = 0$ and $\phi \in \mathcal{D} \cap C_c^\infty(Q)$, and let $w_{N,r}$ be the harmonic interpolant of ϕ on the rational mesh of side $r < \text{dist}(\text{supp } \phi, \partial Q)$. By (8.6) with $\|\phi(z + r \cdot)\|_{C^\beta(\partial \square_0)/\mathbb{R}} \leq Cr \|\nabla \phi\|_\infty$,

$$E_N(w_{N,r}) = \sum_q \Lambda_{N,q}(\phi|_{\partial q}) \leq Cr^{-d} \max_q \Lambda_{\sigma/2,2}(q; A_N) r^{d-2} r^2 \|\nabla \phi\|_\infty^2,$$

which is bounded along the sequence in the Skorokhod representation at fixed r , since only finitely many cells of $\mathcal{C}$ occur. The maximum principle on each cell gives $\|w_{N,r} - \phi\|_\infty \leq Cr \|\nabla \phi\|_\infty$. Hence

$$|\langle f, \phi \rangle| \leq |\langle f, w_{N,r} \rangle| + Cr \|f\|_1 \|\nabla \phi\|_\infty,$$

and the first term tends to zero as $N \rightarrow \infty$, since $\langle f, G_N f \rangle \rightarrow \langle f, Gf \rangle = 0$. Letting $r \downarrow 0$ gives $f \perp C_c^\infty(Q)$, so $f = 0$.

Step 3. The spectral theorem gives $E(u) = \|G^{-1/2}u\|_2^2$ with domain $\text{Ran } G^{1/2}$, a dense closed form domain. For every $\lambda > 0$, Step 1 gives

$$(G_N^{-1} + \lambda I)^{-1} = G_N(I + \lambda G_N)^{-1} \longrightarrow G(I + \lambda G)^{-1} = (G^{-1} + \lambda I)^{-1}$$

in operator norm. The resolvent criterion [Kuwae and Shioya, 2003, Theorem 2.4] therefore gives Mosco convergence on $L^2(Q, dx)$. Applying the liminf inequality to normal contractions of recovery sequences proves the Markov property of E . Finally, $E(u, Gf) = \langle u, f \rangle$; if u is orthogonal in the form norm to $G\mathcal{D}$, then $\langle (I + G)u, f \rangle = 0$ for every $f \in \mathcal{D}$, so $u = 0$. This proves (ii)–(iv). $\square$

Since $\mathcal{C}$ contains countably many cubes, the proposition gives, along the sequence in the Skorokhod representation, a limit form E^Q on every cube Q of $\mathcal{C}$; we write $E = E^Q$ when the cube is fixed.

The comparison arguments of Section 8.6 need the quadratic forms $\langle f, G_N^Q f \rangle$ to be bounded above and below with positive probability, uniformly in N and in the extracted limit. Both bounds follow from moment bounds.

Lemma 8.20 (Positive and negative moments of the quadratic forms). *Fix a cube Q of side at most one and a nonzero function $f \in C_c^\infty(Q)$. For every $p < \infty$, if $\bar{\delta}$ is sufficiently small, then*

$$\sup_N \mathbb{E}[\langle f, G_N^Q f \rangle^p + \langle f, G_N^Q f \rangle^{-p}] < \infty.$$

The same bounds hold for every extracted limit, uniformly over the subsequence and over all values of $\bar{\delta}$ in a fixed, sufficiently small interval.

Proof. The coercivity bound (8.5) gives $\langle f, G_N^Q f \rangle \leq K_N \|f\|_2^2$. For the negative moments, fix a mesh of side r and let $w_{N,r}$ be the harmonic interpolant of f on it, as in Step 2 of the proof of Proposition 8.19. The maximum principle gives $\|w_{N,r} - f\|_\infty \leq Cr \|\nabla f\|_\infty$. Fix r so small that $\langle f, w_{N,r} \rangle \geq \frac{1}{2} \|f\|_2^2$ for every N . The energy of $w_{N,r}$ has a uniformly bounded p th moment by (8.6)–(8.7) on this fixed finite mesh. Optimize the variational formula (8.3) over the multiples of $w_{N,r}$ to obtain

$$\langle f, G_N^Q f \rangle^{-1} \leq \frac{E_N^Q(w_{N,r})}{\langle f, w_{N,r} \rangle^2} \leq 4 \|f\|_2^{-4} E_N^Q(w_{N,r}).$$

Fatou's lemma passes both moment bounds to each extracted limit. $\square$

8.4.2. *Traces and localization.* We will apply the following trace estimate to the energy measures of the harmonic interpolants of cutoff functions (Lemma 8.22).

Lemma 8.21 (Measure trace estimate). *Let $t > d - 1$ and $K < \infty$, and let ν be a finite positive Borel measure on $\mathbb{R}^d$ supported in $\overline{Q}$ such that $\nu(B_r(x)) \leq Kr^t$ for every $x \in \overline{Q}$ and $0 < r \leq 1$. Then the map $v \mapsto v$, from smooth functions on $\overline{Q}$ to $L^2(\nu)$, extends by completion to $H^{1/2}(Q)$, and there exists $C(d, t, Q) < \infty$ such that, for every $v \in H^{1/2}(Q)$,*

$$(8.32) \quad \int |v|^2 d\nu \leq C(d, t, Q)(K + \nu(Q)) \|v\|_{H^{1/2}(Q)}^2.$$

Moreover, if $v_N \rightarrow v$ in $L^2(Q)$ and $\sup_N \|v_N\|_{H^{3/4}(Q)} < \infty$, then $v_N \rightarrow v$ in $H^{1/2}(Q)$.

Proof. The Bessel kernel of order one is bounded by $C|z|^{1-d}$ for $|z| \leq 1$ and is bounded outside the unit ball. The assumption $\nu(B_r(x)) \leq Kr^t$ and $t > d - 1$ therefore give

$$\sup_{x \in \text{supp } \nu} \int (|x - y|^{1-d} \vee 1) d\nu(y) \leq C \left(\nu(\overline{Q}) + K \sum_{n \geq 0} 2^{-n(t-d+1)} \right) \leq C(K + \nu(\overline{Q})).$$

Regard the Bessel potential of order $1/2$ as a map from $L^2(dx)$ to $L^2(\nu)$. The composition of this map with its adjoint has the Bessel kernel of order one as its kernel. This kernel is bounded by the integrand above. By Schur's test, the squared norm of the map is at most $C(K + \nu(\overline{Q}))$. The Bessel-potential description of $H^{1/2}(\mathbb{R}^d)$ and a bounded extension operator from the cube give (8.32) for smooth v .

The boundary of Q is ν -null: cover it with $O(r^{1-d})$ balls. Hence $\nu(\overline{Q}) = \nu(Q)$. Density of smooth functions in $H^{1/2}(Q)$ gives the asserted completion, and convergence in $L^2(dx)$ identifies it with the ordinary integral when ν has a bounded density with respect to Lebesgue measure. Finally, the same extension operator and Sobolev interpolation give

$$\|w\|_{H^{1/2}(Q)} \leq C \|w\|_{L^2(Q)}^{1/3} \|w\|_{H^{3/4}(Q)}^{2/3}.$$

Weak compactness first puts v in $H^{3/4}(Q)$; applying this inequality to $w = v_N - v$ proves the last assertion. $\square$

Along the sequence in the Skorokhod representation, recovery sequences and solutions with zero Dirichlet boundary values are bounded in $H^{3/4}$ by (8.5), and the energy measures of the cutoff functions of Lemma 8.22 below satisfy (8.10) with a common constant. Thus (8.32) controls $\int (v_N - v)^2 d\Gamma_N(\chi_N)$ uniformly in N . We use this estimate when localizing recovery sequences in Proposition 8.24.

8.4.3. *Regularity, locality, and restriction to subcubes.* To obtain energy measures for E , we show that E is regular and strongly local. Both properties are proved by multiplying recovery sequences by cutoff functions. Smooth cutoff functions are not suitable, since A_N has no upper bound uniform in N ; we use instead the harmonic interpolants of cutoff functions, whose energies, constants in the Morrey bounds and Hölder norms are controlled cell by cell by (8.6)–(8.11).

Lemma 8.22 (Harmonic cutoff functions). *Let $K \subseteq O$ be finite unions of closed and open rational cubes, respectively, with K compact and $\overline{O} \subseteq Q$, and let χ_N be the harmonic interpolant, on a mesh of $\mathcal{C}$, of a smooth cutoff function from $\mathcal{D}$, with values in $[0, 1]$, equal to one near K and compactly supported in O . The mesh is so fine that every cell meeting the set where this function lies strictly between 0 and 1 is contained in $O \setminus K$. Then the following statements hold.*

- (i) *The function χ_N belongs to $H_0^1(Q) \cap C(\overline{Q})$, satisfies $0 \leq \chi_N \leq 1$, equals one on a neighbourhood of K and zero outside O , and $\nabla \chi_N = 0$ on every cell on which this function is constant.*

- (ii) Along the sequence in the Skorokhod representation, the energy $E_N(\chi_N)$, the constant in the bound (8.10) for $\Gamma_N(\chi_N)$ at all radii, and the C^α norm of χ_N are bounded. In particular, along a subsequence, χ_N converges uniformly to a continuous function χ that agrees with the original cutoff function on every cell where that function is constant.
- (iii) Let $r > 0$, and let $\chi_{N,r}$ be the harmonic interpolant on the mesh of side r of a smooth function $\mathbb{R}^d \rightarrow [0, 1]$ that vanishes within distance r of ∂Q , equals one at distance at least $3r$ from ∂Q and has gradient at most C/r on $\mathbb{R}^d$. Let K_N be the constant of (8.7). Then $0 \leq \chi_{N,r} \leq 1$ and

$$E_N(\chi_{N,r}) \leq C(K_N + 3^{-\eta N} \|A_N\|_{L^\infty(Q)}) r^{-1-\eta}.$$

Proof. On a cell on which the cutoff function is constant the harmonic extension is that constant, by uniqueness, and the maximum principle gives $0 \leq \chi_N \leq 1$. Continuity across faces follows from the matching traces, and continuity up to the boundary from the Hölder bound for the cell solutions. At a fixed mesh there are finitely many cells, each with bounded $\Lambda_{\sigma/2,2}(q; A_N)$ and bounded constants in the Morrey and Hölder bounds along the sequence in the Skorokhod representation. Sum over the cells to obtain (ii). The Arzelà–Ascoli theorem gives uniform convergence.

For (iii), the maximum principle gives $0 \leq \chi_{N,r} \leq 1$ as in (i). There are $O(r^{-(d-1)})$ cells on which the function defining $\chi_{N,r}$ is not constant. On each such cell the rescaled function has C^β norm at most C . If $r \geq 3^{-N}$, then (8.6) and (8.7) bound the energy on each such cell by $C\Lambda_{\sigma/2,2}(q; A_N)r^{d-2} \leq CK_N r^{d-2-\eta}$. If $r < 3^{-N}$, the smooth function defining $\chi_{N,r}$ is itself a competitor in the minimization on each cell, which gives

$$E_N(\chi_{N,r}) \leq C\|A_N\|_{L^\infty(Q)} r^{-1} \leq C(3^{-\eta N} \|A_N\|_{L^\infty(Q)}) r^{-1-\eta}.$$

Both the factor $3^{-\eta N} \|A_N\|_{L^\infty(Q)}$ and the constant of (8.7) were included in the joint extraction, and their sum bounds the energy for all widths r . $\square$

Proposition 8.23 (Regularity). *The Dirichlet form E of Proposition 8.19 is regular.*

Proof. For $f \in \mathcal{D}$, the Hölder bound (8.11) gives $u_N = G_N f \rightarrow Gf$ uniformly on $\overline{Q}$, with zero boundary values. Let $\chi_{N,r}$ be the cutoff function of Lemma 8.22(iii). Covering the boundary layer $\{x \in Q : \text{dist}(x, \partial Q) < 3r\}$ by $O(r^{1-d})$ balls and using (8.10)–(8.11) gives

$$\begin{aligned} E_N(u_N(1 - \chi_{N,r})) &\leq 2\Gamma_N(u_N)(\{\text{dist}(\cdot, \partial Q) < 3r\}) + 2 \sup_{\text{dist}(\cdot, \partial Q) < 3r} |u_N|^2 E_N(\chi_{N,r}) \\ (8.33) \quad &\leq C(\omega, f)(r^{t-d+1} + r^{2\alpha-1-\eta}). \end{aligned}$$

At fixed r , pass to a subsequence along which $\chi_{N,r} \rightarrow \chi_r$ uniformly. By (8.33) and the liminf inequality, $Gf\chi_r \in D(E) \cap C_c(Q)$ converges to Gf in the form norm as $r \downarrow 0$. Since the functions Gf are dense in $D(E)$ by Proposition 8.19(iv), $D(E) \cap C_c(Q)$ is dense in $D(E)$ for the form norm.

For density in the uniform norm, take $\phi \in \mathcal{D} \cap C_c^\infty(Q)$ and its harmonic interpolants on a mesh of side r . Their bounded energies and Hölder norms give a uniform subsequential limit $w_r \in D(E) \cap C_c(Q)$. The cell maximum principle gives $\|w_r - \phi\|_\infty \leq Cr\|\nabla\phi\|_\infty$. Since these ϕ are dense in $C_0(Q)$, the space $D(E) \cap C_c(Q)$ is uniformly dense as well. $\square$

Proposition 8.24 (Strong locality on $D(E) \cap C_c(Q)$). *Let $u, v \in D(E) \cap C_c(Q)$, and suppose that u equals a constant c on a neighbourhood of $\text{supp } v$. Then $E(u, v) = 0$.*

Proof. Choose two cutoff functions from $\mathcal{D}$ and let ψ_N, χ_N be their harmonic interpolants from Lemma 8.22, with $\psi_N = 1$ near $\text{supp } v$, $\chi_N = 1$ near $\text{supp } \psi_N$, and $\text{supp } \chi_N \subseteq \{u = c\}^\circ$. For recovery sequences u_N, v_N of u and v , consider the sequences $c\chi_N + (1 - \chi_N)u_N$ and $\psi_N v_N$. These are still recovery sequences. Indeed, they converge in L^2 to u and v , and the only extra gradient terms are controlled by

$$\int (u_N - c)^2 d\Gamma_N(\chi_N) + \int v_N^2 d\Gamma_N(\psi_N) \longrightarrow 0.$$

On the support of $\nabla\chi_N$ we have $u = c$, and on that of $\nabla\psi_N$ we have $v = 0$. Apply the trace estimate (8.32) together with the $H^{1/2}$ convergence of the recovery sequences. Young's inequality gives $\limsup E_N(c\chi_N + (1-\chi_N)u_N) \leq (1+\varepsilon)E(u)$ for every $\varepsilon > 0$, and likewise for $\psi_N v_N$. The liminf inequality of Proposition 8.19(iii) gives equality. Since $c\chi_N + (1-\chi_N)u_N = c$ near $\text{supp}\psi_N$, the form E_N of the two sequences vanishes. Applying the liminf inequality to their sum and difference yields $E(u \pm v) \leq E(u) + E(v)$, hence $E(u, v) = 0$. $\square$

By regularity and Proposition 8.24, uniqueness in the Beurling–Deny representation [Fukushima et al., 2011, Section 3.2] makes the jump and killing measures vanish. Thus E is strongly local, and its energy measures $\Gamma_E(u, v)$ satisfy the chain rule, Leibniz rule, Cauchy–Schwarz inequality and locality on open sets [Fukushima et al., 2011, Theorem 3.2.2 and Lemma 3.2.5].

We normalize the energy measure by $\int \phi d\Gamma_E(u) = E(u, u\phi) - \frac{1}{2}E(u^2, \phi)$ for bounded $u \in D(E)$ and $\phi \in D(E) \cap C_c(Q)$, so that the corresponding measure at finite N is $\Gamma_N(u) = A_N |\nabla u|^2 dx$, and $\Gamma_E(u)(Q) = E(u)$ and $\Gamma_E(u, v)(Q) = E(u, v)$. The measure $\Gamma_E(u)$ is supported in $\text{supp} u$.

Proposition 8.25 (Consistency under restriction to subcubes). *Let $q \subseteq Q$ be an open cube of $\mathcal{C}$, and let E^q and E^Q be the limit forms on q and Q along the sequence in the Skorokhod representation. Then extension by zero to Q sends $D(E^q)$ onto the E^Q -closure of $\{w \in D(E^Q) \cap C_c(Q) : \text{supp} w \subseteq q\}$, and $E^q(u)$ equals the E^Q -energy of the extension of u by zero, for every $u \in D(E^q)$.*

Proof. Zero extension of a recovery sequence on q shows that the E^Q -energy of the extension of u by zero is at most $E^q(u)$. Conversely, fix $w \in D(E^Q) \cap C_c(q)$. Localize a recovery sequence on Q by a harmonic cutoff function equal to one near $\text{supp} w$ and supported in q . The trace argument in Proposition 8.24 shows that this is still a recovery sequence. The liminf inequality on q (Proposition 8.19(iii)) gives $E^q(w|_q) \leq E^Q(w)$. Hence equality holds on $D(E^Q) \cap C_c(q)$. Regularity of E^q (Proposition 8.23) and the first inequality extend equality to all of $D(E^q)$. The resulting isometry has closed image, and this image is exactly the stated closure. $\square$

8.4.4. *Continuous weights and minimal energies with given boundary values.* For $\varrho \in C(\overline{Q})$ with $\varrho > 0$, let $E_N^\varrho(u) = \int_Q \varrho d\Gamma_N(u)$ for $u \in H_0^1(Q)$, and let G_N^ϱ be its Dirichlet Green operator.

Lemma 8.26 (Form order implies measure order). *Let E, F be regular strongly local Dirichlet forms on $L^2(Q, dx)$ with $D(E) = D(F)$, let $q \subseteq Q$ be open, and let $C < \infty$ be such that $F(w) \leq CE(w)$ for every $w \in D(E) \cap C_c(Q)$ with $\text{supp} w \subseteq q$. Then $\Gamma_F(u) \leq C\Gamma_E(u)$ on q for every $u \in D(E) \cap C_c(Q)$.*

Proof. Let $v \in D(E) \cap C_c(Q)$ with $\text{supp} v \subseteq q$, and let $k \geq 1$. The functions $k^{-1}v \sin(ku)$ and $k^{-1}v \cos(ku) = k^{-1}v(\cos(ku) - 1) + k^{-1}v$ belong to $D(E) \cap C_c(Q)$ and they are supported in $\text{supp} v$. For either form, the chain and Leibniz rules show that the cross terms in the energy measures of these two functions cancel in the sum, so that

$$E(k^{-1}v \sin ku) + E(k^{-1}v \cos ku) = \int v^2 d\Gamma_E(u) + k^{-2}E(v),$$

and likewise for F . Applying $F \leq CE$ to both functions and letting $k \rightarrow \infty$ gives $\int v^2 d\Gamma_F(u) \leq C \int v^2 d\Gamma_E(u)$. For $0 \leq \phi \in C_c(q)$, choose by regularity a function in $D(E) \cap C_c(Q)$ with values in $[0, 1]$ that equals one on $\text{supp} \phi$ and is supported in q . Also choose functions in $D(E) \cap C_c(Q)$ converging uniformly to $\phi^{1/2}$. Then the products of the latter with the former are admissible and their squares converge uniformly to ϕ . Hence $\int \phi d\Gamma_F(u) \leq C \int \phi d\Gamma_E(u)$. $\square$

Proposition 8.27 (Continuous weights). *Let $\varrho \in C(\overline{Q})$ be positive. Along the sequence in the Skorokhod representation, the forms E_N^ϱ Mosco-converge to a form E^ϱ such that $D(E^\varrho) = D(E)$ and*

$$(8.34) \quad E^\varrho(u, v) = \int_Q \varrho d\Gamma_E(u, v), \quad u, v \in D(E).$$

Proof. The bounds $(\inf \varrho)E_N \leq E_N^g \leq (\sup \varrho)E_N$ show that the forms E_N^g satisfy the $H^{3/4}$ coercivity bound (8.5) up to a constant factor, and that the order reverses for their Green operators: $(\sup \varrho)^{-1}G_N \leq G_N^g \leq (\inf \varrho)^{-1}G_N$. By the finite-rank compactness argument of Proposition 8.19, every subsequence therefore has a further subsequence along which G_N^g converges in norm on all cubes of $\mathcal{C}$. The reversed order makes each limit injective. The same proposition constructs their Mosco limit F , with $D(F) = D(E)$ and equivalent form norms. Therefore F is regular, and $D(F) \cap C_c(Q) = D(E) \cap C_c(Q)$. The proofs of locality and consistency under restriction to smaller cubes also apply: recovery sequences retain their $H^{3/4}$ bound, and the energy measures of the cutoff functions are at most $(\sup \varrho)\Gamma_N(\chi_N)$. Thus F is strongly local and its restrictions to smaller cubes satisfy $(\inf_q \varrho)E^q \leq F^q \leq (\sup_q \varrho)E^q$ for $q \in \mathcal{C}$. By Lemma 8.26, the same comparison holds for energy measures on q . For $u \in D(E) \cap C_c(Q)$, take a finite partition of unity near $\text{supp } u$ subordinate to cubes on which $\text{osc } \varrho \leq \varepsilon$. Integrate against this partition of unity to obtain $|F(u) - \int \varrho d\Gamma_E(u)| \leq \varepsilon E(u)$. Let $\varepsilon \downarrow 0$, extend by density in the form norm and polarize. Every subsequential limit form is therefore the form in (8.34), which proves convergence of the whole sequence in the Skorokhod representation. $\square$

We also obtain convergence in probability of the weighted Dirichlet Green operators along the subsequence extracted above. Indeed, given a subsequence, we represent the weight, the original energy limit and the preceding coordinates jointly on one probability space. The comparison (8.24) absorbs the error in the uniform approximation of the weight. The proof of Proposition 8.27 identifies every limit with (8.34). The subsequence criterion for convergence in probability then gives the assertion, also for random positive continuous weights and for countable families of them.

Lemma 8.28 (Bounded weights and deletion of a field). *Let g be a bounded Borel function on Q . Then $E^g(u, v) = \int e^g d\Gamma_E(u, v)$ defines a regular strongly local Dirichlet form with domain $D(E)$, and $\Gamma_{E^g}(u, v) = e^g \Gamma_E(u, v)$. Moreover, for every field $\mathbf{g}_{-j}$ with $j \geq 0$, the form $e^{-\mathbf{g}_{-j}}E$, given by $(u, v) \mapsto \int e^{-\mathbf{g}_{-j}} d\Gamma_E(u, v)$, is measurable with respect to the remaining fields.*

Proof. The bounds $e^{-\|g\|_\infty}E \leq E^g \leq e^{\|g\|_\infty}E$ show that E^g is closed on $D(E)$, with a form norm equivalent to that of E , so that E^g is regular, like E . The chain rule for energy measures gives the Markov property, and the locality of energy measures gives strong locality [Fukushima et al., 2011, Theorem 3.2.2 and Lemma 3.2.5]. The defining identity of the energy measure and the Leibniz rule give the formula for Γ_{E^g} .

For the deletion, observe that $e^{-\mathbf{g}_{-j}}$ is continuous and bounded above and below on each fixed cube. The coefficient $e^{-\mathbf{g}_{-j}}A_N$ is independent of the deleted field for all sufficiently large N . Proposition 8.27, applied to random weights as explained after its proof, identifies the limit of the Dirichlet Green operators of $e^{-\mathbf{g}_{-j}}A_N$ with the Dirichlet Green operator of $e^{-\mathbf{g}_{-j}}E$. Hence each quadratic form $\langle f, G^Q f \rangle$ of this Dirichlet Green operator, with $f \in \mathcal{D}$ and $Q \in \mathcal{C}$, is a limit in probability of functions of the remaining fields. It is therefore measurable with respect to the remaining fields. These quadratic forms determine the form $e^{-\mathbf{g}_{-j}}E$. $\square$

Therefore, conditionally on the remaining fields, the forms corresponding to coefficients in which the deleted field is multiplied by the cutoff functions of Lemma 8.15 can be defined by weighting the fixed form $e^{-\mathbf{g}_{-j}}E$ with the independent restrictions of the deleted field. The first part of the lemma allows these weights to be discontinuous.

Corollary 8.29 (Energy-measure convergence). *Along the sequence in the Skorokhod representation, the following statements hold, with weak convergence of finite measures on $\overline{Q}$.*

- (a) *If $u_N \in H_0^1(Q)$ is a recovery sequence for $u \in D(E)$, then $\Gamma_N(u_N) \rightarrow \Gamma_E(u)$.*
- (b) *If $u_N \rightarrow u$ in L^2 and $\sup_N E_N(u_N) < \infty$, then every weak subsequential limit of $\Gamma_N(u_N)$ is at least $\Gamma_E(u)$.*

(c) If (u_N) is as in (b) and (v_N) is a recovery sequence for $v \in D(E)$, then $\Gamma_N(u_N, v_N) \rightarrow \Gamma_E(u, v)$.

Proof. Let $0 \leq \phi \in C(\overline{Q})$. For a recovery sequence, the liminf inequalities of Proposition 8.27 for the weighted forms with the positive weights $1 \pm \varepsilon\phi$, where $\varepsilon > 0$ and $\varepsilon\|\phi\|_\infty < 1$, together with $E_N(u_N) \rightarrow E(u)$, give both inequalities needed for (a). For (b), apply the liminf inequality for the weight $\phi + \varepsilon$, and let $\varepsilon \downarrow 0$ along a subsequence on which the measures converge weakly.

For (c), Cauchy–Schwarz bounds the total variations of the mutual energy measures. Pass to a joint subsequential limit of $\Gamma_N(u_N)$ and $\Gamma_N(u_N, v_N)$. Apply (b) to $u_N + sv_N$ and (a) to v_N . For every rational s , this shows that the measure

$$\lim_N \Gamma_N(u_N) - \Gamma_E(u) + 2s(\lim_N \Gamma_N(u_N, v_N) - \Gamma_E(u, v))$$

is nonnegative. Testing against $0 \leq \phi \in C(\overline{Q})$ and letting $s \rightarrow \pm\infty$ identifies the limit of $\Gamma_N(u_N, v_N)$ with $\Gamma_E(u, v)$. Every subsequential limit is therefore the same. $\square$

For the regular strongly local form E , the one-dimensional energy-image-density theorem and the absolutely continuous chain rule [Fitzsimmons, 2004, Proposition 1.16 and its proof] give $\Gamma_E(v)(\{v = c\}) = 0$ and $\Gamma_E(T(v)) = T'(v)^2 \Gamma_E(v)$ for Lipschitz T with $T(0) = 0$, using quasi-continuous representatives. The reference normalizes the energy measure as $2\Gamma_E$, which does not affect either assertion.

Lemma 8.30 (Zero-trace differences lie in $D(E^q)$). *Let $q \subseteq Q$ be a cube of $\mathcal{C}$, and let $v \in D(E^Q) \cap C(\overline{Q})$ vanish on ∂q . Then $v_q = v\mathbf{1}_{\overline{q}}$ belongs to the space $D(E^q) \subseteq D(E^Q)$ of functions extended by zero, and $E^Q(v_q) = \Gamma_E(v)(q)$.*

Proof. Since $v = 0$ on ∂q , for every $\epsilon > 0$ the set $\{|v| \geq \epsilon\} \cap \overline{q}$ is compact in q . Truncate v_q at height ϵ , that is, replace it by $\text{sgn}(v_q)(|v_q| - \epsilon)_+$, and multiply by a cutoff function in $D(E^Q) \cap C_c(Q)$ equal to one near this set and supported in q . The result belongs to $D(E^q)$ by consistency under restriction to smaller cubes (Proposition 8.25). The chain rule and locality show that its energy measure is the restriction of $\Gamma_E(v)$ to $q \cap \{|v| > \epsilon\}$. The energy of the difference of the truncations at two heights is the mass of $\Gamma_E(v)$ on the part of q where $|v|$ lies between these heights. Since the level sets of v carry no energy, the truncations are Cauchy in the form norm and converge uniformly to v_q . Since E^q is closed, $v_q \in D(E^q)$. Letting $\epsilon \downarrow 0$ in the first identity gives $E(v_q) = \Gamma_E(v)(q)$, with $\Gamma_E(v_q)(\partial q) = 0$. $\square$

We now identify the limits of the minimal energies $\Lambda_{N,q}$ with given boundary values on cubes of $\mathcal{C}$. The candidate is the energy in q of the limit U of the harmonic interpolants of a fixed smooth datum on the 3^d cells around q .

Proposition 8.31 (Limits of the minimal energies with given boundary values). *Let $q = z + r\Box_0 \in \mathcal{C}$, let $Q = z + 3r\Box_0$ be the cube formed by the 3^d congruent cells around q , let b be a boundary datum on ∂q from $\mathcal{D}$, and fix a smooth extension of b to a function with compact support in Q . Let U_N be the function equal, on each of the 3^d cells, to the A_N -harmonic extension of the restriction of this extension to the cell boundary. Then, along the sequence in the Skorokhod representation, U_N converges uniformly on $\overline{Q}$ to a function U with the following properties.*

- (i) Boundary values: the function U belongs to $D(E^Q) \cap C(\overline{Q})$, and $U = b$ on ∂q .
- (ii) No energy on the faces: the energy measure of U satisfies $\Gamma_E(U)(\partial q) = 0$.
- (iii) Harmonicity: for every $\phi \in D(E^q)$, extended by zero to Q , we have $E^Q(U, \phi) = 0$.
- (iv) Minimality: the minimal energies converge, and

$$\lim_N \Lambda_{N,q}(b) = \Gamma_E(U)(q) = \min\{\Gamma_E(V)(q) : V \in D(E^Q) \cap C(\overline{Q}), V = b \text{ on } \partial q\}.$$

Proof. The energy, Hölder norm and constant in the Morrey bound for U_N are bounded by summing the corresponding cell bounds. Hence every subsequence has a further subsequence with $U_N \rightarrow U$ uniformly, where $U \in D(E^Q)$ and U equals the fixed extension of b on every cell boundary. We show that (U_N) is a recovery sequence for U .

Let W_N be a recovery sequence for U , let Σ be the union of the cell boundaries, and take on each cell the cutoff function of Lemma 8.22(iii) with $r = \rho$. Their sum $\chi_{N,\rho}$ vanishes near Σ and equals one away from its 3ρ -neighborhood. Use $\chi_{N,\rho}W_N + (1 - \chi_{N,\rho})U_N$, which has the same cell-boundary values as U_N , as a competitor. Cellwise minimality and convexity give, for $\varepsilon > 0$,

$$\begin{aligned} E_N(U_N) &\leq (1 + \varepsilon) [E_N(W_N) + \Gamma_N(U_N)(\{\text{dist}(\cdot, \Sigma) < 3\rho\})] \\ &\quad + (1 + \varepsilon^{-1}) \int (W_N - U_N)^2 d\Gamma_N(\chi_{N,\rho}). \end{aligned}$$

At fixed ρ , the last term tends to zero by (8.32) and convergence in $H^{1/2}$. The $\Gamma_N(U_N)$ -measure of the set of points at distance less than 3ρ from Σ is at most $C\rho^{t-d+1}$ by the Morrey bound at every radius and a covering of the finitely many faces. Let first $N \rightarrow \infty$, then $\rho \downarrow 0$ and $\varepsilon \downarrow 0$. The liminf inequality gives $E_N(U_N) \rightarrow E(U)$.

Corollary 8.29(a) now gives $\Gamma_N(U_N) \rightarrow \Gamma_E(U)$. The common Morrey bound passes to the limit. Hence $\Gamma_E(U)(\Sigma) = 0$ and $\Lambda_{N,q}(b) = \Gamma_N(U_N)(q) \rightarrow \Gamma_E(U)(q)$. For $\phi \in D(E^q)$, a recovery sequence on q , extended by zero, is also a recovery sequence on Q by consistency under restriction to smaller cubes (Proposition 8.25). The identity $E_N(U_N, \phi_N) = 0$ at finite N passes to $E(U, \phi) = 0$ by Corollary 8.29(c).

If $V = b$ on ∂q , then by Lemma 8.30 the function $w = (V - U)\mathbb{1}_{\bar{q}}$ lies in $D(E^q)$. Its energy measure does not charge ∂q . Harmonicity therefore yields $\Gamma_E(V)(q) = \Gamma_E(U)(q) + E(w) \geq \Gamma_E(U)(q)$. The same argument on each cell shows that the difference of two possible limits has zero energy on every cell. Since the energy measures do not charge the faces, coercivity forces this difference to vanish on Q . Thus the uniform convergence and all the conclusions hold along the full sequence in the Skorokhod representation. The argument applies to every fixed finite partition into cubes with a common smooth datum vanishing on the outer boundary. $\square$

We will also use this boundary identification after removing the infrared factor. On the same realization on Q , put $\varrho = e^{-H}$ and let $E^{e,Q}$ be the form obtained from the same padded Dirichlet Green limit by Proposition 8.27. Its energy measure is $\Gamma_{E^e} = \varrho\Gamma_E$, its domain is $D(E^Q)$, and its form norm is equivalent to that of E^Q . Thus regularity and consistency under restriction to subcubes are preserved.

We give the cell bounds for A_N^0 also on cells larger than one. The unit-scale bounds follow from the proof of Proposition 8.7 with the infrared partial sum equal to zero. On a fixed larger cell, join the harmonic extensions of the smooth datum on a finite mesh of cubes of side at most one. Their energy bounds control the energy of this competitor and hence the total energy of the cell minimizer. For a bounded volume source, the same bound follows from the quadratic response identity and the fixed-cube coercivity in Lemma 8.9.

For a cell $q = z + \square_j$, $j \geq 0$, let $f \in L^\infty(q)$ and $\varphi \in C^2(\bar{q})$, and let $u_N \in H^1(q)$ solve $-\nabla \cdot (A_N^0 \nabla u_N) = f$ in q with $u_N = \varphi$ on ∂q . Use the coordinates $y = 3^N x$ and the relabelled fields of Step 1 of the growth proof. The whole cell becomes $3^N z + \square_{N+j}$, and we have $A_N^0(3^{-N}y) = \bar{\mathbf{a}}_N^{-1} \mathbf{a}_N(y)$ and $\mathbf{b}_{N,N+j} = \bar{\mathbf{a}}_N$. Apply Proposition 6.17 with cutoff $L = N$, outer scale $m = N + j$ and the exponent α_M chosen in the growth proof. Its energy and oscillation estimates apply on the whole cell, including windows meeting its faces. The coefficient $\bar{\mathbf{a}}_N$ cancels with the scalar $\bar{\mathbf{a}}_N^{-1}$ when returning to the original energy. Lemma 8.5 supplies the vector source when a volume source is present. Steps 2–3 of the growth proof therefore give, for every $x \in q$, with $\max\{t, d - 2 + 2\alpha\} <$

$$t_1 < d - 2 + 2\alpha_M,$$

$$(8.35) \quad \Gamma_N^0(u_N)(B_r(x) \cap q) \leq K_{N,q}(\|f\|_\infty + \|\varphi\|_{C^2})^2 r^{t_1}, \quad 3^{-N} \leq r \leq 1.$$

Here $\Gamma_N^0(u) = A_N^0 |\nabla u|^2 dx$. The total energy bound treats radii beyond the regularity range, as in Step 3. The tail (6.91) is uniform in N, j and the deterministic translation; choose the moment orders for the total energy and $3^{t_1 \mathcal{L}_N(\alpha_M, N+j)}$ before decreasing $\bar{\delta}$, so that (8.13) holds at these orders. This bounds the listed moments of $K_{N,q}$ uniformly in N .

At radii below 3^{-N} , retain Steps 4–5 of the growth proof, with $H = 0$. Combine the small-oscillation estimate of those steps with (8.35); the positive powers $3^{-(t_1-t)N}$, $3^{-(d+2-t)N}$ and $3^{-(d-t)N}$ absorb the coefficient and log-gradient moments at the finitely many enlarged orders. The odd reflections give the same bound at faces, edges and corners. Step 7, using the whole-cell oscillation estimate above, gives the C^α bound. Together, these steps give the Morrey bound (8.10) for Γ_N^0 at every radius $r \leq 1$ and the Hölder bounds of Proposition 8.7(ii) for A_N^0 on q , with one random constant $K_{N,q}$ that does not depend on f and φ and whose moments of the listed orders are bounded uniformly in N . This is the whole-cell argument used in Step 3 of Lemma 9.6, with α_M in place of $7/8$. Stationarity makes the moment constants uniform over deterministic translations of a fixed cell.

Include these finitely many cell constants in the same joint compactness extraction. The cutoff energy measures are bounded by $(\sup_Q \varrho) \Gamma_N(\chi_{N,\rho})$. The same comparison with a recovery sequence for $E^{\varrho,Q}$ makes the weighted harmonic interpolants a recovery sequence: the face-layer error is $C\rho^{t-d+1}$, and the cutoff term vanishes by (8.32). The weighted energy-measure convergence then gives zero energy on the faces, harmonicity and minimality. In particular,

$$(8.36) \quad \Lambda_{N,q}^0(b) \longrightarrow \min\{\Gamma_{E^{\varrho}}(V)(q) : V \in D(E^{\varrho,Q}) \cap C(\overline{Q}), V = b \text{ on } \partial q\}.$$

This identifies the boundary minimum of the weighted form on the same padded cube as the operator limit. For a random H , include it in the joint representation and use (8.24) to absorb its uniform approximation. The subsequence criterion then gives convergence in probability on the original field space, after uniqueness. This applies in particular to the uses of e^{-H} below.

Proposition 8.32 (Hölder boundary data, gluing, harmonic replacement). *Let β and α be as fixed in Section 8.4, so that $1/2 < \beta < \alpha < 1$, and set $\sigma = \frac{1}{2}(\beta - 1/2)$. Then the following statements hold along the sequence in the Skorokhod representation.*

- (i) *Let $q = z + r\Box_0$ and $Q = z + 3r\Box_0$ be as in Proposition 8.31. For every $b \in C^\alpha(\partial q)$, the limit $\Lambda_{E,q}(b) = \lim_N \Lambda_{N,q}(b)$ exists and satisfies*

$$\Lambda_{E,q}(b) \leq C \left(\sup_N \Lambda_{\sigma/2,2}(q; A_N) \right) r^{d-2} \|b(z + r \cdot)\|_{C^{\beta/\mathbb{R}}}^2,$$

and all conclusions of Proposition 8.31 hold for b , with a harmonic extension $U \in D(E^Q) \cap C(\overline{Q})$.

- (ii) *Let Q be a cube of $\mathcal{C}$, and let $q_1, \dots, q_m \in \mathcal{C}$ be disjoint cells such that the closure of each q_i and the concentric cube of three times its side length lie in Q . Let $V \in D(E^Q) \cap C(\overline{Q})$ satisfy $V|_{\partial q_i} \in C^\alpha$ for every i . Let V' equal V outside $\bigcup_i q_i$ and equal, on each q_i , the harmonic extension U_i of $V|_{\partial q_i}$. Then $V' \in D(E^Q) \cap C(\overline{Q})$ and*

$$E^Q(V') = E^Q(V) - \sum_i [\Gamma_E(V)(q_i) - \Lambda_{E,q_i}(V|_{\partial q_i})] \leq E^Q(V).$$

- (iii) *If moreover $\Gamma_E(V)$ charges no face ∂q_i (for instance, if V is itself a limit of functions that, at finite N , are harmonic in each cell for the cutoff coefficient and share a common smooth datum on the union of the cell boundaries), and the cells q_i partition a cube $Q' \subseteq Q$, then $\Gamma_E(V')(Q') = \sum_i \Lambda_{E,q_i}(V|_{\partial q_i})$.*

Proof. (i) Extend b to a function in $C_c^\alpha(Q)$ and approximate it in C^β by functions in $\mathcal{D}$. On the fixed finite partition, apply the bound (8.6) for Hölder boundary data and the maximum principle to two approximations. The bounds show that the energy E_N of the difference of the corresponding interpolants is at most C times the squared C^β norm of the difference of the approximations. The sup norm of the difference of the interpolants is at most the sup norm of the difference of the approximations. The interpolants converge by Proposition 8.31, and the same bounds hold for their limits by the liminf inequality. Hence the corresponding harmonic extensions converge in energy and uniformly to a limit U with boundary values b . Harmonicity passes to the limit by energy convergence. The energy on the faces also remains zero, since energy measures depend continuously in total variation on the function in the form norm. Minimality follows again from harmonicity and Lemma 8.30. Since $\sqrt{\Lambda_{N,q}}$ is a seminorm of the datum, its reverse triangle inequality and (8.6) transfer the convergence of $\Lambda_{N,q}$ and the asserted bound from the approximations to b . The argument holds for each realization and covers random boundary data.

(ii) Consistency under restriction to subcubes extends each U_i to $D(E^Q)$, and Lemma 8.30 puts $(U_i - V)\mathbb{1}_{\overline{q_i}}$ in $D(E^{q_i})$ with zero energy on ∂q_i . Adding these differences to V shows that V' belongs to the form domain. Locality identifies its energy measure with $\Gamma_E(U_i)$ inside q_i and with $\Gamma_E(V)$ elsewhere, including the faces by Cauchy–Schwarz. Integrating gives the identity. Cell minimality gives the inequality.

(iii) The same identification and the assumption that $\Gamma_E(V)$ charges no face give the sum over the partition. For a common smooth datum, the energy measure of the limit does not charge the faces by Proposition 8.31; for a common C^α datum, this follows from the approximation in energy in (i). $\square$

Lemma 8.33 (Finite partitions with prescribed boundary values). *Let α be as fixed in Section 8.4, let the cubes q_i form a finite partition of a cube Q , and let $b \in C^\alpha(\overline{Q})$ vanish on ∂Q . Let v be the function equal, on each cell q_i , to the limiting harmonic extension of $b|_{\partial q_i}$. Then $v \in D(E^Q) \cap C(\overline{Q})$,*

$$E^Q(v) \leq \sum_i \Lambda_{E,q_i}(b|_{\partial q_i}),$$

and

$$\|v - b\|_\infty \leq \sup_i \text{osc}_{q_i} b.$$

Proof. For smooth data compactly supported in Q , apply the last paragraph of the proof of Proposition 8.31 to this finite partition. The harmonic interpolants converge uniformly and in energy, and their cell energies add. For general b , use β as fixed in Section 8.4, with $\beta < \alpha$. The zero extension of b can be approximated in C^β by $C_c^\infty(Q)$ functions. The bound (8.6) for Hölder boundary data and the maximum principle, as in Proposition 8.32(i), give convergence in energy and uniformly. This proves the energy bound, also for cells touching ∂Q . The cell maximum principle gives $\|v - b\|_\infty \leq \sup_i \text{osc}_{q_i} b$. $\square$

8.5. Estimates along chains of nested cubes. The energy measure $\Gamma_E(u)$ depends on the same fields as the regularity, extension and harmonic-approximation estimates of Lemma 8.34. To control the energy in the cells where these estimates fail, Proposition 8.39 bounds, on a single event that does not depend on u , the number of bad cells, defined below, along every chain of nested cubes. This bound can then be integrated against the energy measure.

Let E be a subsequential limit along the subsequence of cutoffs of Section 8.4, realized on the Skorokhod probability space.

8.5.1. Continuous versions of the conditions. Fix $s \in (0, 1/32]$ before choosing $\overline{\delta}$; the harmonic lemma is applied with parameter $8s \in (0, 1/4]$. Let $F_{m,z}$, $P_{m,z}$ and $R_{m,z}$ be the left sides of (6.8), (6.9) and (6.10), translated to z . Here $P_{m,z}$ is a scalar statistic. For $\overline{\delta} = 0$, they equal 0, 2 and 0,

respectively. We use the quantity $\mathbf{G}_{m,z}(s)$ of (6.49), suppressing its fixed argument s , and the quantity $\mathcal{J}_n(P)$ of (4.1).

To measure failure of the bounds at center z and scale m , apply continuous piecewise linear functions to $\mathbf{F}_{m,z}$, $\mathbf{P}_{m,z}$ and $\mathbf{R}_{m,z}$, and let $Z_{m,z}$ be their sum. Each function increases linearly from zero to one between its lower and upper thresholds, is zero below the lower threshold and one above the upper threshold, and lies between the indicators of exceeding these thresholds. The threshold pairs are $(\epsilon/2, \epsilon)$, $(6, 12)$ and $(\epsilon^2/4, \epsilon^2)$, respectively. Thus $Z_{m,z}$ takes values in $[0, 3]$; it is a continuous version of the number of the three conditions (6.8), (6.9) and (6.10) that fail at center z and scale m . The probability bounds of Lemmas 6.5, 6.6 and 6.7 are applied at the lower thresholds. We call these quantities and $\mathbf{G}$ the primitive quantities. Every response entering $\mathbf{R}$ or $\mathbf{G}$ at stationary scale n is computed for $\bar{\mathbf{a}}_n^{-1}\mathbf{a}_n$, hence for A_n^0 after translation and rescaling to the unit cube. This own cutoff is part of their definition. The normalized quantities on a physical cell, defined next, are different quantities. Where $Z_{m,z} < 1$, each quantity lies below its upper threshold. Apply the deterministic good-scale estimates at these upper thresholds; in particular, they use the threshold 12 for $\mathbf{P}_{m,z}$ in place of 6. Only the constants change, provided $\tau^2 \leq s \log 3/16$: the product bound 12 gives $e^{\|\mathbf{g}_{m,n}\|_\infty} \leq 12 \cdot 3^{s(m-n)/8}$, while (6.22) gives $e^{|\rho_{m,n}|} \leq 3^{s(m-n)/16}$. The squared coefficient ratio in (6.21), multiplied by the bound (6.10) and its weight, therefore still decays like $3^{-s(m-n)}$.

8.5.2. The good event of a cell. We specify the quantities on a cell after dividing out its coefficient scale. Use β fixed in Section 8.4 and set $\sigma = 1/2(\beta - 1/2)$ as in Lemma 8.4. The exponent pairs used below are $(\sigma/2, 2)$ for boundary extension and $(1, 1)$ for the mean-zero Poincaré bound; the homogenization error uses $(s, \infty, 2)$. These exponents are fixed before the moment orders and the disorder threshold.

For a cube $U = w + 3^{-\ell}\square_0$ with integer $\ell \leq N$, including the negative depths of fixed enlargements, put

$$\kappa_j = e^{(j+1)\tau^2}\bar{\mathbf{a}}_j, \quad s_N(U) = \frac{\kappa_{N-\ell}}{\kappa_N} \exp \left(H(w) + \begin{cases} \sum_{j=0}^{\ell-1} \mathbf{g}_{-j}(w), & \ell \geq 0, \\ -\sum_{j=\ell}^{-1} \mathbf{g}_{-j}(w), & \ell < 0. \end{cases} \right).$$

For $\ell \geq 0$ this is exactly the scalar of Lemma 9.13. The negative-depth formula removes the finitely many retained coarse layers. In the unit chart of U , let $\tilde{A}_{N,U}(y) = s_N(U)^{-1}A_N(w + 3^{-\ell}y)$ and define

$$L_N(U) = \frac{\lambda_{\sigma/2,2}(U; A_N)}{s_N(U)}, \quad U_N(U) = \frac{\Lambda_{\sigma/2,2}(U; A_N)}{s_N(U)}, \quad V_N(U) = \frac{s_N(U)}{\lambda_{\sigma/2,2}(U; A_N)}.$$

Also retain the matrices $s_N(U)^{-1}\mathbf{a}(U; A_N)$ and $s_N(U)\mathbf{a}_*^{-1}(U; A_N)$, entry by entry. For scalar comparisons include every cube U of the catalogue fixed below, in particular q , its concentric enlargement $3q$, and the chosen harmonic-comparison cube, among the cubes W . For each such W retain $\mathcal{E}_{s,\infty,2}(\square_0, 0; \tilde{A}_{N,W}, I)$ and $s_N(q)/s_N(W)$. The ellipticity coordinates have value 1, the matrices have value I , the error has value 0 and the ratio has value 1 when $\bar{\delta} = 0$. The finite tests use these normalized coordinates and their deviations from these values. By Lemma 9.13, for $\ell \geq 0$ the coefficient $\tilde{A}_{N,U}$ is the anchored coefficient $A_{N-\ell}$ in the translated and rescaled environment; it includes the centered increments of the fields coarser than U . The signed formula gives the same identification for negative ℓ . In particular, its matrices are not the own-cutoff matrices of $A_{N-\ell}^0$. The estimates for the primitive quantities and for these normalized arrays are proved separately below.

The pair $(1, 1)$ requires no additional random test. Weighted Cauchy–Schwarz in (2.33) gives, for every cube and coefficient,

$$\Lambda_{1,1} \leq C_\sigma \Lambda_{\sigma/2,2}, \quad \lambda_{1,1}^{-1} \leq C_\sigma \lambda_{\sigma/2,2}^{-1}, \quad C_\sigma = \frac{(1 - 3^{-1})^2}{(1 - 3^{-2+\sigma})(1 - 3^{-\sigma})}.$$

Indeed, apply Cauchy–Schwarz to $\sum_{j \geq 0} 3^{-j} M_j^{1/2} = \sum_{j \geq 0} 3^{-(1-\sigma/2)j} (3^{-\sigma j} M_j)^{1/2}$, where M_j is the maximum matrix norm at relative depth j in either definition. This supplies both required exponent pairs on every enlargement from the same normalized ellipticity arrays.

Fix a rational cube Q and its triadic subdivision. For $q = z + 3^{-k} \square_0$, use a fixed finite catalogue of concentric enlargements and the cubes of finitely many translated triadic grids needed for the harmonic comparison. A catalogue cube U has depth $\ell_U = k - i$ for a fixed integer offset i and center $z + 3^{-\ell_U} v$ for a fixed grid shift v . Include q and its concentric enlargement of side $3 \cdot 3^{-k}$. Choose the finite grids so that every centered subcube of q is contained in a catalogue descendant of side at most nine times its own. At descendant depth D , use all descendant centers and the finitely many catalogue centers needed for the scalar comparisons. Their number is at most $C_d 3^{dD}$.

Fix integers $k_1 \geq 0$ and $k_0 \geq \max\{k_1, 1\}$. At each observation center w , the prefix is the sum over the physical depths $j = \ell_U - k_1, \dots, \ell_U + D$, that is, over the depths of U and its descendants down to relative depth D , together with the k_1 coarser depths $\ell_U - k_1, \dots, \ell_U - 1$. The prefix conditions are imposed only for $D \geq k_0$, so the prefix has at most $D + k_1 + 1 \leq 3D$ terms. Its summands are the primitive quantities $Z_{N-j, 3^N w}$ and $\mathbf{G}_{N-j, 3^N w}$ in the stationary coordinates of (8.1). Use this prefix when $\ell_U + D \leq N$, so that every index $N - j$ is nonnegative; otherwise put it equal to zero and use the below-cutoff estimates.

Choose fixed numbers

$$0 < \lambda_{\text{cut}} < \lambda_{\text{lim}}/4 < \lambda_{\text{lim}} < \lambda_{\text{det}} < 1, \quad 0 < \vartheta < 1, \quad \epsilon_{\text{hom}} > 0, \quad c_{\text{det}} > 0.$$

Fix the primitive tolerance $\epsilon \in (0, 1)$ sufficiently small for the deterministic estimates at the chosen regularity exponents, and then choose λ_{det} sufficiently small for those estimates, and λ_{lim} and λ_{cut} satisfying the displayed inequalities. In particular, after reducing $\bar{\delta}$, the quantity $C(\lambda_{\text{det}} + \bar{\delta}^2 + \epsilon^8)$ is smaller than the gap between the iteration exponent and α . Choose c_{det} small enough that the harmonic-comparison hypothesis holds even at error $2\epsilon_{\text{hom}} c_{\text{det}}$; this leaves room for passage from the limiting error to a finite cutoff. The finite-cutoff prefix threshold is $\lambda_{\text{cut}} D$; the limiting prefix threshold is $\lambda_{\text{lim}} D$, strictly below $\lambda_{\text{det}} D$ consumed by that iteration. The **good event** $G_E(q)$ is defined using the limiting arrays of these exact finite-cutoff quantities along the subsequence defining E . It requires, for every catalogue cube U , every $D \geq k_0$ and every observation center, both limiting prefixes to be smaller than $\lambda_{\text{lim}} D$, and the following finite tests:

- (a) On every catalogue enlargement, $\vartheta \leq L_E(U)$ and $U_E(U) \leq \vartheta^{-1}$. Consequently, the normalized $(1, 1)$ lower and upper bounds are ϑ/C_σ and $C_\sigma \vartheta^{-1}$. For $\mathcal{A}_E(U) = \lim_N s_N(U)^{-1} \mathbf{a}(U; A_N)$ these bounds also give

$$\frac{\vartheta^2}{d} \text{tr } \mathcal{A}_E(U) |p|^2 \leq p \cdot \mathcal{A}_E(U) p \quad (p \in \mathbb{R}^d).$$

- (b) On the chosen comparison cube W , the limiting error obeys

$$\lim_N \mathcal{E}_{s, \infty, 2}(\square_0, 0; \widetilde{A}_{N, W}, I) \leq \epsilon_{\text{hom}} c_{\text{det}}.$$

- (c) For every comparison cube W the limiting ratio $s_N(q)/s_N(W)$ belongs to $(1/2, 2)$, strictly inside the closed interval consumed by the deterministic comparison.

At a finite cutoff, write $G_N^{\text{det}}(q)$ for the conditions consumed by the deterministic estimates: prefix threshold $\lambda_{\text{det}} D$, ellipticity bounds $\vartheta/2$ and $2\vartheta^{-1}$, error bound $2\epsilon_{\text{hom}} c_{\text{det}}$ and closed ratio interval $[1/2, 2]$. Each fixed finite set of these conditions follows eventually from $G_E(q)$ under almost-sure

convergence of the arrays. The smaller prefix threshold $\lambda_{\text{cut}}D$ is used to prove the probability bound passed to the limit at the threshold $\lambda_{\text{lim}}D/4$; eventual satisfaction of the threshold $\lambda_{\text{cut}}D$ is not needed. The finite collection is chosen before N , k , the source and the environment. Its ellipticity and matrix conditions are used for extension and comparison on a good cell. The regularity iteration itself uses only the primitive prefixes, together with its reference-ratio estimates; it does not require transfer of the normalized tests (a)–(c).

For continuous tests in (a)–(c), use piecewise linear ramps with the following inner and outer thresholds. For the lower ellipticity coordinate they are $(1 + \vartheta)/2$ and ϑ , with the ramp increasing as the coordinate decreases; for the upper coordinate they are $(1 + \vartheta^{-1})/2$ and ϑ^{-1} ; for the error they are $\epsilon_{\text{hom}}c_{\text{det}}/2$ and $\epsilon_{\text{hom}}c_{\text{det}}$. For the lower ratio test they are $3/4$ and $1/2$, and for the upper ratio test $3/2$ and 2 . The outer thresholds are the bounds defining $G_E(q)$; the inner thresholds place the zero-disorder coordinates in the regions where their respective ramps vanish. The deterministic finite-cutoff bounds are weaker, as specified above. Each ramp is zero at that value and one when the corresponding outer condition fails. Finite sums of these ramps, or of absolute values of the centered normalized coordinates, are fixed nonnegative Lipschitz functions of the finite array. Their constants depend only on the displayed thresholds and catalogue.

Lemma 8.34 (Deterministic consequences of the good event). *Fix N and a cell q in the deterministic event $G_N^{\text{det}}(q)$. Every weak solution satisfies the following statements.*

- (i) *The local Hölder estimate with source holds at exponent α , with a fixed deterministic constant. More precisely, if the iteration of Lemma 6.11 is run with a fixed exponent in $(\alpha, 1)$, then the constant in its conclusion after D scales is at most $3^{Ck_0 + C(\lambda_{\text{det}} + \bar{\delta}^2 + \epsilon^8)D}$, where $C(\lambda_{\text{det}} + \bar{\delta}^2 + \epsilon^8)$ is smaller than the difference between this exponent and α .*
- (ii) *The bound (8.6) for Hölder boundary values holds on q with the factor $\Lambda_{\sigma/2,2}(q; A_N)$ of Lemma 8.4, which is comparable to the finite-cutoff scalar $s_N(q)$ specified above; this scalar is the factor by which the coefficient on q , rescaled to unit size, is a multiple of a copy of A_{N-k}^0 , up to the oscillation factor of Lemma 8.2.*
- (iii) *The interior harmonic comparison of Lemma 6.16 holds with tolerance ϵ_{hom} and the source term normalized by $s_N(W)$ on its comparison cube W .*

Proof. On $G_N^{\text{det}}(q)$, we apply the iteration of Lemma 6.11 with the bad set consisting of the k_1 additional indices and the indices of the scales at which one of the conditions fails. This set has at most $2k_0 + \lambda_{\text{det}}D$ elements. By (6.89) the exponent in the constant of the conclusion is at most $C(\bar{\delta}^2 + \epsilon^8 + \lambda_{\text{det}})D + Ck_0$. This gives (i). Condition (a) gives both the $(\sigma/2, 2)$ bound for extension and, by its weighted Cauchy–Schwarz comparison, the $(1, 1)$ bound for mean-zero Poincaré. In particular, $\text{tr } \mathcal{A}_N \leq dU_N$ and $p \cdot \mathcal{A}_N p \geq L_N |p|^2$ give trace coercivity with constant $\vartheta^2/(4d)$ at a finite cutoff, and ϑ^2/d in the limit; here $\mathcal{A}_N = s_N^{-1} \mathbf{a}(U; A_N)$. Condition (a), the estimate (2.50) and the trace extension constructed in the proof of Lemma 8.4, with $0 < \sigma < \beta - 1/2$, give (ii). To obtain (iii), use the deterministic comparison argument in the proof of Lemma 6.16, with comparison coefficient $s_N(W)I$. On the unit chart, divide both coefficient and source by $s_N(W)$; the comparison coefficient is then I . Use regularity parameter $8s \leq 1/4$, so that the error index is s . The proof after the anchor error estimate uses only that error, positive scalar comparison and the ellipticity bounds, and applies to this scalar as well. Condition (b) and the choice of c_{det} give the tolerance ϵ_{hom} ; scaling back gives the source factor $r^2 s_N(W)^{-1}$. Condition (c) replaces $s_N(W)$ by $s_N(q)$ up to a fixed factor when required. These bounds depend only on the coefficient and the deterministic geometry, so the same bounds apply to every weak solution. $\square$

Lemma 8.35 (Passage to the limit in sums over finitely many scales). *Along the subsequence that defines E , the following statements hold for each fixed finite interval of scales.*

- (i) The quantities $Z_{m,z}$ and $\mathbf{G}_{m,z}$ and each of the finitely many coarse-grained quantities in (a)–(c) converge in L^p , for each of the finitely many orders p fixed at the start of Section 8. The moment bounds are uniform in the deterministic position of the interval of scales.
- (ii) Let $A, D > 0$. If, at every cutoff, the sum of $Z_{m,z}$ or of $\mathbf{G}_{m,z}$ over the interval of scales exceeds a given threshold with probability at most e^{-AD} , then the limit of this sum exceeds any larger threshold with probability at most e^{-AD} .

Proof. We first prove (i). For each requested p , fix an auxiliary order $P \geq 2p$ with d/P below the margin of the geometric weights, and then decrease $\bar{\delta}$ so that $C_P \bar{\delta}^2$ is below the remaining margin, where C_P is the constant in the bound $e^{C_P \bar{\delta}^2 j}$ on the L^P norm of $\mathcal{J}$ obtained below. Split the terms that involve only the fields into finite sums at finitely many depths and geometrically weighted tails. By (1.5) the tails are bounded in L^p by their geometric weights times the moments of the fields. The other parts are geometrically weighted maxima of J . For the $\mathcal{J}$ -terms of $\mathbf{R}$ and $\mathbf{G}$, the cube scale is the stationary cutoff of its coefficient $\bar{\mathbf{a}}_m^{-1} \mathbf{a}_m$. Translation and rescaling to $\square_0$ give the coefficient A_m^0 in its own reindexed environment, with no remaining cell scalar or normalization ratio. On $\square_0$ the exact decomposition is

$$J(\square_0, e, e; B) = 1/2 \Lambda_{\square_0, B}(e \cdot x) + 1/2 e \cdot \mathbf{a}_*^{-1}(\square_0; B) e - |e|^2.$$

Both the affine Dirichlet and constant-flux Neumann responses occur. Include these own-cutoff, translated normalized responses in the countable extraction of Section 8.4. For the finite coordinates in (a)–(c), include their actual normalized matrices, scalar ratios and geometrically weighted descendant arrays in that extraction as well.

We truncate the geometrically weighted sums at a large relative depth. The response extraction gives convergence in measure of the finitely many retained quantities; the uniform bounds at the larger fixed moment orders upgrade this to convergence in L^p . At a relative depth j beyond the truncation there are at most $C 3^{dj}$ cells. By Proposition 4.1 and the bound (3.25) on the ratios $\bar{\mathbf{a}}_m \bar{\mathbf{a}}_n^{-1}$, the quantity $\mathcal{J}$ of each cell has L^P norm at most of order $e^{C_P \bar{\delta}^2 j}$, and taking a spatial maximum at the auxiliary order P introduces a factor $3^{dj/P}$. The tail is then exponentially small in the truncation depth, uniformly in the cutoff.

Below the smallest field scale 3^{-N} , the maximum of the coefficient on a cell whose side equals that scale controls every smaller descendant by the direct bounds (4.11), so the finer depths require no further spatial maximum. The continuous piecewise linear functions and minima are Lipschitz, and square roots satisfy $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$. Hence the quantities built from these are Cauchy in L^p .

For (ii), on the event that the limiting sum exceeds the larger threshold, a subsequence of the approximating sums that converges almost surely eventually exceeds some threshold strictly between the two thresholds, and each approximating sum exceeds that threshold with probability at most e^{-AD} . Fatou's lemma applied to the indicators gives the bound for the limit. $\square$

A cell q is **good** for E when the event $G_E(q)$ specified above holds, and **bad** otherwise. Refine the subsequence defining E jointly for the deterministic reference ratios and the countable family of normalized arrays. In particular, $s_N(q) \rightarrow s_E(q) > 0$, with the scalar of Section 8.6; for a negative-depth enlargement use the signed retained-sum formula above. Lemma 8.35 gives convergence in L^p for each of the fixed orders. A further diagonal refinement gives almost-sure convergence at all the countably many depths and observation centers. Each fixed finite set of conditions defining $G_N^{\text{det}}(q)$ then holds eventually on $G_E(q)$, using $\lambda_{\text{lim}} < \lambda_{\text{det}}$, the weaker finite-cutoff ellipticity and error bounds, and the open limiting ratio interval.

Apply Lemma 8.34 at each fixed depth to the finite-cutoff solutions converging to $u = G_E f$ with smooth f . Pass their inequalities to the limit along recovery sequences, use Corollary 8.29(a) for their local energy measures, and intersect over the depths. This gives, on $G_E(q)$, the local Hölder

estimate with source, the bound (8.6) for Hölder boundary values, and harmonic approximation with tolerance ϵ_{hom} . The error constant was chosen above to absorb the factor two in the finite-cutoff bound. The comparison argument of Section 8.6 applies regularity only to the solutions $G_E f$ with smooth f . For the second limit form F introduced there, the argument uses only the maximum principle for harmonic extensions.

8.5.3. Events depending on finitely many fields. The conditions depend on all the fields. We approximate the quantities in the conditions by functions of the fields with indices in a finite interval around the scale of the cube, and then cover the failure of $G_E(q)$ by events that depend on such intervals and have exponentially small probabilities.

Lemma 8.36 (Approximation by functions of finitely many fields). *Let $X_{n,z}$ be the quantity $\mathbf{G}_{n,z}$, one of the three summands of $Z_{n,z}$, or a fixed finite nonnegative combination of the absolute values of the centered normalized coordinates specified in (a)–(c), or their fixed continuous ramps. There exist $a, c > 0$, depending only on the fixed exponents and catalogue and not on the moment order, such that the following holds for every fixed $1 \leq p < \infty$. Include the needed auxiliary orders in the finite list fixed above. For sufficiently small $\bar{\delta}$, depending on that list and the fixed thresholds, there is $C_p < \infty$. It depends on d , the regularity and iteration exponents, the moment orders, the catalogue and the ramp thresholds, and is uniform in the valid scale, deterministic center and cutoff. For the quantities in (a)–(c), the notation $X_{n,z}$ refers to the cell of physical depth $k = N - n$ and physical center $3^{-N}z$ in the relabeled coordinates; all catalogue depths must be at most N . The window about stationary index n is therefore the window about original bilateral index $n - N = -k$. For every $h \geq 1$, every deterministic n and z for which these quantities are defined at the chosen cutoff, there is a random variable $\widehat{X}_{n,z}^{(h)}$, measurable with respect to the fields with indices in an interval of length at most $C(h+1)$ about n , such that*

$$(8.37) \quad \|X_{n,z} - \widehat{X}_{n,z}^{(h)}\|_{L^p} \leq C_p \bar{\delta}^c 3^{-ah}.$$

Proof. The proof proceeds in four steps, and the quantity we track is the error $\|X_{n,z} - \widehat{X}_{n,z}^{(h)}\|_{L^p}$. In Steps 1–3 we truncate the geometrically weighted sums defining $X_{n,z}$ at a depth proportional to h , so that the retained terms are, or are approximated by, functions of the fields in the window: Step 1 treats the terms that involve only the fields, Step 2 the product term $\mathbf{P}$, and Step 3 the energy terms of $\mathbf{R}$ and $\mathbf{G}$ and the normalized coordinates in (a)–(c). Step 4 obtains the factor $\bar{\delta}^c$ in (8.37). The heart of the matter is Step 3, where Proposition 8.17 approximates each response by its conditional expectation given finitely many fields, and the geometric depth weights absorb the spatial maxima over the $C3^{dr}$ cells at relative depth r .

Step 1. We treat the terms that involve only the fields. Before the geometric weights are applied, the depth- j sum in $\mathbf{F}_{n,z}$ has L^p norm at most $C_p \bar{\delta}^{(j+1)^{3/2}}$: we cover $z + \square_{n+1+j}$ by cubes of the scale of each field, use the sub-Gaussian bound for the maximum from (1.5), and sum the $2j+1$ terms. We discard the depths $j > ch$, whose total contribution is at most $C_p \bar{\delta} 3^{-sch/16}$. The tail terms of $\mathbf{G}$ (see (6.49)) carry the explicit weight 3^{k-i} .

Step 2. We treat the product term. We retain the depths $j \leq \lfloor h/2 \rfloor$ and stop the second product at $n+h$. Both $\mathbf{P}$ and its approximant are at least 2, the approximant is not larger, and the removed depths contribute only their excess over 2. Fix j . The first product factor at depth j is the exponential of $\sup_{x \in z + \square_{n+1+j}} \sum_{i=(n-j) \vee 0}^{n+j} |\mathbf{g}_i(x)|$. We cover the cube by at most $C_d 3^{d(2j+1)}$ cubes of side $3^{(n-j) \vee 0}$. On each such cube, this sum is at most the sum over i of the value of $|\mathbf{g}_i|$ at the center plus $C_d 3^{(n-j) \vee 0}$ times the gradient norm of $\mathbf{g}_i$ on the cube; these terms are independent in i and sub-Gaussian with parameter $C_d \bar{\delta}$. We take the exponential moment before the spatial maximum, which bounds the moment of order u on each cube by $\exp(C_d u \bar{\delta}^{(j+1)} + C_d u^2 \bar{\delta}^2 (j+1))$, and a maximum over the covering cubes, taken in L^q with $q \geq p$, introduces a factor equal to their number

to the power $1/q$. With $e^x - 1 \leq xe^x$, choosing q large depending on d, s, p and then $\bar{\delta}$ small shows that $3^{-sj/8}$ times the L^p norm of the first product factor minus one is at most $C_p \bar{\delta} (j+1) 3^{-sj/16}$.

The second product factor is the exponential of $4 \sup_x \sum_{i \geq n+j} |\mathbf{g}_i(x) - \mathbf{g}_i(z)|$, which is at most a sum, over $i \geq n+j$, of sub-Gaussian variables multiplied by $C_d 3^{n+j-i}$. Hence its excess over one has L^p norm at most $C_p \bar{\delta}$, and the part with $i > n+h$ contributes $C_p \bar{\delta} 3^{j-h}$.

Summing over $j \leq h/2$ gives $C_p \bar{\delta} 3^{-h/2}$, and the discarded depths $j > h/2$ contribute $\sum_{j > h/2} 3^{-sj/8}$ times the L^p norm of the sum of the two excesses at depth j , which is at most $C_p \bar{\delta} 3^{-ah}$.

Step 3. We treat the energy terms, fixing their moment orders first. Choose an auxiliary $P \geq \max\{2, 2p\}$ so large that $d/P \leq \min\{s, \sigma/2\}/32$. Increase it, if necessary, to dominate the fixed spatial and comparison exponents. Fix moment orders $3P$ and $6P$ for the Morrey and smoothed Neumann bounds, the $4P$ and $8P$ centered-response orders, and the finitely many higher product and mesh orders required by those estimates, before reducing $\bar{\delta}$. Reduce $\bar{\delta}$ until twelve times each selected order is at most $C^{-1} \bar{\delta}^{-2} |\log \bar{\delta}|^{-1}$ and the geometric moment conditions below hold. Lemma 8.15, both clauses of Proposition 8.17, and their moment inputs hold for zero infrared and the positive coarse-scale partial infrared sums, uniformly in their truncation. Their proofs use only centered infrared increments on a fixed compact enlargement. Lemma 8.1 bounds these increments uniformly for the zero infrared potential, every positive partial sum and the full potential. Thus the moment inputs of Proposition 8.12 and the subsequent perturbation estimates are unchanged.

For $\mathbf{R}$ and the energy part of $\mathbf{G}$, every $\mathcal{J}$ is an own-cutoff response, as in the preceding proof. Apply both clauses of Proposition 8.17 to the affine Dirichlet and constant-flux Neumann terms, using the finite polarization family $e_i, e_i + e_j$, normalizing these vectors and restoring their quadratic factors by homogeneity. The slower rate gives a nonnegative approximation $\widehat{\mathcal{J}}$ with

$$\|\mathcal{J} - \widehat{\mathcal{J}}\|_{L^p} \leq C_P \bar{\delta}^{1/2} 3^{-b\ell/256}.$$

The approximant can be obtained by conditioning $\mathcal{J}$ itself; nonnegativity is preserved. When ℓ exceeds its own cutoff, the zero-infrared response already lies in the retained window and the error is zero. There is no physical cell scalar in this estimate.

Retain relative depths $r \leq R = \lfloor h/4 \rfloor$ and use $\ell = \lfloor h/2 \rfloor$ for each retained response. At depth r there are at most $C 3^{dr}$ cells, so the maximum over them introduces a factor $C 3^{dr/P}$. The retained errors and the discarded tail of $\mathbf{R}$ are bounded by

$$C_P \bar{\delta}^{1/2} 3^{-bh/512} \sum_{r \geq 2} 3^{-(s/8-d/P)r} + C_P \bar{\delta}^2 \sum_{r > R} 3^{-(s/8-d/P)r}.$$

For $\mathbf{G}$ use $|\sqrt{\min\{x, 1\}} - \sqrt{\min\{y, 1\}}| \leq \sqrt{|x - y|}$ and its stronger depth weight. The square root halves the error exponents. Hence, for these two energy families,

$$\|X - \widehat{X}\|_{L^p} \leq C_P \bar{\delta}^{1/4} 3^{-a_{\text{RG}} h}, \quad a_{\text{RG}} = \min\{b/1024, s/64\} > 0.$$

Small h is absorbed into the constant. The retained original field indices lie in an interval of length $C(h+1)$ about the parent scale. In bilateral physical coordinates, a parent of depth k has this interval about $-k$. Fixed catalogue offsets only enlarge it by a fixed amount.

For the normalized coordinates in (a)–(c), first write the coarse ellipticity sums, both response matrices and the error using the coefficient $\widehat{A}_{N,U}$. Split the sums at relative depth R . On a retained descendant, separate the reference ratio between that descendant and U . This ratio is part of the normalized coefficient decomposition; its positive and inverse moments satisfy $\|c + c^{-1}\|_{L^u} \leq C_u e^{C_u \bar{\delta}^2 r}$. Retain the intervening coarse fields and truncate the remaining centered infrared increments. Apply the unit response bounds at $2P$ and Hölder at $2P$ to these products, using the already selected $6P$ Morrey inputs. After the spatial maximum, the bound is multiplied by $3^{-wr} 3^{dr/P} e^{C_P \bar{\delta}^2 r}$, where $w > 0$ is the fixed depth weight of the quantity being treated. Choose P with $d/P \leq w/4$

and then $\bar{\delta}$ with $C_P \bar{\delta}^2 / \log 3 \leq w/4$. This leaves $3^{-wr/2}$. The omitted infrared increments contribute at most $C_P \bar{\delta} 3^{-ch}$ for a fixed $c > 0$. Below the smallest field scale, the direct bound (4.11) on each terminal cell controls all its finer descendants, so there is one spatial maximum at that terminal depth.

The upper ellipticity sum is approximated first, then the inverse lower ellipticity sum. For the lower coefficient, truncate the reciprocal at $t_h = \frac{1}{2} 3^{-ah/2}$ and use the moments at order $4P$ to control the small-denominator part. This loses at most half the decay exponent a ; it does not change its independence from p . Polarization supplies the two normalized matrices. Apply the same descendant argument to the error with comparison coefficient $s_N(W)I$. The finite scalar ratios use their explicit signed retained-sum formula. These are exactly the three ellipticity coordinates, the two matrices, the error and the ratio specified above. The finite family of exponents and catalogue cubes permits one common field interval, the minimum of their positive decay rates and the minimum of their disorder thresholds. A fixed Lipschitz function of the approximated coordinates then gives the asserted band approximation for every finite test. In particular the $(1, 1)$ consequences require no extra approximation: their bounds follow deterministically from the $(\sigma/2, 2)$ tests on each enlargement.

Step 4. We obtain the factor $\bar{\delta}^c$ in (8.37). The deviations of the quantities defining the conditions from their values at $\bar{\delta} = 0$ satisfy $\|\mathbf{F}\|_p + \|\mathbf{P} - 2\|_p \leq C_P \bar{\delta}$, $\|\mathbf{R}\|_p \leq C_P \bar{\delta}^2$ and $\|\mathbf{G}\|_p \leq C_P \bar{\delta}^{1/2}$. The first bound follows from Steps 1 and 2. For $\mathbf{R}$, group the cubes of (6.10) over which the supremum is taken by $r = m - n$. There are at most $C_d 3^{dr}$ such cubes, the quantity $\mathcal{J}$ of each has L^q norm $C_q \bar{\delta}^2$ by Proposition 4.1, and $\sum_{r \geq 2} 3^{-(\frac{s}{8} - \frac{d}{q})r} < \infty$ for a fixed $q \geq p$ large enough. For $\mathbf{G}$, we take $m = n$ in Lemma 6.12 and obtain $C_P \bar{\delta}(1 + |\log \bar{\delta}|^{1/2}) \leq C_P \bar{\delta}^{1/2}$. If a truncation gives only $C_P 3^{-ah}$, without a positive power of $\bar{\delta}$, we combine it with a bound $C_P \bar{\delta}^b$, for some $b > 0$, on the distance of X from its deterministic value at $\bar{\delta} = 0$. Conditional expectation onto the fields with indices in the interval of length $C(h + 1)$ about n is a contraction in L^p . Thus the distance of X from its conditional expectation is at most twice its distance from the approximation. The same bound holds with the value at $\bar{\delta} = 0$ in place of the approximation. The minimum of the two resulting bounds is at most their geometric mean $2C_P \bar{\delta}^{b/2} 3^{-ah/2}$. For completeness, the centered moment bound for the normalized coordinates is $\max_{\xi} \|\xi - \xi^{(0)}\|_{L^p} \leq C_P \bar{\delta}^{c_1}$ with $c_1 > 0$, where ξ runs over the fixed finite array, $\xi^{(0)}$ is its unit-coefficient value and c_1 is fixed before p . To obtain it, apply (2.38) with comparison I after translation, scaling and removal of $s_N(U)$. The small-error moments of (4.4), together with the centered infrared and coarse-field oscillation bounds, control the right-hand side uniformly in the own cutoff. The order $\lambda \leq \Lambda$ turns the upper bounds for Λ and λ^{-1} into two-sided bounds, so their deviations from 1 are bounded by $C(\mathcal{E}_{\sigma/2, \infty, 2} + \mathcal{E}_{\sigma/2, \infty, 2}^2)$. The spectral estimate in the proof of (2.38) gives the corresponding deviations of both matrices from I . The error uses its own small-error bound; the signed retained-sum formula and the ordering of $\bar{\mathbf{a}}$ give the ratio deviation. All products use the larger orders already selected. Below the smallest field scale, use the direct terminal-cell bound as above. The fixed Lipschitz ramps inherit these bounds. Take the minimum of the finitely many positive decay and disorder-power exponents. These exponents are fixed by the geometric weights and the regularity parameters before p is chosen; only the auxiliary moment orders, disorder threshold and prefactor depend on p . $\square$

Lemma 8.37 (Sums of Z and $\mathbf{G}$ over scales at a fixed center). *Let $\lambda > 0$ and $A < \infty$. If $\bar{\delta}$ is sufficiently small, then, for every deterministic center z , every interval $n, \dots, n + k - 1$ of scales, in the coordinates of (8.1), with all quantities evaluated at a single cutoff, and each $X \in \{Z, \mathbf{G}\}$,*

$$\mathbb{P}\left[\sum_{i=0}^{k-1} X_{n+i,z} > \frac{\lambda k}{4}\right] \leq C e^{-Ak}.$$

Proof. For $X = Z$ we apply Lemmas 6.5, 6.6 and 6.7 at the lower thresholds of the continuous piecewise linear functions. The exponential rates in these lemmas tend to infinity as $\bar{\delta} \downarrow 0$. For $X = \mathbf{G}$ we apply Lemma 6.12. Its estimate (6.50) concerns the average of the approximation errors over the interval, with an $\mathcal{O}_{\Gamma_2}$ fluctuation of size $(m - n + 1)^{-1/2}$. After multiplication by the length $m - n + 1$, the sum exceeds $\lambda(m - n + 1)$ with probability at most $\exp(-c\lambda^2(m - n + 1)\bar{\delta}^{-2}|\log \bar{\delta}|^{-1})$. The bounds are uniform in the position of the interval by the stationarity in Assumption (g1) and by Proposition 6.17. $\square$

For use with the limiting good event, fix a physical depth j , an observation center and a window $\mathcal{B}_{j,h}$ about bilateral index $-j$. Write X_N for any primitive summand or finite ramp above, pulled back from the stationary coordinates at cutoff N , and X_E for its limit. The window $\mathcal{B}_{j,h}$ is the same for all sufficiently large N . Lemma 8.35(i) and contraction give

$$\|\mathbb{E}[X_N | \mathcal{B}_{j,h}] - \mathbb{E}[X_E | \mathcal{B}_{j,h}]\|_{L^p} \leq \|X_N - X_E\|_{L^p} \rightarrow 0.$$

By Lemma 8.36, the distance of X_N from its conditional expectation is at most $2C_p\bar{\delta}^c 3^{-ah}$. Passing to the limit gives the same bound for X_E ; the centered moment bounds pass to the limit as well. Thus the limiting quantities have band approximants $\mathbb{E}[X_E | \mathcal{B}_{j,h}]$, with the same decay exponents and at most twice the constant. For the primitive prefixes, apply Lemma 8.37 with parameter λ_{cut} to the interval of depths $\ell_U - k_1, \dots, \ell_U + D$ of the prefix. Its length is at most $3D$ since $D \geq k_0 \geq \max\{k_1, 1\}$, so the resulting bound applies at threshold $\lambda_{\text{cut}}D < \lambda_{\text{lim}}D/4$. Lemma 8.35(ii) then gives the exponential bound at $\lambda_{\text{lim}}D/4$ for the limiting sum. The rate can be chosen arbitrarily large before the moment orders and disorder threshold are fixed.

Lemma 8.38 (Covering the limiting bad event). *Use the limiting arrays defining $G_E(q)$ and set $\lambda = \lambda_{\text{lim}}$. Let $A < \infty$. If $\bar{\delta}$ is sufficiently small, then, for every cell q at triadic depth n , there exist events $W_{q,h}$ for $h \geq 1$ such that $W_{q,h}$ is measurable with respect to the fields $\mathbf{g}_{-j}$ whose indices j lie between $n - h$ and $n + 2h$, $\mathbb{P}(W_{q,h}) \leq e^{-Ah}$, and, up to a null set,*

$$\{G_E(q) \text{ fails}\} \subseteq \bigcup_{h \geq 1} W_{q,h}.$$

The same conclusion holds for the failure of one additional scalar or matrix condition whose centered L^p norm is at most $C_p\bar{\delta}$ and whose approximation by a function of the fields with indices in an interval of length at most $C(h + 1)$ about n has error at most $C_p\bar{\delta}^c 3^{-ah}$ in L^p , for every $h \geq 1$.

Proof. Outside a common null set, all dyadic conditional-expectation approximants converge to their limiting quantities. This follows from the summable L^p errors and a countable intersection over the cells, centers and finite tests. If $G_E(q)$ fails, then either one of the finite tests (a)–(c) fails, or there are a prefix length $k = D \geq k_0$, one of at most Ce^{Ck} choices of a cube and a center, and a limiting primitive prefix sum S with $S \geq \lambda k$, where $\lambda = \lambda_{\text{lim}}$. The k_1 coarser depths only change the constant in the bounds below. Approximate S by $S^{(\ell)}$, the sum of the conditional expectations of its summands on windows of radius $2^\ell k$. The limiting band bound above gives $\|S - S^{(\ell)}\|_{L^p} \leq C_p k \bar{\delta}^c 3^{-a2^\ell k}$. If $S \geq \lambda k$, then either $S^{(0)} > \lambda k/2$ or, for some $\ell \geq 1$, $|S^{(\ell)} - S^{(\ell-1)}| > \lambda k 2^{-\ell-2}$. Indeed, the increment allowances sum to $\lambda k/4$, and the approximants converge almost surely to S .

The first event implies either $S > \lambda k/4$ or $|S - S^{(0)}| > \lambda k/4$. Use the limiting prefix bound proved before the lemma, with a rate larger than $3A + C$, and Markov's inequality. Its probability is at most $Ce^{-(3A+C)k} + (4C_p\bar{\delta}^c/\lambda)^p 3^{-apk}$. The ℓ th increment event has probability at most $(C_p\bar{\delta}^c)^p 2^{p(\ell+C)} 3^{-ap2^{\ell-1}k}$.

Take the union over the Ce^{Ck} spatial choices. For each h , combine the events for pairs (k, ℓ) with $2^\ell k \asymp h$; there are polynomially many such pairs. Their field windows lie within an interval of radius Ch about n . After relabeling h , enlarge this interval to $[n - h, n + 2h]$. Choose p so that the

decay rate $ap \log 3$ absorbs the prescribed rate, the spatial count and this fixed enlargement, and then decrease $\bar{\delta}$ to absorb the prefactors.

For each finite test (a)–(c), let $T_E \in [0, 1]$ be its specified ramp. Failure of that test implies $T_E = 1$, including either endpoint of the open ratio interval. The limiting centered bound gives $\|T_E\|_{L^p} \leq C_p \bar{\delta}^{c_1}$, and its conditional-expectation error is at most $C_p \bar{\delta}^c 3^{-ah}$. Start at a fixed window large enough to include the catalogue offsets. The probability that this first approximant exceeds $1/2$ is small by its L^p norm; decrease $\bar{\delta}$ so that it has the required exponential bound at that fixed radius. Telescope the approximants with increment allowances $2^{-\ell-2}$ and apply Markov's inequality as for the prefixes. The finite union of these covers gives the required events for (a)–(c), after starting with a slightly larger rate. The same argument applies to the additional centered condition in the statement, after dividing by its fixed positive failure threshold. $\square$

Fix $L = 3^{H_1}$ and a bounded cube Q . A descendant at L -adic generation n has side length L^{-n} times the side length of Q . When Q has triadic side length 3^{-k_Q} , its descendants at generation n have physical triadic depth $k = k_Q + H_1 n$. A chain of length J is a sequence of J cubes of the L -adic subdivision of Q , each a child of the preceding one and the first a child of Q .

Proposition 8.39 (Bad cells along every chain). *Let $L = 3^{H_1}$, fix a bounded cube Q , and let $\theta \in (0, 1)$ and $b_* > 0$. There exists $\bar{\delta}_0 > 0$, depending on d, L, θ, b_* and the parameters of the good event, such that, if $\bar{\delta} \leq \bar{\delta}_0$, then there is a random variable $B_\omega < \infty$ such that, for every $t > 0$,*

$$\mathbb{P}(B_\omega > t) \leq (1 - e^{-b_*})^{-1} \exp\left(-\frac{b_* t}{1 - \theta}\right),$$

and such that, almost surely, for every $J \geq 1$ and every chain π of length J ,

$$(8.38) \quad \#\{\text{bad cells on } \pi\} \leq \theta J + B_\omega.$$

Proof. Apply Lemma 8.38 with rate A . On a fixed chain, successive generations have field-index centers spaced by H_1 ; for a triadic root the center is $-(k_Q + H_1 n)$. The rare-interval estimate holds on every integer interval, so a fixed root offset does not change it. Enlarge the interval of field indices determining $W_{q,h}$ to the symmetric interval of radius $2h$ about its depth, set all events at other depths or odd radii to the empty event, and apply Lemma B.5. In that lemma, the rate is $A/3$ and the density is at least θ/H_1 . Thus, once $A\theta \geq CH_1$, the probability that this chain has at least θJ bad cells is at most $e^{-cA\theta J}$. There are L^{dJ} chains. Choose A with $cA\theta \geq d \log L + b_*$. The probability that some chain of length J has at least θJ bad cells is then at most $e^{-b_* J}$. These choices precede the moment order and the threshold for $\bar{\delta}$ in Lemma 8.38.

Set $B_\omega = \sup_J \max_{|\pi|=J} (\#\{\text{bad cells on } \pi\} - \theta J)_+$. If $B_\omega > t$, then for some $J > t/(1 - \theta)$ there is a chain of length J with more than θJ bad cells. Summing $e^{-b_* J}$ over these J proves the stated tail bound and (8.38). $\square$

Every point of Q lies in exactly one chain of half-open cells. Integrating (8.38) over these chains gives, on the same event, for every finite Borel measure μ on Q ,

$$(8.39) \quad \sum_{j=1}^J \mu(\text{union of the bad cells at } L\text{-adic generation } j) \leq (\theta J + B_\omega) \mu(Q).$$

In particular this holds for the random energy measure $\Gamma_E(u)$.

8.6. Comparison of two subsequential limits. In this subsection we show that two subsequential limit forms differ only by a deterministic scalar factor (Theorem 8.52). Let E and F be two such forms, obtained along two subsequences of cutoffs, realized in the same Skorokhod representation (Section 8.4), so that both are functions of the same fields. Their domains are not assumed to be equal. We compare them on the cubes of $\mathcal{C}$.

The preliminary two-sided comparison (Theorem 8.42) adapts the argument of Ding and Gwynne [2023a, Section 3.4] on the ratios of the normalizations of the two forms. In that argument, comparisons built from paths around annuli, together with upper and lower bounds on crossing distances, exclude ratios of the two normalizations that are too large or too small. Here harmonic extensions play the role of the paths. Proposition 8.41 glues harmonic extensions on cubes at the scales where the ratio r_k of the two normalizations, defined below, is small, and shows that if this happens on a positive density of scales, then F is bounded by a small multiple of E . By the variational formula (8.31), this contradicts the moment bounds of Lemma 8.20. This excludes ratios that are too small. Exchanging E and F excludes ratios that are too large. The proof of Theorem 8.42 then applies the same construction on the remaining scales, where the ratio is bounded above and below, to compare the domains and the energies in both directions.

As in the metric uniqueness proofs of Gwynne and Miller [2021, Section 6] and Ding and Gwynne [2023b, Section 4.5], we consider the optimal upper and lower comparison constants. Those proofs shorten geodesic segments to obtain probability bounds incompatible with a strict difference between the constants. They must control which annuli a field-dependent geodesic visits. The probability comparison and counting argument in Ding and Gwynne [2023b, Section 4.2] provide this control after suitable smooth changes of the Gaussian free field.

Here Proposition 8.39 bounds the number of bad cells along every chain of nested cubes, on a single event that does not depend on the solution. Its integrated form (8.39) therefore applies to the energy measure of the solution being replaced. We use harmonic extensions on disjoint cubes in place of shorter path segments. Since the solution is close to affine on good cells, the comparison of the minimal energies of E and F with affine boundary values transfers to its boundary values. Summing, over these cubes, the amount by which the minimal F -energy falls below the upper constant times the minimal E -energy improves one of the comparison constants for the forms themselves.

In this subsection a cell q is called good only when both $G_E(q)$ and $G_F(q)$ hold. Applying Proposition 8.39 to each form with half the allowed density of bad cells and adding the two counts gives (8.38) for this notion of bad cell. On a good cell we thus have the regularity and coercivity estimates for E and the bound (8.6) for Hölder boundary values for F , and the same with the roles of E and F exchanged.

The comparison is expressed through the scalars $s_E(q)$ and $s_F(q)$ of the two forms. At a cell q of side 3^{-k} and center z , the scalar $s_E(q)$ of E is

$$s_E(q) = \exp\left(H(z) + \sum_{j=0}^{k-1} (\mathbf{g}_{-j}(z) - \tau^2)\right) \lim_{N \rightarrow \infty} \frac{\bar{\mathbf{a}}_{N-k}}{\bar{\mathbf{a}}_N},$$

the limit being taken along the subsequence that defines E . We define $s_F(q)$ in the same way, along the subsequence that defines F . After a further diagonal extraction, these limits exist for all k at once: since $|\bar{\mathbf{a}}_N \bar{\mathbf{a}}_{N-k}^{-1} - 1| \leq C \bar{\delta}^2 k e^{C \bar{\delta}^2 k}$ by (3.25), the ratio $\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N$ lies in a compact subinterval of $(0, \infty)$ that depends only on k . The ratio $r_k = \frac{s_F(q)}{s_E(q)}$ of these two scalars is deterministic and does not depend on z .

Choose the parameters in the following order. First fix the geometric and regularity parameters below, then L , the moment orders, and a first threshold for $\bar{\delta}$, which determines the comparison constant C_0 of Theorem 8.42. With C_0 fixed, we tighten the tolerance ϵ_{hom} of the harmonic comparison and add the comparison of the minimal energies of E and F with affine boundary values, of Corollary 8.47, whose error is proportional to the difference between the optimal comparison constants defined below. This requires only a further decrease of $\bar{\delta}$. The thresholds for sums over intervals of scales and L stay fixed. None of these choices depends on the difference of the optimal constants.

8.6.1. Comparison from a positive density of small ratios. We first show that a positive density of scales at which $r_k \leq a$ forces $F \leq C_* a E$. Fix η_0 with $8(2(1-\alpha) + \eta) < \eta_0 < 1/3$, which is possible by the choice of α and η in Section 8.4, and choose the parameters so that $D \geq 8d/\eta_0$ and $\theta \leq \eta_0/8$. By the choice of η_0 we have $2(1-\alpha) + \eta < \eta_0/8$, which is equivalent to $d-2+2\alpha-\eta > d-\eta_0/8$. Then choose $L = 3^{H_1}$ so large that $L^{\eta_0/3-\eta_0/8} \geq 2C_d^{\eta_0/3}$, where C_d is the constant in the bound below on the number of children whose enlargement is not contained in the parent. Here we use the L -adic grid, translated as in the proof of Proposition 8.41, with cell sides L^{-n} , so n is measured from side length one and the physical triadic depth is $k = H_1 n$. Thus $r_{H_1 n}$ is r_k at that physical depth. In an L -adic subdivision starting from a root Q of side 3^{-k_Q} , generation n instead has physical depth $k_Q + H_1 n$. For a root of side L^{-n_0} this shifts every index by the fixed integer n_0 , which does not affect upper densities. For a child q of side r of a cell p of side Lr , consider its concentric enlargement of side $3r$. At most $C_d L^{d-1}$ children of each cell have an enlargement whose closure is not contained in p . A set $S \subseteq \mathbb{N}$ has **upper density** at least η if $\limsup_J J^{-1} \#(S \cap \{1, \dots, J\}) \geq \eta$.

On a good cell whose enlargement has its closure in the parent, the regularity of E and the bound (8.6) for Hölder boundary values of F combine as follows.

Lemma 8.40 (Extension on good cells). *Let Q be a cube of $\mathcal{C}$ and f a smooth function, and set $u = G_E^Q f$, where G_E^Q is the Dirichlet Green operator of E on Q . The constant C below depends only on d , the fixed regularity and iteration exponents, k_0 , the primitive and trace constants, the displayed good-event thresholds and ratio bounds, and the finite enlargement and grid geometry. It is independent of the cell, depth, source, extracted subsequence and environment. This constant uses only the primitive iteration thresholds, the extension exponent, the ellipticity bounds and the scalar comparisons on q and $3q$; it is independent of ϵ_{hom} and c_{det} . It can therefore be fixed before the harmonic-comparison tolerance is tightened. The random constant K may also depend on Q , f and the environment.*

- (i) *If $q = z + r\Box_0$ is a good child of p at depth k whose concentric enlargement of side $3r$ has its closure in p , then*

$$\inf_{c \in \mathbb{R}} \|u(z + r \cdot) - c\|_{C^\alpha(\Box_0)} \leq C r^{2-d/2} s_E(q)^{-1/2} \sqrt{\Gamma_E(u)(p)} + C r^2 s_E(q)^{-1} \|f\|_\infty,$$

and

$$\Lambda_{F,q}(u|_{\partial q}) \leq C r_k [\Gamma_E(u)(p) + s_E(q)^{-1} r^{d+2} \|f\|_\infty^2].$$

- (ii) *There is a random constant $K < \infty$ such that $\Lambda_{F,q}(u|_{\partial q}) \leq K r^{d-2+2\alpha-\eta}$ for every cell $q \subseteq Q$ of side r .*

Proof. For (i), the local Hölder estimate with source on the good cell bounds the oscillation on q by the normalized L^2 oscillation on the concentric cube of side $3r$ and the source (Section 8.5). The coarse-grained Poincaré inequality (2.46) on that cube bounds its oscillation by its energy. Since its closure lies in p , this energy is at most $\Gamma_E(u)(p)$, giving the factor $r^{2-d/2}$ in unnormalized units. The source is controlled by (8.11): by Lemma 8.5, convolving f with the gradient of the Newtonian kernel gives a vector field with $C^{1/2}$ seminorm at most $C r^{1/2} \|f\|_\infty$ on a cube of side r . Scaling gives the term $r^2 s_E(q)^{-1} \|f\|_\infty$. For the form F on q , the coefficient factor in (8.6) is comparable to $s_F(q) = r_k s_E(q)$. Inserting the first estimate into that bound gives the second estimate: we square the first bound and multiply by $s_F(q) r^{d-2}$.

For (ii), the global bound (8.11) for u gives $\|u(z + r \cdot) - c\|_{C^\beta(\partial\Box_0)} \leq r^\alpha [u]_\alpha$, and the bound (8.7) on grid cubes bounds the factor in (8.6) for F on q by $K r^{-\eta}$. Then (8.6) gives the bound $K r^{-\eta} r^{d-2} r^{2\alpha} [u]_\alpha^2$. $\square$

Proposition 8.41 (Comparison from a positive density of small ratios). *Assume that the parameters are chosen as above. There exists a deterministic $C_* < \infty$, depending on d , η_0 , L and the constants of the good event, such that the following holds for every $a > 0$. Let S be a deterministic*

set of L -adic depths of upper density at least η_0 with $r_{H_1 n} \leq a$ for every $n \in S$. Then, on every cube of $\mathcal{C}$, $D(E) \subseteq D(F)$ and, for every $v \in D(E)$,

$$(8.40) \quad F(v) \leq C_* a E(v).$$

Proof. Fix a cube Q of $\mathcal{C}$. Translate the L -adic grid so that its origin is a corner of Q . Since the side of Q is triadic, every sufficiently fine L -adic mesh partitions Q exactly. Fix $u = G_E f$ with smooth f , fix $c > 0$, and set $\mu = \Gamma_E(u) + c dx$. By (8.7) with $\eta < 2$, the source term in Lemma 8.40 is at most $K r^{d+2-\eta} \|f\|_\infty^2 \leq c r^d = c|q|$ for cells of side below a number h_0 depending on u, c and the environment.

Fix an L -adic mesh of side at most h_0 that partitions Q ; all cells below are subcubes of the finitely many cubes of this mesh, and we apply Proposition 8.39 to each of them.

Step 1. We stop the subdivision at good cells that carry a fixed fraction of the mass of their parents, and bound the sum, over these cells q , of the minimal F -energy $\Lambda_{F,q}(u|_{\partial q})$ with the boundary values of u . Within each cube of the mesh of side at most h_0 we follow the subdivision tree and stop at the first child q that lies at an L -adic depth in S , is good, has an enlargement with closure in its parent, and satisfies $\mu(p) \leq L^D \mu(q)$. By Lemma 8.40, each cell at which the subdivision stops satisfies $\Lambda_{F,q}(u|_{\partial q}) \leq C a [\mu(p) + c|q|] \leq C a L^D \mu(q)$. The interiors of these cells are disjoint. Hence the sum of $\Lambda_{F,q}(u|_{\partial q})$ over them is at most $C_* a \mu(Q)$ with $C_* = C L^D$.

Step 2. We show that the sum of $\Lambda_{F,q}(u|_{\partial q})$ over the remaining cells, those on chains along which the subdivision never stops, tends to zero. There are arbitrarily large depths J at which at least $\frac{3\eta_0}{4}J$ of the L -adic depths below the mesh of side at most h_0 belong to S . Along a chain on which the subdivision has not stopped by such a depth J , every step to a good child with an enlargement in its parent, at an L -adic depth in S , fails the condition $\mu(p) \leq L^D \mu(q)$. Since μ is monotone along the chain and drops by more than the factor L^D at each of these steps, while every leaf satisfies $\mu(\text{leaf}) \geq c|\text{leaf}| = cL^{-dJ}$ on the L -adic mesh of depth J , the number of these steps is at most

$$\frac{d}{D}J + \frac{\log(\mu(Q)/c)}{D \log L} \leq \frac{\eta_0}{8}J + O_{Q,c,u,\omega}(1).$$

By (8.38), at most $\theta J + B_\omega \leq \frac{\eta_0}{8}J + B_\omega$ of the steps at L -adic depths in S are bad. For all sufficiently large such J , the chain therefore has at least $\frac{3\eta_0}{4}J - \frac{\eta_0}{4}J - O(1) \geq \frac{\eta_0}{3}J$ steps at L -adic depths in S to a child whose enlargement is not contained in its parent. At a step to such a child there are at most $C_d L^{d-1}$ choices of the child. At every other step there are at most L^d . Summing over the at most 2^J positions of the steps to such children, we find that the number of leaves on chains along which the subdivision does not stop is at most

$$C_Q 2^J L^{dJ} (C_d/L)^{\eta_0 J/3} \leq C(Q, c, u, \omega) L^{(d-\frac{\eta_0}{8})J},$$

where we used $L^{\eta_0/3-\eta_0/8} \geq 2C_d^{\eta_0/3}$. By Lemma 8.40(ii), the sum of $\Lambda_{F,q}(u|_{\partial q})$ over them is at most $K L^{(d-\frac{\eta_0}{8})J}$ times $L^{-(d-2+2\alpha-\eta)J}$, which tends to zero because $d-2+2\alpha-\eta > d-\eta_0/8$.

Step 3. We show that $u \in D(F)$ and $F(u) \leq C_* a E(u)$. At a finite depth J , the cells at which the subdivision stops and the remaining leaves form a finite partition of Q into cells, on whose boundaries we take the values of u . By Lemma 8.33, the F -harmonic extensions of these boundary values glue to a function $v_J \in D(F)$ with $F(v_J) \leq C_* a \mu(Q) + o(1)$. By the maximum principle, each extension lies between the extreme boundary values of u on its cell, so $\|v_J - u\|_\infty$ is at most the modulus of continuity of u at scale h_0 . For fixed h_0 we take J large and then let $h_0 \rightarrow 0$. Then $v_J \rightarrow u$ uniformly, and the closedness of F gives $F(u) \leq C_* a (E(u) + c|Q|)$. Letting $c \downarrow 0$ gives $u \in D(F)$ and $F(u) \leq C_* a E(u)$.

Step 4. We extend the inequality to $D(E)$. Let $v \in D(E)$. By Proposition 8.19, there are smooth f_n with $G_E f_n \rightarrow v$ in the form norm of E . By Step 3, the sequence $(G_E f_n)$ is Cauchy

in the form norm of F , and the closedness of F identifies its limit with v . This gives $v \in D(F)$ and (8.40). $\square$

Theorem 8.42 (Two-sided comparison). *There exists a deterministic $C_0 \geq 1$ such that, for every pair of subsequential limit forms and every $0 < \bar{\delta} \leq \bar{\delta}_*$, where $\bar{\delta}_*$ is the initial threshold for $\bar{\delta}$ chosen above, on every cube of $\mathcal{C}$,*

$$(8.41) \quad D(E) = D(F) \quad \text{and} \quad C_0^{-1}E \leq F \leq C_0E.$$

Proof. Fix a nonzero smooth source f_0 on the unit cube. By the positive and negative moments of $\langle f_0, G_E f_0 \rangle$ and $\langle f_0, G_F f_0 \rangle$ in Lemma 8.20, choose a deterministic $A \geq 1$, uniform over the subsequential limit forms and over $0 < \bar{\delta} \leq \bar{\delta}_*$, so that $\mathbb{P}[A^{-1} \leq \langle f_0, G_E f_0 \rangle, \langle f_0, G_F f_0 \rangle \leq A] > 0$. It suffices to choose A so that each of the four tail probabilities is less than $1/8$. Set $a_0 = (2C_*A^2)^{-1}$ and let S_- be the set of L -adic depths n with $r_{H_1n} \leq a_0$. Since the ratios r_k are deterministic, so is S_- . If S_- has upper density at least η_0 , then Proposition 8.41 gives $D(E) \subseteq D(F)$ and $F \leq C_*a_0E$ on $D(E)$. The variational formula (8.31) then gives, almost surely,

$$\begin{aligned} \langle f_0, G_F f_0 \rangle &= \sup_{v \in D(F)} \{2\langle f_0, v \rangle - F(v)\} \geq \sup_{v \in D(E)} \{2\langle f_0, v \rangle - C_*a_0E(v)\} \\ &= (C_*a_0)^{-1} \langle f_0, G_E f_0 \rangle = 2A^2 \langle f_0, G_E f_0 \rangle, \end{aligned}$$

which contradicts $\mathbb{P}[A^{-1} \leq \langle f_0, G_E f_0 \rangle, \langle f_0, G_F f_0 \rangle \leq A] > 0$. Hence S_- has upper density less than η_0 . Exchanging E and F gives the same bound for the set S_+ of L -adic depths with $r_{H_1n} \geq a_0^{-1}$. The complement of $S_- \cup S_+$ therefore has lower density greater than $1 - 2\eta_0 > \eta_0$, since $\eta_0 < 1/3$, and on this complement $a_0 < r_{H_1n} < a_0^{-1}$. Proposition 8.41, applied in both directions with $a = a_0^{-1}$, gives $F \leq C_*a_0^{-1}E$ and $E \leq C_*a_0^{-1}F$ with equal domains. Thus (8.41) holds with $C_0 = C_*/a_0 = 2C_*^2A^2$. $\square$

8.6.2. *The optimal constants are deterministic.* For $a > 0$ we write $aE \leq F$ if $aE^Q \leq F^Q$ on $D(E^Q) = D(F^Q)$ for every cube Q of $\mathcal{C}$, and we optimize over all these cubes:

$$m = \sup\{a : aE \leq F\}, \quad M = \inf\{a : F \leq aE\} \quad \text{and} \quad \Delta = M - m.$$

The numbers m and M are the optimal lower and upper comparison constants, and Δ is their difference. By (8.41), $C_0^{-1} \leq m \leq M \leq C_0$, and $mE \leq F \leq ME$ by monotone limits. The comparison constants are defined for each realization; the next lemma shows that they are deterministic, which Theorem 8.52 uses to conclude that the proportionality constant is deterministic.

Lemma 8.43 (The optimal comparison constants are deterministic). *The optimal comparison constants m and M are almost surely constant.*

Proof. The comparison constants are measurable: on each cube of $\mathcal{C}$, $aE \leq F$ is equivalent, by the variational formula (8.31), to $\langle f, G_F f \rangle \leq a^{-1} \langle f, G_E f \rangle$ for all sources f in the countable dense family.

Resampling one field multiplies both energy measures by the same positive continuous weight, by (8.34). By Lemma 8.26, an inequality between the two forms is equivalent to the same inequality between their energy measures. Multiplying both measures by a common weight preserves these inequalities in both directions. Thus resampling one field leaves m and M unchanged. Since these constants are bounded, Lemma 8.16, with no fields retained, gives $\|m - \mathbb{E}m\|_1 = \|M - \mathbb{E}M\|_1 = 0$. $\square$

8.6.3. *Concentration of the difference between the energies.* Assume $\Delta > 0$. We prove a concentration bound for the difference of the two energies, with an error proportional to Δ .

We first check that the minimal energies of E and F with given boundary values on a cell are taken over the same class of admissible functions. Let q be a cell and let $Q \supseteq q$ be a larger cube of $\mathcal{C}$. A boundary value ℓ on ∂q , affine or of class C^β with $\beta > 1/2$, is realized by an E -harmonic

extension through a neighbourhood of ∂q , which belongs to $D(E^Q)$. By (8.41) this extension lies in $D(F^Q)$. Its restriction to q , together with the identification of the minimal energies in Propositions 8.31 and 8.32, places it in the admissible class of both $\Lambda_{E,q}(\ell)$ and $\Lambda_{F,q}(\ell)$. Thus both minimal energies below are taken over the same class of functions with the prescribed boundary values, on which $m\Lambda_{E,q} \leq \Lambda_{F,q} \leq M\Lambda_{E,q}$ as quadratic forms.

Lemma 8.44 (Difference of two minimizers). *Let u_E and u_F be the minimizers of E and F in q with the same affine boundary values, let $c \in [m, M]$, and set $D = F - cE$ and $w = u_F - u_E$. Then $F(w, v) = -D(u_E, v)$ for every $v \in D(E^q)$, and*

$$(8.42) \quad E(w) \leq \left(\frac{\Delta}{m}\right)^2 E(u_E).$$

Moreover, for all u, v in the common domain and every Borel set B ,

$$|\Gamma_D(u, v)|(B) \leq \Delta \sqrt{\Gamma_E(u)(B) \Gamma_E(v)(B)}.$$

Proof. Subtracting the Euler equations $F(u_F, v) = 0 = E(u_E, v)$ gives $F(w, v) = F(u_F, v) - F(u_E, v) = -F(u_E, v) + cE(u_E, v) = -D(u_E, v)$. Since $(m - c)E \leq D \leq (M - c)E$ and $|m - c|, |M - c| \leq \Delta$, the symmetric form D satisfies $|D(u, v)| \leq \Delta \sqrt{E(u)E(v)}$. Taking $v = w$ gives $mE(w) \leq F(w) = -D(u_E, w) \leq \Delta \sqrt{E(u_E)E(w)}$, which is (8.42).

Let Q be the cube on which the forms are considered and let $A \subseteq Q$ be a Borel set. Lemma 8.26, applied to (E, F) with constant M and to (F, E) with constant m^{-1} , gives $m\Gamma_E(h) \leq \Gamma_F(h) \leq M\Gamma_E(h)$ for $h \in D(E) \cap C_c(Q)$. Since $\Gamma_D(h) = \Gamma_F(h) - c\Gamma_E(h)$ and $|m - c|, |M - c| \leq \Delta$, it follows that $|\Gamma_D(h)(A)| \leq \Delta \Gamma_E(h)(A)$. This extends to every h in the common domain, because $D(E) \cap C_c(Q)$ is dense in the form norm, which is equivalent to that of F by (8.41), and the energy measures $\Gamma_E(h)$ and $\Gamma_F(h)$ depend continuously in total variation on h in the form norm. Polarization therefore gives, for every $t > 0$,

$$4t|\Gamma_D(u, v)(A)| \leq 2\Delta[\Gamma_E(u)(A) + t^2\Gamma_E(v)(A)].$$

Optimizing in t gives $|\Gamma_D(u, v)(A)| \leq \Delta \sqrt{\Gamma_E(u)(A) \Gamma_E(v)(A)}$. Sum this bound over a finite Borel partition of B , apply Cauchy–Schwarz, and take the supremum over the partitions to obtain the total-variation bound. $\square$

We normalize the minimal energies with affine boundary values on a cell. Fix $c \in [m, M]$. Let ℓ range over the affine boundary values in a fixed finite set that contains the coordinate functions and determines quadratic forms by polarization. Set $D_q = \sum_{i=1}^d \Lambda_{E,q}(x_i)$ and, for one of these boundary values, $\mathcal{R} = (\Lambda_{F,q}(\ell) - c\Lambda_{E,q}(\ell))/D_q$. By polarization, these quantities give the entries of the matrix B_k in (8.46). Define the normalized measures

$$\nu = \frac{1}{D_q} \sum_{\ell} [\Gamma_E(u_E^\ell) + \Gamma_E(u_F^\ell)], \quad \zeta = \frac{1}{D_q} \sum_{\ell} \Gamma_E(w^\ell),$$

where the sums run over the fixed affine boundary values and $w^\ell = u_F^\ell - u_E^\ell$. Then $\nu(q) \leq C(C_0)$ by (8.41), and $\zeta(q) \leq C(C_0)\Delta^2$ by (8.42). The measure ν inherits the Morrey bound (8.10) of the energy measures of the minimizers. For a bounded g supported in B , let $\mathcal{R}^g$ denote the quantity $\mathcal{R}$ computed for the forms E^g and F^g .

Lemma 8.45 (Perturbation of the normalized difference $\mathcal{R}$). *There exists $C(C_0) < \infty$ such that, for every bounded g supported in B ,*

$$(8.43) \quad |\mathcal{R}^g - \mathcal{R}| \leq C\|g\|_\infty e^{C\|g\|_\infty} [\Delta\nu(B) + \sqrt{\nu(B)\zeta(B)}].$$

Proof. Let u_E^g and u_F^g minimize the weighted forms E^g and F^g with the same boundary values as u_E and u_F , respectively. Put $H_q = \Lambda_{F,q}(\ell) - c\Lambda_{E,q}(\ell)$, $D = F - cE$, $D^g = F^g - cE^g$, $v_E = u_E^g - u_E$

and $v_F = u_F^g - u_F$. Subtract the weighted Euler equations to obtain, for every test function ϕ with zero boundary values on ∂q ,

$$(8.44) \quad F^g(v_F - v_E, \phi) = - \int_B (e^g - 1) [d\Gamma_F(w, \phi) + d\Gamma_D(u_E, \phi)] - D^g(v_E, \phi).$$

Test (8.44) with $v_F - v_E$ and use Lemma 8.44 and the perturbation bound (8.23) for v_E . Cauchy–Schwarz gives

$$(8.45) \quad E^g(v_F - v_E) \leq C \|g\|_\infty^2 e^{C\|g\|_\infty} [\Gamma_E(w)(B) + \Delta^2 \Gamma_E(u_E)(B)].$$

By (8.23),

$$H_q^g - H_q = \int_B (e^g - 1) d[\Gamma_F(u_F) - c\Gamma_E(u_E)] - [F^g(v_F) - cE^g(v_E)].$$

Use the bilinear identities

$$\begin{aligned} \Gamma_F(u_F) - c\Gamma_E(u_E) &= \Gamma_D(u_E) + \Gamma_F(w, u_F + u_E), \\ F^g(v_F) - cE^g(v_E) &= D^g(v_E) + F^g(v_F - v_E, v_F + v_E). \end{aligned}$$

Lemma 8.44 bounds the first line, in total variation on B , by $CD_q[\Delta\nu(B) + \sqrt{\nu(B)\zeta(B)}]$. For the second, (8.23) for each form and (8.45) give

$$E^g(v_E) + E^g(v_F + v_E) \leq C \|g\|_\infty^2 e^{C\|g\|_\infty} D_q \nu(B),$$

and

$$E^g(v_F - v_E) \leq C \|g\|_\infty^2 e^{C\|g\|_\infty} D_q [\zeta(B) + \Delta^2 \nu(B)].$$

Cauchy–Schwarz and $|D^g(v_E)| \leq \Delta E^g(v_E)$ thus bound the correction by $C \|g\|_\infty^2 e^{C\|g\|_\infty} D_q [\Delta\nu(B) + \sqrt{\nu(B)\zeta(B)}]$. Absorbing $\|g\|_\infty^2$ into $\|g\|_\infty e^{C\|g\|_\infty}$ proves the required bound for $|H_q^g - H_q|/D_q$.

Finally, $|H_q| \leq C\Delta D_q$, $e^{-\|g\|_\infty} D_q \leq D_q^g \leq e^{\|g\|_\infty} D_q$, and (8.23) summed over the coordinate data gives $|D_q^g - D_q| \leq C \|g\|_\infty e^{C\|g\|_\infty} D_q \nu(B)$. Apply these bounds to $\mathcal{R}^g - \mathcal{R} = (H_q^g - H_q)/D_q^g + H_q(D_q - D_q^g)/(D_q D_q^g)$. $\square$

Proposition 8.46 (Concentration of the normalized difference B_k). *Let A_E and A_F be the coarse-grained matrices of E and F on a cell q at depth k , defined by $\Lambda_{E,q}(\xi \cdot x) = |q| \xi \cdot A_E \xi$ for $\xi \in \mathbb{R}^d$, and similarly for F . Set $c_k = \mathbb{E}[\text{tr } A_F / \text{tr } A_E]$ and $B_k = (A_F - c_k A_E) / \text{tr } A_E$. Then $c_k \in [m, M]$ and $\mathbb{E} B_k = 0$. Moreover, there exists $a > 0$, depending only on the dimension and on the regularity exponents fixed earlier, and not on C_0 or p , such that the following holds for every $1 \leq p < \infty$. If $\bar{\delta}$ is sufficiently small, depending on p and C_0 , then, for every k and every integer $h \geq 0$,*

$$(8.46) \quad \begin{aligned} \|B_k\|_{L^p} &\leq C_p(C_0) \bar{\delta} \Delta, \\ \|B_k - \mathbb{E}[B_k \mid \text{fields with indices within distance } h \text{ of } k]\|_{L^p} &\leq C_p(C_0) \bar{\delta} \Delta 3^{-ah}, \end{aligned}$$

where $C_p(C_0)$ does not depend on k .

Proof. Step 1. The form order gives $c_k \in [m, M]$. By Assumptions (g1) and (g3), the joint law of A_E and A_F , after their common scalar factor is removed, is invariant under signed permutations of the coordinates. Thus $\mathbb{E} B_k$ is a scalar matrix. Its trace is zero by the definition of c_k , so $\mathbb{E} B_k = 0$.

Step 2. We bound the moments of the constant in the Morrey bound (8.10) for ν , uniformly in k . Rescale a cell to the unit cube and divide E by $s_E(q)$ and F by $s_F(q)$. The fields of scale smaller than the cell then have their original law, and the sum of the fields of larger scale, with its value at the cell center subtracted, has the moments of Lemma 8.1, uniformly in k . At finite cutoff, let u minimize the normalized energy with affine boundary values of slope e_1 . Test the Neumann variational formula for boundary flux $e_1 \cdot n$ with scalar multiples of u . The product of the normalized Dirichlet and Neumann energies is then at least $(\int_{\square_0} e_1 \cdot \nabla u)^2 = 1$. This inequality passes to the extracted limits. The proof of Lemma 8.2, applied to the normalized cell, bounds the positive

moments of the normalized Dirichlet and Neumann energies, uniformly in the cutoff and in k , and Fatou's lemma passes these bounds to the extracted limits. By the product inequality, each of the two energies is at least the reciprocal of the other, so the same bounds control the reciprocals. For some deterministic $A < \infty$, both normalized minimal energies with slope e_1 lie in $[A^{-1}, A]$ with positive probability. On that event, the order of the original minimal energies and their scaling give that r_k times the ratio of the two normalized energies lies in $[m, M]$, so that $mA^{-2} \leq r_k \leq MA^2$. The last bounds hold deterministically because r_k is deterministic. Also the reciprocal of the sum of the normalized minimal energies over the coordinate directions is at most the normalized energy of the Neumann problem for E with boundary flux $e_1 \cdot n$. Hölder's inequality, the Morrey bounds for the two minimizers and these reciprocal moments now give the moments, of the finitely many orders fixed at the start of Section 8, of the constant in the Morrey bound for ν , uniformly in k .

Step 3. Let $p \geq 2$, and consider one of the fields of scale smaller than the cell, of scale r . Use the cutoff functions and strips of Lemma 8.15, with boxes of side $r^{(t-d+1)/(t+1)}$ and strips of width $4\sqrt{d}r$. The restrictions of the field to the separated interior subboxes are independent by (g1), and by Lemma 8.28 the whole field can be removed from the limit forms. With b from Lemma 8.14, the original measure satisfies $\nu(\text{strips}) \leq Kr^b$. By (8.43), the change of $\mathcal{R}$ caused by deleting the field on the strips is at most $C\Delta S e^{CS}(Kr^b + K^{1/2}r^{b/2})$, where S is the sup norm of the field. As in Lemma 8.15, (8.23) controls the changed minimizers, so the energy measures after the cutoff function is applied to the field satisfy $\hat{\nu}(q) \leq C$, $\hat{\zeta}(q) \leq C\Delta^2$ and $\max_B \hat{\nu}(B) \leq Ce^{CS}Kr^b$. The bounds on $\hat{\nu}(q)$ and $\hat{\zeta}(q)$ follow from the order between the two forms, which is preserved when the same cutoff function is applied to the field in both forms. The new ingredient in the Efron–Stein estimate is the bound on the sum of squares

$$\sum_B [\Delta \hat{\nu}(B) + \sqrt{\hat{\nu}(B)\hat{\zeta}(B)}]^2 \leq 2 \max_B \hat{\nu}(B) [\Delta^2 \hat{\nu}(q) + \hat{\zeta}(q)] \leq C\Delta^2 \max_B \hat{\nu}(B).$$

Thus the first and last terms of (8.29) in Lemma 8.15, and the conditional Efron–Stein estimate, all keep the factor Δ . By (8.26), the norms of the fields are absorbed after reducing the exponent b by a fixed fraction. For an entry X of B_k , resampling the field at relative index j consequently gives

$$\|X - X^{(j)}\|_p \leq C_p(C_0)\bar{\delta}\Delta \begin{cases} 3^{-aj}, & j \geq 0, \\ 3^j, & j < 0. \end{cases}$$

For $j < 0$ this follows directly from (8.43) and the bound of Step 2 on the sum of the fields of larger scale, with their value at the cell center subtracted. Choose a below a fixed fraction of b . Apply Lemma 8.16 with no fields retained and then with the fields within distance h of k retained. Centering, polarization and the two geometric sums give (8.46). $\square$

By the concentration bound, we can add to the good event the requirement that the minimal energies of F and $c_k E$ with affine boundary values differ by at most a small multiple of Δ times the minimal energy of E with the same boundary values.

Corollary 8.47 (Minimal energies with affine boundary values on good cells). *Let $\epsilon > 0$. The good event of Section 8.5 can be strengthened by one additional condition, with a threshold for $\bar{\delta}$ and a bad-cell density θ that do not depend on Δ , such that, on every good cell q at depth k and for every affine boundary value ℓ ,*

$$(8.47) \quad |\Lambda_{F,q}(\ell) - c_k \Lambda_{E,q}(\ell)| \leq \epsilon \Delta \Lambda_{E,q}(\ell).$$

Proof. Apply the last assertion of Lemma 8.38 to $X = B_k/\Delta$, with the additional condition $|X| \leq \epsilon c_0$, where $c_0 > 0$ is the deterministic constant of the next paragraph. Its hypotheses are precisely (8.46). Reserve half the allowed density of bad cells for this condition. Choose the rate A of the events $W_{q,h}$ of Lemma 8.38 larger than the exponential growth rate $d \log L$ of the number

of chains. Then choose the required moment order and decrease $\bar{\delta}$. Proposition 8.39 gives the strengthened good event, with choices independent of Δ .

We pass from B_k to (8.47). Condition (a) of the good event bounds the condition number of the normalized matrix A_E : the coarse-grained ellipticity constants (2.33) bound $|A_E|$ and $|A_E^{-1}|$. Hence $\lambda_{\min}(A_E) \geq c_0 \operatorname{tr} A_E$ with a deterministic $c_0 > 0$. For an affine function with slope ξ , we obtain $|\xi \cdot (A_F - c_k A_E) \xi| \leq |B_k| \operatorname{tr} A_E |\xi|^2 \leq \epsilon c_0 \Delta \operatorname{tr} A_E |\xi|^2 \leq \epsilon \Delta \xi \cdot A_E \xi$, which is (8.47). $\square$

8.6.4. Affine approximation of the boundary values.

Lemma 8.48 (Affine approximation of the boundary values). *Let $1/2 < \beta < \alpha < 1$, $0 < \gamma < 1$, $\zeta > 0$ and $\varrho > 0$, and assume that*

$$(8.48) \quad \frac{d + \zeta}{2} - \gamma \left(\frac{d - 2}{2} + \alpha + \frac{\alpha - \beta}{\alpha + d/2} (2 - \alpha) \right) < 0.$$

Let E be a subsequential limit form along the subsequence of Section 8.4, realized in the Skorokhod representation, let $f \in C_c^\infty(Q)$, $u = G_E^Q f$, $c > 0$ and $\mu = \Gamma_E(u) + c dx$. Suppose that $L = 3^{H_1}$ is chosen large, with $\lfloor \gamma H_1 \rfloor + 6 < H_1$, that ϵ_{hom} is then chosen small, and that $\bar{\delta}$ is then chosen small, all independently of f and c . Then the following holds almost surely for cells of side below a number $h_0 > 0$, which may depend on f , c and the environment. Let q be a good child, of side r , of a cell $p \subseteq Q$ of side Lr . If the cube concentric with q of side $3^{\lfloor \gamma H_1 \rfloor + 4} r$, used in conditions (b)–(c), has its closure in p , and $\mu(p) \leq L^{d+\zeta} \mu(q)$, then there is an affine function ℓ_q such that

$$(8.49) \quad \Lambda_{E,q}(u|_{\partial q} - \ell_q) \leq \varrho \mu(q).$$

Proof. The constants C below depend only on d , α , β , γ , ζ and the constants of the good event fixed before L (as in Lemma 8.40); they do not depend on L , ϵ_{hom} , $\bar{\delta}$, f , c or the environment. Take $L = 3^{H_1}$ with $\lfloor \gamma H_1 \rfloor + 6 < H_1$ and put $S = 3^{\lfloor \gamma H_1 \rfloor + 4} r / 729$. Condition (c) of the good event compares s_E at the sides $729S$ and r within a factor two. The mass bound on the parent and the estimate of the oscillation by the energy bound the oscillation of u on the cube of side S concentric with q by $CS^{2-d/2} s_E(q)^{-1/2} L^{d+\zeta/2} \sqrt{\mu(q)}$, together with the source term $CS^2 s_E(q)^{-1} \|f\|_\infty$. Condition (b) and Lemma 6.16, with the harmonic comparison on the concentric cube of side $9S$ and its error measured on the concentric cube of side S , show that the $\underline{L}^2$ distance on the cube of side S between u and the Laplace-harmonic function on the cube of side $9S$ with the boundary values of u is at most

$$C \epsilon_{\text{hom}} S^{(2-d)/2} s_E(q)^{-1/2} L^{(d+\zeta)/2} \sqrt{\mu(q)} + CS^2 s_E(q)^{-1} \|f\|_\infty.$$

The estimates at finite cutoff pass to the extracted solutions by uniform convergence and Corollary 8.29, since the energy measure $\Gamma_E(u)$ does not charge the faces of p . Let ℓ_q be the affine function that best approximates u in $L^2(q)$; we use it for both norms below. Approximating the Laplace-harmonic function by its first-order Taylor polynomial gains a factor $(r/S)^2$ in $L^2(q)$. Restricting the error of the harmonic approximation from the concentric cube of side S to q introduces a factor $(S/r)^{d/2}$ in volume-normalized L^2 . The local Hölder bound on q gains a factor $(r/S)^\alpha$, since the affine projection is bounded on C^α . Since $L^\gamma r / 27 < S \leq L^\gamma r / 9$, after division by $r^{2-d/2} \sqrt{\mu(q) / s_E(q)}$ and reducing ϵ_{hom} after fixing L , the rescaled L^2 norm of $u - \ell_q$ on q is therefore at most

$$CL^{d+\zeta/2-\gamma \frac{d+2}{2}} + C \epsilon_{\text{hom}} L^{d+\zeta/2+\gamma},$$

and its rescaled C^α norm on q is at most $CL^{d+\zeta/2-\gamma(\frac{d-2}{2}+\alpha)}$. Interpolate between these two bounds with the exponent $(\alpha - \beta)/(\alpha + d/2)$ and apply the bound (8.6) for Hölder boundary values on the good cell, where the factor in it is comparable to $s_E(q)$. The spatial and coefficient factors cancel, giving

$$\Lambda_{E,q}(u|_{\partial q} - \ell_q) \leq C \left(L^{d+\zeta/2-\gamma \frac{d+2}{2}} + \epsilon_{\text{hom}} L^{d+\zeta/2+\gamma} \right)^{2(\alpha-\beta)/(\alpha+d/2)} L^{2(1-\frac{\alpha-\beta}{\alpha+d/2})(\frac{d+\zeta}{2}-\gamma(\frac{d-2}{2}+\alpha))} \mu(q).$$

At $\epsilon_{\text{hom}} = 0$, half the exponent of L is the left side of (8.48), which is negative. First choose L to make this contribution smaller than $\frac{\epsilon}{4}$. Next choose ϵ_{hom} so small that the contribution of the error of the harmonic approximation is comparably small. Then decrease $\bar{\delta}$ so that the enlarged good event has the required density of bad cells along every chain. After the same normalization, the source terms are at most $C_L c^{-1/2} s_E(q)^{-1/2} r \|f\|_{C^1} \leq K_{L,\omega} c^{-1/2} r^{1-\eta/2} \|f\|_{C^1}$, by $\mu(q) \geq cr^d$ and (8.7). They tend to zero because $\eta < 2$. A small h_0 makes them smaller than $\frac{\epsilon}{4}$. The left side of (8.48) tends to $-(1-\beta)/(1+d/2) < 0$ as $\alpha, \gamma \uparrow 1$ and $\zeta \downarrow 0$, so the requirement on α in Section 8.4 and the choices of γ and ζ can be met in every dimension. $\square$

8.6.5. Strict improvement of a comparison constant. We now combine concentration for affine boundary values with the affine approximation of the solution's boundary values to rule out $\Delta > 0$. Let $\mathcal{U} = \{n : c_{H_1 n} \leq (m+M)/2\}$. Either this set or its complement has upper density at least $1/2$. Fix $\epsilon = 1/8$ in (8.47). On a good cell q at an L -adic depth in $\mathcal{U}$, for every affine ℓ ,

$$B_q(\ell) := M\Lambda_{E,q}(\ell) - \Lambda_{F,q}(\ell) \geq (M - c_k - \epsilon\Delta)\Lambda_{E,q}(\ell) \geq \frac{3}{8}\Delta\Lambda_{E,q}(\ell).$$

Since both minimal energies are taken over the same class of admissible functions, $0 \leq B_q \leq \Delta\Lambda_{E,q}$ on all admissible boundary values. If u is replaced on q by the F -harmonic function with the same boundary values b , then the new function has F -energy $\Lambda_{F,q}(b) = M\Lambda_{E,q}(b) - B_q(b)$ on q , which is at most $M\Gamma_E(u)(q) - B_q(b)$.

Harmonic replacements will give $F(u) \leq (M - 3\Delta/256)E(u)$ for $u = G_E f$, once the errors they introduce are controlled. Density extends this inequality to the common domain, contradicting the optimality of M . If the complementary set of L -adic depths has upper density at least $1/2$, then we interchange the forms and improve the optimal upper constant $1/m$ instead. Both the energy saved by the replacements and the approximation error are proportional to Δ , so the tolerances can be chosen independently of its size.

With β and α as fixed in Section 8.4, choose γ and ζ satisfying (8.48). For $L = 3^{H_1}$ as in Lemma 8.48, set $b = H_1 - \lfloor \gamma H_1 \rfloor - 4$. Then $b \geq 1$, $H_1 \geq b + 2$ and $3^{-b}Lr$ is the side of the cube concentric with q used for the harmonic comparison in Lemma 8.48. With the dimensional constant $C_d \geq 1$ of the next lemma, the closure of this cube lies in p for every child at distance at least $C_d 3^{-b}$ times the side of p from ∂p , while $C_d 3^{-b} \leq 81C_d L^{\gamma-1}$ tends to zero as L grows. Averaging over finitely many translated grids makes the mass of the children close to the boundary of their parent small, for every measure.

Lemma 8.49 (Shifted rational grids). *Let $L = 3^{H_1}$ with $H_1 \geq b + 2$, and set $M = 3^b + 1$. Translate the nested L -adic grid by the shifts $\mathcal{S} = \{0, 1/M, \dots, (M-1)/M\}^d$. Here $\mathcal{S}$ denotes this finite set of grid shifts; in Definition 4.2 it denotes the induction assertion. Consider the children at depth n whose closure is within distance $C_d 3^{-b} L^{1-n}$ of the boundary of their parent. The dimensional constant in the counting bounds below may be larger than the constant C_d in this distance. Then, for every x and every $n \geq 1$,*

$$\begin{aligned} M^{-d} \#\{\sigma \in \mathcal{S} : \text{the depth-}n \text{ child of } x \text{ in grid } \sigma \text{ is within distance } C_d 3^{-b} L^{1-n} \\ \text{of the boundary of its parent}\} \leq 2d \left(C_d 3^{-b} + \frac{1}{L} + \frac{1}{M} \right) \leq C_d 3^{-b}. \end{aligned}$$

Consequently, for every finite Borel measure μ on Q , the average over $\sigma \in \mathcal{S}$ of the mass of the cells at depth n within distance $C_d 3^{-b} L^{1-n}$ of the boundary of their parent is at most $C_d 3^{-b} \mu(Q)$.

Proof. Fix a coordinate i . Relative to its parent at depth $n-1$, the point x has fractional coordinate $\{L^{n-1}(x_i - \sigma_i)\}$. Since $\gcd(L, M) = 1$, multiplication by L^{n-1} permutes the residues modulo M . As σ_i ranges over the M shifts, these fractional coordinates form an equally spaced set of M points with a common rotation. If the child is within this distance of the boundary of its parent in coordinate i , then x_i is within $(C_d 3^{-b} L + 1)L^{-n}$ of a face of the parent. Its fractional coordinate

then lies in two intervals of total length at most $2C_d3^{-b} + 2/L$. A rotated equally spaced set has at most $2M(C_d3^{-b} + 1/L) + 2$ points in these two intervals. We divide by M , sum over the d coordinates, and integrate the pointwise bound against μ . $\square$

Lemma 8.50 (Good cells carry a fixed fraction of the mass). *Suppose that $\mathcal{U}$ has upper density at least $1/2$. Fix $f \in C_c^\infty(Q)$ and $c > 0$, put $u = G_E f$ and $\mu = \Gamma_E(u) + c dx$, let $\epsilon_{\text{coll}} > 0$, and assume that $\theta + L^{-\zeta} + C_d3^{-b} + \epsilon_{\text{coll}} \leq 1/8$. Then, at arbitrarily large L -adic depths $n \in \mathcal{U}$, there are shifts $\sigma \in \mathcal{S}$ and finite families $\mathcal{Q}$ of cells at depth n in the grid σ such that $\mu(\bigcup \mathcal{Q}) \geq \mu(Q)/8$. Every cell $q \in \mathcal{Q}$, of side r and with parent p , is good, satisfies $\mu(p) \leq L^{d+\zeta}\mu(q)$, and has its concentric enlargement of side $3^{\lceil \gamma_{H_1} \rceil + 4}r$ with closure in p .*

Proof. At one L -adic depth, a child that violates $\mu(p) \leq L^{d+\zeta}\mu(q)$ has mass less than $L^{-(d+\zeta)}\mu(p)$, and each parent has at most L^d children. Thus the total mass of these children is at most $L^{-\zeta}\mu(Q)$. By (8.39) for each of the M^d grids, the mass of the bad cells over J L -adic depths is at most $(\theta J + B_\omega^\sigma)\mu(Q)$. The maximum over the finitely many grids of the random constants B_ω^σ is finite on the intersection of the events of Proposition 8.39 for these grids. After division by J , this maximum is absorbed in the $o(1)$ below. By Lemma 8.49, the mass of the cells within distance $C_d3^{-b}L^{1-n}$ of the boundary of their parent, averaged over σ , is at most $C_d3^{-b}\mu(Q)$ at every depth.

We omit a neighbourhood of ∂Q of mass at most $\epsilon_{\text{coll}}\mu(Q)$. At sufficiently large L -adic depths, the parents of the remaining points lie inside Q . Along depths J with $\#(\mathcal{U} \cap \{1, \dots, J\}) \geq J/2 - o(J)$, we sum the mass of the good cells with enlargement in the parent that satisfy the bound on their parents over $n \in \mathcal{U} \cap \{1, \dots, J\}$ and $\sigma \in \mathcal{S}$, and divide by $J|\mathcal{S}|$. This normalized sum is at least

$$\left[\frac{1}{2} - \epsilon_{\text{coll}} - \theta - L^{-\zeta} - C_d3^{-b} - o(1) \right] \mu(Q) > \frac{\mu(Q)}{4}$$

for large J . A finite average that exceeds $\mu(Q)/4$ has a summand that exceeds it, and since every fixed finite set of depths contributes $o(J)$, such depths can be taken arbitrarily large. $\square$

Lemma 8.51 (Harmonic replacement). *Let $u \in D(E^Q) \cap C(\overline{Q})$ and let $\mathcal{Q}$ be a finite disjoint family of cells of side at most h , whose threefold enlargements lie in Q . Assume that $u|_{\partial q} \in C^\alpha$ for every $q \in \mathcal{Q}$, with α as fixed in Section 8.4, and let $u^\mathcal{Q}$ be the function equal to the E -harmonic extension of $u|_{\partial q}$ on each $q \in \mathcal{Q}$ and to u elsewhere. Then $u - u^\mathcal{Q}$ is the E -orthogonal projection of u onto $\bigoplus_q D(E^q)$, and $E(u) = E(u^\mathcal{Q}) + E(u - u^\mathcal{Q})$. If moreover $u = G_E f$, then $E(u - u^\mathcal{Q}) \leq Kh^{3/2}\|f\|_2^2$, where $K < \infty$ depends on Q and on the environment through (8.5), but not on f , h or $\mathcal{Q}$. In particular, $\sum_{q \in \mathcal{Q}} [\Gamma_E(u)(q) - \Lambda_{E,q}(u|_{\partial q})] = E(u - u^\mathcal{Q}) \rightarrow 0$ as $h \rightarrow 0$, uniformly over the choice of $\mathcal{Q}$.*

Proof. The Euler equations and strong locality identify $w = u - u^\mathcal{Q}$ as the stated orthogonal projection and give both energy identities. Recovery sequences and (8.5) give $\|v\|_{H^{3/4}(Q)}^2 \leq KE(v)$; they also show that $D(E^q) \subseteq H^{3/4}(q)$ on each cell and that its functions have zero trace on ∂q . The fractional Poincaré inequality for zero-trace functions, rescaled to q , gives

$$\|w\|_2^2 \leq Ch^{3/2} \sum_{q \in \mathcal{Q}} [w]_{H^{3/4}(q)}^2 \leq Ch^{3/2} [w]_{H^{3/4}(Q)}^2 \leq Kh^{3/2} E(w).$$

Here the seminorm is the Gagliardo double integral. Finally, for $u = G_E f$, $E(w) = E(u, w) = \langle f, w \rangle \leq \|f\|_2 \|w\|_2$, which proves the estimate. $\square$

Theorem 8.52 (Proportionality). *The optimal comparison constants coincide: $\Delta = 0$. Hence there exists a deterministic $c \in [C_0^{-1}, C_0]$ such that*

$$(8.50) \quad F = cE.$$

Proof. Suppose for a contradiction that $\Delta > 0$.

Step 1. We show that $\mathcal{U}$ having upper density at least $1/2$ contradicts the definition of M . Fix a cube Q in $\mathcal{C}$, $u = G_E f$ with smooth f , $c > 0$, $\mu = \Gamma_E(u) + c dx$ and choose $\varrho \leq 3/1024$ in Lemma 8.48. Choose the parameters in the order above, enlarging L so that $\theta + L^{-\zeta} + C_d 3^{-b} + \epsilon_{\text{coll}} \leq 1/8$ with $b = H_1 - \lfloor \gamma H_1 \rfloor - 4$, before fixing ϵ_{hom} and $\bar{\delta}$. Let $\mathcal{Q}$ be the family of Lemma 8.50 at a sufficiently large L -adic depth, with maximal side h . For each cell q of this family, the triangle inequalities for $\sqrt{B_q}$ and $\sqrt{\Lambda_{E,q}}$, applied to $u|_{\partial q} = \ell_q + (u|_{\partial q} - \ell_q)$, give $B_q(u|_{\partial q}) \geq \Delta[\frac{3}{16}\Lambda_{E,q}(u|_{\partial q}) - 4\varrho\mu(q)]$, by (8.49). Summing over $q \in \mathcal{Q}$ and using Lemma 8.51, we obtain

$$\begin{aligned} \sum_{q \in \mathcal{Q}} B_q(u|_{\partial q}) &\geq \Delta \left[\frac{3}{16} \sum_q \Lambda_{E,q}(u|_{\partial q}) - 4\varrho\mu(Q) \right] \\ &\geq \Delta \left[\frac{3}{16} \Gamma_E(u)(\bigcup \mathcal{Q}) - \frac{3}{16} K h^{3/2} \|f\|_2^2 - 4\varrho(E(u) + c|Q|) \right]. \end{aligned}$$

Since $\mu(\bigcup \mathcal{Q}) \geq \mu(Q)/8$, we have $\Gamma_E(u)(\bigcup \mathcal{Q}) \geq E(u)/8 - 7c|Q|/8$, and with $\varrho \leq 3/1024$ this gives

$$\sum_{q \in \mathcal{Q}} B_q(u|_{\partial q}) \geq \Delta \left[\frac{3}{256} E(u) - Cc|Q| - o_h(1) \right].$$

Now $u \in D(F)$ by (8.41). Let v be the function equal to the F -harmonic extension of $u|_{\partial q}$ on each $q \in \mathcal{Q}$ and to u elsewhere. By strong locality and the bound $\Gamma_F \leq M\Gamma_E$ off the cells,

$$\begin{aligned} F(v) &\leq M\Gamma_E(u)(Q \setminus \bigcup \mathcal{Q}) + \sum_q \Lambda_{F,q}(u|_{\partial q}) \\ &= ME(u) - \sum_q [M\Gamma_E(u)(q) - \Lambda_{F,q}(u|_{\partial q})] \leq ME(u) - \sum_q B_q(u|_{\partial q}), \end{aligned}$$

where the last inequality uses $\Gamma_E(u)(q) \geq \Lambda_{E,q}(u|_{\partial q})$. As $h \rightarrow 0$ along the L -adic depths of Lemma 8.50, the maximum principle bounds $\|v - u\|_\infty$ by the modulus of continuity of u at scale h , which tends to zero. Thus $v \rightarrow u$ in L^2 . Lower semicontinuity of F gives $F(u) \leq (M - 3\Delta/256)E(u) + C\Delta c|Q|$.

The tolerances and ϵ_{coll} do not depend on c . Letting $c \downarrow 0$ gives $F(u) \leq (M - 3\Delta/256)E(u)$ for every $u = G_E^Q f$ with smooth f , on every cube Q in $\mathcal{C}$. Step 4 of the proof of Proposition 8.41 extends the bound to $D(E^Q)$ for every cube Q of $\mathcal{C}$. This contradicts the definition of M as an infimum over all these cubes.

Step 2. We show that the complement of $\mathcal{U}$ having upper density at least $1/2$ contradicts the definition of m . On good cells at L -adic depths outside $\mathcal{U}$, we have $\Lambda_{F,q}(\ell) \geq (c_k - \epsilon\Delta)\Lambda_{E,q}(\ell) \geq (m + 3\Delta/8)\Lambda_{E,q}(\ell)$. We regard F as the first form and E as the second. The comparison constants of E relative to F are $1/M \leq 1/m$, with difference $1/m - 1/M$, and the gain at affine boundary values is at least $(3/8)(1/m - 1/M)$, because $m + 3\Delta/8 \leq M$. The argument of Step 1, with the solutions $G_F f$ for smooth f , improves the optimal upper comparison constant $1/m$ of this pair, which contradicts the optimality of m .

Hence $\Delta = 0$, so $F = mE$, and the constant in (8.50) is m , which is deterministic by Lemma 8.43. $\square$

We now show that the proportionality constant c in Theorem 8.52 is one, using the fact that the homogenized matrix of the stationary coefficients A_N^0 is the identity.

8.7. Determination of the proportionality constant. The coefficients A_N^0 of (8.1) are stationary, have range of dependence at most $\sqrt{d}$, and have homogenized matrix I by (8.1).

Theorem 8.53 (Uniform qualitative homogenization). *For every unit vector e ,*

$$\lim_{k \rightarrow \infty} \sup_N \mathbb{E} J(\square_k, e, e; A_N^0) = 0.$$

Proof. The proof proceeds in three steps, and the quantity we track is the sum $f_N(k)$ of the expectations $\mathbb{E}J(\square_k, e_j, e_j; A_N^0)$ over the coordinate directions, defined in Step 1. Step 1 proves uniform moment bounds and chooses cutoffs N_k for which f_{N_k} changes little between the scales k and $4k$. Step 2 extracts a subsequence of these cutoffs along which the expectations of the coarse-grained matrices $A_{N_k}^0(\square_n)$ converge to a deterministic matrix P , and uses the finite range of dependence to show that, along it, the matrices $(A_{N_k}^0)_*^{-1}(\square_n)$ concentrate around a deterministic matrix R . Step 3 uses the estimates of Section 4 to show that $P = R^{-1}$, so that $P = R = I$. The heart of the matter is Step 3, where the small gaps in the subadditivity inequality obtained in Step 1 force the two limits to agree.

Step 1. We obtain uniform moment bounds and choose cutoffs for which the expected energy changes little across a long interval of scales. Fix a sufficiently large even integer ξ . Proposition 4.1, subadditivity (2.16) and stationarity give $\sup_{N,k \geq 0} \|\sup_{|e|=1} J(\square_k, e, e; A_N^0)\|_{L^\xi} < \infty$. Moreover, the family

$$\{\Lambda_{1/4}(\square_k; A_N^0) + \lambda_{1/4}(\square_k; A_N^0)^{-1} : N, k \geq 0\}$$

is tight. This follows from Lemma 4.5, with $s = 1/8$, $r = 2$ and the initial error $C\xi \log(2 + \xi)\bar{\delta}^2$, followed by (2.38) and weighted Cauchy–Schwarz. Choose ξ large enough for that lemma and then reduce $\bar{\delta}$.

Write

$$f_N(k) = \sum_{j=1}^d \mathbb{E}J(\square_k, e_j, e_j; A_N^0) = \frac{1}{2} \mathbb{E} \operatorname{tr}(A_N^0(\square_k) + (A_N^0)_*^{-1}(\square_k) - 2I).$$

By (2.15) and (3.21), $\mathbb{E}A_N^0(\square_k)$ and $\mathbb{E}(A_N^0)_*^{-1}(\square_k)$ decrease to I . Thus $f_N(k) \geq 0$ decreases in k , and the limit of $\sup_N f_N(k)$ as $k \rightarrow \infty$ exists. Choose N_k with $f_{N_k}(4k) \geq \sup_N f_N(4k) - k^{-1}$. Then

$$\sup_{k \leq n \leq 4k} (f_{N_k}(n) - f_{N_k}(4k)) \leq \sup_N f_N(k) - \sup_N f_N(4k) + k^{-1} \rightarrow 0.$$

Step 2. We show that the coarse-grained matrices $(A_{N_k}^0)_*^{-1}(\square_n)$ concentrate and that the expectations of the coarse-grained matrices $A_{N_k}^0(\square_n)$ converge along these scales. By subadditivity, the average of $(A_{N_k}^0)_*^{-1}(z + \square_k)$ over the scale- k subcubes of $\square_n$, minus $(A_{N_k}^0)_*^{-1}(\square_n)$, is positive semidefinite and has expected trace at most $2(f_{N_k}(k) - f_{N_k}(n))$. This difference tends to zero in L^1 , uniformly for $k \leq n \leq 4k$. The moment bound of Step 1 and (2.29) upgrade this to L^2 . For $n \geq 2k$, the variance of the average is at most $C3^{-d(n-k)}$, because the finite range of dependence splits the subcubes into a dimensional number of classes of independent cubes. Consequently, after extracting a subsequence, there are deterministic matrices $P, R \geq I$ such that

$$\sup_{k \leq n \leq 4k} \|\mathbb{E}A_{N_k}^0(\square_n) - P\| \rightarrow 0,$$

and

$$\sup_{2k \leq n \leq 4k} \|(A_{N_k}^0)_*^{-1}(\square_n) - R\|_{L^2} \rightarrow 0.$$

For the first convergence, extract a limit of $\mathbb{E}A_{N_k}^0(\square_{4k})$ and note that $\mathbb{E}A_{N_k}^0(\square_n) - \mathbb{E}A_{N_k}^0(\square_{4k})$ is positive semidefinite with trace at most $2(f_{N_k}(n) - f_{N_k}(4k))$, which tends to zero by Step 1.

Step 3. We use the estimates of Section 4 to show that both limiting matrices equal the identity. Fix $p \in \mathbb{R}^d$ and let $v = v(\cdot, \square_{4k}, p, R^{-1}p; A_{N_k}^0)$. Apply (4.16) with $s = 1/4$ and $L_0 = 2k$. The expected gaps in the subadditivity inequality (2.15) tend to zero uniformly over $2k \leq n \leq 4k$ by Step 1. The mean-square differences of the coarse-grained matrices $(A_{N_k}^0)_*^{-1}(\square_n)$ tend to zero by Step 2 and stationarity. The weights of the two sums in (4.16) are summable, and $(A_{N_k}^0)_*^{-1}(\square_{4k})R^{-1}p - p \rightarrow 0$ in L^2 . Since the coarse-grained ellipticity constants are tight, we obtain $3^{-4k} \|\nabla v\|_{\hat{H}^{-1}(\square_{4k})} \rightarrow 0$ in

probability. Now (4.15), with the same tight constants and the vanishing gaps in the subadditivity inequality, gives $J(\square_{4k-1}, p, R^{-1}p; A_{N_k}^0) \rightarrow 0$ in probability. The moment bounds make these energies uniformly integrable, so their expectations tend to zero. By (2.11), the limit of these expectations is also $\frac{1}{2}p \cdot (P - R^{-1})p$. Thus $P = R^{-1}$. Since $P, R \geq I$, both equal I , and $f_{N_k}(4k) \rightarrow 0$. The choice of N_k then shows that the limit of $\sup_N f_N(k)$ is zero. $\square$

Let $\Lambda_{N,Q}^0$ denote the minimal energy with given boundary values for A_N^0 . The bound (2.29) and Cauchy–Schwarz turn Theorem 8.53 into

$$(8.51) \quad \lim_{k \rightarrow \infty} \sup_N \mathbb{E} \left| \frac{\Lambda_{N,3^k \square_0}^0(p \cdot x)}{3^{dk}} - |p|^2 \right| = 0 \quad (p \in \mathbb{R}^d).$$

Theorem 8.54 (Identification of the scalar). *The scalar c in (8.50) equals one. Consequently, all subsequential limit forms coincide.*

Proof. Let $E^0 = e^{-H}E$ and $F^0 = e^{-H}F$ be the forms obtained by removing the factor e^H with the rule (8.34) for continuous weights, applied to the random continuous weight e^{-H} on a compact cube. They are subsequential limits of the same normalized sequence, whose coefficient is A_N^0 . Fix R and $p \neq 0$. By (8.36), with $\varrho = e^{-H}$ and $E^{\varrho,Q} = E^0$ on the padded cube $Q = 3R\square_0$, $\Lambda_{N,R\square_0}^0(p \cdot x)$ converges along the subsequence that defines E to $\Lambda_{E^0,R\square_0}(p \cdot x)$. The uniform moments of these energies give uniform integrability. Hence

$$\mathbb{E} \left| \frac{\Lambda_{E^0,R\square_0}(p \cdot x)}{R^d} - |p|^2 \right| \leq \sup_N \mathbb{E} \left| \frac{\Lambda_{N,R\square_0}^0(p \cdot x)}{R^d} - |p|^2 \right|.$$

The same holds for F^0 along its subsequence. Letting $R \rightarrow \infty$ through triadic values, (8.51) shows that both $\mathbb{E}|\Lambda_{E^0,R\square_0}(p \cdot x)/R^d - |p|^2|$ and its analogue for F^0 tend to zero. But $F^0 = cE^0$ gives $\Lambda_{F^0,R\square_0} = c\Lambda_{E^0,R\square_0}$ on every cube, so $c|p|^2 = |p|^2$ and $c = 1$.

Since all subsequential limit forms are therefore equal, and their prescribed-boundary minima and Dirichlet Green quadratic forms are strongly relatively compact by Proposition 8.18, every subsequence of these quantities has a further subsequence converging to the same limit, identified by (8.31) and Proposition 8.31. Hence the prescribed-boundary minima and Green quadratic forms, for the countable family of cubes and data fixed in Section 8.4, converge along the full sequence of cutoffs. Convergence of the constant-flux Neumann responses on general triadic cubes is established separately by the smoothed-load comparison in Proposition 9.21(iv). This comparison uses no Neumann limit form; its uniform higher moments also give convergence in L^p . $\square$

9. CONVERGENCE OF THE PROCESS

In this section we prove Theorem 7.1(i). Combining the unique limit form of Section 8 with the limit of the reversible measures, we prove that the path laws converge in probability, and we then upgrade this convergence to almost-sure convergence.

- (1) After constructing the diffusions from every starting point (Section 9.1), we show that the measures μ_N converge and that their mass in small balls is bounded uniformly in the cutoff (Section 9.2). This bound lets us pass to the limit in the resolvent equations of the processes killed on leaving a cube, even though the sources in these equations are given by the random measures. Mollifying these sources and applying the Hölder estimates of Section 8 gives convergence in probability of the killed resolvents, uniformly in the starting point.
- (2) Together with tightness of the path laws (Section 9.3), the convergence of the killed resolvents identifies the limiting path laws and gives convergence in probability (Section 9.4).
- (3) We then compare the energies at two finite cutoffs. On a small cube, rescaling shows that the two energies differ essentially by the ratio of their deterministic normalizations. If these ratios were bounded away from one on a positive proportion of scales, the energy of

a fixed affine boundary value would stay a fixed factor away from its limit, contradicting the convergence in probability. Once such scales are excluded, the comparison holds with summable error probabilities, and the Borel–Cantelli lemma gives almost-sure convergence of the finite-volume energies (Section 9.5). Bounds that hold for all cutoffs on a single event of probability one then carry this almost-sure convergence over to the Dirichlet forms, the killed resolvents and the path laws (Sections 9.6–9.7).

The conventions of Section 8 remain in force, except that Sections 9.1 and 9.3 work with the diffusions X and $X^{(L)}$ and their coefficients $\mathbf{a}$ and $\mathbf{a}_L$.

9.1. Construction of the diffusions. We construct the diffusions from their elliptic resolvents. By Lemma 6.21, the coefficient $\mathbf{a}$ is positive and locally $C^{1,1}$. The same holds for $\mathbf{a}_L$. The logarithmic derivative bound (6.112) gives at most linear growth of the drift, uniformly in L . For either coefficient, write $\mathcal{L} = \Delta + b \cdot \nabla$, where $b = \nabla \log \mathbf{a}$ and $|b(x)|_2 \leq K(1 + |x|_2)$. The weak solutions of $\lambda \mathbf{a}u - \nabla \cdot (\mathbf{a} \nabla u) = \mathbf{a}f$ on increasing bounded cubes, with zero boundary values, converge to the whole-space resolvent. To see the decay and uniqueness, put $s(x) = 1 + |x|_2^2$. Direct differentiation gives $|\mathcal{L}s^{-\theta}| \leq C\theta s^{-\theta}$ and $\mathcal{L}s^\theta \leq C\theta s^\theta$ for $0 < \theta \leq 1$, where $C = 2d + 8 + 3K$. Fix $\lambda > 0$ and choose $C\theta \leq \lambda/2$. If f is supported in $\{|x|_2 \leq R\}$, then $2\lambda^{-1}\|f\|_\infty(1 + R^2)^\theta s^{-\theta}$ is a supersolution of $\lambda - \mathcal{L}$ with source $|f|$. Comparison on each exhausting cube bounds the absolute value of its zero-boundary solution by this function, which tends to zero at infinity. For uniqueness among bounded solutions, compare the difference with εs^θ on large cubes and then let $\varepsilon \downarrow 0$. Contraction extends the resolvent to $C_0(\mathbb{R}^d)$. For $f \in C_0(\mathbb{R}^d)$, uniform approximation by compactly supported data and the local energy estimates pass the weak equation to $R_\lambda f$, so it is the unique bounded weak solution. Consequently, for any smooth $v \in C_0(\mathbb{R}^d)$ with $\mathcal{L}v \in C_0(\mathbb{R}^d)$, uniqueness gives $R_\lambda(\lambda v - \mathcal{L}v) = v$. The maximum principle gives positivity and $\|R_\lambda\| \leq \lambda^{-1}$; uniqueness gives the resolvent identity. For $\varphi \in C_c^\infty(\mathbb{R}^d)$, uniqueness also gives $R_\lambda(\lambda\varphi - \mathcal{L}\varphi) = \varphi$, so $\lambda R_\lambda\varphi \rightarrow \varphi$. Contraction extends this convergence to $C_0(\mathbb{R}^d)$. Hille–Yosida [Kallenberg, 2002, Theorem 19.11] therefore gives a Feller semigroup P_t , with generator $\mathcal{L}$ on $C_c^\infty(\mathbb{R}^d)$. Its positive operators have transition kernels by the Riesz representation theorem. These kernels have total mass one: the same resolvent identity puts $s^{-\theta}$ in the generator domain, and the bound $\|\mathcal{L}s^{-\theta}\|_\infty \leq C\theta$ gives $|P_t s^{-\theta} - s^{-\theta}| \leq Ct\theta$. Letting $\theta \downarrow 0$ proves $P_t 1 = 1$.

The weak resolvent equation also supplies the bounds needed for continuous paths, before any path realization is used. Put $q(y) = |y - x|_2^2$ and use the concave regularization $\varphi_R(y) = R^2 g(q(y)/R^2)$, where $g(a) = a/(1 + a)$ and $R \geq 1$. Both $\varphi_R - R^2$ and $\varphi_R^2 - R^4$ belong to $C_0(\mathbb{R}^d)$, as do their images under $\mathcal{L}$. The preceding weak-resolvent identity places them in the generator domain; adding back the constants is justified by $P_t 1 = 1$. Concavity and direct differentiation give $\mathcal{L}\varphi_R \leq C(a_x + \varphi_R)$ and $\mathcal{L}(\varphi_R^2) \leq C(a_x \varphi_R + \varphi_R^2)$, with $a_x = 1 + |x|_2^2$ and C independent of R . The semigroup identity and Gronwall’s inequality first bound $P_t \varphi_R(x)$ by $C a_x t e^{Ct}$; integrating this bound in the equation for $P_t(\varphi_R^2)(x)$ gives $C a_x^2 t^2 e^{Ct}$. Letting $R \rightarrow \infty$ and using Fatou’s lemma yields the second-moment bound and the fourth-moment bound $\int_{\mathbb{R}^d} |y - x|_2^4 P_t(x, dy) \leq C(1 + |x|_2^2)^2 t^2 e^{Ct}$. In particular, for every compact $A \subseteq \mathbb{R}^d$ and $r > 0$, $\sup_{x \in A} P_t(x, \{|y - x|_2 \geq r\}) = O(t^2) = o(t)$. Kolmogorov extension constructs the finite-dimensional laws on a countable dense set of times. The semigroup identity and the fourth-moment bound give, on every bounded time interval, $E_x |Z_t - Z_s|_2^4 \leq C(x, T)|t - s|^2$, so the Kolmogorov continuity criterion extends these laws to continuous paths. We denote the resulting conservative diffusions for $\mathbf{a}$ and $\mathbf{a}_L$ by X and $X^{(L)}$, respectively. To obtain the strong Markov property, approximate a finite stopping time T from above by $T_n = 2^{-n} \lceil 2^n T \rceil$. The Markov identity holds at each T_n . Path continuity and continuity of $P_t f$ for $f \in C_0(\mathbb{R}^d)$ allow passage to T ; a monotone-class argument extends the identity to bounded Borel tests. Localization gives the identity on $\{T < \infty\}$.

The semigroup is symmetric on $L^2(\mathbb{R}^d, \mathbf{a} dx)$, and its resolvents agree with those of the closure of the form $f \mapsto \int_{\mathbb{R}^d} \mathbf{a} |\nabla f|^2 dx$ on $C_c^\infty(\mathbb{R}^d)$. This follows by testing the weak equation and exhausting $\mathbb{R}^d$ by cubes. Its generator is $\mathbf{a}^{-1} \nabla \cdot (\mathbf{a} \nabla)$. Restricting the form to a cube with zero boundary values identifies its resolvents with those of the diffusion killed on leaving the cube, initially as $\mathbf{a} dx$ -almost-everywhere versions [Fukushima et al., 2011, Theorems 4.4.2–4.4.3]. The identification holds at every interior starting point as follows. For $f \in C_c^\infty(Q)$, let u be the zero-boundary solution of $(\lambda - \mathcal{L})u = f$. It is C^2 in Q and continuous on $\overline{Q}$, with value zero on ∂Q . Take cubes $Q_j \subseteq Q$ increasing to Q and compactly supported C^2 functions agreeing with u near $\overline{Q}_j$. Smooth approximation in the generator identity gives Dynkin's formula for these functions. Stopping at $t \wedge \tau_{Q_j}$ yields

$$u(x) = E_x \left[\int_0^{t \wedge \tau_{Q_j}} e^{-\lambda s} f(X_s) ds + e^{-\lambda(t \wedge \tau_{Q_j})} u(X_{t \wedge \tau_{Q_j}}) \right], \quad x \in Q_j.$$

Path continuity and the zero boundary value allow $j \rightarrow \infty$; boundedness and the exponential factor then allow $t \rightarrow \infty$. Thus $u(x) = E_x \int_0^\infty e^{-\lambda s} f(X_s) ds$ for every $x \in Q$. Approximation by uniformly bounded smooth data, using local elliptic regularity and dominated convergence, extends this identity to bounded continuous data. The same argument applies to $\mathbf{a}_L$.

These laws also describe the usual SDE construction. Since $\nabla \log \mathbf{a}$ is locally Lipschitz, there is a unique maximal strong solution from every $x \in \mathbb{R}^d$ of

$$(9.1) \quad dX_t = \sqrt{2} dW_t + \nabla \log \mathbf{a}(X_t) dt,$$

where W is standard Brownian motion on $\mathbb{R}^d$. Fix a bounded cube Q , $\lambda > 0$ and $f \in C_c^\infty(Q)$, and let u solve $\lambda \mathbf{a} u - \nabla \cdot (\mathbf{a} \nabla u) = \mathbf{a} f$ in Q , with zero boundary values. Interior elliptic regularity and Itô's formula, stopped first strictly inside Q , give, after exhaustion and removal of the deterministic time cutoff,

$$u(x) = \mathbf{E}_x^{\text{SDE}} \left[\int_0^{\tau_Q} e^{-\lambda t} f(X_t) dt \right], \quad x \in Q.$$

Here $\mathbf{E}_x^{\text{SDE}}$ denotes expectation for the maximal solution of (9.1). Thus its killed resolvents agree with those of the analytic construction. Uniqueness of Laplace transforms identifies the killed semigroups, and the Markov property identifies their path laws. Exhausting $\mathbb{R}^d$ identifies the laws up to their lifetimes, for every starting point. The same argument applies to $\mathbf{a}_L$ and to the space and time rescalings in (7.1); Proposition 9.8 makes these lifetimes infinite. For an open set U , write $\tau_U = \inf\{t \geq 0 : X_t \notin U\}$ and let $\tau_U^{(L)}$ be the exit time of $X^{(L)}$ from U . The diffusion $X^{(L)}$ has generator (1.8).

9.2. The reversible measure and resolvent convergence in probability. Write E for the unique limit form; it is measurable with respect to the random fields. Along each subsequence of the Skorokhod representation, the forms E_N^Q Mosco-converge to E^Q on $L^2(Q, dx)$, and their coercivity constants are bounded.

9.2.1. The chaos martingale and the mass of small balls. Recall the measures μ_N^0 and the reversible measures $\mu_N = e^H \mu_N^0$ from the paragraph following (7.1).

Lemma 9.1 (Positive moments of the chaos). *Let $p \geq 1$ be an integer. There exist $C_p < \infty$ and $c_p > 0$ such that, if $\bar{\delta} \leq c_p$, then, for every $N \geq 0$ and every cube q of side $r \leq 1$ contained in a fixed bounded region,*

$$(9.2) \quad \mathbb{E}[\mu_N(q)^p] \leq C_p r^{dp - C_p \bar{\delta}^2}.$$

Moreover, $N \mapsto \mu_N^0(q)$ is a nonnegative martingale, and $\mu_N(q)$ is a nonnegative martingale conditionally on H .

Proof. Since $\mathbb{E}e^{\mathbf{g}_{-j}(x)} = e^{\tau^2}$ for every x and the fields are independent, integration against fixed nonnegative test functions shows that μ_N^0 is a measure-valued martingale. The sum H of the fields with scales larger than one is independent of the fields at scales at most one, so $\mu_N = e^H \mu_N^0$ is a martingale given H . On a bounded region the factor e^{pH} is controlled by $\mathbb{E}e^{p \sup H} < \infty$, which follows from the exponential moments of the $C^{1,1}$ norm of H on compact cubes in Lemma 8.1. It therefore suffices to prove (9.2) for μ_N^0 .

For $p \geq 2$ we expand

$$\mathbb{E}\mu_N^0(q)^p = \int_{q^p} \mathbb{E} \prod_{i=1}^p \prod_{j=0}^N e^{\mathbf{g}_{-j}(x_i) - \tau^2} dx.$$

The field $\mathbf{g}_{-j}$ has range of dependence $C3^{-j}$ by Assumption (g1). If all pairwise distances $|x_i - x_k|$ exceed that range, then the p factors are independent and the normalized expectation equals one. For each of the remaining fields, Hölder's inequality gives $\mathbb{E} \prod_i e^{\mathbf{g}_{-j}(x_i) - \tau^2} \leq \mathbb{E} e^{p \mathbf{g}_0(0)} e^{-p\tau^2} \leq e^{Cp^2\bar{\delta}^2}$ by (1.5). The number of fields for which this separation condition fails is at most $C + \sum_{i < k} \log_3^+(C/|x_i - x_k|)$, so the integrand is bounded by $C \prod_{i < k} (|x_i - x_k| \wedge 1)^{-a}$ with $a = Cp^2\bar{\delta}^2$. There are $p(p-1)/2$ pair factors. Hölder's inequality with exponent $p(p-1)/2$ and the bound $\int_{q \times q} |x - y|^{-ap(p-1)/2} dx dy \leq Cr^{2d-ap(p-1)/2}$, valid when $ap(p-1)/2 < d$, give $\mathbb{E}\mu_N^0(q)^p \leq C_p r^{dp-ap(p-1)/2}$. Since $ap(p-1)/2 \leq Cp^4\bar{\delta}^2$, the condition $ap(p-1)/2 < d$ holds once $\bar{\delta} \leq c_p$, and (9.2) follows. $\square$

Proposition 9.2 (Mass of small balls and the limiting measure). *Let $\epsilon \in (0, 1)$, let p be an integer with $p\epsilon > d$, and assume that $\bar{\delta} \leq c_p$ and $C_p\bar{\delta}^2 < p\epsilon - d$, with c_p and C_p as in Lemma 9.1. Then, almost surely, μ_N converges locally weakly to a locally finite measure μ , and there is a random constant K_μ with $\mathbb{E}K_\mu^p < \infty$ such that, for every x in the fixed bounded region, every $0 < r \leq 1$ and every $N \geq 0$,*

$$(9.3) \quad \mu_N(B_r(x)) \leq K_\mu r^{d-\epsilon} \quad \text{and} \quad \mu(B_r(x)) \leq K_\mu r^{d-\epsilon}.$$

Almost surely, the measure μ has full support and no atoms, and $\mu(\partial Q) = 0$ for every cube Q .

Proof. For a dyadic cube q of side 2^{-k} , Doob's inequality [Kallenberg, 2002, Proposition 7.16] and (9.2) give $\mathbb{E} \sup_N \mu_N(q)^p \leq (p/(p-1))^p \sup_N \mathbb{E} \mu_N(q)^p \leq C 2^{-k(dp-C_p\bar{\delta}^2)}$. Let C be the constant in the covering below and put

$$K_\mu^p = C^p \sum_{k \geq 0} \sum_{q \text{ of side } 2^{-k}} 2^{kp(d-\epsilon)} \sup_N \mu_N(q)^p.$$

There are $C2^{dk}$ cubes of side 2^{-k} in the region, so $\mathbb{E}K_\mu^p \leq C \sum_k 2^{k(d+C_p\bar{\delta}^2-p\epsilon)} < \infty$ by the choice of p and then of $\bar{\delta}$. A ball $B_r(x)$ with $2^{-k-1} < r \leq 2^{-k}$ is covered by a number of cubes of side 2^{-k} that depends only on d . This gives the first bound in (9.3) with one random constant for all N , x and $r \leq 1$.

Since $p > 1$, the bound (9.2) at this order gives uniform integrability of $\mu_N(\varphi)$ for every $\varphi \in C_c(\mathbb{R}^d)$. The martingale convergence theorem [Kallenberg, 2002, Theorem 7.18 and Corollary 7.22] gives almost-sure and L^p convergence against every function in a countable dense subset of $C_c(\mathbb{R}^d)$. The local mass bound above gives vague relative compactness [Kallenberg, 2002, Theorem A2.3], so these limits determine a single locally finite measure μ , and $\mu_N \rightarrow \mu$ locally weakly almost surely. The bound for μ follows from the portmanteau theorem applied to open balls. The measure μ has no atoms by (9.3). To prove that $\mu(\partial Q) = 0$, cover ∂Q by Cr^{1-d} balls of radius r . This gives $\mu(\partial Q) \leq CK_\mu r^{1-\epsilon} \rightarrow 0$.

It remains to prove that μ has full support. Set $\mu^0 = e^{-H}\mu$. Fix an open cube Q and $k \geq 0$, and write $\mu^0 = e^{\sum_{j=0}^k \mathbf{g}_{-j} - (k+1)\tau^2} \mu^{0,(>k)}$, where $\mu^{0,(>k)}$ is the limiting measure built in the same way from the fields $\mathbf{g}_{-j}$ with $j > k$ only. Since $\mathbb{E}\mu^{0,(>k)}(\partial Q) = |\partial Q| = 0$ and the factor in front of $\mu^{0,(>k)}$ is a

fixed continuous positive function, weak convergence shows that the two sides of this identity give the same mass to Q . Hence the events $\{\mu^0(Q) = 0\}$ and $\{\mu^{0,(>k)}(Q) = 0\}$ coincide up to null sets for every k , so $\{\mu^0(Q) = 0\}$ is a tail event of the independent fields. Its probability is therefore 0 or 1, and it is 0 because $\mathbb{E}\mu^0(Q) = |Q| > 0$ by uniform integrability. The positive continuous factor e^H gives the full support of μ . $\square$

9.2.2. Traces on the reversible measures. We choose $\epsilon < 1$ in (9.3), so that the trace inequality (8.32) applies with $t = d - \epsilon > d - 1$ to every μ_N and to μ , with the constant $C(K_\mu + \mu_N(Q))$. We extend functions in $H_0^1(Q)$ by zero to $\mathbb{R}^d$. For such a function w , the zero extension satisfies $\|w\|_{H^{3/4}(\mathbb{R}^d)} \leq C\|w\|_{H^{3/4}(Q)}$, so (8.5) gives

$$(9.4) \quad \|w\|_{H^{3/4}(\mathbb{R}^d)}^2 \leq K_N E_N^Q(w),$$

for every $w \in H_0^1(Q)$, with a constant that is bounded along the sequence in the Skorokhod representation. Traces on μ are defined by density: by (8.32), the map sending a smooth function on $\overline{Q}$ to its class in $L^2(\mu)$ is bounded from $H^{1/2}$ to $L^2(\mu)$, and it extends by continuity.

Lemma 9.3 (Varying-measure convergence). *Let $u_N \in H_0^1(Q)$ satisfy $\sup_N E_N^Q(u_N) < \infty$ along the sequence in the Skorokhod representation and $u_N \rightarrow u$ in $L^2(Q, dx)$. Then, for every bounded continuous function f on $\overline{Q}$,*

$$(9.5) \quad \int u_N^2 d\mu_N \longrightarrow \int u^2 d\mu \quad \text{and} \quad \int f u_N d\mu_N \longrightarrow \int f u d\mu.$$

Proof. By (9.4) the zero extensions are bounded in $H^{3/4}(\mathbb{R}^d)$, and the interpolation inequality $\|w\|_{H^{1/2}} \leq \|w\|_{L^2}^{1/3} \|w\|_{H^{3/4}}^{2/3}$ upgrades the convergence in L^2 to strong convergence in $H^{1/2}(\mathbb{R}^d)$. We write

$$\int u_N^2 d\mu_N - \int u^2 d\mu = \int (u_N - u)(u_N + u) d\mu_N + \left(\int u^2 d\mu_N - \int u^2 d\mu \right).$$

The first term tends to zero by Cauchy–Schwarz and (8.32). For the second, we approximate u in $H^{1/2}$ by smooth functions φ . By (8.32), the resulting error is small uniformly over all μ_N and μ , and $\int \varphi^2 d\mu_N \rightarrow \int \varphi^2 d\mu$ by weak convergence, since $\mu(\partial Q) = 0$. The proof of the second limit is the same. $\square$

9.2.3. The killed resolvent as a minimizer. At finite cutoff the generator is symmetric in $L^2(\mu_N)$ with form E_N^Q . Since the density of μ_N is bounded on $\overline{Q}$ at every finite N , it follows that, for $\lambda > 0$ and bounded f , the killed resolvent $u_N = R_{N,\lambda}^Q f$ is the unique minimizer over $H_0^1(Q)$ of $E_N^Q(u) + \lambda \int u^2 d\mu_N - 2 \int f u d\mu_N$. It satisfies the sub-Markov bound $\|u_N\|_\infty \leq \|f\|_\infty / \lambda$, the weak equation $E_N^Q(u_N, \varphi) = \int (f - \lambda u_N) \varphi d\mu_N$, and, by testing with u_N , the energy bound $E_N^Q(u_N) \leq \|f\|_\infty^2 \mu_N(Q) / \lambda$.

Proposition 9.4 (Passage of the minimization to the limit). *Let $f \in C(\overline{Q})$ and $\lambda > 0$, and assume that, along the sequence in the Skorokhod representation, E_N^Q Mosco-converges to E^Q and that E^Q satisfies the coercivity bound (8.5) with a finite constant. Then the functional*

$$(9.6) \quad u \longmapsto E^Q(u) + \lambda \int u^2 d\mu - 2 \int f u d\mu$$

has a unique minimizer u^ in the energy Hilbert space $(D(E^Q), E^Q)$, and every $L^2(dx)$ cluster point of the killed resolvents $u_N = R_{N,\lambda}^Q f$ equals u^* .*

Proof. The energy space is complete because $E^Q(u) \geq \|G^Q\|^{-1} \|u\|_{L^2}^2$ by the variational formula (8.31). The trace from $D(E^Q)$ to $L^2(\mu)$ is bounded, since $\int u^2 d\mu \leq C(K_\mu + \mu(Q)) \|u\|_{H^{1/2}}^2 \leq$

$CKE^Q(u)$ by (8.32) and the coercivity of E^Q . The functional (9.6) is therefore continuous, strictly convex and coercive on the energy space, and the Riesz representation theorem gives a unique minimizer u^* , which satisfies $E^Q(u^*, v) + \lambda \int u^* v d\mu = \int f v d\mu$ for every $v \in D(E^Q)$.

To pass to the limit, note that the energies $E_N^Q(u_N)$ are bounded, so, by (8.5), a subsequence converges strongly in $L^2(dx)$. The liminf inequality of Mosco convergence and Lemma 9.3 show that the liminf of the finite-cutoff functionals at u_N is at least the value of (9.6) at the limit. Since u_N minimizes the finite-cutoff functional, recovery sequences for every $v \in D(E^Q)$, together with (9.5), show that this liminf is at most the value of (9.6) at v . Hence the limit minimizes (9.6), so it equals u^* . $\square$

Write $R_\lambda^Q f$ for the minimizer u^* .

9.2.4. Uniform convergence of the killed resolvents. To obtain uniform convergence we need Hölder bounds on the killed resolvents that are uniform in N . The source in the resolvent equation is the measure $(f - \lambda u_N)\mu_N$, whose density is not bounded uniformly in N , so we mollify the source before applying the Hölder estimate for bounded sources.

Proposition 9.5 (Uniform convergence of killed resolvents). *Assume the hypotheses of Proposition 9.4. Assume also that solutions with zero Dirichlet boundary values and bounded sources satisfy the Hölder estimate (8.11) with $\alpha = 1/2$ up to the boundary, with the supremum norm of the source on the right side and with a constant whose moments of the finitely many orders fixed at the start of Section 8 are finite. Let $\epsilon < 1/[8(d+2)]$ in (9.3). Then, for every bounded continuous f and every $\lambda > 0$,*

$$(9.7) \quad \sup_{\overline{Q}} |R_{N,\lambda}^Q f - R_\lambda^Q f| \longrightarrow 0 \quad \text{in probability,}$$

and the limit vanishes on ∂Q . There is a random constant K_Q with finite moments of these orders such that, for all $\lambda > 0$ and $f \in C(\overline{Q})$,

$$(9.8) \quad \|R_\lambda^Q f\|_{C^{1/4}(\overline{Q})} \leq K_Q \|f\|_\infty.$$

Proof. Let $u_N = R_{N,\lambda}^Q f$ and $\zeta_N = (f - \lambda u_N)\mu_N$, so that $|\zeta_N| \leq 2\|f\|_\infty \mu_N$ and $E_N^Q(u_N, \varphi) = \int \varphi d\zeta_N$. Fix $0 < r \leq 1$, put $R = r^{d+2} \leq 1$, and let η_R be a mollifier at scale R .

Step 1. We bound the mollified source and the error of the mollification. By (9.3),

$$|\zeta_N * \eta_R(x)| \leq CR^{-d} |\zeta_N|(B_R(x)) \leq CK_\mu R^{-\epsilon} \|f\|_\infty.$$

For the error, we first use that, by Schur's test, $\|\zeta_N\|_{H^{-1/4}}^2$ is at most $\|\zeta_N\|_{TV}$ times the supremum over x of the Bessel potential of order $1/2$ of $|\zeta_N|$ at x . Splitting the Bessel potential into dyadic annuli and using (9.3) bounds this supremum, since $\epsilon < 1/2$. Hence $\|\zeta_N\|_{H^{-1/4}} \leq K\|f\|_\infty$. Then the bound $\sup_\xi (1 + |\xi|^2)^{-1/2} \min(1, R|\xi|)^2 \leq CR$ gives

$$\|\zeta_N - \zeta_N * \eta_R\|_{H^{-3/4}} \leq CR^{1/2} \|\zeta_N\|_{H^{-1/4}} \leq KR^{1/2} \|f\|_\infty.$$

Step 2. We compare u_N with the solution for the mollified source. Let $v_{N,R} \in H_0^1(Q)$ solve $E_N^Q(v_{N,R}, \varphi) = \int \varphi (\zeta_N * \eta_R) dx$. Its source is a bounded function of size $KR^{-\epsilon} \|f\|_\infty$, so (8.11) with $\alpha = 1/2$ gives $\|v_{N,R}\|_{C^{1/2}(\overline{Q})} \leq KR^{-\epsilon} \|f\|_\infty$. The difference of the two equations and (9.4) give

$$\begin{aligned} E_N^Q(u_N - v_{N,R}) &= \langle \zeta_N - \zeta_N * \eta_R, u_N - v_{N,R} \rangle \leq KR^{1/2} \|f\|_\infty \|u_N - v_{N,R}\|_{H^{3/4}} \\ &\leq KR^{1/2} \|f\|_\infty E_N^Q(u_N - v_{N,R})^{1/2}, \end{aligned}$$

hence $\|u_N - v_{N,R}\|_{L^2} \leq KR^{1/2} \|f\|_\infty$.

Step 3. We prove a Hölder bound for u_N . By Steps 1 and 2, the mean-square oscillation of u_N on $B_r(x) \cap Q$ is at most

$$K(r^{1/2}R^{-\epsilon} + r^{-d/2}R^{1/2})\|f\|_\infty = K(r^{1/2-(d+2)\epsilon} + r)\|f\|_\infty,$$

and $1/2 - (d+2)\epsilon > 3/8 > 1/4$. Using zero extensions across ∂Q , Campanato's criterion gives $\|u_N\|_{C^{1/4}(\overline{Q})} \leq K\|f\|_\infty$ and $u_N = 0$ on ∂Q . Here K is a product of finitely many random constants with finite moments of these orders, and it is therefore tight.

Step 4. We prove convergence in probability. Fix two cutoff subsequences, and apply the Skorokhod representation of Section 8.4 to both of them jointly; the variables represented jointly include the fields $(\mathbf{g}_j)_{j \in \mathbb{Z}}$, the reversible measures, the quadratic forms $\langle f, G_N^Q f \rangle$ with $f \in \mathcal{D}$ and $Q \in \mathcal{C}$, and the constants in Steps 1–3. The constants have tight laws. The limit forms of the two sequences coincide by Theorem 8.54, and their limiting reversible measures are the same. The Arzelà–Ascoli theorem and Proposition 9.4 therefore identify both uniform limits with the unique minimizer of the same functional.

Consequently, every pair of cutoff subsequences has a further pair of subsequences along which the joint laws of the two killed resolvents converge to a law supported on the diagonal of $C(\overline{Q})^2$. Since $C(\overline{Q})$ is Polish, the Gyöngy–Krylov criterion [Gyöngy and Krylov, 1996, Lemma 1.1] gives convergence in probability on the original probability space. The joint representation with these quadratic forms and the reversible measure identifies the limit with $R_\lambda^Q f$. The constant in Step 3 is independent of λ and f , which gives (9.8) and the zero boundary values. $\square$

9.3. Tightness. In this subsection we prove Proposition 9.8 (tightness in expectation and in probability). Section 9.4 uses both forms of tightness, together with the compact containment estimate (9.24), to identify the limiting path law from its resolvents. The early-exit bound (9.22) is used again in Section 10 to bound mean exit times from below.

We first bound the probability of an early exit from a cube. After the rescaling introduced below, a cube of side 3^m becomes $\square_0$ and time is measured in units of $3^{2m}/\bar{\mathbf{a}}_{m \wedge L}$. The process started in $\square_{-2}$ leaves $\square_0$ before time t with probability at most $K\sqrt{t}$, where K has moments bounded uniformly in the cutoff, the cube size, and the position of the cube.

The proof bounds the Laplace transform of the exit time by one minus the solution of a resolvent problem with a cutoff function as source. A local maximum principle for subsolutions, with the Laplace parameter equal to $1/t$, then gives the exit-probability bound. Since pointwise ellipticity is not uniform in the cutoff and the reversible measure becomes singular in the limit, we prove this maximum principle by Moser iteration with respect to the reversible measure. The iteration uses the mass bounds, the $H^{3/4}$ coercivity bound, and cutoff functions of Lemma 9.6. The energy measures of these cutoff functions satisfy an upper bound with the same power of the ball radius as the upper mass bound for the reversible measure. The exit-probability bound shows that, with high probability, the paths stay in a large cube up to a fixed time; applied at stopping times, it also controls their modulus of continuity.

Throughout this subsection we assume (g1)–(g4) and that $\bar{\delta}$ is small enough, depending on the moment orders fixed below. We recall the diffusion $X^{(L)}$ from (1.7)–(1.8) and we set $Z_t^{(L)} := 3^{-L}X_{3^{2L}t/\bar{\mathbf{a}}_L}^{(L)}$ for $t \geq 0$. We use rescaled coordinates for $Z^{(L)}$ and physical coordinates for $X^{(L)}$. All statements of this subsection also hold for the rescaled diffusion $3^{-L}X_{3^{2L}t/\bar{\mathbf{a}}_L}$ without cutoff.

For $L, m \in \mathbb{N}_0$ and $z \in \mathbb{R}^d$, consider the cube $z + \square_m$ in physical coordinates, and set

$$j := m \wedge L, \quad c := \left(\frac{\mathbf{a}_L}{\mathbf{a}_j} \right)(z), \quad d\mu := c^{-1} \mathbf{a}_L(z + 3^m x) dx, \quad A := \bar{\mathbf{a}}_j^{-1} c^{-1} \mathbf{a}_L(z + 3^m x).$$

If $m \geq L$, then $c = 1$. In the local coordinates $x \mapsto z + 3^m x$, which rescale $z + \square_m$ to $\square_0$, and in units of time

$$(9.9) \quad \frac{3^{2m}}{\bar{\mathbf{a}}_{m \wedge L}},$$

the generator of $X^{(L)}$ becomes $\mathcal{A} = (d\mu/dx)^{-1} \nabla \cdot (A \nabla)$. Here $\mathcal{A}$ denotes the generator. The notation $\mathcal{A}(U; \mathbf{a})$ in Section 2.2 denotes a space of solutions, while $\mathcal{A}_E(U)$ and $\mathcal{A}_N$ in Section 8.5 denote normalized coarse-grained matrices. The generator is symmetric on $L^2(\mu)$, with Dirichlet form and local energy

$$\mathcal{E}(v, w) := \int A \nabla v \cdot \nabla w \quad \text{and} \quad \mathcal{E}(v; V) := \int_V A |\nabla v|^2.$$

The indexed quantity $\mathcal{E}_{s, \infty, q}$ of Definition 2.3 denotes the coarse-graining error. The factor c normalizes the product of the factors contributed by $\mathbf{g}_i$, with $j < i \leq L$, to have value one at z . By (3.2), the oscillation of the sum of these fields on the cube has uniformly bounded exponential moments, and therefore the moment bounds for the random constants below are uniform in L , m and z . For the coefficient $\mathbf{a}$ without cutoff we use instead $j = m$ and $c = \mathbf{a}(z)/\mathbf{a}_m(z)$, so that the density of μ is $c^{-1} \mathbf{a}(z + 3^m x)$, and the unit of time is $3^{2m}/\bar{\mathbf{a}}_m$. This density is then $\mathbf{a}_m(z + 3^m x)$ multiplied by $\exp(\sum_{i>m} (\mathbf{g}_i(z + 3^m x) - \mathbf{g}_i(z)))$, and this exponential factor satisfies the same oscillation bound.

Lemma 9.6 (Local mass, coercivity and cutoff-function estimates). *Fix a bounded reference cube and a moment order $q \in [1, \infty)$. Also fix finitely many concentric pairs $V_0 \subseteq V_1$ inside the reference cube. For each pair and each $s \in [0, 1]$, let V_s be the cube with the same center whose side length interpolates affinely between those of V_0 and V_1 ; the nested families below consist of the cubes $V_{k2^{-n}}$, for $n \geq 0$ and $0 \leq k \leq 2^n$. There exists a threshold $\bar{\delta}_0 > 0$, which may depend on q , such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then there is a deterministic $B < \infty$, independent of q , such that, for every $L, m \in \mathbb{N}_0$ and $z \in \mathbb{R}^d$, there is a random constant $K = K_{L, m, z} \geq 1$ satisfying*

$$\sup_{L, m, z} \mathbb{E}[K_{L, m, z}^q] < \infty$$

such that, after rescaling $z + \square_m$ to $\square_0$ as in (9.9), the following statements hold.

(i) Mass bounds: for every ball $B_r(x)$ contained in the reference cube with $0 < r \leq 1$,

$$(9.10) \quad K^{-1} r^{d+1/2} \leq \mu(B_r(x)) \leq K r^{d-1/2},$$

and the analogous bounds hold for cubes.

(ii) Coercivity in $H^{3/4}$: for each fixed interior cube V of the family with $n = 0$ and every $v \in H^1(V)$,

$$(9.11) \quad \|v\|_{H^{3/4}(V)}^2 \leq K (\mathcal{E}(v; V) + \|v\|_{L^2(V, dx)}^2),$$

and the last term can be omitted if $v \in H_0^1(V)$. For $V = V_{k2^{-n}}$, the same assertion holds with $K 2^{Bn}$ in place of K .

(iii) Harmonic cutoff functions: between consecutive cubes $V_{k2^{-n}} \subseteq V_{(k+1)2^{-n}}$ with gap $a \asymp 2^{-n}$ there is a cutoff function χ , equal to one on the smaller cube, vanishing near the boundary of the larger cube, and satisfying $0 \leq \chi \leq 1$, such that, for every ball $B_r(x)$ with $0 < r \leq 1$,

$$(9.12) \quad \Gamma_\chi(B_r(x)) := \int_{B_r(x)} A |\nabla \chi|^2 \leq K a^{-B} r^{d-1/2}.$$

When needed, we replace K by a fixed power of K ; this is possible if the lemma is applied from the outset with a correspondingly larger moment order q . For a finite collection of cubes we take the maximum of their constants; the resulting moment bounds remain uniform in L , m and z .

Proof. The three estimates are local versions of estimates proved in Sections 8 and 9.2 for the rescaled coefficient A_N of (8.1): Lemma 9.1 bounds the positive moments of the mass of small cubes under the multiplicative chaos, Lemma 8.3 bounds the $H^{3/4}$ norm by the energy, Proposition 8.7 bounds the energy measure of balls at every radius for solutions of Dirichlet problems, and Lemma 8.22 constructs cutoff functions as harmonic interpolants. We give the additional arguments needed to pass to the rescaling of $z + \square_m$ to $\square_0$ and to obtain one random constant for all the nested cubes.

Step 1. We prove the mass bounds (9.10). For $m \leq L$, reindexing the fields in reverse order identifies the matching-scale mass with $\mu_m^0(\square_0)$ from the paragraph after (7.1); the fields above scale m contribute the oscillation factor described before the lemma, whose positive and negative moments are uniformly bounded. Lemma 9.1, applied to the martingale $N \mapsto \mu_N^0(\square_0)$, bounds the positive moments of $\mu_N^0(\square_0)$ uniformly in N , including the field $\mathbf{g}_0$ at scale one. The bound on the second moment and $\mathbb{E}\mu_N^0(\square_0) = 1$ give $\inf_N \mathbb{P}(\mu_N^0(\square_0) \geq 1/2) > 0$, by the Cauchy–Schwarz inequality. Fix $J \in \mathbb{N}$ and choose two sufficiently separated triadic subcubes of $\square_0$ of side 3^{-J} . Since the fields are independent and have a finite range of dependence, for every $N \geq J$ we have

$$\mu_N^0(\square_0) \geq 3^{-dJ} Y (\mu_{N-J}^{0,(1)}(\square_0) + \mu_{N-J}^{0,(2)}(\square_0)),$$

where $\mu_{N-J}^{0,(1)}$ and $\mu_{N-J}^{0,(2)}$ are independent copies of μ_{N-J}^0 , independent of the random variable $Y > 0$. Here Y is the infimum over $\square_0$ of the part of the density of μ_N^0 contributed by the J coarsest fields. By Assumption (g2), the variable $-\log Y$ has a Gaussian upper tail. Consequently, the function $G(s) := \sup_N \mathbb{P}(\mu_N^0(\square_0) < e^{-s})$ satisfies $G(s) \leq C e^{-cs^2} + G(3s/4)^2$ for all sufficiently large s , where the finitely many $N < J$ are absorbed into the first term. The lower bound on $\mathbb{P}(\mu_N^0(\square_0) \geq 1/2)$ gives $G(s) \leq \rho < 1$ for large s . Letting $s \rightarrow \infty$ in the recursion gives $G(s) \rightarrow 0$. Iterating the recursion on the intervals $[(4/3)^n s_0, (4/3)^{n+1} s_0]$ then gives $G(s) \leq C \exp(-cs^{3/2})$: since $2(3/4)^{3/2} > 1$, squaring the bound at $3s/4$ gives a stronger bound at s , and the Gaussian term in the recursion is of smaller order. Hence, for every $p < \infty$,

$$(9.13) \quad \sup_N \mathbb{E}[\mu_N^0(\square_0)^{-p}] < \infty.$$

For $m > L$, we have $c = 1$. Scaling by 3^L and using stationarity shows that $\mu(\square_0)$ has the law of $3^{-d(m-L)} \mu_L^0(\square_{m-L})$. Partition $\square_{m-L}$ into its $3^{d(m-L)}$ unit cubes. This normalized mass is the average of their masses, each with the law of $\mu_L^0(\square_0)$. For every $p \geq 1$, convexity of $v \mapsto v^p$ and $v \mapsto v^{-p}$ on $(0, \infty)$ bounds both moments of the average by those of one unit mass. Thus the matching-scale positive and negative moment bounds also hold above the cutoff. Independence of the unit masses is not required.

We now pass to the triadic subcubes of the reference cube. Let P be such a subcube, of side 3^{-n} , and split the density of μ into the factors from fields of scale below and above the physical side of P . The product of the factors from fields of scale between the physical side of P and 3^j , and its reciprocal, have moments of order p bounded by $C_p 3^{C_p \bar{\delta}^2 n}$, and the oscillation of the fields above scale j has a uniformly bounded exponential moment. Combining the positive moments of $\mu_N^0(\square_0)$ with (9.13) therefore gives, for every $p < \infty$,

$$\mathbb{E} \left[\left(\frac{\mu(P)}{|P|} \right)^p + \left(\frac{\mu(P)}{|P|} \right)^{-p} \right] \leq C_p 3^{C_p \bar{\delta}^2 n}.$$

The reference cube contains $O(3^{dn})$ triadic subcubes of side 3^{-n} . Choose $p > q + 2d$, and then decrease $\bar{\delta}$ so that $p/2 > d + C_p \bar{\delta}^2$. Markov's inequality and a summation over n prove (9.10) for triadic cubes, with the asserted moments, and a finite collection of translated triadic grids gives the bounds for balls.

Step 2. We prove the $H^{3/4}$ coercivity bound (9.11). If $m \leq L$, it is the estimate of Lemma 8.3, applied with $N = m$ and with the value at z subtracted from each field of scale larger than 3^m ; the proof is uniform in the number of these fields, by (3.2).

For $m \geq L$, fix a moment order $p \geq \max\{q, 64d\}$, put $\varsigma_0 = 1/16$, $\varsigma_1 = \varsigma_0/2$ and $\xi = 2p$, and choose the disorder threshold afterwards. Write C_{resp} for the constant of Proposition 4.1 and c_1 for the admissibility constant of Lemma 4.5. Set

$$\delta_1 = \left(1 + C_{\text{resp}}\xi \log(2 + \xi) + \frac{\xi}{c_1\varsigma_1}\right)\bar{\delta}^2.$$

For small enough $\bar{\delta}$, we have $\bar{\delta}^2 \leq \delta_1 < 1$ and ξ is in the range of Proposition 4.1. That proposition bounds the coarse-grained defect at every scale by δ_1 , and hence supplies the induction hypothesis of Lemma 4.5 with $m_0 = k$ at each application on a physical cube of scale $k \geq L$. Its lower moment requirement holds because $\xi \geq 128d = 4d/\varsigma_1$, and its upper requirement holds because $\xi \leq c_1\varsigma_1\bar{\delta}^{-2}\delta_1$. Applying it with $(\varsigma_0, 1)$ and $(\varsigma_1, 2)$ gives, uniformly in $k \geq L$ and every deterministic center y ,

$$(9.14) \quad \|\mathcal{E}_{\varsigma_0, \infty, 1}(y + \square_k; \mathbf{a}_L, \bar{\mathbf{a}}_L)\|_{L^{2p}} + \|\mathcal{E}_{\varsigma_0/2, \infty, 2}(y + \square_k; \mathbf{a}_L, \bar{\mathbf{a}}_L)\|_{L^{2p}} \leq C_p \bar{\delta}.$$

Stationarity gives the same constants after translation. By (2.38), this bounds $\bar{\mathbf{a}}_L \lambda_{\varsigma_0, 1}^{-1}(y + \square_k; \mathbf{a}_L)$ in L^p uniformly in L, k, y . The squared coarse Poincaré inequality (2.43) therefore uses this random factor times the energy divided by $\bar{\mathbf{a}}_L$, which is precisely the normalization of A in (9.9). Sobolev embedding gives (9.11); the smaller physical cubes use the existing below-cutoff estimates. For $v \in H_0^1(V)$, use (2.47).

For triadic subcubes of side 3^{-n} in the local coordinates, the fields of scale between the physical side of the subcube and 3^j contribute scalar factors. Together with (3.24), they multiply the constant by a random factor with L^p norm at most $C_p 3^{C_p \bar{\delta}^2 n}$. For subcubes of physical side less than one, the ellipticity bounds on unit cubes give a fixed power of 3^n instead, with a random factor whose moments are bounded by Lemma 8.6. Taking the maximum over the $O(3^{dn})$ triadic cubes of side 3^{-n} and summing over n with a sufficiently large geometric weight gives one constant K for all these meshes. The same summation over the $O(2^n)$ cubes $V_{k2^{-n}}$ of the nested families gives $K2^{Bn}$ after enlarging B . The comparison (2.41) of translated cubes allows fixed enlargements and changes of grid.

Step 3. We construct the cutoff functions of (9.12). We use harmonic interpolants as in Lemma 8.22. Choose a smooth cutoff function f that equals one on a neighborhood of the smaller cube and vanishes on a neighborhood of the boundary of the larger cube, and a mesh of side $h \asymp a$ such that the cells meeting the transition region of f lie between the two prescribed cubes. Let χ be the function which, on each cell, is A -harmonic with boundary values f . The maximum principle gives $0 \leq \chi \leq 1$, and $\chi = f$ on cells where f is constant.

On a physical cell of side 3^k , $k \geq 0$, set $j_1 = k \wedge L$. Apply Proposition 6.17 at exponent $7/8$ and outer scale k . Above the cutoff, the whole-cell Dirichlet energy is bounded by (2.50) and (9.14), with $\varsigma_0 = 1/16$. The tail coefficient is $\bar{\mathbf{a}}_{j_1}$, since $j_1 \leq k$. The gradient estimate then gives energy growth with exponent $d - 1/4$ down to physical radius one; its random factor consists of the squared whole-cell energy constant and $3^{d\mathcal{L}}$, where $\mathcal{L}$ is the regularity stopping depth. To bound this factor at a requested order p , obtain the energy constant at order $4p$ and $3^{d\mathcal{L}}$ at order $2p$ before choosing $\bar{\delta}$.

Below physical radius one, use the small-oscillation estimate and the odd reflections of Steps 4–5 of Proposition 8.7. Join it to the preceding bound at a radius proportional to the inverse log-gradient bound. The resulting coefficient has the form of the preceding random factor plus $C3^{-k/4}$ times a polynomial of degree at most $d + 6$ in that factor, the two-sided local coefficient envelope and the log-gradient bound. For the requested order p , take those bounds at order $p(d + 6)$ with geometric factor $3^{\eta k}$, where $\eta = 1/(8(d + 6))$, and take the energy and stopping moments at the enlarged orders just described. The factor $3^{-k/4}$ absorbs $3^{\eta(d+6)k}$. Hölder's inequality thus gives a

cell constant bounded in L^p uniformly in $j_1 \leq k$, including $k > L$. This proves the cell estimate at all radii, including faces and corners. For $k \leq L$, subtract the values of the remaining fields at the cell center as in Proposition 8.7; their oscillation factors and reciprocals have the same uniform moments.

It remains to express the cell estimates after rescaling $z + \square_m$ to $\square_0$ as in (9.9). Let P be a cell of side $h = 3^{-n}$ with $n \leq m$ and physical center z_P , and put $k = m - n$ and $j_1 = k \wedge L$. In the coordinates of the cell, the coefficient A is the normalized coefficient of the cell multiplied by two factors: the exponential $\exp(\sum_{j < i \leq L} (\mathbf{g}_i(z_P) - \mathbf{g}_i(z)))$, which is controlled by the oscillation bound for these fields, and the scalar $(\bar{\mathbf{a}}_{j_1}/\bar{\mathbf{a}}_j)(\mathbf{a}_j(z_P)/\mathbf{a}_{j_1}(z_P))$. By (3.24) and (3.2), this scalar and its reciprocal have L^p norms at most $C_p 3^{C_p \bar{\delta}^2 n}$. After subtracting a constant from f , its rescaled C^2 norm on the cell is at most $Ch \|\nabla f\|_{C^1}$. The estimate on the cell therefore gives a random constant K_P with $\|K_P\|_{L^p} \leq C_p h^{-C_p \bar{\delta}^2}$ such that, for every $0 < r \leq h$,

$$\Gamma_\chi(B_r(x) \cap P) \leq K_P \|\nabla f\|_{C^1}^2 h^{1/2} r^{d-1/2}.$$

At each mesh size there are $O(h^{-d})$ cells. We take the maximum of the constants K_P over these cells and sum over the meshes with a sufficiently large geometric weight. This proves (9.12) when the cells have physical side at least one, or equivalently $h \geq 3^{-m}$.

We give the microscopic estimate when $h < 3^{-m}$. On a fixed reference cube V containing the cells, put

$$D_m = 3^{-m} \|\nabla \log A\|_{L^\infty(V)}, \quad R_m = (1 + D_m)^d \|A\|_{L^\infty(V)}.$$

The Dirichlet principle on each cell P gives $\int_P A |\nabla \chi|^2 \leq \|A\|_\infty \|\nabla f\|_\infty^2 h^d$. For $w = \chi - f \in H_0^1(P)$, its energy is at most four times this bound and $-\nabla \cdot (A \nabla w) = \nabla \cdot (A \nabla f)$. Choose $\ell = c_0 h / (1 + D_m)$ with a sufficiently small dimensional c_0 . Since $h \|\nabla \log A\|_\infty \leq D_m$, the oscillation of $\log A$ on $B_{2\ell}$ is at most $4c_0$. Near the boundary of P , reflect w and its source oddly and A evenly across every face met by the ball. The reflected weak equation has no additional face source. This includes edges and corners, with at most 2^d reflected copies.

After dividing the coefficient by its value at the center, choose c_0 so that the small-oscillation Meyers estimate (8.14) applies with exponent $4d$. The source term is bounded by $C \|\nabla f\|_{C^1}$ after multiplication by ℓ , where $\|\nabla f\|_{C^1} = \|\nabla f\|_\infty + \|D^2 f\|_\infty$. Hölder's inequality and the whole-cell energy bound give, for $r \leq \ell$,

$$\int_{B_r \cap P} A |\nabla w|^2 \leq C \|A\|_\infty \|\nabla f\|_{C^1}^2 (h^d \ell^{-d+1/2} + \ell^{1/2}) r^{d-1/2} \leq C R_m \|\nabla f\|_{C^1}^2 r^{d-1/2}.$$

The coefficient value used in the normalization cancels against the energy; no exponential of D_m occurs. Add the energy of f to obtain the same estimate for χ . For $\ell < r \leq h$, use the whole-cell energy and $h/r \leq C(1 + D_m)$. A ball of radius $r \leq h$ meets only a dimensional number of cells; for $r > h$, summing their energies gives $C \|A\|_\infty \|\nabla f\|_\infty^2 r^d$. Since $r \leq 1$, we have proved, for all centers and radii,

$$(9.15) \quad \Gamma_\chi(B_r(x)) \leq C R_m \|\nabla f\|_{C^1}^2 r^{d-1/2}.$$

The proof of Lemma 8.6 also applies to this local normalization. The physical log-gradient is a sum of fields at scales 3^i , weighted by 3^{-i} ; the maximum over the $O(3^{d(m-i)})$ unit cells at that scale gives $\|D_m\|_{L^p} \leq C_p \sqrt{1+m}$. For $j = m \wedge L$, the density is the product of the fields through j times the coarse oscillation factor. The deterministic factor $\bar{\mathbf{a}}_j^{-1} e^{-(j+1)\tau^2}$ is at most one. The maximum of the fine sum over the physical cube, estimated as in Step 2 of that lemma, gives

$$\|A\|_{L^\infty(V)} \|A\|_{L^p} \leq C_p \exp((C_d \bar{\delta} + C_p \bar{\delta}^2)m).$$

These bounds are uniform in L, m, z : when $m > L$, there are only $L + 1 \leq m + 1$ fields, and when $m \leq L$, the remaining oscillation factor has uniformly bounded exponential moments. For

the uncut coefficient take $j = m$ and use the same bound on the infinite coarse tail. Fix $q \geq 1$ and $B_0 > 0$ first, use Hölder's inequality at the required larger orders, and then decrease $\bar{\delta}$ so that $C_d \bar{\delta} + C_q \bar{\delta}^2 < \frac{1}{2} B_0 \log 3$. It follows that

$$\|R_m\|_{L^q} \leq C_q (1+m)^{d/2} e^{(C_d \bar{\delta} + C_q \bar{\delta}^2)m}, \quad \sup_{L,m,z} \mathbb{E}[(3^{-B_0 m} R_m)^q] < \infty.$$

The factor R_m is independent of the mesh. Taking $B_0 = 1$, we have $R_m \leq (3^{-m} R_m) h^{-1}$, and $\|\nabla f\|_{C^1} \leq C a^{-2}$ with $h \asymp a$. Thus (9.15) implies (9.12) on this branch with $B \geq 5$, independently of q . Finally, the estimates for $\mathbf{a}$ follow also by locally smooth convergence of the normalized coarse tails and Fatou's lemma, as in Proposition 8.7. $\square$

The regularity estimates of Section 6 concern solutions, and the coefficient A has no ellipticity bounds uniform in L , so we derive the local maximum principle for subsolutions from Lemma 9.6 alone.

Lemma 9.7 (Local bound for subsolutions). *After rescaling a cube $z + \square_m$ to $\square_0$ as in (9.9), let $V' \subseteq V$ be fixed cubes. The estimates of Lemma 9.6 imply that there is a random constant K such that, for every bounded nonnegative weak subsolution w of $\nabla \cdot (A \nabla w) \geq 0$ in V ,*

$$(9.16) \quad \operatorname{ess\,sup}_{V'} w \leq K \left(\int_V w^2 d\mu \right)^{1/2}.$$

For every fixed $q < \infty$, there exists $\bar{\delta}_0 > 0$ such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then the constant K has a finite moment of order q .

Since K does not depend on w , the lemma applies with the same constant to every bounded nonnegative solution of $\nabla \cdot (A \nabla w) = \lambda w d\mu/dx$, for every $\lambda \geq 0$. We use this uniformity in λ in Step 1 of the proof of Proposition 9.8.

Proof. By a finite covering it suffices to treat a fixed concentric pair $V' \subseteq V$. In both iterations of Step 4 we use the nested family $V_{k2^{-n}}$ of Lemma 9.6 associated with this pair. We apply Moser iteration with respect to μ , in which trace inequalities for measures satisfying the upper bound in (9.10) replace the Sobolev inequality. Step 1 proves the trace inequalities, Steps 2 and 3 use them to bound the $L^2(dx)$ norm and the integral against the energy measure of a cutoff function by the energy and the $L^2(\mu)$ norm, and Step 4 iterates the resulting reverse Hölder inequalities for w on nested cubes.

Step 1. We prove two trace inequalities. Let ν be a measure satisfying $\nu(B_r(x)) \leq K r^{d-1/2}$ for every ball $B_r(x)$ with $0 < r \leq 1$, and let $p_0 \geq 2$. We show that, if $s > 1/4$ and $t > d/2 - (d-1/2)/p_0$, then

$$(9.17) \quad \|v\|_{L^2(\nu)}^2 \leq CK \|v\|_{H^s}^2 \quad \text{and} \quad \|v\|_{L^{p_0}(\nu)}^2 \leq CK \|v\|_{H^t}^2.$$

For a Littlewood–Paley decomposition of v , each block v_i at frequency 2^i satisfies $\|v_i\|_{L^2(\nu)}^2 \leq CK 2^{i/2} \|v_i\|_2^2$. To see this, write v_i as its convolution with a rapidly decaying reproducing kernel, apply the Cauchy–Schwarz inequality, and integrate against ν using the bound on the mass of balls. Interpolation with Bernstein's inequality for that block gives

$$\|v_i\|_{L^{p_0}(\nu)} \leq CK^{1/p_0} 2^{i(\frac{d}{2} - \frac{d-1/2}{p_0})} \|v_i\|_2.$$

Summing over i with the Cauchy–Schwarz inequality, after extending v from a cube to the whole space, gives (9.17) on the fixed cubes. Below we take $s = 1/2$, $t = 3/4$ and $p_0 = 2 + 1/d$, for which $d/2 - (d-1/2)/p_0 = d/(2d+1) < 3/4$; Section 10 uses other exponents.

Step 2. We bound the $L^2(dx)$ norm of v by the energy of v and its $L^2(\mu)$ norm. Partition a cube into cubes P of side ρ , and let v_P be the average of v over P with respect to Lebesgue measure.

Apply the first trace inequality, with $3/4$ in place of $1/2$, to $v - v_P$. The fractional Poincaré inequality and the lower mass bound give $\int_P |v - v_P|^2 d\mu \leq CK\rho[v]_{H^{3/4}(P)}^2$ and

$$\|v\|_{L^2(dx)}^2 \leq CK\rho^{-1/2}\|v\|_{L^2(\mu)}^2 + CK^2\rho^{1/2}\|v\|_{H^{3/4}}^2.$$

We apply (9.11) to the last term and choose ρ small enough to absorb the resulting multiple of $\|v\|_{L^2(dx)}^2$ into the left side. For every $0 < \varepsilon \leq 1$ this yields

$$(9.18) \quad \|v\|_{L^2(dx)}^2 \leq \varepsilon \mathcal{E}(v) + CK^C \varepsilon^{-C} \|v\|_{L^2(\mu)}^2.$$

Step 3. We bound the integral of v^2 against the energy measure of a cutoff function by the energy of v and its $L^2(\mu)$ norm. Let χ be a cutoff function across a gap a . We apply the first inequality in (9.17) to $\nu = \Gamma_\chi$, which satisfies its hypothesis with Ka^{-B} in place of K by (9.12), interpolate $H^{1/2}$ between $H^{3/4}$ and $L^2(dx)$, and then use (9.11) and (9.18). For every $0 < \varepsilon \leq 1$ this gives

$$(9.19) \quad \int v^2 d\Gamma_\chi \leq \varepsilon \mathcal{E}(v; V) + CK^C \varepsilon^{-B} a^{-B} \int_V v^2 d\mu.$$

Step 4. We iterate. For $p \geq 2$, test the subsolution inequality with $\chi^2 w^{p-1}$ and put $v = w^{p/2}$. The Cauchy–Schwarz inequality and (9.19), with ε a small multiple of p^{-1} , give, for nested cubes $V_r \subseteq V_R$,

$$\mathcal{E}(v; V_r) \leq 2^{-B-2} \mathcal{E}(v; V_R) + CK^C p^B (R-r)^{-B} \int_{V_R} w^p d\mu.$$

Iterate this inequality over radii increasing from r to R with geometrically decreasing gaps. Hence

$$\mathcal{E}(w^{p/2}; V_r) \leq CK^C p^B (R-r)^{-B} \int_{V_R} w^p d\mu.$$

The second trace inequality, the $H^{3/4}$ coercivity bound (9.11) and (9.18) give, since $p_0/2 > 1$,

$$\|w\|_{L^{p_0 p/2}(V_r, \mu)} \leq (CK^C p^B (R-r)^{-B})^{1/p} \|w\|_{L^p(V_R, \mu)}.$$

We iterate with $p_i = 2(p_0/2)^i$ and geometrically decreasing gaps. The product of the resulting constants converges, which proves (9.16). For continuous functions such as $1 - u$ in the proof of Proposition 9.8, the full support of μ identifies the essential supremum with the pointwise supremum. $\square$

Proposition 9.8 (Tightness). *Let $K_{L,x}^\omega$ be the quenched law of either rescaled diffusion defined above, started at x . Equip $C([0, \infty), \mathbb{R}^d)$ with the topology of locally uniform convergence. Almost surely these processes are conservative and, for every compact set $B \subseteq \mathbb{R}^d$, the following statements hold.*

(i) Tightness in expectation: *for every $\varepsilon > 0$, there is a compact set $\mathcal{K} \subseteq C([0, \infty), \mathbb{R}^d)$ such that*

$$(9.20) \quad \sup_L \mathbb{E} \left[\sup_{x \in B} K_{L,x}^\omega(\mathcal{K}^c) \right] \leq \varepsilon.$$

In particular, the annealed laws $\mathbb{E}[K_{L,x}^\omega]$ with $L \in \mathbb{N}_0$ and $x \in B$ are tight.

(ii) Tightness in probability: *for every $\eta > 0$, there is a compact set $\mathcal{K}_\eta$ of probability measures on $C([0, \infty), \mathbb{R}^d)$ such that*

$$(9.21) \quad \sup_L \mathbb{P}[K_{L,x}^\omega \notin \mathcal{K}_\eta \text{ for some } x \in B] \leq \eta.$$

Proof. The proof proceeds in four steps. Step 1 bounds the probability that the process, started in the concentric cube of one ninth the side, leaves the cube before a short time t , measured after the rescaling of space and time in (9.9), by a random constant times $\sqrt{t}$, with moments of the constant bounded uniformly in the cutoff and in the size and position of the cube; this exit estimate is the heart of the matter. Step 2 applies it to large cubes and gives compact containment and conservativeness, Step 3 applies it at stopping times to bound the modulus of continuity and proves (i), and Step 4 deduces (ii) from (i).

Step 1. We show that, after rescaling every cube $z + \square_m$ to $\square_0$ as in (9.9), the process started in $\square_{-2}$ leaves $\square_0$ before time t with probability at most $K\sqrt{t}$. Let χ be a cutoff function of Lemma 9.6 between $\square_{-1}$ and $\square_0$, so that χ takes values in $[0, 1]$, equals one on $\square_{-1}$, vanishes near $\partial\square_0$, and satisfies $\mathcal{E}(\chi) \leq K$. For $\lambda > 0$, let u solve

$$\lambda u - \mathcal{A}u = \lambda \chi \quad \text{in } \square_0, \quad u = 0 \quad \text{on } \partial\square_0.$$

The maximum principle gives $0 \leq u \leq 1$. Since u minimizes $\mathcal{E}(v) + \lambda \|v - \chi\|_{L^2(\mu)}^2$ over $v \in H_0^1(\square_0)$, comparison with $v = \chi$ gives $\int_{\square_0} |u - \chi|^2 d\mu \leq \mathcal{E}(\chi)/\lambda \leq K/\lambda$. On $\square_{-1}$, the nonnegative function $w = 1 - u$ satisfies $\nabla \cdot (A \nabla w) = \lambda w d\mu/dx \geq 0$, and, since $\chi = 1$ on $\square_{-1}$, this estimate bounds its norm in $L^2(\square_{-1}, \mu)$. Therefore Lemma 9.7, whose constant does not depend on λ , gives $\sup_{\square_{-2}} (1 - u) \leq K\lambda^{-1/2}$. Let $\widehat{X}$ be the process obtained by rescaling $z + \square_m$ to $\square_0$, let P_x^ω and E_x^ω be its law and expectation started at x , and let $\tau_{\square_0}$ be its exit time from $\square_0$. The stopped Itô formula gives, for every $x \in \square_0$,

$$u(x) = E_x^\omega \int_0^{\tau_{\square_0}} \lambda e^{-\lambda s} \chi(\widehat{X}_s) ds \leq 1 - E_x^\omega e^{-\lambda \tau_{\square_0}}.$$

Taking $\lambda = 1/t$, we obtain

$$(9.22) \quad \sup_{x \in \square_{-2}} P_x^\omega(\tau_{\square_0} \leq t) \leq e \sup_{x \in \square_{-2}} E_x^\omega \exp\left(-\frac{\tau_{\square_0}}{t}\right) \leq K\sqrt{t}.$$

The constant K does not depend on t , and $\mathbb{E}K \leq C$. In physical coordinates, (9.22) says that the probability that $X^{(L)}$, started at any point of $z + \square_{m-2}$, leaves $z + \square_m$ before time t is at most

$$(9.23) \quad K\sqrt{t\bar{\mathbf{a}}_{m \wedge L}} 3^{-m},$$

and the same bound holds for X with m in place of $m \wedge L$. These bounds concern the processes killed on leaving a bounded cube and do not presume that the processes are conservative.

Step 2. We prove compact containment and conservativeness. For triadic $R \geq 1$, a cube of side R in rescaled coordinates has time unit $R^2 3^{2L}/\bar{\mathbf{a}}_L$ at finite cutoff, and, by (3.24), at least $R^2 3^{2L}/\bar{\mathbf{a}}_L$ for the diffusion X without cutoff. After the cube sizes are adjusted by fixed factors, (9.22) therefore gives, for every $T > 0$ and every $R \geq 1$ larger than a fixed multiple of $\max_{x \in B} |x|$,

$$(9.24) \quad \sup_L \mathbb{E} \left[\sup_{x \in B} K_{L,x}^\omega \left(\sup_{s \leq T} |Z_s| > R \right) \right] \leq C \frac{\sqrt{T}}{R}.$$

Letting $R \rightarrow \infty$ shows that, for almost every environment, the process does not explode from any starting point in B . The same estimate then gives compact containment for the laws on continuous path space.

Step 3. We bound the modulus of continuity and prove (9.20). Fix $R < \infty$ and $r > 0$, fix $k \in \mathbb{N}_0$ such that cubes of side 3^{-k} have diameter at most $r/4$, and cover the ball B_R by finitely many cubes $y + \square_{-k-2}$. For $L \geq k$, the time unit of each of the cubes $y + \square_{-k}$, measured in units of $3^{2L}/\bar{\mathbf{a}}_L$, satisfies, by (3.24), $3^{-2k} \bar{\mathbf{a}}_L / \bar{\mathbf{a}}_{L-k} \geq 3^{-(2 + \frac{2\tau^2}{\log 3})k} > 0$. We apply (9.22) to each cube of this finite cover and take the maximum of the finitely many random constants. The strong Markov

property then gives a constant C , depending on R and r , such that, for every stopping time $S \leq T$ and every $h > 0$,

$$\sup_{L \geq k} \mathbb{P} \left(\sup_{0 \leq s \leq h} |Z_{S+s}^{(L)} - Z_S^{(L)}| > r/4, S < \tau_{B_R} \right) \leq C\sqrt{h}.$$

To obtain (9.20), in which the supremum over starting points lies inside the expectation, we first restrict to the event on which the maximum of the random constants of this finite cover is at most M . Its complement has probability at most C/M , uniformly in $L \geq k$. On this event, for every starting point in B_R , the probability that the path moves by more than $r/4$ within time h is at most $CM\sqrt{h}$, and, by the strong Markov property, the same bound holds after every stopping time before exit from B_R , conditionally on the path up to that stopping time.

Consider the successive exit times of the path from balls of radius $r/4$, the first centered at the starting point and each later one at the position of the path at the preceding exit time. Choose h_0 so small that this probability bound, with $h = h_0$, is at most $1/16$. If $n > 2T/h_0$ such exits occur before time T , then at least $n/2$ of the gaps between consecutive exit times are shorter than h_0 . The strong Markov property in the fixed environment and a union bound over the choices of these gaps bound the probability of this event by $2^n 16^{-n/2}$. Hence the number of exits before time T is tight, uniformly in the starting point.

For a fixed number of exits, the probability that some gap is shorter than h tends to zero as $h \rightarrow 0$. Since the path remains in a ball of radius $r/4$ between consecutive exit times, this gives a uniform bound on the probability that the path oscillates by more than r on an interval of length h . We choose first M , then the number of exits, then h .

For each of the finitely many $L < k$, the coefficient is locally $C^{1,1}$. Nonexplosion and continuous dependence on the starting point show that, almost surely, the laws of these diffusions started in B are uniformly tight, and dominated convergence gives the corresponding bound for the expected supremum. We apply the preceding construction along a sequence of integer time horizons and of spatial tolerances tending to zero, with summable error probabilities. Compact containment and the Arzelà–Ascoli theorem then prove (9.20).

Step 4. We prove (9.21). By (i), for every $j \geq 1$ there is a compact set $\mathcal{K}_j$ of paths for which the left side of (9.20) is at most $\eta 2^{-2j}$. Set

$$\mathcal{K}_\eta := \{P : P(\mathcal{K}_j^c) \leq 2^{-j} \text{ for every } j \geq 1\}.$$

The set $\mathcal{K}_\eta$ is closed and uniformly tight, hence compact. Markov's inequality and a union bound over j prove (9.21). $\square$

9.4. Convergence in probability of the path laws. We use the killed resolvents to identify the limiting laws on $C([0, \infty), \mathbb{R}^d)$. The ingredients are compact containment, the Markov identity at finite cutoff, and the fact that discounted integrals of finite-dimensional observations determine a continuous path law.

We record the short-time approximation for the killed resolvents. For $f \in C_0(Q)$, extend f by zero. Splitting the occupation integral at h gives

$$(9.25) \quad \sup_N \mathbb{E} \|\lambda R_{N,\lambda}^Q f - f\|_\infty \leq \sup_{|x-y| \leq r} |f(x) - f(y)| + 2\|f\|_\infty \left(e^{-\lambda h} + \sup_N \mathbb{E} \sup_{x \in \overline{Q}} K_{N,x}^\omega \left(\sup_{s \leq h} |Z_s - x| > r \right) \right).$$

The last probability tends to zero as $h \downarrow 0$ by (9.20). If a path is killed before h without moving farther than r , its starting point is within r of the boundary, where f is small. Proposition 9.5 passes (9.25) to the limiting resolvents. In particular, $\mathbb{E} \|\lambda R_\lambda^Q f - f\|_\infty \rightarrow 0$ as $\lambda \rightarrow \infty$.

Proposition 9.9 (Convergence in probability). *Let $B \subseteq \mathbb{R}^d$ be compact and $\varepsilon > 0$. Then*

$$\lim_{N, N' \rightarrow \infty} \mathbb{P} \left[\sup_{x \in B} d_{\text{weak}}(K_{N,x}^\omega, K_{N',x}^\omega) > \varepsilon \right] = 0.$$

Consequently, the kernels $x \mapsto K_{N,x}^\omega$ converge in probability, locally uniformly in x , to a continuous random probability kernel $x \mapsto K_x^\omega$ that is measurable with respect to the fields.

Proof. Use the Prokhorov metric d_P associated with the complete metric on path space

$$\sum_{m \geq 1} 2^{-m} (1 \wedge \sup_{t \leq m} |z(t) - z'(t)|).$$

Write $R_{N,\lambda} f(x) = \iint_0^\infty e^{-\lambda t} f(Z_t) dt dK_{N,x}^\omega$ for the whole-space resolvent. If Q contains B , splitting at the exit time gives

$$(9.26) \quad \sup_{x \in B} |R_{N,\lambda} f(x) - R_{N,\lambda}^Q f(x)| \leq \frac{\|f\|_\infty}{\lambda} \left(e^{-\lambda T} + \sup_{x \in B} K_{N,x}^\omega(\tau_Q \leq T) \right).$$

Choose first T large and then Q large. Compact containment (9.24) makes the expectation of the right side arbitrarily small, uniformly in N . Proposition 9.5 therefore makes the whole-space resolvents Cauchy in probability, locally uniformly in x . Contraction and localization give the same conclusion for measurable random continuous sources g_N converging in probability locally uniformly to g , provided $\|g_N\|_\infty$ and $\|g\|_\infty$ are bounded by one deterministic constant. The intermediate sources in the iterated resolvents below have this bound by contraction.

For bounded continuous $f_1, \dots, f_k$ and positive integers $n_1, \dots, n_k$, consider the bounded continuous path functional

$$\int_{(0,\infty)^k} e^{-\sum_{i=1}^k n_i s_i} \prod_{i=1}^k f_i(z(s_1 + \dots + s_i)) ds.$$

The Markov property and Fubini identify its expectation with $R_{N,n_1}(f_1 R_{N,n_2}(f_2 \cdots R_{N,n_k} f_k))(x)$. Iterating the preceding resolvent convergence makes these expectations Cauchy in probability, uniformly for $x \in B$. These functionals determine continuous path laws. After the substitution $u_i = e^{-s_i}$, varying the positive integers n_i gives the integrals of the finite-dimensional observations against every monomial on $[0,1]^k$. Polynomial approximation and continuity in time then determine these observations. A separating algebra of spatial test functions determines the finite-dimensional laws.

Tightness (9.20) now transfers this Cauchy property to d_P . Indeed, on a compact set of paths carrying all but a small mass, finitely many continuous tests suffice to distinguish laws at any prescribed positive d_P distance; the determining functionals approximate these tests there. Markov's inequality controls the mass outside that set, uniformly over $x \in B$. A finite union bound proves the asserted Cauchy property. The finite-cutoff kernels are continuous in x , by their Feller property and local path tightness. Completeness of the Prokhorov metric and a subsequence with summable error probabilities on a compact exhaustion of $\mathbb{R}^d$ give a measurable continuous limiting kernel.

Finally, on the compact set of laws $\mathcal{K}_\eta$ in (9.21), every metric inducing the weak topology is uniformly equivalent to d_P . With probability at least $1 - 2\eta$, both finite-cutoff kernels take values there for every $x \in B$. Letting $\eta \downarrow 0$ transfers the assertion to d_{weak} . $\square$

9.4.1. The Markov property of the limit.

Proposition 9.10 (Properties of the limit). *Almost surely, the limiting kernel $x \mapsto K_x^\omega$ has the following properties.*

- (i) *The map $x \mapsto K_x^\omega$ is continuous in the weak topology.*
- (ii) *The laws $(K_x^\omega)_x$ are the laws of a continuous, conservative, strong Markov process Z .*

(iii) The process is symmetric with respect to μ : for every $t \geq 0$ and all compactly supported continuous functions f and g ,

$$\int g(x) \int f(Z_t) dK_x^\omega \mu(dx) = \int f(x) \int g(Z_t) dK_x^\omega \mu(dx).$$

Proof. Continuity of the kernel and infinite lifetime hold by Proposition 9.9: the limiting laws are probability measures on $C([0, \infty), \mathbb{R}^d)$. The finite-cutoff Markov identity passes to the limit for bounded continuous path functionals, using locally uniform convergence of the kernels and compact containment. A monotone-class argument and approximation of stopping times from above by dyadic times, which uses the continuity of the paths and of $x \mapsto K_x^\omega$, then give the strong Markov property.

At finite cutoff the same identity holds for $K_{N,x}^\omega$ and μ_N , by the symmetry of the generator in $L^2(\mu_N)$. For compactly supported continuous f, g , locally uniform convergence of the kernels and local weak convergence $\mu_N \rightarrow \mu$ (Proposition 9.2) pass this identity to the limit. $\square$

Lemma 9.11 (Killed resolvents of the limiting process). *Almost surely, for every cube $Q \in \mathcal{C}$ and every $\lambda > 0$, the resolvent of the process of Proposition 9.10 killed on leaving Q is R_λ^Q .*

Proof. By Proposition 9.9 and (9.7), we choose a subsequence along which the whole-space kernels converge almost surely, locally uniformly in the starting point, and the killed resolvents converge uniformly on $\overline{Q}$, for a countable dense family of data and positive parameters. Fix an environment in this event. For nonnegative data, the discounted occupation functional stopped at τ_Q is lower semicontinuous on continuous path space, so the portmanteau theorem gives one inequality between its expectation under K_x^ω and $R_\lambda^Q f(x)$.

For the opposite inequality, first take $f = 1$. Fix $\varepsilon > 0$ and a smaller closed concentric cube $F \Subset Q$ so that, for all sufficiently large N , $\lambda R_{N,\lambda}^Q 1 \leq \varepsilon$ on $Q \setminus F$. Uniform convergence and the continuous zero-boundary representative of $R_\lambda^Q 1$ allow this choice. Let $\sigma_{h,F}$ be the first time on the grid $h\mathbb{N}_0$ at which the path is outside F , with $e^{-\lambda\sigma} = 0$. The finite-cutoff restart identity gives

$$(1 - \varepsilon) E_{N,x}^\omega e^{-\lambda\sigma_{h,F}} \leq E_{N,x}^\omega e^{-\lambda\tau_Q} = 1 - \lambda R_{N,\lambda}^Q 1(x).$$

The grid functional on the left is bounded and lower semicontinuous, since F is closed. Portmanteau passes the inequality to the limiting law. Continuity of paths gives $e^{-\lambda\tau_Q} \leq \liminf_{h \downarrow 0} e^{-\lambda\sigma_{h,F}}$; Fatou's lemma and then $\varepsilon \downarrow 0$ give $R_\lambda^Q 1(x) \leq E_x^\omega \int_0^{\tau_Q} e^{-\lambda t} dt$. Together with the first inequality this proves equality for $f = 1$. Applying the first inequality to $\|f\|_\infty + f$ and $\|f\|_\infty - f$ gives equality for every bounded continuous f . Contraction, continuity in λ , and a countable intersection give the identity for all $\lambda > 0$ and all cubes on one event of probability one.

To allow $f \in C(\overline{Q})$, we multiply it by bounded compactly supported cutoff functions increasing to one in Q . The resulting data converge in $L^2(\mu)$, and the variational identity and the trace bound give the convergence of their resolvents in energy. The common $C^{1/4}$ bound then gives uniform convergence, and dominated convergence for the path integral weighted by $e^{-\lambda t}$ proves the identity for f , in particular for $f = 1$. $\square$

The identification of the killed resolvents also gives the finite mean exit times asserted in Theorem 7.1.

Corollary 9.12 (Finite mean exit times). *Almost surely, $\sup_{x \in U} \int \tau_U dK_x^\omega < \infty$ for every bounded open set U .*

Proof. Choose $Q \in \mathcal{C}$ containing $\overline{U}$. Lemma 9.11 and (9.8), with $f = 1$, give

$$\sup_{x \in Q} \int_0^{\tau_Q} e^{-\lambda t} dt dK_x^\omega = \|R_\lambda^Q 1\|_\infty \leq K_Q \quad (\lambda > 0).$$

Monotone convergence as $\lambda \downarrow 0$ and $\tau_U \leq \tau_Q$ prove the assertion. $\square$

9.5. Almost-sure convergence by comparison of two cutoffs. We compare the finite-volume energies at two cutoffs. On a small cube, rescaling to the unit cube writes the ratio of the two energies as a deterministic ratio of normalizations times a ratio of two unit-cube energies. To combine these local comparisons, we join harmonic extensions on disjoint small cubes, using the finite-cutoff counterpart (Lemma 9.17) of the bound of Proposition 8.39 on the number of bad cells along chains of nested cubes.

Our control of the ratios of the two normalizations follows the argument of Ding and Gwynne [2023a, Section 3.4]. There, two random metrics are compared by joining paths around annuli at a positive proportion of scales into a single path. Upper and lower bounds on crossing distances then exclude ratios of the two normalizations that are too large or too small. Devlin [2024, Lemma 3.11] strengthened this argument by using convergence in probability, which rules out ratios bounded away from one on a positive proportion of scales. We apply the argument of Devlin [2024, Lemma 3.11] to the minimal energy with a fixed nonconstant affine boundary value. If the ratios were at most a fixed number less than one on a positive proportion of scales, then joining harmonic extensions on cubes of those scales would bound this energy at a finite cutoff by a fixed factor less than one times its limit, which is positive. This would contradict the convergence in probability. Interchanging the two cutoffs excludes, in the same way, ratios at least a fixed number greater than one. On the remaining scales the ratios are close to one, and Proposition 9.20 turns the comparison on these scales into bounds with summable error probabilities.

We first record how a minimal energy with given boundary values on a small cube rescales. For an integer $k \geq 0$ and $z \in \mathbb{R}^d$, let $\Theta_{k,z}$ replace the field $\mathbf{g}_i$ by $\mathbf{g}_{i-k}(z + 3^{-k} \cdot)$, for every $i \in \mathbb{Z}$. This translates and rescales the fields; their increments are taken only when forming the infrared field H . We write $\Lambda_{m, \square_0}^{\Theta_{k,z}\omega}$ for the minimal energy on $\square_0$ with given boundary values at cutoff m in this environment.

Lemma 9.13 (Rescaling of minimal energies on small cubes). *For every integer $k \geq 0$, the limit*

$$e(k) = \lim_{N \rightarrow \infty} \frac{\bar{\mathbf{a}}_{N-k}}{\bar{\mathbf{a}}_N}$$

exists and is positive. Let $z \in \mathbb{R}^d$, let $q = z + 3^{-k}\square_0$, and, for $N \geq k$, let

$$s_N(q) = \frac{\bar{\mathbf{a}}_{N-k}}{\bar{\mathbf{a}}_N} \exp\left(H(z) + \sum_{j=0}^{k-1} (\mathbf{g}_{-j}(z) - \tau^2)\right)$$

be the value at the center of q of the factor in A_N coming from the normalization and the fields whose scales exceed the side length of q . Then, for every $N \geq k$ and $g \in H^1(q)$,

$$\Lambda_{N,q}(g) = 3^{-(d-2)k} s_N(q) \Lambda_{N-k, \square_0}^{\Theta_{k,z}\omega}(g(z + 3^{-k} \cdot)).$$

For each fixed z , the same identity holds almost surely for the minimal energies of the limit form, with $\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N$ in $s_N(q)$ replaced by $e(k)$, for every boundary trace on which these are defined.

Proof. Reindexing the fields gives, for $r = 3^{-k}$ and every $N \geq k$,

$$A_N(z + ry) = s_N(q) A_{N-k}^{\Theta_{k,z}\omega}(y).$$

The increments of H and of the fields $\mathbf{g}_{-j}$ with $j < k$ form the infrared field in the new coordinates; the fields with $j \geq k$ retain their values. The energy identity follows by changing variables $x = z + ry$. Stationarity, scaling and independence show that $\Theta_{k,z}$ preserves the joint field law. The same transformed environment serves every cutoff. In particular, dividing the two energies at cutoffs N and M by their respective scalars $s_N(q)$ and $s_M(q)$ cancels those scalars exactly and gives the two unit-cube energies at cutoffs $N - k$ and $M - k$ in this one environment.

To prove that the normalizations converge, fix a nonconstant affine datum and choose z with $q \in \mathcal{C}$, which does not affect $e(k)$. Proposition 8.18 and Theorem 8.54 give convergence in probability, along the full sequence of cutoffs, of the minimal energies on the cubes of $\mathcal{C}$ for the data in $\mathcal{D}$, and the seminorm inequality for $\Lambda^{1/2}$ together with (8.6) extends this to the affine datum. Hence both minimal energies in the identity converge in probability; for the one on $\square_0$ in the environment $\Theta_{k,z}\omega$ this holds because $\Theta_{k,z}$ preserves the law of the fields. Their limits are strictly positive: by Lemma 8.3, each of these minimal energies is at least a positive constant times K_N^{-1} , and $\sup_N \|K_N\|_{L^p} < \infty$. By (3.24), for fixed k the deterministic ratios $\bar{a}_{N-k}/\bar{a}_N$ lie in $[1, e^{2\tau^2 k}]$, a compact subset of $(0, \infty)$. Two subsequential limits of these ratios must coincide, since otherwise the same positive minimal energy of the limit form would equal one random variable multiplied by two different constants. Finally, we pass to the limit in the identity for the countable family of smooth traces of Section 8.4, and the bound (8.6) for Hölder boundary values of Lemma 8.4 gives the identity for every trace on which the minimal energy of the limit form is defined. $\square$

We compare the two cutoffs on a cell q through the trace on ∂q of a single function, which is Hölder continuous but otherwise arbitrary. The next lemma reduces this comparison to finitely many smooth traces.

Lemma 9.14 (Reduction to finitely many smooth boundary values). *Let $1/2 < \beta < \alpha < 1$, $A < \infty$ and $\eta > 0$. There exists a finite family of smooth functions on $\partial\square_0$ with the following property. Let Λ_1 and Λ_2 be minimal energies on $\square_0$ with given boundary values such that $\Lambda_i(g) \leq A\|g\|_{C^\beta/\mathbb{R}}^2$ for $i \in \{1, 2\}$ and every $g \in C^\beta(\partial\square_0)$, and such that $|\Lambda_1(h) - \Lambda_2(h)| \leq \eta/2$ for every function h of the family. Then, for every $g \in C^\alpha(\partial\square_0)$,*

$$|\Lambda_1(g) - \Lambda_2(g)| \leq \eta\|g\|_{C^\alpha/\mathbb{R}}^2.$$

Proof. Since $\Lambda_i^{1/2}$ is a seminorm, the triangle inequality gives, for all $g, h \in C^\beta(\partial\square_0)$,

$$|\Lambda_i(g) - \Lambda_i(h)| \leq A\|g - h\|_{C^\beta/\mathbb{R}}(\|g\|_{C^\beta/\mathbb{R}} + \|h\|_{C^\beta/\mathbb{R}}).$$

The unit ball of $C^\alpha/\mathbb{R}$ is precompact in $C^\beta/\mathbb{R}$, so for every tolerance it has a finite net consisting of smooth functions. Take the family to be a sufficiently fine such net, and apply the last display to Λ_1 and to Λ_2 . This gives the assertion for g in the unit ball, and homogeneity gives it for every g . $\square$

The number of functions in the family and their smooth norms may depend on η ; the exponents and the moment orders do not. The constant A is the one in Lemma 9.15(iii) below, and it is fixed before the family is chosen.

On a good cell we also need, at a finite cutoff, the local Hölder estimate and the bound (8.6) for Hölder boundary values that Lemma 8.40 provides for the limit form. The scalar $s_N(q)$ of Lemma 9.13 is the finite-cutoff analogue of the scalar $s_E(q)$ of Section 8.6.

Lemma 9.15 (Good cells at finite cutoff). *Fix $1/2 < \beta < \alpha < 1$, a subdivision factor L , a distance factor $b_0 \geq 3$ and a rate $B > 0$ for the exponential bounds in (i). The thresholds for the sums over intervals of scales, the finite moment orders and $\bar{\delta}_0 > 0$ can be chosen in this order so that the following holds for $\bar{\delta} \leq \bar{\delta}_0$. There are deterministic constants $C, A < \infty$, independent of the cutoff, the cell, the source and the tolerance η of Lemma 9.14, and good events $G_N(q)$ with the following properties. Here $N \geq k \geq 0$, $q = z + r\square_0$ with $r = 3^{-k}$, and, after q is rescaled to $\square_0$, the coefficient has cutoff $m = N - k$. The conclusions hold on the same good events both for A_N and for $A_N^0 = e^{-H}A_N$, with their respective scalars $s_N(q)$ of Lemma 9.13 and $s_N^0(q) = e^{-H(z)}s_N(q)$. Below, $s(q)$ denotes the scalar of the coefficient under consideration, and $\Gamma(u)$ denotes the energy measure of u for that coefficient.*

- (i) Up to a null set, the complement of $G_N(q)$ is covered by events indexed by integers $h \geq 1$, the event of index h being measurable with respect to $\sigma(\mathbf{g}_{-j} : k - h \leq j \leq k + 2h)$ and having probability at most e^{-Bh} .
- (ii) Suppose $G_N(q)$ holds and q lies in its parent p of side Lr with $\text{dist}(q, \partial p) \geq b_0 r$. For every $f \in L^\infty(p)$ and every weak solution $u \in H^1(p)$ of $-\nabla \cdot (A \nabla u) = f$ on p , there is a constant c_q such that

$$(9.27) \quad \|u(z + r \cdot) - c_q\|_{C^\alpha(\square_0)} \leq C r^{(2-d)/2} s(q)^{-1/2} \Gamma(u)(p)^{1/2} + C r^2 s(q)^{-1} \|f\|_\infty.$$

- (iii) On $G_N(q)$, for every $g \in C^\beta(\partial \square_0)$, the normalized minimal energy satisfies

$$r^{2-d} s(q)^{-1} \Lambda_{N,q}(g((\cdot - z)/r)) \leq A \|g\|_{C^\beta/\mathbb{R}}^2.$$

Proof. The proof proceeds in four steps. Step 1 defines $G_N(q)$ and proves (i) for it. Step 2 bounds the normalized coefficient and its gradient on q and its fixed enlargement. Step 3 proves (ii), by the iteration of Lemma 6.11 above the smallest field scale and an interior Sobolev estimate below it. Step 4 adds condition (a), proves (iii) and intersects the good events of the two coefficients. The main point is Step 3: the growth of the coefficient bounds of Step 2 below the smallest field scale must be absorbed by the gain $\alpha_1 - \alpha$ of the Hölder exponent, which fixes how small σ_* and τ^2 must be.

Step 1. We define the event $G_N(q)$. It imposes the conditions of $G_N^{\text{det}}(q)$ on sums over intervals of scales, down to the smallest field scale, with an exponent $\alpha_1 \in (\alpha, 1)$ in place of α in the iteration of Lemma 6.11. We choose the integrability exponent used in Step 3 so that the Sobolev embedding gives a Hölder exponent larger than α_1 , then a small test exponent $\sigma_* \in (0, 1/32]$, and add the root and enlargement tests

$$F_{m+\iota, 3^N z} \leq \epsilon, \quad P_{m+\iota, 3^N z} \leq 12 \quad (\iota = 0, 1), \quad m = N - k.$$

These are the quantities of Section 8.5, evaluated in the relabelled fields $\mathbf{g}_i^{(N)}(y) = \mathbf{g}_{i-N}(3^{-N}y)$ with $s = \sigma_*$. The index m belongs to q , and $m + 1$ to its fixed concentric enlargement $3q$; their stationary centers are both $3^N z$. In the original bilateral indexing their field windows are centered at $-k$ and $1 - k$, respectively.

For these extra tests use the continuous ramps of Section 8.5, with threshold pairs $(\epsilon/2, \epsilon)$ and $(6, 12)$; they are two of the three summands of $Z_{m+\iota, 3^N z}$. Their zero-disorder values are zero. Lemma 8.36 approximates them by quantities depending on windows about the same root indices, with exponentially decreasing errors, uniformly in N and k . Its proof applies verbatim with $s = \sigma_*$, because the exponent $s \in (0, 1/32]$ of Section 8.5 may be any number fixed before $\bar{\delta}$. The proof also gives an L^p bound of order $C_p \bar{\delta}^c$ for these ramps. The telescoping argument of Lemma 8.38, applied to each ramp at this one cutoff, therefore covers its failure by events with any prescribed exponential rate after the moment orders are fixed and $\bar{\delta}$ is reduced. Enlarge and reindex the windows to absorb their fixed widths and the offset 1, and increase the rate before taking the finite union. All these events then have the windows and the probability bound in (i), centered at the physical root depth k , uniformly in m . For the conditions on sums over intervals of $D \leq m$ scales, we apply, at the single fixed cutoff N , the telescoping argument from the proof of Lemma 8.38, which uses dyadically increasing intervals of field indices. Lemmas 8.37 and 8.36 are uniform at that cutoff, so the union over the at most Ce^{CD} spatial choices has the same bound. Choose the moment order used in the approximation so large that the resulting exponential rate exceeds both B and the exponential growth rate of the number of spatial choices, and then decrease $\bar{\delta}$. This gives (i) directly at finite cutoff.

Step 2. We bound the coefficient on the cell and its fixed enlargement. Write $y_0 = 3^N z$ for the stationary center. The term $j = m$ in the root product test gives

$$\sup_{y \in y_0 + \square_{m+1}} \sum_{i=0}^m |\mathbf{g}_i^{(N)}(y)| \leq \log 12 + \frac{\sigma_* m}{8} \log 3.$$

Indeed, the first product in that term is bounded by $12 \cdot 3^{\sigma_* m/8}$ on the larger cube $y_0 + \square_{2m+1}$, and includes all indices $0, \dots, m$. The term $j = 0$ in the same test bounds

$$\sup_{y \in y_0 + \square_{m+1}} \sum_{i=m+1}^{\infty} |\mathbf{g}_i^{(N)}(y) - \mathbf{g}_i^{(N)}(y_0)| \leq \frac{1}{4} \log 12.$$

For A_N^0 this sum stops at $i = N$, so the same bound applies. The identity of Lemma 9.13, with the respective scalar $s(q)$, expresses the logarithm of the normalized coefficient as the sum of the fine fields $i = 0, \dots, m$ and these coarse-field increments, minus $\log \kappa_m$. Here $\kappa_m = e^{(m+1)\tau^2} \bar{\mathbf{a}}_m$. Since $e^{-2\tau^2} \leq \bar{\mathbf{a}}_0 \leq 1$ by (2.14) and Jensen's inequality, (3.24) gives $e^{-2\tau^2(m+1)} \leq \bar{\mathbf{a}}_m \leq 1$, hence $|\log \kappa_m| \leq \tau^2(m+1)$. Thus its logarithm is bounded by $C + C(\sigma_* + \tau^2)m$, for both coefficients.

The root field test controls the gradients separately. Its term $j = m$ bounds the sum of $3^i |\nabla \mathbf{g}_i^{(N)}|$ for $0 \leq i \leq m$ by $3^{\sigma_* m/8}$. For $i = m + j > m$, its term of depth j gives $3^i \|\nabla \mathbf{g}_i^{(N)}\|_{\infty} \leq 3^{\sigma_* j/8}$ on $y_0 + \square_{m+1}$. Hence

$$\sum_{i=m+1}^{\infty} \|\nabla \mathbf{g}_i^{(N)}\|_{L^{\infty}(y_0 + \square_{m+1})} \leq 3^{-m} \sum_{j \geq 1} 3^{-(1-\sigma_*/8)j}.$$

This series converges, also bounding every partial infrared sum. In unit coordinates x , the stationary coordinate is $y_0 + 3^m x$, and the smallest field scale is 3^{-m} . Denoting the coefficient in these unit coordinates divided by $s(q)$ by $\tilde{A}$, the value and gradient bounds give

$$\|\tilde{A}^{-1}\|_{\infty} + \|\tilde{A}\|_{\infty} + 3^{-m} \|\nabla \log \tilde{A}\|_{\infty} \leq C 3^{C(\sigma_* + \tau^2)m}.$$

The norms here are on $3\square_0$. The same estimate holds with H omitted or replaced by any of its partial sums, since all omitted terms were bounded in absolute value.

Step 3. We prove (ii). Above the scale 3^{-m} , the iteration of Lemma 6.11 on the event controlling sums over intervals of scales gives (9.27) with exponent α_1 . Below 3^{-m} , we use the interior Sobolev estimate, with this exponent, of Step 4 of the proof of Proposition 8.7 on balls of radius $a3^{-m}/[C(1 + 3^{-m}\|\nabla \log \tilde{A}\|_{\infty})]$, on which the oscillation of $\log \tilde{A}$ is small. After matching with the estimate at scale 3^{-m} , the energy term carries the factor $3^{-m(\alpha_1 - \alpha)}$ and the source term the factor $3^{-m(2-\alpha)}$, each multiplied only by fixed powers of the quantities in the last display. We choose σ_* , and then τ^2 , so that $C(\sigma_* + \tau^2) < \alpha_1 - \alpha$, where C depends only on d and the integrability exponent. The net exponents of 3^{-m} then remain positive, and (9.27) holds at all radii with a deterministic constant.

Step 4. We prove (iii) and obtain one good event for both coefficients. First intersect the event of Step 1 with condition (a) on q and its fixed enlargements, using the finite-cutoff ellipticity bounds of $G_N^{\det}(q)$ at the trace exponent of Lemma 8.4. These normalized ellipticity conditions have the centered moment and band bounds of Lemma 8.36, also with zero infrared potential. Their failure is covered at the fixed cutoff by the finite-test telescoping argument of Lemma 8.38. Add these covers to those of Step 1, using a larger rate before the finite union. Thus (i) covers condition (a) as well. On that condition, Lemma 8.4 gives (iii), with deterministic A independent of the trace-net tolerance. Finally, we apply the construction separately to the coefficient with and without its factor e^H , and take the intersection of the two good events. When H is replaced by one of its partial sums, the same bounds hold uniformly in the number of terms, by the summable bounds on the increments of the fields of scale larger than one. $\square$

The comparison of Lemma 9.14 on a cell holds when the normalized minimal energies at the two cutoffs are close on the finite family of smooth boundary values of that lemma. By the convergence in probability, this comparison fails only with small probability, but the counting argument of Section 8.5 requires more: the failure must be covered by events of exponentially small probability, each measurable with respect to the fields with indices in a short interval.

Lemma 9.16 (Rare events and field indices). *For each $m \in \mathbb{N}$, let I_m be a countable index set and $X_{m,i}$, $i \in I_m$, random variables with values in $[0, 1]$. Assume that they tend to zero in probability uniformly in i , that is, $\sup_{i \in I_m} \mathbb{P}(X_{m,i} > \varepsilon) \rightarrow 0$ as $m \rightarrow \infty$, for every $\varepsilon > 0$. For each (m, i) let $n_{m,i}$ be its deterministic field index, and let $\mathcal{B}_{m,i,H}$ be the σ -field generated by the fields with indices in the interval of radius H about $n_{m,i}$. Suppose that, for some $p \in [1, \infty)$, $a > 0$ and $C_* \in (0, \infty)$, uniformly in m, i, H ,*

$$\|X_{m,i} - \mathbb{E}[X_{m,i} \mid \mathcal{B}_{m,i,H}]\|_{L^p} \leq C_* 3^{-aH}.$$

There is an absolute constant C such that, for every B with $a \log 3 > CB$, there are common integers H_0, m_0 with the following property. For every $m \geq m_0$ and $i \in I_m$, up to a null set, $\{X_{m,i} > 1/2\}$ is contained in a union of events indexed by $h \geq H_0$. The event with index h is measurable with respect to fields in an interval of radius at most Ch about $n_{m,i}$ and has probability at most e^{-Bh} . The initial radius H_0 depends only on a, p, C_ and B , not on m or i .*

Proof. Fix (m, i) and suppress i from the notation in this proof. Let H_0 be large, to be chosen uniformly in m, i , and put $X_m^{(\ell)} = \mathbb{E}[X_{m,i} \mid \mathcal{B}_{m,i,2^\ell H_0}]$ for $\ell \geq 0$. Since the L^p errors of these approximations are summable in ℓ , we have $X_m^{(\ell)} \rightarrow X_m$ almost surely. The allowances $2^{-\ell-3}$, $\ell \geq 1$, sum to $1/8$. Hence, outside the null set where this convergence fails, if $X_m > 1/2$, then either $X_m^{(0)} > 1/4$ or the increment $X_m^{(\ell)} - X_m^{(\ell-1)}$ exceeds $2^{-\ell-3}$ for some $\ell \geq 1$. By Markov's inequality, the probability of the first event is at most $\mathbb{P}(X_m > 1/8) + (8C_*)^p 3^{-apH_0}$. We first choose H_0 so large that the second term is smaller than e^{-CBH_0} with the required margin, and then use the uniform convergence in probability to make the first term smaller than e^{-CBH_0} for all $m \geq m_0$. By Markov's inequality, the probability that the ℓ th increment exceeds its allowance is at most $(CC_*)^p 2^{p\ell} 3^{-ap2^{\ell-1}H_0}$, and increasing H_0 absorbs the factor $2^{p\ell}$ uniformly in ℓ . Each of these events is measurable with respect to the fields indexed by the larger of the two intervals that determine the two conditional expectations in its definition. We index them by $h = 2^\ell H_0$ and enlarge their intervals to the form used in Lemma 8.38; the assumed margin in $a \log 3 > CB$ absorbs this fixed enlargement. $\square$

We verify the conditioning and convergence inputs for the finite trace tests. Let $h_1, \dots, h_J$ be the smooth family of Lemma 9.14. Let ψ_η vanish on $[0, \eta/4]$, equal one on $[\eta/2, \infty)$ and be linear in between. For $q = z + r\Box_0$, $r = 3^{-k}$, and $M \geq N \geq k$, put

$$X_{N,M}(q) = \psi_\eta \left(\max_{i \leq J} \left| r^{2-d} s_N(q)^{-1} \Lambda_{N,q}(h_i((\cdot - z)/r)) - r^{2-d} s_M(q)^{-1} \Lambda_{M,q}(h_i((\cdot - z)/r)) \right| \right).$$

By Lemma 9.13, these are the responses at the two unit-cube cutoffs $N - k$ and $M - k$, in the same environment $\Theta_{k,z}\omega$. No moment of either cell scalar enters this test.

Write $\mathcal{F}_\ell = \sigma(\mathbf{g}_j : |j| \leq \ell)$ and $\mathcal{R}_{m,i} = \Lambda_{m,\Box_0}(h_i)$. Extend h_i smoothly to $\overline{\Box_0}$. Proposition 8.7 supplies its Morrey bound, with a factor $\|h_i\|_{C^2}^2$, and (8.6) and Lemma 8.2 supply its uniform response moments. Thus Lemma 8.15 bounds the change caused by resampling $\mathbf{g}_{-j}$ by $C_{p,i} \bar{\delta} 3^{-bj/8}$ in L^p for $0 \leq j \leq m$, where b is the exponent of Lemma 8.14; for $j > m$ that field is absent. For $j \geq 1$, resampling $\mathbf{g}_j$ changes the infrared potential by its increment about the origin, whose supremum has sub-Gaussian size $C \bar{\delta} 3^{-j}$. The coefficient comparison (8.24) and Hölder's inequality at the $3p$

response order bound this change by $C_{p,i}\bar{\delta}3^{-j}$. Lemma 8.16 therefore gives, for every $m, \ell \geq 0$,

$$\|\mathcal{R}_{m,i} - \mathbb{E}[\mathcal{R}_{m,i} \mid \mathcal{F}_\ell]\|_{L^p} \leq C_{p,i}\bar{\delta} \left(\sum_{j=\ell+1}^m 3^{-bj/8} + \sum_{j>\ell} 3^{-j} \right) \leq C_{p,i}\bar{\delta} 3^{-b\ell/8}.$$

The first sum is empty when $\ell \geq m$. This is the proof of the Dirichlet branch of Proposition 8.17, extended to windows beyond the cutoff; its stated restriction $m \geq \ell$ is unnecessary here.

The preimage of $\mathcal{F}_\ell$ under $\Theta_{k,z}$ is $\mathcal{B}_{k,\ell} = \sigma(\mathbf{g}_{-j} : |j-k| \leq \ell)$. Conditional expectation commutes with this law-preserving change of environment. Apply ψ_η to the maximum of the differences of the two vectors of conditional expectations on this common window, obtaining a $\mathcal{B}_{k,\ell}$ -measurable variable Y . The Lipschitz constant $4/\eta$ of ψ_η , the finite-maximum inequality in L^p , and contraction of conditional expectation give

$$\|X_{N,M}(q) - \mathbb{E}[X_{N,M}(q) \mid \mathcal{B}_{k,\ell}]\|_{L^p} \leq \frac{16\bar{\delta}}{\eta} \left(\sum_{i=1}^J C_{p,i}^p \right)^{1/p} 3^{-b\ell/8}.$$

Enlarging the window multiplies this bound by at most two, by another use of contraction. The constants are independent of k, z, N, M . Choose $p \geq 2$ so that $(b/8)p \log 3 > CB$ for the counting rate B , choose the finite higher orders needed for the Morrey, response and product bounds, and then decrease $\bar{\delta}$. These orders and this disorder threshold are independent of η and of the smooth net. Only the norm factors $C_{p,i}$, its size J , and hence C_* and the initial window depend on η .

Proposition 8.18 and Theorem 8.54 give full-sequence convergence of the unit-cube energies on the original field space. Their uniform higher moments give L^p convergence. The seminorm inequality for $\Lambda_{m,\square_0}^{1/2}$ and (8.6) extend it from the dense trace family $\mathcal{D}$ to each h_i . Consequently, for every $\varepsilon > 0$,

$$\sup_{m' \geq m} \mathbb{P} \left[\psi_\eta \left(\max_{i \leq J} |\mathcal{R}_{m,i} - \mathcal{R}_{m',i}| \right) > \varepsilon \right] \longrightarrow 0 \quad (m \rightarrow \infty),$$

since $\psi_\eta(x) > \varepsilon$ forces $x > \eta/4$. The joint rescaling makes this uniform over cells with $N-k \rightarrow \infty$. Apply Lemma 9.16 with $m = N-k$ and with the family indexed by the cells and $M \geq N$. Its uniform conditioning estimate and uniform convergence in probability supply common initial windows and eventual remaining cutoffs for all these tests. It is applied here to the finite-cutoff A_N energies. The comparison for A_N^0 is transferred in Lemma 9.19, on the cells at which the subdivision stops.

For an initial cube Q of the triadic subdivision, write its side length as 3^{-k_Q} and define the eligible generations by $I_Q^{a,b}(N) = \{n \in \mathbb{N}_0 : aN \leq k_Q + H_1 n \leq bN\}$. Thus the interval $[aN, bN]$ always refers to physical triadic depth. Good and bad cells are counted by their generation indices in this set.

Lemma 9.17 (Simultaneous good generations for two cutoffs). *Fix $0 < a < b < 1$, $\theta \in (0, 1)$, $L = 3^{H_1}$ and a bounded initial cube Q of the subdivision. Choose the finite moment orders and the threshold $\bar{\delta}_0$ as in Lemmas 9.15 and 9.16, with the rate B of Lemma 9.15(i) large enough for the counting estimate in the proof. For every $\eta > 0$ there are $N_\eta, C_\eta < \infty$ and $c_\eta > 0$ such that, for each pair $M \geq N \geq N_\eta$, outside an event of probability at most $C_\eta 3^{-c_\eta N}$, every chain in the L -adic subdivision of Q has at most $\theta N/H_1$ bad generations with index in $I_Q^{a,b}(N)$.*

Here a generation $n \in I_Q^{a,b}(N)$ of a chain is good if, on its cell of physical depth $k_Q + H_1 n$, Lemma 9.15 holds at both cutoffs and the two separately normalized minimal energies for A_N satisfy the conclusion of Lemma 9.14 for every C^α trace; otherwise it is bad. The comparison for A_N^0 is obtained by the common-weight transfer in Lemma 9.19.

Proof. The preceding application of Lemma 9.16 and the covers of Lemma 9.15(i) at the two cutoffs cover each bad generation by events of the form in Lemma 9.15(i), with a prescribed rate B , after

increasing that rate by a fixed factor. For a cell of physical depth k , the extra tests use stationary indices $N-k$ and $N-k+1$ at cutoff N , and $M-k$ and $M-k+1$ at cutoff M . Both pairs correspond to the same original bilateral indices $-k$ and $1-k$, so their covers have the same field windows at that cell. Their field windows are centered at $-(k_Q + H_1 n)$, for $n \in I_Q^{a,b}(N)$, and these centers are spaced by H_1 . As in the proof of Proposition 8.39, apply Lemma B.5 on this interval of field indices, with the events at unselected indices set to the empty event. For sufficiently large N , the eligible set has $(b-a)N/H_1 + O(1)$ elements and its last generation is $(bN - k_Q)/H_1 + O(1)$. Thus the number of chains ending there is at most $C_Q L^{dbN/H_1}$, where the fixed root offset is absorbed in C_Q . Choose B so that the exponential rate exceeds this growth and take a union bound. If $\theta \geq b-a$, use the estimate with a smaller positive value of θ ; it implies the stated bound. $\square$

At physical triadic depth $k = k_Q + H_1 n$ the minimal energies at the cutoffs N and M carry the normalizations $\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N$ and $\bar{\mathbf{a}}_{M-k}/\bar{\mathbf{a}}_M$ of Lemma 9.13, while the random factor in the scalars $s_N(q)$ and $s_M(q)$ is common to both. The ratio $r_{N,M}(k) = (\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N)/(\bar{\mathbf{a}}_{M-k}/\bar{\mathbf{a}}_M)$ of these two normalizations is therefore, up to the tolerance η of Lemma 9.14, the factor relating the two minimal energies on a good cell of depth k . The next lemma adapts the stopping argument of Proposition 8.41 to two finite cutoffs.

For comparison on an arbitrary cube Q of side 3^{-k_Q} , first partition Q into triadic cubes R of common depth $k_R = H_1 \lfloor k_Q/H_1 \rfloor$. This uses fewer than H_1 preliminary triadic subdivisions and gives finitely many roots, independently of N and M . Subdivide each R L -adically. All these roots have the same eligible set $I_R^{1/4,3/4}(N)$, and generation n in any of them has physical depth $k_R + H_1 n$. For large N , writing $j = k_R/H_1 + n$ identifies this set exactly with $\{j \in \mathbb{N}_0 : N \leq 4H_1 j \leq 3N\}$. The same preliminary partition and the same eligible set are used in both comparisons below. At these depths $r_{N,M}(k_R + H_1 n) = r_{N,M}(H_1 j)$. For each fixed cube and all sufficiently large N , its eligible generations therefore correspond to the same set of absolute indices j with $N \leq 4H_1 j \leq 3N$.

Lemma 9.18 (Comparison of two cutoffs by stopping). *Let $\vartheta > 0$. The geometry, the exponents and the finitely many moment orders can be chosen so that there exists $\bar{\delta}_0 > 0$ such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then the following holds with constants D , L and C independent of η , c , N and M . Let Q be a fixed cube of triadic side length and g a fixed smooth function on ∂Q . For every $\eta > 0$ there exist $C_\eta < \infty$ and $\gamma_\eta > 0$ with the following property. Let $N \leq M$ be cutoffs and $c \leq 2$. Use the preliminary roots R just described, and assume that a deterministic set $S \subseteq I_R^{1/4,3/4}(N)$ satisfies $\#S \geq \vartheta \#I_R^{1/4,3/4}(N)$ and $r_{N,M}(k_R + H_1 n) \leq c$ for every $n \in S$. Then, outside an event of probability at most $C_\eta 3^{-\gamma_\eta N}$,*

$$\Lambda_{N,Q}(g) \leq c(1 + C_\eta L^D) \Lambda_{M,Q}(g) + C_\eta 3^{-\gamma_\eta N} \|g\|_{C^2}^2.$$

Under the same assumptions with $r_{N,M}^{-1}$ in place of $r_{N,M}$, the reverse inequality holds, with $\Lambda_{N,Q}$ and $\Lambda_{M,Q}$ interchanged.

Proof. Step 1. We define where the subdivision stops and estimate the energy on the cells at which it stops. Let u be the E_M -harmonic extension of g to Q , fix a small $\zeta > 0$, and let $\nu = \Gamma_M(u) + 3^{-\zeta N} dx$, so that $\nu(q) \geq 3^{-\zeta N} |q|$ for every cell q . Use the preliminary partition into roots R and their L -adic subdivisions. Apply Lemma 9.17 with $a = 1/4$, $b = 3/4$ to each of these finitely many roots, and take the union of their exceptional events. Choose the fraction of bad generations below. Put $J = \lfloor N/H_1 \rfloor$ and $I = I_R^{1/4,3/4}(N)$. For large N its first and last generations are

$$n_- = \left\lceil \frac{N/4 - k_R}{H_1} \right\rceil, \quad n_+ = \left\lfloor \frac{3N/4 - k_R}{H_1} \right\rfloor.$$

We follow each root subdivision from generation n_- through generation n_+ . Stop at a child q with parent p if its generation $n(q)$ belongs to S , the cell q is good and at distance at least b_0 times

its side from ∂p , and $\nu(p) \leq L^D \nu(q)$. At such a cell q , put $k = k_R + H_1 n(q)$. The local Hölder estimate (9.27) and the comparison of Lemma 9.14 give

$$\Lambda_{N,q}(u|_{\partial q}) \leq r_{N,M}(k) \Lambda_{M,q}(u|_{\partial q}) + C \eta r_{N,M}(k) \Gamma_M(u)(p).$$

Since $\Lambda_{M,q}(u|_{\partial q})$ is bounded by $\Gamma_M(u)(q)$, the stopping rule and the disjointness of the cells at which the subdivision stops show that the sum of $\Lambda_{N,q}(u|_{\partial q})$ over these cells is at most

$$cE_M(u) + C \eta c L^D (E_M(u) + 3^{-\zeta N} |Q|).$$

This bounds the energy of the competitor of Step 4 on these cells.

Step 2. We count the cells at generation n_+ on chains along which the subdivision never stops. The moment estimates of Section 8.2 are uniform in both cutoffs, so, outside an event of probability at most $C3^{-p\epsilon N}$, each of the finitely many global constants used below is at most $3^{\epsilon N}$. Consider a chain of nested cells along which the subdivision does not stop. At each of its steps at which the generation belongs to S and the child is good and at distance at least b_0 times its side from the boundary of its parent, ν drops by a factor larger than L^D . Since $\nu(Q)$ is controlled by the global energy and $\nu(q) \geq 3^{-\zeta N} |q|$ for every cell q at generation n_+ , the number of such steps is at most

$$\frac{1}{D \log L} \left(\log(3^{\zeta N} \nu(Q)) + \frac{3}{4} d N \log 3 + O_Q(1) \right).$$

We choose D large in terms of d and ϑ , and θ, ϵ and ζ small. Since $\#I = J/2 + O(1)$, each chain has at least $\vartheta J/2 - O(1)$ steps with generation in S . Discarding the steps counted above and the bad steps with generation in S leaves at least $\vartheta J/3 - O(1)$ steps with generation in S at which the child is within distance b_0 times its side of the boundary of its parent. Each cell has at most $C_d L^{d-1}$ children within that distance of its boundary, so, for L large, the number of cells at generation n_+ on chains where the subdivision has not stopped is at most $C_Q L^{(\frac{3d}{4} - \frac{\vartheta}{4})J}$. Thus these remaining cells form an exponentially small fraction of the cells at generation n_+ .

Step 3. We bound the energies on the remaining cells at generation n_+ . Their physical depth is $k_R + H_1 n_+ = 3N/4 + O(H_1)$, so their side length is comparable to $3^{-3N/4}$. On such a cell of side r , the bound (8.6) for Hölder boundary values, together with the bounds (8.7) on grid cubes, bounds the energy by $K_N [u]_\alpha^2 r^{d-2+2\alpha-\eta_0}$. We choose the fixed exponents so that $2 - 2\alpha + \eta_0 < \vartheta/8$. The sum of the energies on these cells is then at most $C3^{3\epsilon N} L^{-\vartheta \frac{J}{8}}$, which decays exponentially in N if ϵ is small.

Step 4. We compare the minimal energies with boundary values g . The cells at which the subdivision stops and the remaining cells at generation n_+ form a finite partition of Q . Joining the E_N -harmonic extensions of the traces of u on the cells of this partition gives a competitor with boundary values g . The Dirichlet principle then gives the asserted upper bound for $\Lambda_{N,Q}(g)$, and all excluded events have exponentially small probability. The reverse inequality follows from the same proof with the cutoffs interchanged. The finitely many smaller cutoffs are covered by taking their exceptional event to be the whole probability space and enlarging C_η . $\square$

The same construction compares the energies of solutions with a volume source.

Lemma 9.19 (Comparison for volume sources). *Assume the hypotheses of Lemma 9.18. There is an event of probability at least $1 - C_\eta 3^{-\gamma_\eta N}$ on which the following holds. Let u_M solve $-\nabla \cdot A_M \nabla u_M = f$ on Q , either with a smooth Dirichlet datum g , in which case we put $B = \|f\|_\infty + \|g\|_{C^2}$, or with zero conormal derivative and mean zero, in which case we assume that $\int_Q f = 0$ and put $B = \|f\|_\infty$. Then there is a trial function v_N , with the same Dirichlet datum in the first case and with mean zero in the second, such that*

$$\|v_N - u_M\|_\infty \leq C_\eta 3^{-\gamma_\eta N} B$$

and

$$E_N^Q(v_N) \leq c(1 + C\eta L^D)E_M^Q(u_M) + C_\eta 3^{-\gamma_\eta N} B^2.$$

The same holds with A_N and A_M replaced by $A_N^0 = e^{-H}A_N$ and $A_M^0 = e^{-H}A_M$.

Proof. We use the partition of the proof of Lemma 9.18, now with $\nu = \Gamma_M(u_M) + 3^{-\zeta N} B^2 dx$ in place of the measure of that proof. On a good cube q of side r whose parent p satisfies $\text{dist}(q, \partial p) \geq b_0 r$, the source term in (9.27) of Lemma 9.15(ii) adds to the local comparison the term $C\eta r_{N,M}(k) s_M(q)^{-1} r^{d+2} \|f\|_\infty^2$. On the event of the preceding proof, the bounds (8.7) on grid cubes give $s_M(q)^{-1} \leq 3^{\epsilon N} r^{-\eta_0}$. Since the side of every cube used in the comparison is at most $C3^{-N/4}$, this source term is bounded by $C\eta c 3^{-\zeta N} B^2 |q|$ if $\epsilon + \zeta < (2 - \eta_0)/4$, so it is absorbed in the lower bound $3^{-\zeta N} B^2 |q|$ for $\nu(q)$. The global estimates of Proposition 8.7 (for cubes of side larger than one, Remark 8.8) and Proposition 8.12 bound the energy by $3^{\epsilon N} B^2$ and the C^α norm by $3^{\epsilon N} B$, after these constants are added to the finite list of constants required to be at most $3^{\epsilon N}$.

On each element of the partition we take the A_N -harmonic extension of the trace of u_M . By the maximum principle, its distance from u_M is at most twice the oscillation of u_M on that element. All elements have side at most $C3^{-N/4}$, so this distance is at most $C3^{-(\frac{\alpha}{4}-\epsilon)N} B$. In the Neumann case we first subtract the mean. This proves both inequalities.

For A_N^0 and A_M^0 , the Dirichlet estimates with source on larger cells are supplied by the whole-cell argument leading to (8.35). For the mean-zero Neumann solutions, Proposition 8.12 applies to A_N^0 and A_M^0 on every fixed cube. Its chains use roots of radius at most R_* , so their outer scales do not exceed the cutoff, and the fixed-cube coercivity of Lemma 8.9 bounds the global energy at the same finitely many moment orders. The local trace comparison transfers separately. On each cube $q = z + r\Box_0$ used in the partition, $r \leq C3^{-N/4}$. Put $\varepsilon_q = \text{osc}_q H$ and $s_j^0(q) = e^{-H(z)} s_j(q)$ for $j = N, M$. The variational formula gives, for every trace g ,

$$e^{-\varepsilon_q} s_j(q)^{-1} \Lambda_{j,q}(g) \leq s_j^0(q)^{-1} \Lambda_{j,q}^0(g) \leq e^{\varepsilon_q} s_j(q)^{-1} \Lambda_{j,q}(g).$$

Both cutoffs use this same weight, and $s_N^0(q)/s_M^0(q) = r_{N,M}(k)$. Add $e^{\|H\|_{C^1(Q)}}$, whose fixed moments are finite, to the constants required to be at most $3^{\epsilon N}$. For large N , $\varepsilon_q \leq C3^{-N/5}$ on this event. The uniform good-cell trace bound then transfers the finite trace test with an additional tolerance $C3^{-N/5}$; choose the original tolerance smaller by a fixed factor. Equivalently, a local energy comparison is multiplied by at most $e^{2\varepsilon_q} = 1 + O(3^{-N/5})$. After summing over the cells at which the subdivision stops, its additional error is at most $C3^{-N/5} E_M^0(u_M)$. It is absorbed by the multiplicative tolerance after increasing the deterministic initial cutoff. The additive source and terminal-cell errors retain a positive exponential rate. This proves the two-cutoff comparison for A_N^0 without a convergence assertion uniform over partial infrared sums. $\square$

Proposition 9.20 (Almost-sure convergence of minimal energies). *Let Q be a cube of triadic side length and g a smooth function on ∂Q . Then $\Lambda_{N,Q}(g) \rightarrow \Lambda_{E,Q}(g)$ almost surely as $N \rightarrow \infty$ along the full sequence of integers.*

Proof. Step 1. We show that the normalizations are close to their limits on most of the eligible generations of the preliminary roots R . Put $I = I_R^{1/4, 3/4}(N)$ and $r_N(k) = (\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N)/e(k)$, and fix $0 < \varepsilon < 1$ and a density $0 < \vartheta < 1/3$. Suppose that, for infinitely many N , the set $S = \{n \in I : r_N(k_R + H_1 n) \leq 1 - \varepsilon\}$ has $\#S \geq \vartheta \#I$. For each such fixed N , the set S is finite. The convergence $\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N \rightarrow e(k) > 0$ at each of its physical depths $k = k_R + H_1 n$ therefore gives one cutoff beyond which $r_{N,M}(k_R + H_1 n) \leq 1 - \varepsilon/2$ for every $n \in S$. Apply Lemma 9.18 on $\Box_0$, using the corresponding set of indices $\{k_R/H_1 + n : n \in S\}$ and a fixed nonconstant affine datum, with η so small that $C\eta L^D < \varepsilon/4$, and let $M \rightarrow \infty$ first. Along these N , the minimal energy at cutoff N is bounded by a fixed factor strictly smaller than one times its limit, up to an error that tends to zero in probability. This contradicts its convergence in probability to a strictly positive

limit. The reverse comparison, using $r_{N,M}^{-1}$ and the finite set $\{n \in I : r_N(k_R + H_1 n) \geq 1 + \varepsilon\}$, excludes in the same way a fraction ϑ of large ratios. Hence, for all sufficiently large N , the set $S = \{n \in I : 1 - \varepsilon \leq r_N(k_R + H_1 n) \leq 1 + \varepsilon\}$ has $\#S > (1 - 2\vartheta)\#I$.

Step 2. We prove an exponential bound and conclude. For each large N , we apply the two comparisons of Lemma 9.18, with $S \subseteq I_R^{1/4, 3/4}(N)$ the set of generation indices found in Step 1, and let $M \rightarrow \infty$. Taking η small relative to ε gives, with constants depending on ε and on the datum,

$$\mathbb{P}[|\Lambda_{N,Q}(g) - \Lambda_{E,Q}(g)| > C\varepsilon\Lambda_{E,Q}(g) + C_\varepsilon 3^{-\gamma_\varepsilon N}] \leq C_\varepsilon 3^{-\gamma_\varepsilon N}.$$

The right side is summable in N , so the Borel–Cantelli lemma for $\varepsilon = 1/j$, $j \in \mathbb{N}$, gives the almost-sure convergence. $\square$

The same argument applies to the quantities (ii)–(iv) of the next proposition.

Proposition 9.21 (Summable error probabilities). *Let Q be a cube of triadic side length, and let Z_N be one of the following quantities on Q , computed for the coefficient A_N or for A_N^0 :*

- (i) *the minimal energy $\Lambda_{N,Q}(g)$ with a fixed smooth boundary value g ;*
- (ii) *the quadratic form $\langle f, G_N^Q f \rangle$ of (8.3) for a fixed bounded source f ;*
- (iii) *the quadratic form $\sup_{v \in H^1(Q)} \{2 \int_Q f v - E_N^Q(v)\}$ for a fixed bounded source f with $\int_Q f = 0$;*
- (iv) *the Neumann energy $|Q|p \cdot a_*^{-1}(Q; A_N)p$ of Section 8.1, for the constant flux given by a fixed vector p .*

There exists $\bar{\delta}_0 > 0$ such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then Z_N converges in probability to a limit Z , and, for every $\varepsilon > 0$, there exist $C_\varepsilon < \infty$, $c_\varepsilon > 0$ and a deterministic N_ε such that, for every $N \geq N_\varepsilon$,

$$(9.28) \quad \mathbb{P}[|Z_N - Z| > C_\varepsilon Z + C_\varepsilon 3^{-c_\varepsilon N}] \leq C_\varepsilon 3^{-c_\varepsilon N}.$$

The finitely many moment orders and the threshold $\bar{\delta}_0$ do not depend on ε . For the coefficients A_N^0 , the constants are uniform over deterministic translations of Q . In particular, for every countable family of such quantities, all of them converge almost surely along the full sequence of cutoffs.

Proof. For the minimal energies of A_N , the bound (9.28) was proved in Proposition 9.20. For A_N^0 , Proposition 8.27 and (8.36), with the weight e^{-H} and the same padded limit form, give convergence in probability. The common-weight transfer in Lemma 9.19 supplies the two-cutoff comparison. Fix before disorder all moment orders needed for the rare tests, the global estimates and the products below; their number and values are independent of the tolerance.

For a volume source f , let u_M be the solution and write $Z_M = E_M(u_M) = \int_Q f u_M$. If a is the factor multiplying $E_M(u_M)$ in Lemma 9.19, insert $a^{-1}v_N$ in the variational formula for Z_N . For $1/2 \leq a \leq 3$, the trial-energy and sup-distance bounds give

$$Z_M \leq aZ_N + C_\varepsilon 3^{-\gamma N} \|f\|_\infty^2.$$

Interchanging the cutoffs gives the reverse comparison. Step 1 of Proposition 9.20 supplies a set of eligible generations, of density larger than $1/3$, on which the normalization ratio is close to one. For every fixed tolerance and all large N , the resulting comparisons therefore hold with $a \leq 1 + C\varepsilon$, uniformly in all sufficiently large M . The exceptional events are independent of the bounded source, and $\gamma > 0$ depends on the tolerance.

We first record the moment bound used in case (iii). The mean-zero coercivity constant $K_{Q,N}$ in the proof of Lemma 8.9 gives

$$0 \leq Z_N \leq K_{Q,N} \|f\|_{L^2(Q)}^2.$$

Only upper moments of this constant are needed. For clarity, its factor on a fixed larger triadic cube $Q = z + R\Box_0$, $R = 3^s > 1$, can be computed exactly. Set $\tilde{\mathbf{g}}_i(y) = \mathbf{g}_{i+s}(z + Ry)$; these fields

have the original joint law. If $\widetilde{A}_{N+s}$ is their anchored coefficient, expansion of the logarithms gives

$$A_N(z + Ry) = c_N(Q) \widetilde{A}_{N+s}(y), \quad c_N(Q) = \frac{\bar{\mathbf{a}}_{N+s}}{\bar{\mathbf{a}}_N} \exp \left(H(z) - \sum_{j=1}^s (\mathbf{g}_j(z) - \tau^2) \right).$$

The energy scales by $R^{d-2}c_N(Q)$ and the squared L^2 norm by R^d . Thus the physical L^2 coercivity constant is $R^2c_N(Q)^{-1}\widetilde{K}_{N+s}$. For the inhomogeneous $H^{3/4}$ norm it is at most

$$\max\{R^2, R^{1/2}\} c_N(Q)^{-1} \widetilde{K}_{N+s}.$$

By (3.25) and Lemma 8.1, both $c_N(Q)$ and its inverse have every fixed finite moment uniformly in N , with constants depending on Q and its anchor. Take $\widetilde{K}_{N+s}$ at twice the requested order before choosing disorder, and apply Hölder's inequality; no independence is needed. For A_N^0 , the bound $E_N^{0,Q} \geq e^{-\sup_Q H} E_N^Q$ introduces one more finite-moment factor. For sides at most one use Lemma 8.3; any finitely many cutoffs below the cell depth are also bounded by $\|A_N^{-1}\|_{L^\infty(Q)}$ and the ordinary cube Poincaré inequality.

Now compare two sufficiently large cutoffs with the same fixed N . These upper moments make the multiplicative error small in probability as $\varepsilon \downarrow 0$, and the two exceptional probabilities vanish as $N \rightarrow \infty$. This proves the Cauchy property in probability in case (iii), without a Neumann limit form. In case (ii), the limit is the quadratic form of the unique Dirichlet limit form. Letting $M \rightarrow \infty$ with N fixed gives (9.28) in both cases.

For case (iv), take the mean-zero source $f_{Q,h}$ constructed in the proof of Lemma 8.9. On $Q = w + R(0,1)^d$ it contains the physical factor R^{-1} and satisfies $\|f_{Q,h}\|_\infty \leq C_{Q,p}h^{-1}$. The fixed-cube load comparison there gives an L^q approximation error at most $C_{q,Q,p}h^{1/4}$ and uniformly bounded L^q moments of the smoothed responses, at the selected orders. In the preceding triadic chart, the physical flux response is $R^d c_N(Q)^{-1}$ times the unit-cube response; the factor R^{-1} in the source gives the same scaling for its smoothed response. That proof also supplies the coercivity constant $K_{Q,N}$ for both infrared conventions on the same fixed cube, with a uniform polynomial tail

$$\mathbb{P}[K_{Q,N} > T] \leq B_Q T^{-p_0}, \quad p_0 > 0.$$

For the required larger triadic cubes, its construction uses the law-preserving shift to cutoff $N+k$, the finitely many retained coarse layers and the two compact infrared restrictions. The unit-cube first-moment bound already suffices for $p_0 = 1/2$. This transfer fixes its disorder threshold before the tolerance.

Let $Y_{N,h}$ be the response to the smoothed volume source $f_{Q,h}$. The load comparison gives

$$|Z_N - Y_{N,h}| \leq C_{Q,p}h^{1/4}K_{Q,N}^{1/2}Z_N^{1/2} + C_{Q,p}h^{1/2}K_{Q,N}.$$

For fixed h , case (iii) and the uniform moment approximation prove that $(Z_N)_N$ is Cauchy in probability. For the rate, use the two-cutoff comparison directly, with $h_N = 3^{-\gamma N/4}$ and $T_N = 3^{\gamma N/32}$. Outside the source-independent comparison events and the two events $\{K_{Q,N} > T_N\}, \{K_{Q,M} > T_N\}$, absorb the mixed terms by $ab \leq \varepsilon a^2 + C_\varepsilon b^2$. Applying this absorption in both directions, and taking $\varepsilon \leq 1/2$, yields

$$|Z_N - Z_M| \leq C_\varepsilon Z_N + C_\varepsilon \left(3^{-\gamma N} h_N^{-2} + T_N h_N^{1/2} \right).$$

The two deterministic remainders decay as $3^{-\gamma N/2}$ and $3^{-3\gamma N/32}$. The probability of either coercivity event is at most $B_Q 3^{-p_0 \gamma N/32}$, uniformly for every sufficiently large M . Hence a common positive rate is $c_\varepsilon = \min\{3\gamma/32, p_0\gamma/32\}$. Let $M \rightarrow \infty$ with N fixed, using convergence in probability and then a further subsequence converging almost surely. The resulting bound first has $C_\varepsilon Z_N$; choosing the tolerance smaller absorbs this term against the limit Z and gives (9.28). No centered estimate on the physical cube is used in this comparison.

The finite list of moment orders is chosen before disorder. Only the comparison geometry, initial field windows, constants and the positive rate γ depend on the tolerance; the coercivity order does not. For A_N^0 , translate the fixed cube and its boundary datum or volume source together, and leave the constant flux vector unchanged. Every finite-cutoff response on the translated cube equals the response on the original cube in the translated bilateral field. Choose its limit by the same translation. Stationarity therefore preserves the joint law of the response and its limit, so the same constants and deterministic initial cutoff work for every deterministic translate of this fixed cube. They may depend on its side and the fixed data. The final assertion follows by Borel–Cantelli at $\varepsilon = 1/j$, $j \in \mathbb{N}$, and a countable intersection. $\square$

9.6. Almost-sure bounds uniform in the cutoff. We now make the local estimates uniform over all cutoffs on one event of probability one. On cells of depth at most θN , Proposition 9.21 compares the finite-cutoff energies with their limits, with error probabilities summable over the cells and over N . For the deeper cells, the moment bounds imply that, almost surely, the random constants in the estimates are smaller than any fixed exponential in N once N is large. The margins in the spatial exponents absorb this growth.

Lemma 9.22 (Coarse-grained ellipticity bounds uniform in the cutoff). *Let Q be a cube of triadic side length and let s be one of the positive fractional orders used above. Then, almost surely, for $q \in \{1, 2\}$,*

$$\sup_N (\Lambda_{s,q}(Q; A_N) + \lambda_{s,q}(Q; A_N)^{-1}) < \infty.$$

In particular, the coercivity bound (8.5) and the bound (8.6) for Hölder boundary values on Q hold with one finite random constant for all N . The same conclusions hold with A_N^0 in place of A_N .

Proof. It suffices to consider a unit cube.

Step 1. We bound the finite-cutoff energies of the cells of depth at most θN by their limits. Denote by $\mathcal{R}_m(z)$ the finite family consisting of the minimal energies with the affine boundary values $x \mapsto e \cdot x$ and the Neumann energies for the fluxes e , for $e \in \{e_i, e_i + e_j\}$, computed for A_m^0 in the environment shifted to z , at cutoff m . We apply (9.28) with one fixed small tolerance to this family on every grid cube of depth $k \leq \theta N$, whose cutoff after rescaling to the unit cube is $m = N - k$. There are at most $C3^{dk}$ such cubes. We choose $\theta > 0$ so small that $d\theta < c(1 - \theta)/2$, where c is the exponent in (9.28) at this tolerance; then the sum over these cubes of the exceptional probabilities is summable in N . Hence, almost surely, for all sufficiently large N , each of these finite-cutoff energies is bounded by $C(1 + \mathcal{R}(\Theta_{k,z}\omega))$, where $\mathcal{R}$ is its limit. All the entries are nonnegative, and polarization recovers the coarse-grained matrices from this finite family.

Step 2. We bound the coarse-grained matrices of the cells of depth at most θN . For a cube $q = z + 3^{-k}\square_0$, separating the scalar $s_N(q)$ of Lemma 9.13 gives

$$A_N(z + 3^{-k}y) = s_N(q) \exp\left(\left[H + \sum_{j=0}^{k-1} \mathbf{g}_{-j}\right](z + 3^{-k}y) - \left[H + \sum_{j=0}^{k-1} \mathbf{g}_{-j}\right](z)\right) A_{N-k}^{0, \Theta_{k,z}\omega}(y).$$

The deterministic ratio in $s_N(q)$ and its inverse are at most $e^{C\tau^2 k}$. Hence the factors in front of $A_{N-k}^{0, \Theta_{k,z}\omega}$ and their inverses are bounded by

$$e^{C\tau^2 k} \exp\left(\left|H(z) + \sum_{j=0}^{k-1} \mathbf{g}_{-j}(z)\right| + \text{osc}_q\left[H + \sum_{j=0}^{k-1} \mathbf{g}_{-j}\right]\right),$$

whose L^p norm is at most $C_p e^{C_p \bar{\delta}^2 k}$, by the independence of the fields and the bound on their gradients weighted by their scales. The limits $\mathcal{R}$ have the same moment bounds as their approximants. We choose $\rho < s/4$, a moment order p with $p\rho > 2d$, and then $\bar{\delta}$ small enough. A union bound over all k and z gives a random $K_\omega < \infty$ such that, for every grid cell of depth k , the product of

the expression in the last display and $1 + \mathcal{R}(\Theta_{k,z}\omega)$ is at most $K_\omega 3^{\rho k}$. Since both coarse-grained matrices are monotone in the coefficient, $a(q; A_N) + a_*^{-1}(q; A_N) \leq CK_\omega 3^{\rho k}$ on the cells of depth at most θN , with the same K_ω for all sufficiently large N .

Step 3. We bound the matrices of the cells of depth greater than θN and sum over all depths. For $\theta N < k \leq N$, the proof of Lemma 8.4, applied only to the matrices of the cells, gives the bound $K_N 3^{\rho k}$, where the moments of K_N of the finitely many orders fixed at the start of Section 8 are bounded uniformly in N . By the Borel–Cantelli lemma, $K_N \leq 3^{\xi N}$ for all large N , for every fixed $\xi > 0$. The contribution of these cells to either sum defining $\Lambda_{s,q}$ or $\lambda_{s,q}$ is at most $CK_N 3^{-(s-\rho)\theta N}$, which is bounded if $\xi < (s-\rho)\theta$. The cells below the smallest field scale contribute at most $C3^{-sN}(\sup_Q A_N + \sup_Q A_N^{-1})$, which is bounded almost surely after the fixed decrease of $\bar{\delta}$, by Lemma 8.6 and Markov’s inequality. The cells of depth at most θN contribute at most $CK_\omega \sum_{k \geq 0} 3^{-(s-\rho)k}$. For $q = 1$, weighted Cauchy–Schwarz bounds the square of the sum of the square roots of the matrices by this sum of matrices; for $q = 2$, the definition is already a sum of matrices, with the smaller weight 3^{-2sk} . Finally, the coercivity bound and the bound (8.6) for Hölder boundary values follow from these bounds as in the proofs of Lemmas 8.3 and 8.4. $\square$

Lemma 9.23 (Regularity uniform in the cutoff). *Let $0 < \alpha < 1$ and $d-1 < t < d$ be strictly smaller than the exponents fixed in the moment estimates of Proposition 8.7 and Proposition 8.12. Then, for every fixed cube Q , almost surely the estimates of Proposition 8.7, extended to larger cubes by Remark 8.8, and Proposition 8.12 with the exponents α and t hold on Q with one finite random constant for all N .*

Proof. We run the iteration of Lemma 6.11 using only the primitive conditions on $\mathbf{F}$, $\mathbf{P}$ and $\mathbf{R}$, and the accumulated error $\mathbf{G}$, defined in Section 8.5. The normalized tests (a)–(c) of $G_N^{\det}(q)$ are not inputs to this iteration. Its reference ratios follow from (6.63); the actual coefficient’s initial energy and ellipticity bounds are supplied separately by Lemma 9.22. We transfer the primitive prefixes above the scale $3^{-\theta N}$ to their limits (Steps 1–5), and use the uniform moment bounds below that scale (Step 6).

Step 1. We pass the primitive quantities to the limit. In $\mathbf{R}$ and $\mathbf{G}$, the response at stationary scale n is computed for $\bar{\mathbf{a}}_n^{-1} \mathbf{a}_n$. In its own translated unit chart it is a response of A_n^0 , with no cell scalar or infrared factor. Define the limits of these affine Dirichlet and constant-flux Neumann responses by Proposition 9.21, and use them to define the limiting primitive quantities. The pure-field parts are taken in the same reindexed environment. The responses converge along the full sequence of cutoffs by Proposition 9.21, so the proof of Lemma 8.35(i) gives their finite-interval L^p limits along that full sequence. Geometric tail bounds pass by Fatou, and the conditional-expectation argument preceding Lemma 8.38 gives their limiting band bounds. The matrices computed for $A_N/s_N(U)$ include the centered infrared field and are not identified with these own-cutoff responses. Their almost-sure uniform ellipticity bounds are those of Lemma 9.22, and no convergence of their normalized tests is used here.

Step 2. We compare the finite-cutoff primitive quantities with their limits. Fix $\eta > 0$ below the margins between the lower and upper ramp thresholds. Truncate only the primitive quantities at relative depth $[\theta'N]$, and consider intervals of length at most $[\theta N]$ at all the centers needed in the regularity proof, including the face meshes and the centers of the initial balls. For a primitive response at physical depth j , the remaining own cutoff is $N-j$; its limit is taken in that same translated and reindexed environment. Thus all the retained responses are unit-cube responses of A_{N-j}^0 , not responses of the normalized anchored coefficient. There are at most $CN^C 3^{C_d(\theta+\theta')N}$ of them, with remaining cutoffs at least $[1-C(\theta+\theta')]N-C$. By (9.28) and a union bound, the probability that any retained response differs from its limit by more than $C\eta$ times the limit plus the exponentially small additive error is at most

$$C_\eta N^C 3^{C_d(\theta+\theta')N - c_\eta [1-C(\theta+\theta')]N}.$$

The discarded primitive tails, at finite cutoff and in the limit, have L^p norm at most $C_p 3^{-a[\theta'N]}$. The probability that one of them exceeds η is at most $C_\eta N^{C_d} 3^{C_d \theta N - a p \theta' N}$. Choose θ' small enough for the first exponent to be negative, and then θ small enough for both exponents to be negative. Both bounds are summable in N . Borel–Cantelli gives these comparisons simultaneously for all sufficiently large cutoffs.

Step 3. We compare the finite and limiting primitive prefixes. The relative error of the unit-cube energies controls the continuous piecewise linear functions: polarization gives $|\mathcal{J}_N - \mathcal{J}_\infty| \leq C\eta(1 + \mathcal{J}_\infty)$ on each retained cell at its own cutoff, and weighting gives the same inequality for the condition on $\mathbf{R}$. A bounded continuous piecewise linear function of this quantity changes by at most $C\eta$, and the square-root terms of the sums over intervals of scales change by at most $C\sqrt{\eta}$. The terms of $\mathbf{G}$ involving the fields $\mathbf{g}_j$ at a retained depth coincide once their finite interval of field indices lies below N , and their omitted tails satisfy the geometric bound of Step 2. Consequently, almost surely for all large N , every retained sum over an interval of scales at finite cutoff is bounded by the corresponding limiting sum plus $C\sqrt{\eta}$ per scale.

Step 4. We bound, uniformly over the choices of center and initial radius, the number of scales after which the limiting sums over intervals of scales satisfy the thresholds. For a fixed choice π of center and initial radius, let B_π be the least integer after which the limiting primitive prefixes satisfy the stricter thresholds of the iteration. Apply Lemma 8.37 to Z and $\mathbf{G}$ at a smaller finite-cutoff threshold and pass the bounds to the limit by Lemma 8.35(ii). Summing over the interval lengths gives $\mathbb{P}(B_\pi > b) \leq Ce^{-Ab}$, uniformly in π . At target depth n , the meshes of centers and initial radii used in the proof of Proposition 8.12 involve at most $C(n+1)^{C_d} 3^{C_d n}$ choices of centers, initial radii and faces, edges and corners. We fix a small $\xi_0 > 0$ before decreasing $\bar{\delta}$, and choose the rate A so that $A\xi_0 > C_d \log 3 + 2$. A union bound over the choices π at depth n gives

$$\mathbb{P}\left[\max_{\pi} B_\pi > \xi_0 n + b\right] \leq C(n+1)^{C_d} 3^{C_d n} e^{-A(\xi_0 n + b)}.$$

Summing over n shows that, almost surely, there is one finite B_ω such that $B_\pi \leq \xi_0 n + B_\omega$ for every choice π at every depth n .

Step 5. We transfer these bounds to finite cutoffs and prove the estimates at radii $r \geq 3^{-\theta N}$. We take η so small that the loss per scale of Step 3 is below the strict margin of the iteration. Then, for finite cutoffs and depths $n \leq \theta N$, the same numbers B_π apply with slightly weaker thresholds. The Z prefix bounds the number of failures of the primitive good-scale conditions, since a failed upper threshold contributes one to Z ; the $\mathbf{G}$ prefix bounds the accumulated comparison error. These are the two inputs of (6.61). The reference-ratio input is (6.63), derived from the same primitive conditions and annealed ordering. Thus the iteration applies without imposing (a)–(c) at these cells. Its initial averages and energy factors are bounded separately as follows. The estimate of Lemma 8.10 transfers the original field bounds to the reflected coefficients near faces, edges and corners. The reflection across faces is applied at most $d+1$ times, and the numbers B_π of these applications add up, so their total contribution is bounded by a common random constant times $3^{C\xi_0 n}$.

We next show that the averages b_i at the initial radius of each application are bounded by a random constant times a small negative power of the initial radius. Let θ_0 be the value of θ in the proof of Lemma 9.22, so that the cells of depth at most $\theta_0 N$ were compared with their limits there, and decrease θ to at most $\theta_0/2$. For an initial radius at depth $k \leq \theta N$, we split its coarse-grained sum at descendant depth $\theta_0 N$. For $0 < \rho < s$, the part with descendant depth at most $\theta_0 N$ is at most $CK_\omega 3^{\rho k}$, and the part with descendant depth greater than $\theta_0 N$ is at most

$$CK_N 3^{\rho k} 3^{-(s-\rho)(\theta_0 N - k)} \leq C 3^{\rho k} 3^{(\xi - (s-\rho)\theta_0/2)N},$$

where $K_N \leq 3^{\xi N}$ for large N , as in the proof of Lemma 9.22. We choose $\xi < (s-\rho)\theta_0/2$; the contribution below the smallest field scale satisfies the same bound by Lemma 8.6. The average at its

own scale involves only the value of $H + \sum_{j < k} \mathbf{g}_{-j}$ at the center and the deterministic ratio $\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N$. Hence the energy in the initial ball, the coercivity constant and the constant in (8.6) are bounded by a random constant times a small negative power of the initial radius.

We apply the iteration of Lemma 6.11 with exponents strictly larger than α and t , and choose ξ_0 and ρ so small that the margins in the exponents absorb both losses. The iteration gives one random constant for the oscillation and energy bounds at all radii $r \geq 3^{-\theta N}$.

Step 6. We treat the radii $r \leq 3^{-\theta N}$. We choose exponents $\alpha_0 > \alpha$ and $t_0 > t$ for which the Morrey bounds at all radii of Proposition 8.7 (on cubes of side larger than one, Remark 8.8) and Proposition 8.12 hold. Their constants satisfy $K_N \leq 3^{\xi N}$ almost surely for all large N , for every fixed $\xi > 0$. For $r \leq 3^{-\theta N}$, we have $K_N r^{\alpha_0 - \alpha} \leq 3^{(\xi - \theta(\alpha_0 - \alpha))N}$ and $K_N r^{t_0 - t} \leq 3^{(\xi - \theta(t_0 - t))N}$. We choose ξ smaller than $\theta(\alpha_0 - \alpha)$ and $\theta(t_0 - t)$; then these quantities are bounded. In the oscillation estimates at radii larger than $3^{-\theta N}$, point values are compared with averages over balls of radius $3^{-\theta N}$, using the estimate at small radii just proved. $\square$

Proposition 9.24 (Almost-sure convergence of energies and resolvents). *Almost surely, for every cube $Q \in \mathcal{C}$, $G_N^Q \rightarrow G^Q$ in operator norm on $L^2(Q, dx)$, E_N^Q Mosco-converges to E^Q , and, for every $f \in C(\bar{Q})$ and $\lambda > 0$,*

$$\|R_{N,\lambda}^Q f - R_\lambda^Q f\|_{C(\bar{Q})} \rightarrow 0.$$

On each fixed cube, the $C^{1/4}(\bar{Q})$ norms of $R_{N,\lambda}^Q f$ are bounded by one finite random constant, uniformly in N , in λ in a compact subset of $(0, \infty)$, and in f with $\|f\|_\infty \leq 1$.

Proof. Intersect the events of Lemmas 9.22 and 9.23, Proposition 9.2, and Proposition 9.21 for a countable dense family of bounded sources and their pairwise sums. Coercivity makes the images of the L^2 unit ball under G_N^Q bounded in $H^{3/4}$ uniformly in N , and polarization gives convergence of the matrix elements of G_N^Q on the dense family. The finite-rank argument and the variational formula (8.3) of Proposition 8.19 give norm and Mosco convergence to the limits already identified in probability.

All the constants in Proposition 9.5 are now bounded along the full sequence. Its variational and Arzelà–Ascoli argument therefore proves uniform convergence and the Hölder bound. $\square$

9.7. Almost-sure convergence of the path laws.

Proposition 9.25 (Almost-sure quenched convergence). *Almost surely, for every compact $B \subseteq \mathbb{R}^d$,*

$$\lim_{N \rightarrow \infty} \sup_{x \in B} d_{\text{weak}}(K_{N,x}^\omega, K_x^\omega) = 0.$$

Proof. Intersect the full-probability events of Proposition 9.24, Proposition 9.10 and Lemma 9.11 over the countable family of cubes, and fix an environment in this event. The killed occupation resolvents of $K_{N,x}^\omega$ converge to those of K_x^ω , uniformly on each closed cube.

We first obtain path tightness in this fixed environment. The image of a compact starting set under the continuous kernel $x \mapsto K_x^\omega$ is a compact family of laws on continuous path space. Its paths have infinite lifetime, so their probability of leaving a large cube before a fixed time, or of moving a fixed distance in a sufficiently short time, tends to zero uniformly in the starting point. Uniform convergence of the killed resolvents transfers these bounds to all sufficiently large cutoffs. For compact containment, compare $R_{N,1}^Q 1$ with its limiting counterpart on a sufficiently large cube: the identity $1 - R_{N,1}^Q 1(x) = E_{N,x}^\omega e^{-\tau_Q}$ bounds exits before a fixed time. For short exits, fix cubes $U \Subset V$, with $V \in \mathcal{C}$, and a compact starting set $A \subseteq U$. Choose $0 \leq f \leq 1$ continuous, supported in V , zero near A and one near ∂U . Continuity of the limiting path kernel, whose image of $A \cup \partial U$ is compact, gives $\lambda R_\lambda^V f \rightarrow f$ uniformly there. Given $\varepsilon > 0$, first choose λ so that $\lambda R_\lambda^V f \leq \varepsilon$ on A and $\lambda R_\lambda^V f \geq 3/4$ on ∂U . Proposition 9.24 and Lemma 9.11 then give, for all

sufficiently large N , $u_N := \lambda R_{N,\lambda}^V f \leq 2\varepsilon$ on A and $u_N \geq 1/2$ on ∂U . The strong Markov property at τ_U , applied to the nonnegative occupation integral killed at τ_V , gives

$$u_N(x) \geq E_{N,x}^\omega \left[e^{-\lambda \tau_U} u_N(Z_{\tau_U}); \tau_U \leq h \right] \geq \frac{1}{2} e^{-\lambda h} P_{N,x}^\omega(\tau_U \leq h), \quad x \in A.$$

Consequently, $\sup_{x \in A} P_{N,x}^\omega(\tau_U \leq h) \leq 8\varepsilon$ for $h \leq (\log 2)/\lambda$. The choices are in the order ε , then λ , then N , then h . Using small cubes U and a finite cover gives the bound for a prescribed displacement, uniformly over a compact starting set.

Apply this short-exit bound at successive exits from small balls, as in Step 3 of the proof of Proposition 9.8. The strong Markov property bounds the number of exits on a finite time interval; it also makes the probability of a very short gap tend to zero. Thus the moduli of continuity are tight. Together with compact containment, Arzelà–Ascoli gives tightness on continuous path space.

Now let $N_j \rightarrow \infty$ and $x_j \rightarrow x$. Localization as in (9.26) and killed-resolvent convergence give convergence of the whole-space discounted occupation integrals. Tightness and uniqueness of the Laplace transform give convergence of the one-time laws at every time. The Markov identity then gives convergence of the finite-dimensional laws, and approximation by cylinder functions on compact sets of paths gives $K_{N_j, x_j}^\omega \rightarrow K_x^\omega$. If convergence were not uniform on a compact B , a sequence of offending starting points would have a convergent subsequence, contradicting this conclusion and continuity of $x \mapsto K_x^\omega$. This proves the result in d_P ; tightness transfers it to every metric inducing the weak topology. $\square$

Proof of Theorem 7.1(i). The relabeling (8.1) and the changes of space and time transform the weak elliptic resolvent equations of the physical process into those defining $K_{N,x}^\omega$. Uniqueness of the weak solutions identifies the resolvents and hence the semigroups; the resulting finite-dimensional distributions identify the continuous path laws.

Proposition 9.25 gives the almost-sure convergence (7.2) of the quenched laws, and Proposition 9.9 shows that the limiting kernel is measurable with respect to the fields. Proposition 9.10 gives a continuous kernel, which is the family of laws of a continuous, conservative, strong Markov process with a μ -symmetric semigroup. Since the kernel is continuous, the semigroup maps bounded continuous functions to continuous functions. Proposition 9.2 shows that $\mu = \lim_N \mu_N$ is locally finite, has no atoms and has full support, and Corollary 9.12 gives the finite mean exit times. Finally, integrating the quenched convergence against $\mathbb{P}$ and using that the path functionals are bounded gives the convergence of the annealed laws. $\square$

10. THE LIMITING PROCESS AND MEASURE

In this section we prove the remaining assertions of Theorem 7.1. The mean-exit-time bound of Lemma 10.2 and power-law decay of the effective diffusivities between arbitrary scales bound the mean time to leave a cube of side 3^{-k} by $C3^{-(2+\eta)k}$, for every cutoff $L \geq k$, with C independent of L and k . These bounds pass to the limit and show that the limiting paths leave small cubes faster than Brownian paths. Consequently, their pointwise Hölder exponent is at most $(2 + \eta)^{-1}$, and the annealed path laws are mutually singular with every Brownian law; for each deterministic starting point, so is the quenched path law in almost every environment.

We then prove that the reversible measure is invariant and singular with respect to Lebesgue measure, and that the transition laws are absolutely continuous with respect to it. Together with the bounds on its mass in small balls, this shows that no transition law is Gaussian.

Throughout, we assume (g1)–(g4) at weak disorder and rescale cubes to $\square_0$ as in Section 9.3, with unit of time (9.9). We write P_x^ω and E_x^ω for the law and expectation, in the environment ω , of the diffusion under consideration ($X^{(L)}$, X , or their rescalings to $\square_0$) started at x .

10.1. Power-law decay and mean exit times. We first extend the power-law decay of the effective diffusivities in Proposition 5.6 to the ratio between two arbitrary scales.

Lemma 10.1 (Power-law decay between arbitrary scales). *There exist constants $C < \infty$ and $\eta > 0$, depending only on d and the law of $\mathbf{g}_0$, such that, for every $\ell, m \in \mathbb{N}_0$ with $\ell \leq m$,*

$$(10.1) \quad \frac{\bar{\mathbf{a}}_m}{\bar{\mathbf{a}}_\ell} \leq C 3^{-\eta(m-\ell)},$$

where in dimension two one can take $\eta = \tau^2 / \log 3$.

Proof. In dimension two, use (5.1). For $d \geq 3$, we start the gluing argument of Proposition 5.6 at scale ℓ .

We first prove for Δ_ℓ^π the energy bound that Proposition 4.1 gives for cubes. The Whitney cubes of Δ_ℓ^π of side $3^{\ell-i}$ occupy at most a proportion $C 3^{-i}$ of its volume. For $i \leq \ell$, use on each cube the minimizer of the energy with coefficient $\mathbf{a}_{\ell-i}$ among functions equal to $p \cdot x$ on the boundary of the cube. The factors contributed by the fields in $\mathbf{a}_\ell$ that are absent from $\mathbf{a}_{\ell-i}$ are independent of this minimizer and have mean one. By (3.24), the expected energy density is bounded by $C \bar{\mathbf{a}}_\ell e^{2\tau^2 i} |p|^2$. For $i > \ell$, the affine competitor gives the same bound since $\bar{\mathbf{a}}_\ell \geq e^{-2\tau^2(\ell+1)}$. Join these competitors, which all equal $p \cdot x$ on the boundaries of the Whitney cubes. Uniformly in ℓ , π and p , we obtain

$$\mathbb{E}[p \cdot \mathbf{a}_\ell(\Delta_\ell^\pi) p] \leq C \bar{\mathbf{a}}_\ell |p|^2 \sum_{i \geq 0} 3^{-i} e^{2\tau^2 i} \leq C \bar{\mathbf{a}}_\ell |p|^2.$$

Take R and $q_R < 1$ from (5.18). Keep the fields with indices at most ℓ and with indices $\ell + R, \ell + 2R, \dots$, and omit the others. The induction (5.19), started at scale ℓ from the bound for Δ_ℓ^π above, and reinsertion of the omitted fields by (5.20) give $\bar{\mathbf{a}}_m \leq C \bar{\mathbf{a}}_\ell q_R^{\lfloor (m-\ell)/R \rfloor}$. This proves (10.1) with $\eta = -\log q_R / (R \log 3)$. $\square$

Lemma 10.2 (Mean exit times). *There exists $C < \infty$ such that, for every $L, m \in \mathbb{N}_0$ and every $z \in \mathbb{R}^d$, the cube $U = z + \square_m$ satisfies*

$$(10.2) \quad \mathbb{E} \left[\sup_{x \in U} E_x^\omega \tau_U^{(L)} \right] \leq C 3^{2m} / \bar{\mathbf{a}}_{m \wedge L}.$$

More generally, there is $c > 0$ such that the following holds. For every $q < \infty$, there exists $\bar{\delta}_0 > 0$ such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then there are random constants $K \geq 1$, one for each $L, m \in \mathbb{N}_0$ and $z \in \mathbb{R}^d$, with $\mathbb{E}[K^q]$ bounded uniformly in L, m and z , such that the process obtained by rescaling $z + \square_m$ to $\square_0$ satisfies

$$\sup_{x \in \square_0} E_x^\omega \tau_{\square_0} \leq K$$

and

$$\inf_{x \in \square_{-2}} E_x^\omega \tau_{\square_0} \geq c K^{-2}.$$

The same assertions hold for the diffusion X without cutoff, with $\bar{\mathbf{a}}_m$ in (10.2).

Proof. We rescale $z + \square_m$ to $\square_0$ as in Section 9.3, for either the diffusion $X^{(L)}$ or the diffusion X ; here μ has a positive, locally $C^{1,1}$ density. The mass bound (9.10), the trace inequality (9.17) and the coercivity bound (9.11) for functions with zero boundary values give, on every cube U that is fixed in the rescaled coordinates,

$$(10.3) \quad \|v\|_{L^{2d/(d-1)}(U, \mu)}^2 \leq K \mathcal{E}(v; U), \quad v \in H_0^1(U).$$

Here K may depend on U and has the asserted uniform moment bounds.

Let u solve $-\nabla \cdot (A\nabla u) = d\mu/dx$ in $\square_0$ with zero boundary values. Testing with $(u - h)_+$ gives, for $k > h \geq 0$,

$$\mu(\{u > k\}) \leq \left(\frac{K}{k - h} \right)^{2d/(d-1)} \mu(\{u > h\})^{(d+1)/(d-1)}.$$

Iterate this inequality along the thresholds equal to $1 - 2^{-i}$ times a suitable multiple of $K\mu(\square_0)^{1/d}$ to obtain $\|u\|_\infty \leq CK\mu(\square_0)^{1/d}$, which by (9.10) is bounded by another random constant K with the asserted moments.

We identify u with the mean exit time. The semigroup of the diffusion killed on leaving $\square_0$ is associated with the part of the Dirichlet form on $\square_0$ [Fukushima et al., 2011, Theorems 4.4.2–4.4.3]. Interpolate (10.3) with $L^1(\mu)$ to obtain a Nash inequality with respect to μ . On each fixed cube, it bounds the killed semigroup from $L^1(\mu)$ to $L^\infty(\mu)$ by a constant times t^{-d} . We adapt the eigenfunction construction for symmetric operators in Bass [1998, Chapter VII, Section 6, Theorem 6.3 and Proposition 6.4] to $L^2(\square_0, \mu)$. Interior elliptic regularity of the eigenfunctions, and uniform convergence of the eigenfunction expansion on compact sets of positive times and interior points, give a kernel $p_{\square_0}(t, x, y)$ that is jointly continuous at positive times and interior points.

Since the drift of the rescaled stochastic differential equation is locally bounded, Itô's formula applied to the squared distance from the starting point, stopped on exit from a small ball, bounds the probability of leaving that ball before time t by a quantity tending to zero as $t \downarrow 0$, uniformly over starting points in a compact subset of a fixed cube.

The local Nash bounds, this exit estimate and continuity of the semigroup on compactly supported continuous functions show, by approximation in $L^1(\mu)$ on an enclosing cube, that the semigroup of the diffusion without killing maps bounded Borel functions to continuous functions at each positive time. The exit estimate also gives continuity in the starting point of the transition probabilities of the diffusion killed on leaving $\square_0$. For each fixed positive time and Borel subset of $\square_0$, the killed transition probability and the integral of $p_{\square_0}$ over that subset are continuous and agree μ -almost everywhere. They therefore agree at every interior starting point, since μ has full support.

Coercivity gives a positive spectral gap. Together with the bound on the killed semigroup, this gives $P_x^\omega(\tau_{\square_0} > t) \leq Ce^{-ct}$ for $t \geq 1$, uniformly in $x \in \square_0$. Integrate the survival probability in time to obtain a function of the starting point that is continuous at interior points; as an element of $L^2(\mu)$, it equals the inverse of the positive killed generator applied to 1. This function is the continuous representative of u at interior points. With this representative,

$$u(x) = \int_0^\infty \int_{\square_0} p_{\square_0}(t, x, y) \mu(dy) dt = \int_0^\infty P_x^\omega(\tau_{\square_0} > t) dt = E_x^\omega \tau_{\square_0}, \quad x \in \square_0.$$

The bound on $\|u\|_\infty$ obtained above therefore holds at every interior point for this representative. This proves the upper bound after rescaling, and restoring the time unit (9.9) gives (10.2). The lower bound follows from (9.22): taking $t = (2K)^{-2}$ gives $P_x^\omega(\tau_{\square_0} > t) \geq 1/2$ for every $x \in \square_{-2}$. $\square$

10.2. Annealed and quenched path laws. For a deterministic starting point x , the annealed law $\mathbb{E}[K_x^\omega]$ of the limiting process is the average over ω of its quenched law K_x^ω .

Let $\mathbf{P}$ be a subsequential limit of the annealed laws of the rescaled diffusions $X^{(L)}$ or X in Proposition 9.8, started at deterministic points that converge in rescaled coordinates; in particular, $\mathbf{P}$ may be $\mathbb{E}[K_x^\omega]$. Here $\mathbf{P}$ is a probability measure on paths. In Section 8.5, $\mathbf{P}_{m,z}$ and its abbreviation $\mathbf{P}$ denote the scalar discounted coefficient-ratio statistic. We use the discrete exit times $\sigma_k = \sigma_{3^{-k}}$ of Theorem 7.1(iii), so σ_1 refers to side 3^{-1} , and write $Z_t^* = \sup_{0 \leq s \leq t} |Z_s - Z_0|$.

Theorem 10.3 (Annealed and quenched singularity). *There exists $C < \infty$ such that every law $\mathbf{P}$ as above satisfies, for every $k \in \mathbb{N}_0$,*

$$(10.4) \quad E_{\mathbf{P}} \sigma_k \leq C 3^{-(2+\eta)k},$$

where η is the exponent of Lemma 10.1. Every such annealed law $\mathbf{P}$ is mutually singular with every mixture of Brownian laws with arbitrary covariance matrices. For each deterministic x , the quenched law K_x^ω has the same singularity for $\mathbb{P}$ -almost every ω . More generally, every subsequential limit in distribution of the random quenched laws of either family in Proposition 9.8, started at deterministic points that converge in rescaled coordinates, is almost surely mutually singular with every such mixture of Brownian laws.

This proves (7.3) and the mutual singularity in Theorem 7.1(iii).

Proof. Step 1. We prove (10.4). For $L \geq k$, the mean exit-time bound (10.2), applied at scale $L - k$ with time divided by $3^{2L}/\bar{\mathbf{a}}_L$, and Lemma 10.1 give, for either family,

$$\mathbb{E}\sigma_k(Z^{(L)}) \leq C 3^{-2k} \frac{\bar{\mathbf{a}}_L}{\bar{\mathbf{a}}_{L-k}} \leq C 3^{-(2+\eta)k}.$$

The exit time σ_k is a lower semicontinuous function of the path for the topology of locally uniform convergence. The portmanteau theorem, applied to this nonnegative lower semicontinuous functional, gives (10.4).

Step 2. We prove annealed singularity. Consider the event that $3^{2k}\sigma_k \rightarrow 0$ as $k \rightarrow \infty$. By (10.4),

$$E_{\mathbf{P}} \sum_{k \geq 0} 3^{2k}\sigma_k \leq C \sum_{k \geq 0} 3^{-\eta k} < \infty,$$

so this event has $\mathbf{P}$ -probability one. For a Brownian law with a nonzero finite covariance matrix, Brownian scaling gives $3^{2k}\sigma_k \stackrel{\text{law}}{=} \sigma_0 > 0$. Apply Fatou's lemma to the events that $3^{2k}\sigma_k \leq \varepsilon$. For every $\varepsilon > 0$, the probability of the event under this law is at most the probability that $\sigma_0 \leq \varepsilon$. Letting $\varepsilon \downarrow 0$ shows that the event has probability zero. When the covariance matrix is zero, the law of the constant path also gives the event probability zero.

Step 3. We prove the quenched assertions. For each deterministic x , Step 2 with $\mathbf{P} = \mathbb{E}[K_x^\omega]$ shows that the average over ω of the K_x^ω -probability of the event is one. Since this probability lies in $[0, 1]$, it equals one for $\mathbb{P}$ -almost every ω .

For the general assertion, convergence in distribution of the random quenched laws implies weak convergence of their averages. The average of the random limiting law is therefore an annealed subsequential limit and gives the event mass one by Step 2. The argument of the preceding paragraph then shows that the random limiting law gives the event mass one, almost surely. $\square$

For every law $\mathbf{P}$ as above, the exit-time bound (10.4) gives lower bounds on displacement at small times and an upper bound on the pointwise Hölder exponent defined in (7.4).

Corollary 10.4 (Small-time behavior of the limiting paths). *The following statements hold.*

(i) Oscillation: for every $\varepsilon > 0$, there is $c_\varepsilon > 0$ such that, $\mathbf{P}$ -almost surely, for all sufficiently small $t > 0$,

$$(10.5) \quad Z_t^* \geq c_\varepsilon \left(\frac{t}{(\log(1/t))^{1+\varepsilon}} \right)^{1/(2+\eta)}.$$

In particular, $\mathbf{P}$ -almost surely, $Z_t^*/\sqrt{t} \rightarrow \infty$ as $t \downarrow 0$.

(ii) Hölder exponent: $\mathbf{P}$ -almost surely, the pointwise Hölder exponent at zero is at most $1/(2+\eta)$, and at most $(2 + \tau^2/\log 3)^{-1}$ in dimension two.

(iii) Exit times: for every $\varepsilon > 0$, $\mathbf{P}$ -almost surely, for all sufficiently large k ,

$$(10.6) \quad \sigma_k \leq k^{1+\varepsilon} 3^{-(2+\eta)k}.$$

(iv) Moments of the running maximum: for every $p > 0$, there is $c_p > 0$ such that, for every $0 < t \leq 1$,

$$(10.7) \quad E_{\mathbf{P}}(Z_t^*)^p \geq c_p t^{p/(2+\eta)}.$$

For each deterministic x , parts (i)–(iii) also hold K_x^ω -almost surely for $\mathbb{P}$ -almost every ω , by Step 3 of the proof of Theorem 10.3. Parts (i) and (ii) give (7.4) in Theorem 7.1.

Proof. Markov's inequality gives $\mathbf{P}(\sigma_k > k^{1+\varepsilon} 3^{-(2+\eta)k}) \leq Ck^{-1-\varepsilon}$, so the Borel–Cantelli lemma proves (iii). For t sufficiently small, choose k with

$$k^{1+\varepsilon} 3^{-(2+\eta)k} \leq t < (k-1)^{1+\varepsilon} 3^{-(2+\eta)(k-1)}.$$

Then (10.6) gives $Z_t^* \geq 3^{-k}/2$, and k is comparable to $\log(1/t)$, proving (i). Part (i) implies (ii), since $|Z_t - Z_0| = O(t^\gamma)$ would give $Z_t^* = O(t^\gamma)$ and contradict (10.5) for $\gamma > 1/(2+\eta)$.

For (iv), enlarge C in (10.4) to at least one and take the smallest $k \geq 0$ with $C3^{-(2+\eta)k} \leq t/2$. Then $\mathbf{P}(\sigma_k \leq t) \geq 1/2$ and 3^{-k} is comparable to $t^{1/(2+\eta)}$, giving the claimed moment bound. $\square$

Corollary 10.5 (Exit moments and small displacements). *For every $p > 0$, there exists $\bar{\delta}_0 > 0$ such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then there are $0 < c_p \leq C_p < \infty$ such that, for every $L, m \in \mathbb{N}_0$ and $z \in \mathbb{R}^d$, the cube $U = z + \square_m$ satisfies*

$$(10.8) \quad \mathbb{E} \left[\sup_{x \in U} E_x^\omega (\tau_U^{(L)})^p \right] \leq C_p 3^{2mp} \bar{\mathbf{a}}_{m \wedge L}^{-p}.$$

For starting points in the inner cube, the matching lower bound is

$$\mathbb{E} \left[\inf_{x \in z + \square_{m-2}} E_x^\omega (\tau_U^{(L)})^p \right] \geq c_p 3^{2mp} \bar{\mathbf{a}}_{m \wedge L}^{-p}.$$

The same bounds hold for X with m in place of $m \wedge L$. Moreover, every $\mathbf{P}$ as above satisfies $E_{\mathbf{P}} \sigma_k^p \leq C_p 3^{-p(2+\eta)k}$ for every $k \in \mathbb{N}_0$, and

$$\mathbf{P} \left(Z_t^* \leq \varepsilon t^{1/(2+\eta)} \right) \leq C_p \varepsilon^{p(2+\eta)}, \quad 0 < t, \varepsilon \leq 1.$$

Consequently, for every $0 < q < p(2+\eta)$ there is $C_{p,q} < \infty$ such that every $\mathbf{P}$ as above satisfies

$$E_{\mathbf{P}} (Z_t^*)^{-q} \leq C_{p,q} t^{-q/(2+\eta)}, \quad 0 < t \leq 1.$$

Proof. After rescaling $z + \square_m$ to $\square_0$ as in the proof of Lemma 10.2, Markov's inequality and the strong Markov property give, for each environment and every $n \in \mathbb{N}_0$, the bound $\sup_{x \in \square_0} P_x^\omega (\tau_{\square_0} > 2nK) \leq 2^{-n}$. Integrating yields $\sup_x E_x^\omega \tau_{\square_0}^p \leq C_p K^p$. For the lower bound, (9.22) and $\mathbb{E} K \leq C$ give, for a sufficiently small deterministic $c > 0$,

$$\mathbb{E} \left[\inf_{x \in \square_{-2}} E_x^\omega \tau_{\square_0}^p \right] \geq c^p \mathbb{E} \left[\inf_{x \in \square_{-2}} P_x^\omega (\tau_{\square_0} > c) \right] \geq \frac{c^p}{2}.$$

Taking expectations in the upper bound and restoring the time unit (9.9) proves both bounds. Rescaling as in Step 1 of the proof of Theorem 10.3, Lemma 10.1 and the portmanteau theorem for σ_k^p give the exit-moment bound for $\mathbf{P}$.

For $0 < r \leq 1/12$, choose $k \geq 0$ with $4r \leq 3^{-k} < 12r$. If $Z_t^* \leq r$, then $\sigma_k > t$, so Markov's inequality gives

$$\mathbf{P}(Z_t^* \leq r) \leq t^{-p} E_{\mathbf{P}} \sigma_k^p \leq C_p t^{-p} r^{p(2+\eta)}.$$

Set $r = \varepsilon t^{1/(2+\eta)}$ to obtain the stated probability bound when $r \leq 1/12$. If $1/12 < r \leq 1$, then $\varepsilon \geq r > 1/12$, so the probability is at most $1 \leq 12^{p(2+\eta)} \varepsilon^{p(2+\eta)}$. Enlarging C_p to at least $12^{p(2+\eta)}$ covers this interval. Integrating the probability bound against $q\varepsilon^{-q-1} d\varepsilon$ on $(0, 1)$ gives a finite contribution because $q < p(2+\eta)$; the range $\varepsilon \geq 1$ contributes at most $\int_1^\infty q\varepsilon^{-q-1} d\varepsilon = 1$. Restoring the factor $t^{-q/(2+\eta)}$ proves the negative-moment bound. $\square$

10.3. Singularity of the measure and of the transition laws. Recall the limiting measures μ^0 and $\mu = e^H \mu^0$ of Theorem 7.1(ii), constructed in Proposition 9.2 and its proof. For Gaussian fields, $\tau^2 = \frac{1}{2} \mathbb{E}[\mathbf{g}_0(0)^2]$, so μ^0 is Gaussian multiplicative chaos, and μ is its product with the positive, locally $C^{1,1}$ function e^H . Existence of the limit, moment bounds, absence of atoms, full support, and singularity with respect to Lebesgue measure are standard properties of Gaussian multiplicative chaos in the subcritical regime, in which the variance of the field is small enough for the limiting measure to be nonzero. Moreover, the densities of the approximating measures diverge at almost every point with respect to the limiting measure; see Kahane [1985] and Rhodes and Vargas [2014, Sections 2 and 4.1]. The proofs in Section 9.2 and below also apply to the non-Gaussian fields allowed by (g1)–(g4).

The measures $\mathbf{a}_N(3^N x) dx$ have the laws of μ_N^0 and therefore converge in distribution to μ^0 . The measures μ^0 and μ are nonatomic and have full support, and Proposition 9.2 with $\epsilon = 1/2$ gives, on each bounded set of centers, a random constant K such that, for every x in that set,

$$(10.9) \quad \mu^0(B_r(x)) + \mu(B_r(x)) \leq K r^{d-1/2}, \quad 0 < r \leq 1.$$

For every finite moment order, K has a finite moment of that order at weak disorder, with a threshold depending on the order, by the same proposition and by Lemma 8.1 for $e^{\pm H}$.

Theorem 10.6 (The limiting measures). *The following statements hold.*

- (i) Translation invariance: *the law of μ^0 is invariant under translations, μ^0 is not deterministic, and it satisfies $\mathbb{E}[\mu^0(dx)] = dx$. Its restrictions to two sets at distance greater than $\sqrt{d}$ are independent.*
- (ii) Singularity: *almost surely, the density of μ_N^0 at x tends to zero as $N \rightarrow \infty$ for Lebesgue-almost every x , whereas it tends to infinity for μ^0 -almost every and μ -almost every x . In particular, both limiting measures are singular with respect to Lebesgue measure.*
- (iii) Invariance: *almost surely, the reversible measure μ is invariant for the limiting process: for every nonnegative Borel function f and every $t \geq 0$,*

$$(10.10) \quad \iint f(Z_t) dK_x^\omega \mu(dx) = \int f d\mu.$$

Proof. Write ρ_N for the density of μ_N^0 . Lemma 9.1 gives uniform integrability of the masses $\mu_N^0(U)$ on bounded cubes U , and hence

$$\mathbb{E}[\mu^0(dx) \mid \mathbf{g}_0, \dots, \mathbf{g}_{-N}] = \rho_N(x) dx, \quad \mathbb{E}\mu^0(dx) = dx.$$

The strong law and Fubini's theorem give, for $\mathbb{P} \otimes dx$ -almost every (ω, x) ,

$$(10.11) \quad \frac{\log \rho_N(x)}{N+1} = \frac{1}{N+1} \sum_{i=0}^N \mathbf{g}_{-i}(x) - \tau^2 \longrightarrow -\tau^2 < 0.$$

For the behavior at a point sampled from μ^0 , fix a bounded cube U and consider the probability measure $|U|^{-1} \mathbb{P}(d\omega) \mu^{0,\omega}(dx)$ on $\Omega \times U$. This is the change of measure used for Gaussian chaos in Rhodes and Vargas [2014, Section 4.1]; here independence of the fields suffices. The conditional-expectation identity above and stationarity show that, under this measure, the variables $\mathbf{g}_{-j}(x)$ for $j \geq 0$ are independent, and each has the law of $\mathbf{g}_0(0)$ tilted by $e^{\mathbf{g}_0(0) - \tau^2}$.

Their common mean is $\mathbb{E}[\mathbf{g}_0(0) e^{\mathbf{g}_0(0) - \tau^2}] > \tau^2$: under the tilted law, $e^{-\mathbf{g}_0(0)}$ has expectation $e^{-\tau^2}$, and Jensen's inequality for the exponential is strict because $\mathbf{g}_0(0)$ is not almost surely constant. The strong law therefore gives

$$\frac{\log \rho_N(x)}{N+1} \longrightarrow \mathbb{E}[\mathbf{g}_0(0) e^{\mathbf{g}_0(0) - \tau^2}] - \tau^2 > 0$$

almost surely under this probability measure. Thus $\rho_N \rightarrow \infty$ at μ^0 -almost every point, almost surely. The positive factor e^H gives the same conclusion for μ . Together with (10.11), this proves (ii). Since $\mathbb{E}\mu^0(dx) = dx$, singularity also shows that μ^0 is not deterministic. Translation invariance of the law and independence of restrictions to sets at distance greater than $\sqrt{d}$ pass from the fields to μ^0 , which is a measurable function of them.

Finally, symmetry and conservativeness from Proposition 9.10 give (10.10). $\square$

The Sobolev estimate (10.3) gives absolute continuity of the transition laws at positive times with respect to μ . The measure μ is carried by a Lebesgue-null set and gives no mass to affine hyperplanes, so these transition laws are singular with respect to every Gaussian law.

Theorem 10.7 (Quenched transition laws). *Almost surely, for every $x \in \mathbb{R}^d$ and every $t > 0$,*

$$(10.12) \quad K_x^\omega(Z_t \in \cdot) \ll \mu$$

and the law of Z_t under K_x^ω is singular with respect to Lebesgue measure and every Gaussian law. For every bounded open set U , there is $C_U < \infty$ such that, for every $x \in U$ and $t > 0$,

$$K_x^\omega(Z_t \in \cdot, t < \tau_U) \leq C_U t^{-d} \mu|_U.$$

Moreover, almost surely, for every starting point, the quenched joint law of the positions at any finite set of positive times is mutually singular with every Gaussian law, and the quenched path law on any interval $[s, t]$, $0 \leq s < t$, is mutually singular with the law of every continuous Gaussian process on that interval.

Proof. Fix a cube U of a deterministic countable exhaustion of $\mathbb{R}^d$ and let $S_{N,t}$ be the symmetric killed semigroup of E_N^U on $L^2(U, \mu_N)$. By the relabeling (8.1), the coefficient of E_N^U and the measure μ_N have the joint law of the coefficient and the reversible measure after rescaling of X at scale N . Hence (10.3) applies on U with constants K_N satisfying $\sup_N \mathbb{E}K_N < \infty$.

We use Nash's argument. For nonzero $f \in L^1(U, \mu_N) \cap L^2(U, \mu_N)$, interpolation of (10.3) with L^1 , the energy identity and L^1 contraction give

$$\frac{d}{dt} \|S_{N,t}f\|_{L^2(U, \mu_N)}^2 \leq -2K_N^{-1} \|S_{N,t}f\|_{L^2(U, \mu_N)}^{2+2/d} \|f\|_{L^1(U, \mu_N)}^{-2/d}.$$

Integration, symmetry and the semigroup property yield

$$(10.13) \quad \|S_{N,t}f\|_\infty \leq C(K_N/t)^d \|f\|_{L^1(U, \mu_N)}, \quad t > 0.$$

For $f \in C_c(U)$, the function $S_{N,t}f$ is continuous, so this bound holds at every point of U .

By Proposition 9.25, the whole-space path laws converge almost surely, locally uniformly in the starting point. Fatou's lemma gives $\liminf_N K_N < \infty$ almost surely. For each such environment, choose a subsequence on which K_N is bounded. If $f \in C_c(U)$ is nonnegative, the path functional $z \mapsto f(z(t))\mathbf{1}_{\{t < \tau_U\}}$ is lower semicontinuous: a path that stays in U up to time t has a positive distance from its complement. The portmanteau theorem, (10.13), and the almost-sure local weak convergence $\mu_N \rightarrow \mu$ of Proposition 9.2 therefore give the same bound for the killed transition law of K_x^ω . Positive compactly supported tests give the asserted measure domination. The subsequence depends only on the environment and U , so the bound holds for every $x \in U$ and $t > 0$.

Enclosing an arbitrary bounded open set in one of these cubes and comparing exit times gives the bound on that set. By Proposition 9.10, the limiting process is conservative and has continuous paths, so each path remains in some cube of the exhaustion up to time t . Monotone convergence therefore proves (10.12) for all x and $t > 0$ on a single event of probability one.

By Theorem 10.6, a Lebesgue-null set carries μ and hence every transition law, proving singularity with respect to Lebesgue measure and every nondegenerate Gaussian. A bounded subset of an affine hyperplane is covered by $O(r^{1-d})$ balls of radius r , so (10.9) bounds its μ -mass by a constant

times $r^{1/2}$, which tends to zero as $r \downarrow 0$. Thus every affine hyperplane is μ -null. Every degenerate Gaussian, including a point mass, is supported on such a hyperplane, which has zero probability under every transition law by (10.12). This proves mutual singularity with every Gaussian law.

Since the marginals of a Gaussian law are Gaussian, and two laws are mutually singular as soon as their marginals at one time are, projecting onto a single positive time gives the assertions for joint laws and path laws. $\square$

APPENDIX A. MOMENTS OF RANDOM VARIABLES WITH LOG-NORMAL TAILS

Stochastic bounds throughout the paper are stated in the Orlicz notation of [Armstrong et al. \[2019, Appendix A\]](#). All references to Appendix A in this appendix refer to the preprint version [arXiv:1705.05300v2](#), including (A.1)–(A.2), Remark A.2 and Lemmas A.1, A.4, A.7 and A.9. The subscript refers to the Orlicz function $\exp(t^s)$, and for $s > 0$ and $A > 0$ we write

$$X \leq \mathcal{O}_{\Gamma_s}(A) \quad \text{if} \quad \mathbb{E}[\exp((A^{-1} \max\{X, 0\})^s)] \leq 2,$$

and

$$(A.1) \quad X = \mathcal{O}_{\Gamma_s}(A) \quad \text{if} \quad X \leq \mathcal{O}_{\Gamma_s}(A) \quad \text{and} \quad -X \leq \mathcal{O}_{\Gamma_s}(A).$$

If $X \leq \mathcal{O}_{\Gamma_s}(A)$, then, by [Armstrong et al. \[2019, Lemma A.1\]](#),

$$(A.2) \quad \mathbb{P}[X \geq tA] \leq 2 \exp(-t^s) \quad (t \geq 0).$$

Conversely, the bound $\mathbb{P}[X \geq tA] \leq \exp(-t^s)$ for every $t \geq 0$ implies $X \leq \mathcal{O}_{\Gamma_s}(2^{1/s}A)$.

By [Armstrong et al. \[2019, Lemma A.4\]](#), for every $s \geq 1$, every finite index set I , and random variables X_k satisfying $X_k \leq \mathcal{O}_{\Gamma_s}(a_k)$ for every $k \in I$,

$$(A.3) \quad \sum_{k \in I} X_k \leq \mathcal{O}_{\Gamma_s} \left(\sum_{k \in I} a_k \right).$$

The same conclusion holds for a countable family of nonnegative X_k with $\sum_k a_k < \infty$, by monotone convergence. The tail bound (A.2), a union bound, and the converse part of [Armstrong et al. \[2019, Lemma A.1\]](#), in the form of [Armstrong et al. \[2019, Remark A.2\]](#) that requires the tail bound only beyond a threshold, give, for every $s > 0$, $\sigma > 0$ and $k \in \mathbb{N}$, and all random variables $X_1, \dots, X_k$ satisfying $X_i \leq \mathcal{O}_{\Gamma_s}(\sigma)$ for every i ,

$$(A.4) \quad \max_{i \in \{1, \dots, k\}} X_i \leq \mathcal{O}_{\Gamma_s}(C_s(\log(2k))^{1/s} \sigma).$$

If, in addition, $s \in (1, 2]$, the variables are independent and mean zero, and $X_i = \mathcal{O}_{\Gamma_s}(\sigma)$, then, by [Armstrong et al. \[2019, Lemmas A.7 and A.9\]](#),

$$(A.5) \quad \sum_{i=1}^k X_i = \mathcal{O}_{\Gamma_s}(C_s k^{1/2} \sigma).$$

Lemma A.1. *There exists a universal constant $C < \infty$ such that, for every $\sigma > 0$ and every random variable X satisfying $X = \mathcal{O}_{\Gamma_2}(\sigma)$, the following estimates hold. For every $k > 0$,*

$$(A.6) \quad \mathbb{E}[|X|^k] \leq C \sigma^k \Gamma\left(\frac{k}{2} + 1\right),$$

for every $t > 0$,

$$(A.7) \quad \mathbb{E}[\exp(tX) - (1 + tX)] \leq C \sigma^2 t^2 \exp(C \sigma^2 t^2),$$

and, for every $m \geq 1$ and $t > 0$,

$$(A.8) \quad \mathbb{E}[\exp(tX) - 1]^m \leq (C m^{1/2} \sigma t)^m \exp(C(m \sigma t)^2).$$

Proof. By (A.2), applied to X and $-X$, we have $\mathbb{P}[|X| \geq r\sigma] \leq 4e^{-r^2}$ for every $r \geq 0$. The layer-cake formula gives

$$\mathbb{E}[|X|^k] \leq 4k\sigma^k \int_0^\infty r^{k-1} e^{-r^2} dr = 4\sigma^k \Gamma\left(\frac{k}{2} + 1\right),$$

which is (A.6). Moreover, Young's inequality and (A.1) give $\mathbb{E}[e^{\lambda|X|}] \leq 4e^{\lambda^2 \sigma^2 / 2}$ for every $\lambda \geq 0$. The elementary inequalities $|e^u - 1 - u| \leq \frac{1}{2}u^2 e^{|u|}$ and $|e^u - 1|^m \leq |u|^m e^{m|u|}$, together with the

Cauchy–Schwarz inequality, (A.6) and the bound on $\mathbb{E}[e^{\lambda|X|}]$, yield $\mathbb{E}[|e^{tX} - 1 - tX|] \leq C\sigma^2 t^2 e^{C\sigma^2 t^2}$ and

$$\mathbb{E}[|e^{tX} - 1|^m] \leq t^m \mathbb{E}[|X|^{2m}]^{1/2} \mathbb{E}[e^{2mt|X|}]^{1/2} \leq (Cm^{1/2}\sigma t)^m e^{C(m\sigma t)^2}.$$

These are (A.7) and (A.8). □

APPENDIX B. CONCENTRATION INEQUALITIES

In this appendix we collect concentration inequalities used in the body of the paper: Rosenthal's inequality for sums of independent random variables, used in Sections 3 and 4, and bounds on the proportion of scales at which independent or finitely dependent random variables are large, used in Sections 6 and 8.5.

The integer-exponent case of the following estimate follows by applying [Boucheron et al. \[2013, Theorem 15.11\]](#) to both signs of the sum. We give a proof which also covers every real exponent $p \geq 2$.

Lemma B.1 (Rosenthal's inequality). *For every $p \in [2, \infty)$, $N \in \mathbb{N}$ and every sequence $X_1, \dots, X_N$ of independent random variables satisfying $\mathbb{E}[X_i] = 0$, we have the estimate*

$$\mathbb{E} \left[\left| \sum_{i=1}^N X_i \right|^p \right]^{1/p} \leq 6p^{1/2} \left(\sum_{i=1}^N \mathbb{E}[X_i^2] \right)^{1/2} + 4p \mathbb{E} \left[\max_{i \in \{1, \dots, N\}} |X_i|^p \right]^{1/p}.$$

Proof. Write $\|Y\|_r = \mathbb{E}[|Y|^r]^{1/r}$. If the right side is infinite, there is nothing to prove; otherwise all variables below have the required moments. For every real $r \geq 1$, put $K_r = 2^{1/r} e^{-1/2} \sqrt{r}$. Independent signs $\varepsilon_i \in \{-1, 1\}$ satisfy $\|\sum_i \varepsilon_i a_i\|_r \leq K_r (\sum_i a_i^2)^{1/2}$. Indeed, their exponential moment is at most $\exp(t^2 \sum_i a_i^2 / 2)$. Use $x^r \leq (r/t)^r e^{-r} e^{tx}$, bound $e^{t|x|}$ by $e^{tx} + e^{-tx}$, and take $t = \sqrt{r / \sum_i a_i^2}$; the zero case is immediate. Symmetrizing an independent family Y_i with an independent copy, applying this sign estimate to the differences, and using the triangle inequality for the Euclidean square function gives $\|\sum_i (Y_i - \mathbb{E}Y_i)\|_r \leq 2K_r \|(\sum_i Y_i^2)^{1/2}\|_r$. Set $A = \sum_i X_i^2$, $M = \max_i |X_i|$, $u = \|A^{1/2}\|_p$, $b = \|M\|_p$ and $\sigma^2 = \sum_i \mathbb{E}X_i^2$. Apply the preceding estimate to $Y_i = X_i^2$ with $r = p/2 \geq 1$. Since $\sum_i X_i^4 \leq M^2 A$, Hölder's inequality gives

$$u^2 = \|A\|_{p/2} \leq \sigma^2 + 2K_{p/2} \|MA^{1/2}\|_{p/2} \leq \sigma^2 + 2K_{p/2} bu.$$

Consequently $u \leq \sigma + 2K_{p/2}b$. Applying the centered estimate once more, now to X_i at exponent p , yields

$$\left\| \sum_i X_i \right\|_p \leq 2K_p \sigma + 4K_p K_{p/2} b \leq 6\sqrt{p} \sigma + 4pb.$$

The last inequalities follow from $2^{1/p} \leq \sqrt{2}$ and $2^{2/p} \leq 2$: $2K_p \leq 2\sqrt{2/e}\sqrt{p} \leq 6\sqrt{p}$ and $4K_p K_{p/2} \leq 8p/e \leq 4p$. $\square$

Proposition B.2 (Concentration across finitely dependent columns). *Let $p \in [1, \infty)$, $r \in \mathbb{N}$ and $s \in [1/p, 1]$. Let $\{X_{k,j} : k \in \mathbb{Z}, j \in \mathbb{Z}\}$ be a collection of nonnegative random variables satisfying, for every $k, j \in \mathbb{Z}$, the moment bound $\mathbb{E}[X_{k,j}^p] \leq 1$, and r -dependence across columns: for every two subsets $I, J \subseteq \mathbb{Z}$ with $\text{dist}(I, J) \geq r$,*

$$(B.1) \quad \{X_{k,j} : k \in \mathbb{Z}, j \in I\} \quad \text{and} \quad \{X_{k,j} : k \in \mathbb{Z}, j \in J\} \quad \text{are independent}.$$

There exists a universal constant $C > 0$ such that, for every $m_0 \in \mathbb{Z}$, $m \in \mathbb{N}_0$ and $\theta \in (0, 1]$,

$$\mathbb{P} \left[\frac{1}{m+1} \sum_{k=m_0}^{m_0+m} \mathbb{1} \left\{ \sum_{j \in \mathbb{Z}} 3^{-s|k-j|} X_{k,j} > 6s^{-1} C^{1/p} \theta^{-1/p} \right\} > \theta \right] \leq \exp \left(-\frac{sp\theta}{16r} (m+1) \right).$$

Proof. This is [Armstrong et al. \[2026, Proposition D.1, version 2\]](#). $\square$

Proposition B.3 (Concentration across scales). *There is a universal constant $C < \infty$ with the following property. Let $\delta_0 \in (0, 1]$, and let $(X_i)_{i \in \mathbb{Z}}$ be independent nonnegative random variables*

satisfying $X_i = \mathcal{O}_{\Gamma_2}(\delta_0)$ for every $i \in \mathbb{Z}$. Then, for every $m_0 \in \mathbb{Z}$, $M \in \mathbb{N}_0$, $\theta \in (0, 1]$, and $s \in (0, 1]$ such that $\delta_0 \leq C^{-1}(s\theta)^{1/2}$,

$$(B.2) \quad \mathbb{P}\left[\frac{1}{M+1} \sum_{k=m_0}^{m_0+M} \mathbb{1}\left\{\sup_{j \geq 0} 3^{-sj} \exp\left(\sum_{i=0}^{\infty} 3^{-i} X_{k+i+j}\right) > 3\right\} \geq \theta\right] \leq \exp\left(-\frac{s\theta}{C\delta_0^2}(M+1)\right).$$

Proposition B.4 (Concentration across scales for random fields). *Let $\delta_0 \in (0, 1]$, and let $(X_i)_{i \in \mathbb{Z}}$ be independent real-valued random fields. Assume that X_i is $3^i \mathbb{Z}^d$ -stationary and $\|X_i\|_{L^\infty(\square_i)} \leq \mathcal{O}_{\Gamma_2}(\delta_0)$ for every $i \in \mathbb{Z}$. There exists $C(d) < \infty$ such that, for every $m_0 \in \mathbb{Z}$, $M \in \mathbb{N}_0$, $h \in \mathbb{N}$, and $s, \theta \in (0, 1]$ satisfying $\delta_0 \leq C^{-1}s(\theta \wedge h^{-1})^{1/2}$,*

$$(B.3) \quad \mathbb{P}\left[\frac{1}{M+1} \sum_{k=m_0}^{m_0+M} \mathbb{1}\left\{\sup_{j \geq 0} 3^{-sj} \left\|\prod_{i=k-j}^{k+j} \exp(|X_i|)\right\|_{L^\infty(\square_{k+h+j})} > 3\right\} \geq \theta\right] \leq \exp\left(-\frac{s^2\theta}{C\delta_0^2}(M+1)\right).$$

Propositions B.3 and B.4 are deduced from the next lemma.

Lemma B.5 (Selection of rare intervals). *There are universal constants $c > 0$ and $C < \infty$ such that the following holds. Let $(\xi_i)_{i \in \mathbb{Z}}$ be independent random elements and let $\lambda > 0$. For $k \in \mathbb{Z}$ and $j \in \mathbb{N}_0$, let $E_{k,j}$ be an event that depends only on the variables ξ_i with indices $i \in [k-j, k+j] \cap \mathbb{Z}$, and assume that $\mathbb{P}[E_{k,j}] \leq e^{-\lambda(j+1)}$. If $0 < \theta \leq 1$ and $\lambda\theta \geq C$, then every integer interval I of cardinality $N \geq 1$ satisfies*

$$\mathbb{P}[\#\{k \in I : E_{k,j} \text{ occurs for some } j\} \geq \theta N] \leq e^{-c\lambda\theta N}.$$

Proof. A union bound over $k \in I$ and $j \geq N$ shows that the probability that $E_{k,j}$ occurs for some $k \in I$ and $j \geq N$ is at most $e^{-c\lambda N}$. On the complementary event, for each $k \in I$ such that $E_{k,j}$ occurs for some j , choose one such j , necessarily $j < N$, and take the corresponding interval of indices, $[k, k+j]$ or $[k-j, k+j]$. Retain these intervals greedily in decreasing order of length, discarding those at distance less than two from a retained interval. Every discarded interval has length at most that of the interval which discards it, and lies in its enlargement by that length on each side. Hence the retained intervals have total cardinality at least $\theta N/3$.

The retained intervals are separated by at least one missing integer and lie in an interval of cardinality at most $3N$. Their union determines the intervals and their starting indices, so there are at most 2^{3N} configurations. For each configuration, independence and the fact that $j+1$ is at least half the cardinality of the corresponding interval give probability at most $e^{-\lambda\theta N/6}$. Thus the required probability is at most $e^{-c\lambda N} + e^{CN-\lambda\theta N/6}$. Taking the constant in the assumption $\lambda\theta \geq C$ large enough absorbs the counting factor e^{CN} and the sum of the two terms. $\square$

Proof of Proposition B.3. Call an index k bad if the event in the indicator in (B.2) occurs. Every bad index k admits some $j \geq 0$ such that

$$(B.4) \quad X_{k+j} > \frac{\log 3}{3}(1 + sj).$$

Indeed, if $X_{k+j} \leq \frac{\log 3}{3}(1 + sj)$ for every $j \geq 0$, then, for every $j \geq 0$,

$$\sum_{i \geq 0} 3^{-i} X_{k+j+i} \leq \frac{\log 3}{3} \left(\frac{3}{2}(1 + sj) + \frac{3}{4}s \right) \leq \frac{3}{4}(1 + sj) \log 3,$$

which excludes the event in (B.2). The event (B.4) depends only on the index $k+j$ and has probability at most $2e^{-c(1+sj)^2/\delta_0^2} \leq e^{-c_1 s(j+1)/\delta_0^2}$, since $(1+sj)^2 \geq s(j+1)$. We apply Lemma B.5 to these events on the $M+1$ indices $m_0, \dots, m_0+M$, with $\lambda = c_1 s/\delta_0^2$, which gives

$$\mathbb{P}[\#\{\text{bad indices}\} \geq \theta(M+1)] \leq \exp\left(-\frac{s\theta}{C\delta_0^2}(M+1)\right).$$

This is (B.2). $\square$

Proof of Proposition B.4. Let $E_{k,j}$ be the event that the expression at index j inside the supremum in (B.3) exceeds three. It depends only on $(X_i)_{k-j \leq i \leq k+j}$. Cover $\square_{k+h+j}$ by its $3^{d(h+2j)}$ triadic cubes of scale $k-j$. On each of these cubes, the logarithm of the product is bounded above by the sum of the suprema of $|X_i|$ on its scale- i parent cubes. These $2j+1$ random variables are independent, have means at most $C\delta_0$, and, after centering, have sub-Gaussian parameters at most $C\delta_0$. By (A.5), a union bound, and $1 + sj \geq s(j+1)$,

$$\mathbb{P}[E_{k,j}] \leq \exp\left(C(h+j+1) - \frac{c(1+sj)^2}{\delta_0^2(j+1)}\right) \leq \exp\left(-\frac{c_1 s^2}{\delta_0^2}(j+1)\right).$$

Here $\delta_0 \leq C^{-1}s$ absorbs the means and the term proportional to $j+1$, while $\delta_0^2 h \leq C^{-2}s^2$ absorbs the factor from the union bound over the covering cubes at $j=0$, and hence at every j . Lemma B.5, with $\lambda = c_1 s^2 / \delta_0^2$ and $I = \{m_0, \dots, m_0 + M\}$, now gives (B.3). $\square$

APPENDIX C. BESOV NORM COMPARISONS

In this appendix we prove two statements about Besov norms recorded in Section 2.1: the duality estimate (2.8), and the comparison (2.5) of Besov and fractional Sobolev seminorms. We also record the fractional Sobolev–Poincaré inequality used in Section 2.

The fractional Sobolev–Poincaré inequality states that, for each $s \in (0, 1]$ and $p \in [1, d/s]$, there exists $C(s, p, d) < \infty$ such that, for every $m \in \mathbb{Z}$ and $f \in W^{s,p}(\square_m)$,

$$\|f - (f)_{\square_m}\|_{\underline{L}^{pd/(d-sp)}(\square_m)} \leq C 3^{sm} [f]_{\underline{W}^{s,p}(\square_m)}.$$

For $0 < s < 1$, put $Q = \square_m$ and $L = 3^m$. Jensen’s inequality gives

$$\|f - (f)_Q\|_{\underline{L}^p(Q)}^p \leq \frac{1}{|Q|^2} \int_Q \int_Q |f(x) - f(y)|^p dx dy \leq \frac{d^{(d+sp)/2}}{s} L^{sp} [f]_{\underline{W}^{s,p}(Q)}^p.$$

For the second inequality, use $|x - y|_\infty \leq L$, hence $|x - y|_2 \leq \sqrt{d}L$, together with $|Q| = L^d$. This keeps the single volume normalization and the factor s in the fractional seminorm; taking the p th root introduces the factor $s^{-1/p}$ and a dimensional factor. On the unit cube, the extension theorem and Sobolev embedding of Di Nezza et al. [2012, Theorems 5.4 and 6.7] apply also at $p = 1$, provided $sp < d$. Apply them to $f - (f)_Q$ and use the preceding inequality to control the L^p term. Rescaling $f(L \cdot)$ multiplies its normalized fractional seminorm by L^s and leaves its normalized $L^{pd/(d-sp)}$ norm unchanged, proving the displayed estimate. For $s = 1$, the seminorm is the classical gradient seminorm, and the classical Sobolev–Poincaré inequality on the cube, including $p = 1$, applies directly.

Lemma C.1 (Besov duality). *Let $m \in \mathbb{Z}$, $s \in (0, 1]$, and $q, r \in [1, \infty]$, with $q^{-1} + (q')^{-1} = r^{-1} + (r')^{-1} = 1$, and assume that either $s < 1$ or $(s, q) = (1, 1)$. Then (2.8) holds for all functions f, g on $\square_m$ such that $fg \in L^1(\square_m)$ and the right side of (2.8) is finite.*

Proof. Let E_n denote averaging on the partition into scale- n triadic cubes. At finite depth $n < m$,

$$(C.1) \quad E_n g = (g)_{\square_m} + \sum_{k=n+1}^m (E_{k-1} g - E_k g).$$

On a scale- k parent Q and its child Q' , one has $|(g)_{Q'} - (g)_Q| \leq 3^d \|g - (g)_Q\|_{\underline{L}^1(Q)}$. Pairing (C.1) with f and applying Hölder’s inequality first to the cube averages and then to the sum over scales gives

$$\left| \int_{\square_m} f E_n g - (f)_{\square_m} (g)_{\square_m} \right| \leq C s^{-1} \|f\|_{\underline{B}_{r,q}^{-s}(\square_m)} [g]_{B_{1,q',r'}^s(\square_m)}.$$

To see this, insert the factor $3^{s(k-1)}3^{-s(k-1)}$ before applying Hölder's inequality to the sum over scales. Replacing the partition averages by the overlapping averages in (2.4) introduces a factor depending only on d , and the normalizations of the two sums over scales give the factor s^{-1} , also at the endpoint exponents 1 and ∞ . The term of the negative norm at the top scale gives

$$|(f)_{\square_m}(g)_{\square_m}| \leq s^{-1} \|f\|_{\dot{B}_{r,q}^{-s}(\square_m)} 3^{-sm} |(g)_{\square_m}|.$$

If g is bounded, then $E_n g \rightarrow g$ almost everywhere and $|f E_n g| \leq \|g\|_{\infty} |f|$; dominated convergence proves (2.8) in this case.

For general g , put $a = (g)_{\square_m}$ and, for $M > 0$, define

$$T_M(t) = (-M) \vee (t \wedge M), \quad g_M = a + T_M(g - a) - (T_M(g - a))_{\square_m}.$$

The mean of g_M is a . Since T_M is one-Lipschitz, every cube Q appearing in (2.4) satisfies

$$\|g_M - (g_M)_Q\|_{\underline{L}^1(Q)} \leq 2 \|T_M(g - a) - T_M((g)_Q - a)\|_{\underline{L}^1(Q)} \leq 2 \|g - (g)_Q\|_{\underline{L}^1(Q)}.$$

Thus $\|g_M\|_{B_{1,q',r'}^s} \leq 2 \|g\|_{B_{1,q',r'}^s}$. Moreover $g_M \rightarrow g$ almost everywhere and $|f g_M| \leq |f g| + (2|a| + \int_{\square_m} |g - a|) |f|$. Applying the bounded case to g_M and then dominated convergence proves (2.8). $\square$

Lemma C.2. *There exists a constant $C(d) < \infty$ such that, for every $s \in (0, 1)$, $p \in [1, \infty)$, $m \in \mathbb{Z}$ and $u \in L^p(\square_m)$,*

$$(C.2) \quad C^{-1}[u]_{\underline{W}^{s,p}(\square_m)} \leq [u]_{\underline{B}_{p,p,p}^s(\square_m)} \leq C[u]_{\underline{W}^{s,p}(\square_m)}.$$

Proof. Let $\mathcal{Q}_n$ be the set of cubes of side 3^n with centers in $3^{n-1}\mathbb{Z}^d$ which are contained in $\square_m$. Their number is between $|\square_m|3^{-nd}$ and $3^d|\square_m|3^{-nd}$, and their overlap multiplicity is bounded by 4^d . For every such cube, Jensen's inequality and the triangle inequality give

$$\int_Q |u - (u)_Q|^p \leq \int_Q \int_Q |u(x) - u(y)|^p dx dy \leq 2^p \int_Q |u - (u)_Q|^p.$$

Average over $\mathcal{Q}_n$, multiply by $s3^{-nsp}$, and sum over $n \leq m$. Tonelli's theorem shows that $[u]_{\underline{B}_{p,p,p}^s(\square_m)}^p$ is comparable, within factors $C(d)^p$, to

$$\frac{s}{|\square_m|} \int_{\square_m} \int_{\square_m} |u(x) - u(y)|^p \sum_{n \leq m} 3^{-n(d+sp)} \sum_{Q \in \mathcal{Q}_n} \mathbb{1}_Q(x) \mathbb{1}_Q(y) dx dy.$$

For distinct $x, y \in \square_m$, write $r = |x - y|_{\infty}$. The points x and y lie in a common cube of scale n only if $3^n \geq r$, so bounded overlap and a geometric sum bound the displayed kernel above by $C(d)r^{-d-sp}$. Conversely, if $r \leq 3^{m-1}$, choose $n \leq m$ with $3r \leq 3^n < 9r$. The mesh spacing 3^{n-1} ensures that x, y lie in a common $Q \in \mathcal{Q}_n$. For $r > 3^{m-1}$, use the top cube $\square_m$ instead. The kernel is therefore bounded below by $9^{-d-sp}r^{-d-sp}$. Since $r \leq |x - y|_2 \leq \sqrt{d}r$, these kernel bounds also apply with Euclidean distance, with a dimensional change in the constants after taking the p th root: $d^{(d+sp)/(2p)} \leq d^{(d+1)/2}$ for $p \geq 1$ and $s \leq 1$. This proves (C.2), with a constant independent of s and p . $\square$

REFERENCES

- S. Armstrong and T. Kuusi. Elliptic homogenization from qualitative to quantitative, 2024. URL <https://arxiv.org/abs/2210.06488v2>. arXiv:2210.06488.
- S. Armstrong and T. Kuusi. Renormalization group and elliptic homogenization in high contrast, 2025. URL <https://arxiv.org/abs/2405.10732v3>. arXiv:2405.10732, version 3.
- S. Armstrong, T. Kuusi, and J.-C. Mourrat. *Quantitative stochastic homogenization and large-scale regularity*, volume 352 of *Grundlehren der mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences]*. Springer, Cham, 2019. ISBN 978-3-030-15544-5; 978-3-030-15545-2; 978-3-030-15547-6. doi: 10.1007/978-3-030-15545-2. URL <https://doi.org/10.1007/978-3-030-15545-2>. Lemma 10.5 (p. 384) is cited from the authors' PDF dated 7 May 2019 (<https://perso.ens-lyon.fr/jean-christophe.mourrat/AKMbook.pdf>); Appendix A (pp. 416–421) is cited from arXiv:1705.05300v2 (<https://arxiv.org/abs/1705.05300v2>).

- S. Armstrong, A. Bou-Rabee, and T. Kuusi. Superdiffusive central limit theorem for a Brownian particle in a critically-correlated incompressible random drift, 2024. URL <https://arxiv.org/abs/2404.01115v2>. arXiv:2404.01115, version 2.
- Scott Armstrong, Ahmed Bou-Rabee, and Tuomo Kuusi. Superdiffusion and anomalous regularization in self-similar random incompressible flows. *arXiv preprint arXiv:2601.22142*, 2026. doi: 10.48550/arXiv.2601.22142. URL <https://arxiv.org/abs/2601.22142v2>. Version 2, 6 September 2026.
- Richard F. Bass. *Diffusions and Elliptic Operators*. Probability and Its Applications. Springer, New York, 1998. doi: 10.1007/b97611.
- N. Berestycki. Diffusion in planar Liouville quantum gravity. *Ann. Inst. Henri Poincaré Probab. Stat.*, 51(3):947–964, 2015. ISSN 0246-0203,1778-7017. doi: 10.1214/14-AIHP605. URL <https://doi.org/10.1214/14-AIHP605>.
- N. Berestycki. An elementary approach to Gaussian multiplicative chaos. *Electron. Commun. Probab.*, 22:Paper No. 27, 12, 2017. ISSN 1083-589X. doi: 10.1214/17-ECP58. URL <https://doi.org/10.1214/17-ECP58>.
- M. Biskup, J. Ding, and S. Goswami. Return probability and recurrence for the random walk driven by two-dimensional Gaussian free field. *Comm. Math. Phys.*, 373(1):45–106, 2020. ISSN 0010-3616,1432-0916. doi: 10.1007/s00220-019-03589-z. URL <https://doi.org/10.1007/s00220-019-03589-z>.
- Marek Biskup. Extrema of the two-dimensional Discrete Gaussian Free Field. In Martin T. Barlow and Gordon Slade, editors, *Random Graphs, Phase Transitions, and the Gaussian Free Field*, volume 304 of *Springer Proceedings in Mathematics & Statistics*, pages 163–407. Springer, Cham, 2020. doi: 10.1007/978-3-030-32011-9_3. URL <https://arxiv.org/abs/1712.09972>.
- J.-P. Bouchaud and A. Georges. Anomalous diffusion in disordered media: statistical mechanisms, models and physical applications. *Phys. Rep.*, 195(4-5):127–293, 1990. ISSN 0370-1573,1873-6270. doi: 10.1016/0370-1573(90)90099-N. URL [https://doi.org/10.1016/0370-1573\(90\)90099-N](https://doi.org/10.1016/0370-1573(90)90099-N).
- J.P. Bouchaud, A. Comtet, A. Georges, and P. Le Doussal. Anomalous diffusion in random media of any dimensionality. *J. Phys. France*, 48(9):1445–1450, 1987.
- S. Boucheron, G. Lugosi, and P. Massart. *Concentration inequalities*. Oxford University Press, Oxford, 2013. ISBN 978-0-19-953525-5. doi: 10.1093/acprof:oso/9780199535255.001.0001. URL <https://doi.org/10.1093/acprof:oso/9780199535255.001.0001>. A nonasymptotic theory of independence, With a foreword by Michel Ledoux.
- Stéphane Boucheron, Olivier Bousquet, Gábor Lugosi, and Pascal Massart. Moment inequalities for functions of independent random variables. *The Annals of Probability*, 33(2):514–560, 2005. doi: 10.1214/009117904000000856.
- Horacio Castillo and Pierre Le Doussal. Freezing of dynamical exponents in low dimensional random media. *Phys. Rev. Lett.*, 86(21):4859–4862, 2001. doi: 10.1103/PhysRevLett.86.4859. URL <https://arxiv.org/abs/cond-mat/0006373>.
- Charles Devlin, VI. Almost sure convergence of Liouville first passage percolation, 2024. URL <https://arxiv.org/abs/2309.08001>. arXiv:2309.08001, version 2.
- Eleonora Di Nezza, Giampiero Palatucci, and Enrico Valdinoci. Hitchhiker’s guide to the fractional Sobolev spaces. *Bull. Sci. Math.*, 136(5):521–573, 2012. doi: 10.1016/j.bulsci.2011.12.004.
- Jian Ding and Ewain Gwynne. Up-to-constants comparison of Liouville first passage percolation and Liouville quantum gravity. *Sci. China Math.*, 66(5):1053–1072, 2023a. doi: 10.1007/s11425-021-1983-0. URL <https://doi.org/10.1007/s11425-021-1983-0>.
- Jian Ding and Ewain Gwynne. Uniqueness of the critical and supercritical Liouville quantum gravity metrics. *Proc. Lond. Math. Soc. (3)*, 126(1):216–333, 2023b. doi: 10.1112/plms.12492. URL <https://doi.org/10.1112/plms.12492>.
- Jian Ding and Jiamin Wang. Tightness for random walks driven by the planar Gaussian free field at high temperature. *Probab. Theory Related Fields*, 195(3-4):1929–2015, 2026. doi: 10.1007/s00440-026-01467-z. URL <https://doi.org/10.1007/s00440-026-01467-z>.
- B. Duplantier, R. Rhodes, S. Sheffield, and V. Vargas. Log-correlated Gaussian fields: an overview. In *Geometry, analysis and probability*, volume 310 of *Progr. Math.*, pages 191–216. Birkhäuser/Springer, Cham, 2017. doi: 10.1007/978-3-319-49638-2_9. URL https://doi.org/10.1007/978-3-319-49638-2_9.
- A.M. Dykhne. Conductivity of a two-dimensional two-phase system. *Sov. Phys. JETP*, 32(1):63–65, 1971.
- D. S. Fisher. Random walks in random environments. *Phys. Rev. A*, 30:960–964, Aug 1984. doi: 10.1103/PhysRevA.30.960. URL <https://link.aps.org/doi/10.1103/PhysRevA.30.960>.
- D. S. Fisher, D. Friedan, Z. Qiu, S. J. Shenker, and S. H. Shenker. Random walks in two-dimensional random environments with constrained drift forces. *Phys. Rev. A*, 31(6):3841–3845, 1985. doi: 10.1103/PhysRevA.31.3841.
- Patrick J. Fitzsimmons. Superposition operators on Dirichlet spaces. *Tohoku Mathematical Journal. Second Series*, 56(3):327–340, 2004. doi: 10.2748/tmj/1113246670. URL <https://doi.org/10.2748/tmj/1113246670>.
- Masatoshi Fukushima, Yoichi Oshima, and Masayuki Takeda. *Dirichlet forms and symmetric Markov processes*, volume 19 of *De Gruyter Studies in Mathematics*. Walter de Gruyter & Co., Berlin, second revised and extended edition, 2011.

- C. Garban, R. Rhodes, and V. Vargas. Liouville Brownian motion. *Ann. Probab.*, 44(4):3076–3110, 2016. ISSN 0091-1798,2168-894X. doi: 10.1214/15-AOP1042. URL <https://doi.org/10.1214/15-AOP1042>.
- E. Gwynne and J. Miller. Existence and uniqueness of the Liouville quantum gravity metric for $\gamma \in (0, 2)$. *Invent. Math.*, 223(1):213–333, 2021. ISSN 0020-9910,1432-1297. doi: 10.1007/s00222-020-00991-6. URL <https://doi.org/10.1007/s00222-020-00991-6>.
- István Gyöngy and Nicolai Krylov. Existence of strong solutions for Itô's stochastic equations via approximations. *Probability Theory and Related Fields*, 105:143–158, 1996. doi: 10.1007/BF01203833. URL <https://doi.org/10.1007/BF01203833>.
- J.-P. Kahane. Sur le chaos multiplicatif. *Ann. Sci. Math. Québec*, 9(2):105–150, 1985. ISSN 0707-9109.
- Olav Kallenberg. *Foundations of modern probability*. Probability and its Applications (New York). Springer-Verlag, New York, second edition, 2002. ISBN 0-387-95313-2. doi: 10.1007/978-1-4757-4015-8. URL <https://doi.org/10.1007/978-1-4757-4015-8>.
- V. E. Kravtsov, I. V. Lerner, and V. I. Yudson. Random walks in media with constrained disorder. *J. Phys. A: Math. Gen.*, 18(12):L703–L707, 1985. doi: 10.1088/0305-4470/18/12/003.
- Kazuhiro Kuwae and Takashi Shioya. Convergence of spectral structures: a functional analytic theory and its applications to spectral geometry. *Communications in Analysis and Geometry*, 11(4):599–673, 2003. doi: 10.4310/CAG.2003.v11.n4.a1. URL <https://doi.org/10.4310/CAG.2003.v11.n4.a1>.
- Norman G. Meyers. An L^p -estimate for the gradient of solutions of second order elliptic divergence equations. *Annali della Scuola Normale Superiore di Pisa. Classe di Scienze. Serie III*, 17(3):189–206, 1963. URL https://www.numdam.org/item/ASNSP_1963_3_17_3_189_0/.
- R. Rhodes and V. Vargas. Gaussian multiplicative chaos and applications: a review. *Probab. Surv.*, 11:315–392, 2014. ISSN 1549-5787. doi: 10.1214/13-PS218. URL <https://doi.org/10.1214/13-PS218>.

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